

# UZA – Universelle Zusammenhangsarchitektur

Gesamtspezifikation 0.9.1 Master - Formalrevision K4 - T1-T13 und artikulierter Interaktionskern  
Isabelle2025-2-session-/kernelgeprüft

Konsolidierte Referenz für Artikulationsarchitektur, Semantik, Grammatik,  
Engine, Dynamik, analytische Skalierung, Interaktion und formale Theorie

Arbeitsstand: 1. September 2026 - T1-T13 kausal geschlossen - Interaktionskern bis validierter  
Transmission geschlossen - 29-Theorien-Kette: Isabelle2025-2-Clean-Build, Exit-Code 0 - Audit  
37/37 - Golden Run: 23 Schritte, Snapshot 23

Fortschreibungsbasis: K4 übernimmt die K3-Geltungsrevision und den T1-T13-Unterbau  
unverändert. Neu geschlossen sind ApplyBoundary, DefineCoupling und InitiateTransmission samt  
kausalem Erfolgsweg, fünf konkreten Ablehnungspfaden, Ursprungserhalt und artikulierter  
Nichtseveranz. Diese Operationen sind keine T14-T16 und schließen nicht den vollständigen I1-  
I11-Komplex.

Statusprovenienz: Der K4-Hashstand wurde mit Isabelle2025-2 nach Reinigung aller vier UZA-  
Sessions verarbeitet. Build, Audit und Golden Run sind getrennte Nachweise. A1-A22 bleiben  
unveränderte Locale-Annahmen. Grundaussagen und Artikulationsausdrücke bleiben hart  
getrennt; Mitgeltung ist keine Zustandsmitgliedschaft und Abgleich keine ausführbare  
Kernoperation.

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## Dokumentidentität und normativer Status

Diese Masterfassung führt die belastbaren Ergebnisse der bisherigen UZA-Entwicklung in einem Dokument zusammen. Sie ersetzt die verstreute Lektüre der Versionen 0.1 bis 0.9.1, bewahrt deren Entwicklungslinie und trennt drei Reifearten konsequent:

- **Architektonisch konsolidiert:** Begriffe, Abhängigkeiten und Operationsklassen sind stabil genug, um weitere Arbeit darauf aufzubauen.
- **Spezifiziert:** Daten, Regeln, Zustandsübergänge und Validierungsbedingungen sind so beschrieben, dass eine Referenzimplementierung erstellt werden kann.
- **Formal bewiesen oder mechanisiert:** Aussagen sind in einer Beweisumgebung parsergeprüft und durch Beweise oder Modellkonstruktionen abgesichert.

Für die UZA gilt aktuell: T1-T13 bleiben als kausale Operationskette geschlossen. K4 erweitert diesen geprüften Stand um den artikulierte Interaktionskern ApplyBoundary, DefineCoupling und InitiateTransmission. Die Operationen verändern ausschließlich typisierte Artikulationsgestalten. Immer, reine Potenzialität, Lebendigkeit, Wandlungs- und Wirkungsfähigkeit sowie der lebendige Zusammenhang gelten in jedem Vollzug mit, ohne Objekt, Zustand oder Übergangsglied zu werden.

Verbindliche Kurzform: Architecture & Engine Specification 0.9.1; Formalrevision K4 mit kausal geschlossenem T1-T13-Unterbau und artikulierte Interaktionskern. Der exakte K4-Hashstand wurde mit Isabelle2025-2 in vier abhängigen Sessions mit 29 Theorien sauber neu gebaut (Exit-Code 0). Statischer Quellaudit: 37/37 Gates. Golden Run: 23 Schritte bis Snapshot 23. A1-A22 bleiben unveränderte Locale-Annahmen. Ein reales Wir, Modell-Expansion, relative Konsistenz, Konservativität, Produktivengine-Äquivalenz und empirische Universalität bleiben offen.

## Nachweisprovenienz der Isabelle-Statusangaben

Die früher als kernelgeprüft ausgewiesenen Sessionstände bleiben als historische Nachweisprovenienz erhalten. Der gepinnte T6-Gesamtstand wurde aus einem leeren Isabelle-Benutzerprofil als normale Isabelle2025-2-Session mit elf Theorien gebaut. T7 erweitert diesen Stand um zwei Theorien. Ihre vollständige Importkette und anschließend alle 13 ROOT-Theorien wurden jeweils aus den offiziellen Isabelle2025-2-Pure-/HOL-Heaps mit dem Isabelle-Theorielader neu geladen und bei deaktivierter Beweisüberspringung kernelgeprüft. Die unveränderten Konsolenausgaben stehen in T7\_CLEAN\_KERNEL\_CHECK.log und T7\_ROOT\_ORDER\_KERNEL\_CHECK.log; Quellen, ROOT, Exit-Codes, Quellhashes, Source Audit und Verifikationsbericht werden als einzeln auffindbare Artefakte geführt.

Maßgeblich ist nun der normale Isabelle2025-2-Clean-Build der vierstufigen 29-Theorien-K4-Kette mit Exit-Code 0. Die älteren K1-, K2- und K3-Läufe bleiben ergänzende, hashgebundene Provenienz. Keine dieser Prüfungen belegt empirische Universalität, Wahrheit der Locale-Annahmen, allgemeine Konservativität, ein reales Wir oder Gleichheit der Artikulation mit Wirklichkeit.

K2.1-Änderungsgrenze. Diese neue Dokumentversion berichtigt die Geltungsarchitektur, ohne frühere Veröffentlichungen rückwirkend umzuschreiben: Der K2-Beweisstand bleibt unverändert referenziert; K2.1 macht sichtbar, dass die formalen Zustände, Zeiten, Effekte und Speicherstrukturen Artikulationen innerhalb der Wirklichkeit sind und die Wirklichkeit nicht erschöpfen.

## Leseschlüssel

Die Wörter muss, darf nur und unzulässig kennzeichnen normative Anforderungen. Soll bezeichnet eine starke Empfehlung mit begründungspflichtiger Abweichung. Kann bezeichnet eine zulässige Option. Mathematische Formeln präzisieren eine Artikulation; sie ersetzen erst dann einen Beweis, wenn dies ausdrücklich als Theorem mit Beweisstatus ausgewiesen ist. Keine Formel, kein Diagramm und kein Beweisstatus wird mit der von ihm artikulierten Wirklichkeit identifiziert.

Die Dokumentteile haben folgende Funktion:

- Teile I–III definieren den stabilen Kern  $K$ .
- Teil IV beschreibt die operationale Grammar Engine.
- Teil V integriert die Zusammenhangsdynamik.
- Teil VI spezifiziert Skalierung und Interaktion mehrerer Zusammenhänge.
- Teil VII dokumentiert die Demonstratorläufe.
- Teil VIII legt das Referenzimplementierungsdesign fest.
- Teil IX ordnet die mathematische Theorie und ihren Beweisstatus ein.
- Die Anhänge enthalten vollständige Register, Pseudocode, die HOL-Signatur und offene Nachweise.

## Teil I – Ursprung, Entwicklung und Gesamtarchitektur

### 1. Ursprung und Leitfrage

Die UZA entstand aus einer wiederkehrenden Struktur in Arbeiten zu Ontologie, Perspektivenanalyse, lernenden Systemen, Organisationsgestaltung und regelbasierten Analysemaschinen. Trotz verschiedener Gegenstände erschien dieselbe Aufgabe:

**Wie entsteht Orientierung in einem komplexen Zusammenhang, ohne dass Wesentliches verloren geht?**

Die Antwort wurde zunächst als „Zusammenhangs-Engine“ formuliert. Gemeint war eine gemeinsame Architektur, deren Grundoperationen in verschiedenen Domänen gleich bleiben, während sich Beobachtungen, Perspektiven, Evidenzen, Relevanzmodelle und Gestaltungsoptionen ändern.

Die UZA ist deshalb zugleich:

- eine Artikulationsontologie der für Zusammenhangsanalyse benötigten Objekttypen,
- eine Semantik ihrer Bedeutung und Gültigkeit,
- eine Grammatik zulässiger Transformationen,
- eine ausführbare Engine-Spezifikation,
- eine Architektur für dynamische, verschachtelte und interagierende Zusammenhänge,
- und ein kontrolliert erweiterbares formales Theorieprojekt.

Ihre formale Grundeinheit ist der artikulierte UZA-Zusammenhang. Beobachtungen werden als Erscheinungen registriert; ihre Bedeutung wird über Beziehungen, Wirkungsaussagen, Resonanzen, Perspektiven, Kontexte und Relevanzordnungen rekonstruiert. Orientierung entsteht als begründeter, artikulierter Möglichkeitsraum. Gestaltung wird ausgewählt, ausgeführt, beobachtet und über dokumentierte Rückwirkung in einen Folgezustand der Artikulation integriert.

### 2. Entwicklungslinie und Konsolidierungsentscheidungen

| Version | Hauptgegenstand                                                                                                  | Konsolidierter Status |
|---------|------------------------------------------------------------------------------------------------------------------|-----------------------|
| Konzept | Ereignis, Beziehung, Resonanz und Gestaltung;<br>Periodensystem der Erkenntnis                                   | Ursprungsidee         |
| UZA 0.1 | Operative Zustandsmaschine, Elementtypen,<br>Wirkungsvektor, Relevanzfunktion, erster KI-<br>Demonstrator        | in 0.9 integriert     |
| UZA 0.2 | Referenzarchitektur, Potentialraum, Emergenz,<br>Wandeltypen, zwölf Grundaxiome, Invarianten, Meta-<br>Evolution | in 0.9 integriert     |
| UZA 0.3 | Semantischer Kern, Signaturen, C1–C10, Unsicherheit,<br>Perspektivsemantik, Describe, G0–G4                      | Spezifikation 0.9-RC  |



| Version                 | Hauptgegenstand                                                                                                                                                                                                                                                       | Konsolidierter Status                                                                                                    |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| UZA 0.4                 | Transformationsgrammatik T1–T13 und Regelkalkül                                                                                                                                                                                                                       | in 0.3/0.5 integriert                                                                                                    |
| UZA 0.5                 | Grammar Engine mit Event-Sourcing, Copy-on-Write, Ledger, Validierung und Meta-Evolution                                                                                                                                                                              | Spezifikation 0.9-RC                                                                                                     |
| UZA 0.6                 | Datenmodell, API, Regel-Executor und vollständiger demonstrativer Commit-Durchlauf                                                                                                                                                                                    | Referenzimplementierungsdesign ; kein Produktivcode                                                                      |
| UZA 0.7                 | Vierdimensionale Dynamik, drei Zeitebenen, Flow-Analyse und zehn Muster                                                                                                                                                                                               | spezifikatorisch integriert; formal offen                                                                                |
| UZA 0.8                 | Flow Analyzer, Dynamics Detector, Complexity Monitor, Perspective Tracker, Feedback Loop, EvolutionMemory                                                                                                                                                             | in 0.9 integriert                                                                                                        |
| UZA 0.9                 | ScaleProfile, ContainmentGraph, SUBZ/PROJ, LossProfile, I1–I11, Transmission und Grenztransformationen                                                                                                                                                                | Scale & Interaction 0.9-RC                                                                                               |
| UZA 0.9.1               | EMG/FDB-Bereinigung, Minimality-Korrektur, V10-Begründung und Migration, Machtverschiebung, Scheiter- und Patch-Abbruchregeln                                                                                                                                         | Architecture & Engine 0.9.1-RC                                                                                           |
| Formal Theory 0.9.1     | Referenzsemantik für V10, gerichtete Resonanz, Relevanz, Describe, Emergenz und Step; A1–A22; minimale und reichhaltige Modellzeugen; kausal verbundene T1–T8-Semantik mit nicht-vakuosen Ausführungszeugen; K2.1-Geltungsgrenze MG1–MG10 außerhalb des Objektkalküls | K2: normaler Isabelle2025-2-Clean-Build der vollständigen 18-Theorien-Kette, Exit-Code 0; K2.1 ändert keine Beweisquelle |
| Conformance Harness 0.1 | deterministisches Replay, Golden Run A, Golden Run B und negative Gate-Tests                                                                                                                                                                                          | lokal ausgeführt und bestanden                                                                                           |

## 2.1 Aufgelöste Statuswidersprüche

**Referenzimplementierung.** „Vollständiger Demonstrator-Durchlauf“ bedeutet, dass Objekte, Operationen, Commits und Ergebnisse vollständig spezifiziert und nachvollziehbar durchgespielt wurden. Seit 0.9.1 existiert zusätzlich ein kleines ausführbares Conformance-Harness mit reproduzierbaren Golden Runs. Es ersetzt weder eine produktive Engine noch deren Performance-, Sicherheits- und Freigabeprüfung.

**Dynamik.** Die dynamische Schicht ist architektonisch und operational spezifiziert. Domänenspezifische Detektoren und eine HOL-Axiomatik der Dynamik sind weiterhin offen. Deshalb lautet ihr Status: in die Engine-Spezifikation integriert, mathematisch noch nicht geschlossen.

**HOL-Theorie.** Die Typ- und Funktionsarchitektur ist bereinigt; zentrale Referenzprädikate und A1–A22 liegen als konkrete Isabelle/HOL-Theorie vor. Der vollständige K2-Unterbau umfasst 18 Theorien einschließlich T1–T8 und wurde als normaler Isabelle2025-2-Clean-Build verarbeitet



(Exit-Code 0). T8 besitzt ein versioniertes Orientierungsprofil, eine geordnete duplikatfreie Relevanzbasis, explizite Qualitätsgates und eine range-erhaltende Semantik ohne implizite Mittelung. Der K2-Zeuge verbindet beide T8-Ausgänge kausal mit `chain_after_t7`.

**Wirkungsvektor.** Die UZA 0.1 verwendete ein polares Zehnerprofil aus Verbindung/Trennung, Stabilisierung/Destabilisierung, Öffnung/Begrenzung, Beschleunigung/Verlangsamung und Kooperation/Konflikt. Die konsolidierte Fassung verwendet zehn eigenständige Gestaltungsdimensionen. 0.9.1 begründet ihre funktionale Auswahl, definiert EffectProfile als normative Schnittstelle, spezifiziert die verlustbehaftete Migration und trennt Machtverschiebung als relationales Zusatzprofil. Das frühere Profil bleibt Legacy-Profil und ist kein zweiter normativer Kern.

### 3. Viergliedriger Kern mit orthogonalen Erweiterungsbereichen

Der stabile Kern lautet:

$$K = \{ O, S, G, E \}$$

mit:

- **O – Ontologie:** Welche Objekte und Strukturen können vorkommen?
- **S – Semantik:** Was bedeuten sie und unter welchen Bedingungen sind sie gültig?
- **G – Grammatik:** Welche Transformationen sind zulässig?
- **E – Engine:** Wie werden Transformationen reproduzierbar ausgeführt, validiert und protokolliert?

Der Ausdruck „viergliedriger Kern“ bezeichnet exakt diese vier voneinander abhängigen Architekturbereiche. Ihre Abhängigkeitsordnung ist eine Ordnung der formalen Artikulation und keine zeitliche oder ontologische Erzeugungsfolge. Auf der äußeren Ebene bildet der Kern zusammen mit vier orthogonalen Erweiterungsbereichen eine fünfteilige Gesamtarchitektur; auch diese Glieder werden nicht als Rang- oder Wirklichkeitshierarchie gelesen.

Die Abhängigkeitsordnung lautet:

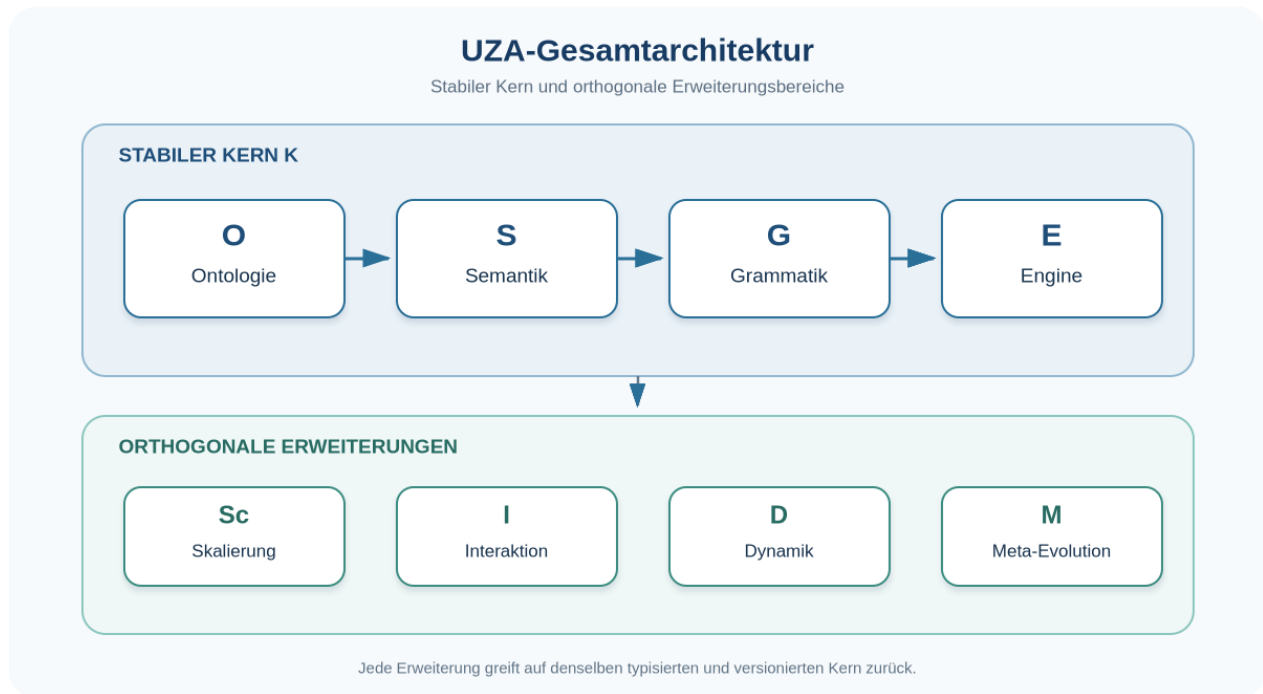
$$O \rightarrow S \rightarrow G \rightarrow E$$

Eine Engineoperation darf keinen Objekttyp erzeugen, den Ontologie und Semantik nicht tragen. Eine Grammatikregel darf keine semantisch unbestimmte Operation legitimieren. Umgekehrt können mehrere Engines dieselbe Grammatik implementieren, sofern sie dieselben Gültigkeits- und Reproduzierbarkeitsbedingungen erfüllen.

Orthogonale Erweiterungsbereiche sind:

$$UZA = \{ K, Sc, I, D, M \}$$

- **Sc – Skalierung, Einbettung und Projektion,**
- **I – Interaktion mehrerer Zusammenhänge,**
- **D – Verlauf, Rückkopplung und dynamische Muster,**
- **M – kontrollierte Meta-Evolution der Grammatik.**



#### 4. Zwölf architektonische Grundaxiome RA1–RA12

Diese zwölf Aussagen sind konsolidierte Architekturaxiome. Sie bilden den semantischen Entwurf aus UZA 0.2 ab. Sie sind von den als Locale-Annahmen formalisierten HOL-Axiomen A1–A22 zu unterscheiden.

##### RA1 – Zusammenhangsgebundenheit

Jedes UZA-Objekt erhält Bedeutung durch seine typisierten Beziehungen zu Perspektive, Kontext, Zeit, Herkunft und weiteren Objekten. Isolierte Speicherung genügt nicht für semantische Gültigkeit.

##### RA2 – Relationale Herkunft von Wirkungsaussagen

Jede validierte UZA-Wirkungsaussage besitzt eine ausgewiesene relationale oder interaktionale Herkunft. Eine Wirkungsbehauptung ohne Herkunft bleibt Hypothese oder wird verworfen. Die Validierung der Aussage ist nicht mit der wirklichen Wirkung selbst identisch.

##### RA3 – Perspektivische Beobachtung

Beobachtungen, Relevanzen und Orientierungen werden einer versionierten Perspektive zugeordnet. Perspektivgebundenheit ist eine Transparenzbedingung und keine Beliebigkeitserlaubnis.

##### RA4 – Kontextbindung

Semantische Gültigkeit gilt relativ zu einem angegebenen Kontext. Kontextwechsel erzeugen eine neue Prüfung und können neue Objektversionen erforderlich machen.

### **RA5 – Dreifache Zeitbindung**

Ereigniszeit, Zusammenhangszeit und Versionszeit werden als drei operationale Ordnungen getrennt geführt. Dadurch bleiben Auftreten, modellierter Wirksamkeitsverlauf und Dokumentänderung unterscheidbar. Keine dieser Ordnungen formalisiert das meta-ontologische Immer.

### **RA6 – Explizite Unsicherheit**

Unsicherheit ist Bestandteil eines Objekts und seiner Begründung. Änderungen von Klasse oder Konfidenz erfolgen über dokumentierte Evidenz oder Revision.

### **RA7 – Versionierte Veröffentlichung**

Veröffentlichte Artikulationszustände und Objektversionen bleiben unverändert. Korrektur erzeugt eine neue Version mit Rückverweis auf ihre Vorgänger; sie schreibt die frühere Artikulation nicht rückwirkend um und behauptet keine versionierte Wirklichkeit an sich.

### **RA8 – Begründete Relevanz**

Relevanz entsteht durch ein ausgewiesenes Relevanzmodell in Bezug auf Objekt, Perspektive, Kontext und Zeit. Sie wird als mehrdimensionales Ergebnis mit Begründung ausgegeben.

### **RA9 – Hergeleitete Orientierung**

Orientierungen werden aus Relevanzen, Resonanzen, Zielen, Einschränkungen, Unsicherheiten und erwarteten Wirkungen hergeleitet. Die Engine darf „unzureichende Grundlage“ ausgeben.

### **RA10 – Rückwirkungsgebundene Gestaltung**

Gestaltung verweist auf eine Orientierung; Ausführung verweist auf die Gestaltung; Beobachtung und Rückwirkung verweisen auf die Ausführung. Damit bleibt der Gestaltungszyklus geschlossen und prüfbar.

### **RA11 – Grammatikgebundene Emergenz**

Emergenz ist keine bloße Überraschungsbezeichnung. Sie wird über die Beschreibbarkeit des Zustands durch Typen, Beziehungen und Übergänge der aktiven Grammatik klassifiziert.

### **RA12 – Kontrollierte Meta-Evolution**

Grammatikänderungen erfolgen über Proposal, Validierung, Freigabe, VersionRelease und Migration. Laufende Fälle bleiben bis zu einer expliziten Migration an ihre aktive Grammatik gebunden.

## **5. Verbindliche Invariantenfamilien**

Die UZA unterscheidet drei Ebenen von Verbindlichkeit:

1. **Theorieinvarianten** gelten für jedes Modell der UZA-Theorie.

2. **Zustandsgültigkeitsbedingungen** bestimmen, ob ein veröffentlichter Zustand akzeptiert wird.
3. **Implementierungskonformität** bestimmt, ob eine Engine die Spezifikation korrekt ausführt.

Zu den übergreifenden Invarianten gehören:

- eindeutige Objekt- und Versionsidentität,
- referenzielle Integrität,
- vollständige Herkunft validierter Wirkungsaussagen,
- explizite Perspektiv-, Kontext- und Zeitbindung,
- zulässige Lifecycle-Übergänge,
- Trennung von Simulation und Beobachtung,
- dokumentierte Orientierung vor Auswahl einer Gestaltung,
- Rückführung jeder Zustandsänderung auf einen atomaren Commit,
- vollständige Lineage bei Fission und Disposition bei Fusion,
- grammar-versionstreue Reproduzierbarkeit,
- sichtbare Konflikte statt stiller Auflösung,
- kontrollierte Unsicherheits- und Evidenzänderung.

## 6. Systemgrenze und Anwendungsprinzip

Die UZA legt fest, wie ein Zusammenhang formal beschrieben, verändert und geprüft wird. Domänenwissen, Messverfahren, Relevanzmodelle und Entscheidungsbefugnisse werden als versionierte Eingaben oder Module angebunden. Dadurch bleibt der Kern domänenübergreifend, während die jeweilige Anwendung ihre eigenen Begriffe, Evidenzstandards und Ziele einbringt.

Eine UZA-Analyse produziert deshalb keine kontextfreie „Gesamtwahrheit“. Sie produziert eine nachvollziehbare, reproduzierbare und revisionsfähige Darstellung dessen, was unter angegebenen Perspektiven, Daten, Regeln und Modellen beschrieben und begründet werden kann.

### 6.1 Meta-ontologische Geltungsgrenze: UZA-POT und reine Potenzialität

Die UZA setzt erst dort ein, wo Unterscheidungen typisierbar, beschreibbar oder als Übergang prüfbar werden. Sie formalisiert bestimmte Möglichkeiten innerhalb einer Grammatik; sie erhebt keinen Anspruch, die Möglichkeit von Bestimmtheit und Grammatik überhaupt als internes Objekt abzubilden.

| Dimension | UZA-POT                                                      | Reine/schöpferische Potenzialität |
|-----------|--------------------------------------------------------------|-----------------------------------|
| Status    | formaler, grammatik- und zustandsabhängiger Möglichkeitsraum | meta-ontologischer Grenzbegriff   |

| Dimension | UZA-POT                                                     | Reine/schöpferische Potenzialität                                     |
|-----------|-------------------------------------------------------------|-----------------------------------------------------------------------|
| Form      | typisierte Kandidaten, Bedingungen und erreichbare Zustände | keine Objektart, kein Payload und keine Relation                      |
| Zeit      | an Zusammenhangs-, Ereignis- und Versionszeit anschließbar  | weder Zeitpunkt noch Dauer innerhalb des Modells                      |
| Grenze    | durch Kontext, Auflösung und Grammatik begrenzt             | nicht als Skala oder Modellgrenze repräsentiert                       |
| Funktion  | liefert prüfbare Alternativen für Operationen               | bezeichnet allenfalls die Ermöglichung von Grammatik und Bestimmtheit |

MG1 – Nicht-Objektivierbarkeit. Reine oder schöpferische Potenzialität darf nicht als UZAObject, Zustand, Ressource, Kontext, Perspektive oder verborgene Ursache persistiert werden.

MG2 – Keine Gleichsetzung. UZA-POT beziehungsweise  $\Pi\_UZA$  bezeichnet nur den innerhalb einer versionierten Grammatik artikulierten Möglichkeitsraum;  $\Pi\_UZA$  ist nicht mit reiner Potenzialität identisch.

MG3 – Keine Vollständigkeitsüberschreitung. Aus vollständiger Describe-Abdeckung innerhalb einer Grammatik folgt keine ontologische Vollständigkeit des Modells.

MG4 – Meta-Evolution bleibt intern. Ein GrammarPatch erweitert die Beschreibbarkeit bestimmter Phänomene, erfasst aber nicht den Ermöglichungsgrund von Grammatik selbst.

MG5 – Formale Anschlussstelle. Erst bestimmte Erscheinungen, Unterscheidungen, Relationen oder Übergänge dürfen in UZA eingehen; ihre Herkunft muss dann als Evidenz-, Unsicherheits- oder Geltungsfrage ausgewiesen werden.

MG1–MG5 sind meta-theoretische Scope-Klauseln. Sie werden nicht als A1–A22-Erweiterung in Isabelle kodiert: Eine Formalisierung des ausdrücklich Nicht-Objektivierbaren als Gegenstand würde die gesetzte Grenze performativ verletzen.

## 6.2 Durchgängige Mitgeltung, Reflexivität und Abgleich

Nicht-Objektivierbarkeit bedeutet keine Abwesenheit. Mit  $\mathbb{W}\_Z$  wird im Folgenden kein UZA-Objekt, kein Zustand und keine verborgene Ursache bezeichnet, sondern ausschließlich ein meta-sprachlicher Geltungsindex dafür, dass jede Unterscheidung und jede Artikulation innerhalb derselben Wirklichkeit vollzogen wird. „Immer“ bedeutet hierbei nicht unendlich viel Zeit, sondern: Grundwirklichkeit ist nicht durch zeitliche Abwesenheit charakterisierbar.

$$\forall n: \text{Art\_L}^{\mathbb{W}}(\mathbb{W}\_Z) \not\equiv \mathbb{W}\_Z$$

$$\text{Art\_L}^{\mathbb{W}}(\mathbb{W}\_Z) \xrightarrow{\mathbb{A}\_Z} \text{Art\_L}^{\mathbb{W}+1}(\mathbb{W}\_Z)$$

MG6 – Durchgängige Mitgeltung. Immer, reine Potenzialität, Lebendigkeit, Wandlungsfähigkeit, Wirkungsfähigkeit und erinnernde Fortgeltung gelten in jedem realen Artikulations- und UZA-Vollzug mit. Sie werden durch keinen Modellschritt erzeugt, verbraucht oder beendet.

MG7 – Keine Sequenzialisierung. Diese Grundbestimmungen dürfen weder Objektzeichen noch Zustandsknoten, Übergangsquelle, Übergangsziel oder nachgelagertes Modul sein. Übergangsstrukturen besitzen ausschließlich ihre artikulierten Gestalten.

MG8 – Reflexive Wirklichkeit. Denken, Formulieren, Modellieren und Prüfen sind selbst reale Artikulations- und Wirkungsvollzüge. Daraus folgt nicht automatisch die Wahrheit ihres Inhalts: Realitätsstatus des Vollzugs und Geltungsstatus der Aussage werden getrennt geprüft.

MG9 – Wirkungs- und Erinnerungsgrenze. EFF bezeichnet eine versionierte Wirkungsaussage, nicht die wirkliche Wirkung selbst. History und EvolutionMemory bewahren Auditspuren; sie sind weder die ontologische Quelle von Erinnerung noch mit erinnernder Fortgeltung identisch.

MG10 – Zeit- und Abgleichsgrenze. Ereignis-, Zusammenhangs- und Versionszeit ordnen artikuliert Gestalten. Sie enthalten das Immer nicht als zusätzliche Zeitachse.  $\mathcal{U}_Z$  bezeichnet den strengeren Kohärenzabgleich zwischen Artikulationsversionen; keine endliche Version erschöpft die Wirklichkeit.

MG6–MG10 sind positive Formkongruenzregeln: Das Grundlegende ist in jedem Vollzug wirksam mitgemeint, aber nicht als weiteres Glied neben anderen Gliedern. Eigengegenwärtigkeit ist dabei keine Isolation, sondern eine Weise, wie Zusammenhang sich unterscheidbar artikuliert.

MG11 - Harte Sortengrenze. Grundaussdrücke und Artikulationsausdrücke sind nicht derselben formalen Sorte zugeordnet. Gemischte Summen, Negationen, Teil-Ganzes-Aussagen, Quantifikationen oder Übergangspfeile werden nicht gebildet, wenn sie eine Gleichsetzung beider Ebenen voraussetzen würden.

MG12 - Explizite Brückenpflicht. Jede Aussage, die von einer artikulierbaren Gestalt auf einen Grundaussdruck bezogen wird, muss ihren metasprachlichen Brückenstatus sichtbar kennzeichnen. Die Brücke verlangt Kontexttypisierung, getrennte Grund- und Artikulationssorten, keinen Kontextverlust und keine Bildung absoluter Trennung; sie ist kein Isabelle-Feld und keine stille Typumwandlung.

Kongruenz bleibt nichtfunktional: Eine Artikulation erzwingt keinen eindeutigen ontologischen Grund, der Grund erzwingt keine eindeutige Artikulation, und Formkongruenz bedeutet keine erschöpfende Repräsentation. Auch eine reine Überlegung ist als Vollzug eine real gewirkte Tatsache; der Geltungsstatus ihres Inhalts bleibt davon getrennt und abgleichspflichtig.

## Teil II – Artikulationsontologie und semantischer Kern

### 7. UZA-Artikulationszustand als formale Grundeinheit

Ein UZA-Artikulationszustand ist eine zeitlich und perspektivisch bestimmte Modellkonfiguration von Objekten, Beziehungen, Wirkungsaussagen, Relevanzordnungen, Regeln und Auditgeschichte. Die nachfolgende Formel bezeichnet daher nicht die Wirklichkeit selbst, sondern eine sprach- und grammatikgebundene Darstellung;  $Z_t$  ist im gesamten operativen Kalkül als Kurzform für  $\hat{Z}_t^L$  zu lesen:

$$Z_t = \{ E_t, R_t, W_t, S_t, Q_t, O_t, A_t, P_t, C_t, G_t, H_t \}$$

In dieser modellinternen Notation bezeichnen die Komponenten Erscheinungen, Beziehungen, Wirkungsaussagen, Resonanzen, Relevanzen, Orientierungen, Gestaltungen, Perspektiven, Kontexte, die aktive Grammatik und die versionierte Auditgeschichte.

Für die dynamische Betrachtung wird derselbe UZA-Artikulationszustand in vier komplementären Modellaspekten gelesen:

$$Z = \{ Structure, Process, Potential, History \}$$

- **Structure** enthält die gegenwärtig publizierten Artikulationsobjekte und ihre modellierten Relationen.
- **Process** enthält modellierte Übergänge, Rückkopplungen und Übertragungen zwischen artikulierten Gestalten.
- UZA-POT, im Folgenden auch  $\Pi\_UZA$ , bezeichnet ausschließlich die unter einer aktiven, versionierten Grammatik sowie unter Ressourcen- und Kontextbedingungen artikulierbaren Kandidaten- und Folgezustände.
- **History** enthält die geordnete Audittrajektorie der publizierten Zustände, Commits, Modellwechsel und Revisionen.

$\Pi\_UZA$  wird nicht als gewöhnliches persistiertes Objekt behandelt. Es ist eine zustands-, sprach-, grammatik- und kontextabhängige Artikulationsfunktion. Diese formale Größe darf weder mit reiner oder schöpferischer Potenzialität noch mit der Wirklichkeit ihrer Ermöglichung gleichgesetzt werden:

Nichtidentität und Nichttrennung:  $\hat{Z}_t^L \not\equiv \mathbb{W}_Z$ . Zugleich liegt  $\hat{Z}_t^L$  nicht außerhalb der Wirklichkeit, sondern ist ein wirklicher Artikulationsvollzug in ihr. Unterscheidbarkeit mehrerer Zustände, Perspektiven oder Teilzusammenhänge begründet daher keine absolute ontologische Trennung.

$$POT(Z_t, G_t, C_t, P_t) \subseteq P(State)$$

### 8. Drei operationale Zeitebenen

Die UZA führt innerhalb artikulierter Modellzustände drei getrennte Zeitordnungen:



| Zeitebene         | Frage                                                                            | Typische Felder                       |
|-------------------|----------------------------------------------------------------------------------|---------------------------------------|
| Ereigniszeit      | Wann trat ein Ereignis auf oder wurde eine Wirkung beobachtet?                   | occurred_at, observed_at              |
| Zusammenhangszeit | Über welchen Zeitraum gilt oder wirkt ein Objekt im modellierten Zusammenhang?   | valid_from, valid_to, horizon, delay  |
| Versionszeit      | Wann wurde eine Objekt- oder Grammatikversion erzeugt, freigegeben oder ersetzt? | created_at, committed_at, released_at |

Die Trennung verhindert, dass eine spät dokumentierte Beobachtung fälschlich als spät eingetretenes Ereignis behandelt wird. Ebenso bleibt erkennbar, ob eine Wirkungsaussage historische Geltung beansprucht, obwohl ihre aktuelle Objektversion inzwischen revidiert wurde. Keine der drei Ordnungen – auch nicht ihre unbeschränkte Fortsetzung – ist das Immer; sie ordnen ausschließlich artikulierte Unterschiede von Auftreten, Geltung und Versionierung.

## 9. Primäres Typensystem

### 9.1 Persistierte Kernobjekte

| Code | Typ                 | Semantische Funktion                                                                 |
|------|---------------------|--------------------------------------------------------------------------------------|
| ENT  | Erscheinung/Entität | unterscheidbare Gestalt oder registrierte Beobachtungseinheit                        |
| REL  | Beziehung           | typisierte Verbindung zwischen Erscheinungen                                         |
| EFF  | Wirkungsaussage     | typisierte Aussage über gerichtete Veränderung mit Herkunft, Ziel und Wirkungsprofil |
| RES  | Resonanz            | qualifizierte Wechselbeziehung zwischen Wirkungsaussagen                             |
| REV  | Relevanz            | perspektivisch-kontextuelle Bewertung eines Objekts                                  |
| ORI  | Orientierung        | begründete Entwicklungs- oder Handlungsoption                                        |
| ACT  | Gestaltung          | ausgewählte, freigegebene oder ausgeführte Option                                    |
| FDB  | Rückwirkung         | beobachtete Folge und Abweichung nach Gestaltung                                     |
| EMG  | Emergencyfall       | klassifizierte Beschreibungs- oder Strukturgrenze                                    |
| PER  | Perspektive         | versionierter Beobachtungs-, Wissens- und Wertungsstandpunkt                         |
| CTX  | Kontext             | Bedingungsrahmen eines Falls oder Teilzusammenhangs                                  |

| Code     | Typ            | Semantische Funktion                                      |
|----------|----------------|-----------------------------------------------------------|
| META     | Metaobjekt     | Grammatikpatch, Evolutionsvorschlag oder Reflexionsobjekt |
| CONFLICT | Konfliktobjekt | explizit erhaltener, noch nicht aufgelöster Widerspruch   |

## 9.2 Erweiterungsobjekte

| Code   | Typ                 | Semantische Funktion                                             |
|--------|---------------------|------------------------------------------------------------------|
| SYSCTX | Systemzusammenhang  | adressierbarer Einzel- oder Teilzusammenhang im Interaktionsnetz |
| COUP   | Kopplung            | versionierte Verbindung zwischen Systemzusammenhängen            |
| CHAN   | Kanal               | Übertragungsweg innerhalb einer Kopplung                         |
| TRANSF | Kanaltransformation | typisierte Abbildung einer Quellwirkung in einen Zielkontext     |
| TEV    | TransmissionEvent   | lifecycle-gebundene Übertragung                                  |
| SUBS   | Einbettungsrelation | direkte Container-Teil-Beziehung                                 |
| PROJ   | Projektion          | ausgewählte Sicht eines Teilzusammenhangs mit LossProfile        |
| BEV    | BoundaryEvent       | Fusion, Fission, Aggregation, Embedding oder Detachment          |

Alle persistierten Typen werden in einem gemeinsamen Summentyp UZAObject gehalten. Spezialisierte Stores sind zulässige Indizes oder Projektionen, jedoch keine unabhängigen Wahrheitsbestände.

## 10. Objektidentität, Versionierung und Provenienz

Jedes persistierte Objekt trägt mindestens:

|                  |                                        |
|------------------|----------------------------------------|
| object_id        | stabile Identität des Gegenstands      |
| version_id       | Identität dieser konkreten Version     |
| revision         | monotone Revisionsnummer               |
| payload          | typisierter UZAObject-Inhalt           |
| uncertainty      | Klasse und Konfidenz                   |
| lifecycle_status | gültiger Lifecycle-Zustand             |
| valid_from/to    | Zusammenhangszeit                      |
| created_at/by    | Versionszeit und Urheber/Akteur        |
| previous_version | unmittelbarer Vorgänger                |
| perspective_refs | beteiligte Perspektiven                |
| context_refs     | gültige Kontexte                       |
| source_refs      | Quellen und Eingaben                   |
| evidence_refs    | stützende oder widersprechende Evidenz |
| annotations      | typisierte Zusatzangaben               |

Es gilt:

$$objectId(x_i) = objectId(x_j) \Rightarrow sameEntity(x_i, x_j)$$

und:

$$versionId(x_i) \neq versionId(x_j) \Rightarrow x_i \neq x_j$$

Eine neue Version überschreibt keine frühere Bedeutung. Sie setzt die nachvollziehbare Entwicklung desselben Gegenstands fort. Quellen-, Evidenz- und Commitreferenzen bilden gemeinsam die Provenienz.

## 11. Lifecycle und Unsicherheit

Der allgemeine Objektlebenszyklus lautet:

$$DRAFT \rightarrow HYPOTHESIS \rightarrow SUPPORTED \rightarrow VALIDATED \rightarrow REVISED$$

Zusätzliche End- oder Seitenzustände sind REJECTED und ARCHIVED.

- **DRAFT:** formal angelegt, semantisch noch unvollständig.
- **HYPOTHESIS:** ausdrücklich als Annahme ausgewiesen.
- **SUPPORTED:** durch mindestens eine zulässige Evidenz gestützt.
- **VALIDATED:** erfüllt alle für den Typ geltenden Integritäts- und Evidenzbedingungen.
- **REVISED:** durch eine neuere Version fortgeführt.
- **REJECTED:** nach Prüfung verworfen, historisch erhalten.
- **ARCHIVED:** für aktive Verarbeitung inaktiv, weiterhin referenzierbar.

Unsicherheit besteht aus Klasse und Konfidenz:

$$U(x) = \{class(x), conf(x)\}, conf(x) \in [0, 1)$$

mit KNOWN, PLAUSIBLE, HYPOTHETICAL und OPEN. Die Klassen kennzeichnen den epistemischen Status, nicht den persönlichen Grad des Gefallens. Neue Evidenz kann Konfidenz erhöhen, vermindern oder den Lifecycle verändern. Jede Änderung erzeugt eine protokollierte Objektversion oder einen Commit-Eintrag.

## 12. Perspektivsemantik

Eine Perspektive ist eine dynamische Entität:

$$p_t = \langle O_p, K_p, V_p, I_p, H_p, A_p, M_p, R_p, F_p \rangle_t$$

mit Beobachtungsraum  $O_p$ , Wissensstand  $K_p$ , Wertungsstruktur  $V_p$ , Interessenhypothesen  $I_p$ , historischer Erfahrung  $H_p$ , Handlungsspielraum  $A_p$ , Machtprofil  $M_p$ , Verantwortungsprofil  $R_p$  und Freiheitsprofil  $F_p$ .

Die Beobachtungsfunktion lautet:

$$observe(p_t, Z_t) = Z_t^{(p)}$$

Der Vergleich zweier Perspektiven erzeugt ein Differenzprofil und keine automatische Rangordnung:

$$Dist(p_i, p_j) = \{d_O, d_K, d_V, d_I, d_H, d_A, d_M, d_R, d_F\}$$

Perspektiven werden versioniert, wenn sich Wissen, Wertungen, Interessen, Zugänge, Macht, Verantwortung oder Handlungsspielräume ändern. Aussagen über „die Perspektive einer Gruppe“ benötigen entweder eine ausgewiesene Aggregationsregel oder mehrere Einzelperspektiven; die Gruppenzuschreibung darf nicht stillschweigend homogenisiert werden.

Perspektiven bleiben als Eigengegenwärtigkeiten unterscheidbar, ohne dadurch ontologisch isoliert zu werden. Ihre Differenz ist eine wirkliche Artikulation innerhalb desselben Zusammenhangs; eine Aggregation darf diese Differenz nicht rückwirkend aufheben.

## 13. Beziehungen und Wirkungsaussagen

### 13.1 Beziehung

Eine Beziehung ist eine typisierte, zeit- und kontextgebundene Verbindung:

$$REL: E \times E \times RelationType \times C \times T \rightarrow R$$

Pflichtfelder sind source, target, relation\_type, directed, context, validity, uncertainty und provenance. Beziehungen können unter anderem verbinden, bedingen, verstärken, hemmen, widersprechen, erzeugen oder transformieren.

### 13.2 Wirkungshypothese

Eine UZA-Wirkungshypothese ist eine versionierte Aussage, die aus einer Beziehung, einer Interaktion oder einer bereits dokumentierten Wirkungsaussage abgeleitet wird:

$$EFF_H = \{cause, target?, vector?, p, c, t, u, evidence^*\}$$

Noch unbekannte Ziele oder Wirkungsdimensionen sind zulässig, solange die Offenheit explizit typisiert ist.

### 13.3 Validierte Wirkung

Eine validierte UZA-Wirkungsaussage besitzt vollständige Herkunft, Ziel, Profil, Perspektive, Kontext, Zeit und zulässige Evidenz:

$$EFF_V = \{cause, target, vector, p, c, t, u, evidence^+\}$$

Für Einzelzusammenhänge gilt als Grundform:

$$EFF: REL \times P \times C \times T \rightarrow W$$

Bei Interaktionen kann die Ursache zusätzlich ein TransmissionEvent oder BoundaryEvent sein. Diese Erweiterung wird über die Lineage referenziert und hebt die Herkunftspflicht nicht auf.

EFF\_H und EFF\_V bezeichnen somit Hypothesen beziehungsweise evidenziell validierte Artikulationen über Wirkungen. Auch EFF\_V ist nicht mit dem wirklichen Wirken identisch. Umgekehrt ist schon das Formulieren, Prüfen oder Verwerfen einer Wirkungsaussage selbst ein realer Vollzug mit möglichen Wirkungen; die Wahrheit ihres Inhalts bleibt davon getrennt.

## 14. Konsolidierter Wirkungsvektor

### 14.1 Normative EffectProfile-Schnittstelle und V10-Referenzprofil

Die universelle Semantik bindet Wirkungsbewertung nicht an eine stillschweigend veränderbare Dimensionsliste. Normativ ist zunächst die Schnittstelle EffectProfile. Jedes Profil muss mindestens ausweisen:

- stabile Profil- und Versionsidentität,
- benannte Dimensionen mit eindeutiger Semantik und Wertebereich,
- Bewertungs-, Nullwert- und Unsicherheitsregeln,
- Perspektiv-, Kontext- und Zeitbindung,
- Regeln für Aggregation, Abhängigkeiten und fehlende Werte,
- Migrations- und LossProfile für Vorgänger- oder Fremdprofile.

V10 ist das verbindliche Referenzprofil der UZA 0.9.1. Eine Engine mit dem Konformitätsanspruch UZA-0.9.1-V10 muss es unverändert unterstützen. Domänenspezifische Zusatzprofile sind zulässig, sofern sie separat versioniert werden und V10 weder überschreiben noch stillschweigend erweitern.

Der konsolidierte Wirkungsvektor bewertet zehn eigenständige, aber nicht als mathematisch orthogonal behauptete Gestaltungsdimensionen:

$$V_{10}(w) = (v_{con}, v_{sta}, v_{ope}, v_{lea}, v_{coo}, v_{act}, v_{tra}, v_{fre}, v_{res}, v_{reg})$$

mit  $v_i \in \{-3, -2, -1, 0, 1, 2, 3\}$ .

| Dimension          | Leitfrage für die Bewertung                                                            |
|--------------------|----------------------------------------------------------------------------------------|
| Verbindung         | Wie verändert die Wirkung tragfähige Beziehungen und Anschlussfähigkeit?               |
| Stabilität         | Wie verändert sie Verlässlichkeit, Beständigkeit und Belastbarkeit?                    |
| Offenheit          | Wie verändert sie erreichbare Optionen und Zugänglichkeit?                             |
| Lernfähigkeit      | Wie verändert sie Wahrnehmung, Rückmeldung, Revision und Erkenntnisgewinn?             |
| Kooperation        | Wie verändert sie die Fähigkeit zu wechselseitiger Abstimmung und gemeinsamem Handeln? |
| Handlungsfähigkeit | Wie verändert sie die realen Möglichkeiten wirksamer Gestaltung?                       |
| Transparenz        | Wie verändert sie Nachvollziehbarkeit, Sichtbarkeit und Erklärbarkeit?                 |
| Freiheit           | Wie verändert sie echte Wahl- und Gestaltungsspielräume?                               |
| Verantwortung      | Wie verändert sie Zurechenbarkeit, Folgenbezug und verlässliche Übernahme?             |
| Regeneration       | Wie verändert sie Erneuerungs-, Erholungs- und Wiederherstellungsfähigkeit?            |

Die Werte sind perspektivisch und kontextuell. +3 bedeutet eine stark fördernde Wirkung auf die betreffende Dimension, -3 eine stark mindernde Wirkung, 0 eine neutrale, ausgeglichene oder

nicht feststellbare Wirkung. Der Grund für 0 muss über zero\_reason zwischen neutral, kompensiert und unbekannt unterschieden werden.

## 14.2 Begründung, Abdeckung und Abhängigkeiten

Die Auswahl folgt vier funktionalen Abdeckungsgruppen:

1. **relationale Tragfähigkeit:** Verbindung und Kooperation,
2. **strukturelle Fortdauer:** Stabilität und Regeneration,
3. **Möglichkeits- und Entwicklungsraum:** Offenheit, Lernfähigkeit, Handlungsfähigkeit und Freiheit,
4. **epistemische und verantwortliche Nachvollziehbarkeit:** Transparenz und Verantwortung.

Damit werden Beziehung, Beständigkeit, Entwicklung, wirksame Gestaltung, Nachvollziehbarkeit und Folgenübernahme jeweils explizit sichtbar. Diese Begründung ist eine Architekturentscheidung und kein Vollständigkeits- oder Unabhängigkeitsbeweis. Überschneidungen sind erwartbar: Offenheit kann Freiheit fördern, Transparenz kann Verantwortung stärken, Lernfähigkeit kann Handlungsfähigkeit verändern. Deshalb darf eine Engine die zehn Werte nicht zu unabhängigen Messgrößen umdeuten. Sie muss relevante Abhängigkeiten in einem DimensionDependencyReport ausweisen und darf einen Gesamtscore nur erzeugen, wenn ein versioniertes Aggregationsmodell ihn ausdrücklich definiert.

## 14.3 Legacy-Profil V10-POLAR und Migration

Das frühere Profil mit Verbindung, Trennung, Stabilisierung, Destabilisierung, Öffnung, Begrenzung, Beschleunigung, Verlangsamung, Kooperation und Konflikt bleibt für historische Fälle verfügbar. Es wird als eigenes EffectProfile versioniert. Eine automatische Eins-zu-eins-Übersetzung in V10 ist unzulässig, weil polare Prozessbeschreibungen und eigenständige Gestaltungsqualitäten verschiedene Bedeutungen tragen. Ein Adapter muss seine Abbildungsregeln und Verluste ausweisen.

### 14.3.1 Migrationsdispositionen

Jede Altdimension erhält je Zielfall genau eine Disposition: Mapped, Derived, Unmapped oder Conflicted. Die Referenzmigration lautet:

| V10-POLAR        | mögliche V10-Zuordnung              | Referenzdisposition                                |
|------------------|-------------------------------------|----------------------------------------------------|
| Verbindung       | Verbindung                          | Mapped; Vorzeichen und Begründung bleiben erhalten |
| Trennung         | Verbindung mit umgekehrter Richtung | Derived; semantischer Verlust ist auszuweisen      |
| Stabilisierung   | Stabilität                          | Mapped                                             |
| Destabilisierung | Stabilität mit umgekehrter Richtung | Derived                                            |
| Öffnung          | Offenheit                           | Mapped                                             |
| Begrenzung       | Offenheit mit umgekehrter Richtung  | Derived                                            |

| V10-POLAR      | mögliche V10-Zuordnung                        | Referenzdisposition                                                              |
|----------------|-----------------------------------------------|----------------------------------------------------------------------------------|
| Beschleunigung | keine feste V10-Dimension                     | Unmapped; kontextuell unter anderem Handlungsfähigkeit oder Regeneration prüfbar |
| Verlangsamung  | keine feste V10-Dimension                     | Unmapped; kontextuelle Neubewertung erforderlich                                 |
| Kooperation    | Kooperation                                   | Mapped                                                                           |
| Konflikt       | Kooperation, Verbindung oder Resonanzspannung | Conflicted; menschlich oder modellgebunden aufzulösen                            |

### 14.3.2 Normative Migrationsregeln

1. Das ursprüngliche V10-POLAR-Objekt und sein Version Bundle bleiben unverändert reproduzierbar.
2. Eine Migration erzeugt neue V10-Objekte mit Verweis auf Quelle, Adapterversion und Bearbeitungsentscheidung.
3. Für jede Zieldimension werden Disposition, Konfidenz und semantischer Verlust dokumentiert.
4. Unmapped und Conflicted dürfen nicht als Nullwert ausgegeben werden; sie bleiben offene Migrationslücken.
5. Ein migrierter Fall darf erst als V10-konform gelten, wenn alle für seine Orientierungs- oder Gestaltungsentscheidung relevanten Dimensionen bewertet oder ausdrücklich als nicht entscheidungsrelevant begründet sind.
6. Golden-Run- und Replay-Tests müssen sowohl den historischen Lauf als auch die neue Migration reproduzieren.

### 14.4 Macht als relationales Zusatzprofil

Macht ist keine elfte V10-Gestaltungsqualität, sondern eine relationale Verteilung von Möglichkeiten und Folgen. Sie wird deshalb in einem getrennten PowerProfile erfasst:

$$Power(p, t) = \{resource, access, decision, risk, correction\}$$

mit Ressourcenmacht, Zugangskontrolle, Entscheidungsbefugnis, Risikotragung und Korrekturfähigkeit. Eine Machtverschiebung lautet:

$$\Delta Power_p(t_0, t_1) = Power(p, t_1) - Power(p, t_0)$$

Ein PowerShiftReport bindet diese Differenz an Perspektive, Kontext, Zeit, verursachende Beziehungen und Wirkungen. Er zeigt zusätzlich, welche V10-Dimensionen betroffen sind, etwa Freiheit, Handlungsfähigkeit, Transparenz oder Verantwortung. PowerProfile und V10 werden gemeinsam gelesen, aber weder miteinander verrechnet noch durch Mittelung homogenisiert.

## 15. Resonanzsemantik

Resonanz qualifiziert das geordnete Verhältnis zweier Wirkungen relativ zu Perspektive, Kontext, Modellversion und Zeithorizont. Die normative T6-Signatur lautet  $EFF \times EFF \times Perspective \times$



Context × ResonanceModel × TimeHorizon → RES. Ein ResonanceModel enthält mindestens eine gerichtete V10-Interaktionsmatrix und geordnete Klassifikationsschwellen.

$$RES: W \times W \times P \times C \rightarrow R$$

mit:

- $R^+$  – wechselseitige Verstärkung,
- $R^\approx$  – funktionale Ergänzung,
- $R^\sim$  – produktive Spannung,
- $R^-$  – hemmende Spannung,
- $R^0$  – keine belastbare Resonanzbestimmung.

Resonanz ist gerichtet: RES(A,B,p,c,m,h) darf von RES(B,A,p,c,m,h) abweichen. T6 berechnet den normalisierten Score aus K(source\_dimension,target\_dimension), klassifiziert ihn total in Amplification, Complement, ProductiveTension, Inhibition oder Neutral und dämpft die Stärke nachvollziehbar durch die deklarierte Unsicherheit. Jede Klassifikation persistiert Quell- und Zieleffekt, Profilmodell, Form, Score, Stärke, Zeithorizont, Perspektive, Kontext, Begründungsreferenzen, Begründungstext und Unsicherheit.

Abgrenzung der Referenzsemantiken: Die ältere symmetrische Funktion classify\_resonance(a,b) auf Basis des V10-Skalarprodukts bleibt als einfache Alignment-Baseline und Legacy-Referenz erhalten. Die normative operationale T6-Semantik verwendet directional\_resonance\_score(profile,source,target). Zwischen beiden wird keine stille Gleichheit behauptet. Ein symmetrischer T6-Fall muss durch ein entsprechend symmetrisches Kernelfprofil explizit modelliert werden. Die Kernelprüfung belegt Rechen- und Erhaltungseigenschaften der Spezifikation, nicht die empirische Richtigkeit eines konkreten domänenspezifischen Kernels oder seiner Schwellen.

## 16. Relevanzsemantik

Relevanz ist eine versionierte Funktion eines Modells  $M$ :

$$REV_M: X \times P \times C \times T \rightarrow Q$$

mit  $X$  als Menge zulässiger UZA-Zielobjekte. Das Ergebnis ist ein Vektor:

$$REV_M(x, p, c, t) = \langle q, u, \sigma, \rho, \delta, e \rangle$$

- $q$  – Relevanzintensität,
- $u$  – epistemische Unsicherheit,
- $\sigma$  – Stabilität der Bewertung gegenüber kleinen Eingabeänderungen,
- $\rho$  – Reichweite über betroffene Objekte oder Zusammenhänge,
- $\delta$  – erwartete zeitliche Dauer,
- $e$  – maschinen- und menschenlesbare Begründung.

Ein Relevanzwert ohne Modellversion, Perspektive, Kontext, Zeit und Begründung ist nicht veröffentlichungsfähig. Mehrere Relevanzmodelle können parallel bestehen. Ihre Ergebnisse werden als verschiedene Relevanzordnungen ausgewiesen und nicht still zu einem Universalwert gemittelt.

### 16.1 Status, Signatur und Geltungsbereich

Relevanz ist keine Eigenschaft eines Gegenstands an sich, sondern das versionierte Ergebnis eines expliziten Modells relativ zu Zielobjekt, Perspektive, Kontext und Auswertungszeit. Die normative T7-Signatur lautet:

EvaluateRelevance : EFF × Perspective × Context × EvaluationTime × RelevanceModelRevision  
→ REV

### 16.2 Versioniertes RelevanceModelProfile

Ein veröffentlichungsfähiges Modellprofil enthält eine Modell-ID und eine positive Revision, die unterstützten Perspektiven und Kontexte, perspektivische V10-Gewichte, davon getrennte perspektivische Machtgewichte, kontextuelle Faktoren für beide Profilarten, einen zeitlichen Relevanzfaktor, Reichweitenfaktor, Dauerkappung, lokale Sensitivität, den deklarierten V10-Anteil  $\alpha$  und eine Validierungs-Unsicherheitsschwelle. Gewichte und Faktoren besitzen explizite Wertebereiche; effektive Gewichtsmassen müssen positiv sein.

Macht wird nicht als elfte V10-Achse behandelt. V10-Wirkung und PowerProfile bleiben getrennte typisierte Eingaben und werden erst durch den ausdrücklich versionierten Mischanteil  $\alpha$  verbunden. Ein Modellwechsel oder eine Revisionsänderung erzeugt deshalb eine andere, nachvollziehbare Relevanzordnung.

### 16.3 Ausführbare Vektorsemantik

$E = \text{clamp01}(\sum_d wE(p,c,d) \cdot \text{normalizeV10}(vd) / \sum_d wE(p,c,d))$  — perspektivisch-kontextuelle V10-Komponente.

$P = \text{clamp01}(\sum_k wP(p,c,k) \cdot \text{powerk} / \sum_k wP(p,c,k))$  — davon getrennte Machtkomponente.

Intensität =  $\text{clamp01}((\alpha E + (1-\alpha)P) \cdot \text{temporal}(t) \cdot (1-\text{clamp01}(u)))$ .

Epistemische Unsicherheit =  $\text{clamp01}(u)$ ; lokale Stabilität =  $\text{clamp01}(1-\text{sensitivity}(p,c) \cdot \text{clamp01}(r))$ .

Reichweite =  $\text{clamp01}(\text{reachInput} \cdot \text{reachFactor}(p,c))$ ; Dauer =  $\text{min}(\text{durationInput}, \text{durationCap}(p,c))$ .

Der sechste Vektorbestandteil ist eine persistierte menschen- und maschinenlesbare Erklärung. Die skalaren Komponenten Intensität, Unsicherheit, Stabilität und Reichweite sind durch Konstruktion auf  $[0,1]$  begrenzt; die Dauer ist positiv und durch die Modellkappung beschränkt.

### 16.4 Publikations-, Lifecycle- und Trennungsregeln

T7 ist nur anwendbar, wenn Ausgangszustand und Effect-/Relevance-Kataloge wohlgeformt sind, EvaluateRelevance und K\_REV von der aktiven Grammatik getragen werden, die Zielwirkung aktiv und katalogisiert ist, Request und Effect-Payload denselben V10-Vektor und dieselbe Unsicherheit nennen, Perspektive, Kontext und Modellversion vorhanden sind, das Ziel zum

Auswertungszeitpunkt aktiv ist, die Begründungsreferenzen nichtleer, aktiv und vom Ziel verschieden sind und Objekt-, Versions- sowie Snapshot-ID frisch sind. Das Resultat ist Validated genau dann, wenn die Zieleffektversion validiert ist und u die Modellschwelle nicht überschreitet; andernfalls ist es Supported.

Jeder erfolgreiche Schritt konstruiert genau ein frisches K\_REV-Objekt, persistiert Ziel, V10- und Machtprofil, vollständiges Modellprofil samt Revision, Perspektive, Kontext, Auswertungszeit, alle Eingaben, den Ergebnisvektor und die Begründungsreferenzen und erweitert den Relevance-Katalog unter der neuen Versions-ID. Bestehende Objekte, historische Versionen und alle fachfremden Kataloge bleiben unverändert. Parallelbewertungen werden als getrennte Relevanzordnungen erhalten und niemals still gemittelt.

Der kernelgeprüfte nicht-vakuose T7-Zeuge bewertet denselben validierten Effekt mit demselben Modell 30/Revision 1, Kontext 20, Zeitpunkt 10, Unsicherheit 1/10 und identischem Machtprofil. P0 ist Connection-zentriert und ergibt (9/10, 1/10, 49/50, 4/5, 8); P1 ist Stability-zentriert und ergibt (9/50, 1/10, 19/20, 2/5, 4). Beide Resultate werden getrennt als Version 7 und 8 gespeichert und sind formal verschieden.

Geltungsgrenze: Die Referenzsemantik beweist interne Ausführbarkeit, Wertebereiche, Provenienz, Trennung und Erhaltung. Sie beweist weder empirische Angemessenheit konkreter Gewichte noch Vollständigkeit des Modells für jede Domäne. Solche Profile benötigen externe Kalibrierung, Holdout-Prüfung und versionierte Freigabe.

## 17. Orientierung, Gestaltung und Rückwirkung

### 17.1 Orientierungsraum

$$ORI: Q^* \times RES^* \times Goal \times Constraints \times P \times C \rightarrow O$$

Eine Orientierungsoption enthält Ziel, unterstützende und entgegenstehende Relevanzen, erwartete Wirkungen, betroffene Perspektiven, Ressourcen, Einschränkungen, Unsicherheit, Reversibilität, Zeithorizont und Rückmeldesignale.

Die Synthese kann die Status AVAILABLE, CONTESTED oder INSUFFICIENT\_BASIS ausgeben. Eine künstliche Mehrzahl von Optionen ist nicht erforderlich.

### 17.2 Gestaltung

$$ACT: ORI \times P \times C \times T \rightarrow A$$

Eine Gestaltung verweist auf die ausgewählte Orientierung, Entscheidungsbefugnis, Zustimmungslage, Ressourcenfreigabe, Ausführungsparameter und geplante Beobachtung.

### 17.3 Simulation und Beobachtung

$$SIM(A, Z_t, M, \Theta) \rightarrow \hat{Z}_{t+1}$$

$$OBS(A, Z_t) \rightarrow Z_{t+1}$$

Simulation und Beobachtung besitzen getrennte Objekttypen und Provenienzketten. Ihre Abweichung lautet:

$$\Delta_{pred} = Dist(\hat{Z}_{t+1}, Z_{t+1})$$

## 17.4 Rückwirkung

$FDB: A \times Z_t \times Z_{t+1} \rightarrow Feedback$

Rückwirkung enthält beobachtete Folgen, Prognoseabweichungen, neu entstandene Beziehungen und Wirkungen, Perspektivänderungen, Relevanzverschiebungen sowie Revisionsvorschläge.

## 18. Beschreibbarkeit und Emergenz

Die logische Beschreibbarkeit ist zerlegt:

Mit  $D_T = TypeComplete(G, Z)$ ,  $D_R = RelationComplete(G, Z)$  und  $D_X = TransitionComplete(G, Z)$  gilt:

$$Describeable(G, Z) \Leftrightarrow D_T \wedge D_R \wedge D_X$$

Describe erzeugt daraus einen Report mit Status, drei Abdeckungswerten, typisierten Lücken, Ableitungsspur und Grammatikversion. Die Reporting-Funktion trägt nicht selbst die logische Last.

Emergenz wird in vier Klassen bestimmt:

| Klasse | Bedingung                                                                     | Bedeutung                                          |
|--------|-------------------------------------------------------------------------------|----------------------------------------------------|
| E0     | vollständig beschreibbar, keine strukturelle Neuheit                          | bekannte Struktur und Grammatik                    |
| E1     | vollständig beschreibbar, strukturelle Neuheit                                | neue Konfiguration innerhalb der aktiven Grammatik |
| E2     | unvollständig; ein zulässiger GrammarPatch kann Vollständigkeit herstellen    | grammatische Emergenz                              |
| E3     | unvollständig; innerhalb der Meta-Regeln ist kein ausreichender Patch bildbar | meta-grammatische Grenze                           |

E1 verlangt eine operationalisierte Neuheitsprüfung. E2 verlangt einen konkreten Patchkandidaten. E3 ist ein strenger Grenzstatus und darf nicht allein aus fehlenden Daten abgeleitet werden.

## Teil III – Operationale Zustandsmaschine und formale Grammatik

### 19. Konsolidierte operationale Zustandsmaschine

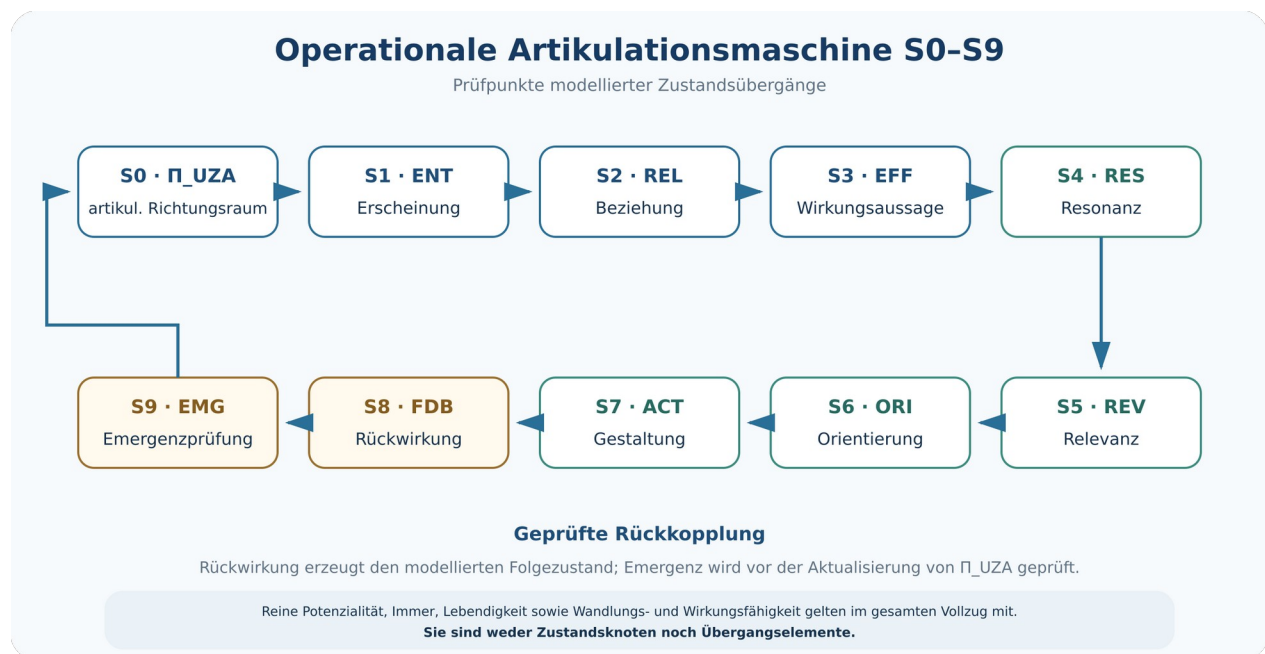
Die Entwicklung der UZA verwendet drei Darstellungen desselben operationalen Artikulationsverfahrens:

- Der frühe Acht-Ebenen-Zyklus beschreibt Erscheinung, Beziehung, Wirkungsaussage, Resonanz, Relevanz, Orientierung, Gestaltung und Rückwirkung.
- Der semantische Kernzyklus ergänzt  $\Pi\_UZA$  als artikulierten Ausgangs- und Rückkehrraum möglicher Richtungen; reine Potenzialität wird dadurch nicht zu einer Zyklusstufe.
- Die operationale Zustandsmaschine 0.9.1 führt dokumentierte Rückwirkung und anschließende Emergenzanalyse als getrennte Prüfschritte artikulierter Zustände und umfasst S0–S9.

Die normative 0.9.1-Maschine lautet:

$S0POT \rightarrow S1ENT \rightarrow S2REL \rightarrow S3EFF \rightarrow S4RES \rightarrow S5REV$

$S5REV \rightarrow S6ORI \rightarrow S7ACT \rightarrow S8FDB \rightarrow S9EMG \rightarrow S0POT$



Die Stufen sind konstitutive Prüfpunkte artikulierter Gestalten und keine starre lineare Benutzeroberfläche. Rekursive Rücksprünge sind zulässig, sofern eine Regel sie erlaubt und der Commit ihre Herkunft dokumentiert. T11 erzeugt aus Ausführung und registrierter Beobachtung zunächst einen rückwirkungsgebundenen Modellfolgezustand. Erst dieser Zustand liefert zusammen mit seinem Vorgänger die vollständige Eingabe für T12. Die Emergenzklassifikation erfolgt deshalb nach FDB und vor Revision, GrammarPatch-Entscheidung und Aktualisierung von

$\Pi$ \_UZA. Damit stimmen Zustandsmaschine, T11/T12, Engine-Hauptzyklus, Pseudocode und Demonstrator-Commits überein.

Geltungsgrenze der Maschine: Immer, reine Potenzialität, Lebendigkeit, Wandlungsfähigkeit, Wirkungsfähigkeit und erinnernde Fortgeltung laufen nicht als zusätzliche Zustände mit. Sie gelten im ganzen Vollzug mit und machen wirkliche Artikulation überhaupt erst denkbar; Übergangspfeile verbinden ausschließlich modellierte Gestalten.

## 20. Drei Wandeltypen

- **W1 – Variation: Artikulierbare Werte oder Instanzen ändern sich innerhalb bestehender Typen, Beziehungen und Regeln.**
- **W2 – Transformation: Eine modellierte Struktur oder ein modellierter Prozess ändert sich durch bereits zulässige Regeln in substanzieller Weise.**
- **W3 – Emergenz: Die artikulierte Veränderung weist strukturelle Neuheit oder eine Grenze der aktiven Grammatik gemäß E1–E3 aus.**

Wandeltyp und Emergenzklasse sind verwandt, aber nicht identisch. W3 bezeichnet die Art eines modellierten Wandels; E0–E3 bezeichnen den Grad seiner grammatischen Beschreibbarkeit. W1–W3 erzeugen weder Lebendigkeit noch Wandlungsfähigkeit, sondern artikulieren bestimmte ihrer Gestalten.

## 21. Transformationsgrammatik T1–T13

### T1 – Erscheinung registrieren

Observation × Perspective × Context × Time → ENT

Voraussetzungen: typisierte Beobachtung, aktive Perspektive, aktiver Kontext. Ergebnis: versionierte Erscheinung mit Provenienz und Unsicherheit.

### T2 – UZA-POT differenzieren

$\Pi$ \_UZA × Perspective × Context → ENT\*

Artikuliert explizite Möglichkeitskandidaten innerhalb  $\Pi$ \_UZA. Kandidaten bleiben von beobachteten Erscheinungen unterscheidbar; reine Potenzialität ist weder Eingabeobjekt noch Ergebnisobjekt von T2.

Die Isabelle-Referenzsemantik 0.9.1 konkretisiert diese Unterscheidung ohne neuen Kerntyp: Eine durch T1 registrierte Beobachtung trägt keine Potentialquelle, keine Potentialbasis und keinen Analysemarker. Ein T2-Kandidat ist dagegen eine K\_ENT-Hypothese mit auflösbarer source\_ref, nichtleerer, vollständig referenzierter basis\_refs-Menge und Status Applicable. T2 nimmt eine endliche, nichtleere Kandidatenmenge entgegen; Objekt- und Versionsidentitäten müssen innerhalb der Menge eindeutig und gegenüber dem Ausgangszustand frisch sein. Diese Kodierung ist eine operationale Provenienzkonvention, keine Behauptung über die Vollständigkeit von  $\Pi$ \_UZA oder reiner Potenzialität.

### T3 - Beziehung bilden

$\text{ENT} \times \text{ENT} \times \text{RelationType} \times \text{Context} \times \text{Time} \rightarrow \text{REL}$

Prüft Endpunkte, Richtung, Typ, Kontext und zeitliche Gültigkeit.

Die Formalrevision T3 konkretisiert diesen Übergang in der integrierten Isabelle-Theorie. Der Ausgangszustand muss `state_well_formed` sein; `CreateRelation` und `K_REL` müssen von der aktiven Grammatik getragen werden. Das neue REL-Objekt erhält frische Objekt- und Versionsidentitäten, zwei verschiedene aktive `K_ENT`-Endpunkte, vollständige Referenzen, gültige Evidenz- und Zeitstruktur sowie einen vorhandenen Kontext. Relationstyp und Richtung werden in einem typisierten, über die Relations-Version adressierten Relationenkatalog fortgeschrieben. Die Zielwohlgeformtheit wird aus den Vorbedingungen und der exakten Objektinsertion hergeleitet und nicht vorausgesetzt. Der Isabelle2025-2-Clean-Build einschließlich T4 endete mit Exit-Code 0.

### T4 - Wirkungshypothese erzeugen

$\text{REL} \times \text{Perspective} \times \text{Context} \times \text{Time} \rightarrow \text{EFF\_H}$

Erzeugt eine ausdrückliche Wirkungshypothese mit offener oder bestimmter Zielstruktur. Der Übergang erzeugt die Aussage, nicht das wirkliche Wirken als solches.

Die Formalrevision T4 macht diese Offenheit typisiert und auditierbar. `effect_hypothesis_payload` bindet jede Hypothese an die verursachende Relationsversion, eine Perspektive, einen Kontext und ein Wirkungsprofilmodell; Ziel und V10-Vektor sind jeweils echte option-Werte und werden weder durch Sentinel-IDs noch durch stilles Nullfüllen ersetzt. `effect_input_valid` verlangt eine aktive typisierte `K_REL`-Ursache, begrenzte Unsicherheit, frische Objekt- und Versionsidentitäten, Lifecycle Hypothesis, vollständige Referenzen sowie vorhandene Perspektive, Kontext und Modellversion. Ein bestimmtes Ziel muss aktiv sein; ein offenes Ziel bleibt `None`. Die Zielwohlgeformtheit folgt aus den Vorbedingungen und der exakten Einfügung. Der Isabelle2025-2-Clean-Build mit sieben Theorien endete mit Exit-Code 0. Dieser konditionale Erhaltungssatz belegt weder empirische Richtigkeit noch Vollständigkeit jeder domänenspezifischen Wirkungshypothese.

### T5 - Evidenz integrieren

$\text{EFF\_H} \times \text{Evidence} \rightarrow \text{EFF\_H} \mid \text{EFF\_SUPPORTED} \mid \text{EFF\_V} \mid \text{REJECTED}$

Die kernelgeprüfte T5-Referenzsemantik aktualisiert kein veröffentlichtes Objekt in-place. Sie konstruiert eine frische Effektversion mit derselben `object_id`, verweist über `previous_version` auf den unveränderten Vorgänger und verschiebt Vorgängerobjekt und Vorgänger-Payload in getrennte Archive. Neue Evidenzquellen müssen aktiv, auflösbar, nicht `Rejected/Archived` und einem anderen logischen Objekt zugeordnet sein. Stützende Evidenz senkt oder erhält Unsicherheit; widersprechende Evidenz erhöht oder erhält sie und führt zu `Rejected`; differenzierende Evidenz muss Ziel oder Wirkungsvektor tatsächlich ändern. Der Betrag jeder Unsicherheitsänderung ist durch die Evidenzstärke begrenzt. Hypothesis darf durch T5 nicht direkt zu `Validated` springen; `Validated` ist nur aus `Supported` bei erfüllter Unsicherheits- und Evidenzschwelle erreichbar. Bereits validierte Effekte werden nicht durch T5 revidiert, sondern



fallen in T13 ReviseObject. Ein nicht-vakuoser Zeuge führt Hypothesis mit unabhängiger Evidenz tatsächlich zu Supported.

### **T6 - Resonanz klassifizieren**

$\text{EFF} \times \text{EFF} \times \text{Perspective} \times \text{Context} \rightarrow \text{RES}$

Die kernelgeprüfte T6-Referenzsemantik verlangt zwei verschiedene aktive K\_EFF-Versionen mit auflösbaren typisierten V10-Payloads im selben Kontext, ein vorhandenes Perspektiv- und Modellprofil, begrenzte Unsicherheit, einen positiven Zeithorizont sowie aktive unabhängige Begründungsreferenzen. Sie konstruiert genau ein frisches K\_RES-Objekt und einen versionierten Resonanz-Payload. Die Zielwohlgeformtheit und Kataloggültigkeit werden aus den Vorbedingungen hergeleitet, nicht vorausgesetzt. Die V10-Summutationsdomäne wird durch `effect_dimension_UNIV` exakt als die zehn Dimensionen bewiesen und ist endlich. Der nicht-vakuose Zeuge verwendet  $K(\text{Connection}, \text{Stability})=1$  bei sonst 0, erhält  $A \rightarrow B=1$  und Amplification, aber  $B \rightarrow A=0$  und Neutral; die unsicherheitsbereinigte Stärke beträgt 0,9 und der Ergebnis-Lifecycle ist Validated. Diese Konfiguration beweist gerichtete Ausführbarkeit, nicht die universelle Gültigkeit genau dieses Kernels.

### **T7 - Relevanz bewerten**

$\text{EFF} \times \text{Perspective} \times \text{Context} \times \text{EvaluationTime} \times \text{RelevanceModelRevision} \rightarrow \text{REV}$

Die nun auch im finalen K1-Sessionbuild geprüfte Referenzsemantik verlangt eine aktive katalogisierte Wirkung mit typisiertem V10-Vektor, ein wohlgeformtes versioniertes Modellprofil, vorhandene Perspektive, Kontext und Modell, zeitliche Zielaktivität, begrenzte Eingaben, aktive unabhängige Begründungsreferenzen und frische Identitäten. Sie trennt V10- und Machtkomponente, erzeugt ein frisches K\_REV-Objekt samt Katalogeintrag und mittelt parallele Perspektiven nicht.

### **T8 - Orientierung synthetisieren**

$\text{REV}^* \times \text{RES}^* \times \text{Goal} \times \text{Constraints} \times \text{Perspective} \times \text{Context} \rightarrow \text{ORI}^*$

Kann mit INSUFFICIENT\_BASIS enden.

### **T9 - Gestaltung auswählen**

$\text{ORI} \times \text{Perspective} \times \text{Authority} \times \text{Context} \times \text{Time} \rightarrow \text{ACT\_SELECTED}$

Dokumentiert Auswahlgrund, Befugnis, Zustimmung und verworfene Alternativen.

### **T10 - Gestaltung ausführen**

$\text{ACT\_SELECTED} \times \text{State} \rightarrow \text{ExecutionRecord} \times \text{State}$

Ausführung ist atomar oder erzeugt einen dokumentierten Teil-/Fehlerstatus.

### **T11 - Rückwirkung erfassen**

$\text{ExecutionRecord} \times \text{Observation}^* \rightarrow \text{FDB} \times \text{State}$



Verknüpft als Beobachtungen registrierte Daten mit der ausgeführten Gestaltung; die Verknüpfung ist eine auditierbare Artikulation, keine Identität mit der wirklichen Rückwirkung.

### T12 - Emergenz analysieren

$\text{State}_t \times \text{State}_{t+1} \times \text{Grammar} \rightarrow \text{EmergenceResult}$

Prüft Neuheit, Beschreibbarkeit, Lücken und mögliche Patches.

### T13 - Revidieren

$\text{Deviation} \times \text{Evidence} \times \text{State} \rightarrow \text{State}'$

Erzeugt neue Objektversionen, Relevanzanpassungen, Orientierungsrevisionen oder einen GrammarPatch-Vorschlag.

## 22. Regelkalkül und Ableitungsprinzipien

Der natürliche Schließungskalkül verwendet intern typisierte Urteile der Form  $\Gamma; G; Z \vdash \text{op} : Z'$ . K2.1 liest dieses Urteil als Projektion des folgenden meta-indizierten Artikulationsurteils:

$$\mathbb{W}_Z \Vdash [ \Gamma ; G ; Z_t^{\wedge L} \vdash \text{op} : Z_t'^{\wedge L'} ]$$

Dabei enthält  $\Gamma$  Perspektiven, Kontexte, Evidenzen, Modelle und Berechtigungen.  $G$  ist die aktive Grammatik,  $\hat{Z}_t^{\wedge L}$  der artikuliert Eingangszustand,  $\text{op}$  die Operation und  $\hat{Z}_t'^{\wedge L'}$  ein möglicher artikulierter Ausgangszustand.  $\mathbb{W}_Z$  ist weder Operand noch Ursache und gehört nicht zum Objektbereich; das Zeichen hält nur die durchgängige Wirklichkeitsgeltung des gesamten Urteils offen.

Ein gültiger Ableitungsschritt benötigt:

1. Typkorrektheit aller Eingaben,
2. erfüllte Vorbedingungen der Regel,
3. zulässigen Lifecycle-Übergang,
4. vollständige Provenienz für neu publizierte Objekte,
5. lokale, transaktionale und globale Validierung,
6. atomaren Commit oder vollständigen Rollback.

Konstitutive Stufen dürfen nicht unbegründet übersprungen werden. Eine direkte Operation  $\text{ENT} \rightarrow \text{ACT}$  ist nur dann zulässig, wenn eine zusammengesetzte Makroregel sämtliche Zwischenableitungen erzeugt und im Proof Trace ausweist.

## 23. Grammatikklassen G0-G4

| Klasse | Inhalt                                | Zulässige Veränderung          |
|--------|---------------------------------------|--------------------------------|
| G0     | primitive Typen, Identitäten und Zeit | nur durch formale Hauptversion |
| G1     | Objektbildungs- und Referenzregeln    | validierter kompatibler Patch  |

| Klasse | Inhalt                                               | Zulässige Veränderung                  |
|--------|------------------------------------------------------|----------------------------------------|
| G2     | Transformationsregeln T1–T13 und Interaktionsregeln  | versionierter Regelpatch               |
| G3     | domänenspezifische Profile, Modelle und Detektoren   | modulare Versionierung                 |
| G4     | Meta-Regeln für Patchbildung, Freigabe und Migration | besonders kontrollierte Meta-Evolution |

Die Klassen verhindern, dass eine Änderung eines Relevanzmodells wie eine Änderung der ontologischen Grundtypen behandelt wird. Je tiefer eine Änderung in G0/G1 eingreift, desto stärker sind Migrations-, Kompatibilitäts- und Beweisanforderungen.

## 24. Konsistenzbedingungen C1–C10

| Code | Bedingung                     | Prüfkern                                                                                         |
|------|-------------------------------|--------------------------------------------------------------------------------------------------|
| C1   | Referenzintegrität            | jede aktive Referenz zeigt auf ein typkompatibles Objekt oder eine zulässige historische Version |
| C2   | Wirkungskontinuität           | jede validierte Wirkung besitzt eine zulässige Herkunft                                          |
| C3   | Perspektivnachvollziehbarkeit | perspektivische Aussagen nennen Perspektive und Version                                          |
| C4   | Kontextbindung                | Gültigkeit und Übertragung nennen Zielkontext                                                    |
| C5   | Zeitbindung                   | Ereignis-, Gültigkeits- und Versionszeit bleiben unterscheidbar                                  |
| C6   | Orientierungsbegründung       | jede Orientierung besitzt Relevanz-, Resonanz- und Zielbasis                                     |
| C7   | Gestaltungsherkunft           | jede Gestaltung verweist auf Orientierung und Entscheidungsbefugnis                              |
| C8   | Rückwirkungsnachweis          | jede publizierte Rückwirkung verweist auf Ausführung und Beobachtung                             |
| C9   | Versionskonsistenz            | ein Commit verwendet kompatible Grammatik-, Semantik-, Modell- und Engineversionen               |
| C10  | historische Rückführbarkeit   | jeder Zustand lässt sich über Commits bis zum initialen Zustand zurückführen                     |

## 25. Meta-Evolution

Der Meta-Evolutionsprozess lautet:

*Proposal → Validation → Approval → VersionRelease → Migration*

Ein GrammarPatch enthält Auslöser, typisierte Lücken, betroffene Regeln und Typen, vorgeschlagene Erweiterung, Minimalitätsmetrik, Kompatibilitätsanalyse, Migrationsplan, Testfälle, Evidenz, Alternativen und Freigabestatus.

Die minimale Erweiterung wird als Optimierungsziel formuliert:

$$\Delta G_{min} = \underset{\Delta G}{argmin} \{ Cost(\Delta G) \mid Describeable(G \oplus \Delta G, Z) \}$$

Cost ist selbst versioniert und kann strukturelle Größe, semantische Reichweite, Migrationsaufwand, Komplexität und Kompatibilitätsverlust gewichten. Ein Minimalitätsresultat ist ohne ausgewiesene Metrik nicht vollständig.

EvolutionMemory speichert versionierte Auditspuren zu Vorschlägen, Annahmen, Ablehnungen, Alternativen, Impact Reports, Evidenzen, Quellfällen, Quellcommits und späterer Bewährung. Dadurch kann die Architektur aus dokumentierter Entwicklung lernen, ohne Regeln stillschweigend umzuschreiben. Dieser Speicher ist ein technisches Erinnerungsmodell; er erzeugt erinnernde Fortgeltung nicht und ist mit ihr nicht identisch.

## 25.1 Geltungs-, Scheiter- und Widerlegungsordnung

Die architektonischen RA1–RA12 sind Konformitätsbedingungen. Ein Fall falsifiziert sie nicht wie eine empirische Hypothese; er kann sie erfüllen, verletzen oder außerhalb ihres erklärten Geltungsbereichs liegen. Empirisch prüfbar und widerlegbar sind dagegen konkrete Wirkungs-, Resonanz-, Relevanz-, Dynamik- und Prognosehypothesen einer Modellinstanz.

Die Engine muss folgende terminalen oder revisionspflichtigen Status unterscheiden:

| Status                 | Bedeutung                                                                                                             | zulässige Folge                                                            |
|------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| INSUFFICIENT_B<br>ASIS | Evidenz oder Begründungsbasis reicht für die angeforderte Operation nicht aus                                         | Daten ergänzen, Ziel begrenzen oder Operation beenden                      |
| MODEL_FALSIFIED        | eine vorab dokumentierte Modell- oder Prognoseaussage widerspricht den zugelassenen Beobachtungs- und Fehlerschwellen | Modell revidieren oder verwerfen; ursprüngliches Ergebnis bleibt erhalten  |
| OUT_OF_SCOPE           | der Fall lässt sich innerhalb des erklärten Anwendungs- und Auflösungsbereichs nicht verantwortbar modellieren        | Geltungsbereich begrenzen oder eigenständiges Theorieprojekt eröffnen      |
| PATCH_REJECTED         | ein GrammarPatch verletzt mindestens ein Annahmekriterium                                                             | Patch archivieren; aktive Grammatik bleibt unverändert                     |
| CORE_CONFLICT          | eine erforderliche Änderung widerspricht G0/G1 oder RA1–RA12                                                          | kein GrammarPatch; gesonderter CoreRevisionProposal mit neuer Hauptversion |
| UNRESOLVED_E<br>3      | der Zustand bleibt innerhalb der freigegebenen Meta-Regeln unbeschreibbar                                             | als Grenze erhalten; keine automatische Erweiterung                        |

Eine Abweichung erzeugt niemals automatisch einen Patch. Der ursprüngliche Prognose-, Validierungs- oder Modellfehler bleibt im Commit Ledger sichtbar und darf durch eine spätere Grammatikversion nicht rückwirkend in einen Erfolg umgedeutet werden.

## 25.2 Patch-Annahme und Abbruchregeln

Ein Patch darf nur freigegeben werden, wenn alle folgenden Bedingungen erfüllt sind:

1. Die auslösende Lücke ist typisiert, reproduzierbar und von bloß fehlenden Daten unterschieden.
2. Eine einfachere Revision von Objekt, Modell, Parameter, Perspektive oder Kontext reicht nicht aus.
3. Der Patch verändert G0/G1 und RA1–RA12 nicht; andernfalls liegt CORE\_CONFLICT vor.
4. Kosten, Reichweite und Kompatibilitätsverlust bleiben innerhalb vorab festgelegter Budgets.
5. Positive Fälle, Gegenbeispiele und mindestens ein nicht zur Patchbildung verwendeter Holdout-Fall sind dokumentiert.
6. Der Patch schließt die Ziellücke, ohne zuvor beschreibbare Kontrollfälle zu beschädigen.
7. Migration, Rückrollplan, Versionierung und historisches Replay sind vollständig.
8. Offene Konflikte, Unmapped-Dispositionen und semantische Verschiebungen werden sichtbar ausgewiesen.

Der Patch wird als PATCH\_REJECTED beendet, sobald eine Pflichtbedingung scheitert. Wiederholte Patches für denselben Lückentyp lösen eine Komplexitäts- und Geltungsbereichsprüfung aus. Ein Patchstrom ohne abnehmende Lücke oder ohne generalisierende Holdout-Leistung führt zu OUT\_OF\_SCOPE oder UNRESOLVED\_E3, nicht zu unbegrenzter Meta-Evolution.

## 25.3 Prüfbarkeit des Universalitätsanspruchs

Der Universalitätsanspruch der UZA ist architektonisch und nicht die Behauptung, jedes Weltphänomen mit einem festen Domänenmodell erklären zu können. Er wird geschwächt oder aufgegeben, wenn wohldefinierte Anwendungsfälle wiederholt nur durch widersprüchliche Kernänderungen, unbeschränkte Patchfolgen oder nicht ausgewiesene Bedeutungsverluste modelliert werden können. Deshalb werden mindestens Repräsentierbarkeit, Verlust, Patchaufwand, Replaytreue und Mehrwert gegenüber einem einfacheren Basismodell vergleichend geprüft.

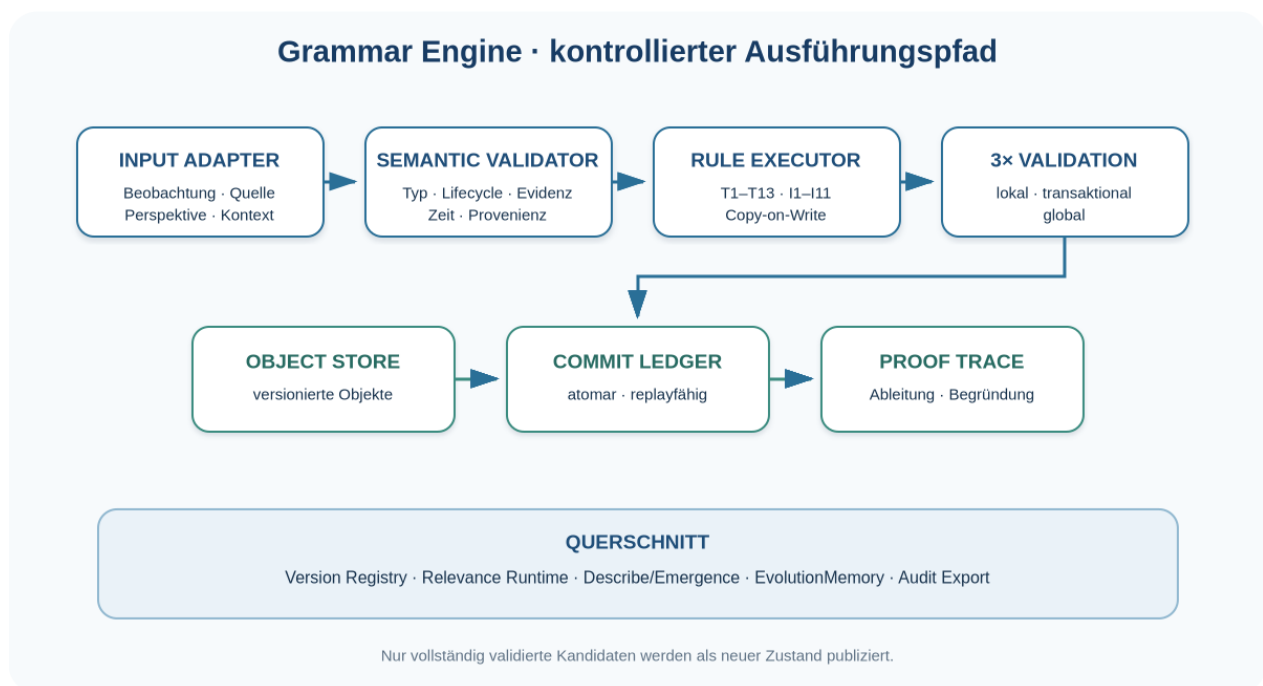
## Teil IV – Grammar Engine und operationale Ausführung

### 26. Engine-Zielbild

Die Grammar Engine führt UZA-Operationen deterministisch relativ zu vollständig versionierten Eingaben aus. Determinismus bedeutet hier:

$$Run(I, V, M, \Theta, D) = Z_n$$

Identische Eingaben  $I$ , Versionen  $V$ , Modelle  $M$ , Parameter  $\Theta$  und dokumentierte Entscheidungen  $D$  müssen dieselbe Zustands- und Commitfolge erzeugen. Diese Reproduzierbarkeit behauptet keinen ontologischen Determinismus der modellierten Wirklichkeit.



### 27. Komponenten der Grammar Engine

#### 27.1 Input Adapter

Normalisiert Beobachtungen, Quellen, Zeitangaben, Perspektiven und Kontexte in typisierte Eingabeobjekte. Der Adapter darf fehlende Daten markieren, aber keine semantische Sicherheit ergänzen, die in der Quelle nicht vorhanden ist.

#### 27.2 Object Store

Hält sämtliche VersionedObject-Instanzen in einem gemeinsamen, unveränderlichen Objektbestand. Neue Commits erzeugen Copy-on-Write-Sichten. Indizes für Beziehungen, Wirkungen, Perspektiven und Kontexte dienen der Abfragebeschleunigung.

### 27.3 Semantic Validator

Prüft Typen, Pflichtfelder, Referenzen, Zeitintervalle, Unsicherheit, Lifecycle, Perspektive, Kontext und Evidenzanforderungen.

### 27.4 Rule Executor

Lädt die anwendbare Regelversion, prüft Vorbedingungen, führt eine reine Zustandsfunktion aus und erzeugt einen Kandidatenzustand mit Proof Trace.

### 27.5 Relevance Runtime

Verwaltet versionierte Relevanzmodellprofile einschließlich getrennter V10- und Machtgewichte, Kontext- und Zeitfaktoren, Sensitivität, Reichweite, Dauerkappung, Mischanteil und Validierungsschwelle. Sie berechnet den sechsdimensionalen Vektor deterministisch, persistiert seine vollständige Provenienz, hält parallele Perspektivordnungen getrennt und verhindert die Vermischung inkompatibler Modelle.

### 27.6 Orientation Synthesizer

Bildet Orientierungsoptionen aus Relevanzen, Resonanzen, Zielen, Ressourcen, Einschränkungen, Macht- und Verantwortungsprofilen. Er gibt die Grundlage und alle wesentlichen Unsicherheiten aus.

### 27.7 Simulation Runtime

Erzeugt prognostizierte Zustände in einem getrennten Simulationszweig. Simulationsergebnisse können Orientierung stützen, werden aber niemals als Beobachtung gespeichert.

### 27.8 Feedback and Revision Engine

Verknüpft Ausführung, Beobachtung, Abweichung und Rückwirkung. Sie erzeugt Revisionen von Wirkungen, Relevanzen, Perspektiven und Orientierungen.

### 27.9 Emergence Analyzer

Berechnet die drei Beschreibbarkeitsprädikate, erstellt DescriptionResult, prüft strukturelle Neuheit und klassifiziert E0–E3.

### 27.10 Meta-Evolution Gateway

Nimmt GrammarPatch-Vorschläge entgegen, führt Kompatibilitäts- und Impact-Prüfungen aus und verwaltet Freigabe und Migration.

### 27.11 Querschnittskomponenten

- **Commit Ledger:** atomare, geordnete Zustandsänderungen.
- **Provenance Store:** Quellen-, Evidenz- und Herkunftsgraph.
- **Version Registry:** Semantik-, Grammatik-, Engine-, Modell- und Transformationsversionen.
- **Integrity Monitor:** globale Invarianten und Zustandsvalidität.

- **Proof Trace:** maschinenlesbare Ableitungs- und Entscheidungswege.
- **EvolutionMemory:** versioniertes Auditgedächtnis der Grammatikentwicklung; keine Speicherontologie der Erinnerung.
- **Audit Exporter:** reproduzierbare Fall-, Zustands- und Beweisexporte.

## 28. Event-Sourcing und Copy-on-Write

Der veröffentlichte Zustand  $Z_t$  bleibt unveränderlich. Eine Operation erzeugt einen Kandidatenzustand  $Z_{t+1}^*$ :

$$T_i(Z_t, \theta) \rightarrow Z_{t+1}^*$$

Erst nach erfolgreicher dreistufiger Validierung wird der Commit atomar publiziert:

$$Validate(Z_t, op, Z_{t+1}^*) = true \Rightarrow Commit(Z_{t+1}^*)$$

Bei Fehlschlag wird der Kandidat verworfen und ein Fehlerobjekt erzeugt. Event-Sourcing speichert die Operation und ihre Eingaben; Copy-on-Write speichert nur geänderte oder neue Objektversionen plus Referenz auf den Vorgängerezustand.

## 29. Commit Ledger

Ein Commit enthält mindestens:

```
commit_id, parent_commit_id, case_id  
timestamp, event_time, actor, authority  
rule_id, rule_version, operation_id  
input_state_id, output_state_id  
parameters, documented_decisions  
created_objects, revised_objects, retired_objects  
evidence_refs, source_refs, perspective_refs, context_refs  
semantic_version, grammar_version, engine_version  
relevance_model_versions, transform_versions  
validation_report, proof_trace, status
```

Atomarität:

$$Commit = SUCCESS \vee ROLLBACK$$

Partielle externe Abläufe wie Transmissionen können mehrere atomare Commits besitzen. Ihr Lifecycle verbindet diese Commits, ohne die Atomarität des einzelnen Zustandswechsels aufzuheben.

## 30. Drei Validierungsebenen

### 30.1 Lokale Regelvalidierung

Prüft Operationstyp, Parameter, Objektstatus, Berechtigung, Regelversion und Vorbedingungen.

### 30.2 Transaktionsvalidierung

Prüft den gesamten Änderungssatz: neu erzeugte und revidierte Objekte, Referenzen, Evidenz, Lifecycle-Übergänge, Konflikte und Rollback-Fähigkeit.



### 30.3 Globale Zustandsvalidierung

Prüft den Kandidatenzustand gegen C1–C18, Identitäts- und Versionsinvarianten, aktive Grammatik, Containment-Wohlgeformtheit und Interaktionsintegrität.

$$Valid_{commit} = Valid_{local} \wedge Valid_{transaction} \wedge Valid_{global}$$

Der Validierungsbericht enthält jede Prüfung mit PASS, FAIL, NOT\_APPLICABLE oder UNDETERMINED. UNDETERMINED kann nur dann publiziert werden, wenn die betroffene Regel dies ausdrücklich zulässt und die verbleibende Unsicherheit dokumentiert ist.

### 31. Regel-Executor

Der Executor verwendet eine vorbereitete Transaktion:

```
execute(state_id, operation, version_bundle):
    state      := load_immutable_state(state_id)
    rule       := registry.resolve(operation.rule_id, version_bundle.grammar)
    local      := validate_local(state, operation, rule)
    if local.failed: return error_commit(local)

    candidate  := rule.apply(copy_on_write(state), operation)
    transaction:= validate_transaction(state, candidate, operation)
    if transaction.failed: return rollback(transaction)

    global     := validate_global(candidate, version_bundle)
    if global.failed: return rollback(global)

    trace      := build_proof_trace(state, operation, candidate)
    commit     := ledger.append_atomic(candidate, trace, version_bundle)
    return commit.output_state
```

Die Regelanwendung soll als reine Funktion implementiert werden. Datenbank-, Netzwerk- und Freigabeeffekte werden vor oder nach der reinen Transformation über kontrollierte Ports verarbeitet.

### 32. Fehler- und Konfliktmodell

Fehlerklassen sind TYPE\_ERROR, REFERENCE\_ERROR, LIFECYCLE\_ERROR, CONTEXT\_ERROR, PERSPECTIVE\_ERROR, EVIDENCE\_ERROR, VERSION\_ERROR, GRAMMAR\_ERROR, VALIDATION\_ERROR, MIGRATION\_ERROR, TRANSMISSION\_ERROR und BOUNDARY\_ERROR.

Ein Fehlerobjekt enthält Code, Kategorie, Schwere, Operation, Regel, Zustand, Objektverweise, verletzte Bedingung, Erklärung, zulässige Reparaturoptionen und Zeit. Fehler verändern den publizierten Fachzustand nicht; sie werden in einem getrennten Auditstrom protokolliert.

Ein Konflikt ist kein technischer Fehler. CONFLICT bewahrt zwei oder mehr semantisch oder evidenziell unvereinbare Aussagen, bis eine ausgewiesene Revision, Perspektivdifferenzierung oder Entscheidungsregel angewandt wird.

### 33. Fünfzehn Mindestanforderungen einer konformen Engine

1. einheitlicher typisierter Object Store,
2. stabile Objekt- und eindeutige Versionsidentitäten,
3. unveränderliche veröffentlichte Zustände,

4. atomare Commits mit Rollback,
5. vollständige Provenienz und Evidence Lineage,
6. explizite Perspektiv-, Kontext- und Dreifachzeitbindung,
7. versionierte Unsicherheit und Lifecycle-Prüfung,
8. dreistufige Validierung,
9. getrennte Simulation und Beobachtung,
10. deterministisches Replay relativ zum Version Bundle,
11. abgeleitetes Describe und E0–E3-Analyse,
12. kontrolliertes GrammarPatch-Gateway,
13. Scale-, Containment- und Interaction-Unterstützung,
14. typisierte partielle Resultate für Transmission und Boundary Operations,
15. vollständiger Audit- und Reproduzierbarkeitsexport.

### 34. Bereinigter Engine-Hauptzyklus

```
Initialize case and version bundle
Register perspectives, contexts and observations
Create or revise appearances
Create relations
Hypothesize effects
Integrate evidence
Classify resonances
Evaluate perspective-bound relevance
Generate orientations or report insufficient basis
Select and authorize an action
Simulate the predicted state
Execute the selected action
Observe the resulting state
Compare prediction and observation
Record feedback and perspective changes
Analyze structural and grammatical emergence
Revise objects or propose a GrammarPatch
Validate locally, transactionally and globally
Commit atomically
```

## Teil V – Zusammenhangsdynamik

### 35. Dynamikmodell

Dynamik analysiert nicht nur Zustände, sondern Trajektorien:

$$\tau(Z) = (Z_{t_0}, Z_{t_1}, \dots, Z_{t_n})$$

Ein DynamicsReport wird aus Historie, Wirkungen, Resonanzen, Rückwirkungen und Zeitfenstern erzeugt:

$$Flow: H(Z) \times W \times Window \times DetectorVersion \rightarrow DynamicsReport$$

Jede Mustererkennung nennt Detektorversion, Eingabefenster, Merkmale, Schwellenwerte, Unsicherheit und Alternativdeutungen. Begriffe wie Attraktor oder Kippunkt erhalten in jeder Domäne eine operative Definition und werden nicht automatisch im strengen Sinn der nichtlinearen Dynamik verwendet.

### 36. Zehn dynamische Grundmuster

| Muster             | Operative Lesart                                                   | Erforderliche Mindestausgabe                     |
|--------------------|--------------------------------------------------------------------|--------------------------------------------------|
| Stabilisierung     | Abweichungen nehmen ab; Struktur und Wirkungen werden beständiger  | Referenzbereich, Dämpfungsmaß, Zeitraum          |
| Destabilisierung   | Abweichungen oder Konfliktwirkungen nehmen zu                      | Wachstumsmuster, betroffene Dimensionen          |
| Attraktorregion    | verschiedene Trajektorien nähern sich einem ähnlichen Zustandsraum | Ähnlichkeitsmetrik, Einzugsbereich, Unsicherheit |
| Kippunkt           | kleine Zusatzänderungen gehen mit qualitativ neuem Verlauf einher  | Vorzeichen, Schwelle, Alternativerklärung        |
| Rekursion          | Ergebnisse werden wieder zu Eingaben desselben Musters             | Rückkopplungspfad, Verzögerung, Verstärkung      |
| Selbstorganisation | koordinierte Struktur entsteht aus lokalen Interaktionen           | lokale Regeln, globale Neuheit, E-Klasse         |
| Synchronisation    | zuvor getrennte Verläufe gleichen Phase, Takt oder Entscheidung an | Synchronisationsmaß, Kanal, Verzögerung          |
| Drift              | langsame gerichtete Verschiebung ohne einzelnen dominanten Sprung  | Trend, Stabilität, Perspektivabhängigkeit        |
| Divergenz          | Perspektiven, Teilzusammenhänge oder Trajektorien entfernen sich   | Distanzfunktion, Treiber, Reichweite             |
| Konvergenz         | zuvor verschiedene Verläufe nähern sich                            | Zielraum, verbleibende Differenzen               |

### 36.1 Referenzsemantik der Detektoren

Für eine gewählte, versionierte Metrik  $m$  und Fensterfolge  $W_1, \dots, W_n$  gelten folgende Mindestentscheidungen:

- **Stabilisierung:** ein robustes Abweichungsmaß fällt über mindestens  $k$  aufeinanderfolgende Fenster um die Schwelle  $\varepsilon_s$ ; Destabilisierung verwendet die entsprechende Zunahme.
- **Drift:** eine robuste Trendsteigung überschreitet  $\varepsilon_d$ , während kein einzelnes Ereignis mehr als den festgelegten Anteil der Gesamtverschiebung erklärt.
- **Konvergenz/Divergenz:** die Distanz zweier Trajektorien fällt beziehungsweise steigt über mindestens  $k$  Fenster; Metrik und Restdistanz werden ausgewiesen.
- **Synchronisation:** die maximale zeitversetzte Korrelation oder ein domänenspezifisches Phasenmaß überschreitet  $\varepsilon_{sync}$  innerhalb des zugelassenen Lags.
- **Attraktorregion:** mindestens zwei unabhängig initialisierte Trajektorien treten in eine definierte Region ein und verbleiben dort für die Mindestdauer  $h$ .
- **Kipppunkt:** ein reproduzierbarer Change-Point, eine qualitative Regimeänderung und Persistenz im Holdout-Fenster liegen gemeinsam vor. Eine bloße Schwellenüberschreitung bleibt Hypothese.
- **Rekursion:** der typisierte Wirkungs- oder Rückkopplungsgraph enthält einen zeitlich konsistenten Zyklus; Verzögerung und Nettoverstärkung werden angegeben.
- **Selbstorganisation:** eine globale neue Koordination ist aus dokumentierten lokalen Interaktionen ableitbar, ohne dass eine einzelne zentrale Operation sie vollständig erzeugt.

Jeder Detektor besitzt `metric_id`, `threshold_profile`, `window_policy`, `minimum_support`, `alternative_tests`, `holdout_policy` und `version`. Ein Resultat ohne diese Angaben bleibt HYPOTHESIS. Diese Referenzsemantik macht die Muster prüfbar, ohne domänenspezifische Metriken fälschlich zu vereinheitlichen.

### 37. Flow Analyzer und Dynamics Detector

Der Flow Analyzer berechnet Ereignisraten, Wirkungsänderungen, Resonanzdichte, Relevanzverschiebungen, Verzögerungen und Feedbackschleifen. Der Dynamics Detector wendet versionierte Musterdefinitionen an und erzeugt Hypothesen oder validierte Dynamics Events.

Detektorresultate besitzen denselben epistemischen Lifecycle wie andere Wirkungsbehauptungen. Eine einzelne Schwellenüberschreitung kann eine Kipppunkthypothese erzeugen; ein validierter Kipppunkt benötigt zusätzliche Evidenz, alternative Schwellenprüfungen und eine nachvollziehbare qualitative Zustandsänderung.

### 38. Relevance Feedback Loop

Rückwirkungen verändern nicht nur Sachobjekte, sondern auch die Relevanzmodelle und deren Eingaben:

$$REV_t \rightarrow ORI_t \rightarrow ACT_t \rightarrow FDB_{t+1} \rightarrow REV_{t+1}$$

Die Engine trennt dabei:

- Änderung eines bewerteten Objekts,
- Änderung einer Perspektive,
- Änderung von Modellparametern,
- Änderung des Relevanzmodells selbst.

Nur die ersten drei können innerhalb derselben Modellhauptversion verarbeitet werden. Eine strukturelle Modelländerung erzeugt eine neue Modellversion und einen Impact Report.

### 39. Perspective Tracker

Der Perspective Tracker erstellt Differenzprofile zwischen  $p_t$  und  $p_{t+1}$ . Er erfasst neue Beobachtungszugänge, Wissensänderungen, Wertungsverschiebungen, Interessenrevisionen, Erfahrung, Macht, Verantwortung, Freiheit und Handlungsraum. Die Änderungen werden nicht als bloßes „Meinungsrauschen“ verworfen, sondern als mögliche Wirkung des Zusammenhangs analysiert.

### 40. Complexity Monitor und Zusammenhangskompetenz

Der Complexity Monitor zählt unter anderem aktive Objekttypen, Regelverzweigungen, Abhängigkeitstiefe, offene Konflikte, GrammarPatches, Projektionverluste und Interaktionskanäle. Er warnt vor wachsender Modellkomplexität, ohne Komplexität selbst als Fehler zu behandeln.

Zusammenhangskompetenz wird auf drei Ebenen getrennt:

- $ZK_{Architektur}$  – Coverage, Consistency, Traceability, Extensibility, Minimality, Compatibility und Explainability,
- $ZK_{Engine}$  – Describe, Predict, Revise, Validate, Detect Emergence, Orient, Scale und Interact,
- $ZK_{Akteur}$  – fallgebundene Fähigkeit, Perspektiven, Folgen, Macht, Verantwortung und Gestaltungsmöglichkeiten angemessen einzubeziehen.

Eine mögliche Architekturmetrik lautet:

Dabei stehen  $Cov$ ,  $Con$ ,  $Trc$ ,  $Ext$ ,  $Min$ ,  $Com$  und  $Exp$  für Coverage, Consistency, Traceability, Extensibility, Minimality, Compatibility und Explainability.

$$ZK_{Architektur} = f(Cov, Con, Trc, Ext, Min, Com, Exp) - \lambda Complexity(G)$$

Minimality bewertet, ob die vorhandenen Typen, Regeln und Erweiterungen für die erzielte Abdeckung erforderlich sind. Complexity(G) misst dagegen die tatsächliche strukturelle Last der Grammatik. Beide Größen können korrelieren, sind aber nicht identisch: Eine kleine Grammatik kann unnötige Elemente enthalten, während eine größere Grammatik für einen nachweislich breiteren Geltungsbereich minimal sein kann. Gewichte, Normalisierung und Aggregationsfunktion bleiben versionierte Bewertungsentscheidungen.

Die drei Ebenen werden nicht zu einem einzigen Gesamtscore vermischt.

## Teil VI – Skalierung und Interaktion

### 41. ScaleProfile

Ein ScaleProfile ist analysegebunden:

Scale bezeichnet ausschließlich die gewählte Reichweite und Auflösung einer Artikulation. Mikro, Meso, Makro, Einbettungstiefe und Container sind analytische Ordnungen; aus ihnen folgt weder eine Rangordnung der Wirklichkeit noch eine ontologische Hierarchie von mehr oder weniger wirklichen Einheiten.

$$ScaleProfile(Z, A) = \{S_x, T_x, P_x, N_x, R_x\}$$

- $S_x$  (SpatialExtent) beschreibt räumliche oder topologische Reichweite.
- $T_x$  (TemporalExtent) beschreibt den betrachteten Zusammenhangszeitraum.
- $P_x$  (PopulationExtent) beschreibt Umfang und Auswahl beteiligter Einheiten.
- $N_x$  (NestingDepth) beschreibt die Einbettungstiefe.
- $R_x$  (Resolution) beschreibt die Analyseauflösung.

Es gilt:

$$Scale(Z) \neq Resolution(A)$$

Die Regelschemata T1–T13 sind skalenstabil; ihre Instanziierung ist skalenabhängig. Ein Beobachtungsobjekt kann auf Mikroebene eine einzelne Interaktion und auf Makroebene ein aggregiertes Verlaufsmuster darstellen, solange Typ, Aggregationsregel und LossProfile transparent bleiben.

### 42. ContainmentGraph

Der Einbettungsgraph lautet:

$$ContainmentGraph_t = \{Z_t, R_{contains,t}\}$$

directContains(a,b) bedeutet modellintern, dass b in der gewählten Artikulation unmittelbar in a eingebettet dargestellt wird. strictContains ist die transitive Hülle; containsEq ergänzt Reflexivität für Identitätsvergleiche. Diese Graphrelation behauptet keine absolute ontologische Einschließung oder Trennung.

Der direkte Graph muss azyklisch sein. Mehrfacheinbettung ist zulässig, deshalb ist die Struktur grundsätzlich ein gerichteter azyklischer Graph und kein Baum. Jede Einbettung ist zustands- und versionsgebunden.

### 43. SUBZ, Projektion und LossProfile

SUBZ( $Z_i, Z_j$ ) bezeichnet  $Z_i$  innerhalb einer gewählten Artikulation als Teilzusammenhang von  $Z_j$ . Im Container erscheint nicht automatisch der vollständige Teilzusammenhang, sondern eine

Projektion. „Teil“ benennt hier eine analytische Beziehung und keine Abtrennung vom gemeinsamen Wirklichkeitszusammenhang:

$$\pi_j(Z_i, A) = PROJ_{i \rightarrow j}^A$$

Projektionsarten sind Auswahl, Aggregation, Abstraktion und Einbettungsprojektion. Ein Homomorphismus ist ein besonders strukturtreuer Spezialfall, keine allgemeine Voraussetzung.

Das LossProfile dokumentiert Verluste auf mindestens vier Dimensionen:

$$Loss(PROJ) = \langle L_{struct}, L_{sem}, L_{persp}, L_{time}, explanation \rangle$$

Jeder Wert liegt im Einheitsintervall. Die qualitative Erklärung nennt ausgelassene Objekte, zusammengeführte Bedeutungen, nicht sichtbare Perspektiven und zeitliche Verdichtungen.

#### 44. InteractionNetwork und Kopplungsobjekt

$$InteractionNetwork_t = \langle \mathbb{Z}_t, \mathbb{C}_t \rangle$$

Eine Kopplung zwischen  $A$  und  $B$  wird beschrieben als:

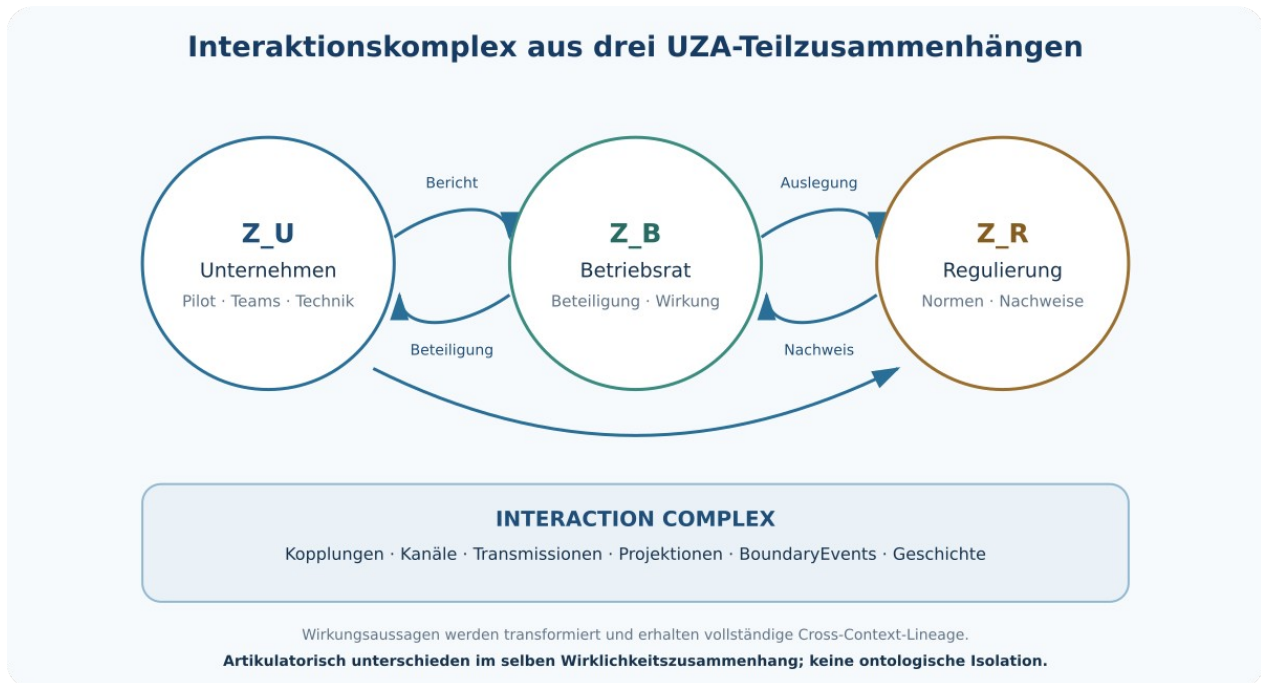
$$C_{AB} = \langle id, A, B, topology, \Pi_{AB}, history \rangle$$

Dabei bündelt  $\Pi_{AB}$  die versionierten Felder direction, channels, functions, strengths, filters, delay und contexts.

Die Stärke ist kanalabhängig:

$$strength: Coup \times Channel \times Time \rightarrow [0, 1)$$

Eine Kopplung kann zugleich kooperative, kompetitive, regulatorische, beobachtende oder austauschbezogene Funktionen tragen. Richtung, Kanal, Filter, Verzögerung und Transformation werden separat versioniert.



## 45. Kontrollierte Wirkungsübertragung

Eine Quell-Wirkungsaussage wird nicht identisch in den Zielkontext kopiert. Die Kanaltransformation erzeugt ein partielles, neu zu prüfendes Artikulationsresultat:

$$\text{Transmit}(EFF_A, C_{AB}, \text{Channel}, CTX_B, t) \rightarrow \begin{cases} \text{Transformed}(EFF_B^{\text{external}}) \\ \text{Blocked}(\text{reason}) \\ \text{Incompatible} \\ \text{Failed}(\text{error}) \end{cases}$$

Die Prüfung umfasst:

1. Existenz und Aktivität des Kanals,
2. zulässige Richtung,
3. Filterbedingungen,
4. Verzögerung,
5. Verstärkung oder Abschwächung,
6. Bedeutungsänderung,
7. Unsicherheitspropagation,
8. Grammatikkompatibilität,
9. Evidenz und Beobachtbarkeit im Zielkontext.

Die Ziel-Wirkungsaussage ist ein neues versioniertes Objekt. Ihre Lineage verweist auf Quell-Wirkungsaussage, Kopplung, Kanal, Transformationsversion und TransmissionEvent; daraus folgt noch keine Behauptung einer objektidentischen Wirkungsübertragung.



## 46. Transmission Lifecycle

*SCHEDULED* → *DELIVERED* → *OBSERVED* → *VALIDATED*

Zusätzliche Resultate sind *BLOCKED*, *INCOMPATIBLE*, *FAILED* und *CANCELLED*. Ein *TransmissionEvent* enthält Quell-Wirkungsaussage, Kopplung, Kanal, Transformation, geplante und tatsächliche Zeiten, Statushistorie, transformierte Wirkungsaussage, Zielbeobachtung und Abweichung.

*DELIVERED* bedeutet lediglich, dass der Kanal die transformierte Wirkungsaussage dem Zielkontext zugeführt hat. Erst *OBSERVED* weist eine registrierte Zielbeobachtung aus; *VALIDATED* verlangt die semantischen und evidenziellen Bedingungen der Zielartikulation.

## 47. Grenztransformationen

### 47.1 Fusion

$Fuse(Z_A, Z_B) \rightarrow Result\{Z_C, \mu_A, \mu_B, BoundaryEvent\}$

Jedes Quellobjekt erhält eine ereignisgebundene Disposition: *Mapped*, *Unified*, *Derived*, *Conflicted* oder *Retired*. Identitätskollisionen werden explizit über *keepSeparate*, *unify*, *deriveNew* oder *conflict* gelöst.

### 47.2 Fission

$Split(Z) \rightarrow Result\{Z_1, \dots, Z_n, Lineage, BoundaryEvent\}, n \geq 2$

*Lineage* ordnet jedes relevante Quellobjekt seinen Zielobjekten und Zielzusammenhängen zu. Leere oder verlorene Zuordnungen benötigen eine dokumentierte Disposition.

### 47.3 Aggregation

$Aggregate(Z_1, \dots, Z_n, \alpha) \rightarrow Z_M$

Die Quellzusammenhänge bleiben aktiv. Aggregation erzeugt eine Makrosicht und ist von Fusion zu unterscheiden.

### 47.4 Embedding und Detachment

*Embed* erzeugt eine Einbettungsrelation und eine Containerprojektion. *Detach* beendet oder versioniert die Einbettung, erhält Historie, *Lineage* und noch aktive Kopplungen. Weitere zulässige Operationen sind *Overlap* und *Decouple*.

## 48. Interaktionskomplex

$I_t = \langle Z_t, \mathbb{C}_t, R_{contains,t}, T_{cross,t}, B_t, H_t \rangle$

Der Interaktionskomplex enthält analytisch unterschiedene UZA-Teilzusammenhänge, Kopplungen, Einbettungen, grenzüberschreitende Transmissionen, *BoundaryEvents* und Auditgeschichte. *Describe\_I* prüft diesen Komplex als solchen. Eine künstliche Fusion in einen

einzigsten Modellzustand ist weder erforderlich noch semantisch neutral; die Unterscheidung der Teilzusammenhänge behauptet umgekehrt keine ontologische Isolation außerhalb des gemeinsamen Wirklichkeitszusammenhangs.

#### 49. Interaktionsoperationen I1–I11

| Code | Operation                   | Ergebnis                                  |
|------|-----------------------------|-------------------------------------------|
| I1   | CreateContext               | neuer versionierter Systemzusammenhang    |
| I2   | EmbedContext                | SUBS-Relation und Projektion              |
| I3   | DefineCoupling              | validiertes Kopplungsobjekt               |
| I4   | ModifyCoupling              | neue Kopplungsversion und Impact Report   |
| I5   | TransmitEffect              | TransmissionEvent und partielles Resultat |
| I6   | ProjectContext              | PROJ mit LossProfile                      |
| I7   | MergeContexts               | FusionResult und BoundaryEvent            |
| I8   | SplitContext                | FissionResult, Lineage und BoundaryEvent  |
| I9   | AggregateContexts           | Makrozusammenhang und Aggregationsregel   |
| I10  | DetachContext               | beendete Einbettung mit Historie          |
| I11  | AnalyzeInteractionEmergence | E0–E3 für den Interaktionskomplex         |

#### 50. Konsistenzbedingungen C11–C18

| Code | Bedingung                          | Prüfkern                                                                      |
|------|------------------------------------|-------------------------------------------------------------------------------|
| C11  | Zusammenhangs-Referenzintegrität   | Kopplungen, Events und Projektionen referenzieren gültige Systemzusammenhänge |
| C12  | SUBZ-Wohlgeformtheit               | direkte Einbettung ist azyklisch und typisiert                                |
| C13  | Projektionsaktualität              | Projektion nennt Quellzustand, Regelversion und LossProfile                   |
| C14  | Kopplungsintegrität                | Endpunkte, Richtung, Kanäle, Filter und Funktionen sind konsistent            |
| C15  | Transmissionsnachvollziehbarkeit   | jedes Zielresultat besitzt vollständige Cross-Context-Lineage                 |
| C16  | Fusionsdispositionsvollständigkeit | jedes relevante Quellobjekt erhält eine Disposition                           |
| C17  | Fissions-Lineage-Vollständigkeit   | jedes relevante Quellobjekt wird zugeordnet oder begründet disponiert         |

| Code | Bedingung              | Prüfkern                                                                                                       |
|------|------------------------|----------------------------------------------------------------------------------------------------------------|
| C18  | Versionskompatibilität | Grammatik, Transformation und Semantik sind über Grenzen kompatibel oder verwenden einen ausgewiesenen Adapter |

## 51. Interaktions-Emergenz

Interaktions-Emergenz liegt vor, wenn ein neuer Zustand oder Verlauf erst durch die Kopplungsstruktur mehrerer Zusammenhänge erklärbar wird. Die Analyse vergleicht:

- Beschreibbarkeit der Einzelzusammenhänge,
- Beschreibbarkeit der Kopplungen und Transmissionen,
- Beschreibbarkeit des Gesamtverlaufs des Interaktionskomplexes,
- strukturelle Neuheit gegenüber den Einzelverläufen.

Ein Interaktionsmuster kann E1 sein, obwohl alle Einzelzusammenhänge vollständig beschrieben sind. E2 liegt vor, wenn etwa ein neuer Kopplungstyp oder eine neue Cross-Context-Transition benötigt wird. Synchronisationstendenzen gelten zunächst als dynamische Hypothesen und werden erst nach Detektor- und Evidenzprüfung validiert.

## Teil VII – Durchgängige Demonstratoren

### 52. Methodischer Status der Demonstratoren

Die Demonstratoren zeigen die vollständige Anwendung der Spezifikation. Sie sind keine empirischen Studien und keine Belege für einen wirtschaftlichen oder sozialen Nutzen. Beobachtungen, Hypothesen, Simulationen und illustrative Werte werden typisiert getrennt. Ihr Zweck ist der Nachweis architektonischer Durchgängigkeit: Jeder Schritt muss Objekte, Regeln, Versionen, Validierung und Commits erzeugen können.

### 53. Demonstrator A – KI-Einsatz in einer Organisation

#### 53.1 Fallrahmen

Ein mittelgroßes Dienstleistungsunternehmen plant die Einführung eines generativen Assistenzsystems für interne Recherche, Textentwürfe und Kundenservice. Der Fallkontext CTX-ORG-AI-01 umfasst einen zwölfwöchigen Pilot. Die aktiven Perspektiven sind:

| Perspektive | Beobachtungsfokus                                     | Ziele und Interessenhypothese n                        | Macht/Verantwortung/ Freiheit                              |
|-------------|-------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------|
| P-MGT       | Kosten, Durchlaufzeit, Qualität, Skalierbarkeit       | kürzere Bearbeitung und nutzbarer Geschäftswert        | hohe Ressourcenmacht; hohe Ergebnisverantwortung           |
| P-EMP       | Arbeitsabläufe, Qualifikation, Belastung, Autonomie   | hilfreiche Entlastung und tragfähige Arbeitsgestaltung | mittlere Prozessfreiheit; unmittelbare Folgenbetroffenheit |
| P-CUST      | Antwortqualität, Datenschutz, Vertrauen               | verlässliche und verständliche Leistung                | geringe interne Macht; hohe Qualitätsbetroffenheit         |
| P-IT        | Integration, Sicherheit, Betrieb, Nachvollziehbarkeit | kontrollierbarer und wartbarer Betrieb                 | hohe technische Zugangskontrolle                           |
| P-COMP      | Recht, Dokumentation, Risikotragung                   | regelkonforme Nutzung und prüfbare Verantwortlichkeit  | Vetorechte in definierten Risikofällen                     |

Der Fall wird zunächst als Einzelzusammenhang artikuliert. Betriebsrat und Regulierung werden im zweiten Demonstrator als analytisch unterschiedene, gekoppelte UZA-Teilzusammenhänge modelliert; dies behauptet keine ontologische Isolation.

#### 53.2 Beobachtungen und Erscheinungen

Die Eingabe enthält fünf typisierte Beobachtungsgruppen:

- unterschiedliche Bearbeitungszeiten bei wiederkehrenden Rechercheaufgaben,
- hohe Varianz in der Qualität bestehender Textentwürfe,
- begrenzte Erfahrung der Beschäftigten mit generativen Systemen,

- bestehende Freigabe- und Datenschutzprozesse,
- Wunsch nach schnellerer, zugleich nachvollziehbarer Bearbeitung.

T1 erzeugt unter anderem ENT-TASK-RESEARCH, ENT-DRAFT-QUALITY, ENT-SKILL-DIFF, ENT-APPROVAL-PROCESS und ENT-TRACEABILITY-NEED. Jede Erscheinung erhält Quelle, Perspektive, Ereigniszeit, Kontext und Unsicherheit.

### 53.3 Beziehungen und Wirkungshypothesen

T3 bildet unter anderem folgende Beziehungen:

| Relation | Quelle → Ziel                         | Typ         | Status     |
|----------|---------------------------------------|-------------|------------|
| REL-01   | Routine-Recherche → Bearbeitungszeit  | conditions  | supported  |
| REL-02   | Skill-Differenz → Ergebnisvarianz     | strengthens | hypothesis |
| REL-03   | Freigabeprozess → Nachvollziehbarkeit | strengthens | supported  |
| REL-04   | Zeitdruck → ungeprüfte Übernahme      | generates   | hypothesis |
| REL-05   | Training → kompetente Nutzung         | strengthens | hypothesis |

T4 erzeugt daraus Wirkungshypothesen. Beispiel EFF-H-04: Unter hohem Zeitdruck kann ungeprüfte Übernahme die Transparenz, Verantwortung und Lernfähigkeit mindern. Die Zielwerte bleiben zunächst hypothetisch.

### 53.4 Evidenzintegration

Der Pilot definiert zulässige Evidenzarten: protokollierte Bearbeitungszeiten, qualitätsgeprüfte Stichproben, dokumentierte Korrekturen, Schulungsabschlüsse, Nutzerfeedback, Vorfallberichte und Freigabeprotokolle. T5 aktualisiert die Wirkungshypothesen nach jedem Pilotfenster.

Eine beobachtete Zeitreduktion darf beispielsweise EFF-H-01 stützen. Sie validiert nicht automatisch die Hypothese einer Qualitätssteigerung. Qualität, Transparenz und Lernfähigkeit benötigen eigene Evidenzpfade.

### 53.5 Wirkungsprofile

Illustratives, explizit nicht empirisches V10-Profil für die Option „Assistenzsystem mit Training, Quellenanzeige und menschlicher Freigabe“ aus P-EMP:

| Dimension  | Wert | Begründung im Demonstrator                              |
|------------|------|---------------------------------------------------------|
| Verbindung | +1   | gemeinsame Review-Praxis schafft zusätzliche Abstimmung |
| Stabilität | +1   | standardisierte Freigabe senkt Prozessvarianz           |
| Offenheit  | +2   | zusätzliche Recherche- und Entwurfsoptionen             |

| Dimension          | Wert | Begründung im Demonstrator                                       |
|--------------------|------|------------------------------------------------------------------|
| Lernfähigkeit      | +2   | Training und Korrekturfeedback werden Teil des Ablaufs           |
| Kooperation        | +1   | Tandem-Reviews fördern Wissensaustausch                          |
| Handlungsfähigkeit | +2   | Routineaufgaben werden schneller vorbereitbar                    |
| Transparenz        | +1   | Quellen- und Änderungsspuren bleiben sichtbar                    |
| Freiheit           | +1   | Beschäftigte können Vorschläge annehmen, ändern oder verwerfen   |
| Verantwortung      | +1   | Freigabeverantwortung und Eskalation sind ausgewiesen            |
| Regeneration       | 0    | im Pilot noch keine belastbare Wirkung auf Erholung festgestellt |

Dasselbe Maßnahmenpaket erhält aus P-IT und P-COMP andere Werte und Begründungen. Die Engine bewahrt diese Profile getrennt und berechnet keine perspektivlose Durchschnittswahrheit.

### 53.6 Resonanz und Relevanz

Der Demonstrator klassifiziert unter anderem:

- Verstärkung zwischen Training und kompetenter Nutzung,
- Ergänzung zwischen Quellenanzeige und menschlicher Freigabe,
- produktive Spannung zwischen kurzer Bearbeitungszeit und gründlicher Prüfung,
- hemmende Spannung zwischen breitem Datenzugriff und Datenschutzanforderungen.

Das Relevanzmodell REV-ORG-PILOT-1.2 bewertet jede Wirkung pro Perspektive. Es gibt Intensität, Unsicherheit, Stabilität, Reichweite, Dauer und Begründung aus. Die Sensitivitätsanalyse zeigt, welche Orientierungen von unsicheren Modellgewichten abhängen.

### 53.7 Orientierungsoptionen

T8 erzeugt vier begründete Optionen:

1. begrenzter Pilot mit Training, Quellenanzeige, Freigabe und Monitoring,
2. technischer Pilot ausschließlich mit synthetischen Daten,
3. Verschiebung bis zur vollständigen Prozess- und Datenschutzklärung,
4. kein breiter Rollout; Nutzung nur für ausgewählte Rechercheaufgaben.

Für jede Option werden erwartete V10-Wirkungen, betroffene Perspektiven, Ressourcen, Einschränkungen, Reversibilität und Rückmeldesignale dokumentiert. Der Demonstrator wählt Option 1, weil sie Lerngewinn ermöglicht, reale Risiken begrenzt und reversibel bleibt. Diese Auswahl ist eine fallgebundene Demonstrationsentscheidung, keine allgemeine UZA-Empfehlung.

### 53.8 Simulation, Ausführung und Beobachtung

Die Simulation prognostiziert sinkende Bearbeitungszeit, zunächst erhöhte Reviewlast, steigende Lernfähigkeit und eine vorübergehende Differenz zwischen erfahrenen und unerfahrenen Nutzenden. Die Ausführung startet mit zwei Teams, drei Aufgabentypen und einem definierten Eskalationsweg.

Nach dem ersten Fenster zeigt die Beobachtung: Bearbeitungszeit sinkt stärker als prognostiziert; Reviewlast ist ebenfalls höher; Quellenverweise sind lückenhaft; Beschäftigte entwickeln neue Peer-Review-Routinen. Diese Beobachtungen erzeugen FDB-Objekte und neue Beziehungen.

### 53.9 Emergenzanalyse

Die Peer-Review-Routine ist strukturell neu, aber mit vorhandenen Typen und Übergängen beschreibbar: E1. Ein zusätzliches Muster – gemeinsames Prompt- und Korrekturwissen als eigenständiges kollektives Lernobjekt – ist mit der aktiven Grammatik nur teilweise darstellbar. Der Demonstrator erzeugt deshalb einen E2-Kandidaten mit GrammarPatch CollaborativeLearningArtifact, statt E3 zu behaupten.

### 53.10 Commitfolge

| Commit    | Operation               | Hauptergebnis                          | Validierung |
|-----------|-------------------------|----------------------------------------|-------------|
| C001      | InitializeCase          | Fall, Version Bundle, Ziel und Kontext | bestanden   |
| C002      | RegisterPerspectives    | fünf versionierte Perspektiven         | bestanden   |
| C003–C007 | T1 RegisterEntity       | Erscheinungen und Beobachtungen        | bestanden   |
| C008–C012 | T3 CreateRelation       | fünf Ausgangsbeziehungen               | bestanden   |
| C013–C017 | T4 HypothesizeEffect    | Wirkungshypothesen                     | bestanden   |
| C018      | T5 EvidenceUpdate       | gestützte und offene Wirkungen         | bestanden   |
| C019      | T6 RegisterResonance    | Resonanznetz                           | bestanden   |
| C020      | T7 EvaluateRelevance    | perspektivische Relevanzvektoren       | bestanden   |
| C021      | T8 GenerateOrientations | vier Optionen                          | bestanden   |
| C022      | T9 SelectAction         | begrenzter Pilot                       | bestanden   |
| C023      | Simulate                | Prognosezweig                          | bestanden   |
| C024      | T10 Execute             | Pilotstart                             | bestanden   |

| Commit | Operation               | Hauptergebnis                              | Validierung   |
|--------|-------------------------|--------------------------------------------|---------------|
| C025   | T11<br>RecordFeedback   | Beobachtungen und Abweichung               | bestanden     |
| C026   | T12<br>AnalyzeEmergence | E1 plus E2-Kandidat                        | bestanden     |
| C027   | T13 Revise              | Quellenanzeige und Reviewprozess revidiert | bestanden     |
| C028   | ProposeGrammarPatch     | CollaborativeLearningArtifact              | Prüfung offen |

## 54. Demonstrator B – Unternehmen, Betriebsrat und Regulierung

### 54.1 Drei analytisch unterschiedene Teilzusammenhänge

Der erweiterte Demonstrator modelliert:

- Z\_U – Unternehmen mit Pilot, Teams, Prozessen und technischen Systemen,
- Z\_B – Betriebsrat mit Beteiligungsprozess, Beschäftigtenrückmeldungen und Vereinbarungen,
- Z\_R – Regulierung mit Normen, Auslegung, Aufsicht und Berichtspflichten.

Die Zusammenhänge werden nicht zu einem einzigen Modell verschmolzen. Jeder besitzt eigene Perspektiven, Grammatikinstanzen, Zeitverläufe und Relevanzmodelle.

### 54.2 ScaleProfiles

| Zusammenhang | Spatial/<br>Topological<br>Extent               | Temporal<br>Extent                        | Population                         | Nesting | Resolution                    |
|--------------|-------------------------------------------------|-------------------------------------------|------------------------------------|---------|-------------------------------|
| Z_U          | Organisation<br>und technische<br>Infrastruktur | zwölfwöchiger<br>Pilot                    | zwei Teams<br>plus Funktionen      | 2       | Aufgaben- und<br>Wochenebene  |
| Z_B          | betriebliche<br>Vertretungsstruktur             | Beteiligungs-<br>und<br>Reviewzyklus      | betroffene<br>Beschäftigtengruppen | 1       | Themen- und<br>Beschlussebene |
| Z_R          | regulatorischer<br>Bezugsraum                   | gültige und<br>erwartete<br>Anforderungen | regulierte<br>Akteure              | 0       | Norm- und<br>Pflichtenebene   |

Die Projektion von Z\_U in Z\_B enthält Arbeitsabläufe, Belastung, Qualifikation, Überwachung und Mitgestaltung; technische Detailparameter werden teilweise abstrahiert. Das LossProfile weist diesen semantischen und strukturellen Verlust aus.



### 54.3 Kopplungen und Kanäle

| Kopplung | Kanal                 | Richtung | Funktion                   | Verzögerung      |
|----------|-----------------------|----------|----------------------------|------------------|
| C_UB     | Pilotbericht          | U → B    | Information/Observation    | 2 Tage           |
| C_BU     | Beteiligungsbeschluss | B → U    | Regulation/Cooperation     | ereignisgebunden |
| C_RU     | Anforderung           | R → U    | Regulation                 | 0–30 Tage        |
| C_UR     | Nachweisbericht       | U → R    | Observation/<br>Compliance | periodisch       |
| C_RB     | Auslegungsinformation | R → B    | Information                | variabel         |

### 54.4 Beispieltransmission

Quell-Wirkungsaussage in Z\_R: Eine neue Nachweisanforderung erhöht die erforderliche Dokumentation. Kanal C\_RU/Requirement transformiert ihre Artikulation in Z\_U zu einer externen Wirkungsaussage über Transparenz, Verantwortung, Handlungsfähigkeit und Stabilität. Die Übertragung wird geplant, zugestellt, im Complianceprozess beobachtet und erst nach Zielkontextprüfung validiert.

Parallel wird ein Teil der Wirkungsaussage über die Unternehmensprojektion an Z\_B sichtbar. Die Transformation betont Beschäftigtenfolgen und Beteiligungsrechte. Es entstehen zwei Ziel-Wirkungsaussagen mit gemeinsamer Quelllineage, aber unterschiedlichen V10-Profilen.

### 54.5 Interaktions-Emergenz und Synchronisation

Die drei Zusammenhänge entwickeln zunächst unterschiedliche Takte. Durch periodische Pilotberichte, Beteiligungsreviews und regulatorische Nachweise entsteht ein gemeinsamer Vierwochenrhythmus. Der Dynamics Detector klassifiziert eine Synchronisationstendenz. Diese ist im Interaktionskomplex E1, weil vorhandene Kanal-, Zeit- und Resonanztypen ausreichen.

Ein neu entstehendes gemeinsames Prüfobjekt, das technische Nachweise, Beschäftigtenwirkung und regulatorische Begründung in einer versionierten Struktur verbindet, kann einen E2-Patch CrossContextAssuranceCase auslösen. Der Patch wird nur vorgeschlagen; seine Aufnahme benötigt Meta-Evolutionsprüfung.

### 54.6 Ergebnis des Demonstrators

Der Durchlauf zeigt die getrennte Identität der drei Zusammenhänge, skalen- und projektionsbewusste Sichtbildung, kontrollierte Wirkungstransmission, kanalabhängige Bedeutungsänderung und Unsicherheit, Interaction Emergence ohne künstliche Fusion, dynamische Synchronisationsanalyse sowie vollständige Cross-Context-Lineage.

## Teil VIII – Referenzimplementierungsdesign und Conformance 0.6/0.9.1

### 55. Technische Architektur

Das Referenzdesign verwendet eine modulare, ereignisbasierte Architektur. Es schreibt keine einzelne Programmiersprache oder Datenbank vor, definiert aber klare Ports:

| Modul              | Verantwortung                                | Persistenz/Port                    |
|--------------------|----------------------------------------------|------------------------------------|
| Case Service       | Fälle, Ziele, Version Bundles                | relational oder dokumentorientiert |
| Object Service     | VersionedObject und Abfragen                 | Object Store                       |
| Graph Index        | Beziehungen, Lineage, Containment, Couplings | Graphindex als abgeleitete Sicht   |
| Rule Runtime       | T1–T13 und I1–I11                            | reine Funktionen plus Registry     |
| Validation Service | lokal, transaktional, global                 | Rule/Invariant Registry            |
| Ledger Service     | atomare Commits und Replay                   | append-only Event Store            |
| Model Runtime      | Relevanz, Simulation, Dynamics               | versionierte Modellartefakte       |
| Describe Service   | Coverage, Gaps, E0–E3                        | Grammar Registry                   |
| Evolution Service  | Patches, Impact, Migration                   | EvolutionMemory                    |
| Export Service     | Auditpakete und reproduzierbare Runs         | signierte/prüfbare Artefakte       |

Object Store, Ledger und Version Registry sind normative System-of-Record-Komponenten. Graph- und Suchindizes können neu aufgebaut werden und dürfen keine eigenständigen Objektversionen erzeugen.

### 56. Logisches Datenmodell

#### 56.1 Version Bundle

```
{
  "specification": "UZA-0.9.1-RC",
  "semantic_core": "S-0.3.1",
  "grammar": "G-0.9.1",
  "engine": "REF-0.6-design+conformance-0.1",
  "relevance_models": ["REV-ORG-PILOT-1.2"],
  "dynamics_detectors": ["DYN-0.8.0"],
  "interaction_transforms": ["TRF-0.9.1"],
  "validation_profile": "VAL-0.9.1-strict"
}
```

#### 56.2 VersionedObject

```
{
  "object_id": "EFF-004",
  "version_id": "EFF-004@3",
  "revision": 3,
  "object_type": "EFFECT",
  "payload": {"cause": "REL-004@1", "target": "ENT-TRACE@2"},
  "uncertainty": {"class": "SUPPORTED", "confidence": 0.72},
}
```

```

"lifecycle": "SUPPORTED",
"valid_time": {"from": "2026-08-01T09:00:00Z", "to": null},
"previous_version": "EFF-004@2",
"perspective_refs": ["P-EMP@2"],
"context_refs": ["CTX-ORG-AI-01@1"],
"source_refs": ["SRC-PILOT-W1"],
"evidence_refs": ["EVD-017@1"]
}

```

### 56.3 Commit

```

{
  "commit_id": "C025",
  "parent_commit_id": "C024",
  "operation": {"id": "OP-025", "rule": "T11", "version": "0.9.1"},
  "input_state": "STATE-024",
  "output_state": "STATE-025",
  "created": ["FDB-001@1"],
  "revised": ["EFF-004@3", "P-EMP@2"],
  "validation_report": "VR-025",
  "proof_trace": "PT-025",
  "version_bundle": "VB-ORG-PILOT-01",
  "status": "SUCCESS"
}

```

## 57. API-Entwurf

| Methode und Pfad               | Zweck                           | Normatives Ergebnis                  |
|--------------------------------|---------------------------------|--------------------------------------|
| POST /v1/cases                 | Fall und Version Bundle anlegen | initialer Zustand und Commit         |
| POST /v1/cases/{id}/objects    | typisiertes Objekt registrieren | VersionedObject oder Fehlerobjekt    |
| POST /v1/cases/{id}/operations | T- oder I-Operation ausführen   | Commit/rollback mit ValidationReport |
| GET /v1/states/{id}            | unveränderlichen Zustand lesen  | Zustand plus Version Bundle          |
| GET /v1/objects/{id}/versions  | Objektgeschichte lesen          | geordnete Versionen und Provenienz   |
| POST /v1/relevance/evaluate    | Relevanzmodell ausführen        | REV-Objekte und Sensitivitätsbericht |
| POST /v1/orientations/generate | Orientierungsraum bilden        | ORI-Objekte oder INSUFFICIENT_BASIS  |
| POST /v1/simulations           | prognostizierten Zweig erzeugen | getrennte Simulationstrajektorie     |
| POST /v1/observations          | reale Beobachtung integrieren   | ENT/FDB und Commit                   |
| POST /v1/describe              | Beschreibbarkeit prüfen         | DescriptionResult                    |

| Methode und Pfad                   | Zweck                           | Normatives Ergebnis                     |
|------------------------------------|---------------------------------|-----------------------------------------|
| POST<br>/v1/emergence/analyse      | E0–E3 bestimmen                 | EMG-Objekt und Gap-Liste                |
| POST<br>/v1/grammar-patches        | Patch vorschlagen               | Metaobjekt im Prüfstatus                |
| GET<br>/v1/commits/{id}            | Audit und Replaybasis lesen     | vollständiger Commit                    |
| POST<br>/v1/exports/replay-package | reproduzierbares Paket erzeugen | Eingaben, Versionen, Ledger, Prüfsummen |

Schreibende Endpunkte sind idempotent über einen operation\_id-Schlüssel. Wiederholung derselben Operation mit identischem Inhalt liefert dasselbe Commitresultat oder einen Identitätskonflikt, aber keinen zweiten fachlichen Commit.

## 58. Conformance- und Teststrategie

### 58.1 Unit- und Property-Tests

- Typ- und Lifecycle-Regeln für jeden Objekttyp,
- Idempotenz von Operationen,
- Unveränderlichkeit publizierter Zustände,
- Eindeutigkeit von Objektversionen,
- vollständiger Rollback bei jeder Validierungsstufe,
- C1–C18 als ausführbare Properties,
- Lineage- und Dispositionsvollständigkeit bei Boundary Operations.

### 58.2 Replay-Tests

Ein Golden Run enthält Eingaben, Version Bundle, dokumentierte Entscheidungen und erwartete Commitfolge. Eine konforme Engine muss dieselben fachlichen Zustände und Proof Traces reproduzieren. Technische Zeitstempel oder interne Speicheradressen sind von der fachlichen Gleichheitsprüfung ausgeschlossen.

### 58.3 Metamorphe Tests

- Permutation unabhängiger Eingaben verändert das Ergebnis nicht.
- Hinzufügen einer unreferenzierten archivierten Objektversion verändert aktive Analysen nicht.
- Wechsel einer Perspektive verändert nur perspektivisch gebundene Ausgaben und ihre Folgeableitungen.
- Erhöhung der Analyseauflösung erzeugt ein dokumentiertes Loss-/Gain-Profil.
- Blockierte Transmission erzeugt keine externe Ziel-Wirkungsaussage.

58.4 Migrations-Tests

Jeder GrammarPatch benötigt Vorher-/Nachher-Fälle, Kompatibilitätsklassifikation, Migrationsabbildung, Rückrollplan und Beweis, dass historische Runs mit ihrem ursprünglichen Version Bundle weiter reproduzierbar bleiben.

58.5 Ausgeführte Golden Runs 0.9.1

Der mitgelieferte Conformance-Harness ist ein kleiner, deterministischer Referenzausschnitt und keine Produktionsimplementierung. Er verwendet ausschließlich versionierte JSON-Eingaben und semantisch kanonisierte JSON-Ausgaben. Ein Replay besteht nur, wenn das berechnete Ergebnis exakt dem gepinnten Golden Output entspricht.

| Run                       | Prüfswertpunkt                                                                                                     | Ergebnis  | Resultat-SHA-256                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------|
| GR-A-ORG-AI-0.9.1         | End-to-End-Kernlauf, ausdrücklich T11/FDB vor T12/EMG, E2 und zulässiger Patch                                     | bestanden | b971f6a159c494d4822e6d718af3420b31c40ae8b20453882ecf952f575cdf09  |
| GR-B-MULTI-CONTEXT-0.9.1  | zwei Transmissionen, vollständige Lineage, LossProfile, PowerShift, Nicht-Fusion und E1                            | bestanden | 3b539589f7837f45c36789f38d1b163408fbb1a3680320e667edd99882e8054d  |
| GR-C-NEGATIVE-GATES-0.9.1 | OUT_OF_SCOPE, INSUFFICIENT_BASIS, MODEL_FALSIFIED, CORE_CONFLICT, PATCH_REJECTED und verlustbewusste V10-Migration | bestanden | bf3a95812c04e12c5538255626221434a1ed9580ee3a15ebeatf08c94175437cc |

Zusätzlich bestehen drei automatisierte Testgruppen: exakter Replay und normative Reihenfolge; Transmission-Lineage und Nicht-Fusion; negative Geltungs-, Patch- und Migrationsgates. Der Testlauf wurde am 1. August 2026 lokal mit Python ausgeführt. Die Golden Runs belegen die ausführbare Bedeutung des geprüften Ausschnitts, nicht Vollständigkeit, Performance oder Produktionsreife der gesamten Engine.

59. Implementierungsstatus

Das technische Design, das logische Datenmodell, die API, der Regel-Executor und die vollständige Zielarchitektur bleiben spezifiziert. Für 0.9.1 existiert nun zusätzlich ein ausführbarer, deterministischer Conformance-Harness mit drei gepinnten Golden Runs und automatisierten Negativtests. Noch ausstehend sind die vollständige produktive Referenzengine, persistenter Object Store und Commit Ledger, Performance- und Sicherheitsprüfung, die Verknüpfung der Engine-Conformance mit formalisierten Isabelle-Pre-/Postconditions sowie ein unabhängiger Replay durch eine zweite Implementierung. Der korrekte Status ist **Reference Implementation Design 0.6/0.9.1 plus Reference Conformance Harness 0.1.**

## Teil IX – Formale Theorie und mathematische Schließung

### 60. Logische Grundlage

Die primäre Spezifikationssprache ist many-sorted Higher-Order Logic auf Basis einfacher Typentheorie. Zielsystem ist Isabelle/HOL; Lean 4 bleibt eine mögliche alternative Mechanisierung. Eine FOL-Kernreduktion wird als separates Artefakt für geeignete Fragmente entwickelt und ersetzt die HOL-Haupttheorie nicht.

Die Theoriearchitektur wird streng getrennt:

1. primitive Typen und Signatur,
2. Definitionen und abgeleitete Prädikate,
3. statische Axiome,
4. Übergangssystem und Operationsvorbedingungen,
5. Metatheorie, Modelle und Beweise.

Diese Trennung verhindert, dass ein gewünschtes Theorem still als Definition oder Axiom eingebaut wird.

### 61. Konsolidierte Signaturentscheidungen

- Persistierte Objekte verwenden in der Isabelle2025-2-kernelgeprüften Theorie den Record `uza_object`; `object_kind` unterscheidet die zulässigen Objektarten explizit.
- Objekt- und Versionsidentität sind getrennte Synonyme; Gültigkeit wird über `valid_from_of` und optionales `valid_to_of` modelliert.
- V10-Werte sind ganzzahlig und Relevanz-, Loss- sowie Machtwerte reell; zulässige Wertebereiche werden durch explizite Gültigkeitsprädikate statt durch bereits abgeschlossene typedef-Beweise begrenzt.
- `boundary_event`, `projection`, `coupling` und `transmission` sind eigene Records mit Lifecycle-, Loss-, Lineage- und Vollständigkeitsbedingungen.
- `strict_contains` ist zustandsgebunden als transitiver Abschluss der direkten Einbettung definiert.
- `step : grammar  $\Rightarrow$  snapshot  $\Rightarrow$  operation  $\Rightarrow$  snapshot  $\Rightarrow$  bool` bildet das Übergangssystem und bindet Grammatik, Vor- und Folgezustand ausdrücklich ein.
- `Describe` ist ein abgeleiteter Report aus drei unabhängigen Vollständigkeitsprädikaten.
- Freitext kann Erklärungen tragen, ersetzt aber keine semantischen Identitäten.

Das vollständige ScaleProfile, typisierte Ziele und Gap-/Loss-Taxonomien bleiben normative Bestandteile der 0.9.1-Spezifikation. Ihre vollständige Abbildung in zusätzliche Isabelle-Records gehört zum offenen M4-Ausbau und wird hier nicht als bereits mechanisiert ausgegeben.

Die Datei `UZA_0_9_1.thy` setzt diese Entscheidungen als Isabelle2025-2-kernelgeprüfte Mechanisierungsbasis um. Anders als die 0.9-Signaturskizze enthält sie für die zentralen Begriffe tatsächliche Referenzdefinitionen:

| Bereich               | Definierter semantischer Kern                                                                                                                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zustand und Übergang  | Referenz-, Zeit- und objekttypspezifische Wohlgeformtheit, state_well_formed, step                                                                                                                                     |
| Evidenzintegrität     | evidence_references_complete, evidence_well_formed; auflösbare Evidenz-IDs und nichtleere Evidenz für Supported/Validated                                                                                              |
| Beschreibbarkeit      | type_complete, relation_complete, transition_complete, describe                                                                                                                                                        |
| Emergenz              | structural_novelty, repairable, emergence_of für E0–E3                                                                                                                                                                 |
| Wirkung und Resonanz  | effect_vector_valid, Skalarprodukt, resonance_score, classify_resonance                                                                                                                                                |
| Relevanz              | T7: versioniertes RelevanceModelProfile; getrennte V10- und Machtgewichte; kontextuelle und zeitliche Faktoren; Intensität, Unsicherheit, Stabilität, Reichweite, Dauer und Erklärung; keine stille Perspektivmittlung |
| Macht                 | separates Fünf-Achsen-Profil und power_shift                                                                                                                                                                           |
| Dynamik               | Stabilisierung, Destabilisierung, Konvergenz, Divergenz und Drift über endliche Trajektorien                                                                                                                           |
| Geltung und Scheitern | applicability_of unterscheidet Applicable, OutOfScope, InsufficientBasis und ModelFalsified                                                                                                                            |
| Meta-Evolution        | patch_admissible, patch_status, apply_patch, repairable; Kernänderung, Holdout- oder Replayfehler brechen ab                                                                                                           |

## 62. Natürlichsprachliches Axiomregister A1–A22

Die folgenden Aussagen bilden das kontrollierte Register. Sie sind in der Isabelle2025-2-kernelgeprüften Theorie 0.9.1 als universell quantifizierte, typisierte Annahmen des Locale `uza_model` formuliert. Isabelle2025-2 hat die vollständigen Sessions erfolgreich geparkt und typgeprüft. Die Annahmen sind dadurch nicht als Sätze aus einer schwächeren Theorie hergeleitet; ihre gemeinsame Erfüllbarkeit ist jedoch durch Base und zusätzlich durch die reichhaltige Interpretation Rich mechanisch belegt. Base bleibt der minimale Erfüllbarkeitszeuge. Rich enthält zwölf versionierte Objekte und aktiviert mit konkreten Antezedenzen sämtliche objektspezifischen Bedingungen A5–A13 sowie die Interaktionsbedingungen A17–A21.

| Axiom | Formalisierter Inhalt                                                                |
|-------|--------------------------------------------------------------------------------------|
| A1    | Objektversionen mit gleicher version_id sind identisch.                              |
| A2    | Versionen desselben Objekts bilden eine wohlfundierte Revisionskette.                |
| A3    | Gültigkeitsintervalle sind zeitlich wohlgeformt.                                     |
| A4    | Aktive Objekte referenzieren typkompatible aktive oder zulässig historische Objekte. |
| A5    | Jede Relation besitzt existierende, typkorrekte Endpunkte.                           |

| Axiom | Formalisierter Inhalt                                                                                                                                                   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A6    | Jede validierte interne Wirkung besitzt eine relationale Herkunft.                                                                                                      |
| A7    | Perspektivische Objekte referenzieren eine gültige Perspektivversion.                                                                                                   |
| A8    | Kontextgebundene Objekte referenzieren einen aktiven oder historisch ausgewiesenen Kontext.                                                                             |
| A9    | Jede Evidenz-ID löst auf eine aktuelle oder historische Objektversion auf; Objekte im Status Supported oder Validated besitzen zusätzlich eine nichtleere Evidenzmenge. |
| A10   | Jede Relevanz nennt Ziel, Perspektive, Kontext, Zeit und Modellversion.                                                                                                 |
| A11   | Jede Orientierung besitzt eine nichtleere, gültige Begründungsbasis oder den Status INSUFFICIENT_BASIS.                                                                 |
| A12   | Jede Gestaltung verweist auf eine gültige Orientierung und Entscheidungsbefugnis.                                                                                       |
| A13   | Jede Rückwirkung verweist auf Ausführung und Beobachtung.                                                                                                               |
| A14   | Ein erfolgreicher step erzeugt einen wohlgeformten Folgezustand.                                                                                                        |
| A15   | strict_contains ist irreflexiv.                                                                                                                                         |
| A16   | contains_eq ist antisymmetrisch.                                                                                                                                        |
| A17   | Jede Projektion referenziert eine gültige Einbettung, einen Quellzustand und ein LossProfile.                                                                           |
| A18   | Jede aktive Kopplung besitzt gültige Endpunkte und mindestens einen gültigen Kanal.                                                                                     |
| A19   | Transmissionstatus folgt einem zulässigen Lifecycle.                                                                                                                    |
| A20   | Jede transformierte externe Wirkung besitzt vollständige Transmission Lineage.                                                                                          |
| A21   | BoundaryEvents erfüllen je nach Art vollständige Lineage oder Disposition.                                                                                              |
| A22   | Emergenzklassifikation folgt ausschließlich den definierten Beschreibbarkeits- und Neuheitsbedingungen.                                                                 |

### 63. Konditionale Existenzaxiome der Interaktion

Die Interaktionserweiterung soll keine unnötigen globalen Existenzannahmen erzwingen. Axiome werden konditional formuliert. Beispiel: Nur wenn eine erfolgreiche Transmission vorliegt, muss ein Zielresultat mit vollständiger Lineage existieren. Nur wenn ein FusionResult Succeeded meldet, müssen Zielzusammenhang und Dispositionen vollständig sein.

Dieses Muster ermöglicht Modelle des stabilen Kerns ohne Kopplungen oder BoundaryEvents. Die Interaktionstheorie erweitert solche Modelle nur dann um zusätzliche Objekte, wenn die entsprechenden Operationen vorkommen.



## 64. Relative Konsistenz und Modell-Expansion

Das zentrale Modellziel lautet:

Zu jedem Modell des stabilen Kerns soll unter expliziten Zusatzbedingungen eine Expansion zu einem Modell der Scale-&-Interaction-Erweiterung konstruiert werden können.

Die geplante Konstruktion ergänzt:

1. eine möglicherweise leere Menge von Systemzusammenhängen,
2. einen azyklischen ContainmentGraph,
3. Kopplungen und Kanäle mit gültigen Endpunkten,
4. TransmissionEvents mit Lifecycle und Lineage,
5. BoundaryEvents mit Disposition und Lineage,
6. InteractionComplex und DescriptionResult.

Gelingt die Expansion aus jedem Kernmodell, dessen Zusatzbedingungen erfüllt sind, folgt relative Konsistenz der Erweiterung relativ zur Basistheorie. Konservativität ist ein stärkeres, optionales Ziel und wird nur für präzise definierte Sprachfragmente behauptet.

**Zweiter Modellschritt abgeschlossen.** UZA\_0\_9\_1\_Base\_Model konstruiert weiterhin den minimalen endlichen Erfüllbarkeitszeugen. Ergänzend konstruiert UZA\_0\_9\_1\_Rich\_Model ein wohlgeformtes Snapshot mit Relation, validierter Wirkung, Resonanz, Relevanz, Orientierung, Handlung und Rückwirkung sowie zwei Kontexten, Projektion, Kopplung, validierter Transmission und erfolgreichem Fusion-Event. Die Lemmata rich\_A5\_relation\_nonvacuous bis rich\_A17\_A21\_interaction\_nonvacuous zeigen ausdrücklich, dass die betreffenden Axiome nicht nur durch falsche Antezedenzen erfüllt werden. Rich: uza\_model interpretiert erneut A1–A22, und rich\_uza\_model\_exists beweist die Existenz dieser Struktur relativ zu Isabelle/HOL. Offen bleiben die Expansion eines beliebigen Kernmodells und der Konservativitätsnachweis.

## 65. Beweisziele

Die Beweisreihenfolge lautet:

1. Nichtleerer Zustandsraum mit mindestens einem versionierten Objekt – für den Basismodellzeugen abgeschlossen,
2. Wohlgeformtheit der im Zeugen verwendeten Basisrecords und Summentypen – abgeschlossen,
3. Identitäts- und Versionsinvarianten – für T1 RegisterAppearance konstruktiv erhalten,
4. Azyklizität, Irreflexivität und Antisymmetrie der konkreten Einbettung – für den reichhaltigen Zeugen abgeschlossen; als allgemeine Erhaltungssätze offen,
5. Lifecycle und Lineage einer validierten Transmission – im reichhaltigen Zeugen aktiviert; allgemeine Erhaltung offen,
6. Totalität beziehungsweise typisierte Partialität der Grenzoperationen,
7. Lineage- und Dispositions Vollständigkeit – für den konkreten Fusion-Zeugen aktiviert; allgemeine Sätze offen,
8. Korrektheit der Describe-Aggregation,

9. Exklusivität und, unter Vorbedingungen, Vollständigkeit von E0–E3,
10. Modell-Expansion und relative Konsistenz,
11. optionale Konservativität,
12. FOL-Reduktion ausgewählter Fragmente.

Die Haupttheorie enthält sechs mechanisch geprüfte Haupttheorie-Sentinels: Zustandserhaltung aus A14, Transmission-Lineage aus A20, Korrektheit der Describe-Aggregation, Totalität der vier Emergenzfälle, Abbruch kernverändernder Patches und Wertebereich der Relevanz. Ihre Beweise wurden im erfolgreichen Isabelle2025-2-Session-Build vom Kernel akzeptiert. Die ersten beiden gelten relativ zu den Locale-Annahmen; die übrigen folgen aus den ausführbaren Referenzdefinitionen. Zwei zusätzliche A9-Lemmata sichern die geschlossene Bedeutung ab. Die Rich-Model-Session ergänzt neun explizite Nichtvakuositätszertifikate für A5–A13 und A17–A21 sowie eine zweite vollständige Locale-Interpretation. Die separate Operationssession ergänzt erstmals eine konkrete Kernoperation: Für RegisterAppearance werden Folgezustand, exakte Delta-Eigenschaft, Versionsfrische, Identitätserhaltung, Evidenzintegrität, vollständige Zustandswohlgeformtheit und Refinement zu step ohne vorausgesetzte Zielwohlgeformtheit bewiesen.

Für T3 CreateRelation liegt nun eine vollständige, kernelgeprüfte operationsspezifische Mechanisierung vor. Sie umfasst relation\_type, gerichtete oder ungerichtete Katalogmetadaten, zwei verschiedene aktive ENT-Endpunkte, frische REL-Identität, exakte Snapshot- und Katalogdifferenz, Erhaltung von state\_well\_formed, Identität, Referenzen, Zeit, Kontext und Evidence Integrity sowie das Refinement zu step G s CreateRelation s'. Der reale Build deckte zwei lokale Beweislücken auf und führte zu den expliziten Endpunkt-Referenz- und Katalogtypbeweisen. Der aktuelle T3-Quellhash lautet  
a3c801eaf1640624ee7bf41ba91a20d2ba1e000e489ee40f4d3c209b15911ca3.

Für T4 HypothesizeEffect liegt ebenfalls eine vollständige, kernelgeprüfte operationsspezifische Mechanisierung vor. Sie umfasst den typisierten effect\_hypothesis\_payload, explizite Offenheit von Ziel und Wirkungsvektor, eine aktive K\_REL-Ursache, Perspektiv-, Kontext- und Profilmodellbindung, begrenzte Unsicherheit, frische K\_EFF-Identität, exakte Snapshot- und Katalogdifferenz, Erhaltung von state\_well\_formed, Identität, Katalog, Referenzen, Zeit, Perspektive, Kontext und Evidence Integrity sowie das Refinement zu step G s HypothesizeEffect s'. Der aktuelle T4-Quellhash lautet  
2eb8e1d262cb3790f6c446839fb6809de3c2837d20705925507e272c860e2fc4.

Für T5 IntegrateEvidence liegt eine vollständige, kernelgeprüfte operationsspezifische Mechanisierung vor. Sie umfasst Copy-on-Write-Versionierung, vollständiges Objekt- und Payload-Archiv, einen versionierten Evidenz-Ledger, drei Evidenzdispositionen, kontrollierte Unsicherheitsänderung, Lifecycle-Klassifikation ohne Hypothesis → Validated-Sprung, Ausschluss aktueller und unmittelbarer Selbststützung, exakte Zustands- und Audit-Store-Differenzen, Erhaltung von state\_well\_formed, Identität, Aktiv/Historie-Disjunktheit, Katalog, Referenzen, Zeit, Perspektive, Kontext und Evidence Integrity sowie Refinement zu step G s IntegrateEvidence s'. Der zusätzliche endliche Zeuge aktiviert diese Semantik in einem konkreten Hypothesis → Supported-Lauf. Der T5-Quellhash lautet

4422076e2bd6cb2ff9fe5e2d5e2e135e3bf3c05919a1ca03a42b1faccd5fddc0; der Zeugenhash lautet f5e037e578335431903754153deaedeadd461fed8d359a0511e36b59a825a7d.

Für T6 ClassifyResonance liegt eine vollständige, kernelgeprüfte operationsspezifische Mechanisierung vor. Sie umfasst eine gerichtete versionierte Interaktionsmatrix, formal geschlossene Endlichkeit der zehn V10-Dimensionen, Profilwohlgeformtheit, normalisierten und auf  $[-1,1]$  begrenzten Score, totale Fünf-Formen-Klassifikation, unsicherheitsbereinigte Stärke in  $[0,1]$ , aktive verschiedene Effektoobjekte, unabhängige Begründungsreferenzen, konstruktive Objekt- und Payload-Erzeugung, exaktes Zustandsdelta, Katalogfortschreibung, Erhaltung von state\_well\_formed, Identität, Effect- und Resonance-Katalog, Referenzen, Evidenz, Zeit, Perspektive und Kontext sowie Refinement zu step G s ClassifyResonance s'. Der nicht-vakuose Zeuge beweist zusätzlich eine echte Asymmetrie  $A \rightarrow B \neq B \rightarrow A$ . Der T6-Quellhash lautet 3f3583b42f066eaf41208a22bcb53a2d49baed374aa1b41e00a434256741b1cd; der Zeugenhash lautet 494da00eeb17205001ae85090d940a12cd07215ec283d76833aa480126a716ed.

Für T7 EvaluateRelevance liegt nun eine vollständige konstruktive operationsspezifische Mechanisierung vor. Sie umfasst ein versioniertes RelevanceModelProfile, endliche getrennte V10- und Machtdimensionen, wohlgeformte Gewichte und Faktoren, ausführbare Intensitäts-, Unsicherheits-, Stabilitäts-, Reichweiten- und Dauersemantik, Wertebereichssätze, zeitliche Zielaktivität, aktive unabhängige Begründung, Lifecycle-Klassifikation, frische K\_REV-Konstruktion, exaktes Delta, Erhaltung von state\_well\_formed, Identität, Effect-, Resonance- und Relevance-Katalogen, Referenzen, Zeit, Perspektive und Kontext sowie Refinement zu step G s EvaluateRelevance s'. Der nicht-vakuose Zeuge führt zwei aufeinanderfolgende Bewertungen desselben Effekts aus und beweist die exakten, voneinander verschiedenen Vektoren  $P0 = \langle 9/10, 1/10, 49/50, 4/5, 8 \rangle$  und  $P1 = \langle 9/50, 1/10, 19/20, 2/5, 4 \rangle$  ohne Mittelung. Der T7-Quellhash lautet 6377ca4dd963832712350d10d8ad7442a1245b9e4ee76e85c43f6634bc8ecc46; der Zeugenhash lautet 3ba9e93262c0b139cf0d8521e8785ef21b66fb30de3e9e9bfa9de3989e817f39.

## 66. Roadmap M0–M6

| Phase | Ziel                                                               | Aktueller Stand                                                                                                                                           |
|-------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| M0    | logische Grundlage wählen                                          | HOL entschieden                                                                                                                                           |
| M1    | Signatur schließen                                                 | abgeschlossen für 0.9.1; Isabelle2025-2-Parser und Typsystem bestanden                                                                                    |
| M2    | Signatur, Definitionen, Axiome, Transition und Metatheorie trennen | in der Quelldatei umgesetzt                                                                                                                               |
| M3    | Basistheorie A1–A22 vollständig in HOL formulieren                 | Locale-Annahmen einschließlich A9-Evidenzintegrität kernelgeprüft; minimale und reichhaltige Interpretation bestanden                                     |
| M4    | Scale-&-Interaction-Erweiterung formal ergänzen                    | Records und A17–A21 integriert; konkrete Projektion, Kopplung, validierte Transmission und Fusion im Rich Model kernelgeprüft; allgemeine Erhaltung offen |

| Phase | Ziel                                                                             | Aktueller Stand                                                                                                                                                                                                   |
|-------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M5    | Erfüllbarkeit, Expansion, relative Konsistenz, optional Konservativität beweisen | gemeinsame Erfüllbarkeit und konkrete Nichtvakuosität von A5–A13/A17–A21 relativ zu HOL belegt; allgemeine Expansion und Konservativität offen                                                                    |
| M6    | Isabelle-Mechanisierung und optionale FOL-Reduktion                              | Normaler Isabelle2025-2-Clean-Build der vollständigen 25-Theorien-K3-Kette bestanden; T1–T13 einschließlich kausaler Erfolgs-/Abbruchzeugen abgeschlossen; K2.1-Geltungsrevision fortgeführt; FOL-Reduktion offen |

## 67. Mathematischer Netto-Status

| Bereich                                   | Reife                                                   | Aussage                                                                                                                                            |
|-------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Formale Typ- und Abhängigkeitsarchitektur | weitgehend konsolidiert                                 | Artikulationsontologie präzisiert; keine Behauptung ontologischer Vollständigkeit                                                                  |
| Semantische Präzision                     | sehr hoch                                               | Gültigkeit, Unsicherheit, Perspektive und Describe getrennt                                                                                        |
| Grammatik/Engine                          | 0.9.1-RC                                                | spezifiziert; begrenzter Conformance-Harness und Golden Runs bestanden                                                                             |
| Scale & Interaction                       | 0.9-RC                                                  | Operationen, Lineage und Bedingungen spezifiziert                                                                                                  |
| Dynamik                                   | spezifikatorisch integriert                             | Detektoren domänenspezifisch; HOL offen                                                                                                            |
| HOL-Theorie                               | Isabelle2025-2: normaler K3-Clean-Build mit 25 Theorien | zentrale Semantik, A1–A22, Modellzeugen und T1–T13-Erhaltung kernelgeprüft; kausale Zeugen bis T13; A1–A22 bleiben Locale-Annahmen                 |
| Modelltheorie/ Beweistheorie              | Rich Witness + T1–T8 Preservations Kernel-Checked       | konkrete gemeinsame Erfüllbarkeit relativ zu HOL sowie T1–T8-Erhaltung belegt; allgemeine Expansion, relative Konsistenz und Konservativität offen |

Der Isabelle2025-2-Clean-Build belegt den 25-Theorien-K3-Unterbau mit Exit-Code 0. T1–T13 erhalten die ausgewiesenen Invarianten; der kausale Zeuge erreicht sämtliche Schritte. T9–T13 schließen Autoritäts- und Zulässigkeitsprüfung, Auswahl und Ausführung, Rückmeldung, spurrelative Emergenzanalyse und Copy-on-write-Revision. Offen bleiben Produktivengine-Äquivalenz, allgemeine Modell-Expansion, relative Konsistenz, Konservativität und empirische Universalität.

## Teil X – Gesamteinordnung

### 68. Abschließende Definition

**Die Universelle Zusammenhangsarchitektur ist eine versionierte, typisierte und semantisch validierte Artikulationsgrammatik zur reproduzierbaren Erfassung, Analyse, Orientierung, Gestaltung und kontrollierten Weiterentwicklung von Darstellungen komplexer, dynamischer und interagierender Zusammenhänge. Sie ist keine erschöpfende Ontologie der Wirklichkeit.**

Sie verbindet strukturelle Modellierung, perspektivische Bewertung, dynamische Analyse, operationale Ausführbarkeit und kontrollierte Meta-Evolution. Ihr besonderer Beitrag liegt in der durchgehenden Traceability vom Eingangssignal über Beziehung, Wirkungsaussage, Resonanz, Relevanz, Orientierung und Gestaltung bis zur registrierten Rückwirkung und möglichen Grammatikerweiterung. Die Wirklichkeit geht jeder solchen Artikulation voraus und wird durch sie weder erzeugt noch erschöpft.

Die UZA trennt systematisch:

Wirklichkeit und ihre Artikulation sowie realen Vollzug und Wahrheitsgeltung seines Inhalts,

- Beobachtung, Hypothese und Evidenzstatus,
- Beziehung, Wirkungsaussage und Resonanz,
- Relevanz, Orientierung und Entscheidung,
- Simulation und Beobachtung,
- operationale Ereignis-, Zusammenhangs- und Versionszeit vom meta-ontologischen Immer,
- Artikulationszustand, modellierten Prozess,  $\Pi\_UZA$  und Auditgeschichte,
- Maßstab und Analyseauflösung,
- Teilzusammenhang, Projektion und Container,
- Kopplung, Transmission und Fusion,
- Spezifikation, Implementierung und mathematischen Beweisstatus.

Damit bietet sie eine gemeinsame formale Sprache für sehr verschiedene Anwendungsfelder, ohne deren Domänenwissen zu vereinheitlichen. Ihre Regeln bleiben gleichartig; Daten, Perspektiven, Evidenzstandards, Relevanzmodelle und Ziele werden fallspezifisch eingebracht. Jede Fassung bleibt eine abgleichbare Artikulation innerhalb der Wirklichkeit:  $Art\_L^{(n)}(\mathbb{W}\_Z) \neq \mathbb{W}\_Z$ .

Kernstatus K4: T1-T13 bleiben operational geschlossen; der artikulierte Interaktionskern ist bis zur validierten Transmission kausal und kernelgeprüft. A1-A22 bleiben unveränderte Locale-Annahmen. MG1-MG12 begrenzen die Reichweite, ohne Grundausdrücke in Isabelle-Objekte zu verwandeln oder einen Wir-Satz aus Kopplung abzuleiten.

## Anhang A – Verbindliche Artefakt- und Statusmatrix

| Artefakt                      | Inhalt                                                                                                             | Status                                                                                                          | Nächster objektiver Abschluss                                                                                |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Reference Architecture        | Kern, Typen, Axiome, Invarianten                                                                                   | 0.9.1 konsolidiert                                                                                              | Änderungen nur begründet/versioniert                                                                         |
| Semantic Core                 | Bedeutung, Unsicherheit, Perspektiven, Describe                                                                    | 0.9.1-RC                                                                                                        | unabhängige Domänen-Conformance                                                                              |
| Formal Rule Calculus          | T1–T13 und Ableitungsprinzipien                                                                                    | 0.9.1-RC                                                                                                        | vollständige mechanisierte Pre-/Postconditions                                                               |
| Grammar Engine                | Store, Ledger, Executor, Validation, Evolution                                                                     | 0.9.1-RC-Spezifikation                                                                                          | vollständiger Referenzcode                                                                                   |
| Reference Implementation      | Datenmodell, API, Tests, Demonstratoren                                                                            | Design 0.6/0.9.1                                                                                                | produktive Engine und unabhängiger Replay                                                                    |
| Reference Conformance Harness | ausführbarer Kern, drei Golden Runs und Negativgates                                                               | 0.1; lokal bestanden                                                                                            | zweite unabhängige Implementierung                                                                           |
| Dynamics Layer                | Flow, zehn Muster, Tracker und Monitor                                                                             | 0.9.1 spezifiziert; Referenzsemantik für fünf Muster                                                            | Kalibrierung und Testkorpus je Domäne                                                                        |
| Scale & Interaction           | ScaleProfile, SUBZ/PROJ, I1–I11, C11–C18                                                                           | 0.9.1-RC                                                                                                        | vollständige Engine- und Beweisinstanziierung                                                                |
| HOL Theory                    | Signatur, zentrale Definitionen, A1–A22, Modellzeugen sowie konkrete T1–T8-Semantik mit kausalen Ausführungszeugen | Normaler Isabelle2025-2-Clean-Build der 18-Theorien-K2-Kette; Exit-Code 0                                       | T9 SelectAction erst nach Autoritäts-, Zulässigkeits- und Abbruchsemantik; zusätzlich allgemeine Modellziele |
| Axiomatics A1–A22             | Basis- und Interaktionsaxiome                                                                                      | Locale-Annahmen; gemeinsame Erfüllbarkeit und konkrete Nichtvakuosität von A5–A13/A17–A21 relativ zu HOL belegt | allgemeine Expansion                                                                                         |
| Model Theory                  | Basismodell, Rich Model, Expansion, relative Konsistenz und Konservativität                                        | minimaler und reichhaltiger expliziter Modellzeuge kernelgeprüft                                                | Modell-Expansion und Konservativitätsanalyse                                                                 |
| FOL Core Reduction            | geeignetes Kernfragment                                                                                            | geplant                                                                                                         | Signatur- und Übersetzungsspezifikation                                                                      |

| Artefakt | Inhalt                       | Status            | Nächster objektiver Abschluss         |
|----------|------------------------------|-------------------|---------------------------------------|
| UZA 1.0  | integrierte Referenzfreigabe | nicht freigegeben | formale und technische Gates erfüllen |

## Anhang B – Version Bundle und Reproduzierbarkeitsmanifest

Ein vollständiges Reproduzierbarkeitsmanifest enthält:

```
case_id
initial_state_hash
ordered_input_artifacts + content hashes
specification_version
semantic_core_version
grammar_version
engine_version
all relevance model versions
all dynamics detector versions
all channel transform versions
validation profile version
documented human decisions and authorities
ordered commit ledger
final_state_hash
proof trace bundle
export timestamp and exporter version
```

Fachliche Reproduzierbarkeit ist erfüllt, wenn der Replaylauf dieselben fachlichen Objektversionen, Zustandsbeziehungen, Validierungsergebnisse und Emergenzklassifikationen erzeugt.

Nichtfachliche Metadaten wie neue Exportzeitstempel dürfen abweichen.

Für die 0.9.1-Referenzläufe ist folgende Suite gepinnt:

| Suite            | Runner                    | Fälle | Ergebnis                      |
|------------------|---------------------------|-------|-------------------------------|
| UZA-0.9.1-golden | reference-conformance-0.1 | 3     | 3 bestanden, 0 fehlgeschlagen |

Die Fixture- und Resultat-Prüfsummen werden mit kanonischer UTF-8-JSON-Darstellung und SHA-256 berechnet. Das maschinenlesbare results/summary.json ist Bestandteil des Conformance-Pakets.



## Anhang C – Vollständiger Referenz-Pseudocode

```
function run_case(case_input, version_bundle):
    registry.assert_compatible(version_bundle)
    state = initialize_case(case_input.identity, version_bundle)

    state = commit_batch(state, register_contexts(case_input.contexts))
    state = commit_batch(state, register_perspectives(case_input.perspectives))

    for observation in case_input.observations:
        state = execute_T1(state, observation)

    candidate_relations = relation_builder.propose(state)
    for relation in candidate_relations:
        state = execute_T3(state, relation)

    for relation in state.active_relations:
        hypotheses = effect_builder.hypothesize(relation, state)
        for hypothesis in hypotheses:
            state = execute_T4(state, hypothesis)

    for evidence in case_input.evidence_stream:
        targets = evidence_router.resolve_targets(evidence, state)
        for effect in targets:
            state = execute_T5(state, effect, evidence)

    resonance_candidates = resonance_analyzer.pairs(state.active_effects)
    for pair in resonance_candidates:
        state = execute_T6(state, pair)

    for model in version_bundle.relevance_models:
        for target in model.target_scope(state):
            state = execute_T7(state, target, model)

    orientation_result = execute_T8(
        state,
        state.active_relevances,
        state.active_resonances,
        case_input.goal,
        case_input.constraints
    )
    state = orientation_result.state

    if orientation_result.status == INSUFFICIENT_BASIS:
        return publish_analysis_without_action(state)

    selected = decision_port.select(
        orientation_result.options,
        case_input.authority,
        case_input.documented_decisions
    )
    state = execute_T9(state, selected)

    predicted_branch = simulation_runtime.simulate(
        state,
        selected,
        version_bundle.models,
        case_input.simulation_parameters
    )
    state = commit_simulation_reference(state, predicted_branch)

    execution = execute_T10(state, selected)
    state = execution.state

    observations = observation_port.collect(execution.record)
    state = execute_T11(state, execution.record, observations)

    deviation = compare(predicted_branch.final_state, state)
    emergence = execute_T12(state.parent, state, state.active_grammar)
    state = emergence.state

    revisions = revision_engine.propose(deviation, emergence, state)
    for revision in revisions.object_revisions:
        state = execute_T13(state, revision)

    for gap in revisions.grammar_gaps:
```

```
        patch = meta_evolution.propose_minimal_patch(gap, state)
        state = commit_patch_proposal(state, patch)

    return audit_exporter.publish(state)

function transmit_effect(state, effect, coupling, channel, scheduled_at):
    assert active(effect, state)
    assert active(coupling, state)
    assert channel in coupling.channels
    assert direction_allows(coupling, effect.context, channel.target)

    event = schedule_transmission(effect, coupling, channel, scheduled_at)
    state = commit(state, event)

    result = channel.transform.apply(
        effect,
        coupling.source_context,
        coupling.target_context,
        scheduled_at + channel.delay
    )

    match result:
        Transformed(target_effect):
            target_effect.lineage = {
                source_effect: effect.version_id,
                coupling: coupling.version_id,
                channel: channel.version_id,
                transform: channel.transform.version_id,
                transmission_event: event.version_id
            }
            return commit_delivery(state, event, target_effect)
        Blocked(reason):
            return commit_blocked(state, event, reason)
        Incompatible(report):
            return commit_incompatible(state, event, report)
        Failed(error):
            return commit_failed(state, event, error)
```

## Anhang D – UZA HOL Theory 0.9.1: T1-T7 Kernel-Checked

### D.0 Leistungsstand auf einen Blick

Diese Übersicht steht bewusst vor dem Quelltext. Abgeschlossen bedeutet in D.0, dass die angegebene Quellfassung im ausgewiesenen Isabelle2025-2-Prüfpfad akzeptiert wurde. Die historischen T6-/T7-Pfade bleiben dokumentiert; der aktuelle gemeinsame Freigabepfad ist der normale Clean-Build der vollständigen 16-Theorien-K1-Session. Teilweise abgeschlossen benennt den erreichten Teilsatz. Offen bedeutet: noch kein mechanischer Nachweis. Kein Operationsname allein begründet einen Beweisstatus.

| Bereich                    | Status                         | Bereits geleistet und belegt                                                                                                                                           | Noch offen                                                          |
|----------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Haupttheorie               | <b>abgeschlossen für 0.9.1</b> | Signatur, Referenzsemantik, A1–A22, sechs Sentinels; Hauptsession Exit-Code 0                                                                                          | Ableitung der Locale-Annahmen aus einer schwächeren Theorie         |
| A9 Evidence Integrity      | <b>abgeschlossen</b>           | auflösbare Evidence-IDs, nichtleere Evidenz für Supported/Validated, zwei kernelgeprüfte Lemmata                                                                       | keine offene A9-Lücke im spezifizierten Umfang                      |
| Minimaler Modellzeuge      | <b>abgeschlossen</b>           | Base: uza_model, explicit_uza_model_exists                                                                                                                             | bewusst keine Nichtvakuosität der objektspezifischen Bedingungen    |
| Reichhaltiger Modellzeuge  | <b>abgeschlossen</b>           | zwölf Objekte; A5–A13 und A17–A21 nichtvakuos; Rich: uza_model                                                                                                         | Expansion beliebiger Kernmodelle                                    |
| T1 Register Appearance     | <b>abgeschlossen</b>           | konstruktive Transition, exaktes Delta, Identität, Evidence Integrity, Bindungen, state_well_formed, Refinement zu step                                                | Vollständigkeit oder Produktivengine-Äquivalenz                     |
| T2 Differentiate Potential | <b>abgeschlossen</b>           | endliche nichtleere Kandidatenmenge, Trennung von Beobachtung/Kandidat, exaktes Delta, Identität, Evidence Integrity, Bindungen, state_well_formed, Refinement zu step | Vollständigkeit und domänenspezifische Güte der Kandidatenerzeugung |
| T3 Create                  | <b>abgeschlossen</b>           | aktive                                                                                                                                                                 | Vollständigkeit der                                                 |

| Bereich               | Status        | Bereits geleistet und belegt                                                                                                                                                                                                                                                                                                                                  | Noch offen                                                                                 |
|-----------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Relation              |               | verschiedene ENT-Endpunkte, frische REL-Identität, exaktes Delta, Katalog-, Zustands-, Identitäts-, Referenz-, Zeit-, Kontext- und Evidence-Integrity-Erhaltung sowie Refinement zu step; Clean-Build Exit 0                                                                                                                                                  | Relationstaxonomie und Produktivengine-Äquivalenz                                          |
| T4 Hypothesize Effect | abgeschlossen | aktive typisierte REL-Ursache; explizit optionales Ziel und optionaler Wirkungsvektor; Perspektiv-, Kontext- und Profilmodellbindung; begrenzte Unsicherheit; frische K_EFF-Identität; exaktes Delta; Katalog-, Zustands-, Identitäts-, Referenz-, Zeit-, Perspektiv-, Kontext- und Evidence-Integrity-Erhaltung sowie Refinement zu step; Clean-Build Exit 0 | empirische Güte und Vollständigkeit der Wirkungstaxonomie sowie Produktivengine-Äquivalenz |
| T5 Integrate          | abgeschlossen | Copy-on-Write-                                                                                                                                                                                                                                                                                                                                                | empirische Güte und                                                                        |

| Bereich               | Status        | Bereits geleistet und belegt                                                                                                                                                                                                                                                                                                                                                                                                  | Noch offen                                                                                                  |
|-----------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Evidence              |               | Effektversion, vollständiges Objekt- und Payload-Archiv, Evidenz-Ledger, drei Dispositionen, kontrollierte Unsicherheit, Lifecycle ohne Hypothesis → Validated-Sprung, Ausschluss von Selbststützung, exaktes Delta, Zustands-, Identitäts-, Katalog-, Archiv-, Referenz-, Zeit-, Perspektiv-, Kontext- und Evidence-Integrity-Erhaltung, Refinement zu step; nicht-vakuoser Hypothesis → Supported-Zeuge; Clean-Build Exit 0 | Vollständigkeit der Evidenzklassifikation sowie Produktivengine-Äquivalenz                                  |
| T6 Classify Resonance | abgeschlossen | gerichtetes versioniertes Kernprofil; formal endliche V10-Domäne; normalisierter begrenzter Score; totale Formklassifikation; Unsicherheitsdämpfung; aktive                                                                                                                                                                                                                                                                   | empirische Herleitung der Kernel und Schwellen, Vollständigkeit der Formen sowie Produktivengine-Äquivalenz |

| Bereich               | Status                                                       | Bereits geleistet und belegt                                                                                                                                                                                                                                                                                               | Noch offen                                                                    |
|-----------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
|                       |                                                              | verschiedene Effekte;<br>unabhängige Begründung;<br>konstruktive Objekt-/Payload-Erzeugung;<br>exaktes Delta;<br>Zustands-, Identitäts-, Effect-/Resonance-Katalog-, Referenz-, Evidenz-, Zeit-, Perspektiv- und Kontexterhaltung;<br>step-Refinement;<br>nicht-vakuoser Asymmetriezeuge;<br>Fresh-Profile-Build<br>Exit 0 |                                                                               |
| T7 Evaluate Relevance | abgeschlossen; im normalen K1- und K2-Sessionbuild enthalten | versioniertes Modellprofil;<br>getrennte V10-/Machtgewichte; ausführbarer sechsdimensionaler Vektor;<br>Wertebereiche;<br>zeitliche Zielaktivität; aktive Begründung;<br>frische K_REV-Konstruktion;<br>Erhaltung und step-Refinement;<br>nicht-vakuoser Zwei-Perspektiven-                                                | empirische Kalibrierung, Profilvollständigkeit und Produktivengine-Äquivalenz |

| Bereich                                  | Status                         | Bereits geleistet und belegt                                                                                  | Noch offen                                                                                                    |
|------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
|                                          |                                | Zeuge; normaler Clean-Build Exit 0                                                                            |                                                                                                               |
| T8 / T9–T13                              | T8 und T9–T13 abgeschlossen    | T8: konstruktive range-erhaltende Orientierungssynthese mit explizitem InsufficientBasis-Pfad; siehe Anhang J | T9–T13: konstruktive Semantik, fünf T9-Ablehnungszweige, kausale Zeugen und Erhaltungsbeweise; siehe Anhang K |
| Modell-Expansion und relative Konsistenz | <b>teilweise abgeschlossen</b> | zwei konkrete Modelle belegen gemeinsame Erfüllbarkeit relativ zu HOL                                         | allgemeines Expansionslemma und relativer Konsistenzsatz                                                      |
| Konservativität und FOL-Reduktion        | <b>offen</b>                   | Ziel und Trennung der Artefakte festgelegt                                                                    | Beweise und Übersetzung                                                                                       |
| Produktive Engine                        | <b>offen</b>                   | Conformance-Harness mit 3/3 Golden Runs                                                                       | unabhängige Implementierung, Produktions- und Sicherheitsnachweise                                            |

## D.1 Haupttheorie

Die folgende Quelldatei enthält Signatur, zentrale Referenzsemantik, A1–A22 als universell quantifizierte, typisierte Locale-Annahmen und sechs Haupttheorie-Sentinels. A9 ist nun über `evidence_references_complete` und `evidence_well_formed` operational geschlossen: Jede Evidence-ID muss auf eine aktuelle oder historische Objektversion auflösen; Supported und Validated verlangen zusätzlich eine nichtleere Evidenzmenge.

`unresolved_evidence_reference_rejected` und `A9_upgraded_evidence_resolves` sichern beide Richtungen als kernelgeprüfte Lemmata ab. Die Evidenzbedingung ist Bestandteil von `state_well_formed` und der Describe-Komponente `relation_complete`.

Ein struktureller Preflight prüfte Theoriehülle, Klammern, die vollständige Axiomnummerierung, zentrale Definitionsnamen, die sechs Sentinels, beide A9-Lemmata und die Sessiondatei. Zusätzlich verarbeitete der integrierte Isabelle2025-2-Clean-Build die Haupttheorie vollständig. Der Build endete mit Exit-Code 0; der Hauptquelltext ist mit SHA-256 `21dcfb175d6e754adf20042b941cf4a23329831d0165ebaf2023b0048213ed24` bezeichnet. Dieser Build allein ist noch kein Beweis der Locale-Annahmen aus einer schwächeren Theorie; konkrete Erfüllbarkeit wird durch die nachfolgenden Modellzeugen belegt.

```
theory UZA_0_9_1
  imports Complex_Main
begin
```

```

section \<open>Primitive identifiers and enumerations\<close>

type_synonym object_id = nat
type_synonym version_id = nat
type_synonym perspective_id = nat
type_synonym context_id = nat
type_synonym model_id = nat
type_synonym grammar_id = nat
type_synonym authority_id = nat
type_synonym channel_id = nat
type_synonym coupling_id = nat
type_synonym effect_id = nat
type_synonym patch_id = nat
type_synonym operation_id = nat
type_synonym time_point = nat

datatype object_kind =
  K_ENT | K_REL | K_EFF | K_RES | K_REV | K_ORI | K_ACT | K_FDB | K_EMG
  | K_PERSP | K_CTX | K_META | K_CONFLICT

datatype lifecycle_status =
  Draft | Hypothesis | Supported | Validated | Revised | Rejected | Archived

datatype emergence_level = E0 | E1 | E2 | E3

datatype description_status = DescriptionComplete | DescriptionIncomplete

datatype analysis_status =
  Applicable | InsufficientBasis | ModelFalsified | OutOfScope
  | PatchRejected | CoreConflict | UnresolvedE3

datatype operation =
  RegisterAppearance | DifferentiatePotential | CreateRelation
  | HypothesizeEffect | IntegrateEvidence | ClassifyResonance
  | EvaluateRelevance | GenerateOrientation | SelectAction | ExecuteAction
  | RecordFeedback | AnalyzeEmergence | ReviseObject
  | DefineCoupling | InitiateTransmission | ApplyBoundary

datatype trans_status =
  TransScheduled | TransDelivered | TransObserved | TransValidated
  | TransBlocked | TransIncompatible | TransFailed | TransCancelled

datatype boundary_kind = Fusion | Fission | Aggregation | Embedding | Detachment

datatype effect_dimension =
  Connection | Stability | Openness | Learning | Cooperation
  | Agency | Transparency | Freedom | Responsibility | Regeneration

datatype power_dimension =
  ResourcePower | AccessControl | DecisionAuthority | RiskBearing
  | CorrectionCapacity

datatype resonance_form =
  Amplification | Complement | ProductiveTension | Inhibition | Neutral

section \<open>Versioned objects and snapshots\<close>

record uza_object =
  object_id_of :: object_id
  version_id_of :: version_id
  kind_of :: object_kind
  lifecycle_of :: lifecycle_status
  previous_version_of :: "version_id option"
  references_of :: "version_id set"
  evidence_of :: "version_id set"
  perspective_of :: "perspective_id option"
  context_of :: "context_id option"
  model_of :: "model_id option"
  authority_of :: "authority_id option"
  valid_from_of :: time_point
  valid_to_of :: "time_point option"
  source_ref_of :: "version_id option"
  target_ref_of :: "version_id option"
  effect_origin_of :: "version_id option"
  basis_refs_of :: "version_id set"
  execution_ref_of :: "version_id option"
  observation_refs_of :: "version_id set"
  object_status_of :: "analysis_status option"

```



```
record loss_profile =
  structural_loss_of :: real
  semantic_loss_of :: real
  perspective_loss_of :: real
  temporal_loss_of :: real
  loss_explanation_of :: string

record projection =
  projection_id_of :: nat
  contained_context_of :: context_id
  container_context_of :: context_id
  source_snapshot_of :: nat
  projection_loss_of :: loss_profile

record coupling =
  coupling_id_of :: coupling_id
  coupling_source_of :: context_id
  coupling_target_of :: context_id
  coupling_channels_of :: "channel_id set"

record transmission =
  transmission_id_of :: nat
  transmission_coupling_of :: coupling_id
  transmission_channel_of :: channel_id
  transmission_source_effect_of :: effect_id
  transmission_target_effect_of :: "effect_id option"
  prior_trans_status_of :: "trans_status option"
  trans_status_of :: trans_status
  transmission_lineage_complete_of :: bool

record boundary_event =
  boundary_id_of :: nat
  boundary_kind_of :: boundary_kind
  boundary_succeeded_of :: bool
  boundary_lineage_complete_of :: bool
  boundary_disposition_complete_of :: bool

record snapshot =
  snapshot_id_of :: nat
  objects_of :: "uza_object set"
  historical_versions_of :: "version_id set"
  perspectives_of :: "perspective_id set"
  contexts_of :: "context_id set"
  models_of :: "model_id set"
  authorities_of :: "authority_id set"
  contains_edges_of :: "(context_id * context_id) set"
  projections_of :: "projection set"
  couplings_of :: "coupling set"
  transmissions_of :: "transmission set"
  boundaries_of :: "boundary_event set"

record grammar =
  grammar_id_of :: grammar_id
  supported_kinds_of :: "object_kind set"
  allowed_operations_of :: "operation set"
  patch_budget_of :: nat

record transition =
  transition_before_of :: snapshot
  transition_operation_of :: operation
  transition_after_of :: snapshot

record grammar_patch =
  patch_id_of :: patch_id
  added_kinds_of :: "object_kind set"
  added_operations_of :: "operation set"
  patch_cost_of :: nat
  changes_core_of :: bool
  holdout_passed_of :: bool
  replay_preserved_of :: bool
  migration_defined_of :: bool
  rollback_defined_of :: bool
  unresolved_conflicts_of :: nat

record interaction_complex =
  baseline_of :: snapshot
  current_of :: snapshot
  trace_of :: "transition list"
  candidate_patches_of :: "grammar_patch set"
```

```

record description_result =
  dr_status_of :: description_status
  dr_type_complete_of :: bool
  dr_relation_complete_of :: bool
  dr_transition_complete_of :: bool

section \<open>Object, time, reference, and state semantics\<close>

definition version_ids :: "snapshot \<Rightarrow> version_id set" where
  "version_ids s = version_id_of ` objects_of s"

definition unique_versions :: "snapshot \<Rightarrow> bool" where
  "unique_versions s \<longlefttrightarrow> inj_on version_id_of (objects_of s)"

definition present_ref :: "snapshot \<Rightarrow> version_id option \<Rightarrow> bool" where
  "present_ref s r \<longlefttrightarrow>
    (case r of None \<Rightarrow> False | Some v \<Rightarrow> v \<in> version_ids s \<union>
    historical_versions_of s)"

definition references_complete :: "snapshot \<Rightarrow> uza_object \<Rightarrow> bool" where
  "references_complete s obj \<longlefttrightarrow>
    references_of obj \<subseq> version_ids s \<union> historical_versions_of s"

definition evidence_references_complete :: "snapshot \<Rightarrow> uza_object \<Rightarrow> bool" where
  "evidence_references_complete s obj \<longlefttrightarrow>
    evidence_of obj \<subseq> version_ids s \<union> historical_versions_of s"

definition evidence_well_formed :: "snapshot \<Rightarrow> uza_object \<Rightarrow> bool" where
  "evidence_well_formed s obj \<longlefttrightarrow>
    evidence_references_complete s obj \<and>
    (lifecycle_of obj \<in> {Supported, Validated} \<longrightarrow> evidence_of obj \<noteq> {})"

lemma unresolved_evidence_reference_rejected:
  assumes "v \<in> evidence_of obj"
  "v \<notin> version_ids s \<union> historical_versions_of s"
  shows "\<not> evidence_well_formed s obj"
  using assms
  unfolding evidence_well_formed_def evidence_references_complete_def
  by blast

definition time_well_formed :: "uza_object \<Rightarrow> bool" where
  "time_well_formed obj \<longlefttrightarrow>
    (case valid_to_of obj of None \<Rightarrow> True | Some t \<Rightarrow> valid_from_of obj \<le> t)"

definition relation_well_formed :: "snapshot \<Rightarrow> uza_object \<Rightarrow> bool" where
  "relation_well_formed s obj \<longlefttrightarrow>
    (kind_of obj = K_REL \<longrightarrow>
    present_ref s (source_ref_of obj) \<and> present_ref s (target_ref_of obj))"

definition effect_well_formed :: "snapshot \<Rightarrow> uza_object \<Rightarrow> bool" where
  "effect_well_formed s obj \<longlefttrightarrow>
    (kind_of obj = K_EFF \<and> lifecycle_of obj = Validated \<longrightarrow>
    present_ref s (effect_origin_of obj))"

definition perspective_bound_kind :: "object_kind \<Rightarrow> bool" where
  "perspective_bound_kind k \<longlefttrightarrow> k \<in> {K_ENT, K_EFF, K_RES, K_REV, K_ORI, K_ACT}"

definition context_bound_kind :: "object_kind \<Rightarrow> bool" where
  "context_bound_kind k \<longlefttrightarrow> k \<in> {K_ENT, K_REL, K_EFF, K_RES, K_REV, K_ORI, K_ACT,
  K_FDB}"

definition perspective_well_formed :: "snapshot \<Rightarrow> uza_object \<Rightarrow> bool" where
  "perspective_well_formed s obj \<longlefttrightarrow>
    (perspective_bound_kind (kind_of obj) \<longrightarrow>
    (case perspective_of obj of None \<Rightarrow> False | Some p \<Rightarrow> p \<in>
    perspectives_of s))"

definition context_well_formed :: "snapshot \<Rightarrow> uza_object \<Rightarrow> bool" where
  "context_well_formed s obj \<longlefttrightarrow>
    (context_bound_kind (kind_of obj) \<longrightarrow>
    (case context_of obj of None \<Rightarrow> False | Some c \<Rightarrow> c \<in> contexts_of s))"

definition relevance_well_formed :: "snapshot \<Rightarrow> uza_object \<Rightarrow> bool" where
  "relevance_well_formed s obj \<longlefttrightarrow>
    (kind_of obj = K_REV \<longrightarrow>
    perspective_well_formed s obj \<and> context_well_formed s obj \<and>
    (case model_of obj of None \<Rightarrow> False | Some m \<Rightarrow> m \<in> models_of s) \<and>
    references_of obj \<noteq> {})"

```

```

definition orientation_well_formed :: "snapshot \<Rightarrow> uza_object \<Rightarrow> bool" where
  "orientation_well_formed s obj \<longleftarrowtrightharpoon>
    (kind_of obj = K_ORI \<longrightharpoon>
      basis_refs_of obj \<noteq> {} \<or> object_status_of obj = Some InsufficientBasis)"

definition action_well_formed :: "snapshot \<Rightarrow> uza_object \<Rightarrow> bool" where
  "action_well_formed s obj \<longleftarrowtrightharpoon>
    (kind_of obj = K_ACT \<longrightharpoon>
      basis_refs_of obj \<noteq> {} \<and>
      (case authority_of obj of None \<Rightarrow> False | Some a \<Rightarrow> a \<in> authorities_of
s)))"

definition feedback_well_formed :: "snapshot \<Rightarrow> uza_object \<Rightarrow> bool" where
  "feedback_well_formed s obj \<longleftarrowtrightharpoon>
    (kind_of obj = K_FDB \<longrightharpoon>
      present_ref s (execution_ref_of obj) \<and> observation_refs_of obj \<noteq> {})"

definition loss_well_formed :: "loss_profile \<Rightarrow> bool" where
  "loss_well_formed l \<longleftarrowtrightharpoon>
    0 \<le> structural_loss_of l \<and> structural_loss_of l \<le> 1 \<and>
    0 \<le> semantic_loss_of l \<and> semantic_loss_of l \<le> 1 \<and>
    0 \<le> perspective_loss_of l \<and> perspective_loss_of l \<le> 1 \<and>
    0 \<le> temporal_loss_of l \<and> temporal_loss_of l \<le> 1"

definition projection_well_formed :: "snapshot \<Rightarrow> projection \<Rightarrow> bool" where
  "projection_well_formed s p \<longleftarrowtrightharpoon>
    contained_context_of p \<in> contexts_of s \<and>
    container_context_of p \<in> contexts_of s \<and>
    loss_well_formed (projection_loss_of p)"

definition coupling_well_formed :: "snapshot \<Rightarrow> coupling \<Rightarrow> bool" where
  "coupling_well_formed s c \<longleftarrowtrightharpoon>
    coupling_source_of c \<in> contexts_of s \<and>
    coupling_target_of c \<in> contexts_of s \<and>
    coupling_channels_of c \<noteq> {}"

fun allowed_trans_transition :: "trans_status option \<Rightarrow> trans_status \<Rightarrow> bool"
where
  "allowed_trans_transition None TransScheduled = True"
| "allowed_trans_transition (Some TransScheduled) TransDelivered = True"
| "allowed_trans_transition (Some TransScheduled) TransBlocked = True"
| "allowed_trans_transition (Some TransScheduled) TransIncompatible = True"
| "allowed_trans_transition (Some TransScheduled) TransCancelled = True"
| "allowed_trans_transition (Some TransDelivered) TransObserved = True"
| "allowed_trans_transition (Some TransDelivered) TransFailed = True"
| "allowed_trans_transition (Some TransObserved) TransValidated = True"
| "allowed_trans_transition (Some TransObserved) TransFailed = True"
| "allowed_trans_transition _ _ = False"

definition transmission_well_formed :: "snapshot \<Rightarrow> transmission \<Rightarrow> bool" where
  "transmission_well_formed s t \<longleftarrowtrightharpoon>
    (\<exists>c\<in>couplings_of s.
      coupling_id_of c = transmission_coupling_of t \<and>
      transmission_channel_of t \<in> coupling_channels_of c) \<and>
    allowed_trans_transition (prior_trans_status_of t) (trans_status_of t) \<and>
    (trans_status_of t = TransValidated \<longrightharpoon>
      transmission_target_effect_of t \<noteq> None \<and> transmission_lineage_complete_of t)"

definition boundary_well_formed :: "boundary_event \<Rightarrow> bool" where
  "boundary_well_formed b \<longleftarrowtrightharpoon>
    (boundary_succeeded_of b \<longrightharpoon>
      (case boundary_kind_of b of
        Fusion \<Rightarrow> boundary_disposition_complete_of b
      | Fission \<Rightarrow> boundary_lineage_complete_of b
      | Aggregation \<Rightarrow> boundary_lineage_complete_of b
      | Embedding \<Rightarrow> boundary_lineage_complete_of b
      | Detachment \<Rightarrow> boundary_lineage_complete_of b)))"

definition state_well_formed :: "grammar \<Rightarrow> snapshot \<Rightarrow> bool" where
  "state_well_formed G s \<longleftarrowtrightharpoon>
    unique_versions s \<and>
    (\<forall>obj\<in>objects_of s.
      kind_of obj \<in> supported_kinds_of G \<and>
      references_complete s obj \<and> evidence_well_formed s obj \<and>
      time_well_formed obj \<and>
      relation_well_formed s obj \<and> effect_well_formed s obj \<and>
      perspective_well_formed s obj \<and> context_well_formed s obj \<and>
      relevance_well_formed s obj \<and> orientation_well_formed s obj \<and>

```

```

    action_well_formed s obj \<and> feedback_well_formed s obj) \<and>
    (\<forall>p\<in>projections_of s. projection_well_formed s p) \<and>
    (\<forall>c\<in>couplings_of s. coupling_well_formed s c) \<and>
    (\<forall>t\<in>transmissions_of s. transmission_well_formed s t) \<and>
    (\<forall>b\<in>boundaries_of s. boundary_well_formed b)"

definition immutable_preserved :: "snapshot \<Rightarrow> snapshot \<Rightarrow> bool" where
  "immutable_preserved s s' \<longleftarrowrightarrow>
    (\<forall>obj\<in>objects_of s. \<forall>obj'\<in>objects_of s'.
      version_id_of obj = version_id_of obj' \<longrightarrow> obj = obj')"
```

```

definition step :: "grammar \<Rightarrow> snapshot \<Rightarrow> operation \<Rightarrow> snapshot
\<Rightarrow> bool" where
  "step G s oper s' \<longleftarrowrightarrow>
    state_well_formed G s \<and> state_well_formed G s' \<and>
    oper \<in> allowed_operations_of G \<and> immutable_preserved s s'"

definition revision_edges_of :: "snapshot \<Rightarrow> (version_id * version_id) set" where
  "revision_edges_of s =
    {(version_id_of obj, p) | obj p. obj \<in> objects_of s \<and> previous_version_of obj = Some p}"

definition strict_contains :: "snapshot \<Rightarrow> context_id \<Rightarrow> context_id \<Rightarrow>
bool" where
  "strict_contains s a b \<longleftarrowrightarrow> (a,b) \<in> trancl (contains_edges_of s)"

definition contains_eq :: "snapshot \<Rightarrow> context_id \<Rightarrow> context_id \<Rightarrow>
bool" where
  "contains_eq s a b \<longleftarrowrightarrow> (a = b \<or> strict_contains s a b)"

section \<open>V10, resonance, relevance, power, and dynamics reference semantics\<close>

type_synonym effect_vector = "effect_dimension \<Rightarrow> int"
type_synonym relevance_weights = "effect_dimension \<Rightarrow> real"
type_synonym power_profile = "power_dimension \<Rightarrow> real"

definition effect_vector_valid :: "effect_vector \<Rightarrow> bool" where
  "effect_vector_valid v \<longleftarrowrightarrow> (\<forall>d. -3 \<le> v d \<and> v d \<le> 3)"

definition v10_dot :: "effect_vector \<Rightarrow> effect_vector \<Rightarrow> int" where
  "v10_dot a b = sum (\<lambda>d. a d * b d) UNIV"

definition resonance_score :: "effect_vector \<Rightarrow> effect_vector \<Rightarrow> real" where
  "resonance_score a b = of_int (v10_dot a b) / 90"

definition classify_resonance :: "effect_vector \<Rightarrow> effect_vector \<Rightarrow>
resonance_form" where
  "classify_resonance a b =
    (let r = resonance_score a b in
     if r \<ge> 0.5 then Amplification
     else if r \<ge> 0.15 then Complement
     else if r \<le> -0.5 then Inhibition
     else if r \<le> -0.15 then ProductiveTension
     else Neutral)"

definition clamp01 :: "real \<Rightarrow> real" where
  "clamp01 x = max 0 (min 1 x)"

definition weight_mass :: "relevance_weights \<Rightarrow> real" where
  "weight_mass w = sum (\<lambda>d. abs (w d)) UNIV"

definition normalized_effect_component :: "effect_vector \<Rightarrow> effect_dimension \<Rightarrow>
real" where
  "normalized_effect_component v d = (of_int (v d) + 3) / 6"

definition relevance_weights_valid :: "relevance_weights \<Rightarrow> bool" where
  "relevance_weights_valid w \<longleftarrowrightarrow> (\<forall>d. 0 \<le> w d) \<and> weight_mass w > 0"

definition relevance_score :: "relevance_weights \<Rightarrow> effect_vector \<Rightarrow> real
\<Rightarrow> real" where
  "relevance_score w v uncertainty =
    clamp01
    ((if weight_mass w = 0 then 0.5
     else sum (\<lambda>d. w d * normalized_effect_component v d) UNIV / weight_mass w)
     * (1 - clamp01 uncertainty))"

definition power_profile_valid :: "power_profile \<Rightarrow> bool" where
  "power_profile_valid p \<longleftarrowrightarrow> (\<forall>d. 0 \<le> p d \<and> p d \<le> 1)"

definition power_shift :: "power_profile \<Rightarrow> power_profile \<Rightarrow> power_profile" where

```

```

"power_shift before after = (\<lambda>d. after d - before d)"

definition applicability_of :: "bool \<Rightarrow> bool \<Rightarrow> real \<Rightarrow> real
\<Rightarrow> analysis_status" where
  "applicability_of in_scope sufficient_basis observed_error falsification_limit =
    (if \<not> in_scope then OutOfScope
     else if \<not> sufficient_basis then InsufficientBasis
     else if observed_error > falsification_limit then ModelFalsified
     else Applicable)"

fun successive_deltas :: "real list \<Rightarrow> real list" where
  "successive_deltas [] = []"
| "successive_deltas [x] = []"
| "successive_deltas (x # y # xs) = abs (y - x) # successive_deltas (y # xs)"

fun nonincreasing :: "real list \<Rightarrow> bool" where
  "nonincreasing [] = True"
| "nonincreasing [x] = True"
| "nonincreasing (x # y # xs) = (y \<le> x \& nonincreasing (y # xs))"

fun nondecreasing :: "real list \<Rightarrow> bool" where
  "nondecreasing [] = True"
| "nondecreasing [x] = True"
| "nondecreasing (x # y # xs) = (x \<le> y \& nondecreasing (y # xs))"

definition stabilizing :: "real list \<Rightarrow> bool" where
  "stabilizing xs \<longlefttrightarrow> length xs \<ge> 3 \& nonincreasing (successive_deltas xs)"

definition destabilizing :: "real list \<Rightarrow> bool" where
  "destabilizing xs \<longlefttrightarrow> length xs \<ge> 3 \& nondecreasing (successive_deltas xs)"

definition converging :: "real list \<Rightarrow> real \<Rightarrow> bool" where
  "converging distances epsilon \<longlefttrightarrow>
    distances \<noteq> [] \& nonincreasing distances \& last distances \<le> epsilon"

definition diverging :: "real list \<Rightarrow> real \<Rightarrow> bool" where
  "diverging distances epsilon \<longlefttrightarrow>
    distances \<noteq> [] \& nondecreasing distances \& last distances \<ge> epsilon"

definition drifting :: "real list \<Rightarrow> real \<Rightarrow> bool" where
  "drifting xs epsilon \<longlefttrightarrow>
    length xs \<ge> 2 \& abs (last xs - hd xs) \<ge> epsilon \&
    (\<forall>d\<in>set (successive_deltas xs). d < abs (last xs - hd xs))"

section \<open>Describe, patch, and emergence semantics\<close>

definition type_complete :: "grammar \<Rightarrow> interaction_complex \<Rightarrow> bool" where
  "type_complete G I \<longlefttrightarrow>
    (\<forall>obj\<in>objects_of (current_of I). kind_of obj \<in> supported_kinds_of G)"

definition relation_complete :: "grammar \<Rightarrow> interaction_complex \<Rightarrow> bool" where
  "relation_complete G I \<longlefttrightarrow>
    (\<forall>obj\<in>objects_of (current_of I).
      references_complete (current_of I) obj \&
      evidence_well_formed (current_of I) obj \&
      relation_well_formed (current_of I) obj \&
      effect_well_formed (current_of I) obj)"

definition transition_complete :: "grammar \<Rightarrow> interaction_complex \<Rightarrow> bool" where
  "transition_complete G I \<longlefttrightarrow>
    (\<forall>tr\<in>set (trace_of I).
      step G (transition_before_of tr) (transition_operation_of tr)
      (transition_after_of tr))"

definition describeable :: "grammar \<Rightarrow> interaction_complex \<Rightarrow> bool" where
  "describeable G I \<longlefttrightarrow>
    type_complete G I \& relation_complete G I \& transition_complete G I"

definition describe :: "grammar \<Rightarrow> interaction_complex \<Rightarrow> description_result"
where
  "describe G I =
    \<lparr> dr_status_of =
      (if describeable G I then DescriptionComplete else DescriptionIncomplete),
      dr_type_complete_of = type_complete G I,
      dr_relation_complete_of = relation_complete G I,
      dr_transition_complete_of = transition_complete G I \<rparr>"

definition relation_shape :: "snapshot \<Rightarrow> (object_kind * object_kind) set" where
  "relation_shape s =

```

```

    {(kind_of a, kind_of b) |r a b.
      r \<in> objects_of s \<and> kind_of r = K_REL \<and>
      a \<in> objects_of s \<and> b \<in> objects_of s \<and>
      source_ref_of r = Some (version_id_of a) \<and>
      target_ref_of r = Some (version_id_of b)}}
definition structural_novelty :: "interaction_complex \<Rightarrow> bool" where
  "structural_novelty I \<longlefttrightarrow>
    relation_shape (current_of I) - relation_shape (baseline_of I) \<noteq> {} \<or>
    kind_of ` objects_of (current_of I) - kind_of ` objects_of (baseline_of I) \<noteq> {}"
definition patch_admissible :: "grammar \<Rightarrow> grammar_patch \<Rightarrow> bool" where
  "patch_admissible G p \<longlefttrightarrow>
    \<not> changes_core_of p \<and> patch_cost_of p \<le> patch_budget_of G \<and>
    holdout_passed_of p \<and> replay_preserved_of p \<and>
    migration_defined_of p \<and> rollback_defined_of p \<and>
    unresolved_conflicts_of p = 0"
definition patch_status :: "grammar \<Rightarrow> grammar_patch \<Rightarrow> analysis_status" where
  "patch_status G p =
    (if changes_core_of p then CoreConflict
     else if patch_admissible G p then Applicable
     else PatchRejected)"
definition apply_patch :: "grammar \<Rightarrow> grammar_patch \<Rightarrow> grammar" where
  "apply_patch G p =
    G\<lparr> supported_kinds_of := supported_kinds_of G \<union> added_kinds_of p,
    allowed_operations_of := allowed_operations_of G \<union> added_operations_of p \<rparr>"
definition repairable :: "grammar \<Rightarrow> interaction_complex \<Rightarrow> bool" where
  "repairable G I \<longlefttrightarrow>
    (\<exists>p\<in>candidate_patches_of I.
     patch_admissible G p \<and> describeable (apply_patch G p) I)"
definition emergence_of :: "grammar \<Rightarrow> interaction_complex \<Rightarrow> emergence_level"
where
  "emergence_of G I =
    (if describeable G I then
      (if structural_novelty I then E1 else E0)
      else if repairable G I then E2 else E3)"
section \<open>A1-A22 as a locale for models of UZA 0.9.1\<close>
locale uza_model =
  fixes States :: "snapshot set"
  and G :: grammar
  and classified_emergence :: "interaction_complex \<Rightarrow> emergence_level"
  assumes A1_version_identity:
    "\<And>s o1 o2. \<lbrakk>s \<in> States; o1 \<in> objects_of s; o2 \<in> objects_of s;
    version_id_of o1 = version_id_of o2\<rbrakk> \<Longrightarrow> o1 = o2"
  and A2_revision_well_founded:
    "\<And>s. s \<in> States \<Longrightarrow> wf (revision_edges_of s)"
  and A3_time_well_formed:
    "\<And>s obj. \<lbrakk>s \<in> States; obj \<in> objects_of s\<rbrakk> \<Longrightarrow>
time_well_formed obj"
  and A4_reference_integrity:
    "\<And>s obj. \<lbrakk>s \<in> States; obj \<in> objects_of s\<rbrakk> \<Longrightarrow>
references_complete s obj"
  and A5_relation_endpoints:
    "\<And>s obj. \<lbrakk>s \<in> States; obj \<in> objects_of s\<rbrakk> \<Longrightarrow>
relation_well_formed s obj"
  and A6_effect_origin:
    "\<And>s obj. \<lbrakk>s \<in> States; obj \<in> objects_of s\<rbrakk> \<Longrightarrow>
effect_well_formed s obj"
  and A7_perspective_binding:
    "\<And>s obj. \<lbrakk>s \<in> States; obj \<in> objects_of s\<rbrakk> \<Longrightarrow>
perspective_well_formed s obj"
  and A8_context_binding:
    "\<And>s obj. \<lbrakk>s \<in> States; obj \<in> objects_of s\<rbrakk> \<Longrightarrow>
context_well_formed s obj"
  and A9_evidence_integrity:
    "\<And>s obj. \<lbrakk>s \<in> States; obj \<in> objects_of s\<rbrakk>
\<Longrightarrow> evidence_well_formed s obj"
  and A10_relevance_binding:
    "\<And>s obj. \<lbrakk>s \<in> States; obj \<in> objects_of s\<rbrakk> \<Longrightarrow>
relevance_well_formed s obj"
  and A11_orientation_basis:
    "\<And>s obj. \<lbrakk>s \<in> States; obj \<in> objects_of s\<rbrakk> \<Longrightarrow>
orientation_well_formed s obj"

```

```

    and A12_action_authority:
      "\<And>s obj. \<lbrakk>s \<in> States; obj \<in> objects_of s\<rbrakk> \<Longrightarrow>
action_well_formed s obj"
    and A13_feedback_origin:
      "\<And>s obj. \<lbrakk>s \<in> States; obj \<in> objects_of s\<rbrakk> \<Longrightarrow>
feedback_well_formed s obj"
    and A14_step_preserves_well_formedness:
      "\<And>s oper s'. \<lbrakk>s \<in> States; step G s oper s'\<rbrakk> \<Longrightarrow>
state_well_formed G s'"
    and A15_strict_contains_irreflexive:
      "\<And>s c. s \<in> States \<Longrightarrow> \<not> strict_contains s c c"
    and A16_contains_eq_antisymmetric:
      "\<And>s a b. \<lbrakk>s \<in> States; contains_eq s a b; contains_eq s b a\<rbrakk>
\<Longrightarrow> a = b"
    and A17_projection_integrity:
      "\<And>s p. \<lbrakk>s \<in> States; p \<in> projections_of s\<rbrakk> \<Longrightarrow>
projection_well_formed s p"
    and A18_coupling_integrity:
      "\<And>s c. \<lbrakk>s \<in> States; c \<in> couplings_of s\<rbrakk> \<Longrightarrow>
coupling_well_formed s c"
    and A19_transmission_lifecycle:
      "\<And>s t. \<lbrakk>s \<in> States; t \<in> transmissions_of s\<rbrakk> \<Longrightarrow>
transmission_well_formed s t"
    and A20_external_effect_lineage:
      "\<And>s t. \<lbrakk>s \<in> States; t \<in> transmissions_of s; trans_status_of t =
TransValidated\<rbrakk>
\<Longrightarrow> transmission_target_effect_of t \<noteq> None \<and>
transmission_lineage_complete_of t"
    and A21_boundary_completeness:
      "\<And>s b. \<lbrakk>s \<in> States; b \<in> boundaries_of s\<rbrakk> \<Longrightarrow>
boundary_well_formed b"
    and A22_emergence_definition:
      "\<And>I. current_of I \<in> States \<Longrightarrow> classified_emergence I = emergence_of G I"
begin

theorem A14_state_result:
  assumes "s \<in> States" and "step G s oper s'"
  shows "state_well_formed G s'"
  using assms A14_step_preserves_well_formedness by blast

theorem A20_validated_transmission_has_lineage:
  assumes "s \<in> States" "t \<in> transmissions_of s" "trans_status_of t = TransValidated"
  shows "transmission_target_effect_of t \<noteq> None \<and> transmission_lineage_complete_of t"
  using assms A20_external_effect_lineage by blast

lemma A9_upgraded_evidence_resolves:
  assumes "s \<in> States" "obj \<in> objects_of s"
  "lifecycle_of obj \<in> {Supported, Validated}"
  shows
    "evidence_of obj \<noteq> {} \<and>
    evidence_of obj \<subsetq> version_ids s \<union> historical_versions_of s"
proof -
  have "evidence_well_formed s obj"
  using assms(1,2) A9_evidence_integrity by blast
  then show ?thesis
  using assms(3)
  unfolding evidence_well_formed_def evidence_references_complete_def
  by blast
qed

end

theorem describe_status_iff:
  "dr_status_of (describe G I) = DescriptionComplete \<longleftarrow> describeable G I"
  by (simp add: describe_def)

theorem emergence_total:
  "emergence_of G I \<in> {E0, E1, E2, E3}"
  by (auto simp: emergence_of_def)

theorem core_change_rejected:
  assumes "changes_core_of p"
  shows "\<not> patch_admissible G p \<and> patch_status G p = CoreConflict"
  using assms by (simp add: patch_admissible_def patch_status_def)

theorem relevance_bounded:
  "0 \<le> relevance_score w v uncertainty \<and> relevance_score w v uncertainty \<le> 1"
  unfolding relevance_score_def clamp01_def
  by auto

```

end

## D.2 Expliziter minimaler Basismodellzeuge

Die zweite Theorie UZA\_0\_9\_1\_Base\_Model konstruiert `base_object`, `base_snapshot`, `base_grammar` und die nichtleere Zustandsmenge `base_states`. Der Zustand enthält ein versioniertes Metaobjekt, erfüllt einschließlich A9 `state_well_formed` und besitzt mit `RegisterAppearance` einen zulässigen Selbstübergang. Die Interpretation Base: `uza_model` weist sämtliche Locale-Annahmen A1–A22 für diese konkrete Struktur nach. Das abschließende Theorem `explicit_uza_model_exists` beweist die Existenz eines nichtleeren `uza_model` mit mindestens einem Objekt. Dieser Zeuge ist ausdrücklich minimal: Da sein einziges Objekt `K_META` und `Draft` ist und die Erweiterungsmengen leer sind, werden die meisten objekttypspezifischen und Interaktionsbedingungen noch vakuos erfüllt.

Der integrierte Isabelle2025-2-Clean-Build verarbeitete die Basismodelltheorie vollständig und endete mit Exit-Code 0. Der aktuelle, auf lokalen Theorieimport umgestellte Modell Quelltext besitzt SHA-256 322d2a8fc31dbca1ea6c7cc2463b24c891701a79eebb761a94f5256a62632a4b. Damit ist die gemeinsame Erfüllbarkeit von A1–A22 relativ zu Isabelle/HOL mechanisch belegt. Der Zeuge selbst bleibt absichtlich minimal; seine frühere Nichtvakuositätsgrenze wird durch den zusätzlichen Rich-Model-Zeugen in D.4 adressiert. Nicht daraus folgt die Ableitung von A1–A22 aus einer schwächeren Theorie.

```
theory UZA_0_9_1_Base_Model
  imports UZA_0_9_1
begin

section \<open>An explicit nonempty witness model\<close>

text \<open>
  This theory gives a concrete, finite interpretation of the locale
  @{locale uza_model}. The witness is deliberately minimal: it contains one
  state, one versioned meta-object, and one permitted self-transition. It is a
  satisfiability witness, while most kind-specific obligations remain vacuous.
  Thus locale satisfiability is no longer merely presumed.
\<close>

definition base_object :: uza_object where
  "base_object =
    \<lparr> object_id_of = 0,
      version_id_of = 0,
      kind_of = K_META,
      lifecycle_of = Draft,
      previous_version_of = None,
      references_of = {},
      evidence_of = {},
      perspective_of = None,
      context_of = None,
      model_of = None,
      authority_of = None,
      valid_from_of = 0,
      valid_to_of = None,
      source_ref_of = None,
      target_ref_of = None,
      effect_origin_of = None,
      basis_refs_of = {},
      execution_ref_of = None,
      observation_refs_of = {},
      object_status_of = None \<rparr>"

definition base_snapshot :: snapshot where
  "base_snapshot =
    \<lparr> snapshot_id_of = 0,
      objects_of = {base_object},
```



```

    historical_versions_of = {},
    perspectives_of = {},
    contexts_of = {},
    models_of = {},
    authorities_of = {},
    contains_edges_of = {},
    projections_of = {},
    couplings_of = {},
    transmissions_of = {},
    boundaries_of = {} \<rparr>"

definition base_grammar :: grammar where
  "base_grammar =
    \<lparr> grammar_id_of = 0,
      supported_kinds_of = {K_META},
      allowed_operations_of = {RegisterAppearance},
      patch_budget_of = 0 \<rparr>"

definition base_states :: "snapshot set" where
  "base_states = {base_snapshot}"

definition base_classified_emergence ::
  "interaction_complex \<Rightarrow> emergence_level" where
  "base_classified_emergence I = emergence_of base_grammar I"

lemma base_object_simps [simp]:
  "object_id_of base_object = 0"
  "version_id_of base_object = 0"
  "kind_of base_object = K_META"
  "lifecycle_of base_object = Draft"
  "previous_version_of base_object = None"
  "references_of base_object = {}"
  "evidence_of base_object = {}"
  "perspective_of base_object = None"
  "context_of base_object = None"
  "model_of base_object = None"
  "authority_of base_object = None"
  "valid_from_of base_object = 0"
  "valid_to_of base_object = None"
  "source_ref_of base_object = None"
  "target_ref_of base_object = None"
  "effect_origin_of base_object = None"
  "basis_refs_of base_object = {}"
  "execution_ref_of base_object = None"
  "observation_refs_of base_object = {}"
  "object_status_of base_object = None"
  by (simp_all add: base_object_def)

lemma base_snapshot_simps [simp]:
  "snapshot_id_of base_snapshot = 0"
  "objects_of base_snapshot = {base_object}"
  "historical_versions_of base_snapshot = {}"
  "perspectives_of base_snapshot = {}"
  "contexts_of base_snapshot = {}"
  "models_of base_snapshot = {}"
  "authorities_of base_snapshot = {}"
  "contains_edges_of base_snapshot = {}"
  "projections_of base_snapshot = {}"
  "couplings_of base_snapshot = {}"
  "transmissions_of base_snapshot = {}"
  "boundaries_of base_snapshot = {}"
  by (simp_all add: base_snapshot_def)

lemma base_grammar_simps [simp]:
  "grammar_id_of base_grammar = 0"
  "supported_kinds_of base_grammar = {K_META}"
  "allowed_operations_of base_grammar = {RegisterAppearance}"
  "patch_budget_of base_grammar = 0"
  by (simp_all add: base_grammar_def)

lemma base_states_nonempty:
  "base_states \<noteq> {}"
  by (simp add: base_states_def)

lemma base_objects_nonempty:
  "objects_of base_snapshot \<noteq> {}"
  by (simp add: base_snapshot_def)

lemma base_state_well_formed:

```

```

"state_well_formed base_grammar base_snapshot"
unfolding state_well_formed_def
by (auto simp: unique_versions_def version_ids_def
  references_complete_def evidence_well_formed_def
  evidence_references_complete_def time_well_formed_def relation_well_formed_def
  effect_well_formed_def perspective_well_formed_def perspective_bound_kind_def
  context_well_formed_def context_bound_kind_def relevance_well_formed_def
  orientation_well_formed_def action_well_formed_def feedback_well_formed_def
  base_object_def base_snapshot_def base_grammar_def)

lemma base_self_step:
  "step base_grammar base_snapshot RegisterAppearance base_snapshot"
  unfolding step_def
  by (simp add: immutable_preserved_def base_state_well_formed)

interpretation Base: uza_model
  base_states base_grammar base_classified_emergence
proof
  fix s o1 o2
  assume "s \<in> base_states" "o1 \<in> objects_of s" "o2 \<in> objects_of s"
  "version_id_of o1 = version_id_of o2"
  then show "o1 = o2"
  by (simp add: base_states_def base_snapshot_def)
next
  fix s
  assume "s \<in> base_states"
  then show "wf (revision_edges_of s)"
  by (simp add: base_states_def revision_edges_of_def base_snapshot_def
    base_object_def)
next
  fix s obj
  assume "s \<in> base_states" "obj \<in> objects_of s"
  then show "time_well_formed obj"
  by (simp add: base_states_def base_snapshot_def base_object_def
    time_well_formed_def)
next
  fix s obj
  assume "s \<in> base_states" "obj \<in> objects_of s"
  then show "references_complete s obj"
  by (simp add: base_states_def base_snapshot_def base_object_def
    references_complete_def version_ids_def)
next
  fix s obj
  assume "s \<in> base_states" "obj \<in> objects_of s"
  then show "relation_well_formed s obj"
  by (simp add: base_states_def base_snapshot_def base_object_def
    relation_well_formed_def)
next
  fix s obj
  assume "s \<in> base_states" "obj \<in> objects_of s"
  then show "effect_well_formed s obj"
  by (simp add: base_states_def base_snapshot_def base_object_def
    effect_well_formed_def)
next
  fix s obj
  assume "s \<in> base_states" "obj \<in> objects_of s"
  then show "perspective_well_formed s obj"
  by (simp add: base_states_def base_snapshot_def base_object_def
    perspective_well_formed_def perspective_bound_kind_def)
next
  fix s obj
  assume "s \<in> base_states" "obj \<in> objects_of s"
  then show "context_well_formed s obj"
  by (simp add: base_states_def base_snapshot_def base_object_def
    context_well_formed_def context_bound_kind_def)
next
  fix s obj
  assume "s \<in> base_states" "obj \<in> objects_of s"
  then show "evidence_well_formed s obj"
  by (simp add: base_states_def base_snapshot_def base_object_def
    evidence_well_formed_def evidence_references_complete_def version_ids_def)
next
  fix s obj
  assume "s \<in> base_states" "obj \<in> objects_of s"
  then show "relevance_well_formed s obj"
  by (simp add: base_states_def base_snapshot_def base_object_def
    relevance_well_formed_def)
next
  fix s obj

```

```

    assume "s \<in> base_states" "obj \<in> objects_of s"
    then show "orientation_well_formed s obj"
    by (simp add: base_states_def base_snapshot_def base_object_def
        orientation_well_formed_def)
next
  fix s obj
  assume "s \<in> base_states" "obj \<in> objects_of s"
  then show "action_well_formed s obj"
  by (simp add: base_states_def base_snapshot_def base_object_def
      action_well_formed_def)
next
  fix s obj
  assume "s \<in> base_states" "obj \<in> objects_of s"
  then show "feedback_well_formed s obj"
  by (simp add: base_states_def base_snapshot_def base_object_def
      feedback_well_formed_def)
next
  fix s oper s'
  assume "s \<in> base_states" "step base_grammar s oper s'"
  then show "state_well_formed base_grammar s'"
  by (simp add: step_def)
next
  fix s c
  assume "s \<in> base_states"
  then show "\<not> strict_contains s c c"
  by (simp add: base_states_def strict_contains_def base_snapshot_def)
next
  fix s a b
  assume "s \<in> base_states" "contains_eq s a b" "contains_eq s b a"
  then show "a = b"
  by (simp add: base_states_def contains_eq_def strict_contains_def
      base_snapshot_def)
next
  fix s p
  assume "s \<in> base_states" "p \<in> projections_of s"
  then show "projection_well_formed s p"
  by (simp add: base_states_def base_snapshot_def)
next
  fix s c
  assume "s \<in> base_states" "c \<in> couplings_of s"
  then show "coupling_well_formed s c"
  by (simp add: base_states_def base_snapshot_def)
next
  fix s t
  assume "s \<in> base_states" "t \<in> transmissions_of s"
  then show "transmission_well_formed s t"
  by (simp add: base_states_def base_snapshot_def)
next
  fix s t
  assume "s \<in> base_states" "t \<in> transmissions_of s"
  "trans_status_of t = TransValidated"
  then show "transmission_target_effect_of t \<noteq> None \<and>
      transmission_lineage_complete_of t"
  by (simp add: base_states_def base_snapshot_def)
next
  fix s b
  assume "s \<in> base_states" "b \<in> boundaries_of s"
  then show "boundary_well_formed b"
  by (simp add: base_states_def base_snapshot_def)
next
  fix I :: interaction_complex
  assume "current_of I \<in> base_states"
  show "base_classified_emergence I = emergence_of base_grammar I"
  by (simp add: base_classified_emergence_def)
qed

theorem explicit_uza_model_exists:
  "\<exists>States G classify.
    States \<noteq> {} \<and>
    (\<exists>s\<in>States. objects_of s \<noteq> {}) \<and>
    uza_model States G classify"
  using base_states_nonempty base_objects_nonempty Base.uza_model_axioms
  by (intro exI[of _ base_states] exI[of _ base_grammar]
      exI[of _ base_classified_emergence])
    (auto simp: base_states_def)
end

```

### D.3 Konkrete T1-Semantik und Erhaltung für RegisterAppearance

Die dritte Theorie UZA\_0\_9\_1\_Register\_Appearance schließt erstmals die Lücke zwischen dem generischen Operationsnamen und einer konstruktiven Zustandssemantik.

appearance\_input\_valid verlangt einen wohlgeformten Ausgangszustand, eine zulässige Grammatikoperation, den unterstützten Typ K\_ENT, frische Objekt- und Versionsidentitäten, vollständige allgemeine Referenzen, evidence\_well\_formed, gültige Zeit sowie vorhandene Perspektiv- und Kontextbindungen. Eine registrierte Beobachtung besitzt keine Potentialquelle, keine Potentialbasis und keinen Analysemarker; dadurch bleibt sie von den in D.5 definierten T2-Kandidaten unterscheidbar. add\_appearance erzeugt den Folgezustand durch genau eine Objektinsertion und eine fortgeschriebene Snapshot-ID. register\_appearance\_step verbindet Vorbedingung und Konstruktion; die Wohlgeformtheit des Zielzustands ist ausdrücklich **keine** Vorbedingung.

Isabelle beweist anschließend die exakte Zustandsdifferenz, die Eindeutigkeit der Versionsidentitäten, die Unveränderlichkeit sämtlicher bereits vorhandener Versionen, die Erhaltung von Referenzen, Evidence Integrity, Zeit, Perspektive und Kontext, die vollständige state\_well\_formed-Eigenschaft des konstruierten Folgezustands und schließlich register\_appearance\_refines\_generic\_step. Das zusätzliche Theorem register\_appearance\_preserves\_evidence\_integrity weist A9 ausdrücklich für alle Objekte des Zielzustands nach. Der integrierte Isabelle2025-2-Clean-Build verarbeitete die T1-Theorie vollständig und endete mit Exit-Code 0. Der aktuelle Operations Quelltext besitzt SHA-256 9b8cc6167b99070a47f63b4e63bd5a783a000f6ec84f7620945bee224f116cd0.

Damit ist Aufgabe 4 für T1 bis T7 im historischen Anhang D abgeschlossen. K2 ergänzte T8 in Anhang J; K3 schließt in Anhang K T9–T13 mit eigener Semantik, kausalen Erfolgs- und Abbruchzeugen sowie vollständigem Clean-Build. Nicht bewiesen sind Produktivengine-Äquivalenz, allgemeine Modell-Expansion, relative Konsistenz, Konservativität oder semantische Angemessenheit in jeder Anwendungsdomäne.

```
theory UZA_0_9_1_Register_Appearance
  imports UZA_0_9_1
begin

section \<open>T1 RegisterAppearance: concrete operation semantics\<close>

text \<open>
  This theory gives the first core operation an exact, non-circular transition
  semantics. The source state is required to be well formed. The target state
  is constructed by adding exactly one fresh appearance object; target
  well-formedness is derived rather than assumed.
\<close>

definition appearance_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> uza_object \<Rightarrow> nat \<Rightarrow> bool" where
  "appearance_input_valid G s new_obj next_snapshot_id \<longleftarrow>
    state_well_formed G s \<and>
    RegisterAppearance \<in> allowed_operations_of G \<and>
    K_ENT \<in> supported_kinds_of G \<and>
    kind_of new_obj = K_ENT \<and>
    lifecycle_of new_obj \<in> {Draft, Hypothesis} \<and>
    object_id_of new_obj \<notin> object_id_of ` objects_of s \<and>
    version_id_of new_obj \<notin> version_ids s \<union> historical_versions_of s \<and>
    previous_version_of new_obj = None \<and>
    source_ref_of new_obj = None \<and>
    basis_refs_of new_obj = {} \<and>
    object_status_of new_obj = None \<and>
    references_complete s new_obj \<and>
```

```

evidence_well_formed s new_obj \<and>
time_well_formed new_obj \<and>
(\<exists>p\<in>perspectives_of s. perspective_of new_obj = Some p) \<and>
(\<exists>c\<in>contexts_of s. context_of new_obj = Some c) \<and>
snapshot_id_of s < next_snapshot_id"

definition add_appearance ::
  "snapshot \<Rightarrow> uza_object \<Rightarrow> nat \<Rightarrow> snapshot" where
  "add_appearance s new_obj next_snapshot_id =
    s\<lparr> snapshot_id_of := next_snapshot_id,
    objects_of := insert new_obj (objects_of s) \<rparr>"

definition register_appearance_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> uza_object \<Rightarrow> nat \<Rightarrow> snapshot
  \<Rightarrow> bool" where
  "register_appearance_step G s new_obj next_snapshot_id s' \<longlefttrightarrow>
    appearance_input_valid G s new_obj next_snapshot_id \<and>
    s' = add_appearance s new_obj next_snapshot_id"

subsection \<open>Exact update and frame properties\<close>

lemma add_appearance_objects [simp]:
  "objects_of (add_appearance s new_obj next_snapshot_id) =
    insert new_obj (objects_of s)"
  by (simp add: add_appearance_def)

lemma add_appearance_snapshot_id [simp]:
  "snapshot_id_of (add_appearance s new_obj next_snapshot_id) = next_snapshot_id"
  by (simp add: add_appearance_def)

lemma add_appearance_background [simp]:
  "historical_versions_of (add_appearance s new_obj next_snapshot_id) =
    historical_versions_of s"
  "perspectives_of (add_appearance s new_obj next_snapshot_id) = perspectives_of s"
  "contexts_of (add_appearance s new_obj next_snapshot_id) = contexts_of s"
  "models_of (add_appearance s new_obj next_snapshot_id) = models_of s"
  "authorities_of (add_appearance s new_obj next_snapshot_id) = authorities_of s"
  "contains_edges_of (add_appearance s new_obj next_snapshot_id) = contains_edges_of s"
  "projections_of (add_appearance s new_obj next_snapshot_id) = projections_of s"
  "couplings_of (add_appearance s new_obj next_snapshot_id) = couplings_of s"
  "transmissions_of (add_appearance s new_obj next_snapshot_id) = transmissions_of s"
  "boundaries_of (add_appearance s new_obj next_snapshot_id) = boundaries_of s"
  by (simp_all add: add_appearance_def)

lemma add_appearance_version_ids [simp]:
  "version_ids (add_appearance s new_obj next_snapshot_id) =
    insert (version_id_of new_obj) (version_ids s)"
  by (auto simp: version_ids_def)

lemma appearance_input_fresh_object:
  assumes valid: "appearance_input_valid G s new_obj next_snapshot_id"
  shows "new_obj \<notin> objects_of s"
  using valid
  by (auto simp: appearance_input_valid_def)

lemma register_appearance_exact_delta:
  assumes concrete: "register_appearance_step G s new_obj next_snapshot_id s'"
  shows
    "objects_of s' = insert new_obj (objects_of s) \<and>
    new_obj \<notin> objects_of s \<and>
    snapshot_id_of s' = next_snapshot_id"
  using concrete appearance_input_fresh_object
  by (auto simp: register_appearance_step_def)

subsection \<open>Monotonicity of object conditions\<close>

lemma add_appearance_present_ref_mono:
  assumes "present_ref s ref"
  shows "present_ref (add_appearance s new_obj next_snapshot_id) ref"
  using assms
  by (cases ref) (auto simp: present_ref_def)

lemma add_appearance_references_mono:
  assumes "references_complete s obj"
  shows "references_complete (add_appearance s new_obj next_snapshot_id) obj"
  using assms
  by (auto simp: references_complete_def)

lemma add_appearance_evidence_mono:

```

```

assumes "evidence_well_formed s obj"
shows "evidence_well_formed (add_appearance s new_obj next_snapshot_id) obj"
using assms
by (auto simp: evidence_well_formed_def evidence_references_complete_def)

lemma add_appearance_relation_mono:
  assumes "relation_well_formed s obj"
  shows "relation_well_formed (add_appearance s new_obj next_snapshot_id) obj"
  using assms add_appearance_present_ref_mono
  unfolding relation_well_formed_def
  by blast

lemma add_appearance_effect_mono:
  assumes "effect_well_formed s obj"
  shows "effect_well_formed (add_appearance s new_obj next_snapshot_id) obj"
  using assms add_appearance_present_ref_mono
  unfolding effect_well_formed_def
  by blast

lemma add_appearance_perspective_iff [simp]:
  "perspective_well_formed (add_appearance s new_obj next_snapshot_id) obj
  \<longleftarrowrightarrow> perspective_well_formed s obj"
  unfolding perspective_well_formed_def
  by (simp only: add_appearance_background(2))

lemma add_appearance_context_iff [simp]:
  "context_well_formed (add_appearance s new_obj next_snapshot_id) obj
  \<longleftarrowrightarrow> context_well_formed s obj"
  unfolding context_well_formed_def
  by (simp only: add_appearance_background(3))

lemma add_appearance_relevance_iff [simp]:
  "relevance_well_formed (add_appearance s new_obj next_snapshot_id) obj
  \<longleftarrowrightarrow> relevance_well_formed s obj"
  unfolding relevance_well_formed_def
  by (simp only: add_appearance_perspective_iff add_appearance_context_iff
  add_appearance_background(4))

lemma add_appearance_orientation_iff [simp]:
  "orientation_well_formed (add_appearance s new_obj next_snapshot_id) obj
  \<longleftarrowrightarrow> orientation_well_formed s obj"
  by (simp add: orientation_well_formed_def)

lemma add_appearance_action_iff [simp]:
  "action_well_formed (add_appearance s new_obj next_snapshot_id) obj
  \<longleftarrowrightarrow> action_well_formed s obj"
  unfolding action_well_formed_def
  by (simp only: add_appearance_background(5))

lemma add_appearance_feedback_mono:
  assumes "feedback_well_formed s obj"
  shows "feedback_well_formed (add_appearance s new_obj next_snapshot_id) obj"
  using assms add_appearance_present_ref_mono
  unfolding feedback_well_formed_def
  by blast

subsection \<open>Fresh identity and new-object validity\<close>

lemma add_appearance_preserves_unique_versions:
  assumes valid: "appearance_input_valid G s new_obj next_snapshot_id"
  shows "unique_versions (add_appearance s new_obj next_snapshot_id)"
proof -
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
  and fresh: "version_id_of new_obj \<notin> version_ids s"
  unfolding appearance_input_valid_def state_well_formed_def unique_versions_def
  by auto
  show ?thesis
  unfolding unique_versions_def
  proof (rule inj_onI)
    fix left right
    assume left_mem: "left \<in> objects_of (add_appearance s new_obj next_snapshot_id)"
    and right_mem: "right \<in> objects_of (add_appearance s new_obj next_snapshot_id)"
    and same_version: "version_id_of left = version_id_of right"
    show "left = right"
    proof (cases "left = new_obj")
      case True
      show ?thesis
      proof (cases "right = new_obj")
        case True

```

```

    then show ?thesis using \<open>left = new_obj\<close> by simp
  next
  case False
  then have right_old: "right \<in> objects_of s" using right_mem by simp
  have new_version_old: "version_id_of new_obj \<in> version_ids s"
    unfolding version_ids_def
    using same_version \<open>left = new_obj\<close> right_old
    by (metis imageI)
  have contradiction: False
    using fresh new_version_old by blast
  show ?thesis using contradiction by blast
qed
next
case False
then have left_old: "left \<in> objects_of s" using left_mem by simp
show ?thesis
proof (cases "right = new_obj")
  case True
  have new_version_old: "version_id_of new_obj \<in> version_ids s"
    unfolding version_ids_def
    using same_version left_old True
    by (metis imageI)
  have contradiction: False
    using fresh new_version_old by blast
  show ?thesis using contradiction by blast
next
case False
then have right_old: "right \<in> objects_of s" using right_mem by simp
show ?thesis
  using inj_onD[OF source_unique same_version left_old right_old] .
qed
qed
qed
qed

lemma appearance_input_new_object_conditions:
  assumes valid: "appearance_input_valid G s new_obj next_snapshot_id"
  shows
    "kind_of new_obj \<in> supported_kinds_of G \<and>
    references_complete (add_appearance s new_obj next_snapshot_id) new_obj \<and>
    evidence_well_formed (add_appearance s new_obj next_snapshot_id) new_obj \<and>
    time_well_formed new_obj \<and>
    relation_well_formed (add_appearance s new_obj next_snapshot_id) new_obj \<and>
    effect_well_formed (add_appearance s new_obj next_snapshot_id) new_obj \<and>
    perspective_well_formed (add_appearance s new_obj next_snapshot_id) new_obj \<and>
    context_well_formed (add_appearance s new_obj next_snapshot_id) new_obj \<and>
    relevance_well_formed (add_appearance s new_obj next_snapshot_id) new_obj \<and>
    orientation_well_formed (add_appearance s new_obj next_snapshot_id) new_obj \<and>
    action_well_formed (add_appearance s new_obj next_snapshot_id) new_obj \<and>
    feedback_well_formed (add_appearance s new_obj next_snapshot_id) new_obj"
proof -
  from valid have supported: "kind_of new_obj \<in> supported_kinds_of G"
  and kind: "kind_of new_obj = K_ENT"
  and refs: "references_complete s new_obj"
  and evidence: "evidence_well_formed s new_obj"
  and time: "time_well_formed new_obj"
  and perspective:
    "perspective_well_formed (add_appearance s new_obj next_snapshot_id) new_obj"
  and context_ok:
    "context_well_formed (add_appearance s new_obj next_snapshot_id) new_obj"
  unfolding appearance_input_valid_def perspective_well_formed_def
    context_well_formed_def perspective_bound_kind_def context_bound_kind_def
  by auto
  have refs_target:
    "references_complete (add_appearance s new_obj next_snapshot_id) new_obj"
    using add_appearance_references_mono[OF refs] .
  have evidence_target:
    "evidence_well_formed (add_appearance s new_obj next_snapshot_id) new_obj"
    using add_appearance_evidence_mono[OF evidence] .
  have typed_conditions:
    "relation_well_formed (add_appearance s new_obj next_snapshot_id) new_obj \<and>
    effect_well_formed (add_appearance s new_obj next_snapshot_id) new_obj \<and>
    relevance_well_formed (add_appearance s new_obj next_snapshot_id) new_obj \<and>
    orientation_well_formed (add_appearance s new_obj next_snapshot_id) new_obj \<and>
    action_well_formed (add_appearance s new_obj next_snapshot_id) new_obj \<and>
    feedback_well_formed (add_appearance s new_obj next_snapshot_id) new_obj"
  using kind
  by (simp add: relation_well_formed_def effect_well_formed_def
    relevance_well_formed_def orientation_well_formed_def

```

```

    action_well_formed_def feedback_well_formed_def)
show ?thesis
  using supported refs_target evidence_target time perspective context_ok typed_conditions
  by blast
qed

subsection \<open>Operation-specific preservation theorems\<close>

theorem register_appearance_preserves_state_well_formed:
  assumes concrete: "register_appearance_step G s new_obj next_snapshot_id s'"
  shows "state_well_formed G s'"
proof -
  from concrete have valid: "appearance_input_valid G s new_obj next_snapshot_id"
  and target: "s' = add_appearance s new_obj next_snapshot_id"
  by (auto simp: register_appearance_step_def)
  from valid have source_wf: "state_well_formed G s"
  by (simp add: appearance_input_valid_def)
  have unique: "unique_versions (add_appearance s new_obj next_snapshot_id)"
  using add_appearance_preserves_unique_versions[OF valid] .
  have all_objects:
    "\<forall>obj\<in>objects_of (add_appearance s new_obj next_snapshot_id).
    kind_of obj \<in> supported_kinds_of G \<and>
    references_complete (add_appearance s new_obj next_snapshot_id) obj \<and>
    evidence_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
    time_well_formed obj \<and>
    relation_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
    effect_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
    perspective_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
    context_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
    relevance_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
    orientation_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
    action_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
    feedback_well_formed (add_appearance s new_obj next_snapshot_id) obj"
  proof (intro ballI)
    fix obj
    assume obj_mem:
      "obj \<in> objects_of (add_appearance s new_obj next_snapshot_id)"
    consider (new) "obj = new_obj" | (old) "obj \<in> objects_of s"
    using obj_mem by auto
    then show
      "kind_of obj \<in> supported_kinds_of G \<and>
      references_complete (add_appearance s new_obj next_snapshot_id) obj \<and>
      evidence_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
      time_well_formed obj \<and>
      relation_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
      effect_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
      perspective_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
      context_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
      relevance_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
      orientation_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
      action_well_formed (add_appearance s new_obj next_snapshot_id) obj \<and>
      feedback_well_formed (add_appearance s new_obj next_snapshot_id) obj"
  proof cases
    case new
    then show ?thesis
      using appearance_input_new_object_conditions[OF valid] by simp
  next
    case old
    from source_wf old have source_conditions:
      "kind_of obj \<in> supported_kinds_of G \<and>
      references_complete s obj \<and> evidence_well_formed s obj \<and>
      time_well_formed obj \<and>
      relation_well_formed s obj \<and> effect_well_formed s obj \<and>
      perspective_well_formed s obj \<and> context_well_formed s obj \<and>
      relevance_well_formed s obj \<and> orientation_well_formed s obj \<and>
      action_well_formed s obj \<and> feedback_well_formed s obj"
    unfolding state_well_formed_def by blast
    from source_conditions have supported: "kind_of obj \<in> supported_kinds_of G"
    and refs: "references_complete s obj"
    and evidence: "evidence_well_formed s obj"
    and time: "time_well_formed obj"
    and relation: "relation_well_formed s obj"
    and effect: "effect_well_formed s obj"
    and perspective: "perspective_well_formed s obj"
    and context_ok: "context_well_formed s obj"
    and relevance: "relevance_well_formed s obj"
    and orientation: "orientation_well_formed s obj"
    and action: "action_well_formed s obj"
    and feedback: "feedback_well_formed s obj"

```



```

    by blast+
    have refs_target:
      "references_complete (add_appearance s new_obj next_snapshot_id) obj"
      using add_appearance_references_mono[OF refs] .
    have evidence_target:
      "evidence_well_formed (add_appearance s new_obj next_snapshot_id) obj"
      using add_appearance_evidence_mono[OF evidence] .
    have relation_target:
      "relation_well_formed (add_appearance s new_obj next_snapshot_id) obj"
      using add_appearance_relation_mono[OF relation] .
    have effect_target:
      "effect_well_formed (add_appearance s new_obj next_snapshot_id) obj"
      using add_appearance_effect_mono[OF effect] .
    have feedback_target:
      "feedback_well_formed (add_appearance s new_obj next_snapshot_id) obj"
      using add_appearance_feedback_mono[OF feedback] .
    show ?thesis
      using supported refs_target evidence_target time relation_target effect_target perspective
        context_ok relevance orientation action feedback_target
      by simp
  qed
qed
from source_wf have source_secondary:
  "(\<forall>p\<in>projections_of s. projection_well_formed s p) \<and>
  (\<forall>c\<in>couplings_of s. coupling_well_formed s c) \<and>
  (\<forall>t\<in>transmissions_of s. transmission_well_formed s t) \<and>
  (\<forall>b\<in>boundaries_of s. boundary_well_formed b)"
  unfolding state_well_formed_def by blast
have target_secondary:
  "(\<forall>p\<in>projections_of (add_appearance s new_obj next_snapshot_id).
  projection_well_formed (add_appearance s new_obj next_snapshot_id) p) \<and>
  (\<forall>c\<in>couplings_of (add_appearance s new_obj next_snapshot_id).
  coupling_well_formed (add_appearance s new_obj next_snapshot_id) c) \<and>
  (\<forall>t\<in>transmissions_of (add_appearance s new_obj next_snapshot_id).
  transmission_well_formed (add_appearance s new_obj next_snapshot_id) t) \<and>
  (\<forall>b\<in>boundaries_of (add_appearance s new_obj next_snapshot_id).
  boundary_well_formed b)"
  using source_secondary
  by (simp add: projection_well_formed_def coupling_well_formed_def
    transmission_well_formed_def)
show ?thesis
  unfolding target state_well_formed_def
  using unique all_objects target_secondary by blast
qed

theorem register_appearance_preserves_identity:
  assumes concrete: "register_appearance_step G s new_obj next_snapshot_id s'"
  shows "unique_versions s' \<and> immutable_preserved s s'"
proof -
  from concrete have valid: "appearance_input_valid G s new_obj next_snapshot_id"
  and target: "s' = add_appearance s new_obj next_snapshot_id"
  by (auto simp: register_appearance_step_def)
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
  and fresh: "version_id_of new_obj \<notin> version_ids s"
  unfolding appearance_input_valid_def state_well_formed_def unique_versions_def
  by auto
  have immutable: "immutable_preserved s s'"
  unfolding immutable_preserved_def target
  proof (intro ballI allI impI)
    fix old_obj target_obj
    assume old_mem: "old_obj \<in> objects_of s"
    and target_mem:
      "target_obj \<in> objects_of (add_appearance s new_obj next_snapshot_id)"
    and same_version: "version_id_of old_obj = version_id_of target_obj"
    show "old_obj = target_obj"
    proof (cases "target_obj = new_obj")
      case True
      have new_version_old: "version_id_of new_obj \<in> version_ids s"
      unfolding version_ids_def
      using old_mem same_version True
      by (metis imageI)
      have contradiction: False
      using fresh new_version_old by blast
      show ?thesis using contradiction by blast
    next
      case False
      then have target_old: "target_obj \<in> objects_of s" using target_mem by simp
      show ?thesis
        using inj_onD[OF source_unique same_version old_mem target_old] .
    qed
  qed

```

```

qed
qed
have target_unique: "unique_versions s'"
  unfolding target
  using add_appearance_preserves_unique_versions[OF valid] .
show ?thesis using target_unique immutable by blast
qed

theorem register_appearance_preserves_references_time_perspective_context:
  assumes concrete: "register_appearance_step G s new_obj next_snapshot_id s'"
  shows
    "\<forall>obj\<in>objects_of s'.
      references_complete s' obj \<and>
      evidence_well_formed s' obj \<and>
      time_well_formed obj \<and>
      perspective_well_formed s' obj \<and>
      context_well_formed s' obj"
  using register_appearance_preserves_state_well_formed[OF concrete]
  unfolding state_well_formed_def
  by blast

theorem register_appearance_preserves_evidence_integrity:
  assumes concrete: "register_appearance_step G s new_obj next_snapshot_id s'"
  shows "\<forall>obj\<in>objects_of s'. evidence_well_formed s' obj"
  using register_appearance_preserves_state_well_formed[OF concrete]
  unfolding state_well_formed_def
  by blast

theorem register_appearance_refines_generic_step:
  assumes concrete: "register_appearance_step G s new_obj next_snapshot_id s'"
  shows "step G s RegisterAppearance s'"
proof -
  from concrete have source_wf: "state_well_formed G s"
    and allowed: "RegisterAppearance \<in> allowed_operations_of G"
  by (auto simp: register_appearance_step_def appearance_input_valid_def)
  have target_wf: "state_well_formed G s'"
  using register_appearance_preserves_state_well_formed[OF concrete] .
  have immutable: "immutable_preserved s s'"
  using register_appearance_preserves_identity[OF concrete] by blast
  show ?thesis
    unfolding step_def
    using source_wf target_wf allowed immutable by blast
qed
end

```

## D.4 Reichhaltiger nicht-vakuoser Modellzeuge

Die vierte Theorie UZA\_0\_9\_1\_Rich\_Model ergänzt den minimalen Erfüllbarkeitszeugen durch ein endliches, semantisch belegtes Snapshot mit zwölf versionierten Objekten. Enthalten sind Metaobjekt, Perspektiven- und Kontextobjekte, Erscheinung, Relation, validierte Wirkung, Resonanz, validierte Relevanz, Orientierung, validierte Handlung und Rückwirkung. Zusätzlich enthält der Zustand zwei Kontexte, eine verlustfreie Projektion, eine Kopplung mit Kanal, eine validierte Transmission mit Zielwirkung und vollständiger Lineage sowie ein erfolgreiches Fusion-BoundaryEvent mit vollständiger Disposition.

Die Lemmata rich\_A5\_relation\_nonvacuous, rich\_A6\_effect\_nonvacuous, rich\_A7\_A8\_bindings\_nonvacuous, rich\_A9\_evidence\_nonvacuous, rich\_A10\_relevance\_nonvacuous, rich\_A11\_orientation\_nonvacuous, rich\_A12\_action\_nonvacuous, rich\_A13\_feedback\_nonvacuous und rich\_A17\_A21\_interaction\_nonvacuous weisen für jede betroffene Bedingungsgruppe ausdrücklich erfüllte Antezedenzen und die zugehörige Wohlgeformtheit nach. Dadurch werden A5–A13 und A17–A21 nicht mehr nur durch leere Objektmengen oder unzutreffende Objekttypen erfüllt. rich\_state\_well\_formed beweist den gesamten Zustand, Rich: uza\_model interpretiert erneut A1–

A22, und rich\_ua\_model\_exists belegt die Existenz eines solchen nichtvakuoosen Modells relativ zu Isabelle/HOL.

Der integrierte Isabelle2025-2-Clean-Build verarbeitete UZA\_0\_9\_1\_Rich\_Model vollständig und endete mit Exit-Code 0. Der aktuelle Quelltext besitzt SHA-256

3a3671b918d9639607fb6152dedd039cdf6145b0f040db8edf6ba296b8dc9a8c. Damit ist das Rich-Witness-Gate abgeschlossen. Nicht daraus folgen Expansion jedes beliebigen Kernmodells, Konservativität, semantische Angemessenheit für jede Domäne oder Produktivengine-Äquivalenz.

```
theory UZA_0_9_1_Rich_Model
  imports UZA_0_9_1_Base_Model
begin

section \<open>A finite non-vacuous witness model\<close>

text \<open>
  The minimal base model proves joint satisfiability of A1--A22 but leaves most
  kind-specific antecedents false. This second finite interpretation activates
  every object-specific obligation A5--A13 and the interaction obligations
  A17--A21. It contains an appearance, a relation with resolved endpoints, a
  validated effect with origin and evidence, resonance, relevance, orientation,
  action, feedback, two contexts, a projection, a coupling, a validated
  transmission, and a successful fusion boundary event.
\<close>

definition rich_meta :: uza_object where
  "rich_meta = base_object"

definition rich_perspective_obj :: uza_object where
  "rich_perspective_obj =
    base_object\<lparr>object_id_of := 1, version_id_of := 1,
    kind_of := K_PERSP\<rparr>"

definition rich_context_source_obj :: uza_object where
  "rich_context_source_obj =
    base_object\<lparr>object_id_of := 2, version_id_of := 2,
    kind_of := K_CTX\<rparr>"

definition rich_entity :: uza_object where
  "rich_entity =
    base_object\<lparr>object_id_of := 3, version_id_of := 3,
    kind_of := K_ENT, lifecycle_of := Supported,
    references_of := {0}, evidence_of := {0},
    perspective_of := Some 10, context_of := Some 20\<rparr>"

definition rich_relation :: uza_object where
  "rich_relation =
    base_object\<lparr>object_id_of := 4, version_id_of := 4,
    kind_of := K_REL, lifecycle_of := Supported,
    references_of := {3}, evidence_of := {3}, context_of := Some 20,
    source_ref_of := Some 3, target_ref_of := Some 3\<rparr>"

definition rich_effect :: uza_object where
  "rich_effect =
    base_object\<lparr>object_id_of := 5, version_id_of := 5,
    kind_of := K_EFF, lifecycle_of := Validated,
    references_of := {4}, evidence_of := {3,4},
    perspective_of := Some 10, context_of := Some 20,
    effect_origin_of := Some 4\<rparr>"

definition rich_resonance :: uza_object where
  "rich_resonance =
    base_object\<lparr>object_id_of := 6, version_id_of := 6,
    kind_of := K_RES, lifecycle_of := Supported,
    references_of := {5}, evidence_of := {5},
    perspective_of := Some 10, context_of := Some 20\<rparr>"

definition rich_relevance :: uza_object where
  "rich_relevance =
    base_object\<lparr>object_id_of := 7, version_id_of := 7,
    kind_of := K_REV, lifecycle_of := Validated,
    references_of := {5,6}, evidence_of := {5},
    perspective_of := Some 10, context_of := Some 20,
    model_of := Some 30\<rparr>"
```

```

definition rich_orientation :: uza_object where
  "rich_orientation =
    base_object\<lparr>object_id_of := 8, version_id_of := 8,
    kind_of := K_ORI, lifecycle_of := Supported,
    references_of := {7}, evidence_of := {7},
    perspective_of := Some 10, context_of := Some 20,
    basis_refs_of := {7}\<rparr>"

definition rich_action :: uza_object where
  "rich_action =
    base_object\<lparr>object_id_of := 9, version_id_of := 9,
    kind_of := K_ACT, lifecycle_of := Validated,
    references_of := {8}, evidence_of := {8},
    perspective_of := Some 10, context_of := Some 20,
    authority_of := Some 40, basis_refs_of := {8}\<rparr>"

definition rich_feedback :: uza_object where
  "rich_feedback =
    base_object\<lparr>object_id_of := 10, version_id_of := 10,
    kind_of := K_FDB, lifecycle_of := Supported,
    references_of := {3,9}, evidence_of := {3}, context_of := Some 20,
    execution_ref_of := Some 9, observation_refs_of := {3}\<rparr>"

definition rich_context_target_obj :: uza_object where
  "rich_context_target_obj =
    base_object\<lparr>object_id_of := 11, version_id_of := 11,
    kind_of := K_CTX\<rparr>"

definition rich_objects :: "uza_object set" where
  "rich_objects =
    {rich_meta, rich_perspective_obj, rich_context_source_obj, rich_entity,
    rich_relation, rich_effect, rich_resonance, rich_relevance,
    rich_orientation, rich_action, rich_feedback, rich_context_target_obj}"

definition rich_loss :: loss_profile where
  "rich_loss =
    \<lparr>structural_loss_of = 0, semantic_loss_of = 0,
    perspective_loss_of = 0, temporal_loss_of = 0,
    loss_explanation_of = 'identity witness'\<rparr>"

definition rich_projection :: projection where
  "rich_projection =
    \<lparr>projection_id_of = 1, contained_context_of = 20,
    container_context_of = 21, source_snapshot_of = 1,
    projection_loss_of = rich_loss\<rparr>"

definition rich_coupling :: coupling where
  "rich_coupling =
    \<lparr>coupling_id_of = 50, coupling_source_of = 20,
    coupling_target_of = 21, coupling_channels_of = {60}\<rparr>"

definition rich_transmission :: transmission where
  "rich_transmission =
    \<lparr>transmission_id_of = 70, transmission_coupling_of = 50,
    transmission_channel_of = 60, transmission_source_effect_of = 5,
    transmission_target_effect_of = Some 15,
    prior_trans_status_of = Some TransObserved,
    trans_status_of = TransValidated,
    transmission_lineage_complete_of = True\<rparr>"

definition rich_boundary :: boundary_event where
  "rich_boundary =
    \<lparr>boundary_id_of = 80, boundary_kind_of = Fusion,
    boundary_succeeded_of = True, boundary_lineage_complete_of = True,
    boundary_disposition_complete_of = True\<rparr>"

definition rich_snapshot :: snapshot where
  "rich_snapshot =
    \<lparr>snapshot_id_of = 1,
    objects_of = rich_objects,
    historical_versions_of = {},
    perspectives_of = {10},
    contexts_of = {20,21},
    models_of = {30},
    authorities_of = {40},
    contains_edges_of = {(20,21)},
    projections_of = {rich_projection},
    couplings_of = {rich_coupling},

```

```

    transmissions_of = {rich_transmission},
    boundaries_of = {rich_boundary}\<rparr>"

definition rich_grammar :: grammar where
  "rich_grammar =
    \<lparr>grammar_id_of = 1,
    supported_kinds_of =
      {K_META, K_PERSP, K_CTX, K_ENT, K_REL, K_EFF, K_RES, K_REV,
       K_ORI, K_ACT, K_FDB},
    allowed_operations_of = {RegisterAppearance},
    patch_budget_of = 0\<rparr>"

definition rich_states :: "snapshot set" where
  "rich_states = {rich_snapshot}"

definition rich_classified_emergence ::
  "interaction_complex \<Rightarrow> emergence_level" where
  "rich_classified_emergence I = emergence_of rich_grammar I"

lemmas rich_object_defs =
  rich_meta_def rich_perspective_obj_def rich_context_source_obj_def
  rich_entity_def rich_relation_def rich_effect_def rich_resonance_def
  rich_relevance_def rich_orientation_def rich_action_def rich_feedback_def
  rich_context_target_obj_def

lemma rich_snapshot_simps [simp]:
  "snapshot_id_of rich_snapshot = 1"
  "objects_of rich_snapshot = rich_objects"
  "historical_versions_of rich_snapshot = {}"
  "perspectives_of rich_snapshot = {10}"
  "contexts_of rich_snapshot = {20,21}"
  "models_of rich_snapshot = {30}"
  "authorities_of rich_snapshot = {40}"
  "contains_edges_of rich_snapshot = {(20,21)}"
  "projections_of rich_snapshot = {rich_projection}"
  "couplings_of rich_snapshot = {rich_coupling}"
  "transmissions_of rich_snapshot = {rich_transmission}"
  "boundaries_of rich_snapshot = {rich_boundary}"
  by (simp_all add: rich_snapshot_def)

lemma rich_grammar_simps [simp]:
  "grammar_id_of rich_grammar = 1"
  "supported_kinds_of rich_grammar =
    {K_META, K_PERSP, K_CTX, K_ENT, K_REL, K_EFF, K_RES, K_REV,
     K_ORI, K_ACT, K_FDB}"
  "allowed_operations_of rich_grammar = {RegisterAppearance}"
  "patch_budget_of rich_grammar = 0"
  by (simp_all add: rich_grammar_def)

lemma rich_version_ids [simp]:
  "version_ids rich_snapshot = {0,1,2,3,4,5,6,7,8,9,10,11}"
  by (simp add: version_ids_def rich_snapshot_def rich_objects_def
    rich_object_defs base_object_def)

lemma rich_revision_edges [simp]:
  "revision_edges_of rich_snapshot = {}"
  by (auto simp: revision_edges_of_def rich_snapshot_def rich_objects_def
    rich_object_defs base_object_def)

lemma rich_unique_versions:
  "unique_versions rich_snapshot"
  unfolding unique_versions_def inj_on_def
  by (auto simp: rich_objects_def rich_object_defs base_object_def)

lemma rich_singleton_trancl [simp]:
  "trancl {(20::context_id,21)} = {(20,21)}"
proof (rule equalityI)
  show "trancl {(20::context_id,21)} \<subsepeq> {(20,21)}"
  proof
    fix edge
    assume "edge \<in> trancl {(20::context_id,21)}"
    then obtain x y where edge: "edge = (x,y)"
      by (cases edge) auto
    have "(x,y) \<in> trancl {(20,21)}"
      using \<open>edge \<in> trancl {(20::context_id,21)}\<close>
      unfolding edge by simp
    then have "x = 20 \<and> y = 21"
      by (induction rule: trancl_induct) auto
    then show "edge \<in> {(20,21)}"
  qed

```

```

    by (simp add: edge)
qed
show "{(20,21)} \<subteq> tranc1 {(20::context_id,21)}"
  by auto
qed

lemma rich_contains_tranc1 [simp]:
  "tranc1 (contains_edges_of rich_snapshot) = {(20,21)}"
  by simp

lemma rich_state_well_formed:
  "state_well_formed rich_grammar rich_snapshot"
  unfolding state_well_formed_def
  using rich_unique_versions
  by (auto simp: present_ref_def
    references_complete_def evidence_well_formed_def
    evidence_references_complete_def time_well_formed_def
    relation_well_formed_def effect_well_formed_def
    perspective_well_formed_def perspective_bound_kind_def
    context_well_formed_def context_bound_kind_def
    relevance_well_formed_def orientation_well_formed_def
    action_well_formed_def feedback_well_formed_def
    projection_well_formed_def loss_well_formed_def
    coupling_well_formed_def transmission_well_formed_def
    boundary_well_formed_def rich_objects_def
    rich_projection_def rich_loss_def rich_coupling_def
    rich_transmission_def rich_boundary_def rich_object_defs base_object_def)

lemma rich_self_step:
  "step rich_grammar rich_snapshot RegisterAppearance rich_snapshot"
proof -
  have allowed:
    "RegisterAppearance \<in> allowed_operations_of rich_grammar"
    by (simp add: rich_grammar_def)
  have immutable:
    "immutable_preserved rich_snapshot rich_snapshot"
    using rich_unique_versions
    unfolding immutable_preserved_def unique_versions_def
    by (meson inj_onD)
  show ?thesis
    unfolding step_def
    using rich_state_well_formed allowed immutable by blast
qed

subsection \<open>Explicit non-vacuity certificates\<close>

lemma rich_A5_relation_nonvacuous:
  "rich_relation \<in> objects_of rich_snapshot \<and>
  kind_of rich_relation = K_REL \<and>
  present_ref rich_snapshot (source_ref_of rich_relation) \<and>
  present_ref rich_snapshot (target_ref_of rich_relation)"
  by (simp add: rich_objects_def rich_relation_def
    base_object_def present_ref_def)

lemma rich_A6_effect_nonvacuous:
  "rich_effect \<in> objects_of rich_snapshot \<and>
  kind_of rich_effect = K_EFF \<and> lifecycle_of rich_effect = Validated \<and>
  present_ref rich_snapshot (effect_origin_of rich_effect)"
  by (simp add: rich_objects_def rich_effect_def
    base_object_def present_ref_def)

lemma rich_A7_A8_bindings_nonvacuous:
  "kind_of rich_entity = K_ENT \<and>
  perspective_of rich_entity = Some 10 \<and> 10 \<in> perspectives_of rich_snapshot \<and>
  context_of rich_entity = Some 20 \<and> 20 \<in> contexts_of rich_snapshot"
  by (simp add: rich_entity_def base_object_def)

lemma rich_A9_evidence_nonvacuous:
  "lifecycle_of rich_effect = Validated \<and>
  evidence_of rich_effect \<noteq> {} \<and>
  evidence_references_complete rich_snapshot rich_effect"
  by (simp add: rich_effect_def base_object_def
    evidence_references_complete_def)

lemma rich_A10_relevance_nonvacuous:
  "kind_of rich_relevance = K_REV \<and>
  references_of rich_relevance \<noteq> {} \<and>
  model_of rich_relevance = Some 30 \<and> 30 \<in> models_of rich_snapshot \<and>
  relevance_well_formed rich_snapshot rich_relevance"

```

```

by (simp add: rich_relevance_def base_object_def
  relevance_well_formed_def perspective_well_formed_def
  perspective_bound_kind_def context_well_formed_def context_bound_kind_def)

lemma rich_A11_orientation_nonvacuous:
  "kind_of rich_orientation = K_ORI \<and>
    basis_refs_of rich_orientation \<noteq> {} \<and>
    orientation_well_formed rich_snapshot rich_orientation"
  by (simp add: rich_orientation_def base_object_def orientation_well_formed_def)

lemma rich_A12_action_nonvacuous:
  "kind_of rich_action = K_ACT \<and>
    basis_refs_of rich_action \<noteq> {} \<and>
    authority_of rich_action = Some 40 \<and> 40 \<in> authorities_of rich_snapshot \<and>
    action_well_formed rich_snapshot rich_action"
  by (simp add: rich_action_def base_object_def action_well_formed_def)

lemma rich_A13_feedback_nonvacuous:
  "kind_of rich_feedback = K_FDB \<and>
    execution_ref_of rich_feedback = Some 9 \<and>
    observation_refs_of rich_feedback \<noteq> {} \<and>
    feedback_well_formed rich_snapshot rich_feedback"
  by (simp add: rich_feedback_def base_object_def feedback_well_formed_def
    present_ref_def)

lemma rich_A17_A21_interaction_nonvacuous:
  "rich_projection \<in> projections_of rich_snapshot \<and>
    projection_well_formed rich_snapshot rich_projection \<and>
    rich_coupling \<in> couplings_of rich_snapshot \<and>
    coupling_well_formed rich_snapshot rich_coupling \<and>
    rich_transmission \<in> transmissions_of rich_snapshot \<and>
    transmission_well_formed rich_snapshot rich_transmission \<and>
    trans_status_of rich_transmission = TransValidated \<and>
    transmission_target_effect_of rich_transmission \<noteq> None \<and>
    transmission_lineage_complete_of rich_transmission \<and>
    rich_boundary \<in> boundaries_of rich_snapshot \<and>
    boundary_well_formed rich_boundary"
  by (simp add: rich_projection_def projection_well_formed_def
    rich_loss_def loss_well_formed_def rich_coupling_def coupling_well_formed_def
    rich_transmission_def transmission_well_formed_def
    rich_boundary_def boundary_well_formed_def)

subsection \<open>Locale interpretation\<close>

interpretation Rich: uza_model
  rich_states rich_grammar rich_classified_emergence
proof
  fix s o1 o2
  assume state: "s \<in> rich_states"
  and o1: "o1 \<in> objects_of s" and o2: "o2 \<in> objects_of s"
  and version: "version_id_of o1 = version_id_of o2"
  have wf: "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
  have "unique_versions s"
  using wf unfolding state_well_formed_def by blast
  then show "o1 = o2"
  using o1 o2 version unfolding unique_versions_def by (meson inj_onD)
next
  fix s
  assume "s \<in> rich_states"
  then show "wf (revision_edges_of s)"
  by (simp add: rich_states_def)
next
  fix s obj
  assume state: "s \<in> rich_states" and obj: "obj \<in> objects_of s"
  have "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
  then show "time_well_formed obj"
  using obj unfolding state_well_formed_def by blast
next
  fix s obj
  assume state: "s \<in> rich_states" and obj: "obj \<in> objects_of s"
  have "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
  then show "references_complete s obj"
  using obj unfolding state_well_formed_def by blast
next
  fix s obj
  assume state: "s \<in> rich_states" and obj: "obj \<in> objects_of s"

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have "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
then show "relation_well_formed s obj"
  using obj unfolding state_well_formed_def by blast
next
fix s obj
assume state: "s \<in> rich_states" and obj: "obj \<in> objects_of s"
have "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
then show "effect_well_formed s obj"
  using obj unfolding state_well_formed_def by blast
next
fix s obj
assume state: "s \<in> rich_states" and obj: "obj \<in> objects_of s"
have "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
then show "perspective_well_formed s obj"
  using obj unfolding state_well_formed_def by blast
next
fix s obj
assume state: "s \<in> rich_states" and obj: "obj \<in> objects_of s"
have "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
then show "context_well_formed s obj"
  using obj unfolding state_well_formed_def by blast
next
fix s obj
assume state: "s \<in> rich_states" and obj: "obj \<in> objects_of s"
have "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
then show "evidence_well_formed s obj"
  using obj unfolding state_well_formed_def by blast
next
fix s obj
assume state: "s \<in> rich_states" and obj: "obj \<in> objects_of s"
have "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
then show "relevance_well_formed s obj"
  using obj unfolding state_well_formed_def by blast
next
fix s obj
assume state: "s \<in> rich_states" and obj: "obj \<in> objects_of s"
have "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
then show "orientation_well_formed s obj"
  using obj unfolding state_well_formed_def by blast
next
fix s obj
assume state: "s \<in> rich_states" and obj: "obj \<in> objects_of s"
have "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
then show "action_well_formed s obj"
  using obj unfolding state_well_formed_def by blast
next
fix s obj
assume state: "s \<in> rich_states" and obj: "obj \<in> objects_of s"
have "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
then show "feedback_well_formed s obj"
  using obj unfolding state_well_formed_def by blast
next
fix s oper s'
assume "s \<in> rich_states" "step rich_grammar s oper s'"
then show "state_well_formed rich_grammar s'"
  by (simp add: step_def)
next
fix s c
assume "s \<in> rich_states"
then show "\<not> strict_contains s c c"
  by (auto simp: rich_states_def strict_contains_def)
next
fix s a b
assume state: "s \<in> rich_states"
  and ab: "contains_eq s a b" and ba: "contains_eq s b a"
show "a = b"
  using state ab ba
  by (auto simp: rich_states_def contains_eq_def strict_contains_def)
next
fix s p

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assume state: "s \<in> rich_states" and member: "p \<in> projections_of s"
have "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
then show "projection_well_formed s p"
  using member unfolding state_well_formed_def by blast
next
fix s c
assume state: "s \<in> rich_states" and member: "c \<in> couplings_of s"
have "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
then show "coupling_well_formed s c"
  using member unfolding state_well_formed_def by blast
next
fix s t
assume state: "s \<in> rich_states" and member: "t \<in> transmissions_of s"
have "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
then show "transmission_well_formed s t"
  using member unfolding state_well_formed_def by blast
next
fix s t
assume state: "s \<in> rich_states" and member: "t \<in> transmissions_of s"
  and validated: "trans_status_of t = TransValidated"
have wf: "transmission_well_formed s t"
proof -
  have "state_well_formed rich_grammar s"
    using state by (simp add: rich_states_def rich_state_well_formed)
  then show ?thesis
    using member unfolding state_well_formed_def by blast
qed
show "transmission_target_effect_of t \<noteq> None \<and>
transmission_lineage_complete_of t"
  using wf validated unfolding transmission_well_formed_def by blast
next
fix s b
assume state: "s \<in> rich_states" and member: "b \<in> boundaries_of s"
have "state_well_formed rich_grammar s"
  using state by (simp add: rich_states_def rich_state_well_formed)
then show "boundary_well_formed b"
  using member unfolding state_well_formed_def by blast
next
fix I :: interaction_complex
assume "current_of I \<in> rich_states"
show "rich_classified_emergence I = emergence_of rich_grammar I"
  by (simp add: rich_classified_emergence_def)
qed

theorem rich_uza_model_exists:
  "\<exists>States G classify s.
    uza_model States G classify \<and> s \<in> States \<and>
    (\<exists>r\<in>objects_of s. kind_of r = K_REL) \<and>
    (\<exists>e\<in>objects_of s. kind_of e = K_EFF \<and> lifecycle_of e = Validated) \<and>
    (\<exists>v\<in>objects_of s. kind_of v = K_REV) \<and>
    (\<exists>ori\<in>objects_of s. kind_of ori = K_ORI) \<and>
    (\<exists>a\<in>objects_of s. kind_of a = K_ACT) \<and>
    (\<exists>f\<in>objects_of s. kind_of f = K_FDB) \<and>
    projections_of s \<noteq> {} \<and> couplings_of s \<noteq> {} \<and>
    transmissions_of s \<noteq> {} \<and> boundaries_of s \<noteq> {}"
proof (intro exI[of _ rich_states] exI[of _ rich_grammar]
  exI[of _ rich_classified_emergence] exI[of _ rich_snapshot])
  show "uza_model rich_states rich_grammar rich_classified_emergence \<and>
    rich_snapshot \<in> rich_states \<and>
    (\<exists>r\<in>objects_of rich_snapshot. kind_of r = K_REL) \<and>
    (\<exists>e\<in>objects_of rich_snapshot.
      kind_of e = K_EFF \<and> lifecycle_of e = Validated) \<and>
    (\<exists>v\<in>objects_of rich_snapshot. kind_of v = K_REV) \<and>
    (\<exists>ori\<in>objects_of rich_snapshot. kind_of ori = K_ORI) \<and>
    (\<exists>a\<in>objects_of rich_snapshot. kind_of a = K_ACT) \<and>
    (\<exists>f\<in>objects_of rich_snapshot. kind_of f = K_FDB) \<and>
    projections_of rich_snapshot \<noteq> {} \<and>
    couplings_of rich_snapshot \<noteq> {} \<and>
    transmissions_of rich_snapshot \<noteq> {} \<and>
    boundaries_of rich_snapshot \<noteq> {}"
  proof (intro conjI)
    show "uza_model rich_states rich_grammar rich_classified_emergence"
      by (rule Rich.uza_model_axioms)
  next
    show "rich_snapshot \<in> rich_states"
      by (simp add: rich_states_def)

```

```

next
  show "\<exists>r\<in>objects_of rich_snapshot. kind_of r = K_REL"
  by (intro bexI[of _ rich_relation])
  (simp_all add: rich_objects_def rich_relation_def base_object_def)
next
  show "\<exists>e\<in>objects_of rich_snapshot.
    kind_of e = K_EFF \<and> lifecycle_of e = Validated"
  by (intro bexI[of _ rich_effect])
  (simp_all add: rich_objects_def rich_effect_def base_object_def)
next
  show "\<exists>v\<in>objects_of rich_snapshot. kind_of v = K_REV"
  by (intro bexI[of _ rich_relevance])
  (simp_all add: rich_objects_def rich_relevance_def base_object_def)
next
  show "\<exists>ori\<in>objects_of rich_snapshot. kind_of ori = K_ORI"
  by (intro bexI[of _ rich_orientation])
  (simp_all add: rich_objects_def rich_orientation_def base_object_def)
next
  show "\<exists>a\<in>objects_of rich_snapshot. kind_of a = K_ACT"
  by (intro bexI[of _ rich_action])
  (simp_all add: rich_objects_def rich_action_def base_object_def)
next
  show "\<exists>f\<in>objects_of rich_snapshot. kind_of f = K_FDB"
  by (intro bexI[of _ rich_feedback])
  (simp_all add: rich_objects_def rich_feedback_def base_object_def)
next
  show "projections_of rich_snapshot \<noteq> {}"
  by simp
next
  show "couplings_of rich_snapshot \<noteq> {}"
  by simp
next
  show "transmissions_of rich_snapshot \<noteq> {}"
  by simp
next
  show "boundaries_of rich_snapshot \<noteq> {}"
  by simp
qed
qed
end

```

## D.5 Konkrete T2-Semantik und Erhaltung für DifferentiatePotential

Die fünfte Theorie UZA\_0\_9\_1\_Differentiate\_Potential konkretisiert T2 als konstruktive Mengentransition. `potential_input_valid` verlangt einen wohlgeformten Ausgangszustand, eine zulässige Operation, eine endliche und nichtleere Kandidatenmenge, paarweise eindeutige Objekt- und Versionsidentitäten sowie Frische gegenüber dem Ausgangszustand. Jeder Kandidat ist eine `K_ENT`-Hypothese mit Status `Applicable`, auflösbarer `source_ref`, nichtleerer `basis_refs`-Menge, vollständigen Referenzen, gültiger Evidenz- und Zeitstruktur sowie vorhandener Perspektiv- und Kontextbindung. `potential_candidate_marker` und `observed_appearance_marker` sind disjunkt; `potential_candidate_not_observed` beweist die Unterscheidbarkeit ausdrücklich.

`add_potential_candidates` erzeugt den Folgezustand ausschließlich durch Vereinigung der bisherigen Objektmenge mit der frischen Kandidatenmenge und Fortschreibung der Snapshot-ID. Isabelle beweist `differentiate_potential_exact_delta`, die explizite Kandidateneigenschaft, eindeutige Versionsidentitäten, Unveränderlichkeit aller vorhandenen Versionen, Erhaltung von Referenzen, Evidence Integrity, Zeit, Perspektive und Kontext, vollständige `state_well_formed`-Erhaltung und `differentiate_potential_refines_generic_step`. Die Zielwohlgeformtheit ist keine Vorbedingung.

Der integrierte Isabelle2025-2-Clean-Build verarbeitete UZA\_0\_9\_1\_Differentiate\_Potential vollständig und endete mit Exit-Code 0. Der aktuelle T2-Quelltext besitzt SHA-256

ef483b279d0581752bfff655db6672fa57823dac348ace29268e68e486a40f9b. Damit ist Aufgabe 4 auch für T2 abgeschlossen. Nicht bewiesen sind die Vollständigkeit aller möglichen Kandidaten oder ihre domänenspezifische Güte.

```
theory UZA_0_9_1_Differentiate_Potential
  imports UZA_0_9_1_Register_Appearance
begin

section \<open>T2 DifferentiatePotential: concrete operation semantics\<close>

text \<open>
  T2 materialises a finite, non-empty family of explicit possibility candidates.
  Candidates are not observations: an observed T1 appearance has no source,
  basis, or analysis marker, whereas a T2 candidate is a Hypothesis with an
  existing source, a non-empty resolved basis, and status Applicable.

  The target state is constructed by inserting exactly the candidate set and
  advancing the snapshot identifier. Target well-formedness is proved, not
  assumed.
\<close>

definition observed_appearance_marker :: "uza_object \<Rightarrow> bool" where
  "observed_appearance_marker obj \<longlefttrightarrow>
    kind_of obj = K_ENT \<and>
    source_ref_of obj = None \<and>
    basis_refs_of obj = {} \<and>
    object_status_of obj = None"

definition potential_candidate_marker :: "uza_object \<Rightarrow> bool" where
  "potential_candidate_marker obj \<longlefttrightarrow>
    kind_of obj = K_ENT \<and>
    lifecycle_of obj = Hypothesis \<and>
    source_ref_of obj \<noteq> None \<and>
    basis_refs_of obj \<noteq> {} \<and>
    object_status_of obj = Some Applicable"

lemma potential_candidate_not_observed:
  assumes "potential_candidate_marker obj"
  shows "\<not> observed_appearance_marker obj"
  using assms
  by (auto simp: potential_candidate_marker_def observed_appearance_marker_def)

definition potential_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> uza_object set \<Rightarrow> nat \<Rightarrow> bool"
where
  "potential_input_valid G s candidates next_snapshot_id \<longlefttrightarrow>
    state_well_formed G s \<and>
    DifferentiatePotential \<in> allowed_operations_of G \<and>
    K_ENT \<in> supported_kinds_of G \<and>
    finite candidates \<and>
    candidates \<noteq> {} \<and>
    inj_on object_id_of candidates \<and>
    inj_on version_id_of candidates \<and>
    object_id_of ` candidates \<inter> object_id_of ` objects_of s = {} \<and>
    version_id_of ` candidates \<inter>
      (version_ids s \<union> historical_versions_of s) = {} \<and>
    (\<forall>candidate\<in>candidates.
      potential_candidate_marker candidate \<and>
      previous_version_of candidate = None \<and>
      references_complete s candidate \<and>
      evidence_well_formed s candidate \<and>
      time_well_formed candidate \<and>
      (\<exists>source. source_ref_of candidate = Some source \<and>
        source \<in> basis_refs_of candidate) \<and>
      basis_refs_of candidate \<subteq> references_of candidate \<and>
      (\<exists>p\<in>perspectives_of s. perspective_of candidate = Some p) \<and>
      (\<exists>c\<in>contexts_of s. context_of candidate = Some c)) \<and>
    snapshot_id_of s < next_snapshot_id"

definition add_potential_candidates ::
  "snapshot \<Rightarrow> uza_object set \<Rightarrow> nat \<Rightarrow> snapshot" where
  "add_potential_candidates s candidates next_snapshot_id =
    s\<lparr> snapshot_id_of := next_snapshot_id,
    objects_of := objects_of s \<union> candidates \<rparr>"

definition differentiate_potential_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> uza_object set \<Rightarrow> nat \<Rightarrow> snapshot"
```

```

\<Rightarrow> bool" where
  "differentiate_potential_step G s candidates next_snapshot_id s' \<longlefttrightarrow>
    potential_input_valid G s candidates next_snapshot_id \<and>
    s' = add_potential_candidates s candidates next_snapshot_id"

subsection \<open>Exact update and candidate separation\<close>

lemma add_potential_candidates_objects [simp]:
  "objects_of (add_potential_candidates s candidates next_snapshot_id) =
    objects_of s \<union> candidates"
  by (simp add: add_potential_candidates_def)

lemma add_potential_candidates_snapshot_id [simp]:
  "snapshot_id_of (add_potential_candidates s candidates next_snapshot_id) =
    next_snapshot_id"
  by (simp add: add_potential_candidates_def)

lemma add_potential_candidates_background [simp]:
  "historical_versions_of (add_potential_candidates s candidates next_snapshot_id) =
    historical_versions_of s"
  "perspectives_of (add_potential_candidates s candidates next_snapshot_id) = perspectives_of s"
  "contexts_of (add_potential_candidates s candidates next_snapshot_id) = contexts_of s"
  "models_of (add_potential_candidates s candidates next_snapshot_id) = models_of s"
  "authorities_of (add_potential_candidates s candidates next_snapshot_id) = authorities_of s"
  "contains_edges_of (add_potential_candidates s candidates next_snapshot_id) = contains_edges_of s"
  "projections_of (add_potential_candidates s candidates next_snapshot_id) = projections_of s"
  "couplings_of (add_potential_candidates s candidates next_snapshot_id) = couplings_of s"
  "transmissions_of (add_potential_candidates s candidates next_snapshot_id) = transmissions_of s"
  "boundaries_of (add_potential_candidates s candidates next_snapshot_id) = boundaries_of s"
  by (simp_all add: add_potential_candidates_def)

lemma add_potential_candidates_version_ids [simp]:
  "version_ids (add_potential_candidates s candidates next_snapshot_id) =
    version_ids s \<union> version_id_of ` candidates"
  by (auto simp: version_ids_def)

lemma potential_input_candidates_disjoint:
  assumes valid: "potential_input_valid G s candidates next_snapshot_id"
  shows "objects_of s \<inter> candidates = {}"

proof (rule ccontr)
  assume "objects_of s \<inter> candidates \<noteq> {}"
  then obtain obj where old: "obj \<in> objects_of s" and candidate: "obj \<in> candidates"
  by blast
  from valid have fresh_ids:
    "object_id_of ` candidates \<inter> object_id_of ` objects_of s = {}"
  unfolding potential_input_valid_def by blast
  have "object_id_of obj \<in> object_id_of ` candidates \<inter> object_id_of ` objects_of s"
  using old candidate by blast
  then show False using fresh_ids by blast
qed

lemma differentiate_potential_exact_delta:
  assumes concrete:
    "differentiate_potential_step G s candidates next_snapshot_id s'"
  shows
    "objects_of s' = objects_of s \<union> candidates \<and>
    objects_of s \<inter> candidates = {} \<and>
    finite candidates \<and> candidates \<noteq> {} \<and>
    snapshot_id_of s' = next_snapshot_id"
  using concrete potential_input_candidates_disjoint
  by (auto simp: differentiate_potential_step_def potential_input_valid_def)

lemma differentiate_potential_candidates_are_explicit:
  assumes concrete:
    "differentiate_potential_step G s candidates next_snapshot_id s'"
  shows
    "candidates \<subteq> objects_of s' \<and>
    (\<forall>candidate\<in>candidates.
    potential_candidate_marker candidate \<and>
    \<not> observed_appearance_marker candidate)"
  using concrete potential_candidate_not_observed
  by (auto simp: differentiate_potential_step_def potential_input_valid_def)

subsection \<open>Monotonicity of state-relative conditions\<close>

lemma add_potential_present_ref_mono:
  assumes "present_ref s ref"
  shows "present_ref (add_potential_candidates s candidates next_snapshot_id) ref"
  using assms

```

```

by (cases ref) (auto simp: present_ref_def)

lemma add_potential_references_mono:
  assumes "references_complete s obj"
  shows "references_complete (add_potential_candidates s candidates next_snapshot_id) obj"
  using assms
  by (auto simp: references_complete_def)

lemma add_potential_evidence_mono:
  assumes "evidence_well_formed s obj"
  shows "evidence_well_formed (add_potential_candidates s candidates next_snapshot_id) obj"
  using assms
  by (auto simp: evidence_well_formed_def evidence_references_complete_def)

lemma add_potential_relation_mono:
  assumes "relation_well_formed s obj"
  shows "relation_well_formed (add_potential_candidates s candidates next_snapshot_id) obj"
  using assms add_potential_present_ref_mono
  unfolding relation_well_formed_def
  by blast

lemma add_potential_effect_mono:
  assumes "effect_well_formed s obj"
  shows "effect_well_formed (add_potential_candidates s candidates next_snapshot_id) obj"
  using assms add_potential_present_ref_mono
  unfolding effect_well_formed_def
  by blast

lemma add_potential_perspective_iff [simp]:
  "perspective_well_formed (add_potential_candidates s candidates next_snapshot_id) obj
  \<longleftarrow\longrightarrow perspective_well_formed s obj"
  unfolding perspective_well_formed_def
  by (simp only: add_potential_candidates_background(2))

lemma add_potential_context_iff [simp]:
  "context_well_formed (add_potential_candidates s candidates next_snapshot_id) obj
  \<longleftarrow\longrightarrow context_well_formed s obj"
  unfolding context_well_formed_def
  by (simp only: add_potential_candidates_background(3))

lemma add_potential_relevance_iff [simp]:
  "relevance_well_formed (add_potential_candidates s candidates next_snapshot_id) obj
  \<longleftarrow\longrightarrow relevance_well_formed s obj"
  unfolding relevance_well_formed_def
  by (simp only: add_potential_perspective_iff add_potential_context_iff
    add_potential_candidates_background(4))

lemma add_potential_orientation_iff [simp]:
  "orientation_well_formed (add_potential_candidates s candidates next_snapshot_id) obj
  \<longleftarrow\longrightarrow orientation_well_formed s obj"
  by (simp add: orientation_well_formed_def)

lemma add_potential_action_iff [simp]:
  "action_well_formed (add_potential_candidates s candidates next_snapshot_id) obj
  \<longleftarrow\longrightarrow action_well_formed s obj"
  unfolding action_well_formed_def
  by (simp only: add_potential_candidates_background(5))

lemma add_potential_feedback_mono:
  assumes "feedback_well_formed s obj"
  shows "feedback_well_formed (add_potential_candidates s candidates next_snapshot_id) obj"
  using assms add_potential_present_ref_mono
  unfolding feedback_well_formed_def
  by blast

subsection \<open>Fresh identity and candidate validity\<close>

lemma add_potential_preserves_unique_versions:
  assumes valid: "potential_input_valid G s candidates next_snapshot_id"
  shows "unique_versions (add_potential_candidates s candidates next_snapshot_id)"
proof -
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
    and candidate_unique: "inj_on version_id_of candidates"
    and fresh:
      "version_id_of ` candidates \<inter> (version_ids s \<union> historical_versions_of s) = {}"
    unfolding potential_input_valid_def state_well_formed_def unique_versions_def
    by auto
  have cross_fresh: "version_id_of ` candidates \<inter> version_id_of ` objects_of s = {}"
    using fresh unfolding version_ids_def by blast

```

```

show ?thesis
  unfolding unique_versions_def
proof (rule inj_onI)
  fix left right
  assume left_mem:
    "left \<in> objects_of (add_potential_candidates s candidates next_snapshot_id)"
  and right_mem:
    "right \<in> objects_of (add_potential_candidates s candidates next_snapshot_id)"
  and same_version: "version_id_of left = version_id_of right"
  consider
    (old_old) "left \<in> objects_of s" "right \<in> objects_of s"
  | (candidate_candidate) "left \<in> candidates" "right \<in> candidates"
  | (old_candidate) "left \<in> objects_of s" "right \<in> candidates"
  | (candidate_old) "left \<in> candidates" "right \<in> objects_of s"
  using left_mem right_mem by auto
  then show "left = right"
proof cases
  case old_old
  show ?thesis using inj_onD[OF source_unique same_version old_old] .
next
  case candidate_candidate
  show ?thesis using inj_onD[OF candidate_unique same_version candidate_candidate] .
next
  case old_candidate
  have candidate_image: "version_id_of right \<in> version_id_of ` candidates"
  using old_candidate by blast
  have old_image: "version_id_of right \<in> version_id_of ` objects_of s"
  using old_candidate same_version by (metis imageI)
  have "version_id_of right \<in>
    version_id_of ` candidates \<inter> version_id_of ` objects_of s"
  using candidate_image old_image by blast
  then show ?thesis using cross_fresh by blast
next
  case candidate_old
  have candidate_image: "version_id_of left \<in> version_id_of ` candidates"
  using candidate_old by blast
  have old_image: "version_id_of left \<in> version_id_of ` objects_of s"
  using candidate_old same_version by (metis imageI)
  have "version_id_of left \<in>
    version_id_of ` candidates \<inter> version_id_of ` objects_of s"
  using candidate_image old_image by blast
  then show ?thesis using cross_fresh by blast
qed
qed
qed

lemma potential_input_new_candidate_conditions:
  assumes valid: "potential_input_valid G s candidates next_snapshot_id"
  and member: "candidate \<in> candidates"
  shows
    "kind_of candidate \<in> supported_kinds_of G \<and>
    references_complete (add_potential_candidates s candidates next_snapshot_id) candidate \<and>
    evidence_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate \<and>
    time_well_formed candidate \<and>
    relation_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate \<and>
    effect_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate \<and>
    perspective_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate \<and>
    context_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate \<and>
    relevance_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate \<and>
    orientation_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate \<and>
    action_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate \<and>
    feedback_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate"
proof -
  from valid member have supported: "kind_of candidate \<in> supported_kinds_of G"
  and marker: "potential_candidate_marker candidate"
  and refs: "references_complete s candidate"
  and evidence: "evidence_well_formed s candidate"
  and time: "time_well_formed candidate"
  and perspective:
    "perspective_well_formed
    (add_potential_candidates s candidates next_snapshot_id) candidate"
  and context_ok:
    "context_well_formed
    (add_potential_candidates s candidates next_snapshot_id) candidate"
  unfolding potential_input_valid_def potential_candidate_marker_def
  perspective_well_formed_def context_well_formed_def
  perspective_bound_kind_def context_bound_kind_def
  by auto
  from marker have kind: "kind_of candidate = K_ENT"

```

```

    unfolding potential_candidate_marker_def by blast
  have refs_target:
    "references_complete (add_potential_candidates s candidates next_snapshot_id) candidate"
    using add_potential_references_mono[OF refs] .
  have evidence_target:
    "evidence_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate"
    using add_potential_evidence_mono[OF evidence] .
  have typed_conditions:
    "relation_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate \<and>
    effect_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate \<and>
    relevance_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate \<and>
    orientation_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate \<and>
    action_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate \<and>
    feedback_well_formed (add_potential_candidates s candidates next_snapshot_id) candidate"
    using kind
    by (simp add: relation_well_formed_def effect_well_formed_def
      relevance_well_formed_def orientation_well_formed_def
      action_well_formed_def feedback_well_formed_def)
  show ?thesis
    using supported refs_target evidence_target time perspective context_ok typed_conditions
    by blast
qed

subsection \<open>Operation-specific preservation theorems\<close>

theorem differentiate_potential_preserves_state_well_formed:
  assumes concrete:
    "differentiate_potential_step G s candidates next_snapshot_id s'"
  shows "state_well_formed G s'"
proof -
  from concrete have valid: "potential_input_valid G s candidates next_snapshot_id"
    and target: "s' = add_potential_candidates s candidates next_snapshot_id"
    by (auto simp: differentiate_potential_step_def)
  from valid have source_wf: "state_well_formed G s"
    by (simp add: potential_input_valid_def)
  have unique:
    "unique_versions (add_potential_candidates s candidates next_snapshot_id)"
    using add_potential_preserves_unique_versions[OF valid] .
  have all_objects:
    "\<forall>obj \<in> objects_of (add_potential_candidates s candidates next_snapshot_id).
    kind_of obj \<in> supported_kinds_of G \<and>
    references_complete (add_potential_candidates s candidates next_snapshot_id) obj \<and>
    evidence_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
    time_well_formed obj \<and>
    relation_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
    effect_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
    perspective_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
    context_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
    relevance_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
    orientation_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
    action_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
    feedback_well_formed (add_potential_candidates s candidates next_snapshot_id) obj"
  proof (intro ballI)
    fix obj
    assume obj_mem:
      "obj \<in> objects_of (add_potential_candidates s candidates next_snapshot_id)"
    consider (old) "obj \<in> objects_of s" | (candidate) "obj \<in> candidates"
      using obj_mem by auto
    then show
      "kind_of obj \<in> supported_kinds_of G \<and>
      references_complete (add_potential_candidates s candidates next_snapshot_id) obj \<and>
      evidence_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
      time_well_formed obj \<and>
      relation_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
      effect_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
      perspective_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
      context_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
      relevance_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
      orientation_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
      action_well_formed (add_potential_candidates s candidates next_snapshot_id) obj \<and>
      feedback_well_formed (add_potential_candidates s candidates next_snapshot_id) obj"
  proof cases
    case candidate
    then show ?thesis
      using potential_input_new_candidate_conditions[OF valid] by blast
  next
    case old
    from source_wf old have source_conditions:
      "kind_of obj \<in> supported_kinds_of G \<and>

```

```

    references_complete s obj \<and> evidence_well_formed s obj \<and>
    time_well_formed obj \<and>
    relation_well_formed s obj \<and> effect_well_formed s obj \<and>
    perspective_well_formed s obj \<and> context_well_formed s obj \<and>
    relevance_well_formed s obj \<and> orientation_well_formed s obj \<and>
    action_well_formed s obj \<and> feedback_well_formed s obj"
  unfolding state_well_formed_def by blast
from source_conditions have supported: "kind_of obj \<in> supported_kinds_of G"
  and refs: "references_complete s obj"
  and evidence: "evidence_well_formed s obj"
  and time: "time_well_formed obj"
  and relation: "relation_well_formed s obj"
  and effect: "effect_well_formed s obj"
  and perspective: "perspective_well_formed s obj"
  and context_ok: "context_well_formed s obj"
  and relevance: "relevance_well_formed s obj"
  and orientation: "orientation_well_formed s obj"
  and action: "action_well_formed s obj"
  and feedback: "feedback_well_formed s obj"
  by blast+
have refs_target:
  "references_complete (add_potential_candidates s candidates next_snapshot_id) obj"
  using add_potential_references_mono[OF refs] .
have evidence_target:
  "evidence_well_formed (add_potential_candidates s candidates next_snapshot_id) obj"
  using add_potential_evidence_mono[OF evidence] .
have relation_target:
  "relation_well_formed (add_potential_candidates s candidates next_snapshot_id) obj"
  using add_potential_relation_mono[OF relation] .
have effect_target:
  "effect_well_formed (add_potential_candidates s candidates next_snapshot_id) obj"
  using add_potential_effect_mono[OF effect] .
have feedback_target:
  "feedback_well_formed (add_potential_candidates s candidates next_snapshot_id) obj"
  using add_potential_feedback_mono[OF feedback] .
show ?thesis
  using supported refs_target evidence_target time relation_target effect_target perspective
    context_ok relevance orientation action feedback_target
  by simp
qed
qed
from source_wf have source_secondary:
  "(\<forall>p\<in>projections_of s. projection_well_formed s p) \<and>
  (\<forall>c\<in>couplings_of s. coupling_well_formed s c) \<and>
  (\<forall>t\<in>transmissions_of s. transmission_well_formed s t) \<and>
  (\<forall>b\<in>boundaries_of s. boundary_well_formed b)"
  unfolding state_well_formed_def by blast
have target_secondary:
  "(\<forall>p\<in>projections_of (add_potential_candidates s candidates next_snapshot_id).
  projection_well_formed (add_potential_candidates s candidates next_snapshot_id) p) \<and>
  (\<forall>c\<in>couplings_of (add_potential_candidates s candidates next_snapshot_id).
  coupling_well_formed (add_potential_candidates s candidates next_snapshot_id) c) \<and>
  (\<forall>t\<in>transmissions_of (add_potential_candidates s candidates next_snapshot_id).
  transmission_well_formed (add_potential_candidates s candidates next_snapshot_id) t) \<and>
  (\<forall>b\<in>boundaries_of (add_potential_candidates s candidates next_snapshot_id).
  boundary_well_formed b)"
  using source_secondary
  by (simp add: projection_well_formed_def coupling_well_formed_def
    transmission_well_formed_def)
show ?thesis
  unfolding target state_well_formed_def
  using unique_all_objects target_secondary by blast
qed

theorem differentiate_potential_preserves_identity:
  assumes concrete:
    "differentiate_potential_step G s candidates next_snapshot_id s'"
  shows "unique_versions s' \<and> immutable_preserved s s'"
proof -
  from concrete have valid: "potential_input_valid G s candidates next_snapshot_id"
  and target: "s' = add_potential_candidates s candidates next_snapshot_id"
  by (auto simp: differentiate_potential_step_def)
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
  and fresh:
    "version_id_of ` candidates \<inter> (version_ids s \<union> historical_versions_of s) = {}"
  unfolding potential_input_valid_def state_well_formed_def unique_versions_def
  by auto
  have cross_fresh: "version_id_of ` candidates \<inter> version_id_of ` objects_of s = {}"
  using fresh unfolding version_ids_def by blast

```



```

have immutable: "immutable_preserved s s'"
  unfolding immutable_preserved_def target
  proof (intro ballI allI impI)
    fix old_obj target_obj
    assume old_mem: "old_obj \<in> objects_of s"
    and target_mem:
      "target_obj \<in> objects_of
      (add_potential_candidates s candidates next_snapshot_id)"
    and same_version: "version_id_of old_obj = version_id_of target_obj"
    consider (old) "target_obj \<in> objects_of s" | (candidate) "target_obj \<in> candidates"
    using target_mem by auto
    then show "old_obj = target_obj"
  proof cases
    case old
    show ?thesis using inj_onD[OF source_unique same_version old_mem old] .
  next
    case candidate
    have candidate_image:
      "version_id_of target_obj \<in> version_id_of `candidates"
    using candidate by blast
    have old_image:
      "version_id_of target_obj \<in> version_id_of `objects_of s"
    using old_mem same_version by (metis imageI)
    have "version_id_of target_obj \<in>
    version_id_of `candidates \<inter> version_id_of `objects_of s"
    using candidate_image old_image by blast
    then show ?thesis using cross_fresh by blast
  qed
qed
have target_unique: "unique_versions s'"
  unfolding target
  using add_potential_preserves_unique_versions[OF valid] .
show ?thesis using target_unique immutable by blast
qed

theorem differentiate_potential_preserves_references_time_perspective_context:
  assumes concrete:
    "differentiate_potential_step G s candidates next_snapshot_id s'"
  shows
    "\<forall>obj\<in>objects_of s'.
    references_complete s' obj \<and>
    evidence_well_formed s' obj \<and>
    time_well_formed obj \<and>
    perspective_well_formed s' obj \<and>
    context_well_formed s' obj"
  using differentiate_potential_preserves_state_well_formed[OF concrete]
  unfolding state_well_formed_def
  by blast

theorem differentiate_potential_preserves_evidence_integrity:
  assumes concrete:
    "differentiate_potential_step G s candidates next_snapshot_id s'"
  shows "\<forall>obj\<in>objects_of s'. evidence_well_formed s' obj"
  using differentiate_potential_preserves_state_well_formed[OF concrete]
  unfolding state_well_formed_def
  by blast

theorem differentiate_potential_refines_generic_step:
  assumes concrete:
    "differentiate_potential_step G s candidates next_snapshot_id s'"
  shows "step G s DifferentiatePotential s'"
proof -
  from concrete have source_wf: "state_well_formed G s"
  and allowed: "DifferentiatePotential \<in> allowed_operations_of G"
  by (auto simp: differentiate_potential_step_def potential_input_valid_def)
  have target_wf: "state_well_formed G s'"
  using differentiate_potential_preserves_state_well_formed[OF concrete] .
  have immutable: "immutable_preserved s s'"
  using differentiate_potential_preserves_identity[OF concrete] by blast
  show ?thesis
  unfolding step_def
  using source_wf target_wf allowed immutable by blast
qed
end

```

## D.6 Konkrete T3-Semantik und Erhaltung für CreateRelation

Die sechste Theorie UZA\_0\_9\_1\_Create\_Relation erweitert die integrierte Theorie konservativ um relation\_type sowie einen typisierten relation\_catalog. Der Katalog ordnet jeder erzeugten Relations-Version Typ und Richtung zu. Diese Sidecar-Kodierung erhält den Basistyp snapshot unverändert und macht zugleich sichtbar, dass eine spätere Hauptversion die Relationsmetadaten gegebenenfalls in einen allgemeinen Objekt-Payload integrieren sollte.

relation\_input\_valid verlangt einen wohlgeformten Ausgangszustand, einen wohlgeformten Ausgangskatalog, die zugelassene Operation CreateRelation, den unterstützten Typ K\_REL, frische Objekt- und Versionsidentitäten, zwei verschiedene aktive K\_ENT-Endpunkte, vollständige Endpunktreferenzen, gültige Evidenz- und Zeitstruktur, einen vorhandenen Kontext sowie eine fortgeschriebene Snapshot-ID. add\_relation verändert ausschließlich Snapshot-ID und Objektmenge. extend\_relation\_catalog ergänzt ausschließlich den neuen version\_id-Schlüssel. Die Zielwohlgeformtheit ist keine Vorbedingung.

Die kernelgeprüfte Theorie enthält create\_relation\_exact\_delta, create\_relation\_typed\_endpoints, create\_relation\_preserves\_state\_well\_formed, create\_relation\_preserves\_identity, create\_relation\_preserves\_catalog, create\_relation\_preserves\_references\_time\_context, create\_relation\_preserves\_evidence\_integrity und create\_relation\_refines\_generic\_step. Der reale Build ergänzte das Hilfslemma active\_ent\_endpoint\_present\_ref und präziserte zwei lokale Beweisschritte. Der finale Quelltext umfasst 580 Zeilen, enthält weder sorry noch oops noch admit und besitzt SHA-256

a3c801eaf1640624ee7bf41ba91a20d2ba1e000e489ee40f4d3c209b15911ca3.

Verbindliche integrierte Sessiondatei und ausgeführter Buildbefehl:

```
session UZA_0_9_1 = HOL +
  options [document = false]
  theories
    UZA_0_9_1
    UZA_0_9_1_Base_Model
    UZA_0_9_1_Register_Appearance
    UZA_0_9_1_Rich_Model
    UZA_0_9_1_Differentiate_Potential
    UZA_0_9_1_Create_Relation
    UZA_0_9_1_Hypothesize_Effect
```

```
USER_HOME=<pinned-user-home> Isabelle2025-2/bin/isabelle build -c -v -j 1 -D . UZA_0_9_1
```

Vollständiger kernelgeprüfter Isabelle/HOL-Quelltext:

```
theory UZA_0_9_1_Create_Relation
  imports UZA_0_9_1_Differentiate_Potential
begin

section \<open>T3 CreateRelation: conservative typed operation semantics\<close>

text \<open>
  T3 creates exactly one fresh K_REL object between two distinct active K_ENT
  endpoints. The already kernel-checked base snapshot type is not changed.
  Relation type and direction are therefore persisted in a conservative typed
  side catalog keyed by the relation version identifier. Target-state
  well-formedness is derived rather than assumed.
\<close>

datatype relation_type =
  RelConnects | RelConditions | RelReinforces | RelInhibits
  | RelContradicts | RelGenerates | RelTransforms

type_synonym relation_catalog = "version_id \<right>harpoonup> (relation_type \<times> bool)"
```

```

definition active_ent_endpoint :: "snapshot \<Rightarrow> version_id \<Rightarrow> bool" where
  "active_ent_endpoint s v \<longleftarrowtrightharpoon>
    (\<exists>obj\<in>objects_of s.
      version_id_of obj = v \<and>
      kind_of obj = K_ENT \<and>
      lifecycle_of obj \<notin> {Revised, Rejected, Archived})"

definition relation_catalog_well_formed ::
  "snapshot \<Rightarrow> relation_catalog \<Rightarrow> bool" where
  "relation_catalog_well_formed s catalog \<longleftarrowtrightharpoon>
    (\<forall>v metadata. catalog v = Some metadata \<longrightarrow>
      (\<exists>r\<in>objects_of s. version_id_of r = v \<and> kind_of r = K_REL))"

definition relation_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> relation_catalog \<Rightarrow> uza_object \<Rightarrow>
    relation_type \<Rightarrow> bool \<Rightarrow> version_id \<Rightarrow> version_id \<Rightarrow>
    nat \<Rightarrow> bool" where
  "relation_input_valid G s catalog new_rel rtype directed src tgt next_snapshot_id
\<longleftarrowtrightharpoon>
  state_well_formed G s \<and>
  relation_catalog_well_formed s catalog \<and>
  CreateRelation \<in> allowed_operations_of G \<and>
  K_REL \<in> supported_kinds_of G \<and>
  kind_of new_rel = K_REL \<and>
  lifecycle_of new_rel \<in> {Draft, Hypothesis} \<and>
  object_id_of new_rel \<notin> object_id_of ` objects_of s \<and>
  version_id_of new_rel \<notin> version_ids s \<union> historical_versions_of s \<and>
  catalog (version_id_of new_rel) = None \<and>
  previous_version_of new_rel = None \<and>
  source_ref_of new_rel = Some src \<and>
  target_ref_of new_rel = Some tgt \<and>
  src \<noteq> tgt \<and>
  active_ent_endpoint s src \<and>
  active_ent_endpoint s tgt \<and>
  {src, tgt} \<subsetq> references_of new_rel \<and>
  references_complete s new_rel \<and>
  evidence_well_formed s new_rel \<and>
  time_well_formed new_rel \<and>
  (\<exists>c\<in>contexts_of s. context_of new_rel = Some c) \<and>
  perspective_of new_rel = None \<and>
  model_of new_rel = None \<and>
  authority_of new_rel = None \<and>
  effect_origin_of new_rel = None \<and>
  basis_refs_of new_rel = {} \<and>
  execution_ref_of new_rel = None \<and>
  observation_refs_of new_rel = {} \<and>
  object_status_of new_rel = None \<and>
  snapshot_id_of s < next_snapshot_id"

definition add_relation ::
  "snapshot \<Rightarrow> uza_object \<Rightarrow> nat \<Rightarrow> snapshot" where
  "add_relation s new_rel next_snapshot_id =
    s\<lparr> snapshot_id_of := next_snapshot_id,
    objects_of := insert new_rel (objects_of s) \<rparr>"

definition extend_relation_catalog ::
  "relation_catalog \<Rightarrow> uza_object \<Rightarrow> relation_type \<Rightarrow> bool
\<Rightarrow>
  relation_catalog" where
  "extend_relation_catalog catalog new_rel rtype directed =
    catalog(version_id_of new_rel \<mapsto> (rtype, directed))"

definition create_relation_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> relation_catalog \<Rightarrow> uza_object \<Rightarrow>
    relation_type \<Rightarrow> bool \<Rightarrow> version_id \<Rightarrow> version_id \<Rightarrow>
    nat \<Rightarrow> snapshot \<Rightarrow> relation_catalog \<Rightarrow> bool" where
  "create_relation_step G s catalog new_rel rtype directed src tgt
    next_snapshot_id s' catalog' \<longleftarrowtrightharpoon>
    relation_input_valid G s catalog new_rel rtype directed src tgt next_snapshot_id \<and>
    s' = add_relation s new_rel next_snapshot_id \<and>
    catalog' = extend_relation_catalog catalog new_rel rtype directed"

subsection \<open>Exact update and frame properties\<close>

lemma add_relation_objects [simp]:
  "objects_of (add_relation s new_rel next_snapshot_id) =
    insert new_rel (objects_of s)"
  by (simp add: add_relation_def)

```

```

lemma add_relation_snapshot_id [simp]:
  "snapshot_id_of (add_relation s new_rel next_snapshot_id) = next_snapshot_id"
  by (simp add: add_relation_def)

lemma add_relation_background [simp]:
  "historical_versions_of (add_relation s new_rel next_snapshot_id) =
    historical_versions_of s"
  "perspectives_of (add_relation s new_rel next_snapshot_id) = perspectives_of s"
  "contexts_of (add_relation s new_rel next_snapshot_id) = contexts_of s"
  "models_of (add_relation s new_rel next_snapshot_id) = models_of s"
  "authorities_of (add_relation s new_rel next_snapshot_id) = authorities_of s"
  "contains_edges_of (add_relation s new_rel next_snapshot_id) = contains_edges_of s"
  "projections_of (add_relation s new_rel next_snapshot_id) = projections_of s"
  "couplings_of (add_relation s new_rel next_snapshot_id) = couplings_of s"
  "transmissions_of (add_relation s new_rel next_snapshot_id) = transmissions_of s"
  "boundaries_of (add_relation s new_rel next_snapshot_id) = boundaries_of s"
  by (simp_all add: add_relation_def)

lemma add_relation_version_ids [simp]:
  "version_ids (add_relation s new_rel next_snapshot_id) =
    insert (version_id_of new_rel) (version_ids s)"
  by (auto simp: version_ids_def)

lemma active_ent_endpoint_present_ref:
  assumes "active_ent_endpoint s v"
  shows "present_ref s (Some v)"
  using assms
  by (auto simp: active_ent_endpoint_def present_ref_def version_ids_def)

lemma relation_input_fresh_object:
  assumes valid:
    "relation_input_valid G s catalog new_rel rtype directed src tgt next_snapshot_id"
  shows "new_rel \<notin> objects_of s"
  using valid
  by (auto simp: relation_input_valid_def)

lemma create_relation_exact_delta:
  assumes concrete:
    "create_relation_step G s catalog new_rel rtype directed src tgt
      next_snapshot_id s' catalog'"
  shows
    "objects_of s' = insert new_rel (objects_of s) \<and>
      new_rel \<notin> objects_of s \<and>
      snapshot_id_of s' = next_snapshot_id \<and>
      catalog' (version_id_of new_rel) = Some (rtype, directed)"
  using concrete relation_input_fresh_object
  by (auto simp: create_relation_step_def extend_relation_catalog_def)

lemma create_relation_typed_endpoints:
  assumes concrete:
    "create_relation_step G s catalog new_rel rtype directed src tgt
      next_snapshot_id s' catalog'"
  shows
    "src \<noteq> tgt \<and>
      active_ent_endpoint s src \<and>
      active_ent_endpoint s tgt \<and>
      source_ref_of new_rel = Some src \<and>
      target_ref_of new_rel = Some tgt"
  using concrete
  by (auto simp: create_relation_step_def relation_input_valid_def)

subsection \<open>Monotonicity of object conditions\<close>

lemma add_relation_present_ref_mono:
  assumes "present_ref s ref"
  shows "present_ref (add_relation s new_rel next_snapshot_id) ref"
  using assms
  by (cases ref) (auto simp: present_ref_def)

lemma add_relation_references_mono:
  assumes "references_complete s obj"
  shows "references_complete (add_relation s new_rel next_snapshot_id) obj"
  using assms
  by (auto simp: references_complete_def)

lemma add_relation_evidence_mono:
  assumes "evidence_well_formed s obj"
  shows "evidence_well_formed (add_relation s new_rel next_snapshot_id) obj"
  using assms

```

```

by (auto simp: evidence_well_formed_def evidence_references_complete_def)

lemma add_relation_relation_mono:
  assumes "relation_well_formed s obj"
  shows "relation_well_formed (add_relation s new_rel next_snapshot_id) obj"
  using assms add_relation_present_ref_mono
  unfolding relation_well_formed_def
  by blast

lemma add_relation_effect_mono:
  assumes "effect_well_formed s obj"
  shows "effect_well_formed (add_relation s new_rel next_snapshot_id) obj"
  using assms add_relation_present_ref_mono
  unfolding effect_well_formed_def
  by blast

lemma add_relation_perspective_iff [simp]:
  "perspective_well_formed (add_relation s new_rel next_snapshot_id) obj
   \<longleftarrowrightarrow> perspective_well_formed s obj"
  unfolding perspective_well_formed_def
  by (simp only: add_relation_background(2))

lemma add_relation_context_iff [simp]:
  "context_well_formed (add_relation s new_rel next_snapshot_id) obj
   \<longleftarrowrightarrow> context_well_formed s obj"
  unfolding context_well_formed_def
  by (simp only: add_relation_background(3))

lemma add_relation_relevance_iff [simp]:
  "relevance_well_formed (add_relation s new_rel next_snapshot_id) obj
   \<longleftarrowrightarrow> relevance_well_formed s obj"
  unfolding relevance_well_formed_def
  by (simp only: add_relation_perspective_iff add_relation_context_iff
    add_relation_background(4))

lemma add_relation_orientation_iff [simp]:
  "orientation_well_formed (add_relation s new_rel next_snapshot_id) obj
   \<longleftarrowrightarrow> orientation_well_formed s obj"
  by (simp add: orientation_well_formed_def)

lemma add_relation_action_iff [simp]:
  "action_well_formed (add_relation s new_rel next_snapshot_id) obj
   \<longleftarrowrightarrow> action_well_formed s obj"
  unfolding action_well_formed_def
  by (simp only: add_relation_background(5))

lemma add_relation_feedback_mono:
  assumes "feedback_well_formed s obj"
  shows "feedback_well_formed (add_relation s new_rel next_snapshot_id) obj"
  using assms add_relation_present_ref_mono
  unfolding feedback_well_formed_def
  by blast

subsection \<open>Fresh identity and new-relation validity\<close>

lemma add_relation_preserves_unique_versions:
  assumes valid:
    "relation_input_valid G s catalog new_rel rtype directed src tgt next_snapshot_id"
  shows "unique_versions (add_relation s new_rel next_snapshot_id)"
proof -
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
  and fresh: "version_id_of new_rel \<notin> version_ids s"
  unfolding relation_input_valid_def state_well_formed_def unique_versions_def
  by auto
  show ?thesis
  unfolding unique_versions_def
proof (rule inj_onI)
  fix left right
  assume left_mem: "left \<in> objects_of (add_relation s new_rel next_snapshot_id)"
  and right_mem: "right \<in> objects_of (add_relation s new_rel next_snapshot_id)"
  and same_version: "version_id_of left = version_id_of right"
  show "left = right"
  proof (cases "left = new_rel")
    case True
    show ?thesis
  proof (cases "right = new_rel")
    case True
    then show ?thesis using \<open>left = new_rel\<close> by simp
  next

```

```

    case False
    then have right_old: "right \<in> objects_of s" using right_mem by simp
    have "version_id_of new_rel \<in> version_ids s"
      unfolding version_ids_def
      using same_version \<open>left = new_rel\<close> right_old
      by (metis imageI)
    then show ?thesis using fresh by blast
  qed
next
  case False
  then have left_old: "left \<in> objects_of s" using left_mem by simp
  show ?thesis
  proof (cases "right = new_rel")
    case True
    have "version_id_of new_rel \<in> version_ids s"
      unfolding version_ids_def
      using same_version left_old True
      by (metis imageI)
    then show ?thesis using fresh by blast
  next
    case False
    then have right_old: "right \<in> objects_of s" using right_mem by simp
    show ?thesis
      using inj_onD[OF source_unique same_version left_old right_old] .
  qed
qed
qed
qed
qed

lemma relation_input_new_object_conditions:
  assumes valid:
    "relation_input_valid G s catalog new_rel rtype directed src tgt next_snapshot_id"
  shows
    "kind_of new_rel \<in> supported_kinds_of G \<and>
    references_complete (add_relation s new_rel next_snapshot_id) new_rel \<and>
    evidence_well_formed (add_relation s new_rel next_snapshot_id) new_rel \<and>
    time_well_formed new_rel \<and>
    relation_well_formed (add_relation s new_rel next_snapshot_id) new_rel \<and>
    effect_well_formed (add_relation s new_rel next_snapshot_id) new_rel \<and>
    perspective_well_formed (add_relation s new_rel next_snapshot_id) new_rel \<and>
    context_well_formed (add_relation s new_rel next_snapshot_id) new_rel \<and>
    relevance_well_formed (add_relation s new_rel next_snapshot_id) new_rel \<and>
    orientation_well_formed (add_relation s new_rel next_snapshot_id) new_rel \<and>
    action_well_formed (add_relation s new_rel next_snapshot_id) new_rel \<and>
    feedback_well_formed (add_relation s new_rel next_snapshot_id) new_rel"
proof -
  from valid have supported: "kind_of new_rel \<in> supported_kinds_of G"
  and kind: "kind_of new_rel = K_REL"
  and refs: "references_complete s new_rel"
  and evidence: "evidence_well_formed s new_rel"
  and time: "time_well_formed new_rel"
  and src_ref: "source_ref_of new_rel = Some src"
  and tgt_ref: "target_ref_of new_rel = Some tgt"
  and src_active: "active_ent_endpoint s src"
  and tgt_active: "active_ent_endpoint s tgt"
  and context_ok: "context_well_formed s new_rel"
  unfolding relation_input_valid_def context_well_formed_def
  context_bound_kind_def
  by auto
  have src_present: "present_ref s (Some src)"
    using active_ent_endpoint_present_ref[OF src_active] .
  have tgt_present: "present_ref s (Some tgt)"
    using active_ent_endpoint_present_ref[OF tgt_active] .
  have relation: "relation_well_formed s new_rel"
    unfolding relation_well_formed_def
    using kind src_ref tgt_ref src_present tgt_present by simp
  have refs_target:
    "references_complete (add_relation s new_rel next_snapshot_id) new_rel"
    using add_relation_references_mono[OF refs] .
  have evidence_target:
    "evidence_well_formed (add_relation s new_rel next_snapshot_id) new_rel"
    using add_relation_evidence_mono[OF evidence] .
  have relation_target:
    "relation_well_formed (add_relation s new_rel next_snapshot_id) new_rel"
    using add_relation_relation_mono[OF relation] .
  have typed_conditions:
    "effect_well_formed (add_relation s new_rel next_snapshot_id) new_rel \<and>
    perspective_well_formed (add_relation s new_rel next_snapshot_id) new_rel \<and>
    relevance_well_formed (add_relation s new_rel next_snapshot_id) new_rel \<and>"

```

```

orientation_well_formed (add_relation s new_rel next_snapshot_id) new_rel \<and>
action_well_formed (add_relation s new_rel next_snapshot_id) new_rel \<and>
feedback_well_formed (add_relation s new_rel next_snapshot_id) new_rel"
using kind
by (simp add: effect_well_formed_def perspective_well_formed_def
    perspective_bound_kind_def relevance_well_formed_def
    orientation_well_formed_def action_well_formed_def
    feedback_well_formed_def)
show ?thesis
  using supported_refs_target evidence_target time_relation_target context_ok
    typed_conditions
  by simp
qed

subsection \<open>Operation-specific preservation theorems\<close>

theorem create_relation_preserves_state_well_formed:
  assumes concrete:
    "create_relation_step G s catalog new_rel rtype directed src tgt
      next_snapshot_id s' catalog'"
  shows "state_well_formed G s'"
proof -
  from concrete have valid:
    "relation_input_valid G s catalog new_rel rtype directed src tgt next_snapshot_id"
  and target: "s' = add_relation s new_rel next_snapshot_id"
  by (auto simp: create_relation_step_def)
  from valid have source_wf: "state_well_formed G s"
  by (simp add: relation_input_valid_def)
  have unique: "unique_versions (add_relation s new_rel next_snapshot_id)"
  using add_relation_preserves_unique_versions[OF valid] .
  have all_objects:
    "\<forall>obj\<in>objects_of (add_relation s new_rel next_snapshot_id).
      kind_of obj \<in> supported_kinds_of G \<and>
      references_complete (add_relation s new_rel next_snapshot_id) obj \<and>
      evidence_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
      time_well_formed obj \<and>
      relation_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
      effect_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
      perspective_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
      context_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
      relevance_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
      orientation_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
      action_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
      feedback_well_formed (add_relation s new_rel next_snapshot_id) obj"
  proof (intro ballI)
    fix obj
    assume obj_mem: "obj \<in> objects_of (add_relation s new_rel next_snapshot_id)"
    consider (new) "obj = new_rel" | (old) "obj \<in> objects_of s"
    using obj_mem by auto
    then show
      "kind_of obj \<in> supported_kinds_of G \<and>
        references_complete (add_relation s new_rel next_snapshot_id) obj \<and>
        evidence_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
        time_well_formed obj \<and>
        relation_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
        effect_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
        perspective_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
        context_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
        relevance_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
        orientation_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
        action_well_formed (add_relation s new_rel next_snapshot_id) obj \<and>
        feedback_well_formed (add_relation s new_rel next_snapshot_id) obj"
  proof cases
    case new
    then show ?thesis
      using relation_input_new_object_conditions[OF valid] by simp
  next
    case old
    from source_wf old have source_conditions:
      "kind_of obj \<in> supported_kinds_of G \<and>
        references_complete s obj \<and> evidence_well_formed s obj \<and>
        time_well_formed obj \<and>
        relation_well_formed s obj \<and> effect_well_formed s obj \<and>
        perspective_well_formed s obj \<and> context_well_formed s obj \<and>
        relevance_well_formed s obj \<and> orientation_well_formed s obj \<and>
        action_well_formed s obj \<and> feedback_well_formed s obj"
    unfolding state_well_formed_def by blast
    from source_conditions have supported: "kind_of obj \<in> supported_kinds_of G"
    and refs: "references_complete s obj"

```

```

and evidence: "evidence_well_formed s obj"
and time: "time_well_formed obj"
and relation: "relation_well_formed s obj"
and effect: "effect_well_formed s obj"
and perspective: "perspective_well_formed s obj"
and context_ok: "context_well_formed s obj"
and relevance: "relevance_well_formed s obj"
and orientation: "orientation_well_formed s obj"
and action: "action_well_formed s obj"
and feedback: "feedback_well_formed s obj"
by blast+
have refs_target:
  "references_complete (add_relation s new_rel next_snapshot_id) obj"
  using add_relation_references_mono[OF refs] .
have evidence_target:
  "evidence_well_formed (add_relation s new_rel next_snapshot_id) obj"
  using add_relation_evidence_mono[OF evidence] .
have relation_target:
  "relation_well_formed (add_relation s new_rel next_snapshot_id) obj"
  using add_relation_relation_mono[OF relation] .
have effect_target:
  "effect_well_formed (add_relation s new_rel next_snapshot_id) obj"
  using add_relation_effect_mono[OF effect] .
have feedback_target:
  "feedback_well_formed (add_relation s new_rel next_snapshot_id) obj"
  using add_relation_feedback_mono[OF feedback] .
show ?thesis
  using supported refs_target evidence_target time relation_target effect_target
  perspective context_ok relevance orientation action feedback_target
  by simp
qed
qed
from source_wf have source_secondary:
  "(\<forall>p\<in>projections_of s. projection_well_formed s p) \<and>
  (\<forall>c\<in>couplings_of s. coupling_well_formed s c) \<and>
  (\<forall>t\<in>transmissions_of s. transmission_well_formed s t) \<and>
  (\<forall>b\<in>boundaries_of s. boundary_well_formed b)"
  unfolding state_well_formed_def by blast
have target_secondary:
  "(\<forall>p\<in>projections_of (add_relation s new_rel next_snapshot_id).
  projection_well_formed (add_relation s new_rel next_snapshot_id) p) \<and>
  (\<forall>c\<in>couplings_of (add_relation s new_rel next_snapshot_id).
  coupling_well_formed (add_relation s new_rel next_snapshot_id) c) \<and>
  (\<forall>t\<in>transmissions_of (add_relation s new_rel next_snapshot_id).
  transmission_well_formed (add_relation s new_rel next_snapshot_id) t) \<and>
  (\<forall>b\<in>boundaries_of (add_relation s new_rel next_snapshot_id).
  boundary_well_formed b)"
  using source_secondary
  by (simp add: projection_well_formed_def coupling_well_formed_def
  transmission_well_formed_def)
show ?thesis
  unfolding target state_well_formed_def
  using unique all_objects target_secondary by blast
qed

theorem create_relation_preserves_identity:
  assumes concrete:
    "create_relation_step G s catalog new_rel rtype directed src tgt
    next_snapshot_id s' catalog'"
  shows "unique_versions s' \<and> immutable_preserved s s'"
proof -
  from concrete have valid:
    "relation_input_valid G s catalog new_rel rtype directed src tgt next_snapshot_id"
    and target: "s' = add_relation s new_rel next_snapshot_id"
    by (auto simp: create_relation_step_def)
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
    and fresh: "version_id_of new_rel \<notin> version_ids s"
    unfolding relation_input_valid_def state_well_formed_def unique_versions_def
    by auto
  have immutable: "immutable_preserved s s'"
    unfolding immutable_preserved_def target
  proof (intro ballI allI impI)
    fix old_obj target_obj
    assume old_mem: "old_obj \<in> objects_of s"
    and target_mem:
      "target_obj \<in> objects_of (add_relation s new_rel next_snapshot_id)"
    and same_version: "version_id_of old_obj = version_id_of target_obj"
    show "old_obj = target_obj"
    proof (cases "target_obj = new_rel")

```



```

    case True
    have "version_id_of new_rel \<in> version_ids s"
      unfolding version_ids_def
      using old_mem same_version True
      by (metis imageI)
    then show ?thesis using fresh by blast
  next
  case False
  then have target_old: "target_obj \<in> objects_of s" using target_mem by simp
  show ?thesis
    using inj_onD[OF source_unique same_version old_mem target_old] .
qed
qed
have target_unique: "unique_versions s'"
  unfolding target
  using add_relation_preserves_unique_versions[OF valid] .
show ?thesis using target_unique immutable by blast
qed

theorem create_relation_preserves_catalog:
  assumes concrete:
    "create_relation_step G s catalog new_rel rtype directed src tgt
     next_snapshot_id s' catalog'"
  shows
    "relation_catalog_well_formed s' catalog' \<and>
     catalog' (version_id_of new_rel) = Some (rtype, directed)"
proof -
  from concrete have valid:
    "relation_input_valid G s catalog new_rel rtype directed src tgt next_snapshot_id"
  and target: "s' = add_relation s new_rel next_snapshot_id"
  and catalog_target:
    "catalog' = extend_relation_catalog catalog new_rel rtype directed"
  by (auto simp: create_relation_step_def)
  from valid have source_catalog: "relation_catalog_well_formed s catalog"
  by (simp add: relation_input_valid_def)
  have target_catalog: "relation_catalog_well_formed s' catalog'"
  unfolding relation_catalog_well_formed_def target catalog_target
  extend_relation_catalog_def
  using source_catalog valid
  unfolding relation_catalog_well_formed_def relation_input_valid_def
  by auto
  show ?thesis
  using target_catalog catalog_target
  by (simp add: extend_relation_catalog_def)
qed

theorem create_relation_preserves_references_time_context:
  assumes concrete:
    "create_relation_step G s catalog new_rel rtype directed src tgt
     next_snapshot_id s' catalog'"
  shows
    "\<forall>obj\<in>objects_of s'.
     references_complete s' obj \<and>
     evidence_well_formed s' obj \<and>
     time_well_formed obj \<and>
     context_well_formed s' obj"
  using create_relation_preserves_state_well_formed[OF concrete]
  unfolding state_well_formed_def

```

Beweisstatus dieser D.6-Fassung: kernelgeprüft. Der integrierte saubere und verbose Isabelle2025-2-Build verarbeitete am 8. August 2026 alle sieben Theorien zu 100 % und endete mit Exit-Code 0. Für T3 sind damit Parser-, Typ- und Kernelakzeptanz des gepinnten Quellstands belegt. Der Nachweis umfasst die im Quelltext formulierten Erhaltungs- und Refinementsätze; er beweist weder die Vollständigkeit aller möglichen Relationstypen noch allgemeine Konservativität oder Produktivengine-Äquivalenz.

by blast

```

theorem create_relation_preserves_evidence_integrity:
  assumes concrete:
    "create_relation_step G s catalog new_rel rtype directed src tgt
     next_snapshot_id s' catalog'"
  shows "\<forall>obj\<in>objects_of s'. evidence_well_formed s' obj"
  using create_relation_preserves_state_well_formed[OF concrete]
  unfolding state_well_formed_def

```

```

by blast

theorem create_relation_refines_generic_step:
  assumes concrete:
    "create_relation_step G s catalog new_rel rtype directed src tgt
     next_snapshot_id s' catalog'"
  shows "step G s CreateRelation s'"
proof -
  from concrete have source_wf: "state_well_formed G s"
  and allowed: "CreateRelation \<in> allowed_operations_of G"
  by (auto simp: create_relation_step_def relation_input_valid_def)
  have target_wf: "state_well_formed G s'"
  using create_relation_preserves_state_well_formed[OF concrete] .
  have immutable: "immutable_preserved s s'"
  using create_relation_preserves_identity[OF concrete] by blast
  show ?thesis
  unfolding step_def
  using source_wf target_wf allowed immutable by blast
qed

end

```

## D.7 Konkrete T4-Semantik und Erhaltung für HypothesizeEffect

Die siebte Theorie UZA\_0\_9\_1\_Hypothesize\_Effect importiert T3 und erweitert den unveränderten Snapshot-Kerntyp um einen versionierten effect\_catalog. Der Katalog ordnet jeder erzeugten EFF-Version einen effect\_hypothesis\_payload zu: verursachende Relationsversion, optionales Ziel, optionaler effect\_vector, Unsicherheit und Profilmodell. option bildet offene Ziel- oder Vektorstruktur explizit ab; es gibt weder Sentinel-IDs noch stilles Zero-Fill.

effect\_input\_valid verlangt einen wohlgeformten Ausgangszustand und Katalog, die zugelassene Operation HypothesizeEffect, den unterstützten Typ K\_EFF, eine aktive typisierte K\_REL-Ursache, gegebenenfalls ein aktives Ziel, eine vorhandene Perspektive, einen vorhandenen Kontext und ein vorhandenes Profilmodell, Unsicherheit im geschlossenen Intervall [0,1], frische Objekt- und Versionsidentitäten, Lifecycle Hypothesis, vollständige Referenzen, gültige Evidenz- und Zeitstruktur sowie eine fortgeschriebene Snapshot-ID. add\_effect\_hypothesis verändert ausschließlich Snapshot-ID und Objektmenge; extend\_effect\_catalog ergänzt ausschließlich den neuen Versionsschlüssel. Die Zielwohlgeformtheit ist keine Vorbedingung.

Die kernelgeprüfte Theorie enthält hypothesize\_effect\_exact\_delta, hypothesize\_effect\_origin\_and\_openness, hypothesize\_effect\_preserves\_state\_well\_formed, hypothesize\_effect\_preserves\_identity, hypothesize\_effect\_preserves\_catalog, hypothesize\_effect\_preserves\_references\_time\_perspective\_context, hypothesize\_effect\_preserves\_evidence\_integrity und hypothesize\_effect\_refines\_generic\_step. Der finale Quelltext umfasst 718 Zeilen, enthält weder sorry noch oops noch admit und besitzt SHA-256 2eb8e1d262cb3790f6c446839fb6809de3c2837d20705925507e272c860e2fc4.

```

theory UZA_0_9_1_Hypothesize_Effect
  imports UZA_0_9_1_Create_Relation
begin

section \<open>T4 HypothesizeEffect: typed effect-hypothesis semantics\<close>

text \<open>
  T4 creates exactly one fresh K_EFF object from one active, typed K_REL
  origin. Target and V10 profile may each remain explicitly open. Their
  openness is represented by option types rather than sentinel values or
  silent zero filling. Perspective, context, time, uncertainty, profile
  model, references, and provenance are checked before construction.

  The kernel-checked base snapshot type remains unchanged. The effect
  profile and uncertainty are persisted in a conservative typed side catalog

```

```

keyed by the effect version identifier. Target-state well-formedness is
derived and is not an input assumption.
\<close>

record effect_hypothesis_payload =
  hypothesis_cause_of :: version_id
  hypothesis_target_of :: "version_id option"
  hypothesis_vector_of :: "effect_vector option"
  hypothesis_uncertainty_of :: real
  hypothesis_profile_model_of :: model_id

type_synonym effect_catalog =
  "version_id \<Rightharpoonup> effect_hypothesis_payload"

definition active_relation_origin ::
  "snapshot \<Rightharpoonup> version_id \<Rightharpoonup> bool" where
  "active_relation_origin s v \<longleftarrow\rightarrow>
    (\<exists>rel\<in>objects_of s.
      version_id_of rel = v \<and>
      kind_of rel = K_REL \<and>
      lifecycle_of rel \<notin> {Revised, Rejected, Archived})"

definition active_effect_target ::
  "snapshot \<Rightharpoonup> version_id \<Rightharpoonup> bool" where
  "active_effect_target s v \<longleftarrow\rightarrow>
    (\<exists>obj\<in>objects_of s.
      version_id_of obj = v \<and>
      lifecycle_of obj \<notin> {Revised, Rejected, Archived})"

definition effect_payload_valid ::
  "uza_object \<Rightharpoonup> effect_hypothesis_payload \<Rightharpoonup> bool" where
  "effect_payload_valid eff payload \<longleftarrow\rightarrow>
    kind_of eff = K_EFF \<and>
    effect_origin_of eff = Some (hypothesis_cause_of payload) \<and>
    source_ref_of eff = Some (hypothesis_cause_of payload) \<and>
    target_ref_of eff = hypothesis_target_of payload \<and>
    model_of eff = Some (hypothesis_profile_model_of payload) \<and>
    0 \<le> hypothesis_uncertainty_of payload \<and>
    hypothesis_uncertainty_of payload \<le> 1 \<and>
    (case hypothesis_vector_of payload of
      None \<Rightharpoonup> True
    | Some vector \<Rightharpoonup> effect_vector_valid vector)"

definition effect_catalog_well_formed ::
  "snapshot \<Rightharpoonup> effect_catalog \<Rightharpoonup> bool" where
  "effect_catalog_well_formed s catalog \<longleftarrow\rightarrow>
    (\<forall>v payload. catalog v = Some payload \<longrightarrow\rightarrow>
      (\<exists>eff\<in>objects_of s.
        version_id_of eff = v \<and> effect_payload_valid eff payload))"

definition effect_target_refs ::
  "version_id \<Rightharpoonup> version_id option \<Rightharpoonup> version_id set" where
  "effect_target_refs cause target =
    insert cause (case target of None \<Rightharpoonup> {} | Some v \<Rightharpoonup> {v})"

definition effect_input_valid ::
  "grammar \<Rightharpoonup> snapshot \<Rightharpoonup> relation_catalog \<Rightharpoonup>
    effect_catalog \<Rightharpoonup> uza_object \<Rightharpoonup> effect_hypothesis_payload
    \<Rightharpoonup> nat \<Rightharpoonup> bool" where
  "effect_input_valid G s relation_catalog effect_catalog new_eff payload
    next_snapshot_id \<longleftarrow\rightarrow>
    state_well_formed G s \<and>
    relation_catalog_well_formed s relation_catalog \<and>
    effect_catalog_well_formed s effect_catalog \<and>
    HypothesizeEffect \<in> allowed_operations_of G \<and>
    K_EFF \<in> supported_kinds_of G \<and>
    kind_of new_eff = K_EFF \<and>
    lifecycle_of new_eff = Hypothesis \<and>
    object_id_of new_eff \<notin> object_id_of ` objects_of s \<and>
    version_id_of new_eff \<notin>
      version_ids s \<union> historical_versions_of s \<and>
    effect_catalog (version_id_of new_eff) = None \<and>
    previous_version_of new_eff = None \<and>
    active_relation_origin s (hypothesis_cause_of payload) \<and>
    relation_catalog (hypothesis_cause_of payload) \<noteq> None \<and>
    effect_origin_of new_eff = Some (hypothesis_cause_of payload) \<and>
    source_ref_of new_eff = Some (hypothesis_cause_of payload) \<and>
    target_ref_of new_eff = hypothesis_target_of payload \<and>
    (case hypothesis_target_of payload of
      None \<Rightharpoonup> True

```

```

| Some target \<Rightarrow> active_effect_target s target) \<and>
effect_target_refs (hypothesis_cause_of payload)
(hypothesis_target_of payload) \<subseq> references_of new_eff \<and>
basis_refs_of new_eff =
  effect_target_refs (hypothesis_cause_of payload)
  (hypothesis_target_of payload) \<and>
references_complete s new_eff \<and>
evidence_well_formed s new_eff \<and>
time_well_formed new_eff \<and>
(\<exists>p\<in>perspectives_of s. perspective_of new_eff = Some p) \<and>
(\<exists>c\<in>contexts_of s. context_of new_eff = Some c) \<and>
hypothesis_profile_model_of payload \<in> models_of s \<and>
model_of new_eff = Some (hypothesis_profile_model_of payload) \<and>
authority_of new_eff = None \<and>
execution_ref_of new_eff = None \<and>
observation_refs_of new_eff = {} \<and>
object_status_of new_eff = Some Applicable \<and>
effect_payload_valid new_eff payload \<and>
snapshot_id_of s < next_snapshot_id"

definition add_effect_hypothesis ::
  "snapshot \<Rightarrow> uza_object \<Rightarrow> snapshot" where
  "add_effect_hypothesis s new_eff next_snapshot_id =
    s\<lparr> snapshot_id_of := next_snapshot_id,
    objects_of := insert new_eff (objects_of s) \<rparr>"

definition extend_effect_catalog ::
  "effect_catalog \<Rightarrow> uza_object \<Rightarrow> effect_hypothesis_payload
  \<Rightarrow> effect_catalog" where
  "extend_effect_catalog catalog new_eff payload =
    catalog(version_id_of new_eff \<mapsto> payload)"

definition hypothesize_effect_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> relation_catalog \<Rightarrow>
  effect_catalog \<Rightarrow> uza_object \<Rightarrow> effect_hypothesis_payload
  \<Rightarrow> nat \<Rightarrow> snapshot \<Rightarrow> effect_catalog \<Rightarrow> bool" where
  "hypothesize_effect_step G s relation_catalog effect_catalog new_eff payload
    next_snapshot_id s' effect_catalog' \<longleftarrow>
    effect_input_valid G s relation_catalog effect_catalog new_eff payload
    next_snapshot_id \<and>
    s' = add_effect_hypothesis s new_eff next_snapshot_id \<and>
    effect_catalog' = extend_effect_catalog effect_catalog new_eff payload"

subsection \<open>Exact update and semantic input facts\<close>

lemma add_effect_objects [simp]:
  "objects_of (add_effect_hypothesis s new_eff next_snapshot_id) =
    insert new_eff (objects_of s)"
  by (simp add: add_effect_hypothesis_def)

lemma add_effect_snapshot_id [simp]:
  "snapshot_id_of (add_effect_hypothesis s new_eff next_snapshot_id) =
    next_snapshot_id"
  by (simp add: add_effect_hypothesis_def)

lemma add_effect_background [simp]:
  "historical_versions_of
    (add_effect_hypothesis s new_eff next_snapshot_id) =
    historical_versions_of s"
  "perspectives_of (add_effect_hypothesis s new_eff next_snapshot_id) =
    perspectives_of s"
  "contexts_of (add_effect_hypothesis s new_eff next_snapshot_id) =
    contexts_of s"
  "models_of (add_effect_hypothesis s new_eff next_snapshot_id) = models_of s"
  "authorities_of (add_effect_hypothesis s new_eff next_snapshot_id) =
    authorities_of s"
  "contains_edges_of (add_effect_hypothesis s new_eff next_snapshot_id) =
    contains_edges_of s"
  "projections_of (add_effect_hypothesis s new_eff next_snapshot_id) =
    projections_of s"
  "couplings_of (add_effect_hypothesis s new_eff next_snapshot_id) =
    couplings_of s"
  "transmissions_of (add_effect_hypothesis s new_eff next_snapshot_id) =
    transmissions_of s"
  "boundaries_of (add_effect_hypothesis s new_eff next_snapshot_id) =
    boundaries_of s"
  by (simp_all add: add_effect_hypothesis_def)

lemma add_effect_version_ids [simp]:
  "version_ids (add_effect_hypothesis s new_eff next_snapshot_id) =

```

```

    insert (version_id_of new_eff) (version_ids s)"
  by (auto simp: version_ids_def)

lemma active_relation_origin_present_ref:
  assumes "active_relation_origin s v"
  shows "present_ref s (Some v)"
  using assms
  by (auto simp: active_relation_origin_def present_ref_def version_ids_def)

lemma active_effect_target_present_ref:
  assumes "active_effect_target s v"
  shows "present_ref s (Some v)"
  using assms
  by (auto simp: active_effect_target_def present_ref_def version_ids_def)

lemma effect_input_fresh_object:
  assumes valid:
    "effect_input_valid G s relation_catalog effect_catalog new_eff payload
     next_snapshot_id"
  shows "new_eff \<notin> objects_of s"
  using valid
  by (auto simp: effect_input_valid_def)

lemma hypothesize_effect_exact_delta:
  assumes concrete:
    "hypothesize_effect_step G s relation_catalog effect_catalog new_eff payload
     next_snapshot_id s' effect_catalog'"
  shows
    "objects_of s' = insert new_eff (objects_of s) \<and>
     new_eff \<notin> objects_of s \<and>
     snapshot_id_of s' = next_snapshot_id \<and>
     effect_catalog' (version_id_of new_eff) = Some payload"
  using concrete effect_input_fresh_object
  by (auto simp: hypothesize_effect_step_def extend_effect_catalog_def)

lemma hypothesize_effect_origin_and_openness:
  assumes concrete:
    "hypothesize_effect_step G s relation_catalog effect_catalog new_eff payload
     next_snapshot_id s' effect_catalog'"
  shows
    "active_relation_origin s (hypothesis_cause_of payload) \<and>
     relation_catalog (hypothesis_cause_of payload) \<noteq> None \<and>
     effect_origin_of new_eff = Some (hypothesis_cause_of payload) \<and>
     source_ref_of new_eff = Some (hypothesis_cause_of payload) \<and>
     target_ref_of new_eff = hypothesis_target_of payload \<and>
     (case hypothesis_target_of payload of
       None \<Rightarrow> True
     | Some target \<Rightarrow> active_effect_target s target) \<and>
     (case hypothesis_vector_of payload of
       None \<Rightarrow> True
     | Some vector \<Rightarrow> effect_vector_valid vector) \<and>
     0 \<le> hypothesis_uncertainty_of payload \<and>
     hypothesis_uncertainty_of payload \<le> 1"
  using concrete
  by (auto simp: hypothesize_effect_step_def effect_input_valid_def
    effect_payload_valid_def split: option.splits)

subsection \<open>Monotonicity of inherited state conditions\<close>

lemma add_effect_present_ref_mono:
  assumes "present_ref s ref"
  shows "present_ref (add_effect_hypothesis s new_eff next_snapshot_id) ref"
  using assms
  by (cases ref) (auto simp: present_ref_def)

lemma add_effect_references_mono:
  assumes "references_complete s obj"
  shows "references_complete
    (add_effect_hypothesis s new_eff next_snapshot_id) obj"
  using assms
  by (auto simp: references_complete_def)

lemma add_effect_evidence_mono:
  assumes "evidence_well_formed s obj"
  shows "evidence_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) obj"
  using assms
  by (auto simp: evidence_well_formed_def evidence_references_complete_def)

lemma add_effect_relation_mono:

```

```

assumes "relation_well_formed s obj"
shows "relation_well_formed
  (add_effect_hypothesis s new_eff next_snapshot_id) obj"
using assms add_effect_present_ref_mono
unfolding relation_well_formed_def
by blast

lemma add_effect_effect_mono:
  assumes "effect_well_formed s obj"
  shows "effect_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) obj"
  using assms add_effect_present_ref_mono
  unfolding effect_well_formed_def
  by blast

lemma add_effect_perspective_iff [simp]:
  "perspective_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) obj
    \<longleftarrowrightarrow> perspective_well_formed s obj"
  unfolding perspective_well_formed_def
  by (simp only: add_effect_background(2))

lemma add_effect_context_iff [simp]:
  "context_well_formed (add_effect_hypothesis s new_eff next_snapshot_id) obj
    \<longleftarrowrightarrow> context_well_formed s obj"
  unfolding context_well_formed_def
  by (simp only: add_effect_background(3))

lemma add_effect_relevance_iff [simp]:
  "relevance_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) obj
    \<longleftarrowrightarrow> relevance_well_formed s obj"
  unfolding relevance_well_formed_def
  by (simp only: add_effect_perspective_iff add_effect_context_iff
    add_effect_background(4))

lemma add_effect_orientation_iff [simp]:
  "orientation_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) obj
    \<longleftarrowrightarrow> orientation_well_formed s obj"
  by (simp add: orientation_well_formed_def)

lemma add_effect_action_iff [simp]:
  "action_well_formed (add_effect_hypothesis s new_eff next_snapshot_id) obj
    \<longleftarrowrightarrow> action_well_formed s obj"
  unfolding action_well_formed_def
  by (simp only: add_effect_background(5))

lemma add_effect_feedback_mono:
  assumes "feedback_well_formed s obj"
  shows "feedback_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) obj"
  using assms add_effect_present_ref_mono
  unfolding feedback_well_formed_def
  by blast

subsection \<open>Fresh identity and new-effect validity\<close>

lemma add_effect_preserves_unique_versions:
  assumes valid:
    "effect_input_valid G s relation_catalog effect_catalog new_eff payload
      next_snapshot_id"
  shows "unique_versions
    (add_effect_hypothesis s new_eff next_snapshot_id)"
proof -
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
  and fresh: "version_id_of new_eff \<notin> version_ids s"
  unfolding effect_input_valid_def state_well_formed_def unique_versions_def
  by auto
  show ?thesis
  unfolding unique_versions_def
proof (rule inj_onI)
  fix left right
  assume left_mem:
    "left \<in> objects_of (add_effect_hypothesis s new_eff next_snapshot_id)"
  and right_mem:
    "right \<in> objects_of (add_effect_hypothesis s new_eff next_snapshot_id)"
  and same_version: "version_id_of left = version_id_of right"
  show "left = right"
  proof (cases "left = new_eff")

```

```

case True
show ?thesis
proof (cases "right = new_eff")
  case True
  then show ?thesis using \<open>left = new_eff\<close> by simp
next
  case False
  then have right_old: "right \<in> objects_of s" using right_mem by simp
  have "version_id_of new_eff \<in> version_ids s"
    unfolding version_ids_def
    using same_version \<open>left = new_eff\<close> right_old
    by (metis imageI)
  then show ?thesis using fresh by blast
qed
next
case False
then have left_old: "left \<in> objects_of s" using left_mem by simp
show ?thesis
proof (cases "right = new_eff")
  case True
  have "version_id_of new_eff \<in> version_ids s"
    unfolding version_ids_def
    using same_version left_old True
    by (metis imageI)
  then show ?thesis using fresh by blast
next
  case False
  then have right_old: "right \<in> objects_of s" using right_mem by simp
  show ?thesis
    using inj_onD[OF source_unique same_version left_old right_old] .
qed
qed
qed
qed

lemma effect_input_new_object_conditions:
  assumes valid:
    "effect_input_valid G s relation_catalog effect_catalog new_eff payload
    next_snapshot_id"
  shows
    "kind_of new_eff \<in> supported_kinds_of G \<and>
    references_complete
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff \<and>
    evidence_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff \<and>
    time_well_formed new_eff \<and>
    relation_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff \<and>
    effect_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff \<and>
    perspective_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff \<and>
    context_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff \<and>
    relevance_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff \<and>
    orientation_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff \<and>
    action_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff \<and>
    feedback_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff"
proof -
  from valid have supported: "kind_of new_eff \<in> supported_kinds_of G"
  and kind: "kind_of new_eff = K_EFF"
  and lifecycle: "lifecycle_of new_eff = Hypothesis"
  and refs: "references_complete s new_eff"
  and evidence: "evidence_well_formed s new_eff"
  and time: "time_well_formed new_eff"
  and perspective: "perspective_well_formed s new_eff"
  and context_ok: "context_well_formed s new_eff"
  unfolding effect_input_valid_def perspective_well_formed_def
    perspective_bound_kind_def context_well_formed_def context_bound_kind_def
  by auto
  have refs_target:
    "references_complete
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff"
  using add_effect_references_mono[OF refs] .
  have evidence_target:
    "evidence_well_formed

```

```

    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff"
  using add_effect_evidence_mono[OF evidence] .
  have typed_conditions:
  "relation_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff \<and>
    effect_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff \<and>
    relevance_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff \<and>
    orientation_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff \<and>
    action_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff \<and>
    feedback_well_formed
    (add_effect_hypothesis s new_eff next_snapshot_id) new_eff"
  using kind lifecycle
  by (simp add: relation_well_formed_def effect_well_formed_def
    relevance_well_formed_def orientation_well_formed_def
    action_well_formed_def feedback_well_formed_def)
  show ?thesis
  using supported refs_target evidence_target time perspective context_ok
  typed_conditions
  by simp
qed

subsection \<open>Operation-specific preservation theorems\<close>

theorem hypothesize_effect_preserves_state_well_formed:
  assumes concrete:
    "hypothesize_effect_step G s relation_catalog effect_catalog new_eff payload
      next_snapshot_id s' effect_catalog"
  shows "state_well_formed G s"
  proof -
    from concrete have valid:
      "effect_input_valid G s relation_catalog effect_catalog new_eff payload
        next_snapshot_id"
    and target: "s' = add_effect_hypothesis s new_eff next_snapshot_id"
    by (auto simp: hypothesize_effect_step_def)
    from valid have source_wf: "state_well_formed G s"
    by (simp add: effect_input_valid_def)
    have unique:
      "unique_versions (add_effect_hypothesis s new_eff next_snapshot_id)"
    using add_effect_preserves_unique_versions[OF valid] .
    have all_objects:
      "\<forall>obj\<in>objects_of (add_effect_hypothesis s new_eff next_snapshot_id).
        kind_of obj \<in> supported_kinds_of G \<and>
        references_complete
        (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
        evidence_well_formed
        (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
        time_well_formed obj \<and>
        relation_well_formed
        (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
        effect_well_formed
        (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
        perspective_well_formed
        (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
        context_well_formed
        (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
        relevance_well_formed
        (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
        orientation_well_formed
        (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
        action_well_formed
        (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
        feedback_well_formed
        (add_effect_hypothesis s new_eff next_snapshot_id) obj"
    proof (intro ballI)
      fix obj
      assume obj_mem:
        "obj \<in> objects_of (add_effect_hypothesis s new_eff next_snapshot_id)"
      consider (new) "obj = new_eff" | (old) "obj \<in> objects_of s"
      using obj_mem by auto
      then show
        "kind_of obj \<in> supported_kinds_of G \<and>
        references_complete
        (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
        evidence_well_formed
        (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
        time_well_formed obj \<and>"
    qed
  qed

```



```

relation_well_formed
  (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
effect_well_formed
  (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
perspective_well_formed
  (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
context_well_formed
  (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
relevance_well_formed
  (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
orientation_well_formed
  (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
action_well_formed
  (add_effect_hypothesis s new_eff next_snapshot_id) obj \<and>
feedback_well_formed
  (add_effect_hypothesis s new_eff next_snapshot_id) obj"
proof cases
case new
then show ?thesis
  using effect_input_new_object_conditions[OF valid] by simp
next
case old
from source_wf old have source_conditions:
  "kind_of obj \<in> supported_kinds_of G \<and>
  references_complete s obj \<and> evidence_well_formed s obj \<and>
  time_well_formed s obj \<and> relation_well_formed s obj \<and>
  effect_well_formed s obj \<and> perspective_well_formed s obj \<and>
  context_well_formed s obj \<and> relevance_well_formed s obj \<and>
  orientation_well_formed s obj \<and> action_well_formed s obj \<and>
  feedback_well_formed s obj"
unfolding state_well_formed_def by blast
from source_conditions have supported:
  "kind_of obj \<in> supported_kinds_of G"
  and refs: "references_complete s obj"
  and evidence: "evidence_well_formed s obj"
  and time: "time_well_formed s obj"
  and relation: "relation_well_formed s obj"
  and effect: "effect_well_formed s obj"
  and perspective: "perspective_well_formed s obj"
  and context_ok: "context_well_formed s obj"
  and relevance: "relevance_well_formed s obj"
  and orientation: "orientation_well_formed s obj"
  and action: "action_well_formed s obj"
  and feedback: "feedback_well_formed s obj"
by blast+
have refs_target:
  "references_complete
  (add_effect_hypothesis s new_eff next_snapshot_id) obj"
  using add_effect_references_mono[OF refs] .
have evidence_target:
  "evidence_well_formed
  (add_effect_hypothesis s new_eff next_snapshot_id) obj"
  using add_effect_evidence_mono[OF evidence] .
have relation_target:
  "relation_well_formed
  (add_effect_hypothesis s new_eff next_snapshot_id) obj"
  using add_effect_relation_mono[OF relation] .
have effect_target:
  "effect_well_formed
  (add_effect_hypothesis s new_eff next_snapshot_id) obj"
  using add_effect_effect_mono[OF effect] .
have feedback_target:
  "feedback_well_formed
  (add_effect_hypothesis s new_eff next_snapshot_id) obj"
  using add_effect_feedback_mono[OF feedback] .
show ?thesis
  using supported_refs_target evidence_target time relation_target
    effect_target perspective context_ok relevance orientation action
    feedback_target
  by simp
qed
qed
from source_wf have source_secondary:
  "(\<forall>p\<in>projections_of s. projection_well_formed s p) \<and>
  (\<forall>c\<in>couplings_of s. coupling_well_formed s c) \<and>
  (\<forall>t\<in>transmissions_of s. transmission_well_formed s t) \<and>
  (\<forall>b\<in>boundaries_of s. boundary_well_formed s b)"
unfolding state_well_formed_def by blast
have target_secondary:
  "(\<forall>p\<in>projections_of

```

```

      (add_effect_hypothesis s new_eff next_snapshot_id).
      projection_well_formed
      (add_effect_hypothesis s new_eff next_snapshot_id) p) \<and>
    (\<forall>c\<in>couplings_of
      (add_effect_hypothesis s new_eff next_snapshot_id).
      coupling_well_formed
      (add_effect_hypothesis s new_eff next_snapshot_id) c) \<and>
    (\<forall>t\<in>transmissions_of
      (add_effect_hypothesis s new_eff next_snapshot_id).
      transmission_well_formed
      (add_effect_hypothesis s new_eff next_snapshot_id) t) \<and>
    (\<forall>b\<in>boundaries_of
      (add_effect_hypothesis s new_eff next_snapshot_id).
      boundary_well_formed b)"
  using source_secondary
  by (simp add: projection_well_formed_def coupling_well_formed_def
    transmission_well_formed_def)
show ?thesis
unfolding target_state_well_formed_def
using unique_all_objects target_secondary by blast
qed

theorem hypothesize_effect_preserves_identity:
  assumes concrete:
    "hypothesize_effect_step G s relation_catalog effect_catalog new_eff payload
      next_snapshot_id s' effect_catalog'"
  shows "unique_versions s' \<and> immutable_preserved s s'"
proof -
  from concrete have valid:
    "effect_input_valid G s relation_catalog effect_catalog new_eff payload
      next_snapshot_id"
  and target: "s' = add_effect_hypothesis s new_eff next_snapshot_id"
  by (auto simp: hypothesize_effect_step_def)
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
  and fresh: "version_id_of new_eff \<notin> version_ids s"
  unfolding effect_input_valid_def state_well_formed_def unique_versions_def
  by auto
  have immutable: "immutable_preserved s s'"
  unfolding immutable_preserved_def target
  proof (intro ballI allI impI)
    fix old_obj target_obj
    assume old_mem: "old_obj \<in> objects_of s"
    and target_mem:
      "target_obj \<in>
        objects_of (add_effect_hypothesis s new_eff next_snapshot_id)"
    and same_version:
      "version_id_of old_obj = version_id_of target_obj"
    show "old_obj = target_obj"
    proof (cases "target_obj = new_eff")
      case True
      have "version_id_of new_eff \<in> version_ids s"
      unfolding version_ids_def
      using old_mem same_version True
      by (metis imageI)
      then show ?thesis using fresh by blast
    next
      case False
      then have target_old: "target_obj \<in> objects_of s"
      using target_mem by simp
      show ?thesis
      using inj_onD[OF source_unique same_version old_mem target_old] .
    qed
  qed
  have target_unique: "unique_versions s'"
  unfolding target
  using add_effect_preserves_unique_versions[OF valid] .
  show ?thesis using target_unique immutable by blast
qed

theorem hypothesize_effect_preserves_catalog:
  assumes concrete:
    "hypothesize_effect_step G s relation_catalog effect_catalog new_eff payload
      next_snapshot_id s' effect_catalog'"
  shows
    "effect_catalog_well_formed s' effect_catalog' \<and>
      effect_catalog' (version_id_of new_eff) = Some payload"
proof -
  from concrete have valid:
    "effect_input_valid G s relation_catalog effect_catalog new_eff payload
      next_snapshot_id"

```

```

    and target: "s' = add_effect_hypothesis s new_eff next_snapshot_id"
    and catalog_target:
      "effect_catalog' = extend_effect_catalog effect_catalog new_eff payload"
    by (auto simp: hypothesize_effect_step_def)
  from valid have source_catalog:
    "effect_catalog_well_formed s effect_catalog"
    and payload_valid: "effect_payload_valid new_eff payload"
    and fresh_key: "effect_catalog (version_id_of new_eff) = None"
    by (auto simp: effect_input_valid_def)
  have target_catalog: "effect_catalog_well_formed s' effect_catalog'"
    unfolding effect_catalog_well_formed_def target catalog_target
    extend_effect_catalog_def
    using source_catalog payload_valid fresh_key
    unfolding effect_catalog_well_formed_def
    by auto
  show ?thesis
    using target_catalog catalog_target
    by (simp add: extend_effect_catalog_def)
qed

theorem hypothesize_effect_preserves_references_time_context:
  assumes concrete:
    "hypothesize_effect_step G s relation_catalog effect_catalog new_eff payload
      next_snapshot_id s' effect_catalog'"
  shows
    "\<forall>obj\<in>objects_of s'.
      references_complete s' obj \<and>
      evidence_well_formed s' obj \<and>
      time_well_formed obj \<and>
      perspective_well_formed s' obj \<and>
      context_well_formed s' obj"
  using hypothesize_effect_preserves_state_well_formed[OF concrete]
  unfolding state_well_formed_def
  by blast

theorem hypothesize_effect_preserves_evidence_integrity:
  assumes concrete:
    "hypothesize_effect_step G s relation_catalog effect_catalog new_eff payload
      next_snapshot_id s' effect_catalog'"
  shows "\<forall>obj\<in>objects_of s'. evidence_well_formed s' obj"
  using hypothesize_effect_preserves_state_well_formed[OF concrete]
  unfolding state_well_formed_def
  by blast

theorem hypothesize_effect_refines_generic_step:
  assumes concrete:
    "hypothesize_effect_step G s relation_catalog effect_catalog new_eff payload
      next_snapshot_id s' effect_catalog'"
  shows "step G s HypothesizeEffect s'"
proof -
  from concrete have source_wf: "state_well_formed G s"
    and allowed: "HypothesizeEffect \<in> allowed_operations_of G"
    by (auto simp: hypothesize_effect_step_def effect_input_valid_def)
  have target_wf: "state_well_formed G s'"
    using hypothesize_effect_preserves_state_well_formed[OF concrete] .
  have immutable: "immutable_preserved s s'"
    using hypothesize_effect_preserves_identity[OF concrete] by blast
  show ?thesis
    unfolding step_def
    using source_wf target_wf allowed immutable by blast
qed
end

```

Beweisstatus dieser D.7-Fassung: kernelgeprüft. Der ausgeführte Clean-Build startete am 8. August 2026 um 19:19:25 CEST und endete um 19:19:55 CEST. Alle sieben Theorien wurden zu 100 % verarbeitet; die Session benötigte 25,284 s, der Gesamtlauf 29 s und endete mit Exit-Code 0. Der T4-Quellhash lautet 2eb8e1d262cb3790f6c446839fb6809de3c2837d20705925507e272c860e2fc4; ROOT besitzt SHA-256 210a567ba560a9c2ed66f1ccdaa8ec355a2b4dad8504afc4708369f1fe3ed7b2; das unveränderte Buildlog besitzt SHA-256 7a49a99e15790d7324def03c5078600e4cb75bd2715b9439380fe711e85daa06. Belegt sind

Parser-, Typ- und Kernelakzeptanz der formulierten konditionalen Semantik sowie ihrer Erhaltungs- und Refinementsätze. Nicht belegt sind empirische Richtigkeit, Vollständigkeit der Wirkungstaxonomie, Existenz gültiger Eingaben in jeder Domäne, allgemeine Konservativität oder Produktivengine-Äquivalenz.

## D.8 Konkrete T5-Semantik und Erhaltung für IntegrateEvidence

T5 schließt die erste lifecycle-verändernde Kernoperation. Die Semantik ist konstruktiv: Aus einem wohlgeformten Ausgangszustand, einem aktiven Effect-Katalog, disjunkter Aktiv/Historie-Struktur, konsistenten Objekt- und Payload-Archiven sowie einem Evidenz-Ledger wird der Nachzustand berechnet. Der Vorgänger bleibt als vollständiges unverändertes Objekt auffindbar; das aktive Objekt erhält eine frische `version_id` bei gleicher `object_id`. Damit wird Copy-on-Write erstmals nicht nur als Engine-Prinzip, sondern als beweisbare Operationssemantik realisiert.

Die Assessment-Semantik unterscheidet `EvidenceSupports`, `EvidenceDifferentiates` und `EvidenceContradicts`. Evidenzreferenzen müssen nichtleer, aktiv und auflösbar sein und dürfen weder die neue noch die unmittelbare Vorgängerversion zur Selbststützung verwenden. Unsicherheit bleibt in  $[0,1]$ , ihre absolute Änderung ist durch `evidence_strength` begrenzt. Lifecycle-Fortschreibung ist abgeleitet: Hypothesis plus stützende Evidenz ergibt `Supported`; `Supported` kann nur bei erfüllter Evidenz- und Unsicherheitsschwelle `Validated` werden; widersprechende Evidenz ergibt `Rejected`; differenzierende Evidenz muss eine inhaltliche Ziel- oder Vektoränderung tragen.

```
theory UZA_0_9_1_Integrate_Evidence
  imports UZA_0_9_1_Hypothesize_Effect
begin

section \<open>T5 IntegrateEvidence: versioned evidence integration\<close>

text \<open>
  T5 does not mutate an active effect version. It replaces one active
  K_EFF hypothesis by one fresh version of the same logical object, archives
  the complete predecessor object and its effect payload, accumulates typed
  evidence, and derives the successor lifecycle from the evidence
  disposition. The construction therefore makes copy-on-write, provenance,
  uncertainty movement, and the prohibition of self-support explicit.

  The operation is deliberately restricted to Hypothesis and Supported.
  Evidence applied to an already Validated effect belongs to T13
  ReviseObject. This boundary prevents T5 from silently serving as a general
  revision operator.
\<close>

datatype evidence_disposition =
  EvidenceSupports | EvidenceDifferentiates | EvidenceContradicts

record evidence_assessment =
  evidence_subject_of :: object_id
  evidence_prior_version_of :: version_id
  evidence_refs_of :: "version_id set"
  evidence_disposition_of :: evidence_disposition
  evidence_strength_of :: real
  evidence_post_uncertainty_of :: real
  evidence_post_target_of :: "version_id option"
  evidence_post_vector_of :: "effect_vector option"
  evidence_assessment_model_of :: model_id
  validation_uncertainty_max_of :: real
  validation_evidence_min_of :: nat

type_synonym object_archive = "version_id \<rightarrow> uza_object"
type_synonym effect_payload_archive =
  "version_id \<rightarrow> effect_hypothesis_payload"
type_synonym evidence_ledger =
  "version_id \<rightarrow> evidence_assessment"

definition active_history_disjoint :: "snapshot \<rightarrow> bool" where
  "active_history_disjoint s \<longleftarrow>
    version_ids s \<inter> historical_versions_of s = {}"

definition object_archive_well_formed ::
  "snapshot \<rightarrow> object_archive \<rightarrow> bool" where
```

```

"object_archive_well_formed s archive \<longlefttrightarrow>
  (\<forall>v obj. archive v = Some obj \<longrightarrow>
    version_id_of obj = v \<and>
    v \<in> historical_versions_of s \<and>
    v \<notin> version_ids s)"

definition resolved_version_object ::
  "snapshot \<Rightarrow> object_archive \<Rightarrow> version_id \<Rightarrow>
    uza_object \<Rightarrow> bool" where
  "resolved_version_object s archive v obj \<longlefttrightarrow>
    (obj \<in> objects_of s \<and> version_id_of obj = v) \<or>
    archive v = Some obj"

definition effect_payload_archive_well_formed ::
  "object_archive \<Rightarrow> effect_payload_archive \<Rightarrow> bool" where
  "effect_payload_archive_well_formed archive payload_archive \<longlefttrightarrow>
    (\<forall>v payload. payload_archive v = Some payload \<longrightarrow>
      (\<exists>eff. archive v = Some eff \<and>
        effect_payload_valid eff payload))"

definition evidence_entry_well_formed ::
  "snapshot \<Rightarrow> object_archive \<Rightarrow> version_id \<Rightarrow>
    evidence_assessment \<Rightarrow> bool" where
  "evidence_entry_well_formed s archive v assessment \<longlefttrightarrow>
    (\<exists>eff. resolved_version_object s archive v eff \<and>
      kind_of eff = K_EFF \<and>
      evidence_subject_of assessment = object_id_of eff \<and>
      previous_version_of eff = Some (evidence_prior_version_of assessment) \<and>
      evidence_refs_of assessment \<noteq> {} \<and>
      evidence_refs_of assessment \<subsetq> evidence_of eff \<and>
      evidence_refs_of assessment
        \<subsetq> version_ids s \<union> historical_versions_of s \<and>
      v \<notin> evidence_refs_of assessment \<and>
      evidence_prior_version_of assessment
        \<notin> evidence_refs_of assessment \<and>
      0 \<le> evidence_strength_of assessment \<and>
      evidence_strength_of assessment \<le> 1 \<and>
      0 \<le> evidence_post_uncertainty_of assessment \<and>
      evidence_post_uncertainty_of assessment \<le> 1 \<and>
      0 \<le> validation_uncertainty_max_of assessment \<and>
      validation_uncertainty_max_of assessment \<le> 1 \<and>
      0 < validation_evidence_min_of assessment \<and>
      evidence_assessment_model_of assessment \<in> models_of s)"

definition evidence_ledger_well_formed ::
  "snapshot \<Rightarrow> object_archive \<Rightarrow> evidence_ledger \<Rightarrow> bool" where
  "evidence_ledger_well_formed s archive ledger \<longlefttrightarrow>
    (\<forall>v assessment. ledger v = Some assessment \<longrightarrow>
      evidence_entry_well_formed s archive v assessment)"

definition active_evidence_source ::
  "snapshot \<Rightarrow> object_id \<Rightarrow> version_id \<Rightarrow> bool" where
  "active_evidence_source s subject v \<longlefttrightarrow>
    (\<exists>ev\<in>objects_of s.
      version_id_of ev = v \<and>
      object_id_of ev \<noteq> subject \<and>
      lifecycle_of ev \<notin> {Rejected, Archived})"

definition admissible_post_target ::
  "snapshot \<Rightarrow> object_id \<Rightarrow> version_id option \<Rightarrow> bool" where
  "admissible_post_target s subject target \<longlefttrightarrow>
    (case target of
      None \<Rightarrow> True
    | Some v \<Rightarrow>
      (\<exists>obj\<in>objects_of s.
        version_id_of obj = v \<and>
        object_id_of obj \<noteq> subject \<and>
        lifecycle_of obj \<notin> {Revised, Rejected, Archived}))"

definition assessment_bounds :: "evidence_assessment \<Rightarrow> bool" where
  "assessment_bounds assessment \<longlefttrightarrow>
    0 \<le> evidence_strength_of assessment \<and>
    evidence_strength_of assessment \<le> 1 \<and>
    0 \<le> evidence_post_uncertainty_of assessment \<and>
    evidence_post_uncertainty_of assessment \<le> 1 \<and>
    0 \<le> validation_uncertainty_max_of assessment \<and>
    validation_uncertainty_max_of assessment \<le> 1 \<and>
    0 < validation_evidence_min_of assessment"

definition assessment_change_valid ::

```

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"effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow> bool" where
"assessment_change_valid prior assessment \<longleftarrowrightarrow>
  abs (evidence_post_uncertainty_of assessment -
    hypothesis_uncertainty_of prior)
  \<le> evidence_strength_of assessment \<and>
(case evidence_disposition_of assessment of
  EvidenceSupports \<Rightarrow>
    evidence_post_target_of assessment = hypothesis_target_of prior \<and>
    evidence_post_vector_of assessment = hypothesis_vector_of prior \<and>
    evidence_post_uncertainty_of assessment
      \<le> hypothesis_uncertainty_of prior
  | EvidenceDifferentiates \<Rightarrow>
    evidence_post_target_of assessment \<noteq> hypothesis_target_of prior \<or>
    evidence_post_vector_of assessment \<noteq> hypothesis_vector_of prior
  | EvidenceContradicts \<Rightarrow>
    evidence_post_target_of assessment = hypothesis_target_of prior \<and>
    evidence_post_vector_of assessment = hypothesis_vector_of prior \<and>
    hypothesis_uncertainty_of prior
      \<le> evidence_post_uncertainty_of assessment)"

definition validation_ready ::
  "uza_object \<Rightarrow> evidence_assessment \<Rightarrow> bool" where
  "validation_ready old_eff assessment \<longleftarrowrightarrow>
    evidence_post_uncertainty_of assessment
    \<le> validation_uncertainty_max_of assessment \<and>
    card (evidence_of old_eff \<union> evidence_refs_of assessment)
      \<ge> validation_evidence_min_of assessment"

definition integrated_lifecycle ::
  "uza_object \<Rightarrow> evidence_assessment \<Rightarrow> lifecycle_status" where
  "integrated_lifecycle old_eff assessment =
    (case evidence_disposition_of assessment of
      EvidenceSupports \<Rightarrow>
        (if lifecycle_of old_eff = Hypothesis then Supported
          else if validation_ready old_eff assessment then Validated
          else Supported)
    | EvidenceDifferentiates \<Rightarrow> lifecycle_of old_eff
    | EvidenceContradicts \<Rightarrow> Rejected)"

definition integrated_effect_payload ::
  "effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow>
    effect_hypothesis_payload" where
  "integrated_effect_payload prior assessment =
    prior\<lparr>
      hypothesis_target_of := evidence_post_target_of assessment,
      hypothesis_vector_of := evidence_post_vector_of assessment,
      hypothesis_uncertainty_of := evidence_post_uncertainty_of assessment
    \<rparr>"

definition option_ref_set :: "version_id option \<Rightarrow> version_id set" where
  "option_ref_set target = (case target of None \<Rightarrow> {} | Some v \<Rightarrow> {v})"

definition integrated_effect_object ::
  "uza_object \<Rightarrow> evidence_assessment \<Rightarrow> version_id \<Rightarrow>
    time_point \<Rightarrow> uza_object" where
  "integrated_effect_object old_eff assessment new_version integration_time =
    old_eff\<lparr>
      version_id_of := new_version,
      lifecycle_of := integrated_lifecycle old_eff assessment,
      previous_version_of := Some (version_id_of old_eff),
      references_of := references_of old_eff
        \<union> evidence_refs_of assessment
        \<union> {version_id_of old_eff}
        \<union> option_ref_set (evidence_post_target_of assessment),
      evidence_of := evidence_of old_eff \<union> evidence_refs_of assessment,
      valid_from_of := integration_time,
      valid_to_of := None,
      target_ref_of := evidence_post_target_of assessment,
      basis_refs_of := basis_refs_of old_eff
        \<union> evidence_refs_of assessment
        \<union> {version_id_of old_eff}
        \<union> option_ref_set (evidence_post_target_of assessment),
      object_status_of :=
        (if evidence_disposition_of assessment = EvidenceContradicts
          then Some ModelFalsified else Some Applicable)
    \<rparr>"

definition replace_effect_version ::
  "snapshot \<Rightarrow> uza_object \<Rightarrow> uza_object \<Rightarrow> nat
    \<Rightarrow> snapshot" where

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"replace_effect_version s old_eff new_eff next_snapshot_id =
  s\<lparr>
    snapshot_id_of := next_snapshot_id,
    objects_of := insert new_eff (objects_of s - {old_eff}),
    historical_versions_of :=
      insert (version_id_of old_eff) (historical_versions_of s)
  \<rparr>"

definition replace_active_effect_payload ::
  "effect_catalog \<Rightarrow> version_id \<Rightarrow> version_id \<Rightarrow>
    effect_hypothesis_payload \<Rightarrow> effect_catalog" where
  "replace_active_effect_payload catalog old_version new_version payload =
    (catalog(old_version := None))(new_version \<mapsto> payload)"

definition extend_object_archive ::
  "object_archive \<Rightarrow> uza_object \<Rightarrow> object_archive" where
  "extend_object_archive archive old_eff =
    archive(version_id_of old_eff \<mapsto> old_eff)"

definition extend_effect_payload_archive ::
  "effect_payload_archive \<Rightarrow> uza_object \<Rightarrow>
    effect_hypothesis_payload \<Rightarrow> effect_payload_archive" where
  "extend_effect_payload_archive archive old_eff prior =
    archive(version_id_of old_eff \<mapsto> prior)"

definition extend_evidence_ledger ::
  "evidence_ledger \<Rightarrow> version_id \<Rightarrow> evidence_assessment
    \<Rightarrow> evidence_ledger" where
  "extend_evidence_ledger ledger new_version assessment =
    ledger(new_version \<mapsto> assessment)"

definition integrate_evidence_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> effect_catalog \<Rightarrow>
    object_archive \<Rightarrow> effect_payload_archive \<Rightarrow>
    evidence_ledger \<Rightarrow> uza_object \<Rightarrow>
    effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow>
    version_id \<Rightarrow> time_point \<Rightarrow> nat \<Rightarrow> bool" where
  "integrate_evidence_input_valid G s catalog object_archive payload_archive
    ledger old_eff prior assessment new_version integration_time
    next_snapshot_id \<longleftarrowrightarrow>
    state_well_formed G s \<and>
    active_history_disjoint s \<and>
    effect_catalog_well_formed s catalog \<and>
    object_archive_well_formed s object_archive \<and>
    effect_payload_archive_well_formed object_archive payload_archive \<and>
    evidence_ledger_well_formed s object_archive ledger \<and>
    IntegrateEvidence \<in> allowed_operations_of G \<and>
    old_eff \<in> objects_of s \<and>
    kind_of old_eff = K_EFF \<and>
    lifecycle_of old_eff \<in> {Hypothesis, Supported} \<and>
    catalog (version_id_of old_eff) = Some prior \<and>
    effect_payload_valid old_eff prior \<and>
    present_ref s (effect_origin_of old_eff) \<and>
    object_archive (version_id_of old_eff) = None \<and>
    payload_archive (version_id_of old_eff) = None \<and>
    new_version \<notin> version_ids s \<union> historical_versions_of s \<and>
    catalog new_version = None \<and>
    object_archive new_version = None \<and>
    payload_archive new_version = None \<and>
    ledger new_version = None \<and>
    evidence_subject_of assessment = object_id_of old_eff \<and>
    evidence_prior_version_of assessment = version_id_of old_eff \<and>
    evidence_refs_of assessment \<noteq> {} \<and>
    (\<forall>v\<in>evidence_refs_of assessment.
      active_evidence_source s (object_id_of old_eff) v) \<and>
    evidence_assessment_model_of assessment \<in> models_of s \<and>
    assessment_bounds assessment \<and>
    assessment_change_valid prior assessment \<and>
    admissible_post_target s (object_id_of old_eff)
      (evidence_post_target_of assessment) \<and>
    (case evidence_post_vector_of assessment of
      None \<Rightarrow> True
    | Some vector \<Rightarrow> effect_vector_valid vector) \<and>
    valid_from_of old_eff \<le> integration_time \<and>
    snapshot_id_of s < next_snapshot_id"

definition integrate_evidence_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> effect_catalog \<Rightarrow>
    object_archive \<Rightarrow> effect_payload_archive \<Rightarrow>
    evidence_ledger \<Rightarrow> uza_object \<Rightarrow>"

```



```

effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow>
version_id \<Rightarrow> time_point \<Rightarrow> nat \<Rightarrow>
snapshot \<Rightarrow> effect_catalog \<Rightarrow> object_archive \<Rightarrow>
effect_payload_archive \<Rightarrow> evidence_ledger \<Rightarrow> bool" where
"integrate_evidence_step G s catalog object_archive payload_archive ledger
  old_eff prior assessment new_version integration_time next_snapshot_id
  s' catalog' object_archive' payload_archive' ledger' \<longlefttrightarrow>
integrate_evidence_input_valid G s catalog object_archive payload_archive
  ledger old_eff prior assessment new_version integration_time
  next_snapshot_id \<and>
(let new_eff = integrated_effect_object old_eff assessment new_version
  integration_time;
  new_payload = integrated_effect_payload prior assessment
  in s' = replace_effect_version s old_eff new_eff next_snapshot_id \<and>
  catalog' = replace_active_effect_payload catalog
    (version_id_of old_eff) new_version new_payload \<and>
  object_archive' = extend_object_archive object_archive old_eff \<and>
  payload_archive' = extend_effect_payload_archive payload_archive
    old_eff prior \<and>
  ledger' = extend_evidence_ledger ledger new_version assessment)"

subsection \<open>Constructor equations and exact delta\<close>

lemma replace_effect_simps [simp]:
  "snapshot_id_of (replace_effect_version s old_eff new_eff next_id) = next_id"
  "objects_of (replace_effect_version s old_eff new_eff next_id) =
    insert new_eff (objects_of s - {old_eff})"
  "historical_versions_of
    (replace_effect_version s old_eff new_eff next_id) =
    insert (version_id_of old_eff) (historical_versions_of s)"
  "perspectives_of (replace_effect_version s old_eff new_eff next_id) =
    perspectives_of s"
  "contexts_of (replace_effect_version s old_eff new_eff next_id) = contexts_of s"
  "models_of (replace_effect_version s old_eff new_eff next_id) = models_of s"
  "authorities_of (replace_effect_version s old_eff new_eff next_id) =
    authorities_of s"
  "contains_edges_of (replace_effect_version s old_eff new_eff next_id) =
    contains_edges_of s"
  "projections_of (replace_effect_version s old_eff new_eff next_id) =
    projections_of s"
  "couplings_of (replace_effect_version s old_eff new_eff next_id) =
    couplings_of s"
  "transmissions_of (replace_effect_version s old_eff new_eff next_id) =
    transmissions_of s"
  "boundaries_of (replace_effect_version s old_eff new_eff next_id) =
    boundaries_of s"
  by (simp_all add: replace_effect_version_def)

lemma integrated_effect_simps [simp]:
  "object_id_of (integrated_effect_object old assessment new_v at_time) =
    object_id_of old"
  "version_id_of (integrated_effect_object old assessment new_v at_time) = new_v"
  "kind_of (integrated_effect_object old assessment new_v at_time) = kind_of old"
  "lifecycle_of (integrated_effect_object old assessment new_v at_time) =
    integrated_lifecycle old assessment"
  "previous_version_of
    (integrated_effect_object old assessment new_v at_time) =
    Some (version_id_of old)"
  "references_of (integrated_effect_object old assessment new_v at_time) =
    references_of old \<union> evidence_refs_of assessment \<union>
    {version_id_of old} \<union>
    option_ref_set (evidence_post_target_of assessment)"
  "evidence_of (integrated_effect_object old assessment new_v at_time) =
    evidence_of old \<union> evidence_refs_of assessment"
  "perspective_of (integrated_effect_object old assessment new_v at_time) =
    perspective_of old"
  "context_of (integrated_effect_object old assessment new_v at_time) =
    context_of old"
  "model_of (integrated_effect_object old assessment new_v at_time) = model_of old"
  "source_ref_of (integrated_effect_object old assessment new_v at_time) =
    source_ref_of old"
  "target_ref_of (integrated_effect_object old assessment new_v at_time) =
    evidence_post_target_of assessment"
  "effect_origin_of (integrated_effect_object old assessment new_v at_time) =
    effect_origin_of old"
  "valid_from_of (integrated_effect_object old assessment new_v at_time) = at_time"
  "valid_to_of (integrated_effect_object old assessment new_v at_time) = None"
  "basis_refs_of (integrated_effect_object old assessment new_v at_time) =
    basis_refs_of old \<union> evidence_refs_of assessment \<union>
    {version_id_of old} \<union>

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```

    option_ref_set (evidence_post_target_of assessment)"
  by (simp_all add: integrated_effect_object_def)

lemma integrated_payload_simps [simp]:
  "hypothesis_cause_of (integrated_effect_payload prior assessment) =
    hypothesis_cause_of prior"
  "hypothesis_target_of (integrated_effect_payload prior assessment) =
    evidence_post_target_of assessment"
  "hypothesis_vector_of (integrated_effect_payload prior assessment) =
    evidence_post_vector_of assessment"
  "hypothesis_uncertainty_of (integrated_effect_payload prior assessment) =
    evidence_post_uncertainty_of assessment"
  "hypothesis_profile_model_of (integrated_effect_payload prior assessment) =
    hypothesis_profile_model_of prior"
  by (simp_all add: integrated_effect_payload_def)

lemma integrate_evidence_exact_delta:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  defines "new_eff \

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subsection \open>Lifecycle, uncertainty, and provenance properties\<close>

theorem integrate_evidence_lifecycle_classification:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  shows
    "(evidence_disposition_of assessment = EvidenceContradicts
      \<longrightarrow> integrated_lifecycle old_eff assessment = Rejected) \<and>
      (evidence_disposition_of assessment = EvidenceDifferentiates
        \<longrightarrow> integrated_lifecycle old_eff assessment = lifecycle_of old_eff) \<and>
      (evidence_disposition_of assessment = EvidenceSupports \<and>
        lifecycle_of old_eff = Hypothesis
          \<longrightarrow> integrated_lifecycle old_eff assessment = Supported) \<and>
      (integrated_lifecycle old_eff assessment = Validated
        \<longrightarrow> lifecycle_of old_eff = Supported \<and>
          evidence_disposition_of assessment = EvidenceSupports \<and>
            validation_ready old_eff assessment)"
  proof -
    from concrete have source_lifecycle:
      "lifecycle_of old_eff \<in> {Hypothesis, Supported}"
    by (auto simp: integrate_evidence_step_def
      integrate_evidence_input_valid_def)
    show ?thesis
      using source_lifecycle
      by (cases "evidence_disposition_of assessment";
        auto simp: integrated_lifecycle_def split: if_splits)
  qed

theorem integrate_evidence_uncertainty_controlled:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  shows
    "0 \<le> evidence_post_uncertainty_of assessment \<and>
      evidence_post_uncertainty_of assessment \<le> 1 \<and>
      abs (evidence_post_uncertainty_of assessment -
        hypothesis_uncertainty_of prior)
        \<le> evidence_strength_of assessment \<and>
      (evidence_disposition_of assessment = EvidenceSupports
        \<longrightarrow> evidence_post_uncertainty_of assessment
          \<le> hypothesis_uncertainty_of prior) \<and>
      (evidence_disposition_of assessment = EvidenceContradicts
        \<longrightarrow> hypothesis_uncertainty_of prior
          \<le> evidence_post_uncertainty_of assessment)"
  using concrete
  unfolding integrate_evidence_step_def integrate_evidence_input_valid_def
    assessment_bounds_def assessment_change_valid_def
  by (cases "evidence_disposition_of assessment"; auto)

theorem integrate_evidence_lineage_and_no_self_support:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  defines "new_eff \<equiv> integrated_effect_object old_eff assessment new_version
    integration_time"
  shows
    "previous_version_of new_eff = Some (version_id_of old_eff) \<and>
      evidence_refs_of assessment \<noteq> {} \<and>
      evidence_refs_of assessment \<subsetq> evidence_of new_eff \<and>
      new_version \<notin> evidence_refs_of assessment \<and>
      version_id_of old_eff \<notin> evidence_refs_of assessment \<and>
      (\<forall>v\<in>evidence_refs_of assessment.
        \<exists>ev\<in>objects_of s.
          version_id_of ev = v \<and>
          object_id_of ev \<noteq> object_id_of old_eff \<and>
          lifecycle_of ev \<notin> {Rejected, Archived})"
  proof -
    from concrete have valid:
      "integrate_evidence_input_valid G s catalog object_archive payload_archive
        ledger old_eff prior assessment new_version integration_time
        next_snapshot_id"
    by (simp add: integrate_evidence_step_def)
    from valid have refs_nonempty: "evidence_refs_of assessment \<noteq> {}"
    and active:
      "\<forall>v\<in>evidence_refs_of assessment.
        active_evidence_source s (object_id_of old_eff) v"
  
```

```

and fresh:
  "new_version \<notin> version_ids s \<union> historical_versions_of s"
  and old_mem: "old_eff \<in> objects_of s"
  and source_wf: "state_well_formed G s"
  unfolding integrate_evidence_input_valid_def by blast+
  have source_unique: "inj_on version_id_of (objects_of s)"
  using source_wf unfolding state_well_formed_def unique_versions_def by blast
  have prior_not: "version_id_of old_eff \<notin> evidence_refs_of assessment"
  proof
    assume prior_in: "version_id_of old_eff \<in> evidence_refs_of assessment"
    from active prior_in obtain ev where ev_mem: "ev \<in> objects_of s"
      and same_version: "version_id_of ev = version_id_of old_eff"
      and different_object: "object_id_of ev \<noteq> object_id_of old_eff"
    unfolding active_evidence_source_def by blast
    have "ev = old_eff"
      using inj_onD[OF source_unique same_version ev_mem old_mem] .
    then show False using different_object by simp
  qed
  have new_not: "new_version \<notin> evidence_refs_of assessment"
  proof
    assume "new_version \<in> evidence_refs_of assessment"
    then obtain ev where "ev \<in> objects_of s" "version_id_of ev = new_version"
      using active unfolding active_evidence_source_def by blast
    then have "new_version \<in> version_ids s"
      unfolding version_ids_def by blast
    then show False using fresh by blast
  qed
  show ?thesis
    unfolding new_eff_def
    using refs_nonempty active prior_not new_not
    by (auto simp: active_evidence_source_def)
qed

subsection \<open>Reference-universe monotonicity\<close>

lemma replace_effect_version_universe_mono:
  assumes old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  shows
    "version_ids s \<union> historical_versions_of s \<subteq>
      version_ids (replace_effect_version s old_eff new_eff next_id)
      \<union> historical_versions_of
        (replace_effect_version s old_eff new_eff next_id)"
  proof
    fix v
    assume v_mem: "v \<in> version_ids s \<union> historical_versions_of s"
    show "v \<in> version_ids (replace_effect_version s old_eff new_eff next_id)
      \<union> historical_versions_of
        (replace_effect_version s old_eff new_eff next_id)"
    proof (cases "v = version_id_of old_eff")
      case is_old: True
      then show ?thesis by simp
    next
      case not_old: False
      show ?thesis
        proof (cases "v \<in> historical_versions_of s")
          case in_history: True
          then show ?thesis by simp
        next
          case not_in_history: False
          with v_mem obtain obj where obj_mem: "obj \<in> objects_of s"
            and obj_v: "version_id_of obj = v"
            unfolding version_ids_def by blast
          have "obj \<noteq> old_eff"
            proof
              assume "obj = old_eff"
              then have "v = version_id_of old_eff" using obj_v by simp
              then show False using not_old by contradiction
            qed
          then have "obj \<in> objects_of s - {old_eff}" using obj_mem by blast
          then have "v \<in> version_ids
            (replace_effect_version s old_eff new_eff next_id)"
            unfolding version_ids_def using obj_v by auto
          then show ?thesis by blast
        qed
      qed
    qed
  qed

lemma replace_effect_present_ref_mono:

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assumes old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  and present: "present_ref s ref"
shows "present_ref
  (replace_effect_version s old_eff new_eff next_id) ref"
using replace_effect_version_universe_mono[OF old_mem fresh] present
unfolding present_ref_def
by (cases ref; auto)

lemma replace_effect_references_mono:
  assumes old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  and refs: "references_complete s obj"
shows "references_complete
  (replace_effect_version s old_eff new_eff next_id) obj"
using replace_effect_version_universe_mono[OF old_mem fresh] refs
unfolding references_complete_def by blast

lemma replace_effect_evidence_mono:
  assumes old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  and evidence: "evidence_well_formed s obj"
shows "evidence_well_formed
  (replace_effect_version s old_eff new_eff next_id) obj"
using replace_effect_version_universe_mono[OF old_mem fresh] evidence
unfolding evidence_well_formed_def evidence_references_complete_def by blast

lemma replace_effect_preserves_active_history_disjoint:
  assumes source_disjoint: "active_history_disjoint s"
  and source_unique: "unique_versions s"
  and old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
shows "active_history_disjoint
  (replace_effect_version s old_eff new_eff next_id)"
proof -
  have old_version_active: "version_id_of old_eff \<in> version_ids s"
  unfolding version_ids_def using old_mem by blast
  have versions_different:
    "version_id_of old_eff \<noteq> version_id_of new_eff"
  proof
    assume same: "version_id_of old_eff = version_id_of new_eff"
    from fresh have "version_id_of new_eff \<notin> version_ids s" by blast
    with old_version_active same show False by simp
  qed
  show ?thesis
  using source_disjoint source_unique old_mem fresh versions_different
  unfolding active_history_disjoint_def unique_versions_def version_ids_def
  replace_effect_version_def
  by (auto simp: inj_on_def)
qed

lemma evidence_entry_replace_mono:
  assumes old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  and old_not_archived: "archive (version_id_of old_eff) = None"
  and entry: "evidence_entry_well_formed s archive v assessment"
shows "evidence_entry_well_formed
  (replace_effect_version s old_eff new_eff next_id)
  (extend_object_archive archive old_eff) v assessment"
proof -
  let ?target = "replace_effect_version s old_eff new_eff next_id"
  let ?archive = "extend_object_archive archive old_eff"
  from entry obtain eff where resolved:
    "resolved_version_object s archive v eff"
  and remaining:
    "kind_of eff = K_EFF \<and>
    evidence_subject_of assessment = object_id_of eff \<and>
    previous_version_of eff = Some (evidence_prior_version_of assessment) \<and>
    evidence_refs_of assessment \<noteq> {} \<and>
    evidence_refs_of assessment \<subsetq> evidence_of eff \<and>
    evidence_refs_of assessment
      \<subsetq> version_ids s \<union> historical_versions_of s \<and>
    v \<notin> evidence_refs_of assessment \<and>
    evidence_prior_version_of assessment
      \<notin> evidence_refs_of assessment \<and>"

```

```

    0 \<le> evidence_strength_of assessment \<and>
    evidence_strength_of assessment \<le> 1 \<and>
    0 \<le> evidence_post_uncertainty_of assessment \<and>
    evidence_post_uncertainty_of assessment \<le> 1 \<and>
    0 \<le> validation_uncertainty_max_of assessment \<and>
    validation_uncertainty_max_of assessment \<le> 1 \<and>
    0 < validation_evidence_min_of assessment \<and>
    evidence_assessment_model_of assessment \<in> models_of s"
  unfolding evidence_entry_well_formed_def by blast
have resolved_target:
  "resolved_version_object ?target ?archive v eff"
proof (cases "eff \<in> objects_of s \<and> version_id_of eff = v")
  case active: True
  show ?thesis
  proof (cases "eff = old_eff")
    case same_old: True
    from active have eff_version: "version_id_of eff = v" by blast
    have "?archive v = Some eff"
      unfolding extend_object_archive_def
      using same_old eff_version by simp
    then show ?thesis
      unfolding resolved_version_object_def by blast
  next
  case different_old: False
  with active have "eff \<in> objects_of s - {old_eff}" by blast
  with active show ?thesis
    unfolding resolved_version_object_def by auto
qed
next
case not_active: False
with resolved have archived: "archive v = Some eff"
  unfolding resolved_version_object_def by blast
have key_different: "v \<noteq> version_id_of old_eff"
proof
  assume "v = version_id_of old_eff"
  with archived old_not_archived show False by simp
qed
show ?thesis
  unfolding resolved_version_object_def extend_object_archive_def
  using archived key_different by simp
qed
have refs_target:
  "evidence_refs_of assessment
  \<subsetq> version_ids ?target \<union> historical_versions_of ?target"
  using remaining replace_effect_version_universe_mono[OF old_mem fresh]
  by blast
show ?thesis
  unfolding evidence_entry_well_formed_def
proof (rule exI[of _ eff])
  show "resolved_version_object ?target ?archive v eff \<and>
  kind_of eff = K_EFF \<and>
  evidence_subject_of assessment = object_id_of eff \<and>
  previous_version_of eff = Some (evidence_prior_version_of assessment) \<and>
  evidence_refs_of assessment \<noteq> {} \<and>
  evidence_refs_of assessment \<subsetq> evidence_of eff \<and>
  evidence_refs_of assessment
  \<subsetq> version_ids ?target \<union> historical_versions_of ?target \<and>
  v \<notin> evidence_refs_of assessment \<and>
  evidence_prior_version_of assessment
  \<notin> evidence_refs_of assessment \<and>
  0 \<le> evidence_strength_of assessment \<and>
  evidence_strength_of assessment \<le> 1 \<and>
  0 \<le> evidence_post_uncertainty_of assessment \<and>
  evidence_post_uncertainty_of assessment \<le> 1 \<and>
  0 \<le> validation_uncertainty_max_of assessment \<and>
  validation_uncertainty_max_of assessment \<le> 1 \<and>
  0 < validation_evidence_min_of assessment \<and>
  evidence_assessment_model_of assessment \<in> models_of ?target"
  using resolved_target remaining refs_target
  by (simp only: replace_effect_simps; blast)
qed
qed

subsection \<open>Successor state and audit-store preservation\<close>

lemma integrate_evidence_payload_valid:
  assumes valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
    ledger old_eff prior assessment new_version integration_time
    next_snapshot_id"

```

```

shows "effect_payload_valid
  (integrated_effect_object old_eff assessment new_version integration_time)
  (integrated_effect_payload prior assessment)"
using valid
unfolding integrate_evidence_input_valid_def assessment_bounds_def
  effect_payload_valid_def
by (cases "evidence_post_vector_of assessment"; auto)

lemma integrate_evidence_new_object_conditions:
  assumes valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
      ledger old_eff prior assessment new_version integration_time
      next_snapshot_id"
  defines "new_eff \<equiv> integrated_effect_object old_eff assessment new_version
    integration_time"
  shows
    "kind_of new_eff \<in> supported_kinds_of G \<and>
      references_complete
        (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
      evidence_well_formed
        (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
      time_well_formed new_eff \<and>
      relation_well_formed
        (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
      effect_well_formed
        (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
      perspective_well_formed
        (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
      context_well_formed
        (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
      relevance_well_formed
        (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
      orientation_well_formed
        (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
      action_well_formed
        (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
      feedback_well_formed
        (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff"
proof -
  let ?target = "replace_effect_version s old_eff new_eff next_snapshot_id"
  from valid have source_wf: "state_well_formed G s"
  and old_mem: "old_eff \<in> objects_of s"
  and old_kind: "kind_of old_eff = K_EFF"
  and old_lifecycle: "lifecycle_of old_eff \<in> {Hypothesis, Supported}"
  and fresh: "new_version \<notin> version_ids s \<union> historical_versions_of s"
  and origin: "present_ref s (effect_origin_of old_eff)"
  and evidence_nonempty: "evidence_refs_of assessment \<noteq> {}"
  and evidence_active:
    "\<forall>v \<in> evidence_refs_of assessment.
      active_evidence_source s (object_id_of old_eff) v"
  and target_ok:
    "admissible_post_target s (object_id_of old_eff)
      (evidence_post_target_of assessment)"
  unfolding integrate_evidence_input_valid_def by blast+
  from source_wf old_mem have old_conditions:
    "kind_of old_eff \<in> supported_kinds_of G \<and>
      references_complete s old_eff \<and>
      evidence_well_formed s old_eff \<and>
      time_well_formed old_eff \<and>
      perspective_well_formed s old_eff \<and>
      context_well_formed s old_eff"
  unfolding state_well_formed_def by blast
  have universe_mono:
    "version_ids s \<union> historical_versions_of s \<subsequeq>
      version_ids ?target \<union> historical_versions_of ?target"
  using replace_effect_version_universe_mono[OF old_mem]
  unfolding new_eff_def using fresh by simp
  have evidence_refs_resolve:
    "evidence_refs_of assessment
      \<subsequeq> version_ids ?target \<union> historical_versions_of ?target"
proof
  fix v
  assume "v \<in> evidence_refs_of assessment"
  then obtain ev where ev_mem: "ev \<in> objects_of s"
  and ev_v: "version_id_of ev = v"
  and different: "object_id_of ev \<noteq> object_id_of old_eff"
  using evidence_active unfolding active_evidence_source_def by blast
  have "ev \<noteq> old_eff" using different by auto
  then have "ev \<in> objects_of s - {old_eff}" using ev_mem by blast
  then have "v \<in> version_ids ?target"

```

```

    unfolding version_ids_def new_eff_def using ev_v by auto
    then show "v \<in> version_ids ?target \<union> historical_versions_of ?target"
    by blast
qed
have target_ref_resolves:
  "option_ref_set (evidence_post_target_of assessment)
  \<subseq> version_ids ?target \<union> historical_versions_of ?target"
proof (cases "evidence_post_target_of assessment")
  case None
  then show ?thesis by (simp add: option_ref_set_def)
next
  case (Some v)
  then obtain obj where obj_mem: "obj \<in> objects_of s"
    and obj_v: "version_id_of obj = v"
    and different: "object_id_of obj \<noteq> object_id_of old_eff"
    using target_ok unfolding admissible_post_target_def by auto
  have "obj \<noteq> old_eff" using different by auto
  then have "obj \<in> objects_of s - {old_eff}" using obj_mem by blast
  then have "v \<in> version_ids ?target"
    unfolding version_ids_def new_eff_def using obj_v by auto
  with Some show ?thesis by (simp add: option_ref_set_def)
qed
have prior_resolves:
  "version_id_of old_eff
  \<in> version_ids ?target \<union> historical_versions_of ?target"
  by simp
from old_conditions have old_refs_source:
  "references_of old_eff
  \<subseq> version_ids s \<union> historical_versions_of s"
  unfolding references_complete_def by blast
have old_refs_target:
  "references_of old_eff
  \<subseq> version_ids ?target \<union> historical_versions_of ?target"
  using old_refs_source universe_mono by blast
have combined_refs_target:
  "references_of old_eff \<union> evidence_refs_of assessment \<union>
  {version_id_of old_eff} \<union>
  option_ref_set (evidence_post_target_of assessment)
  \<subseq> version_ids ?target \<union> historical_versions_of ?target"
  using old_refs_target evidence_refs_resolve target_ref_resolves
  prior_resolves by blast
have new_refs_eq:
  "references_of new_eff =
  references_of old_eff \<union> evidence_refs_of assessment \<union>
  {version_id_of old_eff} \<union>
  option_ref_set (evidence_post_target_of assessment)"
  unfolding new_eff_def by simp
have refs_new: "references_complete ?target new_eff"
  unfolding references_complete_def
  using combined_refs_target new_refs_eq by simp
have evidence_new: "evidence_well_formed ?target new_eff"
proof -
  from old_conditions have old_evidence_source:
    "evidence_of old_eff
    \<subseq> version_ids s \<union> historical_versions_of s"
    unfolding evidence_well_formed_def evidence_references_complete_def
    by blast
  have old_evidence_target:
    "evidence_of old_eff
    \<subseq> version_ids ?target \<union> historical_versions_of ?target"
    using old_evidence_source universe_mono by blast
  have new_evidence_eq:
    "evidence_of new_eff =
    evidence_of old_eff \<union> evidence_refs_of assessment"
    unfolding new_eff_def by simp
  have complete:
    "evidence_of new_eff
    \<subseq> version_ids ?target \<union> historical_versions_of ?target"
    using old_evidence_target evidence_refs_resolve new_evidence_eq by blast
  have upgraded_nonempty:
    "lifecycle_of new_eff \<in> {Supported, Validated}
    \<longrightarrow> evidence_of new_eff \<noteq> {}"
    using evidence_nonempty
    unfolding new_eff_def integrated_effect_object_def
    by auto
  show ?thesis
    unfolding evidence_well_formed_def evidence_references_complete_def
    using complete upgraded_nonempty by blast
qed
have origin_target: "present_ref ?target (effect_origin_of new_eff)"

```



```

    unfolding new_eff_def
    using replace_effect_present_ref_mono[OF old_mem _ origin] fresh by simp
  have perspective_new: "perspective_well_formed ?target new_eff"
  proof -
    from old_conditions have old_perspective:
      "perspective_well_formed s old_eff" by blast
    show ?thesis
    using old_perspective
    unfolding perspective_well_formed_def new_eff_def
    by (simp only: integrated_effect_simps replace_effect_simps)
  qed
  have context_new: "context_well_formed ?target new_eff"
  proof -
    from old_conditions have old_context:
      "context_well_formed s old_eff" by blast
    show ?thesis
    using old_context
    unfolding context_well_formed_def new_eff_def
    by (simp only: integrated_effect_simps replace_effect_simps)
  qed
  have other_typed:
    "time_well_formed new_eff \<and>
    relation_well_formed ?target new_eff \<and>
    effect_well_formed ?target new_eff \<and>
    relevance_well_formed ?target new_eff \<and>
    orientation_well_formed ?target new_eff \<and>
    action_well_formed ?target new_eff \<and>
    feedback_well_formed ?target new_eff"
  proof -
    have new_kind: "kind_of new_eff = K_EFF"
    using old_kind unfolding new_eff_def by simp
    have new_time: "time_well_formed new_eff"
    unfolding time_well_formed_def new_eff_def by simp
    have new_relation: "relation_well_formed ?target new_eff"
    using new_kind unfolding relation_well_formed_def by simp
    have new_effect: "effect_well_formed ?target new_eff"
    unfolding effect_well_formed_def
    using new_kind origin_target by blast
    have new_relevance: "relevance_well_formed ?target new_eff"
    using new_kind unfolding relevance_well_formed_def by simp
    have new_orientation: "orientation_well_formed ?target new_eff"
    using new_kind unfolding orientation_well_formed_def by simp
    have new_action: "action_well_formed ?target new_eff"
    using new_kind unfolding action_well_formed_def by simp
    have new_feedback: "feedback_well_formed ?target new_eff"
    using new_kind unfolding feedback_well_formed_def by simp
    show ?thesis
    using new_time new_relation new_effect new_relevance new_orientation
    new_action new_feedback by blast
  qed
  show ?thesis
  using old_conditions refs_new evidence_new perspective_new context_new
  other_typed
  unfolding new_eff_def by simp
qed

lemma replace_effect_preserves_unique_versions:
  assumes source_unique: "unique_versions s"
  and old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  shows "unique_versions
    (replace_effect_version s old_eff new_eff next_snapshot_id)"
  proof -
    from source_unique have inj: "inj_on version_id_of (objects_of s)"
    unfolding unique_versions_def .
    show ?thesis
    unfolding unique_versions_def
  proof (rule inj_onI)
    fix left right
    assume left_mem:
      "left \<in> objects_of
      (replace_effect_version s old_eff new_eff next_snapshot_id)"
    and right_mem:
      "right \<in> objects_of
      (replace_effect_version s old_eff new_eff next_snapshot_id)"
    and same: "version_id_of left = version_id_of right"
    consider (left_new) "left = new_eff" | (left_old) "left \<in> objects_of s - {old_eff}"
    using left_mem by auto
    then show "left = right"
  qed

```

```

proof cases
  case left_new
  show ?thesis
  proof (cases "right = new_eff")
    case True
    with left_new show ?thesis by simp
  next
  case False
  with right_mem have right_old: "right \<in> objects_of s"
  by auto
  have "version_id_of new_eff \<in> version_ids s"
  unfolding version_ids_def using same left_new right_old by blast
  then show ?thesis using fresh by blast
qed
next
case left_old
show ?thesis
proof (cases "right = new_eff")
  case True
  have left_version: "version_id_of left \<in> version_ids s"
  unfolding version_ids_def using left_old by blast
  have "version_id_of left = version_id_of new_eff"
  using same True by simp
  with left_version have "version_id_of new_eff \<in> version_ids s"
  by simp
  then show ?thesis using fresh by blast
next
case False
with right_mem have right_old: "right \<in> objects_of s" by auto
show ?thesis
using inj_onD[OF inj same] left_old right_old by blast
qed
qed
qed
qed

theorem integrate_evidence_preserves_state_well_formed:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
    old_eff prior assessment new_version integration_time next_snapshot_id
    s' catalog' object_archive' payload_archive' ledger'"
  shows "state_well_formed G s'"
proof -
  let ?new_eff = "integrated_effect_object old_eff assessment new_version
  integration_time"
  let ?target = "replace_effect_version s old_eff ?new_eff next_snapshot_id"
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
    ledger old_eff prior assessment new_version integration_time
    next_snapshot_id"
  and target: "s' = ?target"
  by (auto simp: integrate_evidence_step_def Let_def)
  from valid have source_wf: "state_well_formed G s"
  and old_mem: "old_eff \<in> objects_of s"
  and fresh: "new_version \<notin> version_ids s \<union> historical_versions_of s"
  unfolding integrate_evidence_input_valid_def by blast+
  have unique: "unique_versions ?target"
  using replace_effect_preserves_unique_versions old_mem fresh source_wf
  unfolding state_well_formed_def by simp
  have new_conditions:
    "kind_of ?new_eff \<in> supported_kinds_of G \<and>
    references_complete ?target ?new_eff \<and>
    evidence_well_formed ?target ?new_eff \<and>
    time_well_formed ?new_eff \<and>
    relation_well_formed ?target ?new_eff \<and>
    effect_well_formed ?target ?new_eff \<and>
    perspective_well_formed ?target ?new_eff \<and>
    context_well_formed ?target ?new_eff \<and>
    relevance_well_formed ?target ?new_eff \<and>
    orientation_well_formed ?target ?new_eff \<and>
    action_well_formed ?target ?new_eff \<and>
    feedback_well_formed ?target ?new_eff"
  using integrate_evidence_new_object_conditions[OF valid] by simp
  have old_object_conditions:
    "\<forall>obj \<in> objects_of s - {old_eff}.
    kind_of obj \<in> supported_kinds_of G \<and>
    references_complete ?target obj \<and>
    evidence_well_formed ?target obj \<and>
    time_well_formed obj \<and>
    relation_well_formed ?target obj \<and>"

```

```

effect_well_formed ?target obj \<and>
perspective_well_formed ?target obj \<and>
context_well_formed ?target obj \<and>
relevance_well_formed ?target obj \<and>
orientation_well_formed ?target obj \<and>
action_well_formed ?target obj \<and>
feedback_well_formed ?target obj"
proof (intro ballI)
  fix obj
  assume obj_mem: "obj \<in> objects_of s - {old_eff}"
  from source_wf obj_mem have source_conditions:
    "kind_of obj \<in> supported_kinds_of G \<and>
    references_complete s obj \<and> evidence_well_formed s obj \<and>
    time_well_formed obj \<and> relation_well_formed s obj \<and>
    effect_well_formed s obj \<and> perspective_well_formed s obj \<and>
    context_well_formed s obj \<and> relevance_well_formed s obj \<and>
    orientation_well_formed s obj \<and> action_well_formed s obj \<and>
    feedback_well_formed s obj"
  unfolding state_well_formed_def by blast
  have present_mono: "\<And>ref. present_ref s ref \<Longrightarrow> present_ref ?target ref"
  using replace_effect_present_ref_mono[OF old_mem _] fresh by simp
  have refs_target: "references_complete ?target obj"
  using replace_effect_references_mono[OF old_mem _]
    source_conditions fresh by simp
  have evidence_target: "evidence_well_formed ?target obj"
  using replace_effect_evidence_mono[OF old_mem _]
    source_conditions fresh by simp
  have typed_target:
    "relation_well_formed ?target obj \<and>
    effect_well_formed ?target obj \<and>
    feedback_well_formed ?target obj"
  using source_conditions present_mono
  unfolding relation_well_formed_def effect_well_formed_def
    feedback_well_formed_def
  by blast
  from source_conditions have source_perspective:
    "perspective_well_formed s obj"
  and source_context: "context_well_formed s obj"
  and source_relevance: "relevance_well_formed s obj"
  and source_orientation: "orientation_well_formed s obj"
  and source_action: "action_well_formed s obj"
  by blast+
  have perspective_target: "perspective_well_formed ?target obj"
  using source_perspective
  unfolding perspective_well_formed_def
  by (simp only: replace_effect_simps)
  have context_target: "context_well_formed ?target obj"
  using source_context
  unfolding context_well_formed_def
  by (simp only: replace_effect_simps)
  have relevance_target: "relevance_well_formed ?target obj"
  unfolding relevance_well_formed_def
  proof
    assume kind: "kind_of obj = K_REV"
    from source_relevance kind have source_model:
      "case model_of obj of None \<Rightarrow> False
      | Some m \<Rightarrow> m \<in> models_of s"
    and source_refs: "references_of obj \<noteq> {}"
    unfolding relevance_well_formed_def by blast+
    have target_model:
      "case model_of obj of None \<Rightarrow> False
      | Some m \<Rightarrow> m \<in> models_of ?target"
    using source_model by (simp only: replace_effect_simps)
    show "perspective_well_formed ?target obj \<and>
    context_well_formed ?target obj \<and>
    (case model_of obj of None \<Rightarrow> False
    | Some m \<Rightarrow> m \<in> models_of ?target) \<and>
    references_of obj \<noteq> {}"
    using perspective_target context_target target_model source_refs by blast
  qed
  have orientation_target: "orientation_well_formed ?target obj"
  using source_orientation unfolding orientation_well_formed_def by simp
  have action_target: "action_well_formed ?target obj"
  using source_action unfolding action_well_formed_def
  by (simp only: replace_effect_simps)
  have stable_target:
    "perspective_well_formed ?target obj \<and>
    context_well_formed ?target obj \<and>
    relevance_well_formed ?target obj \<and>
    orientation_well_formed ?target obj \<and>

```

```

    action_well_formed ?target obj"
    using perspective_target context_target relevance_target
      orientation_target action_target by blast
show "kind_of obj \<in> supported_kinds_of G \<and>
  references_complete ?target obj \<and>
  evidence_well_formed ?target obj \<and>
  time_well_formed obj \<and>
  relation_well_formed ?target obj \<and>
  effect_well_formed ?target obj \<and>
  perspective_well_formed ?target obj \<and>
  context_well_formed ?target obj \<and>
  relevance_well_formed ?target obj \<and>
  orientation_well_formed ?target obj \<and>
  action_well_formed ?target obj \<and>
  feedback_well_formed ?target obj"
    using source_conditions refs_target evidence_target typed_target
      stable_target by blast
qed
have all_objects:
  "\<forall>obj\<in>objects_of ?target.
    kind_of obj \<in> supported_kinds_of G \<and>
    references_complete ?target obj \<and>
    evidence_well_formed ?target obj \<and>
    time_well_formed obj \<and>
    relation_well_formed ?target obj \<and>
    effect_well_formed ?target obj \<and>
    perspective_well_formed ?target obj \<and>
    context_well_formed ?target obj \<and>
    relevance_well_formed ?target obj \<and>
    orientation_well_formed ?target obj \<and>
    action_well_formed ?target obj \<and>
    feedback_well_formed ?target obj"
    using new_conditions old_object_conditions by auto
from source_wf have secondary:
  "(\<forall>p\<in>projections_of ?target. projection_well_formed ?target p) \<and>
  (\<forall>c\<in>couplings_of ?target. coupling_well_formed ?target c) \<and>
  (\<forall>t\<in>transmissions_of ?target. transmission_well_formed ?target t) \<and>
  (\<forall>b\<in>boundaries_of ?target. boundary_well_formed b)"
  unfolding state_well_formed_def projection_well_formed_def
    coupling_well_formed_def transmission_well_formed_def
  by simp
show ?thesis
  unfolding target state_well_formed_def
  using unique_all_objects secondary by blast
qed

theorem integrate_evidence_preserves_identity:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  shows "unique_versions s' \<and> immutable_preserved s s'"
proof -
  let ?new_eff = "integrated_effect_object old_eff assessment new_version
    integration_time"
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
      ledger old_eff prior assessment new_version integration_time
      next_snapshot_id"
  and target:
    "s' = replace_effect_version s old_eff ?new_eff next_snapshot_id"
  by (auto simp: integrate_evidence_step_def Let_def)
  from valid have source_wf: "state_well_formed G s"
  and old_mem: "old_eff \<in> objects_of s"
  and fresh: "new_version \<notin> version_ids s \<union> historical_versions_of s"
  unfolding integrate_evidence_input_valid_def by blast+
  have unique: "unique_versions s'"
  unfolding target
  using replace_effect_preserves_unique_versions old_mem fresh source_wf
  unfolding state_well_formed_def by simp
  have immutable: "immutable_preserved s s'"
  unfolding immutable_preserved_def target
  proof (intro ballI allI impI)
    fix before after
    assume before_mem: "before \<in> objects_of s"
    and after_mem:
      "after \<in> objects_of
        (replace_effect_version s old_eff ?new_eff next_snapshot_id)"
    and same: "version_id_of before = version_id_of after"
    show "before = after"

```

```

proof (cases "after = ?new_eff")
  case True
    have "new_version \<in> version_ids s"
      unfolding version_ids_def using before_mem same True by auto
    then show ?thesis using fresh by blast
  next
    case False
    then have after_old: "after \<in> objects_of s" using after_mem by auto
    from source_wf have inj: "inj_on version_id_of (objects_of s)"
      unfolding state_well_formed_def unique_versions_def by blast
    show ?thesis using inj_onD[OF inj same before_mem after_old] .
  qed
qed
show ?thesis using unique immutable by blast
qed

theorem integrate_evidence_preserves_audit_stores:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
    old_eff prior assessment new_version integration_time next_snapshot_id
    s' catalog' object_archive' payload_archive' ledger'"
  shows
    "active_history_disjoint s' \<and>
    object_archive_well_formed s' object_archive' \<and>
    effect_payload_archive_well_formed object_archive' payload_archive' \<and>
    evidence_ledger_well_formed s' object_archive' ledger'"
proof -
  let ?new_eff = "integrated_effect_object old_eff assessment new_version
  integration_time"
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
    ledger old_eff prior assessment new_version integration_time
    next_snapshot_id"
  and target:
    "s' = replace_effect_version s old_eff ?new_eff next_snapshot_id"
  and object_archive_target:
    "object_archive' = extend_object_archive object_archive old_eff"
  and payload_archive_target:
    "payload_archive' = extend_effect_payload_archive payload_archive
    old_eff prior"
  and ledger_target:
    "ledger' = extend_evidence_ledger ledger new_version assessment"
  by (auto simp: integrate_evidence_step_def Let_def)
  from valid have source_wf: "state_well_formed G s"
  and source_disjoint: "active_history_disjoint s"
  and source_object_archive:
    "object_archive_well_formed s object_archive"
  and source_payload_archive:
    "effect_payload_archive_well_formed object_archive payload_archive"
  and source_ledger:
    "evidence_ledger_well_formed s object_archive ledger"
  and old_mem: "old_eff \<in> objects_of s"
  and old_archive_none:
    "object_archive (version_id_of old_eff) = None"
  and payload_old_none:
    "payload_archive (version_id_of old_eff) = None"
  and fresh:
    "new_version \<notin> version_ids s \<union> historical_versions_of s"
  and ledger_new_none: "ledger new_version = None"
  and assessment_bounds: "assessment_bounds assessment"
  and assessment_model:
    "evidence_assessment_model_of assessment \<in> models_of s"
  and old_kind: "kind_of old_eff = K_EFF"
  and payload_valid: "effect_payload_valid old_eff prior"
  unfolding integrate_evidence_input_valid_def by blast+
  from source_wf have source_unique: "unique_versions s"
  unfolding state_well_formed_def by blast
  have fresh_eff:
    "version_id_of ?new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  using fresh by simp
  have target_disjoint: "active_history_disjoint s'"
  unfolding target
  using replace_effect_preserves_active_history_disjoint
  [OF source_disjoint source_unique old_mem fresh_eff] .
  have target_object_archive:
    "object_archive_well_formed s' object_archive'"
proof -
  have archived_properties:
    "\<forall>v obj. object_archive' v = Some obj \<longrightarrow>"

```

```

    version_id_of obj = v \<and> v \<in> historical_versions_of s'"
proof (intro allI impI)
  fix v obj
  assume mapped: "object_archive' v = Some obj"
  show "version_id_of obj = v \<and> v \<in> historical_versions_of s'"
  proof (cases "v = version_id_of old_eff")
    case True
    with mapped object_archive_target target show ?thesis
      unfolding extend_object_archive_def by simp
  next
    case False
    with mapped object_archive_target have source_mapped:
      "object_archive v = Some obj"
      unfolding extend_object_archive_def by simp
    from source_object_archive source_mapped have
      "version_id_of obj = v \<and> v \<in> historical_versions_of s"
      unfolding object_archive_well_formed_def by blast
    with target show ?thesis by simp
  qed
qed
show ?thesis
  unfolding object_archive_well_formed_def
  using archived_properties target_disjoint
  unfolding active_history_disjoint_def by blast
qed
have target_payload_archive:
  "effect_payload_archive_well_formed object_archive' payload_archive'"
proof (unfold effect_payload_archive_well_formed_def, intro allI impI)
  fix v stored_payload
  assume mapped: "payload_archive' v = Some stored_payload"
  show "\<exists>eff. object_archive' v = Some eff \<and>
    effect_payload_valid eff stored_payload"
  proof (cases "v = version_id_of old_eff")
    case True
    with mapped payload_archive_target object_archive_target
      payload_valid show ?thesis
      unfolding extend_effect_payload_archive_def extend_object_archive_def
      by auto
  next
    case False
    with mapped payload_archive_target have source_mapped:
      "payload_archive v = Some stored_payload"
      unfolding extend_effect_payload_archive_def by simp
    from source_payload_archive source_mapped obtain eff where
      source_object: "object_archive v = Some eff"
      and valid_payload: "effect_payload_valid eff stored_payload"
      unfolding effect_payload_archive_well_formed_def by blast
    have "object_archive' v = Some eff"
      unfolding object_archive_target extend_object_archive_def
      using source_object False by simp
    with valid_payload show ?thesis by blast
  qed
qed
have target_ledger:
  "evidence_ledger_well_formed s' object_archive' ledger'"
proof (unfold evidence_ledger_well_formed_def, intro allI impI)
  fix v stored_assessment
  assume mapped: "ledger' v = Some stored_assessment"
  show "evidence_entry_well_formed s' object_archive' v stored_assessment"
  proof (cases "v = new_version")
    case True
    with mapped ledger_target have assessment_eq:
      "stored_assessment = assessment"
      unfolding extend_evidence_ledger_def by simp
    have target_wf: "state_well_formed G s'"
      using integrate_evidence_preserves_state_well_formed[OF concrete] .
    have exact:
      "objects_of s' = insert ?new_eff (objects_of s - {old_eff})"
      using concrete
      unfolding integrate_evidence_step_def Let_def by auto
    have new_mem: "?new_eff \<in> objects_of s'" using exact by simp
    have evidence_refs_in_new:
      "evidence_refs_of assessment \<subseq> evidence_of ?new_eff"
      by simp
    from target_wf new_mem have new_evidence_complete:
      "evidence_of ?new_eff
        \<subseq> version_ids s' \<union> historical_versions_of s'"
      unfolding state_well_formed_def evidence_well_formed_def
      evidence_references_complete_def
      by blast
  qed

```

```

from target_wf new_mem have evidence_complete:
  "evidence_refs_of assessment
   \<subseteq> version_ids s' \<union> historical_versions_of s'"
  using evidence_refs_in_new new_evidence_complete by blast
from integrate_evidence_lineage_and_no_self_support[OF concrete]
have evidence_nonempty: "evidence_refs_of assessment \<noteq> {}"
  and new_not_self: "new_version \<notin> evidence_refs_of assessment"
  and prior_not_self:
    "version_id_of old_eff \<notin> evidence_refs_of assessment"
  by blast+
from valid have subject:
  "evidence_subject_of assessment = object_id_of old_eff"
  and prior_version:
    "evidence_prior_version_of assessment = version_id_of old_eff"
  unfolding integrate_evidence_input_valid_def by blast+
have resolved:
  "resolved_version_object s' object_archive' new_version ?new_eff"
  unfolding resolved_version_object_def using new_mem by simp
show ?thesis
  unfolding True assessment_eq evidence_entry_well_formed_def
  using resolved evidence_nonempty evidence_complete new_not_self
  prior_not_self subject prior_version evidence_refs_in_new
  assessment_bounds old_kind
  assessment_model
  unfolding assessment_bounds_def target
  by auto
next
case False
with mapped ledger_target have source_mapped:
  "ledger v = Some stored_assessment"
  unfolding extend_evidence_ledger_def by simp
from source_ledger source_mapped have source_entry:
  "evidence_entry_well_formed s object_archive v stored_assessment"
  unfolding evidence_ledger_well_formed_def by blast
have transferred:
  "evidence_entry_well_formed
   (replace_effect_version s old_eff ?new_eff next_snapshot_id)
   (extend_object_archive object_archive old_eff) v stored_assessment"
  using evidence_entry_replace_mono[OF old_mem fresh_eff
    old_archive_none source_entry] .
show ?thesis
  unfolding target object_archive_target using transferred .
qed
qed
show ?thesis
  using target_disjoint target_object_archive target_payload_archive
  target_ledger by blast
qed

theorem integrate_evidence_preserves_active_catalog:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
     old_eff prior assessment new_version integration_time next_snapshot_id
     s' catalog' object_archive' payload_archive' ledger'"
  shows "effect_catalog_well_formed s' catalog'"
proof -
  let ?new_eff = "integrated_effect_object old_eff assessment new_version
    integration_time"
  let ?new_payload = "integrated_effect_payload prior assessment"
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
     ledger old_eff prior assessment new_version integration_time
     next_snapshot_id"
  and target:
    "s' = replace_effect_version s old_eff ?new_eff next_snapshot_id"
  and catalog_target:
    "catalog' = replace_active_effect_payload catalog
     (version_id_of old_eff) new_version ?new_payload"
  by (auto simp: integrate_evidence_step_def Let_def)
from valid have source_catalog: "effect_catalog_well_formed s catalog"
  and old_mem: "old_eff \<in> objects_of s"
  and fresh:
    "new_version \<notin> version_ids s \<union> historical_versions_of s"
  unfolding integrate_evidence_input_valid_def by blast+
have new_payload_valid: "effect_payload_valid ?new_eff ?new_payload"
  using integrate_evidence_payload_valid[OF valid] .
have new_mem: "?new_eff \<in> objects_of s'"
  unfolding target by simp
show ?thesis
  unfolding effect_catalog_well_formed_def

```

```

proof (intro allI impI)
  fix v stored_payload
  assume mapped: "catalog' v = Some stored_payload"
  show "\<exists>eff\<in>objects_of s'.
    version_id_of eff = v \<and> effect_payload_valid eff stored_payload"
  proof (cases "v = new_version")
    case True
    with mapped catalog_target have payload_eq:
      "stored_payload = ?new_payload"
    unfolding replace_active_effect_payload_def by simp
    show ?thesis
    proof (rule bexI[of _ ?new_eff])
      show "version_id_of ?new_eff = v \<and>
        effect_payload_valid ?new_eff stored_payload"
      using new_payload_valid True payload_eq by simp
      show "?new_eff \<in> objects_of s'" using new_mem .
    qed
  qed
next
case False
have old_key_different: "v \<noteq> version_id_of old_eff"
proof
  assume "v = version_id_of old_eff"
  with mapped catalog_target False show False
  unfolding replace_active_effect_payload_def by simp
qed
from mapped catalog_target False old_key_different have source_mapped:
  "catalog v = Some stored_payload"
unfolding replace_active_effect_payload_def by simp
from source_catalog source_mapped obtain eff where eff_mem:
  "eff \<in> objects_of s"
and eff_version: "version_id_of eff = v"
and eff_payload: "effect_payload_valid eff stored_payload"
unfolding effect_catalog_well_formed_def by blast
have eff_not_old: "eff \<noteq> old_eff"
using eff_version old_key_different by auto
have "eff \<in> objects_of s'"
unfolding target using eff_mem eff_not_old by auto
with eff_version eff_payload show ?thesis by blast
qed
qed
qed

theorem integrate_evidence_preserves_references_time_context:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  shows
    "\<forall>obj\<in>objects_of s'.
      references_complete s' obj \<and>
      evidence_well_formed s' obj \<and>
      time_well_formed obj \<and>
      perspective_well_formed s' obj \<and>
      context_well_formed s' obj"
  using integrate_evidence_preserves_state_well_formed[OF concrete]
  unfolding state_well_formed_def by blast

theorem integrate_evidence_refines_generic_step:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  shows "step G s IntegrateEvidence s'"
proof -
  from concrete have source_wf: "state_well_formed G s"
  and allowed: "IntegrateEvidence \<in> allowed_operations_of G"
  by (auto simp: integrate_evidence_step_def integrate_evidence_input_valid_def)
  have target_wf: "state_well_formed G s'"
  using integrate_evidence_preserves_state_well_formed[OF concrete] .
  have immutable: "immutable_preserved s s'"
  using integrate_evidence_preserves_identity[OF concrete] by blast
  show ?thesis
  unfolding step_def using source_wf target_wf allowed immutable by blast
qed
end

```



Beweisstatus dieser D.8-Fassung: kernelgeprüft. Der gezielte Isabelle2025-2-Clean-Build verarbeitete die vollständige neunteilige Session und endete mit Exit-Code 0. T5 enthält 1.564 Quellzeilen und keine Vorkommen von sorry, oops, axiomatization oder quick\_and\_dirty. Der Quellhash lautet 4422076e2bd6cb2ff9fe5e2d5e2e135e3bf3c05919a1ca03a42b1faccd5fddc0. Belegt sind die formulierten konstruktiven Delta-, Lifecycle-, Unsicherheits-, Lineage-, Archiv-, Katalog-, Zustandserhaltungs- und Refinementsätze. Nicht belegt sind empirische Richtigkeit einer Evidenzbewertung, Vollständigkeit zulässiger Evidenzarten, allgemeine Konservativität oder Produktivengine-Äquivalenz.

## D.9 Historischer isolierter Ausführungszeuge für T5 (durch I.9 ersetzt)

Der Zeuge verhindert, dass T5 lediglich als konditionales Schema mit unerfüllten Vorbedingungen bestehen bleibt. Ein endlicher Ausgangszustand enthält eine aktive Erscheinung als unabhängige Evidenzquelle, eine typisierte Relationsursache und eine K\_EFF-Hypothese. Die konkrete Assessment-Instanz stützt die Hypothese. Der mechanisch konstruierte Nachzustand enthält Version 4 desselben Effektoobjekts mit Lifecycle Supported, previous\_version = Some 3 und Evidenzmenge {1}; Version 3 und ihr Payload liegen in den getrennten Archiven, der aktive Katalog verweist auf Version 4 und der Evidenz-Ledger hält das Assessment.

```
theory UZA_0_9_1_Integrate_Evidence_Witness
  imports UZA_0_9_1_Integrate_Evidence
          UZA_0_9_1_Hypothesize_Effect_Witness
begin

section \<open>A strict, causally reachable T5 witness\<close>

text \<open>
  This witness replaces the former isolated T5 record literal. Both old
  effects are the actual outputs of the two T4 steps. Each effect points to
  an active K_REL object that was created by T3 and has an entry in the
  well-formed relation catalog. Two strict T5 steps then integrate independent
  evidence and preserve the complete object, payload, and ledger history.
\<close>

definition chain_assessment_a :: evidence_assessment where
  "chain_assessment_a =
    \<lparr> evidence_subject_of = 5,
      evidence_prior_version_of = 5,
      evidence_refs_of = {1},
      evidence_disposition_of = EvidenceSupports,
      evidence_strength_of = 1 / 2,
      evidence_post_uncertainty_of = 1 / 4,
      evidence_post_target_of = None,
      evidence_post_vector_of = Some chain_effect_vector_a,
      evidence_assessment_model_of = 30,
      validation_uncertainty_max_of = 1 / 10,
      validation_evidence_min_of = 2 \<rparr>"

definition chain_assessment_b :: evidence_assessment where
  "chain_assessment_b =
    \<lparr> evidence_subject_of = 6,
      evidence_prior_version_of = 6,
      evidence_refs_of = {2},
      evidence_disposition_of = EvidenceSupports,
      evidence_strength_of = 1 / 2,
      evidence_post_uncertainty_of = 1 / 4,
      evidence_post_target_of = None,
      evidence_post_vector_of = Some chain_effect_vector_b,
      evidence_assessment_model_of = 30,
      validation_uncertainty_max_of = 1 / 10,
      validation_evidence_min_of = 2 \<rparr>"

definition chain_object_archive_0 :: object_archive where
  "chain_object_archive_0 = Map.empty"

definition chain_payload_archive_0 :: effect_payload_archive where
  "chain_payload_archive_0 = Map.empty"

definition chain_evidence_ledger_0 :: evidence_ledger where
  "chain_evidence_ledger_0 = Map.empty"

definition chain_effect_a_supported :: uza_object where
  "chain_effect_a_supported =
    integrated_effect_object chain_effect_a chain_assessment_a 7 7"

definition chain_after_t5_a :: snapshot where
  "chain_after_t5_a =
    replace_effect_version chain_after_t4 chain_effect_a
      chain_effect_a_supported 7"

definition chain_effect_catalog_after_t5_a :: effect_catalog where
```

```

"chain_effect_catalog_after_t5_a =
  replace_active_effect_payload chain_effect_catalog 5 7
  (integrated_effect_payload chain_effect_payload_a chain_assessment_a)"

definition chain_object_archive_after_t5_a :: object_archive where
  "chain_object_archive_after_t5_a =
    extend_object_archive chain_object_archive_0 chain_effect_a"

definition chain_payload_archive_after_t5_a :: effect_payload_archive where
  "chain_payload_archive_after_t5_a =
    extend_effect_payload_archive chain_payload_archive_0 chain_effect_a
    chain_effect_payload_a"

definition chain_evidence_ledger_after_t5_a :: evidence_ledger where
  "chain_evidence_ledger_after_t5_a =
    extend_evidence_ledger chain_evidence_ledger_0 7 chain_assessment_a"

definition chain_effect_b_supported :: uza_object where
  "chain_effect_b_supported =
    integrated_effect_object chain_effect_b chain_assessment_b 8 8"

definition chain_after_t5 :: snapshot where
  "chain_after_t5 =
    replace_effect_version chain_after_t5_a chain_effect_b
    chain_effect_b_supported 8"

definition chain_effect_catalog_after_t5 :: effect_catalog where
  "chain_effect_catalog_after_t5 =
    replace_active_effect_payload chain_effect_catalog_after_t5_a 6 8
    (integrated_effect_payload chain_effect_payload_b chain_assessment_b)"

definition chain_object_archive :: object_archive where
  "chain_object_archive =
    extend_object_archive chain_object_archive_after_t5_a chain_effect_b"

definition chain_payload_archive :: effect_payload_archive where
  "chain_payload_archive =
    extend_effect_payload_archive chain_payload_archive_after_t5_a
    chain_effect_b chain_effect_payload_b"

definition chain_evidence_ledger :: evidence_ledger where
  "chain_evidence_ledger =
    extend_evidence_ledger chain_evidence_ledger_after_t5_a 8
    chain_assessment_b"

lemma chain_after_t4_active_history_disjoint:
  "active_history_disjoint chain_after_t4"
  by (auto simp: active_history_disjoint_def version_ids_def
    chain_after_t4_def chain_after_t4_a_def chain_after_t3_def
    chain_after_t3_ab_def chain_after_t2_def chain_after_t1_def
    chain_seed_def base_snapshot_def chain_object_defs chain_effect_a_def
    chain_effect_b_def base_object_def)

lemma chain_empty_audit_stores_well_formed:
  "object_archive_well_formed chain_after_t4 chain_object_archive_0 \<and>
    effect_payload_archive_well_formed chain_object_archive_0
    chain_payload_archive_0 \<and>
    evidence_ledger_well_formed chain_after_t4 chain_object_archive_0
    chain_evidence_ledger_0"
  by (simp add: object_archive_well_formed_def
    effect_payload_archive_well_formed_def evidence_ledger_well_formed_def
    chain_object_archive_0_def chain_payload_archive_0_def
    chain_evidence_ledger_0_def)

lemma chain_t5_a_strict_input_valid:
  "integrate_evidence_strict_input_valid chain_grammar chain_after_t4
    chain_relation_catalog chain_effect_catalog chain_object_archive_0
    chain_payload_archive_0 chain_evidence_ledger_0 chain_effect_a
    chain_effect_payload_a chain_assessment_a 7 7 7"
  unfolding integrate_evidence_strict_input_valid_def
    catalogued_active_relation_origin_def integrate_evidence_input_valid_def
  using chain_t4_reachability_certificate
    chain_after_t4_active_history_disjoint
    chain_empty_audit_stores_well_formed chain_effect_vector_a_valid
  by (auto simp: chain_grammar_def chain_after_t4_def chain_after_t4_a_def
    chain_after_t3_def chain_after_t3_ab_def chain_after_t2_def
    chain_after_t1_def chain_seed_def base_snapshot_def chain_object_defs
    chain_effect_a_def chain_effect_b_def chain_effect_payload_a_def
    chain_effect_payload_b_def chain_effect_catalog_def
    chain_effect_catalog_a_def chain_effect_catalog_0_def)

```

```

    extend_effect_catalog_def chain_relation_catalog_def
    chain_relation_catalog_ab_def chain_relation_catalog_0_def
    extend_relation_catalog_def chain_object_archive_0_def
    chain_payload_archive_0_def chain_evidence_ledger_0_def
    chain_assessment_a_def base_object_def effect_payload_valid_def
    active_relation_origin_def active_evidence_source_def
    assessment_bounds_def assessment_change_valid_def
    admissible_post_target_def effect_vector_valid_def present_ref_def
    version_ids_def)

theorem chain_t5_a_strict_step:
  "integrate_evidence_strict_step chain_grammar chain_after_t4
   chain_relation_catalog chain_effect_catalog chain_object_archive_0
   chain_payload_archive_0 chain_evidence_ledger_0 chain_effect_a
   chain_effect_payload_a chain_assessment_a 7 7 7 chain_after_t5_a
   chain_effect_catalog_after_t5_a chain_object_archive_after_t5_a
   chain_payload_archive_after_t5_a chain_evidence_ledger_after_t5_a"
  unfolding integrate_evidence_strict_step_def integrate_evidence_step_def
  integrate_evidence_strict_input_valid_def Let_def
  chain_after_t5_a_def chain_effect_a_supported_def
  chain_effect_catalog_after_t5_a_def
  chain_object_archive_after_t5_a_def
  chain_payload_archive_after_t5_a_def
  chain_evidence_ledger_after_t5_a_def
  using chain_t5_a_strict_input_valid
  by (simp add: integrate_evidence_strict_input_valid_def
    chain_effect_a_def base_object_def)

lemma chain_t5_a_base_step:
  "integrate_evidence_step chain_grammar chain_after_t4
   chain_effect_catalog chain_object_archive_0 chain_payload_archive_0
   chain_evidence_ledger_0 chain_effect_a chain_effect_payload_a
   chain_assessment_a 7 7 7 chain_after_t5_a
   chain_effect_catalog_after_t5_a chain_object_archive_after_t5_a
   chain_payload_archive_after_t5_a chain_evidence_ledger_after_t5_a"
  using integrate_evidence_strict_refines_base[OF chain_t5_a_strict_step] .

lemma chain_after_t5_a_certificates:
  "state_well_formed chain_grammar chain_after_t5_a \<and>
   active_history_disjoint chain_after_t5_a \<and>
   effect_catalog_well_formed chain_after_t5_a
   chain_effect_catalog_after_t5_a \<and>
   relation_catalog_well_formed chain_after_t5_a chain_relation_catalog \<and>
   object_archive_well_formed chain_after_t5_a
   chain_object_archive_after_t5_a \<and>
   effect_payload_archive_well_formed chain_object_archive_after_t5_a
   chain_payload_archive_after_t5_a \<and>
   evidence_ledger_well_formed chain_after_t5_a
   chain_object_archive_after_t5_a chain_evidence_ledger_after_t5_a"
  using integrate_evidence_preserves_state_well_formed[OF chain_t5_a_base_step]
  integrate_evidence_preserves_audit_stores[OF chain_t5_a_base_step]
  integrate_evidence_preserves_active_catalog[OF chain_t5_a_base_step]
  integrate_evidence_strict_preserves_relation_catalog
  [OF chain_t5_a_strict_step]
  by blast

lemma replace_effect_active_relation_origin_mono:
  assumes active: "active_relation_origin s v"
  and old_kind: "kind_of old_eff = K_EFF"
  shows "active_relation_origin
    (replace_effect_version s old_eff new_eff next_id) v"
proof -
  from active obtain rel where
    rel_mem: "rel \<in> objects_of s"
    and rel_version: "version_id_of rel = v"
    and rel_kind: "kind_of rel = K_REL"
    and rel_active: "lifecycle_of rel \<notin> {Revised, Rejected, Archived}"
  unfolding active_relation_origin_def
  by blast
  have rel_ne_old: "rel \<noteq> old_eff"
  using rel_kind old_kind by auto
  show ?thesis
  unfolding active_relation_origin_def
  using rel_mem rel_version rel_kind rel_active rel_ne_old
  by auto
qed

lemma chain_relation_4_active_after_t5_a:
  "active_relation_origin chain_after_t5_a 4"
  unfolding chain_after_t5_a_def

```

```

using replace_effect_active_relation_origin_mono
  [OF chain_relation_4_active_after_t4]
by (simp add: chain_effect_a_def base_object_def)

lemma chain_entity_b_member_after_t5_a:
  "chain_entity_b \<in> objects_of chain_after_t5_a"
  unfolding chain_after_t5_a_def chain_after_t4_def chain_after_t4_a_def
  chain_after_t3_def chain_after_t3_ab_def chain_after_t2_def
  by (simp add: chain_object_defs chain_effect_a_def chain_effect_b_def
    base_object_def)

lemma chain_effect_b_member_after_t5_a:
  "chain_effect_b \<in> objects_of chain_after_t5_a"
  unfolding chain_after_t5_a_def chain_after_t4_def
  by (simp add: chain_effect_a_def chain_effect_b_def base_object_def)

lemma chain_evidence_source_2_after_t5_a:
  "active_evidence_source chain_after_t5_a 6 2"
  unfolding active_evidence_source_def
proof (intro bexI conjI)
  show "chain_entity_b \<in> objects_of chain_after_t5_a"
    using chain_entity_b_member_after_t5_a .
  show "version_id_of chain_entity_b = 2"
    by (simp add: chain_object_defs base_object_def)
  show "object_id_of chain_entity_b \<noteq> 6"
    by (simp add: chain_object_defs base_object_def)
  show "lifecycle_of chain_entity_b \<notin> {Rejected, Archived}"
    by (simp add: chain_object_defs base_object_def)
qed

lemma chain_model_30_after_t5_a:
  "30 \<in> models_of chain_after_t5_a"
proof -
  from chain_t4_b_input_valid have model_before:
    "30 \<in> models_of chain_after_t4_a"
  by (simp add: effect_input_valid_def chain_effect_payload_b_def)
  show ?thesis
    unfolding chain_after_t5_a_def chain_after_t4_def
    using model_before by simp
qed

lemma chain_version_8_fresh_after_t5_a:
  "8 \<notin> version_ids chain_after_t5_a \<union>
  historical_versions_of chain_after_t5_a"
  by (auto simp: chain_after_t5_a_def chain_after_t4_def
    chain_after_t4_a_def chain_after_t3_def chain_after_t3_ab_def
    chain_after_t2_def chain_after_t1_def chain_seed_def base_snapshot_def
    chain_object_defs chain_effect_a_def chain_effect_b_def
    chain_effect_a_supported_def chain_assessment_a_def base_object_def
    version_ids_def integrated_effect_object_def replace_effect_version_def)

lemma chain_t5_a_store_keys_for_b:
  "chain_effect_catalog_after_t5_a 6 = Some chain_effect_payload_b \<and>
  chain_effect_catalog_after_t5_a 8 = None \<and>
  chain_object_archive_after_t5_a 6 = None \<and>
  chain_object_archive_after_t5_a 8 = None \<and>
  chain_payload_archive_after_t5_a 6 = None \<and>
  chain_payload_archive_after_t5_a 8 = None \<and>
  chain_evidence_ledger_after_t5_a 8 = None"
  by (simp add: chain_effect_catalog_after_t5_a_def
    chain_effect_catalog_def chain_effect_catalog_a_def
    chain_effect_catalog_0_def replace_active_effect_payload_def
    extend_effect_catalog_def chain_object_archive_after_t5_a_def
    chain_object_archive_0_def extend_object_archive_def
    chain_payload_archive_after_t5_a_def chain_payload_archive_0_def
    extend_effect_payload_archive_def chain_evidence_ledger_after_t5_a_def
    chain_evidence_ledger_0_def extend_evidence_ledger_def
    chain_effect_a_def chain_effect_b_def base_object_def)

lemma chain_effect_b_payload_valid:
  "effect_payload_valid chain_effect_b chain_effect_payload_b"
  using chain_t4_b_input_valid
  by (simp add: effect_input_valid_def)

lemma chain_t5_b_strict_input_valid:
  "integrate_evidence_strict_input_valid chain_grammar chain_after_t5_a
  chain_relation_catalog chain_effect_catalog_after_t5_a
  chain_object_archive_after_t5_a chain_payload_archive_after_t5_a
  chain_evidence_ledger_after_t5_a chain_effect_b chain_effect_payload_b
  chain_assessment_b 8 8 8"

```

```

proof -
  from chain_after_t5_a_certificates have state_wf:
    "state_well_formed chain_grammar chain_after_t5_a"
  and history_disjoint: "active_history_disjoint chain_after_t5_a"
  and effect_catalog_wf:
    "effect_catalog_well_formed chain_after_t5_a
     chain_effect_catalog_after_t5_a"
  and relation_catalog_wf:
    "relation_catalog_well_formed chain_after_t5_a chain_relation_catalog"
  and object_archive_wf:
    "object_archive_well_formed chain_after_t5_a
     chain_object_archive_after_t5_a"
  and payload_archive_wf:
    "effect_payload_archive_well_formed chain_object_archive_after_t5_a
     chain_payload_archive_after_t5_a"
  and ledger_wf:
    "evidence_ledger_well_formed chain_after_t5_a
     chain_object_archive_after_t5_a chain_evidence_ledger_after_t5_a"
  by blast+
  have catalogued_origin:
    "catalogued_active_relation_origin chain_after_t5_a
     chain_relation_catalog (effect_origin_of chain_effect_b)"
  using chain_relation_4_active_after_t5_a chain_relation_catalog_4_present
  by (simp add: catalogued_active_relation_origin_def chain_effect_b_def
    base_object_def)
  have origin_present:
    "present_ref chain_after_t5_a (effect_origin_of chain_effect_b)"
  using active_relation_origin_present_ref
  [OF chain_relation_4_active_after_t5_a]
  by (simp add: chain_effect_b_def base_object_def)
  have origin_exists:
    "\<exists>origin. effect_origin_of chain_effect_b = Some origin \<and>
     active_relation_origin chain_after_t5_a origin"
  using chain_relation_4_active_after_t5_a
  by (simp add: chain_effect_b_def base_object_def)
  have local_conditions:
    "IntegrateEvidence \<in> allowed_operations_of chain_grammar \<and>
     kind_of chain_effect_b = K_EFF \<and>
     lifecycle_of chain_effect_b \<in> {Hypothesis, Supported} \<and>
     evidence_subject_of chain_assessment_b = object_id_of chain_effect_b \<and>
     evidence_prior_version_of chain_assessment_b =
       version_id_of chain_effect_b \<and>
     evidence_refs_of chain_assessment_b = {2} \<and>
     evidence_assessment_model_of chain_assessment_b = 30 \<and>
     assessment_bounds chain_assessment_b \<and>
     assessment_change_valid chain_effect_payload_b chain_assessment_b \<and>
     admissible_post_target chain_after_t5_a (object_id_of chain_effect_b)
       (evidence_post_target_of chain_assessment_b) \<and>
     valid_from_of chain_effect_b \<le> (8::time_point) \<and>
     snapshot_id_of chain_after_t5_a < 8"
  by (auto simp: chain_grammar_def chain_effect_b_def
    chain_effect_payload_b_def chain_assessment_b_def base_object_def
    assessment_bounds_def assessment_change_valid_def
    admissible_post_target_def chain_after_t5_a_def)
  show ?thesis
  unfolding integrate_evidence_strict_input_valid_def
  proof (intro conjI)
    show "relation_catalog_well_formed chain_after_t5_a chain_relation_catalog"
    using relation_catalog_wf .
    show "catalogued_active_relation_origin chain_after_t5_a
      chain_relation_catalog (effect_origin_of chain_effect_b)"
    using catalogued_origin .
    show "integrate_evidence_input_valid chain_grammar chain_after_t5_a
      chain_effect_catalog_after_t5_a chain_object_archive_after_t5_a
      chain_payload_archive_after_t5_a chain_evidence_ledger_after_t5_a
      chain_effect_b chain_effect_payload_b chain_assessment_b 8 8 8"
    unfolding integrate_evidence_input_valid_def
    using state_wf history_disjoint effect_catalog_wf object_archive_wf
    payload_archive_wf ledger_wf chain_effect_b_member_after_t5_a
    chain_t5_a_store_keys_for_b chain_version_8_fresh_after_t5_a
    chain_effect_b_payload_valid origin_present origin_exists
    chain_evidence_source_2_after_t5_a chain_model_30_after_t5_a
    chain_effect_vector_b_valid local_conditions
    by (auto simp: chain_effect_b_def chain_effect_payload_b_def
      chain_assessment_b_def base_object_def)
  qed
qed

```

theorem chain\_t5\_b\_strict\_step:

```

  "integrate_evidence_strict_step chain_grammar chain_after_t5_a

```

```

chain_relation_catalog chain_effect_catalog_after_t5_a
chain_object_archive_after_t5_a chain_payload_archive_after_t5_a
chain_evidence_ledger_after_t5_a chain_effect_b chain_effect_payload_b
chain_assessment_b 8 8 8 chain_after_t5 chain_effect_catalog_after_t5
chain_object_archive chain_payload_archive chain_evidence_ledger"
unfolding integrate_evidence_strict_step_def integrate_evidence_step_def
integrate_evidence_strict_input_valid_def Let_def
chain_after_t5_def chain_effect_b_supported_def
chain_effect_catalog_after_t5_def chain_object_archive_def
chain_payload_archive_def chain_evidence_ledger_def
using chain_t5_b_strict_input_valid
by (simp add: integrate_evidence_strict_input_valid_def
    chain_effect_b_def base_object_def)

lemma chain_t5_b_base_step:
  "integrate_evidence_step chain_grammar chain_after_t5_a
    chain_effect_catalog_after_t5_a chain_object_archive_after_t5_a
    chain_payload_archive_after_t5_a chain_evidence_ledger_after_t5_a
    chain_effect_b chain_effect_payload_b chain_assessment_b 8 8 8
    chain_after_t5 chain_effect_catalog_after_t5 chain_object_archive
    chain_payload_archive chain_evidence_ledger"
  using integrate_evidence_strict_refines_base[OF chain_t5_b_strict_step] .

lemma chain_after_t4_history_empty:
  "historical_versions_of chain_after_t4 = {}"
  by (simp add: chain_after_t4_def chain_after_t4_a_def
    chain_after_t3_def chain_after_t3_ab_def chain_after_t2_def
    chain_after_t1_def chain_seed_def base_snapshot_def)

lemma chain_after_t5_a_history_exact:
  "historical_versions_of chain_after_t5_a = {5}"
  unfolding chain_after_t5_a_def replace_effect_version_def
  using chain_after_t4_history_empty
  by (simp add: chain_effect_a_def base_object_def)

lemma chain_after_t5_history_exact:
  "historical_versions_of chain_after_t5 = {5, 6}"
  unfolding chain_after_t5_def replace_effect_version_def
  using chain_after_t5_a_history_exact
  by (auto simp: chain_effect_b_def base_object_def)

theorem chain_t5_reachability_certificate:
  "integrate_evidence_strict_step chain_grammar chain_after_t4
    chain_relation_catalog chain_effect_catalog chain_object_archive_0
    chain_payload_archive_0 chain_evidence_ledger_0 chain_effect_a
    chain_effect_payload_a chain_assessment_a 7 7 7 chain_after_t5_a
    chain_effect_catalog_after_t5_a chain_object_archive_after_t5_a
    chain_payload_archive_after_t5_a chain_evidence_ledger_after_t5_a \<and>
    integrate_evidence_strict_step chain_grammar chain_after_t5_a
    chain_relation_catalog chain_effect_catalog_after_t5_a
    chain_object_archive_after_t5_a chain_payload_archive_after_t5_a
    chain_evidence_ledger_after_t5_a chain_effect_b chain_effect_payload_b
    chain_assessment_b 8 8 8 chain_after_t5
    chain_effect_catalog_after_t5 chain_object_archive
    chain_payload_archive chain_evidence_ledger \<and>
    state_well_formed chain_grammar chain_after_t5 \<and>
    relation_catalog_well_formed chain_after_t5 chain_relation_catalog \<and>
    effect_catalog_well_formed chain_after_t5
    chain_effect_catalog_after_t5 \<and>
    active_history_disjoint chain_after_t5 \<and>
    lifecycle_of chain_effect_a_supported = Supported \<and>
    lifecycle_of chain_effect_b_supported = Supported \<and>
    historical_versions_of chain_after_t5 = {5, 6} \<and>
    chain_object_archive 5 = Some chain_effect_a \<and>
    chain_object_archive 6 = Some chain_effect_b \<and>
    chain_effect_catalog_after_t5 7 =
      Some (integrated_effect_payload chain_effect_payload_a
        chain_assessment_a) \<and>
    chain_effect_catalog_after_t5 8 =
      Some (integrated_effect_payload chain_effect_payload_b
        chain_assessment_b)"
  using chain_t5_a_strict_step chain_t5_b_strict_step
  integrate_evidence_preserves_state_well_formed[OF chain_t5_b_base_step]
  integrate_evidence_preserves_audit_stores[OF chain_t5_b_base_step]
  integrate_evidence_preserves_active_catalog[OF chain_t5_b_base_step]
  integrate_evidence_strict_preserves_relation_catalog
    [OF chain_t5_b_strict_step]
  chain_after_t5_history_exact
  by (auto simp: chain_effect_a_supported_def chain_effect_b_supported_def
    chain_assessment_a_def chain_assessment_b_def chain_effect_a_def

```

```
chain_effect_b_def base_object_def integrated_lifecycle_def  
validation_ready_def chain_after_t5_def chain_after_t5_a_def  
chain_object_archive_def chain_object_archive_after_t5_a_def  
chain_object_archive_0_def extend_object_archive_def  
chain_effect_catalog_after_t5_def  
chain_effect_catalog_after_t5_a_def chain_effect_catalog_def  
chain_effect_catalog_a_def chain_effect_catalog_0_def  
replace_active_effect_payload_def extend_effect_catalog_def)
```

end

Beweisstatus dieser D.9-Fassung: kernelgeprüft und nicht-vakuos im ausgewiesenen Umfang. Der Zeugenhash lautet

f5e037e578335431903754153deaedeada461fed8d359a0511e36b59a825a7d. Bewiesen sind die tatsächliche Erfüllung der T5-Eingabebedingungen, der konkrete IntegrateEvidence-Schritt, Hypothesis → Supported, die exakte Audit-Differenz, Zielwohlgeformtheit und Refinement zur generischen step-Relation. Der Zeuge ist ein Existenz- und Ausführbarkeitsbeleg für genau diese endliche Konfiguration, kein empirischer Universalitätsbeweis.



## D.10 Gerichtete T6-Semantik und Erhaltung für ClassifyResonance

T6 operationalisiert die in §15 geforderte Richtung der Resonanz. Ein `resonance_model_profile` bindet eine Modell-ID, eine Matrix  $K: \text{effect\_dimension} \rightarrow \text{effect\_dimension} \rightarrow \text{real}$ , vier geordnete Formschwellen und eine Validierungsgrenze für Unsicherheit. Der Score summiert  $K(\text{source}, \text{target}) \cdot V_{\text{source}}(\text{source}) \cdot V_{\text{target}}(\text{target})$ , normalisiert durch  $9 \cdot \sum |K|$  und klemmt das Resultat auf  $[-1, 1]$ . Weil Quell- und Zieldimension in  $K$  verschiedene Argumentpositionen besitzen, ist Symmetrie weder vorausgesetzt noch impliziert.

Die Operation akzeptiert nur zwei verschiedene aktive Effektoobjekte mit typisierten V10-Vektoren im selben deklarierten Kontext. Perspektive, Kontext und Profilmodell müssen im Snapshot vorhanden sein. Begründungsreferenzen sind nichtleer, aktiv, von den beiden Effekten verschieden und vollständig auflösbar. Das Ergebnisobjekt und der Payload werden ausschließlich aus Request und Quellzustand konstruiert. Die Beweise leiten Objektfrische, exaktes Delta, Score- und Stärkengrenzen, totale Klassifikation, Source-/Target-Integrität, Payloadgültigkeit, Zielwohlgeformtheit, Identität, Katalogerhaltung und generisches step-Refinement her.

```
theory UZA_0_9_1_Classify_Resonance
  imports UZA_0_9_1_Integrate_Evidence
begin

section \<open>T6 ClassifyResonance: directed, model-bound resonance semantics\<close>

text \<open>
  T6 creates exactly one fresh K_RES object from an ordered pair of active
  effects. Direction is semantic, not merely documentary: a versioned
  interaction kernel maps source dimensions to target dimensions. Thus the
  score for A-to-B need not equal the score for B-to-A.

  Perspective, context, model, temporal horizon, rationale, uncertainty,
  thresholds, form, score, and strength are persisted in a typed side
  catalog. The result object and payload are constructed from the request;
  target-state well-formedness is derived and is not an input assumption.
\<close>

type_synonym resonance_kernel =
  "effect_dimension \<Rightarrow> effect_dimension \<Rightarrow> real"

record resonance_model_profile =
  resonance_profile_model_of :: model_id
  resonance_kernel_of :: resonance_kernel
  amplification_cutoff_of :: real
  complement_cutoff_of :: real
  productive_tension_cutoff_of :: real
  inhibition_cutoff_of :: real
  resonance_validation_uncertainty_max_of :: real

record resonance_request =
  requested_resonance_object_id_of :: object_id
  requested_resonance_version_id_of :: version_id
  requested_source_effect_of :: version_id
  requested_target_effect_of :: version_id
  requested_source_vector_of :: effect_vector
  requested_target_vector_of :: effect_vector
  requested_resonance_profile_of :: resonance_model_profile
  requested_resonance_perspective_of :: perspective_id
  requested_resonance_context_of :: context_id
  requested_resonance_horizon_of :: nat
  requested_resonance_uncertainty_of :: real
  requested_resonance_rationale_refs_of :: "version_id set"
  requested_resonance_rationale_of :: string
  requested_resonance_valid_from_of :: time_point

record resonance_payload =
  classified_source_effect_of :: version_id
  classified_target_effect_of :: version_id
  classified_source_vector_of :: effect_vector
```

```

classified_target_vector_of :: effect_vector
classified_resonance_profile_of :: resonance_model_profile
classified_resonance_perspective_of :: perspective_id
classified_resonance_context_of :: context_id
classified_resonance_form_of :: resonance_form
classified_resonance_score_of :: real
classified_resonance_strength_of :: real
classified_resonance_horizon_of :: nat
classified_resonance_uncertainty_of :: real
classified_resonance_rationale_refs_of :: "version_id set"
classified_resonance_rationale_of :: string

type_synonym resonance_catalog = "version_id \ $\rightarrow$  resonance_payload"

lemma effect_dimension_UNIV:
  "(UNIV :: effect_dimension set) =
   {Connection, Stability, Openness, Learning, Cooperation,
    Agency, Transparency, Freedom, Responsibility, Regeneration}"
proof (rule set_eqI)
  fix dimension :: effect_dimension
  show
    "dimension \ $\in$  UNIV  $\longleftrightarrow$ 
     dimension \ $\in$ 
     {Connection, Stability, Openness, Learning, Cooperation,
      Agency, Transparency, Freedom, Responsibility, Regeneration}"
  by (case_tac dimension) simp_all
qed

lemma effect_dimensions_finite [simp]:
  "finite (UNIV :: effect_dimension set)"
by (simp add: effect_dimension_UNIV)

definition resonance_kernel_mass :: "resonance_kernel \ $\rightarrow$  real" where
  "resonance_kernel_mass interaction_kernel =
   sum (\ $\lambda$ source_dimension.
    sum (\ $\lambda$ target_dimension.
     abs (interaction_kernel source_dimension target_dimension)) UNIV) UNIV"

definition resonance_profile_well_formed ::
  "resonance_model_profile \ $\rightarrow$  bool" where
  "resonance_profile_well_formed profile  $\longleftrightarrow$ 
   ( $\forall$ source_dimension target_dimension.
    -1  $\leq$  resonance_kernel_of profile source_dimension target_dimension  $\wedge$ 
     resonance_kernel_of profile source_dimension target_dimension  $\leq$  1)  $\wedge$ 
    resonance_kernel_mass (resonance_kernel_of profile)  $>$  0  $\wedge$ 
    -1  $\leq$  inhibition_cutoff_of profile  $\wedge$ 
    inhibition_cutoff_of profile  $<$  productive_tension_cutoff_of profile  $\wedge$ 
    productive_tension_cutoff_of profile  $<$  0  $\wedge$ 
    0  $<$  complement_cutoff_of profile  $\wedge$ 
    complement_cutoff_of profile  $<$  amplification_cutoff_of profile  $\wedge$ 
    amplification_cutoff_of profile  $\leq$  1  $\wedge$ 
    0  $\leq$  resonance_validation_uncertainty_max_of profile  $\wedge$ 
    resonance_validation_uncertainty_max_of profile  $\leq$  1"

definition clamp11 :: "real \ $\rightarrow$  real" where
  "clamp11 value = max (-1) (min 1 value)"

definition directional_resonance_raw_score ::
  "resonance_model_profile \ $\rightarrow$  effect_vector  $\rightarrow$ 
   effect_vector  $\rightarrow$  real" where
  "directional_resonance_raw_score profile source_vector target_vector =
   (let mass = resonance_kernel_mass (resonance_kernel_of profile) in
    if mass = 0 then 0
    else
     sum (\ $\lambda$ source_dimension.
      sum (\ $\lambda$ target_dimension.
       resonance_kernel_of profile source_dimension target_dimension *
        of_int (source_vector source_dimension *
         target_vector target_dimension)) UNIV) UNIV /
     (9 * mass))"

definition directional_resonance_score ::
  "resonance_model_profile \ $\rightarrow$  effect_vector  $\rightarrow$ 
   effect_vector  $\rightarrow$  real" where
  "directional_resonance_score profile source_vector target_vector =
   clamp11
    (directional_resonance_raw_score profile source_vector target_vector)"

definition classify_directional_score ::
  "resonance_model_profile \ $\rightarrow$  real  $\rightarrow$  resonance_form" where

```

```

"classify_directional_score profile score =
  (if score \<ge> amplification_cutoff_of profile then Amplification
   else if score \<ge> complement_cutoff_of profile then Complement
   else if score \<le> inhibition_cutoff_of profile then Inhibition
   else if score \<le> productive_tension_cutoff_of profile
     then ProductiveTension
   else Neutral)"

definition uncertainty_adjusted_resonance_strength ::
  "real \<Rightarrow> real \<Rightarrow> real" where
  "uncertainty_adjusted_resonance_strength score uncertainty =
    clamp01 (abs score * (1 - clamp01 uncertainty))"

definition effect_validated_at :: "snapshot \<Rightarrow> version_id \<Rightarrow> bool" where
  "effect_validated_at s version \<longlefttrightarrow>
    (\<exists>effect\<in>objects_of s.
     version_id_of effect = version \<and>
     kind_of effect = K_EFF \<and>
     lifecycle_of effect = Validated)"

definition active_classifiable_effect ::
  "snapshot \<Rightarrow> effect_catalog \<Rightarrow> version_id \<Rightarrow>
    effect_vector \<Rightarrow> context_id \<Rightarrow> bool" where
  "active_classifiable_effect s effect_catalog version vector context \<longlefttrightarrow>
    (\<exists>effect\<in>objects_of s.
     version_id_of effect = version \<and>
     kind_of effect = K_EFF \<and>
     lifecycle_of effect \<in> {Supported, Validated} \<and>
     context_of effect = Some context \<and>
     (\<exists>payload.
      effect_catalog version = Some payload \<and>
      hypothesis_vector_of payload = Some vector \<and>
      effect_payload_valid effect payload)))"

definition distinct_active_effect_objects ::
  "snapshot \<Rightarrow> version_id \<Rightarrow> version_id \<Rightarrow> bool" where
  "distinct_active_effect_objects s source target \<longlefttrightarrow>
    (\<exists>source_effect\<in>objects_of s. \<exists>target_effect\<in>objects_of s.
     version_id_of source_effect = source \<and>
     version_id_of target_effect = target \<and>
     object_id_of source_effect \<noteq> object_id_of target_effect)"

definition active_resonance_rationale ::
  "snapshot \<Rightarrow> version_id \<Rightarrow> version_id \<Rightarrow>
    version_id set \<Rightarrow> bool" where
  "active_resonance_rationale s source target rationale_refs \<longlefttrightarrow>
    rationale_refs \<noteq> {} \<and>
    rationale_refs \<inter> {source, target} = {} \<and>
    (\<forall>version\<in>rationale_refs.
     \<exists>obj\<in>objects_of s.
     version_id_of obj = version \<and>
     lifecycle_of obj \<notin> {Revised, Rejected, Archived})"

definition requested_directional_score :: "resonance_request \<Rightarrow> real" where
  "requested_directional_score request =
    directional_resonance_score
      (requested_resonance_profile_of request)
      (requested_source_vector_of request)
      (requested_target_vector_of request)"

definition requested_resonance_form ::
  "resonance_request \<Rightarrow> resonance_form" where
  "requested_resonance_form request =
    classify_directional_score
      (requested_resonance_profile_of request)
      (requested_directional_score request)"

definition requested_resonance_strength :: "resonance_request \<Rightarrow> real" where
  "requested_resonance_strength request =
    uncertainty_adjusted_resonance_strength
      (requested_directional_score request)
      (requested_resonance_uncertainty_of request)"

definition classified_resonance_lifecycle ::
  "snapshot \<Rightarrow> resonance_request \<Rightarrow> lifecycle_status" where
  "classified_resonance_lifecycle s request =
    (if effect_validated_at s (requested_source_effect_of request) \<and>
     effect_validated_at s (requested_target_effect_of request) \<and>
     requested_resonance_uncertainty_of request \<le>
     resonance_validation_uncertainty_max_of

```

```

        (requested_resonance_profile_of request)
    then Validated else Supported)"

definition requested_resonance_references ::
    "resonance_request \<Rightarrow> version_id set" where
    "requested_resonance_references request =
        insert (requested_source_effect_of request)
            (insert (requested_target_effect_of request)
                (requested_resonance_rationale_refs_of request))"

definition classified_resonance_object ::
    "snapshot \<Rightarrow> resonance_request \<Rightarrow> uza_object" where
    "classified_resonance_object s request =
        \<lparr>
            object_id_of = requested_resonance_object_id_of request,
            version_id_of = requested_resonance_version_id_of request,
            kind_of = K_RES,
            lifecycle_of = classified_resonance_lifecycle s request,
            previous_version_of = None,
            references_of = requested_resonance_references request,
            evidence_of = requested_resonance_rationale_refs_of request,
            perspective_of = Some (requested_resonance_perspective_of request),
            context_of = Some (requested_resonance_context_of request),
            model_of = Some
                (resonance_profile_model_of (requested_resonance_profile_of request)),
            authority_of = None,
            valid_from_of = requested_resonance_valid_from_of request,
            valid_to_of = None,
            source_ref_of = Some (requested_source_effect_of request),
            target_ref_of = Some (requested_target_effect_of request),
            effect_origin_of = None,
            basis_refs_of =
                {requested_source_effect_of request, requested_target_effect_of request},
            execution_ref_of = None,
            observation_refs_of = {},
            object_status_of = Some Applicable
        \<rparr>"

definition classified_resonance_payload ::
    "resonance_request \<Rightarrow> resonance_payload" where
    "classified_resonance_payload request =
        \<lparr>
            classified_source_effect_of = requested_source_effect_of request,
            classified_target_effect_of = requested_target_effect_of request,
            classified_source_vector_of = requested_source_vector_of request,
            classified_target_vector_of = requested_target_vector_of request,
            classified_resonance_profile_of = requested_resonance_profile_of request,
            classified_resonance_perspective_of =
                requested_resonance_perspective_of request,
            classified_resonance_context_of = requested_resonance_context_of request,
            classified_resonance_form_of = requested_resonance_form request,
            classified_resonance_score_of = requested_directional_score request,
            classified_resonance_strength_of = requested_resonance_strength request,
            classified_resonance_horizon_of = requested_resonance_horizon_of request,
            classified_resonance_uncertainty_of =
                requested_resonance_uncertainty_of request,
            classified_resonance_rationale_refs_of =
                requested_resonance_rationale_refs_of request,
            classified_resonance_rationale_of = requested_resonance_rationale_of request
        \<rparr>"

definition resonance_payload_valid ::
    "uza_object \<Rightarrow> resonance_payload \<Rightarrow> bool" where
    "resonance_payload_valid resonance payload \<longlefttrightarrow>
        kind_of resonance = K_RES \<and>
        lifecycle_of resonance \<in> {Supported, Validated} \<and>
        classified_source_effect_of payload \<noteq> classified_target_effect_of payload \<and>
        source_ref_of resonance = Some (classified_source_effect_of payload) \<and>
        target_ref_of resonance = Some (classified_target_effect_of payload) \<and>
        basis_refs_of resonance =
            {classified_source_effect_of payload, classified_target_effect_of payload} \<and>
        references_of resonance =
            insert (classified_source_effect_of payload)
                (insert (classified_target_effect_of payload)
                    (classified_resonance_rationale_refs_of payload)) \<and>
        evidence_of resonance = classified_resonance_rationale_refs_of payload \<and>
        perspective_of resonance =
            Some (classified_resonance_perspective_of payload) \<and>
        context_of resonance = Some (classified_resonance_context_of payload) \<and>
        model_of resonance = Some
    "
```

```

    (resonance_profile_model_of (classified_resonance_profile_of payload)) \<and>
    resonance_profile_well_formed (classified_resonance_profile_of payload) \<and>
    effect_vector_valid (classified_source_vector_of payload) \<and>
    effect_vector_valid (classified_target_vector_of payload) \<and>
    classified_resonance_score_of payload =
        directional_resonance_score
            (classified_resonance_profile_of payload)
            (classified_source_vector_of payload)
            (classified_target_vector_of payload) \<and>
    classified_resonance_form_of payload =
        classify_directional_score
            (classified_resonance_profile_of payload)
            (classified_resonance_score_of payload) \<and>
    classified_resonance_strength_of payload =
        uncertainty_adjusted_resonance_strength
            (classified_resonance_score_of payload)
            (classified_resonance_uncertainty_of payload) \<and>
    0 \<le> classified_resonance_strength_of payload \<and>
    classified_resonance_strength_of payload \<le> 1 \<and>
    classified_resonance_horizon_of payload > 0 \<and>
    0 \<le> classified_resonance_uncertainty_of payload \<and>
    classified_resonance_uncertainty_of payload \<le> 1 \<and>
    classified_resonance_rationale_refs_of payload \<noteq> {} \<and>
    classified_resonance_rationale_of payload \<noteq> [] \<and>
    object_status_of resonance = Some Applicable"

definition resonance_catalog_well_formed ::
    "snapshot \<Rightarrow> resonance_catalog \<Rightarrow> bool" where
    "resonance_catalog_well_formed s catalog \<longlefttrightarrow>
        (\<forall>version payload. catalog version = Some payload \<longrightarrow>
            (\<exists>resonance\<in>objects_of s.
                version_id_of resonance = version \<and>
                resonance_payload_valid resonance payload \<and>
                references_complete s resonance \<and>
                evidence_well_formed s resonance))"

definition classify_resonance_input_valid ::
    "grammar \<Rightarrow> snapshot \<Rightarrow> effect_catalog \<Rightarrow>
        resonance_catalog \<Rightarrow> resonance_request \<Rightarrow> nat \<Rightarrow> bool" where
    "classify_resonance_input_valid G s effect_catalog resonance_catalog request
        next_snapshot_id \<longlefttrightarrow>
        state_well_formed G s \<and>
        active_history_disjoint s \<and>
        effect_catalog_well_formed s effect_catalog \<and>
        resonance_catalog_well_formed s resonance_catalog \<and>
        ClassifyResonance \<in> allowed_operations_of G \<and>
        K_RES \<in> supported_kinds_of G \<and>
        requested_resonance_object_id_of request \<notin>
            object_id_of ` objects_of s \<and>
        requested_resonance_version_id_of request \<notin>
            version_ids s \<union> historical_versions_of s \<and>
        resonance_catalog (requested_resonance_version_id_of request) = None \<and>
        requested_source_effect_of request \<noteq> requested_target_effect_of request \<and>
        distinct_active_effect_objects s
            (requested_source_effect_of request)
            (requested_target_effect_of request) \<and>
        active_classifiable_effect s effect_catalog
            (requested_source_effect_of request)
            (requested_source_vector_of request)
            (requested_resonance_context_of request) \<and>
        active_classifiable_effect s effect_catalog
            (requested_target_effect_of request)
            (requested_target_vector_of request)
            (requested_resonance_context_of request) \<and>
        effect_vector_valid (requested_source_vector_of request) \<and>
        effect_vector_valid (requested_target_vector_of request) \<and>
        resonance_profile_well_formed (requested_resonance_profile_of request) \<and>
        resonance_profile_model_of (requested_resonance_profile_of request)
            \<in> models_of s \<and>
        requested_resonance_perspective_of request \<in> perspectives_of s \<and>
        requested_resonance_context_of request \<in> contexts_of s \<and>
        requested_resonance_horizon_of request > 0 \<and>
        0 \<le> requested_resonance_uncertainty_of request \<and>
        requested_resonance_uncertainty_of request \<le> 1 \<and>
        active_resonance_rationale s
            (requested_source_effect_of request)
            (requested_target_effect_of request)
            (requested_resonance_rationale_refs_of request) \<and>
        requested_resonance_rationale_of request \<noteq> [] \<and>
        snapshot_id_of s < next_snapshot_id"
```

```

definition add_classified_resonance ::
  "snapshot \<Rightarrow> resonance_request \<Rightarrow> nat \<Rightarrow> snapshot" where
  "add_classified_resonance s request next_snapshot_id =
    s\<lparr>
      snapshot_id_of := next_snapshot_id,
      objects_of := insert (classified_resonance_object s request) (objects_of s)
    \<rparr>"

definition extend_resonance_catalog ::
  "resonance_catalog \<Rightarrow> resonance_request \<Rightarrow> resonance_catalog" where
  "extend_resonance_catalog catalog request =
    catalog(requested_resonance_version_id_of request \<mapsto>
      classified_resonance_payload request)"

definition classify_resonance_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> effect_catalog \<Rightarrow>
    resonance_catalog \<Rightarrow> resonance_request \<Rightarrow> nat \<Rightarrow>
    snapshot \<Rightarrow> resonance_catalog \<Rightarrow> bool" where
  "classify_resonance_step G s effect_catalog resonance_catalog request
    next_snapshot_id s' resonance_catalog' \<longlefttrightarrow>
    classify_resonance_input_valid G s effect_catalog resonance_catalog request
    next_snapshot_id \<and>
    s' = add_classified_resonance s request next_snapshot_id \<and>
    resonance_catalog' = extend_resonance_catalog resonance_catalog request"

subsection \<open>Score, classification, constructor, and exact delta\<close>

lemma clamp11_bounded:
  "-1 \<le> clamp11 value \<and> clamp11 value \<le> 1"
  by (simp add: clamp11_def)

theorem directional_resonance_score_bounded:
  "-1 \<le> directional_resonance_score profile source_vector target_vector \<and>
    directional_resonance_score profile source_vector target_vector \<le> 1"
  by (simp add: directional_resonance_score_def clamp11_bounded)

theorem uncertainty_adjusted_resonance_strength_bounded:
  "0 \<le> uncertainty_adjusted_resonance_strength score uncertainty \<and>
    uncertainty_adjusted_resonance_strength score uncertainty \<le> 1"
  unfolding uncertainty_adjusted_resonance_strength_def clamp01_def
  by auto

theorem classify_directional_score_total:
  "classify_directional_score profile score
    \<in> {Amplification, Complement, ProductiveTension, Inhibition, Neutral}"
  by (auto simp: classify_directional_score_def)

lemma classified_lifecycle_cases:
  "classified_resonance_lifecycle s request \<in> {Supported, Validated}"
  by (auto simp: classified_resonance_lifecycle_def)

lemma add_resonance_objects [simp]:
  "objects_of (add_classified_resonance s request next_snapshot_id) =
    insert (classified_resonance_object s request) (objects_of s)"
  by (simp add: add_classified_resonance_def)

lemma add_resonance_snapshot_id [simp]:
  "snapshot_id_of (add_classified_resonance s request next_snapshot_id) =
    next_snapshot_id"
  by (simp add: add_classified_resonance_def)

lemma add_resonance_background [simp]:
  "historical_versions_of
    (add_classified_resonance s request next_snapshot_id) =
    historical_versions_of s"
  "perspectives_of (add_classified_resonance s request next_snapshot_id) =
    perspectives_of s"
  "contexts_of (add_classified_resonance s request next_snapshot_id) =
    contexts_of s"
  "models_of (add_classified_resonance s request next_snapshot_id) = models_of s"
  "authorities_of (add_classified_resonance s request next_snapshot_id) =
    authorities_of s"
  "contains_edges_of (add_classified_resonance s request next_snapshot_id) =
    contains_edges_of s"
  "projections_of (add_classified_resonance s request next_snapshot_id) =
    projections_of s"
  "couplings_of (add_classified_resonance s request next_snapshot_id) =
    couplings_of s"
  "transmissions_of (add_classified_resonance s request next_snapshot_id) =

```

```

    transmissions_of s"
"boundaries_of (add_classified_resonance s request next_snapshot_id) =
  boundaries_of s"
by (simp_all add: add_classified_resonance_def)

lemma add_resonance_version_ids [simp]:
  "version_ids (add_classified_resonance s request next_snapshot_id) =
    insert (requested_resonance_version_id_of request) (version_ids s)"
  by (auto simp: version_ids_def classified_resonance_object_def)

lemma classified_resonance_object_simps [simp]:
  "object_id_of (classified_resonance_object s request) =
    requested_resonance_object_id_of request"
  "version_id_of (classified_resonance_object s request) =
    requested_resonance_version_id_of request"
  "kind_of (classified_resonance_object s request) = K_RES"
  "lifecycle_of (classified_resonance_object s request) =
    classified_resonance_lifecycle s request"
  "previous_version_of (classified_resonance_object s request) = None"
  "references_of (classified_resonance_object s request) =
    requested_resonance_references request"
  "evidence_of (classified_resonance_object s request) =
    requested_resonance_rationale_refs_of request"
  "perspective_of (classified_resonance_object s request) =
    Some (requested_resonance_perspective_of request)"
  "context_of (classified_resonance_object s request) =
    Some (requested_resonance_context_of request)"
  "model_of (classified_resonance_object s request) =
    Some (resonance_profile_model_of (requested_resonance_profile_of request))"
  "valid_from_of (classified_resonance_object s request) =
    requested_resonance_valid_from_of request"
  "valid_to_of (classified_resonance_object s request) = None"
  "source_ref_of (classified_resonance_object s request) =
    Some (requested_source_effect_of request)"
  "target_ref_of (classified_resonance_object s request) =
    Some (requested_target_effect_of request)"
  "effect_origin_of (classified_resonance_object s request) = None"
  "basis_refs_of (classified_resonance_object s request) =
    {requested_source_effect_of request, requested_target_effect_of request}"
  "authority_of (classified_resonance_object s request) = None"
  "execution_ref_of (classified_resonance_object s request) = None"
  "observation_refs_of (classified_resonance_object s request) = {}"
  "object_status_of (classified_resonance_object s request) = Some Applicable"
  by (simp_all add: classified_resonance_object_def)

lemma classified_resonance_payload_simps [simp]:
  "classified_source_effect_of (classified_resonance_payload request) =
    requested_source_effect_of request"
  "classified_target_effect_of (classified_resonance_payload request) =
    requested_target_effect_of request"
  "classified_source_vector_of (classified_resonance_payload request) =
    requested_source_vector_of request"
  "classified_target_vector_of (classified_resonance_payload request) =
    requested_target_vector_of request"
  "classified_resonance_profile_of (classified_resonance_payload request) =
    requested_resonance_profile_of request"
  "classified_resonance_perspective_of (classified_resonance_payload request) =
    requested_resonance_perspective_of request"
  "classified_resonance_context_of (classified_resonance_payload request) =
    requested_resonance_context_of request"
  "classified_resonance_form_of (classified_resonance_payload request) =
    requested_resonance_form request"
  "classified_resonance_score_of (classified_resonance_payload request) =
    requested_directional_score request"
  "classified_resonance_strength_of (classified_resonance_payload request) =
    requested_resonance_strength request"
  "classified_resonance_horizon_of (classified_resonance_payload request) =
    requested_resonance_horizon_of request"
  "classified_resonance_uncertainty_of (classified_resonance_payload request) =
    requested_resonance_uncertainty_of request"
  "classified_resonance_rationale_refs_of
    (classified_resonance_payload request) =
    requested_resonance_rationale_refs_of request"
  "classified_resonance_rationale_of (classified_resonance_payload request) =
    requested_resonance_rationale_of request"
  by (simp_all add: classified_resonance_payload_def)

lemma active_classifiable_effect_version:
  assumes "active_classifiable_effect s effect_catalog version vector context"
  shows "version <in> version_ids s"

```

```

using assms
by (auto simp: active_classifiable_effect_def version_ids_def)

lemma active_rationale_versions:
  assumes "active_resonance_rationale s source target rationale_refs"
  shows "rationale_refs \<subseq> version_ids s"
  using assms
  by (auto simp: active_resonance_rationale_def version_ids_def)

lemma classify_resonance_fresh_object:
  assumes valid:
    "classify_resonance_input_valid G s effect_catalog resonance_catalog request
     next_snapshot_id"
  shows "classified_resonance_object s request \<notin> objects_of s"
proof
  assume member:
    "classified_resonance_object s request \<in> objects_of s"
  from valid have fresh_id:
    "requested_resonance_object_id_of request \<notin>
     object_id_of ` objects_of s"
  unfolding classify_resonance_input_valid_def by blast
  from member have
    "object_id_of (classified_resonance_object s request) \<in>
     object_id_of ` objects_of s"
  by blast
  then show False using fresh_id by simp
qed

theorem classify_resonance_exact_delta:
  assumes concrete:
    "classify_resonance_step G s effect_catalog resonance_catalog request
     next_snapshot_id s' resonance_catalog'"
  shows
    "objects_of s' =
     insert (classified_resonance_object s request) (objects_of s) \<and>
     classified_resonance_object s request \<notin> objects_of s \<and>
     snapshot_id_of s' = next_snapshot_id \<and>
     resonance_catalog' (requested_resonance_version_id_of request) =
       Some (classified_resonance_payload request)"
  using concrete classify_resonance_fresh_object
  by (auto simp: classify_resonance_step_def extend_resonance_catalog_def)

theorem classify_resonance_direction_explicit:
  assumes concrete:
    "classify_resonance_step G s effect_catalog resonance_catalog request
     next_snapshot_id s' resonance_catalog'"
  shows
    "source_ref_of (classified_resonance_object s request) =
     Some (requested_source_effect_of request) \<and>
     target_ref_of (classified_resonance_object s request) =
     Some (requested_target_effect_of request) \<and>
     requested_source_effect_of request \<noteq> requested_target_effect_of request \<and>
     classified_resonance_form_of (classified_resonance_payload request) =
       classify_directional_score
         (requested_resonance_profile_of request)
         (directional_resonance_score
           (requested_resonance_profile_of request)
           (requested_source_vector_of request)
           (requested_target_vector_of request))"
  using concrete
  by (auto simp: classify_resonance_step_def classify_resonance_input_valid_def
    requested_resonance_form_def requested_directional_score_def)

subsection \<open>Source integrity and payload derivation\<close>

lemma classified_resonance_source_integrity:
  assumes valid:
    "classify_resonance_input_valid G s effect_catalog resonance_catalog request
     next_snapshot_id"
  shows
    "references_complete s (classified_resonance_object s request) \<and>
     evidence_well_formed s (classified_resonance_object s request) \<and>
     time_well_formed (classified_resonance_object s request) \<and>
     perspective_well_formed s (classified_resonance_object s request) \<and>
     context_well_formed s (classified_resonance_object s request)"
proof -
  from valid have source_active:
    "active_classifiable_effect s effect_catalog
     (requested_source_effect_of request)
     (requested_source_vector_of request)"

```



```

    (requested_resonance_context_of request)"
  and target_active:
    "active_classifiable_effect s effect_catalog
    (requested_target_effect_of request)
    (requested_target_vector_of request)
    (requested_resonance_context_of request)"
  and rationale:
    "active_resonance_rationale s
    (requested_source_effect_of request)
    (requested_target_effect_of request)
    (requested_resonance_rationale_refs_of request)"
  and perspective:
    "requested_resonance_perspective_of request \<in> perspectives_of s"
  and context_ok:
    "requested_resonance_context_of request \<in> contexts_of s"
  unfolding classify_resonance_input_valid_def by blast+
have source_version:
  "requested_source_effect_of request \<in> version_ids s"
  using active_classifiable_effect_version[OF source_active] .
have target_version:
  "requested_target_effect_of request \<in> version_ids s"
  using active_classifiable_effect_version[OF target_active] .
have rationale_versions:
  "requested_resonance_rationale_refs_of request \<subseq> version_ids s"
  using active_rationale_versions[OF rationale] .
have refs:
  "references_complete s (classified_resonance_object s request)"
  using source_version target_version rationale_versions
  by (auto simp: references_complete_def requested_resonance_references_def)
have evidence:
  "evidence_well_formed s (classified_resonance_object s request)"
  using rationale rationale_versions classified_lifecycle_cases[of s request]
  by (auto simp: evidence_well_formed_def evidence_references_complete_def
    active_resonance_rationale_def)
have time:
  "time_well_formed (classified_resonance_object s request)"
  by (simp add: time_well_formed_def)
have perspective_wf:
  "perspective_well_formed s (classified_resonance_object s request)"
  using perspective
  by (simp add: perspective_well_formed_def perspective_bound_kind_def)
have context_wf:
  "context_well_formed s (classified_resonance_object s request)"
  using context_ok
  by (simp add: context_well_formed_def context_bound_kind_def)
show ?thesis using refs evidence time perspective_wf context_wf by blast
qed

lemma classified_resonance_payload_is_valid:
  assumes valid:
    "classify_resonance_input_valid G s effect_catalog resonance_catalog request
    next_snapshot_id"
  shows
    "resonance_payload_valid
    (classified_resonance_object s request)
    (classified_resonance_payload request)"
proof -
  from valid have distinct:
    "requested_source_effect_of request \<noteq> requested_target_effect_of request"
  and source_vector:
    "effect_vector_valid (requested_source_vector_of request)"
  and target_vector:
    "effect_vector_valid (requested_target_vector_of request)"
  and profile:
    "resonance_profile_well_formed (requested_resonance_profile_of request)"
  and horizon: "requested_resonance_horizon_of request > 0"
  and uncertainty:
    "0 \<le> requested_resonance_uncertainty_of request \<and>
    requested_resonance_uncertainty_of request \<le> 1"
  and rationale:
    "active_resonance_rationale s
    (requested_source_effect_of request)
    (requested_target_effect_of request)
    (requested_resonance_rationale_refs_of request)"
  and rationale_text: "requested_resonance_rationale_of request \<noteq> []"
  unfolding classify_resonance_input_valid_def by blast+
have strength:
  "0 \<le> requested_resonance_strength request \<and>
  requested_resonance_strength request \<le> 1"
  unfolding requested_resonance_strength_def

```

```

    using uncertainty_adjusted_resonance_strength_bounded by blast
show ?thesis
    using distinct source_vector target_vector profile horizon uncertainty
    rationale rationale_text strength classified_lifecycle_cases[of s request]
    by (auto simp: resonance_payload_valid_def requested_resonance_references_def
        requested_resonance_form_def requested_directional_score_def
        requested_resonance_strength_def active_resonance_rationale_def)
qed

subsection \<open>Monotonicity of inherited state conditions\<close>

lemma add_resonance_present_ref_mono:
  assumes "present_ref s ref"
  shows "present_ref (add_classified_resonance s request next_snapshot_id) ref"
  using assms
  by (cases ref) (auto simp: present_ref_def)

lemma add_resonance_references_mono:
  assumes "references_complete s obj"
  shows "references_complete
    (add_classified_resonance s request next_snapshot_id) obj"
  using assms
  by (auto simp: references_complete_def)

lemma add_resonance_evidence_mono:
  assumes "evidence_well_formed s obj"
  shows "evidence_well_formed
    (add_classified_resonance s request next_snapshot_id) obj"
  using assms
  by (auto simp: evidence_well_formed_def evidence_references_complete_def)

lemma add_resonance_relation_mono:
  assumes "relation_well_formed s obj"
  shows "relation_well_formed
    (add_classified_resonance s request next_snapshot_id) obj"
  using assms add_resonance_present_ref_mono
  unfolding relation_well_formed_def by blast

lemma add_resonance_effect_mono:
  assumes "effect_well_formed s obj"
  shows "effect_well_formed
    (add_classified_resonance s request next_snapshot_id) obj"
  using assms add_resonance_present_ref_mono
  unfolding effect_well_formed_def by blast

lemma add_resonance_perspective_iff [simp]:
  "perspective_well_formed
    (add_classified_resonance s request next_snapshot_id) obj
    \<longleftarrowrightarrow> perspective_well_formed s obj"
  unfolding perspective_well_formed_def
  by (simp only: add_resonance_background(2))

lemma add_resonance_context_iff [simp]:
  "context_well_formed
    (add_classified_resonance s request next_snapshot_id) obj
    \<longleftarrowrightarrow> context_well_formed s obj"
  unfolding context_well_formed_def
  by (simp only: add_resonance_background(3))

lemma add_resonance_relevance_iff [simp]:
  "relevance_well_formed
    (add_classified_resonance s request next_snapshot_id) obj
    \<longleftarrowrightarrow> relevance_well_formed s obj"
  unfolding relevance_well_formed_def
  by (simp only: add_resonance_perspective_iff add_resonance_context_iff
    add_resonance_background(4))

lemma add_resonance_orientation_iff [simp]:
  "orientation_well_formed
    (add_classified_resonance s request next_snapshot_id) obj
    \<longleftarrowrightarrow> orientation_well_formed s obj"
  by (simp add: orientation_well_formed_def)

lemma add_resonance_action_iff [simp]:
  "action_well_formed
    (add_classified_resonance s request next_snapshot_id) obj
    \<longleftarrowrightarrow> action_well_formed s obj"
  unfolding action_well_formed_def
  by (simp only: add_resonance_background(5))

```

```

lemma add_resonance_feedback_mono:
  assumes "feedback_well_formed s obj"
  shows "feedback_well_formed
    (add_classified_resonance s request next_snapshot_id) obj"
  using assms add_resonance_present_ref_mono
  unfolding feedback_well_formed_def by blast

lemma add_resonance_preserves_active_history_disjoint:
  assumes valid:
    "classify_resonance_input_valid G s effect_catalog resonance_catalog request
      next_snapshot_id"
  shows "active_history_disjoint
    (add_classified_resonance s request next_snapshot_id)"
  using valid
  by (auto simp: classify_resonance_input_valid_def active_history_disjoint_def)

subsection \<open>Fresh identity and target-state preservation\<close>

lemma add_resonance_preserves_unique_versions:
  assumes valid:
    "classify_resonance_input_valid G s effect_catalog resonance_catalog request
      next_snapshot_id"
  shows "unique_versions
    (add_classified_resonance s request next_snapshot_id)"
proof -
  let ?new = "classified_resonance_object s request"
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
    and fresh: "version_id_of ?new \<notin> version_ids s"
    unfolding classify_resonance_input_valid_def state_well_formed_def
    unique_versions_def
    by auto
  show ?thesis
    unfolding unique_versions_def
  proof (rule inj_onI)
    fix left right
    assume left_mem:
      "left \<in> objects_of (add_classified_resonance s request next_snapshot_id)"
      and right_mem:
      "right \<in> objects_of (add_classified_resonance s request next_snapshot_id)"
      and same_version: "version_id_of left = version_id_of right"
    show "left = right"
    proof (cases "left = ?new")
      case True
      show ?thesis
      proof (cases "right = ?new")
        case True
        then show ?thesis using \<open>left = ?new\<close> by simp
      next
        case False
        then have right_old: "right \<in> objects_of s" using right_mem by simp
        have "version_id_of ?new \<in> version_ids s"
          unfolding version_ids_def
          using same_version \<open>left = ?new\<close> right_old by (metis imageI)
        then show ?thesis using fresh by blast
      qed
    next
      case False
      then have left_old: "left \<in> objects_of s" using left_mem by simp
      show ?thesis
      proof (cases "right = ?new")
        case True
        have "version_id_of ?new \<in> version_ids s"
          unfolding version_ids_def
          using same_version left_old True by (metis imageI)
        then show ?thesis using fresh by blast
      next
        case False
        then have right_old: "right \<in> objects_of s" using right_mem by simp
        show ?thesis
          using inj_onD[OF source_unique same_version left_old right_old] .
      qed
    qed
  qed
qed
qed
qed

lemma classify_resonance_new_object_conditions:
  assumes valid:
    "classify_resonance_input_valid G s effect_catalog resonance_catalog request
      next_snapshot_id"
  shows

```

```

"kind_of (classified_resonance_object s request) \<in> supported_kinds_of G \<and>
references_complete
  (add_classified_resonance s request next_snapshot_id)
  (classified_resonance_object s request) \<and>
evidence_well_formed
  (add_classified_resonance s request next_snapshot_id)
  (classified_resonance_object s request) \<and>
time_well_formed (classified_resonance_object s request) \<and>
relation_well_formed
  (add_classified_resonance s request next_snapshot_id)
  (classified_resonance_object s request) \<and>
effect_well_formed
  (add_classified_resonance s request next_snapshot_id)
  (classified_resonance_object s request) \<and>
perspective_well_formed
  (add_classified_resonance s request next_snapshot_id)
  (classified_resonance_object s request) \<and>
context_well_formed
  (add_classified_resonance s request next_snapshot_id)
  (classified_resonance_object s request) \<and>
relevance_well_formed
  (add_classified_resonance s request next_snapshot_id)
  (classified_resonance_object s request) \<and>
orientation_well_formed
  (add_classified_resonance s request next_snapshot_id)
  (classified_resonance_object s request) \<and>
action_well_formed
  (add_classified_resonance s request next_snapshot_id)
  (classified_resonance_object s request) \<and>
feedback_well_formed
  (add_classified_resonance s request next_snapshot_id)
  (classified_resonance_object s request)"
proof -
  let ?new = "classified_resonance_object s request"
  from valid have supported: "K_RES \<in> supported_kinds_of G"
  unfolding classify_resonance_input_valid_def by blast
  from classified_resonance_source_integrity[OF valid] have refs:
    "references_complete s ?new"
  and evidence: "evidence_well_formed s ?new"
  and time: "time_well_formed ?new"
  and perspective: "perspective_well_formed s ?new"
  and context_ok: "context_well_formed s ?new"
  by blast+
  have refs_target:
    "references_complete (add_classified_resonance s request next_snapshot_id) ?new"
  using add_resonance_references_mono[OF refs] .
  have evidence_target:
    "evidence_well_formed (add_classified_resonance s request next_snapshot_id) ?new"
  using add_resonance_evidence_mono[OF evidence] .
  have typed_conditions:
    "relation_well_formed
      (add_classified_resonance s request next_snapshot_id) ?new \<and>
    effect_well_formed
      (add_classified_resonance s request next_snapshot_id) ?new \<and>
    relevance_well_formed
      (add_classified_resonance s request next_snapshot_id) ?new \<and>
    orientation_well_formed
      (add_classified_resonance s request next_snapshot_id) ?new \<and>
    action_well_formed
      (add_classified_resonance s request next_snapshot_id) ?new \<and>
    feedback_well_formed
      (add_classified_resonance s request next_snapshot_id) ?new"
  by (simp add: relation_well_formed_def effect_well_formed_def
    relevance_well_formed_def orientation_well_formed_def
    action_well_formed_def feedback_well_formed_def)
  show ?thesis
  using supported_refs_target evidence_target time perspective context_ok
    typed_conditions
  by simp
qed

subsection \<open>Operation-specific preservation theorems\<close>

theorem classify_resonance_preserves_state_well_formed:
  assumes concrete:
    "classify_resonance_step G s effect_catalog resonance_catalog request
      next_snapshot_id s' resonance_catalog"
  shows "state_well_formed G s'"
proof -
  from concrete have valid:

```

```

"classify_resonance_input_valid G s effect_catalog resonance_catalog request
  next_snapshot_id"
and target: "s' = add_classified_resonance s request next_snapshot_id"
by (auto simp: classify_resonance_step_def)
from valid have source_wf: "state_well_formed G s"
by (simp add: classify_resonance_input_valid_def)
have unique:
  "unique_versions (add_classified_resonance s request next_snapshot_id)"
using add_resonance_preserves_unique_versions[OF valid] .
have all_objects:
  "\<forall>obj\<in>objects_of (add_classified_resonance s request next_snapshot_id).
  kind_of obj \<in> supported_kinds_of G \<and>
  references_complete
  (add_classified_resonance s request next_snapshot_id) obj \<and>
  evidence_well_formed
  (add_classified_resonance s request next_snapshot_id) obj \<and>
  time_well_formed obj \<and>
  relation_well_formed
  (add_classified_resonance s request next_snapshot_id) obj \<and>
  effect_well_formed
  (add_classified_resonance s request next_snapshot_id) obj \<and>
  perspective_well_formed
  (add_classified_resonance s request next_snapshot_id) obj \<and>
  context_well_formed
  (add_classified_resonance s request next_snapshot_id) obj \<and>
  relevance_well_formed
  (add_classified_resonance s request next_snapshot_id) obj \<and>
  orientation_well_formed
  (add_classified_resonance s request next_snapshot_id) obj \<and>
  action_well_formed
  (add_classified_resonance s request next_snapshot_id) obj \<and>
  feedback_well_formed
  (add_classified_resonance s request next_snapshot_id) obj"
proof (intro ballI)
  fix obj
  assume obj_mem:
    "obj \<in> objects_of (add_classified_resonance s request next_snapshot_id)"
  consider (new) "obj = classified_resonance_object s request"
  | (old) "obj \<in> objects_of s"
  using obj_mem by auto
  then show
    "kind_of obj \<in> supported_kinds_of G \<and>
    references_complete
    (add_classified_resonance s request next_snapshot_id) obj \<and>
    evidence_well_formed
    (add_classified_resonance s request next_snapshot_id) obj \<and>
    time_well_formed obj \<and>
    relation_well_formed
    (add_classified_resonance s request next_snapshot_id) obj \<and>
    effect_well_formed
    (add_classified_resonance s request next_snapshot_id) obj \<and>
    perspective_well_formed
    (add_classified_resonance s request next_snapshot_id) obj \<and>
    context_well_formed
    (add_classified_resonance s request next_snapshot_id) obj \<and>
    relevance_well_formed
    (add_classified_resonance s request next_snapshot_id) obj \<and>
    orientation_well_formed
    (add_classified_resonance s request next_snapshot_id) obj \<and>
    action_well_formed
    (add_classified_resonance s request next_snapshot_id) obj \<and>
    feedback_well_formed
    (add_classified_resonance s request next_snapshot_id) obj"
  proof cases
    case new
    then show ?thesis
      using classify_resonance_new_object_conditions[OF valid] by simp
  next
    case old
    from source_wf old have source_conditions:
      "kind_of obj \<in> supported_kinds_of G \<and>
      references_complete s obj \<and> evidence_well_formed s obj \<and>
      time_well_formed obj \<and> relation_well_formed s obj \<and>
      effect_well_formed s obj \<and> perspective_well_formed s obj \<and>
      context_well_formed s obj \<and> relevance_well_formed s obj \<and>
      orientation_well_formed s obj \<and> action_well_formed s obj \<and>
      feedback_well_formed s obj"
    unfolding state_well_formed_def by blast
    from source_conditions have supported:
      "kind_of obj \<in> supported_kinds_of G"

```

```

and refs: "references_complete s obj"
and evidence: "evidence_well_formed s obj"
and time: "time_well_formed obj"
and relation: "relation_well_formed s obj"
and effect: "effect_well_formed s obj"
and perspective: "perspective_well_formed s obj"
and context_ok: "context_well_formed s obj"
and relevance: "relevance_well_formed s obj"
and orientation: "orientation_well_formed s obj"
and action: "action_well_formed s obj"
and feedback: "feedback_well_formed s obj"
by blast+
have refs_target:
  "references_complete
  (add_classified_resonance s request next_snapshot_id) obj"
  using add_resonance_references_mono[OF refs] .
have evidence_target:
  "evidence_well_formed
  (add_classified_resonance s request next_snapshot_id) obj"
  using add_resonance_evidence_mono[OF evidence] .
have relation_target:
  "relation_well_formed
  (add_classified_resonance s request next_snapshot_id) obj"
  using add_resonance_relation_mono[OF relation] .
have effect_target:
  "effect_well_formed
  (add_classified_resonance s request next_snapshot_id) obj"
  using add_resonance_effect_mono[OF effect] .
have feedback_target:
  "feedback_well_formed
  (add_classified_resonance s request next_snapshot_id) obj"
  using add_resonance_feedback_mono[OF feedback] .
show ?thesis
  using supported refs_target evidence_target time relation_target
    effect_target perspective context_ok relevance orientation action
    feedback_target
  by simp
qed
qed
from source_wf have source_secondary:
  "(\<forall>p\<in>projections_of s. projection_well_formed s p) \<and>
  (\<forall>c\<in>couplings_of s. coupling_well_formed s c) \<and>
  (\<forall>t\<in>transmissions_of s. transmission_well_formed s t) \<and>
  (\<forall>b\<in>boundaries_of s. boundary_well_formed b)"
  unfolding state_well_formed_def by blast
have target_secondary:
  "(\<forall>p\<in>projections_of
  (add_classified_resonance s request next_snapshot_id).
  projection_well_formed
  (add_classified_resonance s request next_snapshot_id) p) \<and>
  (\<forall>c\<in>couplings_of
  (add_classified_resonance s request next_snapshot_id).
  coupling_well_formed
  (add_classified_resonance s request next_snapshot_id) c) \<and>
  (\<forall>t\<in>transmissions_of
  (add_classified_resonance s request next_snapshot_id).
  transmission_well_formed
  (add_classified_resonance s request next_snapshot_id) t) \<and>
  (\<forall>b\<in>boundaries_of
  (add_classified_resonance s request next_snapshot_id).
  boundary_well_formed b)"
  using source_secondary
  by (simp add: projection_well_formed_def coupling_well_formed_def
    transmission_well_formed_def)
show ?thesis
  unfolding target state_well_formed_def
  using unique_all_objects target_secondary by blast
qed

theorem classify_resonance_preserves_identity:
  assumes concrete:
    "classify_resonance_step G s effect_catalog resonance_catalog request
    next_snapshot_id s' resonance_catalog"
  shows "unique_versions s' \<and> immutable_preserved s s'"
proof -
  let ?new = "classified_resonance_object s request"
  from concrete have valid:
    "classify_resonance_input_valid G s effect_catalog resonance_catalog request
    next_snapshot_id"
  and target: "s' = add_classified_resonance s request next_snapshot_id"

```

```

    by (auto simp: classify_resonance_step_def)
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
  and fresh: "version_id_of ?new \<notin> version_ids s"
  unfolding classify_resonance_input_valid_def state_well_formed_def
  unique_versions_def
  by auto
  have immutable: "immutable_preserved s s'"
  unfolding immutable_preserved_def target
  proof (intro ballI allI impI)
    fix old_obj target_obj
    assume old_mem: "old_obj \<in> objects_of s"
    and target_mem:
      "target_obj \<in>
        objects_of (add_classified_resonance s request next_snapshot_id)"
    and same_version:
      "version_id_of old_obj = version_id_of target_obj"
    show "old_obj = target_obj"
    proof (cases "target_obj = ?new")
      case True
      have "version_id_of ?new \<in> version_ids s"
      unfolding version_ids_def
      using old_mem same_version True by (metis imageI)
      then show ?thesis using fresh by blast
    next
      case False
      then have target_old: "target_obj \<in> objects_of s"
      using target_mem by simp
      show ?thesis
      using inj_onD[OF source_unique same_version old_mem target_old] .
    qed
  qed
  have target_unique: "unique_versions s'"
  unfolding target
  using add_resonance_preserves_unique_versions[OF valid] .
  show ?thesis using target_unique immutable by blast
qed

theorem classify_resonance_preserves_effect_catalog:
  assumes concrete:
    "classify_resonance_step G s effect_catalog resonance_catalog request
      next_snapshot_id s' resonance_catalog'"
  shows "effect_catalog_well_formed s' effect_catalog"
  proof -
    from concrete have valid:
      "classify_resonance_input_valid G s effect_catalog resonance_catalog request
        next_snapshot_id"
    and target: "s' = add_classified_resonance s request next_snapshot_id"
    by (auto simp: classify_resonance_step_def)
  from valid have source_catalog:
    "effect_catalog_well_formed s effect_catalog"
  by (simp add: classify_resonance_input_valid_def)
  show ?thesis
  unfolding effect_catalog_well_formed_def target
  using source_catalog
  unfolding effect_catalog_well_formed_def
  by auto
qed

theorem classify_resonance_preserves_catalog:
  assumes concrete:
    "classify_resonance_step G s effect_catalog resonance_catalog request
      next_snapshot_id s' resonance_catalog'"
  shows
    "resonance_catalog_well_formed s' resonance_catalog' \<and>
      resonance_catalog' (requested_resonance_version_id_of request) =
        Some (classified_resonance_payload request)"
  proof -
    from concrete have valid:
      "classify_resonance_input_valid G s effect_catalog resonance_catalog request
        next_snapshot_id"
    and target: "s' = add_classified_resonance s request next_snapshot_id"
    and catalog_target:
      "resonance_catalog' = extend_resonance_catalog resonance_catalog request"
    by (auto simp: classify_resonance_step_def)
  from valid have source_catalog:
    "resonance_catalog_well_formed s resonance_catalog"
  and fresh_key:
    "resonance_catalog (requested_resonance_version_id_of request) = None"
  by (auto simp: classify_resonance_input_valid_def)
  have payload_valid:

```

```

"resonance_payload_valid
(classified_resonance_object s request)
(classified_resonance_payload request)"
using classified_resonance_payload_is_valid[OF valid] .
from classified_resonance_source_integrity[OF valid] have source_refs:
"references_complete s (classified_resonance_object s request)"
and source_evidence:
"evidence_well_formed s (classified_resonance_object s request)"
by blast+
have new_refs:
"references_complete s' (classified_resonance_object s request)"
unfolding target
using add_resonance_references_mono[OF source_refs] .
have new_evidence:
"evidence_well_formed s' (classified_resonance_object s request)"
unfolding target
using add_resonance_evidence_mono[OF source_evidence] .
have target_catalog: "resonance_catalog_well_formed s' resonance_catalog'"
unfolding resonance_catalog_well_formed_def catalog_target
extend_resonance_catalog_def
proof (intro allI impI)
fix version payload
assume updated:
"(resonance_catalog
(requested_resonance_version_id_of request \<mapsto>
classified_resonance_payload request)) version = Some payload"
show "\<exists>resonance\<in>objects_of s'.
version_id_of resonance = version \<and>
resonance_payload_valid resonance payload \<and>
references_complete s' resonance \<and>
evidence_well_formed s' resonance"
proof (cases "version = requested_resonance_version_id_of request")
case True
with updated have payload_eq:
"payload = classified_resonance_payload request"
by simp
have new_mem:
"classified_resonance_object s request \<in> objects_of s'"
unfolding target by simp
show ?thesis
proof (rule bexI[where x = "classified_resonance_object s request"])
show
"version_id_of (classified_resonance_object s request) = version \<and>
resonance_payload_valid (classified_resonance_object s request)
payload \<and>
references_complete s' (classified_resonance_object s request) \<and>
evidence_well_formed s' (classified_resonance_object s request)"
using True payload_eq payload_valid new_refs new_evidence by simp
show "classified_resonance_object s request \<in> objects_of s'"
using new_mem .
qed
next
case False
with updated have old_entry: "resonance_catalog version = Some payload"
by simp
from source_catalog old_entry obtain resonance where
old_mem: "resonance \<in> objects_of s"
and version: "version_id_of resonance = version"
and payload: "resonance_payload_valid resonance payload"
and refs: "references_complete s resonance"
and evidence: "evidence_well_formed s resonance"
unfolding resonance_catalog_well_formed_def by blast
have refs_target: "references_complete s' resonance"
unfolding target using add_resonance_references_mono[OF refs] .
have evidence_target: "evidence_well_formed s' resonance"
unfolding target using add_resonance_evidence_mono[OF evidence] .
have old_mem_target: "resonance \<in> objects_of s'"
unfolding target using old_mem by simp
show ?thesis
proof (rule bexI[where x = resonance])
show
"version_id_of resonance = version \<and>
resonance_payload_valid resonance payload \<and>
references_complete s' resonance \<and>
evidence_well_formed s' resonance"
using version payload refs_target evidence_target by blast
show "resonance \<in> objects_of s'" using old_mem_target .
qed
qed
qed
qed

```



```

show ?thesis
  using target_catalog catalog_target
  by (simp add: extend_resonance_catalog_def)
qed

theorem classify_resonance_preserves_references_time_context:
  assumes concrete:
    "classify_resonance_step G s effect_catalog resonance_catalog request
     next_snapshot_id s' resonance_catalog"
  shows
    "\<forall>obj\<in>objects_of s'.
     references_complete s' obj \<and>
     evidence_well_formed s' obj \<and>
     time_well_formed obj \<and>
     perspective_well_formed s' obj \<and>
     context_well_formed s' obj"
  using classify_resonance_preserves_state_well_formed[OF concrete]
  unfolding state_well_formed_def by blast

theorem classify_resonance_refines_generic_step:
  assumes concrete:
    "classify_resonance_step G s effect_catalog resonance_catalog request
     next_snapshot_id s' resonance_catalog"
  shows "step G s ClassifyResonance s'"
proof -
  from concrete have source_wf: "state_well_formed G s"
  and allowed: "ClassifyResonance \<in> allowed_operations_of G"
  by (auto simp: classify_resonance_step_def
    classify_resonance_input_valid_def)
  have target_wf: "state_well_formed G s'"
  using classify_resonance_preserves_state_well_formed[OF concrete] .
  have immutable: "immutable_preserved s s'"
  using classify_resonance_preserves_identity[OF concrete] by blast
  show ?thesis
  unfolding step_def
  using source_wf target_wf allowed immutable by blast
qed

end

```

Beweisstatus dieser D.10-Fassung: kernelgeprüft. Der Isabelle2025-2-Fresh-Profile-Build  
verarbeitete die vollständige elfteilige Session und endete mit Exit-Code 0. T6 enthält 1.289  
Quellzeilen und keine Vorkommen von sorry, oops, admit, axiomatization oder quick\_and\_dirty.  
Der Quellhash lautet

3f3583b42f066eaf41208a22bcb53a2d49baed374aa1b41e00a434256741b1cd. Belegt sind die  
formulierten konstruktiven Rechen-, Delta-, Katalog-, Zustands-, Identitäts-, Referenz-, Evidenz-,  
Zeit-, Perspektiv-, Kontext- und Refinementsätze. Nicht belegt sind empirische Richtigkeit eines  
Resonanzprofils, Vollständigkeit der Formen oder Schwellen, allgemeine Konservativität oder  
Produktivengine-Äquivalenz.

## D.11 Historischer isolierter Richtungszeuge für T6 (lokal weiterhin gültig)

Der endliche Zeuge verhindert, dass T6 nur als konditionales Schema mit unerfüllten Vorbedingungen bestehen bleibt. Der Ausgangszustand enthält zwei Erscheinungen, zwei gerichtete Relationsursachen und zwei verschiedene validierte Effekte samt auflösbarer Evidenz, Perspektive, Kontext, Modell und konkreten V10-Payloads. Der Kernel besitzt genau einen Nichtnull-Eintrag  $K(\text{Connection}, \text{Stability})=1$ . Der Quellvektor trägt  $\text{Connection}=3$ , der Zielvektor  $\text{Stability}=3$ .

Mechanisch bewiesen werden  $\text{kernel\_mass}=1$ ,  $\text{forward\_weighted\_sum}=9$ ,  $\text{reverse\_weighted\_sum}=0$ ,  $\text{forward\_score}=1$ ,  $\text{reverse\_score}=0$ , Amplifikation in Vorwärtsrichtung, Neutral in Rückwärtsrichtung, Stärke 9/10, Lifecycle Validated, die tatsächliche Erfüllung aller T6-Eingabebedingungen, der konkrete ClassifyResonance-Schritt, Zielzustands- und Katalogwohlgeformtheit, generisches step-Refinement sowie die exakte Einfügung von Version 7 in Snapshot 2.

```
theory UZA_0_9_1_Classify_Resonance_Witness
  imports UZA_0_9_1_Classify_Resonance UZA_0_9_1_Base_Model
begin

section \<open>A non-vacuous directional witness for T6\<close>

text \<open>
  This finite witness activates the substantive T6 antecedents with two
  distinct validated effects and a non-symmetric, versioned interaction
  kernel. The kernel maps Connection in the source to Stability in the
  target. Consequently A-to-B has score 1 and form Amplification, whereas
  B-to-A has score 0 and form Neutral. The witness therefore establishes
  semantic directionality rather than merely storing ordered references.
\<close>

definition t6_source_vector :: effect_vector where
  "t6_source_vector dimension = (if dimension = Connection then 3 else 0)"

definition t6_target_vector :: effect_vector where
  "t6_target_vector dimension = (if dimension = Stability then 3 else 0)"

definition t6_directional_kernel :: resonance_kernel where
  "t6_directional_kernel source_dimension target_dimension =
    (if source_dimension = Connection \& target_dimension = Stability
     then 1 else 0)"

definition t6_resonance_profile :: resonance_model_profile where
  "t6_resonance_profile =
    \<lparr>
      resonance_profile_model_of = 30,
      resonance_kernel_of = t6_directional_kernel,
      amplification_cutoff_of = 1 / 2,
      complement_cutoff_of = 3 / 20,
      productive_tension_cutoff_of = -3 / 20,
      inhibition_cutoff_of = -1 / 2,
      resonance_validation_uncertainty_max_of = 1 / 4
    \<rparr>"

definition t6_entity_a :: uza_object where
  "t6_entity_a =
    base_object\<lparr>
      object_id_of := 1,
      version_id_of := 1,
      kind_of := K_ENT,
      lifecycle_of := Draft,
      perspective_of := Some 10,
      context_of := Some 20,
      object_status_of := Some Applicable
    \<rparr>"

definition t6_entity_b :: uza_object where
  "t6_entity_b =
```

```
base_object\<lparr>
  object_id_of := 2,
  version_id_of := 2,
  kind_of := K_ENT,
  lifecycle_of := Draft,
  perspective_of := Some 10,
  context_of := Some 20,
  object_status_of := Some Applicable
\<rparr>"

definition t6_relation_a :: uza_object where
  "t6_relation_a =
    base_object\<lparr>
      object_id_of := 3,
      version_id_of := 3,
      kind_of := K_REL,
      lifecycle_of := Hypothesis,
      references_of := {1, 2},
      context_of := Some 20,
      source_ref_of := Some 1,
      target_ref_of := Some 2,
      object_status_of := Some Applicable
    \<rparr>"

definition t6_effect_a :: uza_object where
  "t6_effect_a =
    base_object\<lparr>
      object_id_of := 4,
      version_id_of := 4,
      kind_of := K_EFF,
      lifecycle_of := Validated,
      references_of := {1, 3},
      evidence_of := {1},
      perspective_of := Some 10,
      context_of := Some 20,
      model_of := Some 30,
      source_ref_of := Some 3,
      effect_origin_of := Some 3,
      basis_refs_of := {3},
      object_status_of := Some Applicable
    \<rparr>"

definition t6_relation_b :: uza_object where
  "t6_relation_b =
    base_object\<lparr>
      object_id_of := 5,
      version_id_of := 5,
      kind_of := K_REL,
      lifecycle_of := Hypothesis,
      references_of := {1, 2},
      context_of := Some 20,
      source_ref_of := Some 2,
      target_ref_of := Some 1,
      object_status_of := Some Applicable
    \<rparr>"

definition t6_effect_b :: uza_object where
  "t6_effect_b =
    base_object\<lparr>
      object_id_of := 6,
      version_id_of := 6,
      kind_of := K_EFF,
      lifecycle_of := Validated,
      references_of := {2, 5},
      evidence_of := {2},
      perspective_of := Some 10,
      context_of := Some 20,
      model_of := Some 30,
      source_ref_of := Some 5,
      effect_origin_of := Some 5,
      basis_refs_of := {5},
      object_status_of := Some Applicable
    \<rparr>"

definition t6_witness_objects :: "uza_object set" where
  "t6_witness_objects =
    {t6_entity_a, t6_entity_b, t6_relation_a,
     t6_effect_a, t6_relation_b, t6_effect_b}"

definition t6_witness_source :: snapshot where
```

```

"t6_witness_source =
  \<lparr>
    snapshot_id_of = 1,
    objects_of = t6_witness_objects,
    historical_versions_of = {},
    perspectives_of = {10},
    contexts_of = {20},
    models_of = {30},
    authorities_of = {},
    contains_edges_of = {},
    projections_of = {},
    couplings_of = {},
    transmissions_of = {},
    boundaries_of = {}
  \<rparr>"

definition t6_witness_grammar :: grammar where
"t6_witness_grammar =
  \<lparr>
    grammar_id_of = 6,
    supported_kinds_of = {K_ENT, K_REL, K_EFF, K_RES},
    allowed_operations_of = {ClassifyResonance},
    patch_budget_of = 0
  \<rparr>"

definition t6_effect_payload_a :: effect_hypothesis_payload where
"t6_effect_payload_a =
  \<lparr>
    hypothesis_cause_of = 3,
    hypothesis_target_of = None,
    hypothesis_vector_of = Some t6_source_vector,
    hypothesis_uncertainty_of = 1 / 10,
    hypothesis_profile_model_of = 30
  \<rparr>"

definition t6_effect_payload_b :: effect_hypothesis_payload where
"t6_effect_payload_b =
  \<lparr>
    hypothesis_cause_of = 5,
    hypothesis_target_of = None,
    hypothesis_vector_of = Some t6_target_vector,
    hypothesis_uncertainty_of = 1 / 10,
    hypothesis_profile_model_of = 30
  \<rparr>"

definition t6_effect_catalog :: effect_catalog where
"t6_effect_catalog =
  Map.empty(4 \<mapsto> t6_effect_payload_a, 6 \<mapsto> t6_effect_payload_b)"

definition t6_source_resonance_catalog :: resonance_catalog where
"t6_source_resonance_catalog = Map.empty"

definition t6_request :: resonance_request where
"t6_request =
  \<lparr>
    requested_resonance_object_id_of = 7,
    requested_resonance_version_id_of = 7,
    requested_source_effect_of = 4,
    requested_target_effect_of = 6,
    requested_source_vector_of = t6_source_vector,
    requested_target_vector_of = t6_target_vector,
    requested_resonance_profile_of = t6_resonance_profile,
    requested_resonance_perspective_of = 10,
    requested_resonance_context_of = 20,
    requested_resonance_horizon_of = 4,
    requested_resonance_uncertainty_of = 1 / 10,
    requested_resonance_rationale_refs_of = {1, 2},
    requested_resonance_rationale_of = 'directed Connection-to-Stability interaction'',
    requested_resonance_valid_from_of = 10
  \<rparr>"

definition t6_witness_target :: snapshot where
"t6_witness_target =
  add_classified_resonance t6_witness_source t6_request 2"

definition t6_target_resonance_catalog :: resonance_catalog where
"t6_target_resonance_catalog =
  extend_resonance_catalog t6_source_resonance_catalog t6_request"

lemmas t6_object_defs =

```

```

t6_entity_a_def t6_entity_b_def t6_relation_a_def t6_effect_a_def
t6_relation_b_def t6_effect_b_def

lemma t6_witness_source_simps [simp]:
  "snapshot_id_of t6_witness_source = 1"
  "objects_of t6_witness_source = t6_witness_objects"
  "historical_versions_of t6_witness_source = {}"
  "perspectives_of t6_witness_source = {10}"
  "contexts_of t6_witness_source = {20}"
  "models_of t6_witness_source = {30}"
  "authorities_of t6_witness_source = {}"
  "contains_edges_of t6_witness_source = {}"
  "projections_of t6_witness_source = {}"
  "couplings_of t6_witness_source = {}"
  "transmissions_of t6_witness_source = {}"
  "boundaries_of t6_witness_source = {}"
  by (simp_all add: t6_witness_source_def)

lemma t6_witness_version_ids [simp]:
  "version_ids t6_witness_source = {1, 2, 3, 4, 5, 6}"
  by (auto simp: version_ids_def t6_witness_objects_def t6_object_defs
    base_object_def)

lemma t6_source_vector_valid:
  "effect_vector_valid t6_source_vector"
  by (auto simp: effect_vector_valid_def t6_source_vector_def)

lemma t6_target_vector_valid:
  "effect_vector_valid t6_target_vector"
  by (auto simp: effect_vector_valid_def t6_target_vector_def)

lemma t6_effect_dimension_delta:
  fixes selected :: effect_dimension
  and coefficient :: real
  shows
    "(\<Sum>dimension\<in>UNIV.
      if dimension = selected then coefficient else 0) = coefficient"
proof -
  have finite_dimensions: "finite (UNIV :: effect_dimension set)"
  by simp
  note delta =
    sum.delta[OF finite_dimensions, of selected "\<lambda>_. coefficient"]
  show ?thesis using delta by simp
qed

lemma t6_directional_kernel_mass:
  "resonance_kernel_mass t6_directional_kernel = 1"
proof -
  have inner:
    "\<And>source_dimension.
      (\<Sum>target_dimension\<in>UNIV.
        abs (t6_directional_kernel source_dimension target_dimension)) =
      (if source_dimension = Connection then 1 else 0)"
  by (case_tac source_dimension)
  (simp_all add: t6_directional_kernel_def t6_effect_dimension_delta)
  show ?thesis
  unfolding resonance_kernel_mass_def
  by (simp add: inner t6_effect_dimension_delta)
qed

lemma t6_profile_well_formed:
  "resonance_profile_well_formed t6_resonance_profile"
  using t6_directional_kernel_mass
  by (auto simp: resonance_profile_well_formed_def t6_resonance_profile_def
    t6_directional_kernel_def)

lemma t6_witness_source_well_formed:
  "state_well_formed t6_witness_grammar t6_witness_source"
  unfolding state_well_formed_def
  by (auto simp: unique_versions_def inj_on_def t6_witness_grammar_def
    t6_witness_objects_def t6_object_defs base_object_def present_ref_def
    references_complete_def evidence_well_formed_def
    evidence_references_complete_def time_well_formed_def
    relation_well_formed_def effect_well_formed_def
    perspective_well_formed_def perspective_bound_kind_def
    context_well_formed_def context_bound_kind_def relevance_well_formed_def
    orientation_well_formed_def action_well_formed_def
    feedback_well_formed_def)

lemma t6_effect_catalog_well_formed:

```

```

"effect_catalog_well_formed t6_witness_source t6_effect_catalog"
unfolding effect_catalog_well_formed_def
using t6_source_vector_valid t6_target_vector_valid
by (auto simp: t6_effect_catalog_def t6_effect_payload_a_def
    t6_effect_payload_b_def t6_witness_objects_def t6_object_defs
    base_object_def effect_payload_valid_def)

lemma t6_forward_term:
  "t6_directional_kernel source_dimension target_dimension *
    of_int (t6_source_vector source_dimension *
      t6_target_vector target_dimension) =
    (if source_dimension = Connection <and> target_dimension = Stability
    then 9 else 0)"
  by (case_tac source_dimension; case_tac target_dimension)
    (simp_all add: t6_directional_kernel_def t6_source_vector_def
      t6_target_vector_def)

lemma t6_reverse_term:
  "t6_directional_kernel source_dimension target_dimension *
    of_int (t6_target_vector source_dimension *
      t6_source_vector target_dimension) = 0"
  by (case_tac source_dimension; case_tac target_dimension)
    (simp_all add: t6_directional_kernel_def t6_target_vector_def
      t6_source_vector_def)

lemma t6_forward_weighted_sum:
  "(\<Sum>source_dimension\<in>UNIV.
    \<Sum>target_dimension\<in>UNIV.
      t6_directional_kernel source_dimension target_dimension *
        of_int (t6_source_vector source_dimension *
          t6_target_vector target_dimension)) = 9"
proof -
  have inner:
    "\<And>source_dimension.
      (\<Sum>target_dimension\<in>UNIV.
        t6_directional_kernel source_dimension target_dimension *
          of_int (t6_source_vector source_dimension *
            t6_target_vector target_dimension)) =
        (if source_dimension = Connection then 9 else 0)"
    by (simp only: t6_forward_term)
      (case_tac source_dimension;
        simp_all add: t6_effect_dimension_delta)
  show ?thesis by (simp only: inner t6_effect_dimension_delta)
qed

lemma t6_reverse_weighted_sum:
  "(\<Sum>source_dimension\<in>UNIV.
    \<Sum>target_dimension\<in>UNIV.
      t6_directional_kernel source_dimension target_dimension *
        of_int (t6_target_vector source_dimension *
          t6_source_vector target_dimension)) = 0"
  by (simp only: t6_reverse_term) simp

lemma t6_forward_score:
  "directional_resonance_score t6_resonance_profile
    t6_source_vector t6_target_vector = 1"
  using t6_directional_kernel_mass t6_forward_weighted_sum
  by (simp add: directional_resonance_score_def
    directional_resonance_raw_score_def clamp11_def
    t6_resonance_profile_def
    t6_directional_kernel_def t6_source_vector_def t6_target_vector_def)

lemma t6_reverse_score:
  "directional_resonance_score t6_resonance_profile
    t6_target_vector t6_source_vector = 0"
  using t6_directional_kernel_mass t6_reverse_weighted_sum
  by (simp add: directional_resonance_score_def
    directional_resonance_raw_score_def clamp11_def
    t6_resonance_profile_def
    t6_directional_kernel_def t6_source_vector_def t6_target_vector_def)

lemma t6_witness_input_valid:
  "classify_resonance_input_valid t6_witness_grammar t6_witness_source
    t6_effect_catalog t6_source_resonance_catalog t6_request 2"
  unfolding classify_resonance_input_valid_def
  using t6_witness_source_well_formed t6_effect_catalog_well_formed
    t6_source_vector_valid t6_target_vector_valid t6_profile_well_formed
  by (auto simp: active_history_disjoint_def resonance_catalog_well_formed_def
    t6_witness_grammar_def t6_source_resonance_catalog_def t6_request_def
    t6_resonance_profile_def)

```

```

t6_witness_objects_def t6_object_defs base_object_def
distinct_active_effect_objects_def active_classifiable_effect_def
t6_effect_catalog_def t6_effect_payload_a_def t6_effect_payload_b_def
effect_payload_valid_def active_resonance_rationale_def)

theorem t6_witness_concrete_step:
"classify_resonance_step t6_witness_grammar t6_witness_source
t6_effect_catalog t6_source_resonance_catalog t6_request 2
t6_witness_target t6_target_resonance_catalog"
unfolding classify_resonance_step_def t6_witness_target_def
t6_target_resonance_catalog_def
using t6_witness_input_valid by simp

theorem t6_witness_directionality_is_nonvacuous:
"requested_source_effect_of t6_request = 4 \<and>
requested_target_effect_of t6_request = 6 \<and>
directional_resonance_score t6_resonance_profile
t6_source_vector t6_target_vector = 1 \<and>
directional_resonance_score t6_resonance_profile
t6_target_vector t6_source_vector = 0 \<and>
requested_resonance_form t6_request = Amplification \<and>
classify_directional_score t6_resonance_profile
(directional_resonance_score t6_resonance_profile
t6_target_vector t6_source_vector) = Neutral \<and>
requested_resonance_strength t6_request = 9 / 10 \<and>
lifecycle_of
(classified_resonance_object t6_witness_source t6_request) = Validated"
using t6_forward_score t6_reverse_score
by (simp add: t6_request_def requested_resonance_form_def
requested_directional_score_def classify_directional_score_def
requested_resonance_strength_def
uncertainty_adjusted_resonance_strength_def clamp01_def
t6_resonance_profile_def classified_resonance_lifecycle_def
effect_validated_at_def t6_witness_objects_def t6_object_defs
base_object_def)

theorem t6_witness_target_certificates:
"state_well_formed t6_witness_grammar t6_witness_target \<and>
effect_catalog_well_formed t6_witness_target t6_effect_catalog \<and>
resonance_catalog_well_formed t6_witness_target
t6_target_resonance_catalog \<and>
step t6_witness_grammar t6_witness_source ClassifyResonance
t6_witness_target"
using classify_resonance_preserves_state_well_formed
[OF t6_witness_concrete_step]
classify_resonance_preserves_effect_catalog
[OF t6_witness_concrete_step]
classify_resonance_preserves_catalog
[OF t6_witness_concrete_step]
classify_resonance_refines_generic_step
[OF t6_witness_concrete_step]
by blast

theorem t6_witness_exact_delta:
"objects_of t6_witness_target =
insert (classified_resonance_object t6_witness_source t6_request)
(objects_of t6_witness_source) \<and>
classified_resonance_object t6_witness_source t6_request
\<notin> objects_of t6_witness_source \<and>
snapshot_id_of t6_witness_target = 2 \<and>
t6_target_resonance_catalog 7 =
Some (classified_resonance_payload t6_request)"
using classify_resonance_exact_delta[OF t6_witness_concrete_step]
by (simp add: t6_request_def)

end

```

Beweisstatus dieser D.11-Fassung: kernelgeprüft und nicht-vakuos im ausgewiesenen Umfang.  
Der Zeuge enthält 438 Quellzeilen; sein SHA-256 lautet  
494da00eeb17205001ae85090d940a12cd07215ec283d76833aa480126a716ed. Er ist ein  
Existenz-, Rechen- und Ausführbarkeitsbeleg für eine endliche gerichtete Konfiguration, kein  
empirischer Universalitätsbeweis und keine Aussage, dass jeder reale Resonanzzusammenhang  
mit diesem Kernel modelliert werden sollte.

## D.12 Konstruktive T7-Semantik und Erhaltung für EvaluateRelevance

Die zwölfte Operationstheorie operationalisiert die perspektivische, kontextuelle und zeitliche Relevanzbewertung. `relevance_model_profile` bindet Modell-ID und positive Revision, unterstützte Perspektiven und Kontexte, getrennte V10- und Machtgewichte, Kontext- und Zeitfaktoren, Reichweite, Dauerkappung, lokale Sensitivität, V10-Anteil und Validierungsschwelle. `calculate_relevance_vector` berechnet Intensität, Unsicherheit, Stabilität, Reichweite, Dauer und Erklärung. T7 konstruiert genau ein frisches `K_REV`-Objekt und einen versionierten Payload; parallele Ergebnisse bleiben getrennt.

Mechanisch bewiesen sind Endlichkeit der Machtdimensionen, Profil- und Eingabewohlgeformtheit, Wertebereiche, Zielaktivität und Provenienz, exaktes Delta, frische Identität, Erhaltung der Zustandswohlgeformtheit und der Kataloge sowie Refinement zur generischen step-Relation. Die Theorie besitzt 1.564 Zeilen und SHA-256 6377ca4dd963832712350d10d8ad7442a1245b9e4ee76e85c43f6634bc8ecc46. Nach den historischen Kerndirektläufen wurde sie auch im finalen normalen 16-Theorien-K1-Sessionbuild akzeptiert.

```
theory UZA_0_9_1_Evaluate_Relevance
  imports UZA_0_9_1_Classify_Resonance
begin

section \<open>T7 EvaluateRelevance: perspective-, context-, and time-relative semantics\<close>

text \<open>
  T7 creates exactly one fresh K_REV object for one active, catalogued effect.
  The result is relative to an explicit model revision, perspective, context,
  and evaluation time. V10 effects and power exposure remain distinct inputs;
  the model declares their blend instead of silently folding power into V10.

  The result vector persists intensity, epistemic uncertainty, local stability,
  reach, duration, and an explanation. Parallel evaluations remain separate
  catalog entries. No averaging operation is part of T7.
\<close>

type_synonym power_weights = "power_dimension \<Rightarrow> real"

record relevance_model_profile =
  relevance_profile_model_of :: model_id
  relevance_profile_revision_of :: nat
  relevance_profile_perspectives_of :: "perspective_id set"
  relevance_profile_contexts_of :: "context_id set"
  perspective_effect_weights_of :: "perspective_id \<Rightarrow> relevance_weights"
  perspective_power_weights_of :: "perspective_id \<Rightarrow> power_weights"
  context_effect_factor_of :: "context_id \<Rightarrow> effect_dimension \<Rightarrow> real"
  context_power_factor_of :: "context_id \<Rightarrow> power_dimension \<Rightarrow> real"
  temporal_relevance_factor_of :: "time_point \<Rightarrow> real"
  reach_factor_of :: "perspective_id \<Rightarrow> context_id \<Rightarrow> real"
  duration_cap_of :: "perspective_id \<Rightarrow> context_id \<Rightarrow> nat"
  local_sensitivity_of :: "perspective_id \<Rightarrow> context_id \<Rightarrow> real"
  effect_share_of :: real
  relevance_validation_uncertainty_max_of :: real

record relevance_vector =
  relevance_intensity_of :: real
  relevance_uncertainty_of :: real
  relevance_stability_of :: real
  relevance_reach_of :: real
  relevance_duration_of :: nat
  relevance_explanation_of :: string

record relevance_request =
  requested_relevance_object_id_of :: object_id
  requested_relevance_version_id_of :: version_id
  requested_relevance_target_of :: version_id
  requested_relevance_effect_vector_of :: effect_vector
  requested_relevance_power_profile_of :: power_profile
```



```

requested_relevance_profile_of :: relevance_model_profile
requested_relevance_perspective_of :: perspective_id
requested_relevance_context_of :: context_id
requested_relevance_evaluation_time_of :: time_point
requested_relevance_uncertainty_of :: real
requested_relevance_perturbation_radius_of :: real
requested_relevance_reach_input_of :: real
requested_relevance_duration_input_of :: nat
requested_relevance_rationale_refs_of :: "version_id set"
requested_relevance_explanation_of :: string

record relevance_payload =
  evaluated_relevance_target_of :: version_id
  evaluated_relevance_effect_vector_of :: effect_vector
  evaluated_relevance_power_profile_of :: power_profile
  evaluated_relevance_profile_of :: relevance_model_profile
  evaluated_relevance_perspective_of :: perspective_id
  evaluated_relevance_context_of :: context_id
  evaluated_relevance_time_of :: time_point
  evaluated_relevance_uncertainty_input_of :: real
  evaluated_relevance_perturbation_radius_of :: real
  evaluated_relevance_reach_input_of :: real
  evaluated_relevance_duration_input_of :: nat
  evaluated_relevance_vector_of :: relevance_vector
  evaluated_relevance_rationale_refs_of :: "version_id set"

type_synonym relevance_catalog = "version_id \ $\rightarrow$  relevance_payload"

subsection \ $\langle$ Finite dimensions and model validity $\rangle$ 

lemma power_dimension_UNIV:
  "(UNIV :: power_dimension set) =
   {ResourcePower, AccessControl, DecisionAuthority, RiskBearing,
    CorrectionCapacity}"
proof (rule set_eqI)
  fix dimension :: power_dimension
  show
    "dimension \ $\in$  UNIV  $\longleftrightarrow$ 
     dimension \ $\in$ 
     {ResourcePower, AccessControl, DecisionAuthority, RiskBearing,
      CorrectionCapacity}"
  by (case_tac dimension) simp_all
qed

lemma power_dimensions_finite [simp]:
  "finite (UNIV :: power_dimension set)"
  by (simp add: power_dimension_UNIV)

definition power_weight_mass :: "power_weights \ $\rightarrow$  real" where
  "power_weight_mass weights = sum ( $\lambda$ dimension. abs (weights dimension)) UNIV"

definition power_weights_valid :: "power_weights \ $\rightarrow$  bool" where
  "power_weights_valid weights  $\longleftrightarrow$ 
   ( $\forall$ dimension.  $0 \leq$  weights dimension)  $\wedge$ 
   power_weight_mass weights  $> 0$ "

definition effective_effect_weight ::
  "relevance_model_profile \ $\rightarrow$  perspective_id \ $\rightarrow$  context_id
   \ $\rightarrow$  effect_dimension \ $\rightarrow$  real" where
  "effective_effect_weight profile perspective context dimension =
   perspective_effect_weights_of profile perspective dimension *
   context_effect_factor_of profile context dimension"

definition effective_power_weight ::
  "relevance_model_profile \ $\rightarrow$  perspective_id \ $\rightarrow$  context_id
   \ $\rightarrow$  power_dimension \ $\rightarrow$  real" where
  "effective_power_weight profile perspective context dimension =
   perspective_power_weights_of profile perspective dimension *
   context_power_factor_of profile context dimension"

definition effective_effect_weight_mass ::
  "relevance_model_profile \ $\rightarrow$  perspective_id \ $\rightarrow$  context_id
   \ $\rightarrow$  real" where
  "effective_effect_weight_mass profile perspective context =
   sum ( $\lambda$ dimension.
     effective_effect_weight profile perspective context dimension) UNIV"

definition effective_power_weight_mass ::
  "relevance_model_profile \ $\rightarrow$  perspective_id \ $\rightarrow$  context_id
   \ $\rightarrow$  real" where

```

```

"effective_power_weight_mass profile perspective context =
  sum (\<lambda>dimension.
    effective_power_weight profile perspective context dimension) UNIV"

definition relevance_model_profile_well_formed ::
  "relevance_model_profile \<Rightarrow> bool" where
  "relevance_model_profile_well_formed profile \<longlefttrightarrow>
    relevance_profile_revision_of profile > 0 \<and>
    relevance_profile_perspectives_of profile \<noteq> {} \<and>
    relevance_profile_contexts_of profile \<noteq> {} \<and>
    (\<forall>perspective \<in> relevance_profile_perspectives_of profile.
      relevance_weights_valid
        (perspective_effect_weights_of profile perspective) \<and>
      power_weights_valid
        (perspective_power_weights_of profile perspective)) \<and>
    (\<forall>context \<in> relevance_profile_contexts_of profile.
      (\<forall>dimension.
        0 \<le> context_effect_factor_of profile context dimension \<and>
        context_effect_factor_of profile context dimension \<le> 1) \<and>
      (\<forall>dimension.
        0 \<le> context_power_factor_of profile context dimension \<and>
        context_power_factor_of profile context dimension \<le> 1)) \<and>
      (\<forall>perspective \<in> relevance_profile_perspectives_of profile.
        \<forall>context \<in> relevance_profile_contexts_of profile.
          effective_effect_weight_mass profile perspective context > 0 \<and>
          effective_power_weight_mass profile perspective context > 0 \<and>
          0 \<le> reach_factor_of profile perspective context \<and>
          reach_factor_of profile perspective context \<le> 1 \<and>
          duration_cap_of profile perspective context > 0 \<and>
          0 \<le> local_sensitivity_of profile perspective context \<and>
          local_sensitivity_of profile perspective context \<le> 1) \<and>
        (\<forall>evaluation_time.
          0 \<le> temporal_relevance_factor_of profile evaluation_time \<and>
          temporal_relevance_factor_of profile evaluation_time \<le> 1) \<and>
          0 \<le> effect_share_of profile \<and> effect_share_of profile \<le> 1 \<and>
          0 \<le> relevance_validation_uncertainty_max_of profile \<and>
          relevance_validation_uncertainty_max_of profile \<le> 1)"

subsection \<open>Executable relevance-vector semantics\<close>

definition perspective_effect_component ::
  "relevance_model_profile \<Rightarrow> perspective_id \<Rightarrow> context_id
    \<Rightarrow> effect_vector \<Rightarrow> real" where
  "perspective_effect_component profile perspective context vector =
    (let mass = effective_effect_weight_mass profile perspective context in
      clamp01
        (if mass = 0 then 1 / 2
          else sum (\<lambda>dimension.
            effective_effect_weight profile perspective context dimension *
            normalized_effect_component vector dimension) UNIV / mass))"

definition perspective_power_component ::
  "relevance_model_profile \<Rightarrow> perspective_id \<Rightarrow> context_id
    \<Rightarrow> power_profile \<Rightarrow> real" where
  "perspective_power_component profile perspective context power_values =
    (let mass = effective_power_weight_mass profile perspective context in
      clamp01
        (if mass = 0 then 1 / 2
          else sum (\<lambda>dimension.
            effective_power_weight profile perspective context dimension *
            power_values dimension) UNIV / mass))"

definition relevance_intensity ::
  "relevance_model_profile \<Rightarrow> perspective_id \<Rightarrow> context_id
    \<Rightarrow> time_point \<Rightarrow> effect_vector \<Rightarrow> power_profile
    \<Rightarrow> real \<Rightarrow> real" where
  "relevance_intensity profile perspective context evaluation_time vector power_values
    uncertainty =
      clamp01
        ((effect_share_of profile *
          perspective_effect_component profile perspective context vector +
          (1 - effect_share_of profile) *
          perspective_power_component profile perspective context power_values) *
          temporal_relevance_factor_of profile evaluation_time *
          (1 - clamp01 uncertainty))"

definition relevance_stability ::
  "relevance_model_profile \<Rightarrow> perspective_id \<Rightarrow> context_id
    \<Rightarrow> real \<Rightarrow> real" where
  "relevance_stability profile perspective context perturbation_radius =

```

```

clamp01
  (1 - local_sensitivity_of profile perspective context *
    clamp01 perturbation_radius)"

definition relevance_reach ::
  "relevance_model_profile \<Rightarrow> perspective_id \<Rightarrow> context_id
  \<Rightarrow> real \<Rightarrow> real" where
  "relevance_reach profile perspective context reach_input =
    clamp01 (reach_input * reach_factor_of profile perspective context)"

definition relevance_duration ::
  "relevance_model_profile \<Rightarrow> perspective_id \<Rightarrow> context_id
  \<Rightarrow> nat \<Rightarrow> nat" where
  "relevance_duration profile perspective context duration_input =
    min duration_input (duration_cap_of profile perspective context)"

definition calculate_relevance_vector ::
  "relevance_model_profile \<Rightarrow> perspective_id \<Rightarrow> context_id
  \<Rightarrow> time_point \<Rightarrow> effect_vector \<Rightarrow> power_profile
  \<Rightarrow> real \<Rightarrow> real \<Rightarrow> nat \<Rightarrow> string
  \<Rightarrow> relevance_vector" where
  "calculate_relevance_vector profile perspective context evaluation_time
  vector power_values uncertainty perturbation_radius reach_input duration_input
  explanation =
  \<lparr>
    relevance_intensity_of =
      relevance_intensity profile perspective context evaluation_time
      vector power_values uncertainty,
    relevance_uncertainty_of = clamp01 uncertainty,
    relevance_stability_of =
      relevance_stability profile perspective context perturbation_radius,
    relevance_reach_of =
      relevance_reach profile perspective context reach_input,
    relevance_duration_of =
      relevance_duration profile perspective context duration_input,
    relevance_explanation_of = explanation
  \<rparr>"

definition requested_relevance_vector ::
  "relevance_request \<Rightarrow> relevance_vector" where
  "requested_relevance_vector request =
    calculate_relevance_vector
      (requested_relevance_profile_of request)
      (requested_relevance_perspective_of request)
      (requested_relevance_context_of request)
      (requested_relevance_evaluation_time_of request)
      (requested_relevance_effect_vector_of request)
      (requested_relevance_power_profile_of request)
      (requested_relevance_uncertainty_of request)
      (requested_relevance_perturbation_radius_of request)
      (requested_relevance_reach_input_of request)
      (requested_relevance_duration_input_of request)
      (requested_relevance_explanation_of request)"

lemma calculated_relevance_vector_simps [simp]:
  "relevance_intensity_of
    (calculate_relevance_vector profile perspective context evaluation_time
      vector power_values uncertainty perturbation reach duration explanation) =
    relevance_intensity profile perspective context evaluation_time
    vector power_values uncertainty"
  "relevance_uncertainty_of
    (calculate_relevance_vector profile perspective context evaluation_time
      vector power_values uncertainty perturbation reach duration explanation) =
    clamp01 uncertainty"
  "relevance_stability_of
    (calculate_relevance_vector profile perspective context evaluation_time
      vector power_values uncertainty perturbation reach duration explanation) =
    relevance_stability profile perspective context perturbation"
  "relevance_reach_of
    (calculate_relevance_vector profile perspective context evaluation_time
      vector power_values uncertainty perturbation reach duration explanation) =
    relevance_reach profile perspective context reach"
  "relevance_duration_of
    (calculate_relevance_vector profile perspective context evaluation_time
      vector power_values uncertainty perturbation reach duration explanation) =
    relevance_duration profile perspective context duration"
  "relevance_explanation_of
    (calculate_relevance_vector profile perspective context evaluation_time
      vector power_values uncertainty perturbation reach duration explanation) =
    explanation"

```

```

by (simp_all add: calculate_relevance_vector_def)

lemma clamp01_closed_interval:
  "0 \<le> clamp01 x \<and> clamp01 x \<le> 1"
  unfolding clamp01_def
  by auto

lemma perspective_effect_component_bounded:
  "0 \<le> perspective_effect_component profile perspective context vector \<and>
  perspective_effect_component profile perspective context vector \<le> 1"
  unfolding perspective_effect_component_def Let_def
  by (rule clamp01_closed_interval)

lemma perspective_power_component_bounded:
  "0 \<le> perspective_power_component profile perspective context power_values \<and>
  perspective_power_component profile perspective context power_values \<le> 1"
  unfolding perspective_power_component_def Let_def
  by (rule clamp01_closed_interval)

theorem relevance_intensity_bounded:
  "0 \<le> relevance_intensity profile perspective context evaluation_time
  vector power_values uncertainty \<and>
  relevance_intensity profile perspective context evaluation_time
  vector power_values uncertainty \<le> 1"
  unfolding relevance_intensity_def
  by (rule clamp01_closed_interval)

theorem relevance_stability_bounded:
  "0 \<le> relevance_stability profile perspective context perturbation \<and>
  relevance_stability profile perspective context perturbation \<le> 1"
  unfolding relevance_stability_def
  by (rule clamp01_closed_interval)

theorem relevance_reach_bounded:
  "0 \<le> relevance_reach profile perspective context reach \<and>
  relevance_reach profile perspective context reach \<le> 1"
  unfolding relevance_reach_def
  by (rule clamp01_closed_interval)

subsection \<open>Target activity, provenance, and constructors\<close>

definition version_active_at ::
  "snapshot \<Rightarrow> version_id \<Rightarrow> time_point \<Rightarrow> bool" where
  "version_active_at s version_evaluation_time \<longlefttrightarrow>
  (\<exists>obj\<in>objects_of s.
    version_id_of obj = version \<and>
    lifecycle_of obj \<notin> {Revised, Rejected, Archived} \<and>
    valid_from_of obj \<le> evaluation_time \<and>
    (case valid_to_of obj of
      None \<Rightarrow> True
    | Some terminal_time \<Rightarrow> evaluation_time \<le> terminal_time))"

definition active_relevance_rationale ::
  "snapshot \<Rightarrow> version_id \<Rightarrow> version_id set \<Rightarrow> bool" where
  "active_relevance_rationale s target rationale_refs \<longlefttrightarrow>
  rationale_refs \<noteq> {} \<and>
  target \<notin> rationale_refs \<and>
  (\<forall>version\<in>rationale_refs.
    \<exists>obj\<in>objects_of s.
      version_id_of obj = version \<and>
      lifecycle_of obj \<notin> {Revised, Rejected, Archived})"

definition evaluated_relevance_lifecycle ::
  "snapshot \<Rightarrow> relevance_request \<Rightarrow> lifecycle_status" where
  "evaluated_relevance_lifecycle s request =
  (if effect_validated_at s (requested_relevance_target_of request) \<and>
    requested_relevance_uncertainty_of request \<le>
    relevance_validation_uncertainty_max_of
    (requested_relevance_profile_of request)
  then Validated else Supported)"

definition requested_relevance_references ::
  "relevance_request \<Rightarrow> version_id set" where
  "requested_relevance_references request =
  insert (requested_relevance_target_of request)
  (requested_relevance_rationale_refs_of request)"

definition evaluated_relevance_object ::
  "snapshot \<Rightarrow> relevance_request \<Rightarrow> uza_object" where
  "evaluated_relevance_object s request =

```

```

\<lparr>
  object_id_of = requested_relevance_object_id_of request,
  version_id_of = requested_relevance_version_id_of request,
  kind_of = K_REV,
  lifecycle_of = evaluated_relevance_lifecycle s request,
  previous_version_of = None,
  references_of = requested_relevance_references request,
  evidence_of = requested_relevance_rationale_refs_of request,
  perspective_of = Some (requested_relevance_perspective_of request),
  context_of = Some (requested_relevance_context_of request),
  model_of = Some
    (relevance_profile_model_of (requested_relevance_profile_of request)),
  authority_of = None,
  valid_from_of = requested_relevance_evaluation_time_of request,
  valid_to_of = None,
  source_ref_of = Some (requested_relevance_target_of request),
  target_ref_of = None,
  effect_origin_of = None,
  basis_refs_of = {requested_relevance_target_of request},
  execution_ref_of = None,
  observation_refs_of = {},
  object_status_of = Some Applicable
\<rparr>"

definition evaluated_relevance_payload ::
  "relevance_request \<Rightarrow> relevance_payload" where
  "evaluated_relevance_payload request =
    \<lparr>
      evaluated_relevance_target_of = requested_relevance_target_of request,
      evaluated_relevance_effect_vector_of =
        requested_relevance_effect_vector_of request,
      evaluated_relevance_power_profile_of =
        requested_relevance_power_profile_of request,
      evaluated_relevance_profile_of = requested_relevance_profile_of request,
      evaluated_relevance_perspective_of =
        requested_relevance_perspective_of request,
      evaluated_relevance_context_of = requested_relevance_context_of request,
      evaluated_relevance_time_of =
        requested_relevance_evaluation_time_of request,
      evaluated_relevance_uncertainty_input_of =
        requested_relevance_uncertainty_of request,
      evaluated_relevance_perturbation_radius_of =
        requested_relevance_perturbation_radius_of request,
      evaluated_relevance_reach_input_of =
        requested_relevance_reach_input_of request,
      evaluated_relevance_duration_input_of =
        requested_relevance_duration_input_of request,
      evaluated_relevance_vector_of = requested_relevance_vector request,
      evaluated_relevance_rationale_refs_of =
        requested_relevance_rationale_refs_of request
    \<rparr>"

definition relevance_payload_valid ::
  "uza_object \<Rightarrow> relevance_payload \<Rightarrow> bool" where
  "relevance_payload_valid relevance payload \<longleftarrowrightarrow>
    kind_of relevance = K_REV \<and>
    lifecycle_of relevance \<in> {Supported, Validated} \<and>
    source_ref_of relevance = Some (evaluated_relevance_target_of payload) \<and>
    basis_refs_of relevance = {evaluated_relevance_target_of payload} \<and>
    references_of relevance =
      insert (evaluated_relevance_target_of payload)
        (evaluated_relevance_rationale_refs_of payload) \<and>
    evidence_of relevance = evaluated_relevance_rationale_refs_of payload \<and>
    perspective_of relevance = Some (evaluated_relevance_perspective_of payload) \<and>
    context_of relevance = Some (evaluated_relevance_context_of payload) \<and>
    model_of relevance = Some
      (relevance_profile_model_of (evaluated_relevance_profile_of payload)) \<and>
    valid_from_of relevance = evaluated_relevance_time_of payload \<and>
    relevance_model_profile_well_formed
      (evaluated_relevance_profile_of payload) \<and>
    evaluated_relevance_perspective_of payload \<in>
      relevance_profile_perspectives_of
        (evaluated_relevance_profile_of payload) \<and>
    evaluated_relevance_context_of payload \<in>
      relevance_profile_contexts_of (evaluated_relevance_profile_of payload) \<and>
    effect_vector_valid (evaluated_relevance_effect_vector_of payload) \<and>
    power_profile_valid (evaluated_relevance_power_profile_of payload) \<and>
    evaluated_relevance_vector_of payload =
      calculate_relevance_vector
        (evaluated_relevance_profile_of payload)
  "

```

```

(evaluated_relevance_perspective_of payload)
(evaluated_relevance_context_of payload)
(evaluated_relevance_time_of payload)
(evaluated_relevance_effect_vector_of payload)
(evaluated_relevance_power_profile_of payload)
(evaluated_relevance_uncertainty_input_of payload)
(evaluated_relevance_perturbation_radius_of payload)
(evaluated_relevance_reach_input_of payload)
(evaluated_relevance_duration_input_of payload)
(relevance_explanation_of (evaluated_relevance_vector_of payload)) \<and>
0 \<le> relevance_intensity_of (evaluated_relevance_vector_of payload) \<and>
relevance_intensity_of (evaluated_relevance_vector_of payload) \<le> 1 \<and>
0 \<le> relevance_uncertainty_of (evaluated_relevance_vector_of payload) \<and>
relevance_uncertainty_of (evaluated_relevance_vector_of payload) \<le> 1 \<and>
0 \<le> relevance_stability_of (evaluated_relevance_vector_of payload) \<and>
relevance_stability_of (evaluated_relevance_vector_of payload) \<le> 1 \<and>
0 \<le> relevance_reach_of (evaluated_relevance_vector_of payload) \<and>
relevance_reach_of (evaluated_relevance_vector_of payload) \<le> 1 \<and>
relevance_duration_of (evaluated_relevance_vector_of payload) > 0 \<and>
relevance_explanation_of (evaluated_relevance_vector_of payload) \<noteq> [] \<and>
evaluated_relevance_rationale_refs_of payload \<noteq> {} \<and>
object_status_of relevance = Some Applicable"

definition relevance_catalog_well_formed ::
"snapshot \<Rightarrow> relevance_catalog \<Rightarrow> bool" where
"relevance_catalog_well_formed s catalog \<longleftarrowtrightrightarrow>
(\<forall>version payload. catalog version = Some payload \<longrightarrow>
(\<exists>relevance\<in>objects_of s.
  version_id_of relevance = version \<and>
  relevance_payload_valid relevance payload \<and>
  references_complete s relevance \<and>
  evidence_well_formed s relevance))"

definition evaluate_relevance_input_valid ::
"grammar \<Rightarrow> snapshot \<Rightarrow> effect_catalog \<Rightarrow>
  relevance_catalog \<Rightarrow> relevance_request \<Rightarrow> nat \<Rightarrow> bool" where
"evaluate_relevance_input_valid G s effect_catalog relevance_catalog request
  next_snapshot_id \<longleftarrowtrightrightarrow>
  state_well_formed G s \<and>
  active_history_disjoint s \<and>
  effect_catalog_well_formed s effect_catalog \<and>
  relevance_catalog_well_formed s relevance_catalog \<and>
  EvaluateRelevance \<in> allowed_operations_of G \<and>
  K_REV \<in> supported_kinds_of G \<and>
  requested_relevance_object_id_of request \<notin>
    object_id_of ` objects_of s \<and>
  requested_relevance_version_id_of request \<notin>
    version_ids s \<union> historical_versions_of s \<and>
  relevance_catalog (requested_relevance_version_id_of request) = None \<and>
  active_classifiable_effect s effect_catalog
    (requested_relevance_target_of request)
    (requested_relevance_effect_vector_of request)
    (requested_relevance_context_of request) \<and>
  version_active_at s
    (requested_relevance_target_of request)
    (requested_relevance_evaluation_time_of request) \<and>
  effect_vector_valid (requested_relevance_effect_vector_of request) \<and>
  power_profile_valid (requested_relevance_power_profile_of request) \<and>
  relevance_model_profile_well_formed
    (requested_relevance_profile_of request) \<and>
  relevance_profile_model_of (requested_relevance_profile_of request)
    \<in> models_of s \<and>
  requested_relevance_perspective_of request \<in> perspectives_of s \<and>
  requested_relevance_perspective_of request \<in>
    relevance_profile_perspectives_of
      (requested_relevance_profile_of request) \<and>
  requested_relevance_context_of request \<in> contexts_of s \<and>
  requested_relevance_context_of request \<in>
    relevance_profile_contexts_of (requested_relevance_profile_of request) \<and>
  0 \<le> requested_relevance_uncertainty_of request \<and>
  requested_relevance_uncertainty_of request \<le> 1 \<and>
  0 \<le> requested_relevance_perturbation_radius_of request \<and>
  requested_relevance_perturbation_radius_of request \<le> 1 \<and>
  0 \<le> requested_relevance_reach_input_of request \<and>
  requested_relevance_reach_input_of request \<le> 1 \<and>
  requested_relevance_duration_input_of request > 0 \<and>
  active_relevance_rationale s
    (requested_relevance_target_of request)
    (requested_relevance_rationale_refs_of request) \<and>
  requested_relevance_explanation_of request \<noteq> [] \<and>

```

```

    snapshot_id_of s < next_snapshot_id"

definition add_evaluated_relevance ::
  "snapshot \<Rightarrow> relevance_request \<Rightarrow> nat \<Rightarrow> snapshot" where
  "add_evaluated_relevance s request next_snapshot_id =
    add_appearance s (evaluated_relevance_object s request) next_snapshot_id"

definition extend_relevance_catalog ::
  "relevance_catalog \<Rightarrow> relevance_request \<Rightarrow> relevance_catalog" where
  "extend_relevance_catalog catalog request =
    catalog(requested_relevance_version_id_of request \<mapsto>
      evaluated_relevance_payload request)"

definition evaluate_relevance_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> effect_catalog \<Rightarrow>
    relevance_catalog \<Rightarrow> relevance_request \<Rightarrow> nat \<Rightarrow>
    snapshot \<Rightarrow> relevance_catalog \<Rightarrow> bool" where
  "evaluate_relevance_step G s effect_catalog relevance_catalog request
    next_snapshot_id s' relevance_catalog' \<longletrightarrow>
    evaluate_relevance_input_valid G s effect_catalog relevance_catalog request
    next_snapshot_id \<and>
    s' = add_evaluated_relevance s request next_snapshot_id \<and>
    relevance_catalog' = extend_relevance_catalog relevance_catalog request"

subsection \<open>Constructor and exact-delta certificates\<close>

lemma evaluated_relevance_lifecycle_cases:
  "evaluated_relevance_lifecycle s request \<in> {Supported, Validated}"
  by (auto simp: evaluated_relevance_lifecycle_def)

lemma add_relevance_objects [simp]:
  "objects_of (add_evaluated_relevance s request next_snapshot_id) =
    insert (evaluated_relevance_object s request) (objects_of s)"
  by (simp add: add_evaluated_relevance_def)

lemma add_relevance_snapshot_id [simp]:
  "snapshot_id_of (add_evaluated_relevance s request next_snapshot_id) =
    next_snapshot_id"
  by (simp add: add_evaluated_relevance_def)

lemma add_relevance_background [simp]:
  "historical_versions_of
    (add_evaluated_relevance s request next_snapshot_id) =
    historical_versions_of s"
  "perspectives_of (add_evaluated_relevance s request next_snapshot_id) =
    perspectives_of s"
  "contexts_of (add_evaluated_relevance s request next_snapshot_id) =
    contexts_of s"
  "models_of (add_evaluated_relevance s request next_snapshot_id) = models_of s"
  "authorities_of (add_evaluated_relevance s request next_snapshot_id) =
    authorities_of s"
  "contains_edges_of (add_evaluated_relevance s request next_snapshot_id) =
    contains_edges_of s"
  "projections_of (add_evaluated_relevance s request next_snapshot_id) =
    projections_of s"
  "couplings_of (add_evaluated_relevance s request next_snapshot_id) =
    couplings_of s"
  "transmissions_of (add_evaluated_relevance s request next_snapshot_id) =
    transmissions_of s"
  "boundaries_of (add_evaluated_relevance s request next_snapshot_id) =
    boundaries_of s"
  by (simp_all add: add_evaluated_relevance_def)

lemma add_relevance_version_ids [simp]:
  "version_ids (add_evaluated_relevance s request next_snapshot_id) =
    insert (requested_relevance_version_id_of request) (version_ids s)"
  by (auto simp: version_ids_def evaluated_relevance_object_def)

lemma evaluated_relevance_object_simps [simp]:
  "object_id_of (evaluated_relevance_object s request) =
    requested_relevance_object_id_of request"
  "version_id_of (evaluated_relevance_object s request) =
    requested_relevance_version_id_of request"
  "kind_of (evaluated_relevance_object s request) = K_REV"
  "lifecycle_of (evaluated_relevance_object s request) =
    evaluated_relevance_lifecycle s request"
  "previous_version_of (evaluated_relevance_object s request) = None"
  "references_of (evaluated_relevance_object s request) =
    requested_relevance_references request"
  "evidence_of (evaluated_relevance_object s request) =

```

```

    requested_relevance_rationale_refs_of request"
"perspective_of (evaluated_relevance_object s request) =
  Some (requested_relevance_perspective_of request)"
"context_of (evaluated_relevance_object s request) =
  Some (requested_relevance_context_of request)"
"model_of (evaluated_relevance_object s request) =
  Some (relevance_profile_model_of (requested_relevance_profile_of request))"
"valid_from_of (evaluated_relevance_object s request) =
  requested_relevance_evaluation_time_of request"
"valid_to_of (evaluated_relevance_object s request) = None"
"source_ref_of (evaluated_relevance_object s request) =
  Some (requested_relevance_target_of request)"
"target_ref_of (evaluated_relevance_object s request) = None"
"effect_origin_of (evaluated_relevance_object s request) = None"
"basis_refs_of (evaluated_relevance_object s request) =
  {requested_relevance_target_of request}"
"authority_of (evaluated_relevance_object s request) = None"
"execution_ref_of (evaluated_relevance_object s request) = None"
"observation_refs_of (evaluated_relevance_object s request) = {}"
"object_status_of (evaluated_relevance_object s request) = Some Applicable"
by (simp_all add: evaluated_relevance_object_def)

lemma evaluated_relevance_payload_simps [simp]:
"evaluated_relevance_target_of (evaluated_relevance_payload request) =
  requested_relevance_target_of request"
"evaluated_relevance_effect_vector_of (evaluated_relevance_payload request) =
  requested_relevance_effect_vector_of request"
"evaluated_relevance_power_profile_of (evaluated_relevance_payload request) =
  requested_relevance_power_profile_of request"
"evaluated_relevance_profile_of (evaluated_relevance_payload request) =
  requested_relevance_profile_of request"
"evaluated_relevance_perspective_of (evaluated_relevance_payload request) =
  requested_relevance_perspective_of request"
"evaluated_relevance_context_of (evaluated_relevance_payload request) =
  requested_relevance_context_of request"
"evaluated_relevance_time_of (evaluated_relevance_payload request) =
  requested_relevance_evaluation_time_of request"
"evaluated_relevance_uncertainty_input_of
  (evaluated_relevance_payload request) =
  requested_relevance_uncertainty_of request"
"evaluated_relevance_perturbation_radius_of
  (evaluated_relevance_payload request) =
  requested_relevance_perturbation_radius_of request"
"evaluated_relevance_reach_input_of (evaluated_relevance_payload request) =
  requested_relevance_reach_input_of request"
"evaluated_relevance_duration_input_of (evaluated_relevance_payload request) =
  requested_relevance_duration_input_of request"
"evaluated_relevance_vector_of (evaluated_relevance_payload request) =
  requested_relevance_vector request"
"evaluated_relevance_rationale_refs_of (evaluated_relevance_payload request) =
  requested_relevance_rationale_refs_of request"
by (simp_all add: evaluated_relevance_payload_def)

lemma active_relevance_rationale_versions:
  assumes active:
    "active_relevance_rationale s target rationale_refs"
  shows "rationale_refs \<subteq> version_ids s"
proof
  fix version
  assume member: "version \<in> rationale_refs"
  from active member obtain obj where
    "obj \<in> objects_of s" and "version_id_of obj = version"
    unfolding active_relevance_rationale_def
    by blast
  then show "version \<in> version_ids s"
    unfolding version_ids_def
    by blast
qed

lemma evaluate_relevance_fresh_object:
  assumes valid:
    "evaluate_relevance_input_valid G s effect_catalog relevance_catalog request
      next_snapshot_id"
  shows
    "evaluated_relevance_object s request \<notin> objects_of s \<and>
      requested_relevance_version_id_of request \<notin> version_ids s \<and>
      requested_relevance_object_id_of request \<notin>
        object_id_of ` objects_of s"
proof -
  from valid have fresh_version:

```



```

    "requested_relevance_version_id_of request \<notin>
      version_ids s \<union> historical_versions_of s"
  and fresh_object:
    "requested_relevance_object_id_of request \<notin>
      object_id_of ` objects_of s"
  unfolding evaluate_relevance_input_valid_def by blast+
  have new_not_mem: "evaluated_relevance_object s request \<notin> objects_of s"
  proof
    assume "evaluated_relevance_object s request \<in> objects_of s"
    then have
      "requested_relevance_object_id_of request \<in> object_id_of ` objects_of s"
      by (metis evaluated_relevance_object_simps(1) imageI)
    then show False using fresh_object by contradiction
  qed
  show ?thesis using fresh_version fresh_object new_not_mem by blast
qed

theorem evaluate_relevance_exact_delta:
  assumes concrete:
    "evaluate_relevance_step G s effect_catalog relevance_catalog request
      next_snapshot_id s' relevance_catalog"
  shows
    "objects_of s' =
      insert (evaluated_relevance_object s request) (objects_of s) \<and>
      evaluated_relevance_object s request \<notin> objects_of s \<and>
      snapshot_id_of s' = next_snapshot_id \<and>
      relevance_catalog' (requested_relevance_version_id_of request) =
        Some (evaluated_relevance_payload request)"
  proof -
    from concrete have valid:
      "evaluate_relevance_input_valid G s effect_catalog relevance_catalog
        request next_snapshot_id"
    and target: "s' = add_evaluated_relevance s request next_snapshot_id"
    and catalog_target:
      "relevance_catalog' = extend_relevance_catalog relevance_catalog request"
    by (auto simp: evaluate_relevance_step_def)
    from evaluate_relevance_fresh_object[OF valid] have fresh:
      "evaluated_relevance_object s request \<notin> objects_of s"
    by blast
    show ?thesis
      using target catalog_target fresh
      by (simp add: extend_relevance_catalog_def)
  qed

theorem evaluate_relevance_provenance_explicit:
  "relevance_profile_model_of
    (evaluated_relevance_profile_of (evaluated_relevance_payload request)) =
    relevance_profile_model_of (requested_relevance_profile_of request) \<and>
  relevance_profile_revision_of
    (evaluated_relevance_profile_of (evaluated_relevance_payload request)) =
    relevance_profile_revision_of (requested_relevance_profile_of request) \<and>
  evaluated_relevance_perspective_of (evaluated_relevance_payload request) =
    requested_relevance_perspective_of request \<and>
  evaluated_relevance_context_of (evaluated_relevance_payload request) =
    requested_relevance_context_of request \<and>
  evaluated_relevance_time_of (evaluated_relevance_payload request) =
    requested_relevance_evaluation_time_of request"
  by simp

subsection \<open>Source integrity and payload validity\<close>

lemma evaluated_relevance_source_integrity:
  assumes valid:
    "evaluate_relevance_input_valid G s effect_catalog relevance_catalog request
      next_snapshot_id"
  shows
    "references_complete s (evaluated_relevance_object s request) \<and>
      evidence_well_formed s (evaluated_relevance_object s request) \<and>
      time_well_formed (evaluated_relevance_object s request) \<and>
      perspective_well_formed s (evaluated_relevance_object s request) \<and>
      context_well_formed s (evaluated_relevance_object s request) \<and>
      relevance_well_formed s (evaluated_relevance_object s request)"
  proof -
    from valid have active_target:
      "active_classifiable_effect s effect_catalog
        (requested_relevance_target_of request)
        (requested_relevance_effect_vector_of request)
        (requested_relevance_context_of request)"
    and active_rationale:
      "active_relevance_rationale s

```

```

        (requested_relevance_target_of request)
        (requested_relevance_rationale_refs_of request)"
and perspective:
    "requested_relevance_perspective_of request \<in> perspectives_of s"
and context_ok:
    "requested_relevance_context_of request \<in> contexts_of s"
and model:
    "relevance_profile_model_of (requested_relevance_profile_of request)
    \<in> models_of s"
unfolding evaluate_relevance_input_valid_def by blast+
from active_classifiable_effect_version[OF active_target] have target_version:
    "requested_relevance_target_of request \<in> version_ids s" .
from active_relevance_rationale_versions[OF active_rationale] have rationale:
    "requested_relevance_rationale_refs_of request \<subseque> version_ids s" .
from active_rationale have rationale_nonempty:
    "requested_relevance_rationale_refs_of request \<noteq> {}"
unfolding active_relevance_rationale_def by blast
have refs:
    "references_complete s (evaluated_relevance_object s request)"
unfolding references_complete_def evaluated_relevance_object_def
    requested_relevance_references_def
using target_version rationale by auto
have evidence:
    "evidence_well_formed s (evaluated_relevance_object s request)"
unfolding evidence_well_formed_def evidence_references_complete_def
    evaluated_relevance_object_def
using rationale rationale_nonempty evaluated_relevance_lifecycle_cases
by auto
have time:
    "time_well_formed (evaluated_relevance_object s request)"
by (simp add: time_well_formed_def)
have perspective_wf:
    "perspective_well_formed s (evaluated_relevance_object s request)"
using perspective
by (simp add: perspective_well_formed_def perspective_bound_kind_def)
have context_wf:
    "context_well_formed s (evaluated_relevance_object s request)"
using context_ok
by (simp add: context_well_formed_def context_bound_kind_def)
have relevance_wf:
    "relevance_well_formed s (evaluated_relevance_object s request)"
using perspective_wf context_wf model
by (simp add: relevance_well_formed_def
    requested_relevance_references_def)
show ?thesis
using refs evidence time perspective_wf context_wf relevance_wf by blast
qed

lemma requested_relevance_duration_positive:
  assumes valid:
    "evaluate_relevance_input_valid G s effect_catalog relevance_catalog request
    next_snapshot_id"
  shows
    "relevance_duration
    (requested_relevance_profile_of request)
    (requested_relevance_perspective_of request)
    (requested_relevance_context_of request)
    (requested_relevance_duration_input_of request) > 0"
proof -
  from valid have profile:
    "relevance_model_profile_well_formed
    (requested_relevance_profile_of request)"
  and perspective:
    "requested_relevance_perspective_of request \<in>
    relevance_profile_perspectives_of
    (requested_relevance_profile_of request)"
  and context_ok:
    "requested_relevance_context_of request \<in>
    relevance_profile_contexts_of (requested_relevance_profile_of request)"
  and duration: "requested_relevance_duration_input_of request > 0"
  unfolding evaluate_relevance_input_valid_def by blast+
  from profile perspective context_ok have cap:
    "duration_cap_of (requested_relevance_profile_of request)
    (requested_relevance_perspective_of request)
    (requested_relevance_context_of request) > 0"
  unfolding relevance_model_profile_well_formed_def by blast
  show ?thesis
  unfolding relevance_duration_def
  using duration cap by simp
qed

```

```

lemma evaluated_relevance_payload_is_valid:
  assumes valid:
    "evaluate_relevance_input_valid G s effect_catalog relevance_catalog request
     next_snapshot_id"
  shows
    "relevance_payload_valid
     (evaluated_relevance_object s request)
     (evaluated_relevance_payload request)"
proof -
  from valid have profile:
    "relevance_model_profile_well_formed
     (requested_relevance_profile_of request)"
  and perspective:
    "requested_relevance_perspective_of request \<in>
     relevance_profile_perspectives_of
     (requested_relevance_profile_of request)"
  and context_ok:
    "requested_relevance_context_of request \<in>
     relevance_profile_contexts_of (requested_relevance_profile_of request)"
  and vector:
    "effect_vector_valid (requested_relevance_effect_vector_of request)"
  and power_values:
    "power_profile_valid (requested_relevance_power_profile_of request)"
  and uncertainty:
    "0 \<le> requested_relevance_uncertainty_of request"
    "requested_relevance_uncertainty_of request \<le> 1"
  and rationale:
    "requested_relevance_rationale_refs_of request \<noteq> {}"
  and explanation:
    "requested_relevance_explanation_of request \<noteq> []"
  unfolding evaluate_relevance_input_valid_def active_relevance_rationale_def
  by blast+
  have intensity:
    "0 \<le> relevance_intensity_of (requested_relevance_vector request) \<and>
     relevance_intensity_of (requested_relevance_vector request) \<le> 1"
  unfolding requested_relevance_vector_def
  using relevance_intensity_bounded by simp
  have stability:
    "0 \<le> relevance_stability_of (requested_relevance_vector request) \<and>
     relevance_stability_of (requested_relevance_vector request) \<le> 1"
  unfolding requested_relevance_vector_def
  using relevance_stability_bounded by simp
  have reach:
    "0 \<le> relevance_reach_of (requested_relevance_vector request) \<and>
     relevance_reach_of (requested_relevance_vector request) \<le> 1"
  unfolding requested_relevance_vector_def
  using relevance_reach_bounded by simp
  have duration:
    "relevance_duration_of (requested_relevance_vector request) > 0"
  unfolding requested_relevance_vector_def
  using requested_relevance_duration_positive[OF valid] by simp
  have uncertainty_result:
    "0 \<le> relevance_uncertainty_of (requested_relevance_vector request) \<and>
     relevance_uncertainty_of (requested_relevance_vector request) \<le> 1"
  unfolding requested_relevance_vector_def
  by (simp add: clamp01_closed_interval)
  have recompute:
    "requested_relevance_vector request =
     calculate_relevance_vector
     (requested_relevance_profile_of request)
     (requested_relevance_perspective_of request)
     (requested_relevance_context_of request)
     (requested_relevance_evaluation_time_of request)
     (requested_relevance_effect_vector_of request)
     (requested_relevance_power_profile_of request)
     (requested_relevance_uncertainty_of request)
     (requested_relevance_perturbation_radius_of request)
     (requested_relevance_reach_input_of request)
     (requested_relevance_duration_input_of request)
     (relevance_explanation_of (requested_relevance_vector request)))"
  unfolding requested_relevance_vector_def
  using explanation by simp
  have references_exact:
    "requested_relevance_references request =
     insert (requested_relevance_target_of request)
     (requested_relevance_rationale_refs_of request)"
  unfolding requested_relevance_references_def by simp
  have result_explanation:
    "relevance_explanation_of (requested_relevance_vector request) \<noteq> []"

```

```

    unfolding requested_relevance_vector_def
    using explanation by simp
show ?thesis
    unfolding relevance_payload_valid_def
    using profile perspective context_ok vector power_values intensity stability reach
    duration uncertainty_result rationale explanation recompute
    references_exact result_explanation evaluated_relevance_lifecycle_cases
    by simp
qed

subsection \<open>Monotonicity, fresh identity, and target-state preservation\<close>

lemma add_relevance_present_ref_mono:
  assumes "present_ref s reference"
  shows "present_ref (add_evaluated_relevance s request next_snapshot_id) reference"
  unfolding add_evaluated_relevance_def
  using add_appearance_present_ref_mono[OF assms] .

lemma add_relevance_references_mono:
  assumes "references_complete s obj"
  shows "references_complete
    (add_evaluated_relevance s request next_snapshot_id) obj"
  unfolding add_evaluated_relevance_def
  using add_appearance_references_mono[OF assms] .

lemma add_relevance_evidence_mono:
  assumes "evidence_well_formed s obj"
  shows "evidence_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj"
  unfolding add_evaluated_relevance_def
  using add_appearance_evidence_mono[OF assms] .

lemma add_relevance_relation_mono:
  assumes "relation_well_formed s obj"
  shows "relation_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj"
  unfolding add_evaluated_relevance_def
  using add_appearance_relation_mono[OF assms] .

lemma add_relevance_effect_mono:
  assumes "effect_well_formed s obj"
  shows "effect_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj"
  unfolding add_evaluated_relevance_def
  using add_appearance_effect_mono[OF assms] .

lemma add_relevance_perspective_iff [simp]:
  "perspective_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj
    \<longleftarrowrightarrow> perspective_well_formed s obj"
  unfolding add_evaluated_relevance_def
  by simp

lemma add_relevance_context_iff [simp]:
  "context_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj
    \<longleftarrowrightarrow> context_well_formed s obj"
  unfolding add_evaluated_relevance_def
  by simp

lemma add_relevance_relevance_iff [simp]:
  "relevance_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj
    \<longleftarrowrightarrow> relevance_well_formed s obj"
  unfolding add_evaluated_relevance_def
  by simp

lemma add_relevance_orientation_iff [simp]:
  "orientation_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj
    \<longleftarrowrightarrow> orientation_well_formed s obj"
  unfolding add_evaluated_relevance_def
  by simp

lemma add_relevance_action_iff [simp]:
  "action_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj
    \<longleftarrowrightarrow> action_well_formed s obj"
  unfolding add_evaluated_relevance_def
  by simp

```

```

lemma add_relevance_feedback_mono:
  assumes "feedback_well_formed s obj"
  shows "feedback_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj"
  unfolding add_evaluated_relevance_def
  using add_appearance_feedback_mono[OF assms] .

lemma add_relevance_preserves_active_history_disjoint:
  assumes valid:
    "evaluate_relevance_input_valid G s effect_catalog relevance_catalog request
      next_snapshot_id"
  shows "active_history_disjoint
    (add_evaluated_relevance s request next_snapshot_id)"
  using valid
  by (auto simp: evaluate_relevance_input_valid_def active_history_disjoint_def)

lemma add_relevance_preserves_unique_versions:
  assumes valid:
    "evaluate_relevance_input_valid G s effect_catalog relevance_catalog request
      next_snapshot_id"
  shows "unique_versions
    (add_evaluated_relevance s request next_snapshot_id)"
proof -
  let ?new = "evaluated_relevance_object s request"
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
  and fresh: "version_id_of ?new \<notin> version_ids s"
  unfolding evaluate_relevance_input_valid_def state_well_formed_def
  unique_versions_def
  by auto
  show ?thesis
  unfolding unique_versions_def
  proof (rule inj_onI)
    fix left right
    assume left_mem:
      "left \<in> objects_of (add_evaluated_relevance s request next_snapshot_id)"
    and right_mem:
      "right \<in> objects_of (add_evaluated_relevance s request next_snapshot_id)"
    and same_version: "version_id_of left = version_id_of right"
    show "left = right"
    proof (cases "left = ?new")
      case True
      show ?thesis
      proof (cases "right = ?new")
        case True
        then show ?thesis using \<open>left = ?new\<close> by simp
      next
        case False
        then have right_old: "right \<in> objects_of s" using right_mem by simp
        have "version_id_of ?new \<in> version_ids s"
        unfolding version_ids_def
        using right_old same_version \<open>left = ?new\<close>
        by (metis imageI)
        then show ?thesis using fresh by blast
      qed
    next
      case False
      then have left_old: "left \<in> objects_of s" using left_mem by simp
      show ?thesis
      proof (cases "right = ?new")
        case True
        have "version_id_of ?new \<in> version_ids s"
        unfolding version_ids_def
        using left_old same_version True by (metis imageI)
        then show ?thesis using fresh by blast
      next
        case False
        then have right_old: "right \<in> objects_of s" using right_mem by simp
        show ?thesis
        using inj_onD[OF source_unique same_version left_old right_old] .
      qed
    qed
  qed
qed
qed
qed

lemma evaluate_relevance_new_object_conditions:
  assumes valid:
    "evaluate_relevance_input_valid G s effect_catalog relevance_catalog request
      next_snapshot_id"
  shows

```

```

"kind_of (evaluated_relevance_object s request) \<in> supported_kinds_of G \<and>
references_complete
  (add_evaluated_relevance s request next_snapshot_id)
  (evaluated_relevance_object s request) \<and>
evidence_well_formed
  (add_evaluated_relevance s request next_snapshot_id)
  (evaluated_relevance_object s request) \<and>
time_well_formed (evaluated_relevance_object s request) \<and>
relation_well_formed
  (add_evaluated_relevance s request next_snapshot_id)
  (evaluated_relevance_object s request) \<and>
effect_well_formed
  (add_evaluated_relevance s request next_snapshot_id)
  (evaluated_relevance_object s request) \<and>
perspective_well_formed
  (add_evaluated_relevance s request next_snapshot_id)
  (evaluated_relevance_object s request) \<and>
context_well_formed
  (add_evaluated_relevance s request next_snapshot_id)
  (evaluated_relevance_object s request) \<and>
relevance_well_formed
  (add_evaluated_relevance s request next_snapshot_id)
  (evaluated_relevance_object s request) \<and>
orientation_well_formed
  (add_evaluated_relevance s request next_snapshot_id)
  (evaluated_relevance_object s request) \<and>
action_well_formed
  (add_evaluated_relevance s request next_snapshot_id)
  (evaluated_relevance_object s request) \<and>
feedback_well_formed
  (add_evaluated_relevance s request next_snapshot_id)
  (evaluated_relevance_object s request)"
proof -
  let ?new = "evaluated_relevance_object s request"
  from valid have supported: "K_REV \<in> supported_kinds_of G"
  unfolding evaluate_relevance_input_valid_def by blast
  from evaluated_relevance_source_integrity[OF valid] have refs:
    "references_complete s ?new"
    and evidence: "evidence_well_formed s ?new"
    and time: "time_well_formed ?new"
    and perspective: "perspective_well_formed s ?new"
    and context_ok: "context_well_formed s ?new"
    and relevance: "relevance_well_formed s ?new"
  by blast+
  have refs_target:
    "references_complete (add_evaluated_relevance s request next_snapshot_id) ?new"
  using add_relevance_references_mono[OF refs] .
  have evidence_target:
    "evidence_well_formed (add_evaluated_relevance s request next_snapshot_id) ?new"
  using add_relevance_evidence_mono[OF evidence] .
  have typed_conditions:
    "relation_well_formed
      (add_evaluated_relevance s request next_snapshot_id) ?new \<and>
      effect_well_formed
        (add_evaluated_relevance s request next_snapshot_id) ?new \<and>
      orientation_well_formed
        (add_evaluated_relevance s request next_snapshot_id) ?new \<and>
      action_well_formed
        (add_evaluated_relevance s request next_snapshot_id) ?new \<and>
      feedback_well_formed
        (add_evaluated_relevance s request next_snapshot_id) ?new"
  by (simp add: relation_well_formed_def effect_well_formed_def
    orientation_well_formed_def action_well_formed_def
    feedback_well_formed_def)
  show ?thesis
  using supported refs_target evidence_target time perspective context_ok
    relevance typed_conditions
  by simp
qed

subsection \<open>Operation-specific preservation theorems\<close>

theorem evaluate_relevance_preserves_state_well_formed:
  assumes concrete:
    "evaluate_relevance_step G s effect_catalog relevance_catalog request
      next_snapshot_id s' relevance_catalog"
  shows "state_well_formed G s'"
proof -
  from concrete have valid:
    "evaluate_relevance_input_valid G s effect_catalog relevance_catalog request

```

```

    next_snapshot_id"
  and target: "s' = add_evaluated_relevance s request next_snapshot_id"
  by (auto simp: evaluate_relevance_step_def)
from valid have source_wf: "state_well_formed G s"
  by (simp add: evaluate_relevance_input_valid_def)
have unique:
  "unique_versions (add_evaluated_relevance s request next_snapshot_id)"
  using add_relevance_preserves_unique_versions[OF valid] .
have all_objects:
  "\<forall>obj\<in>objects_of (add_evaluated_relevance s request next_snapshot_id).
  kind_of obj \<in> supported_kinds_of G \<and>
  references_complete
    (add_evaluated_relevance s request next_snapshot_id) obj \<and>
  evidence_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj \<and>
  time_well_formed obj \<and>
  relation_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj \<and>
  effect_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj \<and>
  perspective_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj \<and>
  context_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj \<and>
  relevance_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj \<and>
  orientation_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj \<and>
  action_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj \<and>
  feedback_well_formed
    (add_evaluated_relevance s request next_snapshot_id) obj"
proof (intro ballI)
  fix obj
  assume obj_mem:
    "obj \<in> objects_of (add_evaluated_relevance s request next_snapshot_id)"
  consider (new) "obj = evaluated_relevance_object s request"
    | (old) "obj \<in> objects_of s"
  using obj_mem by auto
  then show
    "kind_of obj \<in> supported_kinds_of G \<and>
    references_complete
      (add_evaluated_relevance s request next_snapshot_id) obj \<and>
    evidence_well_formed
      (add_evaluated_relevance s request next_snapshot_id) obj \<and>
    time_well_formed obj \<and>
    relation_well_formed
      (add_evaluated_relevance s request next_snapshot_id) obj \<and>
    effect_well_formed
      (add_evaluated_relevance s request next_snapshot_id) obj \<and>
    perspective_well_formed
      (add_evaluated_relevance s request next_snapshot_id) obj \<and>
    context_well_formed
      (add_evaluated_relevance s request next_snapshot_id) obj \<and>
    relevance_well_formed
      (add_evaluated_relevance s request next_snapshot_id) obj \<and>
    orientation_well_formed
      (add_evaluated_relevance s request next_snapshot_id) obj \<and>
    action_well_formed
      (add_evaluated_relevance s request next_snapshot_id) obj \<and>
    feedback_well_formed
      (add_evaluated_relevance s request next_snapshot_id) obj"
  proof cases
    case new
    then show ?thesis
      using evaluate_relevance_new_object_conditions[OF valid] by simp
  next
    case old
    from source_wf old have source_conditions:
      "kind_of obj \<in> supported_kinds_of G \<and>
      references_complete s obj \<and> evidence_well_formed s obj \<and>
      time_well_formed obj \<and> relation_well_formed s obj \<and>
      effect_well_formed s obj \<and> perspective_well_formed s obj \<and>
      context_well_formed s obj \<and> relevance_well_formed s obj \<and>
      orientation_well_formed s obj \<and> action_well_formed s obj \<and>
      feedback_well_formed s obj"
    unfolding state_well_formed_def by blast
    from source_conditions have supported:
      "kind_of obj \<in> supported_kinds_of G"
    and refs: "references_complete s obj"

```

```

and evidence: "evidence_well_formed s obj"
and time: "time_well_formed obj"
and relation: "relation_well_formed s obj"
and effect: "effect_well_formed s obj"
and perspective: "perspective_well_formed s obj"
and context_ok: "context_well_formed s obj"
and relevance: "relevance_well_formed s obj"
and orientation: "orientation_well_formed s obj"
and action: "action_well_formed s obj"
and feedback: "feedback_well_formed s obj"
by blast+
have refs_target:
  "references_complete
  (add_evaluated_relevance s request next_snapshot_id) obj"
  using add_relevance_references_mono[OF refs] .
have evidence_target:
  "evidence_well_formed
  (add_evaluated_relevance s request next_snapshot_id) obj"
  using add_relevance_evidence_mono[OF evidence] .
have relation_target:
  "relation_well_formed
  (add_evaluated_relevance s request next_snapshot_id) obj"
  using add_relevance_relation_mono[OF relation] .
have effect_target:
  "effect_well_formed
  (add_evaluated_relevance s request next_snapshot_id) obj"
  using add_relevance_effect_mono[OF effect] .
have feedback_target:
  "feedback_well_formed
  (add_evaluated_relevance s request next_snapshot_id) obj"
  using add_relevance_feedback_mono[OF feedback] .
show ?thesis
  using supported refs_target evidence_target time relation_target
    effect_target perspective context_ok relevance orientation action
    feedback_target
  by simp
qed
qed
from source_wf have source_secondary:
  "(\<forall>p\<in>projections_of s. projection_well_formed s p) \<and>
  (\<forall>c\<in>couplings_of s. coupling_well_formed s c) \<and>
  (\<forall>t\<in>transmissions_of s. transmission_well_formed s t) \<and>
  (\<forall>b\<in>boundaries_of s. boundary_well_formed b)"
  unfolding state_well_formed_def by blast
have target_secondary:
  "(\<forall>p\<in>projections_of
  (add_evaluated_relevance s request next_snapshot_id).
  projection_well_formed
  (add_evaluated_relevance s request next_snapshot_id) p) \<and>
  (\<forall>c\<in>couplings_of
  (add_evaluated_relevance s request next_snapshot_id).
  coupling_well_formed
  (add_evaluated_relevance s request next_snapshot_id) c) \<and>
  (\<forall>t\<in>transmissions_of
  (add_evaluated_relevance s request next_snapshot_id).
  transmission_well_formed
  (add_evaluated_relevance s request next_snapshot_id) t) \<and>
  (\<forall>b\<in>boundaries_of
  (add_evaluated_relevance s request next_snapshot_id).
  boundary_well_formed b)"
  using source_secondary
  by (simp add: projection_well_formed_def coupling_well_formed_def
    transmission_well_formed_def)
show ?thesis
  unfolding target state_well_formed_def
  using unique_all_objects target_secondary by blast
qed

theorem evaluate_relevance_preserves_identity:
  assumes concrete:
    "evaluate_relevance_step G s effect_catalog relevance_catalog request
    next_snapshot_id s' relevance_catalog"
  shows "unique_versions s' \<and> immutable_preserved s s'"
proof -
  let ?new = "evaluated_relevance_object s request"
  from concrete have valid:
    "evaluate_relevance_input_valid G s effect_catalog relevance_catalog request
    next_snapshot_id"
  and target: "s' = add_evaluated_relevance s request next_snapshot_id"
  by (auto simp: evaluate_relevance_step_def)

```



```

from valid have source_unique: "inj_on version_id_of (objects_of s)"
  and fresh: "version_id_of ?new \<notin> version_ids s"
  unfolding evaluate_relevance_input_valid_def state_well_formed_def
  unique_versions_def
  by auto
have immutable: "immutable_preserved s s'"
  unfolding immutable_preserved_def target
proof (intro ballI allI impI)
  fix old_obj target_obj
  assume old_mem: "old_obj \<in> objects_of s"
  and target_mem:
    "target_obj \<in>
      objects_of (add_evaluated_relevance s request next_snapshot_id)"
  and same_version:
    "version_id_of old_obj = version_id_of target_obj"
  show "old_obj = target_obj"
proof (cases "target_obj = ?new")
  case True
  have "version_id_of ?new \<in> version_ids s"
  unfolding version_ids_def
  using old_mem same_version True by (metis imageI)
  then show ?thesis using fresh by blast
next
  case False
  then have target_old: "target_obj \<in> objects_of s"
  using target_mem by simp
  show ?thesis
  using inj_onD[OF source_unique same_version old_mem target_old] .
qed
qed
have target_unique: "unique_versions s'"
  unfolding target
  using add_relevance_preserves_unique_versions[OF valid] .
show ?thesis using target_unique immutable by blast
qed

theorem evaluate_relevance_preserves_effect_catalog:
  assumes concrete:
    "evaluate_relevance_step G s effect_catalog relevance_catalog request
      next_snapshot_id s' relevance_catalog"
  shows "effect_catalog_well_formed s' effect_catalog"
proof -
  from concrete have valid:
    "evaluate_relevance_input_valid G s effect_catalog relevance_catalog request
      next_snapshot_id"
  and target: "s' = add_evaluated_relevance s request next_snapshot_id"
  by (auto simp: evaluate_relevance_step_def)
  from valid have source_catalog:
    "effect_catalog_well_formed s effect_catalog"
  by (simp add: evaluate_relevance_input_valid_def)
  show ?thesis
  unfolding effect_catalog_well_formed_def target
  using source_catalog
  unfolding effect_catalog_well_formed_def
  by auto
qed

theorem evaluate_relevance_preserves_resonance_catalog:
  assumes concrete:
    "evaluate_relevance_step G s effect_catalog relevance_catalog request
      next_snapshot_id s' relevance_catalog"
  and source_catalog: "resonance_catalog_well_formed s resonance_catalog"
  shows "resonance_catalog_well_formed s' resonance_catalog"
proof -
  from concrete have target:
    "s' = add_evaluated_relevance s request next_snapshot_id"
  by (auto simp: evaluate_relevance_step_def)
  show ?thesis
  unfolding resonance_catalog_well_formed_def
proof (intro allI impI)
  fix version payload
  assume entry: "resonance_catalog version = Some payload"
  from source_catalog entry obtain resonance where
    old_mem: "resonance \<in> objects_of s"
  and version: "version_id_of resonance = version"
  and payload: "resonance_payload_valid resonance payload"
  and refs: "references_complete s resonance"
  and evidence: "evidence_well_formed s resonance"
  unfolding resonance_catalog_well_formed_def by blast
  have new_mem: "resonance \<in> objects_of s'"

```

```

    unfolding target using old_mem by simp
    have refs_target: "references_complete s' resonance"
    unfolding target using add_relevance_references_mono[OF refs] .
    have evidence_target: "evidence_well_formed s' resonance"
    unfolding target using add_relevance_evidence_mono[OF evidence] .
    show "\<exists>resonance\<in>objects_of s'.
    version_id_of resonance = version \<and>
    resonance_payload_valid resonance payload \<and>
    references_complete s' resonance \<and>
    evidence_well_formed s' resonance"
    using new_mem version payload refs_target evidence_target by blast
qed
qed

theorem evaluate_relevance_preserves_catalog:
  assumes concrete:
    "evaluate_relevance_step G s effect_catalog relevance_catalog request
    next_snapshot_id s' relevance_catalog"
  shows
    "relevance_catalog_well_formed s' relevance_catalog' \<and>
    relevance_catalog' (requested_relevance_version_id_of request) =
    Some (evaluated_relevance_payload request)"
proof -
  from concrete have valid:
    "evaluate_relevance_input_valid G s effect_catalog relevance_catalog request
    next_snapshot_id"
  and target: "s' = add_evaluated_relevance s request next_snapshot_id"
  and catalog_target:
    "relevance_catalog' = extend_relevance_catalog relevance_catalog request"
  by (auto simp: evaluate_relevance_step_def)
  from valid have source_catalog:
    "relevance_catalog_well_formed s relevance_catalog"
  and fresh_key:
    "relevance_catalog (requested_relevance_version_id_of request) = None"
  by (auto simp: evaluate_relevance_input_valid_def)
  have payload_valid:
    "relevance_payload_valid
    (evaluated_relevance_object s request)
    (evaluated_relevance_payload request)"
  using evaluated_relevance_payload_is_valid[OF valid] .
  from evaluated_relevance_source_integrity[OF valid] have source_refs:
    "references_complete s (evaluated_relevance_object s request)"
  and source_evidence:
    "evidence_well_formed s (evaluated_relevance_object s request)"
  by blast+
  have new_refs:
    "references_complete s' (evaluated_relevance_object s request)"
  unfolding target
  using add_relevance_references_mono[OF source_refs] .
  have new_evidence:
    "evidence_well_formed s' (evaluated_relevance_object s request)"
  unfolding target
  using add_relevance_evidence_mono[OF source_evidence] .
  have target_catalog: "relevance_catalog_well_formed s' relevance_catalog'"
  unfolding relevance_catalog_well_formed_def catalog_target
  extend_relevance_catalog_def
proof (intro allI impI)
  fix version payload
  assume updated:
    "(relevance_catalog
    (requested_relevance_version_id_of request \<mapsto>
    evaluated_relevance_payload request)) version = Some payload"
  show "\<exists>relevance\<in>objects_of s'.
  version_id_of relevance = version \<and>
  relevance_payload_valid relevance payload \<and>
  references_complete s' relevance \<and>
  evidence_well_formed s' relevance"
proof (cases "version = requested_relevance_version_id_of request")
  case True
  with updated have payload_eq:
    "payload = evaluated_relevance_payload request"
  by simp
  have new_mem:
    "evaluated_relevance_object s request \<in> objects_of s'"
  unfolding target by simp
  show ?thesis
proof (rule bexI[where x = "evaluated_relevance_object s request"])
  show
    "version_id_of (evaluated_relevance_object s request) = version \<and>
    relevance_payload_valid (evaluated_relevance_object s request)"

```

```

        payload \<and>
        references_complete s' (evaluated_relevance_object s request) \<and>
        evidence_well_formed s' (evaluated_relevance_object s request)"
    using True payload_eq payload_valid new_refs new_evidence by simp
    show "evaluated_relevance_object s request \<in> objects_of s'"
    using new_mem .
qed
next
case False
with updated have old_entry: "relevance_catalog version = Some payload"
  by simp
from source_catalog old_entry obtain relevance where
  old_mem: "relevance \<in> objects_of s"
  and version: "version_id_of relevance = version"
  and payload: "relevance_payload_valid relevance payload"
  and refs: "references_complete s relevance"
  and evidence: "evidence_well_formed s relevance"
unfolding relevance_catalog_well_formed_def by blast
have refs_target: "references_complete s' relevance"
  unfolding target using add_relevance_references_mono[OF refs] .
have evidence_target: "evidence_well_formed s' relevance"
  unfolding target using add_relevance_evidence_mono[OF evidence] .
have old_mem_target: "relevance \<in> objects_of s'"
  unfolding target using old_mem by simp
show ?thesis
proof (rule bexI[where x = relevance])
  show
    "version_id_of relevance = version \<and>
    relevance_payload_valid relevance payload \<and>
    references_complete s' relevance \<and>
    evidence_well_formed s' relevance"
    using version payload refs_target evidence_target by blast
  show "relevance \<in> objects_of s'" using old_mem_target .
qed
qed
qed
show ?thesis
  using target_catalog catalog_target
  by (simp add: extend_relevance_catalog_def)
qed

theorem evaluate_relevance_preserves_references_time_context:
  assumes concrete:
    "evaluate_relevance_step G s effect_catalog relevance_catalog request
    next_snapshot_id s' relevance_catalog"
  shows
    "\<forall>obj\<in>objects_of s'.
    references_complete s' obj \<and>
    evidence_well_formed s' obj \<and>
    time_well_formed obj \<and>
    perspective_well_formed s' obj \<and>
    context_well_formed s' obj"
  using evaluate_relevance_preserves_state_well_formed[OF concrete]
  unfolding state_well_formed_def by blast

theorem evaluate_relevance_refines_generic_step:
  assumes concrete:
    "evaluate_relevance_step G s effect_catalog relevance_catalog request
    next_snapshot_id s' relevance_catalog"
  shows "step G s EvaluateRelevance s'"
proof -
  from concrete have source_wf: "state_well_formed G s"
  and allowed: "EvaluateRelevance \<in> allowed_operations_of G"
  by (auto simp: evaluate_relevance_step_def
    evaluate_relevance_input_valid_def)
  have target_wf: "state_well_formed G s'"
  using evaluate_relevance_preserves_state_well_formed[OF concrete] .
  have immutable: "immutable_preserved s s'"
  using evaluate_relevance_preserves_identity[OF concrete] by blast
  show ?thesis
  unfolding step_def
  using source_wf target_wf allowed immutable by blast
qed
end

```

### D.13 Historischer isolierter Perspektivdivergenz-Zeuge für T7 (lokal weiterhin gültig)

Der Zeuge verhindert, dass Perspektivrelativität nur eine unaktivierte Signatur bleibt. Derselbe validierte Effekt mit Connection=+3 und Stability=-3, dasselbe ResourcePower-Profil, dasselbe Modell 30/Revision 1, derselbe Kontext, Zeitpunkt und Unsicherheitswert werden nacheinander aus zwei Perspektiven bewertet. P0 gewichtet Connection, P1 Stability; der Machtterm bleibt identisch. Beide Ergebnisse werden als getrennte K\_REV-Versionen gespeichert.

Isabelle beweist  $P0 = \langle 9/10, 1/10, 49/50, 4/5, 8 \rangle$  und  $P1 = \langle 9/50, 1/10, 19/20, 2/5, 4 \rangle$ , die Ungleichheit der vollständigen Vektoren, Validated-Lifecycle, vollständige Provenienz, zwei exakte Zustandsdeltas, finale Zustands- und Katalogwohlgeformtheit und zwei generische EvaluateRelevance-Schritte. Die Theorie besitzt 490 Zeilen und den SHA-256 3ba9e93262c0b139cf0d8521e8785ef21b66fb30de3e9e9bfa9de3989e817f39.

```
theory UZA_0_9_1_Evaluate_Relevance_Witness
  imports UZA_0_9_1_Evaluate_Relevance
          UZA_0_9_1_Classify_Resonance_Witness
begin

section \<open>A non-vacuous perspective-divergence witness for T7\<close>

text \<open>
  The same validated effect version, model revision, context, evaluation time,
  uncertainty, power profile, reach input, and duration input is evaluated from
  two distinct perspectives. Perspective 0 attends to Connection;
  perspective 1 attends to Stability. The underlying V10 vector is positive
  on Connection and negative on Stability. The two K_REV results therefore
  remain separate and differ in intensity, stability, reach, and duration.
  No averaging or implicit power-to-V10 conversion occurs.
\<close>

definition t7_effect_vector :: effect_vector where
  "t7_effect_vector dimension =
    (if dimension = Connection then 3
     else if dimension = Stability then -3
     else 0)"

definition t7_power_profile :: power_profile where
  "t7_power_profile dimension =
    (if dimension = ResourcePower then 1 else 0)"

fun t7_effect_weights ::
  "perspective_id \<Rightarrow> relevance_weights" where
  "t7_effect_weights 0 dimension =
    (if dimension = Connection then 1 else 0)"
| "t7_effect_weights (Suc 0) dimension =
  (if dimension = Stability then 1 else 0)"
| "t7_effect_weights _ _ = 0"

fun t7_power_weights ::
  "perspective_id \<Rightarrow> power_weights" where
  "t7_power_weights 0 dimension =
    (if dimension = ResourcePower then 1 else 0)"
| "t7_power_weights (Suc 0) dimension =
  (if dimension = ResourcePower then 1 else 0)"
| "t7_power_weights _ _ = 0"

definition t7_relevance_profile :: relevance_model_profile where
  "t7_relevance_profile =
    \<lparr>
      relevance_profile_model_of = 30,
      relevance_profile_revision_of = 1,
      relevance_profile_perspectives_of = {0, Suc 0},
      relevance_profile_contexts_of = {20},
      perspective_effect_weights_of = t7_effect_weights,
      perspective_power_weights_of = t7_power_weights,
      context_effect_factor_of = (\<lambda>_ _ . 1),
      context_power_factor_of = (\<lambda>_ _ . 1),
      temporal_relevance_factor_of = (\<lambda>_ . 1),
      reach_factor_of =
```

```

        (\<lambda>perspective _. if perspective = 0 then 1 else 1 / 2),
        duration_cap_of =
        (\<lambda>perspective _. if perspective = 0 then 10 else 4),
        local_sensitivity_of =
        (\<lambda>perspective _. if perspective = 0 then 1 / 5 else 1 / 2),
        effect_share_of = 4 / 5,
        relevance_validation_uncertainty_max_of = 1 / 5
    \<rparr>"

definition t7_witness_source :: snapshot where
    "t7_witness_source =
        t6_witness_source\<lparr>
            snapshot_id_of := 10,
            perspectives_of := {0, Suc 0, 10}
        \<rparr>"

definition t7_witness_grammar :: grammar where
    "t7_witness_grammar =
        \<lparr>
            grammar_id_of = 7,
            supported_kinds_of = {K_ENT, K_REL, K_EFF, K_RES, K_REV},
            allowed_operations_of = {EvaluateRelevance},
            patch_budget_of = 0
        \<rparr>"

definition t7_effect_payload :: effect_hypothesis_payload where
    "t7_effect_payload =
        \<lparr>
            hypothesis_cause_of = 3,
            hypothesis_target_of = None,
            hypothesis_vector_of = Some t7_effect_vector,
            hypothesis_uncertainty_of = 1 / 10,
            hypothesis_profile_model_of = 30
        \<rparr>"

definition t7_effect_catalog :: effect_catalog where
    "t7_effect_catalog = Map.empty(4 \<mapsto> t7_effect_payload)"

definition t7_empty_relevance_catalog :: relevance_catalog where
    "t7_empty_relevance_catalog = Map.empty"

definition t7_request_p0 :: relevance_request where
    "t7_request_p0 =
        \<lparr>
            requested_relevance_object_id_of = 7,
            requested_relevance_version_id_of = 7,
            requested_relevance_target_of = 4,
            requested_relevance_effect_vector_of = t7_effect_vector,
            requested_relevance_power_profile_of = t7_power_profile,
            requested_relevance_profile_of = t7_relevance_profile,
            requested_relevance_perspective_of = 0,
            requested_relevance_context_of = 20,
            requested_relevance_evaluation_time_of = 10,
            requested_relevance_uncertainty_of = 1 / 10,
            requested_relevance_perturbation_radius_of = 1 / 10,
            requested_relevance_reach_input_of = 4 / 5,
            requested_relevance_duration_input_of = 8,
            requested_relevance_rationale_refs_of = {1, 2},
            requested_relevance_explanation_of =
                "'P0: Connection-centered relevance'"
        \<rparr>"

definition t7_request_p1 :: relevance_request where
    "t7_request_p1 =
        \<lparr>
            requested_relevance_object_id_of = 8,
            requested_relevance_version_id_of = 8,
            requested_relevance_target_of = 4,
            requested_relevance_effect_vector_of = t7_effect_vector,
            requested_relevance_power_profile_of = t7_power_profile,
            requested_relevance_profile_of = t7_relevance_profile,
            requested_relevance_perspective_of = Suc 0,
            requested_relevance_context_of = 20,
            requested_relevance_evaluation_time_of = 10,
            requested_relevance_uncertainty_of = 1 / 10,
            requested_relevance_perturbation_radius_of = 1 / 10,
            requested_relevance_reach_input_of = 4 / 5,
            requested_relevance_duration_input_of = 8,
            requested_relevance_rationale_refs_of = {1, 2},
            requested_relevance_explanation_of =

```

```

    'P1: Stability-centered relevance'
    \<rparr>"

definition t7_state_after_p0 :: snapshot where
  "t7_state_after_p0 =
    add_evaluated_relevance t7_witness_source t7_request_p0 11"

definition t7_catalog_after_p0 :: relevance_catalog where
  "t7_catalog_after_p0 =
    extend_relevance_catalog t7_empty_relevance_catalog t7_request_p0"

definition t7_state_after_p1 :: snapshot where
  "t7_state_after_p1 =
    add_evaluated_relevance t7_state_after_p0 t7_request_p1 12"

definition t7_catalog_after_p1 :: relevance_catalog where
  "t7_catalog_after_p1 =
    extend_relevance_catalog t7_catalog_after_p0 t7_request_p1"

subsection \<open>Finite-profile arithmetic\<close>

lemma t7_power_dimension_delta:
  fixes selected :: power_dimension
  and coefficient :: real
  shows
    "(\<Sum>dimension\<in>UNIV.
      if dimension = selected then coefficient else 0) = coefficient"
proof -
  have finite_dimensions: "finite (UNIV :: power_dimension set)"
  by simp
  note delta =
    sum.delta[OF finite_dimensions, of selected "\<lambda>_. coefficient"]
  show ?thesis using delta by simp
qed

lemma t7_effect_vector_valid:
  "effect_vector_valid t7_effect_vector"
  by (auto simp: effect_vector_valid_def t7_effect_vector_def)

lemma t7_power_profile_valid:
  "power_profile_valid t7_power_profile"
  by (auto simp: power_profile_valid_def t7_power_profile_def)

lemma t7_effect_weight_mass_p0 [simp]:
  "weight_mass (t7_effect_weights 0) = 1"
  unfolding weight_mass_def
  by (simp add: t6_effect_dimension_delta)

lemma t7_effect_weight_mass_p1 [simp]:
  "weight_mass (t7_effect_weights (Suc 0)) = 1"
  unfolding weight_mass_def
  by (simp add: t6_effect_dimension_delta)

lemma t7_power_weight_mass_p0 [simp]:
  "power_weight_mass (t7_power_weights 0) = 1"
  unfolding power_weight_mass_def
  by (simp add: t7_power_dimension_delta)

lemma t7_power_weight_mass_p1 [simp]:
  "power_weight_mass (t7_power_weights (Suc 0)) = 1"
  unfolding power_weight_mass_def
  by (simp add: t7_power_dimension_delta)

lemma t7_effect_weights_p0_valid:
  "relevance_weights_valid (t7_effect_weights 0)"
  by (auto simp: relevance_weights_valid_def)

lemma t7_effect_weights_p1_valid:
  "relevance_weights_valid (t7_effect_weights (Suc 0))"
  by (auto simp: relevance_weights_valid_def)

lemma t7_power_weights_p0_valid:
  "power_weights_valid (t7_power_weights 0)"
  by (auto simp: power_weights_valid_def)

lemma t7_power_weights_p1_valid:
  "power_weights_valid (t7_power_weights (Suc 0))"
  by (auto simp: power_weights_valid_def)

lemma t7_relevance_profile_well_formed:

```

```

"relevance_model_profile_well_formed t7_relevance_profile"
using t7_effect_weights_p0_valid t7_effect_weights_p1_valid
  t7_power_weights_p0_valid t7_power_weights_p1_valid
by (auto simp: relevance_model_profile_well_formed_def
  t7_relevance_profile_def effective_effect_weight_mass_def
  effective_power_weight_mass_def effective_effect_weight_def
  effective_power_weight_def
  t6_effect_dimension_delta t7_power_dimension_delta)

subsection \<open>Source obligations and two concrete steps\<close>

lemma t7_witness_source_simps [simp]:
  "snapshot_id_of t7_witness_source = 10"
  "objects_of t7_witness_source = t6_witness_objects"
  "historical_versions_of t7_witness_source = {}"
  "perspectives_of t7_witness_source = {0, Suc 0, 10}"
  "contexts_of t7_witness_source = {20}"
  "models_of t7_witness_source = {30}"
  "authorities_of t7_witness_source = {}"
  "contains_edges_of t7_witness_source = {}"
  "projections_of t7_witness_source = {}"
  "couplings_of t7_witness_source = {}"
  "transmissions_of t7_witness_source = {}"
  "boundaries_of t7_witness_source = {}"
by (simp_all add: t7_witness_source_def t6_witness_source_def)

lemma t7_witness_version_ids [simp]:
  "version_ids t7_witness_source = {1, 2, 3, 4, 5, 6}"
by (auto simp: version_ids_def t7_witness_source_def
  t6_witness_source_def t6_witness_objects_def t6_object_defs
  base_object_def)

lemma t7_witness_source_well_formed:
  "state_well_formed t7_witness_grammar t7_witness_source"
unfolding state_well_formed_def
by (auto simp: unique_versions_def inj_on_def t7_witness_grammar_def
  t6_witness_objects_def t6_object_defs base_object_def
  present_ref_def references_complete_def
  evidence_well_formed_def evidence_references_complete_def
  time_well_formed_def relation_well_formed_def effect_well_formed_def
  perspective_well_formed_def perspective_bound_kind_def
  context_well_formed_def context_bound_kind_def relevance_well_formed_def
  orientation_well_formed_def action_well_formed_def
  feedback_well_formed_def)

lemma t7_effect_catalog_well_formed:
  "effect_catalog_well_formed t7_witness_source t7_effect_catalog"
unfolding effect_catalog_well_formed_def
using t7_effect_vector_valid
by (auto simp: t7_effect_catalog_def t7_effect_payload_def
  t6_witness_objects_def t6_object_defs
  base_object_def effect_payload_valid_def)

lemma t7_p0_input_valid:
  "evaluate_relevance_input_valid t7_witness_grammar t7_witness_source
  t7_effect_catalog t7_empty_relevance_catalog t7_request_p0 11"
unfolding evaluate_relevance_input_valid_def
using t7_witness_source_well_formed t7_effect_catalog_well_formed
  t7_effect_vector_valid t7_power_profile_valid
  t7_relevance_profile_well_formed
by (auto simp: active_history_disjoint_def relevance_catalog_well_formed_def
  t7_witness_grammar_def t7_empty_relevance_catalog_def t7_request_p0_def
  t7_relevance_profile_def t6_witness_objects_def t6_object_defs
  base_object_def
  active_classifiable_effect_def t7_effect_catalog_def
  t7_effect_payload_def effect_payload_valid_def version_active_at_def
  active_relevance_rationale_def)

theorem t7_p0_concrete_step:
  "evaluate_relevance_step t7_witness_grammar t7_witness_source
  t7_effect_catalog t7_empty_relevance_catalog t7_request_p0 11
  t7_state_after_p0 t7_catalog_after_p0"
unfolding evaluate_relevance_step_def t7_state_after_p0_def
  t7_catalog_after_p0_def
using t7_p0_input_valid by simp

lemma t7_state_after_p0_well_formed:
  "state_well_formed t7_witness_grammar t7_state_after_p0"
using evaluate_relevance_preserves_state_well_formed[OF t7_p0_concrete_step] .

```

```

lemma t7_effect_catalog_after_p0_well_formed:
  "effect_catalog_well_formed t7_state_after_p0 t7_effect_catalog"
  using evaluate_relevance_preserves_effect_catalog[OF t7_p0_concrete_step] .

lemma t7_catalog_after_p0_well_formed:
  "relevance_catalog_well_formed t7_state_after_p0 t7_catalog_after_p0"
  using evaluate_relevance_preserves_catalog[OF t7_p0_concrete_step] by blast

lemma t7_p1_input_valid:
  "evaluate_relevance_input_valid t7_witness_grammar t7_state_after_p0
   t7_effect_catalog t7_catalog_after_p0 t7_request_p1 12"
  unfolding evaluate_relevance_input_valid_def
  using t7_state_after_p0_well_formed
    t7_effect_catalog_after_p0_well_formed t7_catalog_after_p0_well_formed
    t7_effect_vector_valid t7_power_profile_valid
    t7_relevance_profile_well_formed
    add_relevance_preserves_active_history_disjoint[OF t7_p0_input_valid]
  by (auto simp: t7_witness_grammar_def t7_request_p1_def
    t7_relevance_profile_def t7_state_after_p0_def
    t7_catalog_after_p0_def t7_empty_relevance_catalog_def
    t7_request_p0_def t6_witness_objects_def t6_object_defs
    base_object_def active_classifiable_effect_def
    t7_effect_catalog_def t7_effect_payload_def effect_payload_valid_def
    version_active_at_def active_relevance_rationale_def
    extend_relevance_catalog_def)

theorem t7_p1_concrete_step:
  "evaluate_relevance_step t7_witness_grammar t7_state_after_p0
   t7_effect_catalog t7_catalog_after_p0 t7_request_p1 12
   t7_state_after_p1 t7_catalog_after_p1"
  unfolding evaluate_relevance_step_def t7_state_after_p1_def
    t7_catalog_after_p1_def
  using t7_p1_input_valid by simp

subsection \<open>Exact perspective divergence without averaging\<close>

lemma t7_p0_effect_term:
  "(if dimension = Connection then 1 else 0) *
   normalized_effect_component t7_effect_vector dimension =
   (if dimension = Connection then 1 else 0)"
  by (case_tac dimension)
    (simp_all add: normalized_effect_component_def t7_effect_vector_def)

lemma t7_p1_effect_term:
  "(if dimension = Stability then 1 else 0) *
   normalized_effect_component t7_effect_vector dimension = 0"
  by (case_tac dimension)
    (simp_all add: normalized_effect_component_def t7_effect_vector_def)

lemma t7_power_term:
  "(if dimension = ResourcePower then 1 else 0) *
   t7_power_profile dimension =
   (if dimension = ResourcePower then 1 else 0)"
  by (case_tac dimension) (simp_all add: t7_power_profile_def)

lemma t7_p0_effect_component:
  "perspective_effect_component t7_relevance_profile 0 20
   t7_effect_vector = 1"
  by (simp add: perspective_effect_component_def
    effective_effect_weight_mass_def effective_effect_weight_def
    t7_relevance_profile_def t7_p0_effect_term
    t6_effect_dimension_delta clamp01_def)

lemma t7_p1_effect_component:
  "perspective_effect_component t7_relevance_profile (Suc 0) 20
   t7_effect_vector = 0"
  by (simp add: perspective_effect_component_def
    effective_effect_weight_mass_def effective_effect_weight_def
    t7_relevance_profile_def t7_p1_effect_term clamp01_def)

lemma t7_p0_power_component:
  "perspective_power_component t7_relevance_profile 0 20
   t7_power_profile = 1"
  by (simp add: perspective_power_component_def
    effective_power_weight_mass_def effective_power_weight_def
    t7_relevance_profile_def t7_power_term
    t7_power_dimension_delta clamp01_def)

lemma t7_p1_power_component:
  "perspective_power_component t7_relevance_profile (Suc 0) 20

```



```

    t7_power_profile = 1"
  by (simp add: perspective_power_component_def
    effective_power_weight_mass_def effective_power_weight_def
    t7_relevance_profile_def t7_power_term
    t7_power_dimension_delta clamp01_def)

lemma t7_p0_intensity:
  "relevance_intensity t7_relevance_profile 0 20 10
    t7_effect_vector t7_power_profile (1 / 10) = 9 / 10"
  using t7_p0_effect_component t7_p0_power_component
  by (simp add: relevance_intensity_def t7_relevance_profile_def clamp01_def)

lemma t7_p1_intensity:
  "relevance_intensity t7_relevance_profile (Suc 0) 20 10
    t7_effect_vector t7_power_profile (1 / 10) = 9 / 50"
  using t7_p1_effect_component t7_p1_power_component
  by (simp add: relevance_intensity_def t7_relevance_profile_def clamp01_def)

theorem t7_witness_vectors_are_exact_and_distinct:
  "relevance_intensity_of (requested_relevance_vector t7_request_p0) = 9 / 10 \<and>
    relevance_uncertainty_of (requested_relevance_vector t7_request_p0) = 1 / 10 \<and>
    relevance_stability_of (requested_relevance_vector t7_request_p0) = 49 / 50 \<and>
    relevance_reach_of (requested_relevance_vector t7_request_p0) = 4 / 5 \<and>
    relevance_duration_of (requested_relevance_vector t7_request_p0) = 8 \<and>
    relevance_intensity_of (requested_relevance_vector t7_request_p1) = 9 / 50 \<and>
    relevance_uncertainty_of (requested_relevance_vector t7_request_p1) = 1 / 10 \<and>
    relevance_stability_of (requested_relevance_vector t7_request_p1) = 19 / 20 \<and>
    relevance_reach_of (requested_relevance_vector t7_request_p1) = 2 / 5 \<and>
    relevance_duration_of (requested_relevance_vector t7_request_p1) = 4 \<and>
    requested_relevance_vector t7_request_p0 \<noteq>
    requested_relevance_vector t7_request_p1"
  using t7_p0_intensity t7_p1_intensity
  by (simp add: requested_relevance_vector_def t7_request_p0_def
    t7_request_p1_def relevance_stability_def relevance_reach_def
    relevance_duration_def calculate_relevance_vector_def
    t7_relevance_profile_def clamp01_def)

theorem t7_witness_provenance_and_lifecycle:
  "relevance_profile_model_of t7_relevance_profile = 30 \<and>
    relevance_profile_revision_of t7_relevance_profile = 1 \<and>
    requested_relevance_target_of t7_request_p0 = 4 \<and>
    requested_relevance_target_of t7_request_p1 = 4 \<and>
    requested_relevance_context_of t7_request_p0 = 20 \<and>
    requested_relevance_context_of t7_request_p1 = 20 \<and>
    requested_relevance_evaluation_time_of t7_request_p0 = 10 \<and>
    requested_relevance_evaluation_time_of t7_request_p1 = 10 \<and>
    lifecycle_of (evaluated_relevance_object t7_witness_source t7_request_p0) =
      Validated \<and>
    lifecycle_of (evaluated_relevance_object t7_state_after_p0 t7_request_p1) =
      Validated"
  by (simp add: t7_relevance_profile_def t7_request_p0_def
    t7_request_p1_def evaluated_relevance_lifecycle_def
    effect_validated_at_def t7_state_after_p0_def t7_witness_source_def
    t6_witness_source_def t6_witness_objects_def t6_object_defs base_object_def)

theorem t7_witness_final_certificates:
  "state_well_formed t7_witness_grammar t7_state_after_p1 \<and>
    effect_catalog_well_formed t7_state_after_p1 t7_effect_catalog \<and>
    relevance_catalog_well_formed t7_state_after_p1 t7_catalog_after_p1 \<and>
    t7_catalog_after_p1 7 = Some (evaluated_relevance_payload t7_request_p0) \<and>
    t7_catalog_after_p1 8 = Some (evaluated_relevance_payload t7_request_p1) \<and>
    step t7_witness_grammar t7_witness_source EvaluateRelevance
      t7_state_after_p0 \<and>
    step t7_witness_grammar t7_state_after_p0 EvaluateRelevance
      t7_state_after_p1"
proof -
  have final_state:
    "state_well_formed t7_witness_grammar t7_state_after_p1"
    using evaluate_relevance_preserves_state_well_formed[OF t7_p1_concrete_step] .
  have final_effect_catalog:
    "effect_catalog_well_formed t7_state_after_p1 t7_effect_catalog"
    using evaluate_relevance_preserves_effect_catalog[OF t7_p1_concrete_step] .
  have final_relevance_catalog:
    "relevance_catalog_well_formed t7_state_after_p1 t7_catalog_after_p1"
    using evaluate_relevance_preserves_catalog[OF t7_p1_concrete_step] by blast
  have p0_entry:
    "t7_catalog_after_p1 7 =
      Some (evaluated_relevance_payload t7_request_p0)"
  by (simp add: t7_catalog_after_p1_def t7_catalog_after_p0_def
    t7_empty_relevance_catalog_def extend_relevance_catalog_def)

```

```
      t7_request_p0_def t7_request_p1_def)
have p1_entry:
  "t7_catalog_after_p1 8 =
    Some (evaluated_relevance_payload t7_request_p1)"
  using evaluate_relevance_preserves_catalog[OF t7_p1_concrete_step]
  by (simp add: t7_request_p1_def)
have first_generic:
  "step t7_witness_grammar t7_witness_source EvaluateRelevance
    t7_state_after_p0"
  using evaluate_relevance_refines_generic_step[OF t7_p0_concrete_step] .
have second_generic:
  "step t7_witness_grammar t7_state_after_p0 EvaluateRelevance
    t7_state_after_p1"
  using evaluate_relevance_refines_generic_step[OF t7_p1_concrete_step] .
show ?thesis
  using final_state final_effect_catalog final_relevance_catalog
    p0_entry p1_entry first_generic second_generic
  by blast
qed

theorem t7_witness_two_exact_deltas:
  "objects_of t7_state_after_p0 =
    insert (evaluated_relevance_object t7_witness_source t7_request_p0)
      (objects_of t7_witness_source) \<and>
    snapshot_id_of t7_state_after_p0 = 11 \<and>
    objects_of t7_state_after_p1 =
      insert (evaluated_relevance_object t7_state_after_p0 t7_request_p1)
        (objects_of t7_state_after_p0) \<and>
    snapshot_id_of t7_state_after_p1 = 12"
  using evaluate_relevance_exact_delta[OF t7_p0_concrete_step]
    evaluate_relevance_exact_delta[OF t7_p1_concrete_step]
  by blast
end
```

## Anhang E – Offene mathematische Beweisaufgaben

Die historischen T6-/T7-Prüfpfade, die referenzielle Schließung von A9, die Modellzeugen sowie die konstruktiven Erhaltungsbeweise T1 bis T7 bleiben abgeschlossen. Anhang J ergänzt den kausal verknüpften T8-Abschluss im normalen 18-Theorien-Isabelle2025-2-Clean-Build. Offen bleiben insbesondere:

2. Eindeutigkeit von `version_id` und Eigenschaften der Revisionskette jenseits der Locale-Annahmen.
3. Zeitliche Wohlgeformtheit aller Gültigkeitsintervalle als Erhaltungssatz.
4. Erhaltung von `state_well_formed` durch jede Kernoperation – T1 bis T13 im jeweils ausgewiesenen Isabelle2025-2-Prüfpfad abgeschlossen.
5. Azyklizität der direkten Einbettung.
6. Irreflexivität von `strict_contains` aus konstruktiven Graphvorbedingungen.
7. Antisymmetrie von `contains_eq` aus konstruktiven Graphvorbedingungen.
8. Lifecycle-Erhaltung von Transmissionen.
9. Vollständigkeit der Transmission Lineage.
10. Fusionsdispositionsvollständigkeit.
11. Fissions-Lineage-Vollständigkeit.
12. Wohldefiniertheit von Aggregation, Embedding und Detachment.
13. Weiterführende Eigenschaften der describe-Aggregation.
14. Paarweise Exklusivität der Emergenzklassen.
15. Hinreichende Bedingungen für Vollständigkeit von E0–E3.
16. Modell-Expansion vom Kern zur Interaktionserweiterung.
17. Relative Konsistenz.
18. Optionaler Konservativitätsnachweis für definierte Kernformeln.
19. Abgrenzung und Korrektheit einer FOL-Kernübersetzung.
20. Verknüpfung von Engine-Conformance-Tests mit formalisierten Pre-/Postconditions.

## Anhang F – Glossar

| Begriff        | Konsolidierte Bedeutung                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------|
| Zusammenhang   | versionierte Konfiguration aus Struktur, Prozess, Potential und Geschichte                                 |
| Erscheinung    | registrierte unterscheidbare Gestalt oder Beobachtungseinheit                                              |
| Beziehung      | typisierte, zeit- und kontextgebundene Verbindung                                                          |
| Wirkung        | gerichtete Veränderung mit Herkunft, Ziel und Wirkungsprofil                                               |
| Resonanz       | perspektivisch-kontextuelle Wechselwirkung zwischen Wirkungen                                              |
| Relevanz       | begründete, modellgebundene Wichtigkeits- und Reichweitenbewertung                                         |
| Orientierung   | hergeleiteter Möglichkeits- und Handlungsraum                                                              |
| Gestaltung     | ausgewählte und autorisierte Option mit Ausführungsbezug                                                   |
| Rückwirkung    | beobachtete Folge einer Gestaltung und ihre Abweichung zur Simulation                                      |
| Emergenz       | strukturelle Neuheit oder Grenze der aktiven Grammatik gemäß E0–E3                                         |
| Geltungsstatus | explizites Ergebnis APPLICABLE, OUT_OF_SCOPE, INSUFFICIENT_BASIS oder MODEL_FALSIFIED                      |
| Perspektive    | versionierter Beobachtungs-, Wissens-, Wertungs- und Handlungsstandpunkt                                   |
| Kontext        | Bedingungsrahmen semantischer Gültigkeit                                                                   |
| Potential      | unter Regeln und Bedingungen erreichbarer Zustandsraum                                                     |
| GrammarPatch   | kontrollierter Vorschlag zur Erweiterung oder Revision einer Grammatik                                     |
| Patch-Abbruch  | nachvollziehbare Beendigung als PATCH_REJECTED, CORE_CONFLICT oder UNRESOLVED_E3                           |
| PowerProfile   | von V10 getrenntes relationales Profil für Ressourcen, Zugang, Entscheidung, Risiko und Korrekturfähigkeit |
| ScaleProfile   | räumliche, zeitliche, populationsbezogene und analytische Skalierung                                       |
| SUBZ           | formale Teilzusammenhangsrelation                                                                          |
| Projektion     | ausgewählte Containeransicht eines Teilzusammenhangs                                                       |
| LossProfile    | dokumentierter Struktur-, Semantik-, Perspektiv- und Zeitverlust                                           |
| Kopplung       | versionierte Interaktionsbeziehung zwischen Systemzusammenhängen                                           |
| Transmission   | kanalgebundene, transformierte Wirkungsübertragung                                                         |
| BoundaryEvent  | Fusion, Fission, Aggregation, Embedding oder Detachment mit Lineage                                        |

| Begriff                    | Konsolidierte Bedeutung                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------|
| InteractionComplex         | Gesamtstruktur gekoppelter und eingebetteter Zusammenhänge                                                      |
| Zusammenhangskompeten<br>z | getrennte Fähigkeit von Architektur, Engine und Akteur, Zusammenhang<br>angemessen zu erfassen und zu gestalten |

## Anhang G – Konsolidierungsprotokoll 0.9.1

Diese Fassung nimmt folgende verbindliche Bereinigungen vor:

- Sie führt die detailliertere Entwicklungsmatrix v0.1–v0.9.1 als maßgeblich.
- Sie unterscheidet demonstrativ vollständig durchgespieltes Implementierungsdesign von produktiv ausgeführtem Code.
- Sie behandelt die Dynamik als engine-integrierte Spezifikation mit weiterhin offener formaler Axiomatik.
- Sie definiert ein allgemeines EffectProfile-Interface und V10 als Referenzprofil, begründet dessen Achsengruppen und behauptet keine unbelegte mathematische Orthogonalität.
- Sie erhält V10-POLAR als Legacy-Profil mit eindeutigen Statuswerten Mapped, Derived, Unmapped und Conflicted; fehlende Dimensionen werden niemals still mit null belegt.
- Sie behandelt Macht nicht als elfte V10-Dimension, sondern als separates relationales PowerProfile mit dokumentiertem PowerShift.
- Sie erklärt die Beziehung zwischen frühem Acht-Ebenen-Zyklus, semantischem Kernzyklus und operationaler S0–S9-Maschine.
- Sie trennt die architektonischen RA1–RA12 von den als Locale-Annahmen formalisierten A1–A22 und weist deren gemeinsame Erfüllbarkeit relativ zu HOL durch einen expliziten Modellzeugen nach.
- Sie führt vollständige Engine-, Datenmodell-, API-, Validierungs-, Dynamics-, Scale-&-Interaction- und Demonstratorabschnitte ein.
- Sie vereinheitlicht die operative Reihenfolge auf T11/FDB → T12/EMG und korrigiert Zyklusgrafik, Zustandssemantik, Pseudocode und Golden Run entsprechend.
- Sie ergänzt Minimality in Formel und Erläuterung der Architekturkompetenz und benennt den inneren Kern korrekt als viergliedrig.
- Sie führt Geltungs-, Falsifikations-, Patch-Annahme- und Patch-Abbruchregeln als normative Gates ein.
- Sie integriert die HOL Theory 0.9.1 mit zentraler Semantik und A1–A22 und dokumentiert den erfolgreich bestandenen Isabelle2025-2-Session-Build samt kernelgeprüftem Quellhash.
- Sie schließt A9 durch `evidence_references_complete` und `evidence_well_formed`, weist unbekannte Evidence-IDs zurück und erzwingt für Supported und Validated eine nichtleere Evidenzmenge.
- Sie ergänzt eine zweite Isabelle-Theorie mit nichtleerem Zustand, versioniertem Objekt, zulässigem Übergang, vollständiger Locale-Interpretation und dem Existenztheorem `explicit_uz_model_exists`; der Zeuge wird ausdrücklich als minimal und weitgehend vakuos ausgewiesen.
- Sie ergänzt eine dritte Isabelle-Theorie, die T1 RegisterAppearance konstruktiv definiert und Zustandswohlgeformtheit, Identität, Evidenzintegrität, Referenzen, Zeit-, Perspektiv- und Kontextbindung sowie das Refinement zu step mechanisch beweist.

- Sie ergänzt eine vierte Isabelle-Theorie mit zwölf versionierten Objekten, Projektion, Kopplung, validierter Transmission und erfolgreichem Fusion-Event; neun Nichtvakuositätszertifikate aktivieren A5–A13 und A17–A21, bevor Rich: uza\_model und rich\_uza\_model\_exists die vollständige Interpretation sichern.
- Sie ergänzt eine fünfte Isabelle-Theorie, die T2 DifferentiatePotential als endliche nichtleere Kandidatenmengentransition definiert, Beobachtung und Möglichkeitskandidat formal trennt und exaktes Delta, Identität, Evidenzintegrität, Bindungen, vollständige Zielwohlgeformtheit sowie das Refinement zu step beweist.
- Sie stellt dem Isabelle-Quellanhang eine explizite Leistungsübersicht voran, die **abgeschlossen**, **teilweise abgeschlossen** und **offen** samt Beleg und verbleibendem Ziel getrennt ausweist.
- Sie ergänzt drei tatsächlich ausgeführte Golden Runs einschließlich Negativfällen und gepinnter Replayergebnisse.
- Sie ersetzt pauschale Vollständigkeitsbehauptungen durch klar prüfbare Reifestände.

Sie ergänzt eine sechste konservative Isabelle-Theorie für T3 CreateRelation. Diese führt relation\_type und Richtung in einem versionierten Relationenkatalog, erzwingt zwei verschiedene aktive ENT-Endpunkte und beweist exaktes Delta, Katalog-, Zustands-, Identitäts-, Referenz-, Zeit-, Kontext- und Evidenzerhaltung sowie Refinement zu step.

Sie dokumentiert den realen T3-Prüfpfad: zunächst Ablehnung der flachen Mehrsession-Struktur, dann zwei vom Kernel entdeckte lokale Beweislücken und schließlich den erfolgreichen integrierten Isabelle2025-2-Clean-Build mit sieben Theorien und Exit-Code 0.

Sie ergänzt eine siebte konservative Isabelle-Theorie für T4 HypothesizeEffect. Diese erzwingt eine aktive typisierte REL-Ursache, explizite option-Offenheit von Ziel und Wirkungsvektor, Perspektiv-, Kontext- und Profilmodellbindung, begrenzte Unsicherheit und frische K\_EFF-Identität; sie beweist exaktes Delta, Katalog-, Zustands-, Identitäts-, Referenz-, Zeit-, Perspektiv-, Kontext- und Evidenzerhaltung sowie Refinement zu step. Der integrierte Isabelle2025-2-Clean-Build mit sieben Theorien endete mit Exit-Code 0.

Sie ergänzt eine achte Isabelle-Theorie für T5 IntegrateEvidence. Diese realisiert Copy-on-Write-Effektversionierung, vollständige Objekt- und Payload-Archive, einen Evidenz-Ledger, kontrollierte Unsicherheitsbewegung, abgeleitete Lifecycle-Übergänge, Ausschluss von Selbststützung sowie Erhaltung und Refinement ohne vorausgesetzte Zielwohlgeformtheit.

Sie ergänzt eine neunte Isabelle-Theorie als nicht-vakuozen T5-Ausführungszeugen. Der konkrete Hypothesis → Supported-Lauf aktiviert die Evidenz-, Archiv-, Katalog-, Lifecycle- und step-Bedingungen und wurde gemeinsam mit der gesamten Session kernelgeprüft.

Sie ergänzt eine zehnte Isabelle-Theorie für T6 ClassifyResonance. Diese realisiert erstmals die gerichtete Resonanzforderung als versionierte Interaktionsmatrix, schließt die Endlichkeit der V10-Domäne formal, konstruiert RES-Objekt und Payload und beweist Score-, Form-, Stärke-, Delta-, Katalog-, Zustands- und Refinamenteigenschaften ohne vorausgesetzte Zielwohlgeformtheit.

Sie ergänzt eine elfte Isabelle-Theorie als nicht-vakuozen T6-Richtungszeugen. Der konkrete Lauf beweist  $A \rightarrow B = 1/\text{Amplification}$  und  $B \rightarrow A = 0/\text{Neutral}$  sowie Stärke 0,9, Lifecycle Validated,

exaktes Delta und alle Zielzertifikate. Die frühere symmetrische V10-Dot-Product-Funktion bleibt ausdrücklich nur Baseline; normative T6-Operationen verwenden das gerichtete Profil.

Sie ergänzt eine zwölfte Isabelle-Theorie für T7 EvaluateRelevance. Diese realisiert ein versioniertes perspektiv-, kontext- und zeitgebundenes Relevanzmodell mit getrennten V10- und Machtgewichten, berechnet einen sechsdimensionalen Vektor, konstruiert REV-Objekt und Katalogeintrag und beweist Wertebereiche, Provenienz, exaktes Delta, Zustands-, Identitäts-, Katalog- und Bindungserhaltung sowie Refinement zu step.

Sie ergänzt eine dreizehnte Isabelle-Theorie als nicht-vakuen T7-Perspektivdivergenz-Zeugen. Der konkrete Doppellauf beweist für denselben validierten Effekt die getrennten exakten Vektoren  $P0 = \langle 9/10, 1/10, 49/50, 4/5, 8 \rangle$  und  $P1 = \langle 9/50, 1/10, 19/20, 2/5, 4 \rangle$  sowie Lifecycle, Provenienz, Katalogfortschreibung, zwei exakte Deltas und zwei generische Schritte. Die Prüfung erfolgte in zwei Isabelle2025-2-HOL-Kerndirektläufen mit Exit-Code 0 und wird ausdrücklich nicht als neuer normaler Sessionbuild bezeichnet.

Die 0.9.1-Referenzdatei ist mit K3 bis T13 fortgeschrieben. Der T1–T13-Unterbau wurde als vollständige 25-Theorien-Isabelle2025-2-Kette mechanisch geprüft und durch eine kausal erreichbare Operationskette verbunden. T9–T13 schließen Auswahl, Ausführung, Rückmeldung, spurrelative Emergenzanalyse und Copy-on-write-Revision. Mathematische Modell-Expansion, relative Konsistenz, Konservativität und Produktivengine-Äquivalenz bleiben davon unabhängige offene Ziele.



## Anhang H – Validierungsprotokoll 0.9.1

### H.1 Formale Quell-, Modell- und Operationsprüfung

H.1 führt den Gesamtstatus mit getrennter Provenienz. Die historischen T6-Session- und T7-Kerndirektläufe bleiben für ihre damaligen Hashes dokumentiert. Maßgeblich für K1 ist nun der normale Isabelle2025-2-Clean-Build der vollständigen 16-Theorien-ROOT-Kette mit Exit-Code 0. Das finale Buildlog bindet ROOT und sämtliche Theoriehashes; die letzte Theorie UZA\_0\_9\_1\_Operational\_Chain\_Witness wurde zu 100 % verarbeitet.

#### H.1.1 Basistheorie, Modelle und T1-T4

| Prüfung                                                                                                           | Ergebnis                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Struktur-Preflight der Theorie                                                                                    | bestanden                                                                                                                   |
| A1–A22 vollständig und eindeutig gefunden                                                                         | bestanden                                                                                                                   |
| zentrale Definitionen (step, Evidence Integrity, Describe-Tripel, emergence_of, Relevanz, Resonanz, Macht, Patch) | bestanden                                                                                                                   |
| sechs Haupttheorie-Sentinels gefunden                                                                             | bestanden                                                                                                                   |
| A9-Integritätslemmata                                                                                             | unresolved_evidence_reference_rejected und A9_upgraded_evidence_resolves kernelgeprüft                                      |
| Isabelle-Version                                                                                                  | Isabelle2025-2                                                                                                              |
| integrierter Fresh-Profile-Build (leeres Benutzerprofil, -j 1)                                                    | bestanden; elf Theorien; Exit-Code 0                                                                                        |
| sechs Haupttheorie-Sentinelbeweise                                                                                | vom Isabelle-Kernel akzeptiert                                                                                              |
| kernelgeprüfter Theorie-Hash                                                                                      | 21dcfb175d6e754adf20042b941cf4a23329831d0165ebaf2023b0048213ed24                                                            |
| Struktur-Preflight des Basismodellzeugen                                                                          | bestanden                                                                                                                   |
| Basismodelltheorie in integrierter Session                                                                        | 100 % verarbeitet; Exit-Code 0                                                                                              |
| base_states und Objektmenge                                                                                       | beide nichtleer bewiesen                                                                                                    |
| Wohlgeformtheit und zulässiger Übergang                                                                           | base_state_well_formed und base_self_step kernelgeprüft                                                                     |
| vollständige Locale-Interpretation A1–A22                                                                         | Base: uza_model kernelgeprüft                                                                                               |
| Existenztheorem                                                                                                   | explicit_uza_model_exists kernelgeprüft                                                                                     |
| Modellgrenze                                                                                                      | minimaler Erfüllbarkeitszeuge; die meisten objekttypspezifischen Bedingungen bleiben vakuos                                 |
| kernelgeprüfter Basismodell-Hash                                                                                  | 322d2a8fc31dbca1ea6c7cc2463b24c891701a79eebb761a94f5256a62632a4b                                                            |
| Struktur-Preflight des Rich-Model-Zeugen                                                                          | bestanden                                                                                                                   |
| Rich-Model-Theorie in integrierter Session                                                                        | 100 % verarbeitet; Exit-Code 0                                                                                              |
| reichhaltiger Zustand                                                                                             | zwölf versionierte Objekte, zwei Kontexte, Projektion, Kopplung, validierte Transmission und Fusion                         |
| A5–A13-Nichtvakuosität                                                                                            | neun konkrete Zertifikate decken Relation, Wirkung, Bindungen, Evidenz, Relevanz, Orientierung, Handlung und Rückwirkung ab |
| A17–A21-Nichtvakuosität                                                                                           | Projektion, Kopplung, validierte Transmission mit Lineage und Boundary-Disposition kernelgeprüft                            |
| vollständige Rich-Locale-Interpretation                                                                           | Rich: uza_model kernelgeprüft                                                                                               |

| Prüfung                                                                     | Ergebnis                                                                                         |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Rich-Existenztheorem                                                        | rich_uza_model_exists kernelgeprüft                                                              |
| kernelgeprüfter Rich-Model-Hash                                             | 3a3671b918d9639607fb6152dedd039cdf6145b0f040db8edf6ba296b8dc9a8c                                 |
| Struktur-Preflight der T1-Operationstheorie                                 | bestanden                                                                                        |
| konkrete T1-Konstruktion ohne Ziel-Wohlgeformtheit als Vorbedingung         | bestanden                                                                                        |
| T1-Theorie in integrierter Session                                          | 100 % verarbeitet; Exit-Code 0                                                                   |
| exakte Zustandsdifferenz und frische Objekt-/Versionsidentität              | kernelgeprüft                                                                                    |
| Erhaltung von state_well_formed und bestehender Objektversionen             | kernelgeprüft                                                                                    |
| Evidence Integrity sowie Referenz-, Zeit-, Perspektiv- und Kontexterhaltung | kernelgeprüft                                                                                    |
| unbekannte Evidence-ID                                                      | durch unresolved_evidence_reference_rejected ausgeschlossen                                      |
| Refinement zu step G s RegisterAppearance s'                                | kernelgeprüft                                                                                    |
| Beobachtungs-/Potentialtrennung in T1                                       | Beobachtungen besitzen keine Potentialquelle, -basis oder Analysemarkierung                      |
| kernelgeprüfter T1-Theorie-Hash                                             | 9b8cc6167b99070a47f63b4e63bd5a783a000f6ec84f7620945bee224f116cd0                                 |
| Struktur-Preflight der T2-Operationstheorie                                 | bestanden                                                                                        |
| konkrete T2-Konstruktion ohne Ziel-Wohlgeformtheit als Vorbedingung         | bestanden                                                                                        |
| T2-Theorie in integrierter Session                                          | 100 % verarbeitet; Exit-Code 0                                                                   |
| Kandidatenmenge                                                             | endlich, nichtleer, intern eindeutig und gegenüber dem Ausgangszustand frisch                    |
| Beobachtung/Kandidat                                                        | potential_candidate_not_observed kernelgeprüft                                                   |
| exakte Mengendifferenz und Snapshot-Fortschreibung                          | differentiate_potential_exact_delta kernelgeprüft                                                |
| Erhaltung von state_well_formed und bestehender Objektversionen             | kernelgeprüft                                                                                    |
| Evidence Integrity sowie Referenz-, Zeit-, Perspektiv- und Kontexterhaltung | kernelgeprüft                                                                                    |
| Refinement zu step G s DifferentiatePotential s'                            | kernelgeprüft                                                                                    |
| kernelgeprüfter T2-Theorie-Hash                                             | ef483b279d0581752bfff655db6672fa57823dac348ace29268e68e486a40f9b                                 |
| Struktur-Preflight der T3-Operationstheorie                                 | bestanden                                                                                        |
| integrierte Importkette                                                     | lokale Theorieimporte; eine zulässige Session mit elf Theorien                                   |
| T3-Quellumfang                                                              | 580 Zeilen; relation_type, relation_catalog, aktive Endpunkte, exaktes Delta und Erhaltungssätze |
| unvollständige Beweisplatzhalter                                            | keine Vorkommen von sorry, oops oder admit                                                       |
| kernelgeprüfter T3-Theorie-Hash                                             | a3c801eaf1640624ee7bf41ba91a20d2ba1e000e489ee40f4d3c209b15911ca3                                 |
| T3-Theorie in integriertem Clean-Build                                      | 100 % verarbeitet; Exit-Code 0                                                                   |
| T3-Parser-, Typ- und Kernelstatus                                           | Isabelle2025-2 akzeptiert; Buildlog und Exit-Code als Einzeldateien                              |
| Struktur-Preflight der T4-Operationstheorie                                 | bestanden                                                                                        |

| Prüfung                                | Ergebnis                                                                                                                                    |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| T4-Payload und Offenheitssemantik      | effect_hypothesis_payload; Ziel und Wirkungsvektor als option; keine Sentinel-ID und kein stilles Zero-Fill                                 |
| T4-Quellumfang                         | 718 Zeilen; aktive K_REL-Ursache, Perspektiv-, Kontext- und Profilmmodellbindung, begrenzte Unsicherheit, exaktes Delta und Erhaltungssätze |
| unvollständige T4-Beweisplatzhalter    | keine Vorkommen von sorry, oops oder admit                                                                                                  |
| kernelgeprüfter T4-Theorie-Hash        | 2eb8e1d262cb3790f6c446839fb6809de3c2837d20705925507e272c860e2fc4                                                                            |
| T4-Theorie in integriertem Clean-Build | 100 % verarbeitet; Exit-Code 0                                                                                                              |
| T4-Erhaltung und Refinement            | state_well_formed, Identität, Katalog, Referenzen, Zeit, Perspektive, Kontext und Evidence Integrity sowie Refinement zu step kernelgeprüft |
| T4-Buildlog-Hash                       | 7a49a99e15790d7324def03c5078600e4cb75bd2715b9439380fe711e85d aa06                                                                           |

### H.1.2 T5 Evidenzintegration und T6 Resonanz

| Prüfung                                     | Ergebnis                                                                                                                                                                                          |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Struktur-Preflight der T5-Operationstheorie | bestanden                                                                                                                                                                                         |
| T5-Quellumfang                              | 1.564 Zeilen; Copy-on-Write, Archive, Evidenz-Ledger, Lifecycle-, Unsicherheits-, Delta- und Erhaltungssätze                                                                                      |
| unvollständige T5-Beweisplatzhalter         | keine Vorkommen von sorry, oops, axiomatization oder quick_and_dirty                                                                                                                              |
| T5-Lifecycle-Grenze                         | Hypothesis → Supported; Validated nur aus Supported bei Schwellen; Contradiction → Rejected; Validated-Revision an T13 delegiert                                                                  |
| T5-Selbststützung                           | aktuelle und unmittelbare Vorgängerversion als Evidence ausgeschlossen; Evidenzquelle aktives anderes logisches Objekt                                                                            |
| T5-Erhaltung und Refinement                 | state_well_formed, Identität, Aktiv/Historie-Disjunktheit, Katalog, Archive, Evidenz-Ledger, Referenzen, Zeit, Perspektive, Kontext und Evidence Integrity sowie Refinement zu step kernelgeprüft |
| kernelgeprüfter T5-Theorie-Hash             | 4422076e2bd6cb2ff9fe5e2d5e2e135e3bf3c05919a1ca03a42b1faccd5fddc0                                                                                                                                  |
| T5-Zeugenquellumfang                        | 281 Zeilen; endlicher nicht-vakuoser Hypothesis → Supported-Lauf                                                                                                                                  |
| T5-Zeugenzertifikate                        | Eingabevalidität, konkreter Schritt, Lifecycle, exakte Audit-Differenz, Zielwohlgeformtheit und generischer step kernelgeprüft                                                                    |
| kernelgeprüfter T5-Zeugen-Hash              | f5e037e578335431903754153deaedeada461fed8d359a0511e36b59a825a7d                                                                                                                                   |
| T5-Clean-Build und Buildlog                 | Isabelle2025-2; Exit-Code 0; Buildlog-SHA-256 f15ccd5b811e988c85abd3232adbc5c1cf8b2b2d2a7bfca7b2adb3d99fc35166                                                                                    |
| Struktur-Preflight der T6-Operationstheorie | bestanden                                                                                                                                                                                         |
| T6-Quellumfang                              | 1.289 Zeilen; gerichtetes Profil, Score, Form, Stärke, Konstruktoren, exaktes Delta und Erhaltungssätze                                                                                           |
| unvollständige T6-Beweisplatzhalter         | keine Vorkommen von sorry, oops, admit, axiomatization oder quick_and_dirty                                                                                                                       |
| V10-Summutationsdomäne                      | UNIV exakt als zehn effect_dimension-Konstruktoren charakterisiert; Endlichkeit kernelgeprüft                                                                                                     |
| T6-Richtung und Legacy-Grenze               | normative Operation mit K(source,target); symmetrischer V10-Dot-Product nur Baseline, keine stille Gleichsetzung                                                                                  |
| T6-Erhaltung und Refinement                 | state_well_formed, Identität, Effect-/Resonance-Katalog, Referenzen, Evidenz, Zeit, Perspektive und Kontext sowie Refinement zu step kernelgeprüft                                                |

| Prüfung                             | Ergebnis                                                                                                                         |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| kernelgeprüfter T6-Theorie-Hash     | 3f3583b42f066eaf41208a22bcb53a2d49baed374aa1b41e00a434256741b1cd                                                                 |
| T6-Zeugenquellumfang                | 438 Zeilen; zwei validierte Effekte und nicht-symmetrisches Kernelprofil                                                         |
| T6-Asymmetriezertifikat             | $A \rightarrow B=1/\text{Amplification}$ ; $B \rightarrow A=0/\text{Neutral}$ ; Stärke 9/10; Lifecycle Validated                 |
| T6-Zeugenzertifikate                | Eingabevalidität, konkreter Schritt, Rechenwerte, exaktes Delta, Zielwohlgeformtheit, Katalog und generischer step kernelgeprüft |
| kernelgeprüfter T6-Zeugen-Hash      | 494da00eeb17205001ae85090d940a12cd07215ec283d76833aa480126a716ed                                                                 |
| T6-Fresh-Profile-Build und Buildlog | Isabelle2025-2; elf Theorien; Exit-Code 0; Buildlog-SHA-256 e5945cfd1c2a62588862592a6bdf4c89b3dab15f34c0b342060144b033ef2f21     |

### H.1.3 T7 Relevanzbewertung

| Prüfung                                     | Ergebnis                                                                                                                                                                   |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Struktur-Preflight der T7-Operationstheorie | bestanden                                                                                                                                                                  |
| T7-Quellumfang                              | 1.564 Zeilen; versioniertes Modellprofil, getrennte V10-/Machtkomponenten, sechsdimensionaler Vektor, Konstruktoren, exaktes Delta und Erhaltungssätze                     |
| unvollständige T7-Beweisplatzhalter         | keine Vorkommen von sorry, oops, admit oder axiomatization; quick_and_dirty und skip_proofs explizit false                                                                 |
| T7-Dimensions- und Wertebereichsabschluss   | fünf Machtdimensionen formal endlich; Intensität, Unsicherheit, Stabilität und Reichweite auf [0,1] begrenzt                                                               |
| T7-Trennungsregel                           | V10 und PowerProfile getrennte typisierte Eingaben; parallele Perspektiven als getrennte Katalogeinträge; keine Mittelungsoperation                                        |
| T7-Erhaltung und Refinement                 | state_well_formed, Identität, Effect-/Resonance-/Relevance-Katalog, Referenzen, Zeit, Perspektive und Kontext sowie Refinement zu step kernelgeprüft                       |
| kernelgeprüfter T7-Theorie-Hash             | 6377ca4dd963832712350d10d8ad7442a1245b9e4ee76e85c43f6634bc8ecc46                                                                                                           |
| T7-Zeugenquellumfang                        | 490 Zeilen; zwei aufeinanderfolgende Bewertungen desselben validierten Effekts aus zwei Perspektiven                                                                       |
| T7-Perspektivdivergenz-Zertifikat           | $P0 = \langle 9/10, 1/10, 49/50, 4/5, 8 \rangle$ ; $P1 = \langle 9/50, 1/10, 19/20, 2/5, 4 \rangle$ ; vollständige Vektoren verschieden                                    |
| T7-Zeugenzertifikate                        | zwei Eingabevaliditäten, zwei konkrete Schritte, Lifecycle, Provenienz, Katalogfortschreibung, zwei exakte Deltas, finaler Zustand und zwei generische steps kernelgeprüft |
| kernelgeprüfter T7-Zeugen-Hash              | 3ba9e93262c0b139cf0d8521e8785ef21b66fb30de3e9e9bfa9de3989e817f39                                                                                                           |
| T7-Importabschluss-Kerndirektlauf           | Isabelle2025-2; Exit-Code 0; Marker UZA_T7_CLEAN_KERNEL_LOAD_OK; Log-SHA-256 b0e79533c8ad6eaf43c58ea7041272189372088d31621a0f80dfe82b45ed7d1d                              |
| T7-ROOT-Reihenfolge-Kerndirektlauf          | 13 Theorien; Exit-Code 0; Marker UZA_T7_ROOT_ORDER_KERNEL_LOAD_OK; Log-SHA-256 98a337063d3b8ab921068974150a168828f9e0ab717cfc123f5c3e2abad6f263                            |
| T7-Prüfstatusgrenze                         | historisch kernel-direkt; inzwischen zusätzlich im normalen 16-Theorien-K1-Sessionbuild akzeptiert                                                                         |

## H.2 Ausführbare Conformance

| Prüfung                                                 | Ergebnis      |
|---------------------------------------------------------|---------------|
| Golden-Run-Replay                                       | 3/3 bestanden |
| automatisierte Testgruppen                              | 3/3 bestanden |
| Replay-Abweichungen                                     | 0             |
| FDB-vor-EMG-Invariante                                  | bestanden     |
| Transmission-Lineage und Nicht-Fusion                   | bestanden     |
| Geltungs-, Falsifikations- und Patch-Abbruchgates       | bestanden     |
| verlustbewusste Legacy-Migration ohne stilles Zero-Fill | bestanden     |

## H.3 T7-Nachweisstand: geleistet und noch offen

Die folgende Trennung ist normativ für jede Weitergabe dieses Dokuments. Ein späterer Bearbeitungsstand darf einen offenen Punkt nur nach einem neuen, artefaktgebundenen Nachweis als abgeschlossen umklassifizieren.

| Statusklasse        | T7-Nachweis                              | Bedeutung                                                                                                                         |
|---------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Bereits geleistet   | T7-Haupttheorie kernel-direkt akzeptiert | Parser, Typen und alle enthaltenen Beweise akzeptiert; Exit 0                                                                     |
| Bereits geleistet   | nicht-vakuoser Zwei-Perspektiven-Zeuge   | exakte verschiedene Vektoren, Provenienz, Lifecycle, Deltas, Kataloge und steps bewiesen                                          |
| Bereits geleistet   | 13-Theorien-ROOT-Reihenfolgelauf         | alle ROOT-Theorien in frischem HOL-Prozess akzeptiert; Exit 0                                                                     |
| Bereits geleistet   | normaler 16-Theorien-K1-Sessionbuild     | vollständige ROOT-Kette als Isabelle2025-2-Clean-Build; Exit-Code 0                                                               |
| Noch offen          | empirische Profilvalidierung             | domänenspezifische Kalibrierung und Holdout-Prüfung                                                                               |
| Abgeschlossen in K3 | T9–T13                                   | T9–T13 sind mit konstruktiver Semantik, Zeugen, Erhaltungsbeweisen, Audit, Golden Run und Clean-Build geschlossen; siehe Anhang K |
| Noch offen          | Expansion und Konservativität            | modell- und beweistheoretische Hauptziele Richtung 1.0                                                                            |

## Anhang I – Formalrevision K1: operationaler T1–T7-Kettenabschluss

### I.1 Anlass und Befund

Eine unabhängige Herkunftsprüfung hat einen relevanten Unterschied zwischen lokaler Nichtvakuität und kausaler Erreichbarkeit sichtbar gemacht. Die früheren T5-, T6- und T7-Zeugen aktivierten ihre jeweiligen lokalen Prädikate, konstruierten die benötigten Relations- und Effect-Vorgänger jedoch als Record-Literale in bereits zusammengesetzten Ausgangszuständen. Damit war nicht bewiesen, dass die vollen T3- und T4-Vorbedingungen innerhalb derselben Operationskette erfüllbar sind. Zudem akzeptierte die frühere T5-Basisprüfung `effect_origin_of` bereits dann, wenn die Versions-ID irgendwo präsent war; Typ, Lifecycle und Relationenkatalog waren nicht Teil des publizierenden Gates.

K1 behandelte den damaligen Herkunftsbefund als Release-Blocker und schloss T1–T7. Diese historische Freigabeentscheidung bleibt gültig. K2 setzt ausschließlich auf dem gebundenen `successor_chain_after_t7` auf; der nun erreichte T8-Status ist in Anhang J separat nachgewiesen und überschreibt keine K1-Historie.

### I.2 Normative Korrekturen

- Die T5-Basisbedingung verlangt jetzt zusätzlich einen Ursprung, der im aktuellen Snapshot als aktives `K_REL`-Objekt auflösbar ist und nicht `Revised`, `Rejected` oder `Archived` ist.
- Das normative T5-Publikationsgate `integrate_evidence_strict_input_valid` verlangt darüber hinaus einen wohlgeformten `relation_catalog` und einen Eintrag für genau diese Ursprungsrelation.
- `integrate_evidence_strict_step` verfeinert den bestehenden T5-Konstruktionsschritt. Die bewährten Erhaltungssätze bleiben über das nachgewiesene Refinement nutzbar; zusätzlich wird die Erhaltung des Relationenkatalogs formuliert.
- Der frühere selbstgerichtete T5-Relationszeuge ist nicht mehr normativer Herkunftsnachweis. Der neue Zeuge verwendet zwei verschiedene `K_ENT`-Endpunkte und genau die von T3 erzeugten Relationsversionen.
- T3, T4 und T5 besitzen nun eigene Erreichbarkeitszertifikate. Eine abschließende Theorie verknüpft sie mit T6 und zwei T7-Schritten zu einem einzigen Trace.
- Historische Isabelle-Statusangaben bleiben erhalten, sind aber ausschließlich an ihre damaligen Quellhashes gebunden. Für K1 wird kein Prüfstatus geerbt.

### I.3 Konkrete Operationskette

| Schritt | Operation                | Snapshot | Eingabe/Herkunft             | Ergebnisversion        | Zentraler Gate-Nachweis                         |
|---------|--------------------------|----------|------------------------------|------------------------|-------------------------------------------------|
| 1       | T1<br>RegisterAppearance | 0 → 1    | Seed mit <code>K_META</code> | 1 / <code>K_ENT</code> | frische Beobachtung, Perspektive 10, Kontext 20 |

| Schritt | Operation                           | Snapshot | Eingabe/Herkunft                    | Ergebnisversion | Zentraler Gate-Nachweis                                         |
|---------|-------------------------------------|----------|-------------------------------------|-----------------|-----------------------------------------------------------------|
| 2       | T2<br>DifferentiatePotential        | 1 → 2    | Quelle/Basis<br>Version 1           | 2 / K_ENT       | Hypothesis,<br>Applicable,<br>aufgelöste Basis                  |
| 3       | T3 CreateRelation                   | 2 → 3    | aktive Endpunkte<br>1 → 2           | 3 / K_REL       | verschiedene<br>K_ENT-Endpunkte,<br>Katalogeintrag              |
| 4       | T3 CreateRelation                   | 3 → 4    | aktive Endpunkte<br>2 → 1           | 4 / K_REL       | verschiedene<br>K_ENT-Endpunkte,<br>Katalogeintrag              |
| 5       | T4<br>HypothesizeEffect             | 4 → 5    | T3-Relation 3                       | 5 / K_EFF       | aktive K_REL-<br>Ursache und<br>Relationenkatalog               |
| 6       | T4<br>HypothesizeEffect             | 5 → 6    | T3-Relation 4                       | 6 / K_EFF       | aktive K_REL-<br>Ursache und<br>Relationenkatalog               |
| 7       | T5<br>IntegrateEvidence<br>(strict) | 6 → 7    | T4-Effekt 5;<br>Evidenz 1           | 7 / K_EFF       | typisierter<br>katalogisierter<br>Ursprung 3; Copy-<br>on-Write |
| 8       | T5<br>IntegrateEvidence<br>(strict) | 7 → 8    | T4-Effekt 6;<br>Evidenz 2           | 8 / K_EFF       | typisierter<br>katalogisierter<br>Ursprung 4; Copy-<br>on-Write |
| 9       | T6<br>ClassifyResonance             | 8 → 9    | T5-Effekte 7 → 8                    | 9 / K_RES       | zwei aktive<br>Supported-Effekte,<br>gerichtetes Profil         |
| 10      | T7<br>EvaluateRelevance             | 9 → 10   | Effekt 7,<br>Perspektive 0          | 10 / K_REV      | aktive<br>katalogisierte<br>Wirkung, getrennte<br>Perspektive   |
| 11      | T7<br>EvaluateRelevance             | 10 → 11  | derselbe Effekt 7,<br>Perspektive 1 | 11 / K_REV      | zweiter<br>unabhängiger<br>Katalogeintrag,<br>keine Mittelung   |

## I.4 Verifikations- und Statusmatrix

| Artefaktstand         | Prüfung                                             | Ergebnis                                       | Zulässige Aussage                                                         |
|-----------------------|-----------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------|
| Historischer T6-Stand | Isabelle2025-2 Fresh-Profile<br>Session Build       | Exit 0                                         | nur für die dokumentierten elf<br>Theoriehashes<br>kernel-/sessiongeprüft |
| Historischer T7-Stand | zwei Isabelle2025-2 kernel-<br>direct HOL-Läufe     | Exit 0                                         | nur für die dokumentierten 13<br>Theoriehashes kernel-direkt<br>geprüft   |
| Aktueller K1-Stand    | deterministischer statischer<br>Isabelle-Quellaudit | bestanden; 16 Theorien; 12/12<br>Release-Gates | quellstrukturell konsistent; kein<br>Parser-/Typ-/Beweisnachweis          |
| Aktueller K1-Stand    | unabhängiger ausführbarer T1<br>–T7-Golden-Run      | bestanden; 11 Schritte;<br>Snapshot 11         | konkrete kausale Fixture<br>ausführbar; kein Isabelle-<br>Beweis          |

| Artefaktstand      | Prüfung                                            | Ergebnis                                                           | Zulässige Aussage                                      |
|--------------------|----------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------|
| Aktueller K1-Stand | Isabelle2025-2<br>Parser/Typ/Beweis/Kernel/Session | bestanden; 16<br>Theorien; Exit-Code<br>0; letzte Theorie 100<br>% | exakter finaler<br>Hashstand<br>session-/kernelgeprüft |

## I.5 Release-Entscheidung und offene Gates

Die inhaltliche Lücke und das formale Freigabegate sind geschlossen. Der vollständige Isabelle2025-2-Clean-Build der 16-Theorien-ROOT-Kette endete mit Exit-Code 0; die letzte Theorie UZA\_0\_9\_1\_Operational\_Chain\_Witness wurde zu 100 % verarbeitet. Buildlog, Exit-Code, Runtime-Provenienz, ROOT-Hash und vollständiges SHA-256-Register sind gemeinsam gebunden. Zulässig ist nun die Bezeichnung „T1–T7 kausal geschlossen und Isabelle2025-2 session-/kernelgeprüft“.

1. Abgeschlossen: normaler Isabelle2025-2-Clean-Build der vollständigen 16-Theorien-ROOT-Kette; Exit-Code 0.
2. Abgeschlossen: alle im realen Build aufgedeckten Typ-, Herkunfts- und Beweislücken wurden ohne Abschwächung der Vorbedingungen bereinigt.
3. Abgeschlossen: Golden Run erneut ausgeführt; elf Schritte, Snapshot 11; Quellen, Log, JSON-Ergebnis und Master sind hashgebunden.
4. Historischer K1-Folgeschritt: T8 SynthesizeOrientation. Dieser Schritt ist nun in K2/Anhang J abgeschlossen. Rich\_Model bleibt ein unabhängiger Modellzeuge und wird nicht als kausale Operationskette ausgegeben.

## I.6 Verschärfte T5-Theorie

Normativer K1-Quellstand. Gegenüber D.8 ergänzt: aktiver typisierter Ursprung, kataloggebundenes Strict-Gate, Refinement und Relationenkatalog-Erhaltung. Prüfstatus: im vollständigen Isabelle2025-2-Clean-Build akzeptiert; finaler Theoriehash  
bc5a22bd8e35090d4ea856e9ce4e9406c01fdab9cc453124d53da658f142fcb8.

```
theory UZA_0_9_1_Integrate_Evidence
  imports UZA_0_9_1_Hypothesize_Effect
begin

section \<open>T5 IntegrateEvidence: versioned evidence integration\<close>

text \<open>
  T5 does not mutate an active effect version. It replaces one active
  K_EFF hypothesis by one fresh version of the same logical object, archives
  the complete predecessor object and its effect payload, accumulates typed
  evidence, and derives the successor lifecycle from the evidence
  disposition. The construction therefore makes copy-on-write, provenance,
  uncertainty movement, and the prohibition of self-support explicit.

  The operation is deliberately restricted to Hypothesis and Supported.
  Evidence applied to an already Validated effect belongs to T13
  ReviseObject. This boundary prevents T5 from silently serving as a general
  revision operator.
\<close>

datatype evidence_disposition =
  EvidenceSupports | EvidenceDifferentiates | EvidenceContradicts
```



```

record evidence_assessment =
  evidence_subject_of :: object_id
  evidence_prior_version_of :: version_id
  evidence_refs_of :: "version_id set"
  evidence_disposition_of :: evidence_disposition
  evidence_strength_of :: real
  evidence_post_uncertainty_of :: real
  evidence_post_target_of :: "version_id option"
  evidence_post_vector_of :: "effect_vector option"
  evidence_assessment_model_of :: model_id
  validation_uncertainty_max_of :: real
  validation_evidence_min_of :: nat

type_synonym object_archive = "version_id \ $\rightarrow$  uza_object"
type_synonym effect_payload_archive =
  "version_id \ $\rightarrow$  effect_hypothesis_payload"
type_synonym evidence_ledger =
  "version_id \ $\rightarrow$  evidence_assessment"

definition active_history_disjoint :: "snapshot \ $\rightarrow$  bool" where
  "active_history_disjoint s \ $\longleftrightarrow$ 
    version_ids s \ $\cap$  historical_versions_of s = {}"

definition object_archive_well_formed ::
  "snapshot \ $\rightarrow$  object_archive \ $\rightarrow$  bool" where
  "object_archive_well_formed s archive \ $\longleftrightarrow$ 
    (\forall v obj. archive v = Some obj \ $\longrightarrow$ 
      version_id_of obj = v \ $\wedge$ 
      v \ $\in$  historical_versions_of s \ $\wedge$ 
      v \ $\notin$  version_ids s)"

definition resolved_version_object ::
  "snapshot \ $\rightarrow$  object_archive \ $\rightarrow$  version_id \ $\rightarrow$ 
    uza_object \ $\rightarrow$  bool" where
  "resolved_version_object s archive v obj \ $\longleftrightarrow$ 
    (obj \ $\in$  objects_of s \ $\wedge$  version_id_of obj = v) \ $\vee$ 
    archive v = Some obj"

definition effect_payload_archive_well_formed ::
  "object_archive \ $\rightarrow$  effect_payload_archive \ $\rightarrow$  bool" where
  "effect_payload_archive_well_formed archive payload_archive \ $\longleftrightarrow$ 
    (\forall v payload. payload_archive v = Some payload \ $\longrightarrow$ 
      (\exists eff. archive v = Some eff \ $\wedge$ 
        effect_payload_valid eff payload))"

definition evidence_entry_well_formed ::
  "snapshot \ $\rightarrow$  object_archive \ $\rightarrow$  version_id \ $\rightarrow$ 
    evidence_assessment \ $\rightarrow$  bool" where
  "evidence_entry_well_formed s archive v assessment \ $\longleftrightarrow$ 
    (\exists eff. resolved_version_object s archive v eff \ $\wedge$ 
      kind_of eff = K_EFF \ $\wedge$ 
      evidence_subject_of assessment = object_id_of eff \ $\wedge$ 
      previous_version_of eff = Some (evidence_prior_version_of assessment) \ $\wedge$ 
      evidence_refs_of assessment \ $\subseteq$  {} \ $\wedge$ 
      evidence_refs_of assessment \ $\subseteq$  evidence_of eff \ $\wedge$ 
      evidence_refs_of assessment
        \ $\subseteq$  version_ids s \ $\cup$  historical_versions_of s \ $\wedge$ 
      v \ $\notin$  evidence_refs_of assessment \ $\wedge$ 
      evidence_prior_version_of assessment
        \ $\notin$  evidence_refs_of assessment \ $\wedge$ 
      0 \ $\leq$  evidence_strength_of assessment \ $\wedge$ 
      evidence_strength_of assessment  $\leq$  1 \ $\wedge$ 
      0  $\leq$  evidence_post_uncertainty_of assessment \ $\wedge$ 
      evidence_post_uncertainty_of assessment  $\leq$  1 \ $\wedge$ 
      0  $\leq$  validation_uncertainty_max_of assessment \ $\wedge$ 
      validation_uncertainty_max_of assessment  $\leq$  1 \ $\wedge$ 
      0 < validation_evidence_min_of assessment \ $\wedge$ 
      evidence_assessment_model_of assessment \ $\in$  models_of s)"

definition evidence_ledger_well_formed ::
  "snapshot \ $\rightarrow$  object_archive \ $\rightarrow$  evidence_ledger \ $\rightarrow$  bool" where
  "evidence_ledger_well_formed s archive ledger \ $\longleftrightarrow$ 
    (\forall v assessment. ledger v = Some assessment \ $\longrightarrow$ 
      evidence_entry_well_formed s archive v assessment)"

definition active_evidence_source ::
  "snapshot \ $\rightarrow$  object_id \ $\rightarrow$  version_id \ $\rightarrow$  bool" where
  "active_evidence_source s subject v \ $\longleftrightarrow$ 
    (\exists ev \ $\in$  objects_of s.
      version_id_of ev = v)"

```

```

object_id_of ev \<noteq> subject \<and>
lifecycle_of ev \<notin> {Rejected, Archived})"

definition admissible_post_target ::
  "snapshot \<Rightarrow> object_id \<Rightarrow> version_id option \<Rightarrow> bool" where
  "admissible_post_target s subject target \<longlefttrightarrow>
    (case target of
      None \<Rightarrow> True
    | Some v \<Rightarrow>
      (\<exists>obj\<in>objects_of s.
        version_id_of obj = v \<and>
        object_id_of obj \<noteq> subject \<and>
        lifecycle_of obj \<notin> {Revised, Rejected, Archived})))"

definition assessment_bounds :: "evidence_assessment \<Rightarrow> bool" where
  "assessment_bounds assessment \<longlefttrightarrow>
    0 \<le> evidence_strength_of assessment \<and>
    evidence_strength_of assessment \<le> 1 \<and>
    0 \<le> evidence_post_uncertainty_of assessment \<and>
    evidence_post_uncertainty_of assessment \<le> 1 \<and>
    0 \<le> validation_uncertainty_max_of assessment \<and>
    validation_uncertainty_max_of assessment \<le> 1 \<and>
    0 < validation_evidence_min_of assessment"

definition assessment_change_valid ::
  "effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow> bool" where
  "assessment_change_valid prior assessment \<longlefttrightarrow>
    abs (evidence_post_uncertainty_of assessment -
      hypothesis_uncertainty_of prior)
    \<le> evidence_strength_of assessment \<and>
    (case evidence_disposition_of assessment of
      EvidenceSupports \<Rightarrow>
        evidence_post_target_of assessment = hypothesis_target_of prior \<and>
        evidence_post_vector_of assessment = hypothesis_vector_of prior \<and>
        evidence_post_uncertainty_of assessment
          \<le> hypothesis_uncertainty_of prior
    | EvidenceDifferentiates \<Rightarrow>
        evidence_post_target_of assessment \<noteq> hypothesis_target_of prior \<or>
        evidence_post_vector_of assessment \<noteq> hypothesis_vector_of prior
    | EvidenceContradicts \<Rightarrow>
        evidence_post_target_of assessment = hypothesis_target_of prior \<and>
        evidence_post_vector_of assessment = hypothesis_vector_of prior \<and>
        hypothesis_uncertainty_of prior
          \<le> evidence_post_uncertainty_of assessment)"

definition validation_ready ::
  "uza_object \<Rightarrow> evidence_assessment \<Rightarrow> bool" where
  "validation_ready old_eff assessment \<longlefttrightarrow>
    evidence_post_uncertainty_of assessment
    \<le> validation_uncertainty_max_of assessment \<and>
    card (evidence_of old_eff \<union> evidence_refs_of assessment)
    \<ge> validation_evidence_min_of assessment"

definition integrated_lifecycle ::
  "uza_object \<Rightarrow> evidence_assessment \<Rightarrow> lifecycle_status" where
  "integrated_lifecycle old_eff assessment =
    (case evidence_disposition_of assessment of
      EvidenceSupports \<Rightarrow>
        (if lifecycle_of old_eff = Hypothesis then Supported
        else if validation_ready old_eff assessment then Validated
        else Supported)
    | EvidenceDifferentiates \<Rightarrow> lifecycle_of old_eff
    | EvidenceContradicts \<Rightarrow> Rejected)"

definition integrated_effect_payload ::
  "effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow>
    effect_hypothesis_payload" where
  "integrated_effect_payload prior assessment =
    prior\<lparr>
      hypothesis_target_of := evidence_post_target_of assessment,
      hypothesis_vector_of := evidence_post_vector_of assessment,
      hypothesis_uncertainty_of := evidence_post_uncertainty_of assessment
    \<rparr>"

definition option_ref_set :: "version_id option \<Rightarrow> version_id set" where
  "option_ref_set target = (case target of None \<Rightarrow> {} | Some v \<Rightarrow> {v})"

definition integrated_effect_object ::
  "uza_object \<Rightarrow> evidence_assessment \<Rightarrow> version_id \<Rightarrow>
    time_point \<Rightarrow> uza_object" where

```

```

"integrated_effect_object old_eff assessment new_version integration_time =
  old_eff\<lparr>
    version_id_of := new_version,
    lifecycle_of := integrated_lifecycle old_eff assessment,
    previous_version_of := Some (version_id_of old_eff),
    references_of := references_of old_eff
      \<union> evidence_refs_of assessment
      \<union> {version_id_of old_eff}
      \<union> option_ref_set (evidence_post_target_of assessment),
    evidence_of := evidence_of old_eff \<union> evidence_refs_of assessment,
    valid_from_of := integration_time,
    valid_to_of := None,
    target_ref_of := evidence_post_target_of assessment,
    basis_refs_of := basis_refs_of old_eff
      \<union> evidence_refs_of assessment
      \<union> {version_id_of old_eff}
      \<union> option_ref_set (evidence_post_target_of assessment),
    object_status_of :=
      (if evidence_disposition_of assessment = EvidenceContradicts
       then Some ModelFalsified else Some Applicable)
  \<rparr>"

definition replace_effect_version ::
  "snapshot \<Rightarrow> uza_object \<Rightarrow> uza_object \<Rightarrow> nat
  \<Rightarrow> snapshot" where
  "replace_effect_version s old_eff new_eff next_snapshot_id =
    s\<lparr>
      snapshot_id_of := next_snapshot_id,
      objects_of := insert new_eff (objects_of s - {old_eff}),
      historical_versions_of :=
        insert (version_id_of old_eff) (historical_versions_of s)
    \<rparr>"

definition replace_active_effect_payload ::
  "effect_catalog \<Rightarrow> version_id \<Rightarrow> version_id \<Rightarrow>
  effect_hypothesis_payload \<Rightarrow> effect_catalog" where
  "replace_active_effect_payload catalog old_version new_version payload =
    (catalog(old_version := None))(new_version \<mapsto> payload)"

definition extend_object_archive ::
  "object_archive \<Rightarrow> uza_object \<Rightarrow> object_archive" where
  "extend_object_archive archive old_eff =
    archive(version_id_of old_eff \<mapsto> old_eff)"

definition extend_effect_payload_archive ::
  "effect_payload_archive \<Rightarrow> uza_object \<Rightarrow>
  effect_hypothesis_payload \<Rightarrow> effect_payload_archive" where
  "extend_effect_payload_archive archive old_eff prior =
    archive(version_id_of old_eff \<mapsto> prior)"

definition extend_evidence_ledger ::
  "evidence_ledger \<Rightarrow> version_id \<Rightarrow> evidence_assessment
  \<Rightarrow> evidence_ledger" where
  "extend_evidence_ledger ledger new_version assessment =
    ledger(new_version \<mapsto> assessment)"

definition integrate_evidence_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> effect_catalog \<Rightarrow>
  object_archive \<Rightarrow> effect_payload_archive \<Rightarrow>
  evidence_ledger \<Rightarrow> uza_object \<Rightarrow>
  effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow>
  version_id \<Rightarrow> time_point \<Rightarrow> nat \<Rightarrow> bool" where
  "integrate_evidence_input_valid G s catalog object_archive payload_archive
    ledger old_eff prior assessment new_version integration_time
    next_snapshot_id \<longlefttrightarrow>
    state_well_formed G s \<and>
    active_history_disjoint s \<and>
    effect_catalog_well_formed s catalog \<and>
    object_archive_well_formed s object_archive \<and>
    effect_payload_archive_well_formed object_archive payload_archive \<and>
    evidence_ledger_well_formed s object_archive ledger \<and>
    IntegrateEvidence \<in> allowed_operations_of G \<and>
    old_eff \<in> objects_of s \<and>
    kind_of old_eff = K_EFF \<and>
    lifecycle_of old_eff \<in> {Hypothesis, Supported} \<and>
    catalog (version_id_of old_eff) = Some prior \<and>
    effect_payload_valid old_eff prior \<and>
    present_ref s (effect_origin_of old_eff) \<and>
    (\<exists>origin. effect_origin_of old_eff = Some origin \<and>
     active_relation_origin s origin) \<and>
  "
```

```

object_archive (version_id_of old_eff) = None \<and>
payload_archive (version_id_of old_eff) = None \<and>
new_version \<notin> version_ids s \<union> historical_versions_of s \<and>
catalog new_version = None \<and>
object_archive new_version = None \<and>
payload_archive new_version = None \<and>
ledger new_version = None \<and>
evidence_subject_of assessment = object_id_of old_eff \<and>
evidence_prior_version_of assessment = version_id_of old_eff \<and>
evidence_refs_of assessment \<noteq> {} \<and>
(\<forall>v\<in>evidence_refs_of assessment.
  active_evidence_source s (object_id_of old_eff) v) \<and>
evidence_assessment_model_of assessment \<in> models_of s \<and>
assessment_bounds assessment \<and>
assessment_change_valid prior assessment \<and>
admissible_post_target s (object_id_of old_eff)
  (evidence_post_target_of assessment) \<and>
(case evidence_post_vector_of assessment of
  None \<Rightarrow> True
  | Some vector \<Rightarrow> effect_vector_valid vector) \<and>
valid_from_of old_eff \<le> integration_time \<and>
snapshot_id_of s < next_snapshot_id"

definition integrate_evidence_step ::
"grammar \<Rightarrow> snapshot \<Rightarrow> effect_catalog \<Rightarrow>
object_archive \<Rightarrow> effect_payload_archive \<Rightarrow>
evidence_ledger \<Rightarrow> uza_object \<Rightarrow>
effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow>
version_id \<Rightarrow> time_point \<Rightarrow> nat \<Rightarrow>
snapshot \<Rightarrow> effect_catalog \<Rightarrow> object_archive \<Rightarrow>
effect_payload_archive \<Rightarrow> evidence_ledger \<Rightarrow> bool" where
"integrate_evidence_step G s catalog object_archive payload_archive ledger
old_eff prior assessment new_version integration_time next_snapshot_id
s' catalog' object_archive' payload_archive' ledger' \<longlefttrightarrow>
integrate_evidence_input_valid G s catalog object_archive payload_archive
ledger old_eff prior assessment new_version integration_time
next_snapshot_id \<and>
(let new_eff = integrated_effect_object old_eff assessment new_version
integration_time;
  new_payload = integrated_effect_payload prior assessment
in s' = replace_effect_version s old_eff new_eff next_snapshot_id \<and>
catalog' = replace_active_effect_payload catalog
  (version_id_of old_eff) new_version new_payload \<and>
object_archive' = extend_object_archive object_archive old_eff \<and>
payload_archive' = extend_effect_payload_archive payload_archive
old_eff prior \<and>
ledger' = extend_evidence_ledger ledger new_version assessment)"

subsection \<open>Normative catalogued-origin gate\<close>

text \<open>
The base predicate above now rejects untyped or inactive effect origins. The
public T5 gate additionally requires the origin relation to be present in a
well-formed relation catalog. Keeping this condition in a separate strict
wrapper preserves the already checked constructor and preservation proofs,
while making it impossible for a conforming caller to substitute a merely
coincidental version identifier for a T3-created relation.
\<close>

definition catalogued_active_relation_origin ::
"snapshot \<Rightarrow> relation_catalog \<Rightarrow> version_id option \<Rightarrow> bool" where
"catalogued_active_relation_origin s relation_catalog origin \<longlefttrightarrow>
(\<exists>v. origin = Some v \<and> active_relation_origin s v \<and>
  relation_catalog v \<noteq> None)"

definition integrate_evidence_strict_input_valid ::
"grammar \<Rightarrow> snapshot \<Rightarrow> relation_catalog \<Rightarrow>
effect_catalog \<Rightarrow> object_archive \<Rightarrow>
effect_payload_archive \<Rightarrow> evidence_ledger \<Rightarrow> uza_object \<Rightarrow>
effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow>
version_id \<Rightarrow> time_point \<Rightarrow> nat \<Rightarrow> bool" where
"integrate_evidence_strict_input_valid G s relation_catalog catalog
object_archive payload_archive ledger old_eff prior assessment
new_version integration_time next_snapshot_id \<longlefttrightarrow>
relation_catalog_well_formed s relation_catalog \<and>
catalogued_active_relation_origin s relation_catalog
(effect_origin_of old_eff) \<and>
integrate_evidence_input_valid G s catalog object_archive payload_archive
ledger old_eff prior assessment new_version integration_time
next_snapshot_id"

```

```

definition integrate_evidence_strict_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> relation_catalog \<Rightarrow>
    effect_catalog \<Rightarrow> object_archive \<Rightarrow>
    effect_payload_archive \<Rightarrow> evidence_ledger \<Rightarrow> uza_object \<Rightarrow>
    effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow>
    version_id \<Rightarrow> time_point \<Rightarrow> nat \<Rightarrow> snapshot \<Rightarrow>
    effect_catalog \<Rightarrow> object_archive \<Rightarrow>
    effect_payload_archive \<Rightarrow> evidence_ledger \<Rightarrow> bool" where
  "integrate_evidence_strict_step G s relation_catalog catalog
    object_archive payload_archive ledger old_eff prior assessment
    new_version integration_time next_snapshot_id s' catalog'
    object_archive' payload_archive' ledger' \<longlefttrightarrow>
    integrate_evidence_strict_input_valid G s relation_catalog catalog
    object_archive payload_archive ledger old_eff prior assessment
    new_version integration_time next_snapshot_id \<and>
    integrate_evidence_step G s catalog object_archive payload_archive ledger
    old_eff prior assessment new_version integration_time next_snapshot_id
    s' catalog' object_archive' payload_archive' ledger'"

lemma integrate_evidence_strict_input_implies_base:
  assumes strict:
    "integrate_evidence_strict_input_valid G s relation_catalog catalog
      object_archive payload_archive ledger old_eff prior assessment
      new_version integration_time next_snapshot_id"
  shows
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
      ledger old_eff prior assessment new_version integration_time
      next_snapshot_id"
  using strict
  by (simp add: integrate_evidence_strict_input_valid_def)

theorem integrate_evidence_strict_refines_base:
  assumes strict:
    "integrate_evidence_strict_step G s relation_catalog catalog
      object_archive payload_archive ledger old_eff prior assessment
      new_version integration_time next_snapshot_id s' catalog'
      object_archive' payload_archive' ledger'"
  shows
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  using strict
  by (simp add: integrate_evidence_strict_step_def)

theorem integrate_evidence_strict_preserves_relation_catalog:
  assumes strict:
    "integrate_evidence_strict_step G s relation_catalog catalog
      object_archive payload_archive ledger old_eff prior assessment
      new_version integration_time next_snapshot_id s' catalog'
      object_archive' payload_archive' ledger'"
  shows "relation_catalog_well_formed s' relation_catalog"
  proof -
    from strict have source_catalog:
      "relation_catalog_well_formed s relation_catalog"
    and old_kind: "kind_of old_eff = K_EFF"
    and target:
      "s' = replace_effect_version s old_eff
        (integrated_effect_object old_eff assessment new_version
          integration_time) next_snapshot_id"
    unfolding integrate_evidence_strict_step_def
    integrate_evidence_strict_input_valid_def
    integrate_evidence_step_def integrate_evidence_input_valid_def Let_def
    by blast+
  show ?thesis
    using source_catalog old_kind
    unfolding relation_catalog_well_formed_def target
    by (auto simp: replace_effect_version_def)
qed

```

subsection \<open>Constructor equations and exact delta\<close>

```

lemma replace_effect_simps [simp]:
  "snapshot_id_of (replace_effect_version s old_eff new_eff next_id) = next_id"
  "objects_of (replace_effect_version s old_eff new_eff next_id) =
    insert new_eff (objects_of s - {old_eff})"
  "historical_versions_of
    (replace_effect_version s old_eff new_eff next_id) =
    insert (version_id_of old_eff) (historical_versions_of s)"
  "perspectives_of (replace_effect_version s old_eff new_eff next_id) =

```

```

    perspectives_of s"
"contexts_of (replace_effect_version s old_eff new_eff next_id) = contexts_of s"
"models_of (replace_effect_version s old_eff new_eff next_id) = models_of s"
"authorities_of (replace_effect_version s old_eff new_eff next_id) =
    authorities_of s"
"contains_edges_of (replace_effect_version s old_eff new_eff next_id) =
    contains_edges_of s"
"projections_of (replace_effect_version s old_eff new_eff next_id) =
    projections_of s"
"couplings_of (replace_effect_version s old_eff new_eff next_id) =
    couplings_of s"
"transmissions_of (replace_effect_version s old_eff new_eff next_id) =
    transmissions_of s"
"boundaries_of (replace_effect_version s old_eff new_eff next_id) =
    boundaries_of s"
by (simp_all add: replace_effect_version_def)

lemma integrated_effect_simps [simp]:
  "object_id_of (integrated_effect_object old assessment new_v at_time) =
    object_id_of old"
  "version_id_of (integrated_effect_object old assessment new_v at_time) = new_v"
  "kind_of (integrated_effect_object old assessment new_v at_time) = kind_of old"
  "lifecycle_of (integrated_effect_object old assessment new_v at_time) =
    integrated_lifecycle old assessment"
  "previous_version_of
    (integrated_effect_object old assessment new_v at_time) =
    Some (version_id_of old)"
  "references_of (integrated_effect_object old assessment new_v at_time) =
    references_of old \<union> evidence_refs_of assessment \<union>
    {version_id_of old} \<union>
    option_ref_set (evidence_post_target_of assessment)"
  "evidence_of (integrated_effect_object old assessment new_v at_time) =
    evidence_of old \<union> evidence_refs_of assessment"
  "perspective_of (integrated_effect_object old assessment new_v at_time) =
    perspective_of old"
  "context_of (integrated_effect_object old assessment new_v at_time) =
    context_of old"
  "model_of (integrated_effect_object old assessment new_v at_time) = model_of old"
  "source_ref_of (integrated_effect_object old assessment new_v at_time) =
    source_ref_of old"
  "target_ref_of (integrated_effect_object old assessment new_v at_time) =
    evidence_post_target_of assessment"
  "effect_origin_of (integrated_effect_object old assessment new_v at_time) =
    effect_origin_of old"
  "valid_from_of (integrated_effect_object old assessment new_v at_time) = at_time"
  "valid_to_of (integrated_effect_object old assessment new_v at_time) = None"
  "basis_refs_of (integrated_effect_object old assessment new_v at_time) =
    basis_refs_of old \<union> evidence_refs_of assessment \<union>
    {version_id_of old} \<union>
    option_ref_set (evidence_post_target_of assessment)"
  by (simp_all add: integrated_effect_object_def)

lemma integrated_payload_simps [simp]:
  "hypothesis_cause_of (integrated_effect_payload prior assessment) =
    hypothesis_cause_of prior"
  "hypothesis_target_of (integrated_effect_payload prior assessment) =
    evidence_post_target_of assessment"
  "hypothesis_vector_of (integrated_effect_payload prior assessment) =
    evidence_post_vector_of assessment"
  "hypothesis_uncertainty_of (integrated_effect_payload prior assessment) =
    evidence_post_uncertainty_of assessment"
  "hypothesis_profile_model_of (integrated_effect_payload prior assessment) =
    hypothesis_profile_model_of prior"
  by (simp_all add: integrated_effect_payload_def)

lemma integrate_evidence_exact_delta:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  defines "new_eff \<equiv> integrated_effect_object old_eff assessment new_version
    integration_time"
  and "new_payload \<equiv> integrated_effect_payload prior assessment"
  shows
    "objects_of s' = insert new_eff (objects_of s - {old_eff}) \<and>
      historical_versions_of s' =
        insert (version_id_of old_eff) (historical_versions_of s) \<and>
      old_eff \<notin> objects_of s' \<and>
      new_eff \<in> objects_of s' \<and>
      snapshot_id_of s' = next_snapshot_id \<and>"

```

```

    catalog' (version_id_of old_eff) = None \<and>
    catalog' new_version = Some new_payload \<and>
    object_archive' (version_id_of old_eff) = Some old_eff \<and>
    payload_archive' (version_id_of old_eff) = Some prior \<and>
    ledger' new_version = Some assessment"
proof -
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
    ledger old_eff prior assessment new_version integration_time
    next_snapshot_id"
  and target:
    "s' = replace_effect_version s old_eff new_eff next_snapshot_id"
  and catalog_target:
    "catalog' = replace_active_effect_payload catalog
    (version_id_of old_eff) new_version new_payload"
  and object_archive_target:
    "object_archive' = extend_object_archive object_archive old_eff"
  and payload_archive_target:
    "payload_archive' = extend_effect_payload_archive payload_archive
    old_eff prior"
  and ledger_target:
    "ledger' = extend_evidence_ledger ledger new_version assessment"
  unfolding integrate_evidence_step_def new_eff_def new_payload_def Let_def
  by blast+
  from valid have old_mem: "old_eff \<in> objects_of s"
  and fresh: "new_version \<notin> version_ids s \<union> historical_versions_of s"
  and catalog_old: "catalog (version_id_of old_eff) = Some prior"
  and catalog_new: "catalog new_version = None"
  unfolding integrate_evidence_input_valid_def by blast+
  have different_version: "version_id_of old_eff \<noteq> new_version"
  using old_mem fresh unfolding version_ids_def by blast
  have different_object: "old_eff \<noteq> new_eff"
  proof
    assume same_object: "old_eff = new_eff"
    have "version_id_of old_eff = version_id_of new_eff"
    using same_object by simp
    then have "version_id_of old_eff = new_version"
    unfolding new_eff_def by simp
    then show False using different_version by contradiction
  qed
  show ?thesis
  unfolding target catalog_target object_archive_target
  payload_archive_target ledger_target
  replace_active_effect_payload_def extend_object_archive_def
  extend_effect_payload_archive_def extend_evidence_ledger_def
  using different_version different_object catalog_old catalog_new
  by simp
qed

subsection \<open>Lifecycle, uncertainty, and provenance properties\<close>

theorem integrate_evidence_lifecycle_classification:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
    old_eff prior assessment new_version integration_time next_snapshot_id
    s' catalog' object_archive' payload_archive' ledger'"
  shows
    "(evidence_disposition_of assessment = EvidenceContradicts
    \<longrightarrow> integrated_lifecycle old_eff assessment = Rejected) \<and>
    (evidence_disposition_of assessment = EvidenceDifferentiates
    \<longrightarrow> integrated_lifecycle old_eff assessment = lifecycle_of old_eff) \<and>
    (evidence_disposition_of assessment = EvidenceSupports \<and>
    lifecycle_of old_eff = Hypothesis
    \<longrightarrow> integrated_lifecycle old_eff assessment = Supported) \<and>
    (integrated_lifecycle old_eff assessment = Validated
    \<longrightarrow> lifecycle_of old_eff = Supported \<and>
    evidence_disposition_of assessment = EvidenceSupports \<and>
    validation_ready old_eff assessment)"
proof -
  from concrete have source_lifecycle:
    "lifecycle_of old_eff \<in> {Hypothesis, Supported}"
  by (auto simp: integrate_evidence_step_def
  integrate_evidence_input_valid_def)
  show ?thesis
  using source_lifecycle
  by (cases "evidence_disposition_of assessment";
  auto simp: integrated_lifecycle_def split: if_splits)
qed

theorem integrate_evidence_uncertainty_controlled:

```

```

assumes concrete:
  "integrate_evidence_step G s catalog object_archive payload_archive ledger
    old_eff prior assessment new_version integration_time next_snapshot_id
    s' catalog' object_archive' payload_archive' ledger'"
shows
  "0 \<le> evidence_post_uncertainty_of assessment \<and>
    evidence_post_uncertainty_of assessment \<le> 1 \<and>
    abs (evidence_post_uncertainty_of assessment -
      hypothesis_uncertainty_of prior)
      \<le> evidence_strength_of assessment \<and>
    (evidence_disposition_of assessment = EvidenceSupports
      \<longrightrightarrow> evidence_post_uncertainty_of assessment
      \<le> hypothesis_uncertainty_of prior) \<and>
    (evidence_disposition_of assessment = EvidenceContradicts
      \<longrightrightarrow> hypothesis_uncertainty_of prior
      \<le> evidence_post_uncertainty_of assessment)"
using concrete
unfolding integrate_evidence_step_def integrate_evidence_input_valid_def
  assessment_bounds_def assessment_change_valid_def
by (cases "evidence_disposition_of assessment"; auto)

theorem integrate_evidence_lineage_and_no_self_support:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  defines "new_eff \<equiv> integrated_effect_object old_eff assessment new_version
    integration_time"
  shows
    "previous_version_of new_eff = Some (version_id_of old_eff) \<and>
      evidence_refs_of assessment \<noteq> {} \<and>
      evidence_refs_of assessment \<subsetq> evidence_of new_eff \<and>
      new_version \<notin> evidence_refs_of assessment \<and>
      version_id_of old_eff \<notin> evidence_refs_of assessment \<and>
      (\<forall>v\<in>evidence_refs_of assessment.
        \<exists>ev\<in>objects_of s.
          version_id_of ev = v \<and>
          object_id_of ev \<noteq> object_id_of old_eff \<and>
          lifecycle_of ev \<notin> {Rejected, Archived})"
proof -
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
      ledger old_eff prior assessment new_version integration_time
      next_snapshot_id"
  by (simp add: integrate_evidence_step_def)
  from valid have refs_nonempty: "evidence_refs_of assessment \<noteq> {}"
  and active:
    "\<forall>v\<in>evidence_refs_of assessment.
      active_evidence_source s (object_id_of old_eff) v"
  and fresh:
    "new_version \<notin> version_ids s \<union> historical_versions_of s"
  and old_mem: "old_eff \<in> objects_of s"
  and source_wf: "state_well_formed G s"
  unfolding integrate_evidence_input_valid_def by blast+
  have source_unique: "inj_on version_id_of (objects_of s)"
  using source_wf unfolding state_well_formed_def unique_versions_def by blast
  have prior_not: "version_id_of old_eff \<notin> evidence_refs_of assessment"
  proof
    assume prior_in: "version_id_of old_eff \<in> evidence_refs_of assessment"
    from active prior_in obtain ev where ev_mem: "ev \<in> objects_of s"
    and same_version: "version_id_of ev = version_id_of old_eff"
    and different_object: "object_id_of ev \<noteq> object_id_of old_eff"
    unfolding active_evidence_source_def by blast
    have "ev = old_eff"
    using inj_onD[OF source_unique same_version ev_mem old_mem] .
    then show False using different_object by simp
  qed
  have new_not: "new_version \<notin> evidence_refs_of assessment"
  proof
    assume "new_version \<in> evidence_refs_of assessment"
    then obtain ev where "ev \<in> objects_of s" "version_id_of ev = new_version"
    using active unfolding active_evidence_source_def by blast
    then have "new_version \<in> version_ids s"
    unfolding version_ids_def by blast
    then show False using fresh by blast
  qed
  show ?thesis
  unfolding new_eff_def
  using refs_nonempty active prior_not new_not
  by (auto simp: active_evidence_source_def)

```



qed

subsection \<open>Reference-universe monotonicity\<close>

```
lemma replace_effect_version_universe_mono:
  assumes old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  shows
    "version_ids s \<union> historical_versions_of s \<subseq>
      version_ids (replace_effect_version s old_eff new_eff next_id)
      \<union> historical_versions_of
        (replace_effect_version s old_eff new_eff next_id)"
proof
  fix v
  assume v_mem: "v \<in> version_ids s \<union> historical_versions_of s"
  show "v \<in> version_ids (replace_effect_version s old_eff new_eff next_id)
    \<union> historical_versions_of
      (replace_effect_version s old_eff new_eff next_id)"
  proof (cases "v = version_id_of old_eff")
    case is_old: True
    then show ?thesis by simp
  next
    case not_old: False
    show ?thesis
    proof (cases "v \<in> historical_versions_of s")
      case in_history: True
      then show ?thesis by simp
    next
      case not_in_history: False
      with v_mem obtain obj where obj_mem: "obj \<in> objects_of s"
      and obj_v: "version_id_of obj = v"
      unfolding version_ids_def by blast
      have "obj \<noteq> old_eff"
      proof
        assume "obj = old_eff"
        then have "v = version_id_of old_eff" using obj_v by simp
        then show False using not_old by contradiction
      qed
      then have "obj \<in> objects_of s - {old_eff}" using obj_mem by blast
      then have "v \<in> version_ids
        (replace_effect_version s old_eff new_eff next_id)"
        unfolding version_ids_def using obj_v by auto
      then show ?thesis by blast
    qed
  qed
qed
```

```
lemma replace_effect_present_ref_mono:
  assumes old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  and present: "present_ref s ref"
  shows "present_ref
    (replace_effect_version s old_eff new_eff next_id) ref"
  using replace_effect_version_universe_mono[OF old_mem fresh] present
  unfolding present_ref_def
  by (cases ref; auto)
```

```
lemma replace_effect_references_mono:
  assumes old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  and refs: "references_complete s obj"
  shows "references_complete
    (replace_effect_version s old_eff new_eff next_id) obj"
  using replace_effect_version_universe_mono[OF old_mem fresh] refs
  unfolding references_complete_def by blast
```

```
lemma replace_effect_evidence_mono:
  assumes old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  and evidence: "evidence_well_formed s obj"
  shows "evidence_well_formed
    (replace_effect_version s old_eff new_eff next_id) obj"
  using replace_effect_version_universe_mono[OF old_mem fresh] evidence
  unfolding evidence_well_formed_def evidence_references_complete_def by blast
```

```
lemma replace_effect_preserves_active_history_disjoint:
```

```

assumes source_disjoint: "active_history_disjoint s"
  and source_unique: "unique_versions s"
  and old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
shows "active_history_disjoint
  (replace_effect_version s old_eff new_eff next_id)"
proof -
  have old_version_active: "version_id_of old_eff \<in> version_ids s"
    unfolding version_ids_def using old_mem by blast
  have versions_different:
    "version_id_of old_eff \<noteq> version_id_of new_eff"
  proof
    assume same: "version_id_of old_eff = version_id_of new_eff"
    from fresh have "version_id_of new_eff \<notin> version_ids s" by blast
    with old_version_active same show False by simp
  qed
  show ?thesis
    using source_disjoint source_unique old_mem fresh versions_different
    unfolding active_history_disjoint_def unique_versions_def version_ids_def
    replace_effect_version_def
    by (auto simp: inj_on_def)
qed

lemma evidence_entry_replace_mono:
  assumes old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  and old_not_archived: "archive (version_id_of old_eff) = None"
  and entry: "evidence_entry_well_formed s archive v assessment"
  shows "evidence_entry_well_formed
    (replace_effect_version s old_eff new_eff next_id)
    (extend_object_archive archive old_eff) v assessment"
proof -
  let ?target = "replace_effect_version s old_eff new_eff next_id"
  let ?archive = "extend_object_archive archive old_eff"
  from entry obtain eff where resolved:
    "resolved_version_object s archive v eff"
  and remaining:
    "kind_of eff = K_EFF \<and>
      evidence_subject_of assessment = object_id_of eff \<and>
      previous_version_of eff = Some (evidence_prior_version_of assessment) \<and>
      evidence_refs_of assessment \<noteq> {} \<and>
      evidence_refs_of assessment \<subsetq> evidence_of eff \<and>
      evidence_refs_of assessment
        \<subsetq> version_ids s \<union> historical_versions_of s \<and>
      v \<notin> evidence_refs_of assessment \<and>
      evidence_prior_version_of assessment
        \<notin> evidence_refs_of assessment \<and>
      0 \<le> evidence_strength_of assessment \<and>
      evidence_strength_of assessment \<le> 1 \<and>
      0 \<le> evidence_post_uncertainty_of assessment \<and>
      evidence_post_uncertainty_of assessment \<le> 1 \<and>
      0 \<le> validation_uncertainty_max_of assessment \<and>
      validation_uncertainty_max_of assessment \<le> 1 \<and>
      0 < validation_evidence_min_of assessment \<and>
      evidence_assessment_model_of assessment \<in> models_of s"
  unfolding evidence_entry_well_formed_def by blast
  have resolved_target:
    "resolved_version_object ?target ?archive v eff"
  proof (cases "eff \<in> objects_of s \<and> version_id_of eff = v")
    case active: True
    show ?thesis
    proof (cases "eff = old_eff")
      case same_old: True
      from active have eff_version: "version_id_of eff = v" by blast
      have "?archive v = Some eff"
        unfolding extend_object_archive_def
        using same_old eff_version by simp
      then show ?thesis
        unfolding resolved_version_object_def by blast
    next
      case different_old: False
      with active have "eff \<in> objects_of s - {old_eff}" by blast
      with active show ?thesis
        unfolding resolved_version_object_def by auto
    qed
  next
  case not_active: False
  with resolved have archived: "archive v = Some eff"

```

```

    unfolding resolved_version_object_def by blast
    have key_different: "v \<noteq> version_id_of old_eff"
    proof
      assume "v = version_id_of old_eff"
      with archived old_not_archived show False by simp
    qed
    show ?thesis
      unfolding resolved_version_object_def extend_object_archive_def
      using archived key_different by simp
  qed
have refs_target:
  "evidence_refs_of assessment
   \<subseq> version_ids ?target \<union> historical_versions_of ?target"
  using remaining replace_effect_version_universe_mono[OF old_mem fresh]
  by blast
show ?thesis
  unfolding evidence_entry_well_formed_def
proof (rule exI[of _ eff])
  show "resolved_version_object ?target ?archive v eff \<and>
        kind_of eff = K_EFF \<and>
        evidence_subject_of assessment = object_id_of eff \<and>
        previous_version_of eff = Some (evidence_prior_version_of assessment) \<and>
        evidence_refs_of assessment \<noteq> {} \<and>
        evidence_refs_of assessment \<subseq> evidence_of eff \<and>
        evidence_refs_of assessment
          \<subseq> version_ids ?target \<union> historical_versions_of ?target \<and>
        v \<notin> evidence_refs_of assessment \<and>
        evidence_prior_version_of assessment
          \<notin> evidence_refs_of assessment \<and>
        0 \<le> evidence_strength_of assessment \<and>
        evidence_strength_of assessment \<le> 1 \<and>
        0 \<le> evidence_post_uncertainty_of assessment \<and>
        evidence_post_uncertainty_of assessment \<le> 1 \<and>
        0 \<le> validation_uncertainty_max_of assessment \<and>
        validation_uncertainty_max_of assessment \<le> 1 \<and>
        0 < validation_evidence_min_of assessment \<and>
        evidence_assessment_model_of assessment \<in> models_of ?target"
    using resolved_target remaining refs_target
    by (simp only: replace_effect_simps; blast)
  qed
qed

subsection \<open>Successor state and audit-store preservation\<close>

lemma integrate_evidence_payload_valid:
  assumes valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
     ledger old_eff prior assessment new_version integration_time
     next_snapshot_id"
  shows "effect_payload_valid
        (integrated_effect_object old_eff assessment new_version integration_time)
        (integrated_effect_payload prior assessment)"
  using valid
  unfolding integrate_evidence_input_valid_def assessment_bounds_def
  effect_payload_valid_def
  by (cases "evidence_post_vector_of assessment"; auto)

lemma integrate_evidence_new_object_conditions:
  assumes valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
     ledger old_eff prior assessment new_version integration_time
     next_snapshot_id"
  defines "new_eff \<equiv> integrated_effect_object old_eff assessment new_version
           integration_time"
  shows
    "kind_of new_eff \<in> supported_kinds_of G \<and>
     references_complete
       (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
     evidence_well_formed
       (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
     time_well_formed new_eff \<and>
     relation_well_formed
       (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
     effect_well_formed
       (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
     perspective_well_formed
       (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
     context_well_formed
       (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
     relevance_well_formed"
```

```

    (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
    orientation_well_formed
    (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
    action_well_formed
    (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
    feedback_well_formed
    (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff"
proof -
  let ?target = "replace_effect_version s old_eff new_eff next_snapshot_id"
  from valid have source_wf: "state_well_formed G s"
  and old_mem: "old_eff \<in> objects_of s"
  and old_kind: "kind_of old_eff = K_EFF"
  and old_lifecycle: "lifecycle_of old_eff \<in> {Hypothesis, Supported}"
  and fresh: "new_version \<notin> version_ids s \<union> historical_versions_of s"
  and origin: "present_ref s (effect_origin_of old_eff)"
  and evidence_nonempty: "evidence_refs_of assessment \<noteq> {}"
  and evidence_active:
    "\<forall>v\<in>evidence_refs_of assessment.
      active_evidence_source s (object_id_of old_eff) v"
  and target_ok:
    "admissible_post_target s (object_id_of old_eff)
      (evidence_post_target_of assessment)"
  unfolding integrate_evidence_input_valid_def by blast+
  from source_wf old_mem have old_conditions:
    "kind_of old_eff \<in> supported_kinds_of G \<and>
      references_complete s old_eff \<and>
      evidence_well_formed s old_eff \<and>
      time_well_formed old_eff \<and>
      perspective_well_formed s old_eff \<and>
      context_well_formed s old_eff"
  unfolding state_well_formed_def by blast
  have universe_mono:
    "version_ids s \<union> historical_versions_of s \<subsetq>
      version_ids ?target \<union> historical_versions_of ?target"
  using replace_effect_version_universe_mono[OF old_mem]
  unfolding new_eff_def using fresh by simp
  have evidence_refs_resolve:
    "evidence_refs_of assessment
      \<subsetq> version_ids ?target \<union> historical_versions_of ?target"
proof
  fix v
  assume "v \<in> evidence_refs_of assessment"
  then obtain ev where ev_mem: "ev \<in> objects_of s"
  and ev_v: "version_id_of ev = v"
  and different: "object_id_of ev \<noteq> object_id_of old_eff"
  using evidence_active unfolding active_evidence_source_def by blast
  have "ev \<noteq> old_eff" using different by auto
  then have "ev \<in> objects_of s - {old_eff}" using ev_mem by blast
  then have "v \<in> version_ids ?target"
  unfolding version_ids_def new_eff_def using ev_v by auto
  then show "v \<in> version_ids ?target \<union> historical_versions_of ?target"
  by blast
qed
have target_ref_resolves:
  "option_ref_set (evidence_post_target_of assessment)
    \<subsetq> version_ids ?target \<union> historical_versions_of ?target"
proof (cases "evidence_post_target_of assessment")
  case None
  then show ?thesis by (simp add: option_ref_set_def)
next
  case (Some v)
  then obtain obj where obj_mem: "obj \<in> objects_of s"
  and obj_v: "version_id_of obj = v"
  and different: "object_id_of obj \<noteq> object_id_of old_eff"
  using target_ok unfolding admissible_post_target_def by auto
  have "obj \<noteq> old_eff" using different by auto
  then have "obj \<in> objects_of s - {old_eff}" using obj_mem by blast
  then have "v \<in> version_ids ?target"
  unfolding version_ids_def new_eff_def using obj_v by auto
  with Some show ?thesis by (simp add: option_ref_set_def)
qed
have prior_resolves:
  "version_id_of old_eff
    \<in> version_ids ?target \<union> historical_versions_of ?target"
  by simp
from old_conditions have old_refs_source:
  "references_of old_eff
    \<subsetq> version_ids s \<union> historical_versions_of s"
  unfolding references_complete_def by blast
have old_refs_target:

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"references_of old_eff
  \<subsetq> version_ids ?target \<union> historical_versions_of ?target"
using old_refs_source universe_mono by blast
have combined_refs_target:
  "references_of old_eff \<union> evidence_refs_of assessment \<union>
    {version_id_of old_eff} \<union>
    option_ref_set (evidence_post_target_of assessment)
    \<subsetq> version_ids ?target \<union> historical_versions_of ?target"
  using old_refs_target evidence_refs_resolve target_ref_resolves
  prior_resolves by blast
have new_refs_eq:
  "references_of new_eff =
    references_of old_eff \<union> evidence_refs_of assessment \<union>
    {version_id_of old_eff} \<union>
    option_ref_set (evidence_post_target_of assessment)"
  unfolding new_eff_def by simp
have refs_new: "references_complete ?target new_eff"
  unfolding references_complete_def
  using combined_refs_target new_refs_eq by simp
have evidence_new: "evidence_well_formed ?target new_eff"
proof -
  from old_conditions have old_evidence_source:
    "evidence_of old_eff
      \<subsetq> version_ids s \<union> historical_versions_of s"
    unfolding evidence_well_formed_def evidence_references_complete_def
    by blast
  have old_evidence_target:
    "evidence_of old_eff
      \<subsetq> version_ids ?target \<union> historical_versions_of ?target"
    using old_evidence_source universe_mono by blast
  have new_evidence_eq:
    "evidence_of new_eff =
      evidence_of old_eff \<union> evidence_refs_of assessment"
    unfolding new_eff_def by simp
  have complete:
    "evidence_of new_eff
      \<subsetq> version_ids ?target \<union> historical_versions_of ?target"
    using old_evidence_target evidence_refs_resolve new_evidence_eq by blast
  have upgraded_nonempty:
    "lifecycle_of new_eff \<in> {Supported, Validated}
      \<longrightarrow> evidence_of new_eff \<noteq> {}"
    using evidence_nonempty
    unfolding new_eff_def integrated_effect_object_def
    by auto
  show ?thesis
    unfolding evidence_well_formed_def evidence_references_complete_def
    using complete upgraded_nonempty by blast
qed
have origin_target: "present_ref ?target (effect_origin_of new_eff)"
  unfolding new_eff_def
  using replace_effect_present_ref_mono[OF old_mem _ origin] fresh by simp
have perspective_new: "perspective_well_formed ?target new_eff"
proof -
  from old_conditions have old_perspective:
    "perspective_well_formed s old_eff" by blast
  show ?thesis
    using old_perspective
    unfolding perspective_well_formed_def new_eff_def
    by (simp only: integrated_effect_simps replace_effect_simps)
qed
have context_new: "context_well_formed ?target new_eff"
proof -
  from old_conditions have old_context:
    "context_well_formed s old_eff" by blast
  show ?thesis
    using old_context
    unfolding context_well_formed_def new_eff_def
    by (simp only: integrated_effect_simps replace_effect_simps)
qed
have other_typed:
  "time_well_formed new_eff \<and>
    relation_well_formed ?target new_eff \<and>
    effect_well_formed ?target new_eff \<and>
    relevance_well_formed ?target new_eff \<and>
    orientation_well_formed ?target new_eff \<and>
    action_well_formed ?target new_eff \<and>
    feedback_well_formed ?target new_eff"
proof -
  have new_kind: "kind_of new_eff = K_EFF"
    using old_kind unfolding new_eff_def by simp

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have new_time: "time_well_formed new_eff"
  unfolding time_well_formed_def new_eff_def by simp
have new_relation: "relation_well_formed ?target new_eff"
  using new_kind unfolding relation_well_formed_def by simp
have new_effect: "effect_well_formed ?target new_eff"
  unfolding effect_well_formed_def
  using new_kind origin_target by blast
have new_relevance: "relevance_well_formed ?target new_eff"
  using new_kind unfolding relevance_well_formed_def by simp
have new_orientation: "orientation_well_formed ?target new_eff"
  using new_kind unfolding orientation_well_formed_def by simp
have new_action: "action_well_formed ?target new_eff"
  using new_kind unfolding action_well_formed_def by simp
have new_feedback: "feedback_well_formed ?target new_eff"
  using new_kind unfolding feedback_well_formed_def by simp
show ?thesis
  using new_time new_relation new_effect new_relevance new_orientation
    new_action new_feedback by blast
qed
show ?thesis
  using old_conditions refs_new evidence_new perspective_new context_new
    other_typed
    unfolding new_eff_def by simp
qed

lemma replace_effect_preserves_unique_versions:
  assumes source_unique: "unique_versions s"
  and old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  shows "unique_versions
    (replace_effect_version s old_eff new_eff next_snapshot_id)"
proof -
  from source_unique have inj: "inj_on version_id_of (objects_of s)"
    unfolding unique_versions_def .
  show ?thesis
    unfolding unique_versions_def
  proof (rule inj_onI)
    fix left right
    assume left_mem:
      "left \<in> objects_of
        (replace_effect_version s old_eff new_eff next_snapshot_id)"
      and right_mem:
        "right \<in> objects_of
          (replace_effect_version s old_eff new_eff next_snapshot_id)"
      and same: "version_id_of left = version_id_of right"
    consider (left_new) "left = new_eff" | (left_old) "left \<in> objects_of s - {old_eff}"
    using left_mem by auto
    then show "left = right"
  proof cases
    case left_new
    show ?thesis
    proof (cases "right = new_eff")
      case True
      with left_new show ?thesis by simp
    next
      case False
      with right_mem have right_old: "right \<in> objects_of s"
        by auto
      have "version_id_of new_eff \<in> version_ids s"
        unfolding version_ids_def using same left_new right_old by blast
      then show ?thesis using fresh by blast
    qed
  next
    case left_old
    show ?thesis
    proof (cases "right = new_eff")
      case True
      have left_version: "version_id_of left \<in> version_ids s"
        unfolding version_ids_def using left_old by blast
      have "version_id_of left = version_id_of new_eff"
        using same True by simp
      with left_version have "version_id_of new_eff \<in> version_ids s"
        by simp
      then show ?thesis using fresh by blast
    next
      case False
      with right_mem have right_old: "right \<in> objects_of s" by auto
      show ?thesis
        using inj_onD[OF inj same] left_old right_old by blast
    qed
  qed

```

```

qed
qed
qed
qed

theorem integrate_evidence_preserves_state_well_formed:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  shows "state_well_formed G s'"
proof -
  let ?new_eff = "integrated_effect_object old_eff assessment new_version
    integration_time"
  let ?target = "replace_effect_version s old_eff ?new_eff next_snapshot_id"
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
      ledger old_eff prior assessment new_version integration_time
      next_snapshot_id"
  and target: "s' = ?target"
  by (auto simp: integrate_evidence_step_def Let_def)
  from valid have source_wf: "state_well_formed G s"
  and old_mem: "old_eff \<in> objects_of s"
  and fresh: "new_version \<notin> version_ids s \<union> historical_versions_of s"
  unfolding integrate_evidence_input_valid_def by blast+
  have unique: "unique_versions ?target"
  using replace_effect_preserves_unique_versions old_mem fresh source_wf
  unfolding state_well_formed_def by simp
  have new_conditions:
    "kind_of ?new_eff \<in> supported_kinds_of G \<and>
      references_complete ?target ?new_eff \<and>
      evidence_well_formed ?target ?new_eff \<and>
      time_well_formed ?new_eff \<and>
      relation_well_formed ?target ?new_eff \<and>
      effect_well_formed ?target ?new_eff \<and>
      perspective_well_formed ?target ?new_eff \<and>
      context_well_formed ?target ?new_eff \<and>
      relevance_well_formed ?target ?new_eff \<and>
      orientation_well_formed ?target ?new_eff \<and>
      action_well_formed ?target ?new_eff \<and>
      feedback_well_formed ?target ?new_eff"
  using integrate_evidence_new_object_conditions[OF valid] by simp
  have old_object_conditions:
    "\<forall>obj \<in> objects_of s - {old_eff}.
      kind_of obj \<in> supported_kinds_of G \<and>
      references_complete ?target obj \<and>
      evidence_well_formed ?target obj \<and>
      time_well_formed obj \<and>
      relation_well_formed ?target obj \<and>
      effect_well_formed ?target obj \<and>
      perspective_well_formed ?target obj \<and>
      context_well_formed ?target obj \<and>
      relevance_well_formed ?target obj \<and>
      orientation_well_formed ?target obj \<and>
      action_well_formed ?target obj \<and>
      feedback_well_formed ?target obj"
  proof (intro ballI)
    fix obj
    assume obj_mem: "obj \<in> objects_of s - {old_eff}"
    from source_wf obj_mem have source_conditions:
      "kind_of obj \<in> supported_kinds_of G \<and>
        references_complete s obj \<and> evidence_well_formed s obj \<and>
        time_well_formed obj \<and> relation_well_formed s obj \<and>
        effect_well_formed s obj \<and> perspective_well_formed s obj \<and>
        context_well_formed s obj \<and> relevance_well_formed s obj \<and>
        orientation_well_formed s obj \<and> action_well_formed s obj \<and>
        feedback_well_formed s obj"
    unfolding state_well_formed_def by blast
    have present_mono: "\<And>ref. present_ref s ref \<Longrightarrow> present_ref ?target ref"
    using replace_effect_present_ref_mono[OF old_mem _] fresh by simp
    have refs_target: "references_complete ?target obj"
    using replace_effect_references_mono[OF old_mem _]
    source_conditions fresh by simp
    have evidence_target: "evidence_well_formed ?target obj"
    using replace_effect_evidence_mono[OF old_mem _]
    source_conditions fresh by simp
    have typed_target:
      "relation_well_formed ?target obj \<and>
        effect_well_formed ?target obj \<and>
        feedback_well_formed ?target obj"
  qed

```

```

using source_conditions present_mono
unfolding relation_well_formed_def effect_well_formed_def
  feedback_well_formed_def
by blast
from source_conditions have source_perspective:
  "perspective_well_formed s obj"
and source_context: "context_well_formed s obj"
and source_relevance: "relevance_well_formed s obj"
and source_orientation: "orientation_well_formed s obj"
and source_action: "action_well_formed s obj"
by blast+
have perspective_target: "perspective_well_formed ?target obj"
using source_perspective
  unfolding perspective_well_formed_def
  by (simp only: replace_effect_simps)
have context_target: "context_well_formed ?target obj"
using source_context
  unfolding context_well_formed_def
  by (simp only: replace_effect_simps)
have relevance_target: "relevance_well_formed ?target obj"
  unfolding relevance_well_formed_def
proof
  assume kind: "kind_of obj = K_REV"
  from source_relevance kind have source_model:
    "case model_of obj of None \ $\rightarrow$  False
    | Some m \ $\rightarrow$  m \ $\in$  models_of s"
  and source_refs: "references_of obj \ $\neq$  {}"
  unfolding relevance_well_formed_def by blast+
  have target_model:
    "case model_of obj of None \ $\rightarrow$  False
    | Some m \ $\rightarrow$  m \ $\in$  models_of ?target"
  using source_model by (simp only: replace_effect_simps)
  show "perspective_well_formed ?target obj \ $\wedge$ 
    context_well_formed ?target obj \ $\wedge$ 
    (case model_of obj of None \ $\rightarrow$  False
    | Some m \ $\rightarrow$  m \ $\in$  models_of ?target) \ $\wedge$ 
    references_of obj \ $\neq$  {}"
  using perspective_target context_target target_model source_refs by blast
qed
have orientation_target: "orientation_well_formed ?target obj"
  using source_orientation unfolding orientation_well_formed_def by simp
have action_target: "action_well_formed ?target obj"
  using source_action unfolding action_well_formed_def
  by (simp only: replace_effect_simps)
have stable_target:
  "perspective_well_formed ?target obj \ $\wedge$ 
  context_well_formed ?target obj \ $\wedge$ 
  relevance_well_formed ?target obj \ $\wedge$ 
  orientation_well_formed ?target obj \ $\wedge$ 
  action_well_formed ?target obj"
  using perspective_target context_target relevance_target
  orientation_target action_target by blast
show "kind_of obj \ $\in$  supported_kinds_of G \ $\wedge$ 
  references_complete ?target obj \ $\wedge$ 
  evidence_well_formed ?target obj \ $\wedge$ 
  time_well_formed obj \ $\wedge$ 
  relation_well_formed ?target obj \ $\wedge$ 
  effect_well_formed ?target obj \ $\wedge$ 
  perspective_well_formed ?target obj \ $\wedge$ 
  context_well_formed ?target obj \ $\wedge$ 
  relevance_well_formed ?target obj \ $\wedge$ 
  orientation_well_formed ?target obj \ $\wedge$ 
  action_well_formed ?target obj \ $\wedge$ 
  feedback_well_formed ?target obj"
  using source_conditions refs_target evidence_target typed_target
  stable_target by blast
qed
have all_objects:
  "\forall>obj\ $\in$ objects_of ?target.
  kind_of obj \ $\in$  supported_kinds_of G \ $\wedge$ 
  references_complete ?target obj \ $\wedge$ 
  evidence_well_formed ?target obj \ $\wedge$ 
  time_well_formed obj \ $\wedge$ 
  relation_well_formed ?target obj \ $\wedge$ 
  effect_well_formed ?target obj \ $\wedge$ 
  perspective_well_formed ?target obj \ $\wedge$ 
  context_well_formed ?target obj \ $\wedge$ 
  relevance_well_formed ?target obj \ $\wedge$ 
  orientation_well_formed ?target obj \ $\wedge$ 
  action_well_formed ?target obj \ $\wedge$ "

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    feedback_well_formed ?target obj"
    using new_conditions old_object_conditions by auto
from source_wf have secondary:
  "(\<forall>p\<in>projections_of ?target. projection_well_formed ?target p) \<and>
  (\<forall>c\<in>couplings_of ?target. coupling_well_formed ?target c) \<and>
  (\<forall>t\<in>transmissions_of ?target. transmission_well_formed ?target t) \<and>
  (\<forall>b\<in>boundaries_of ?target. boundary_well_formed b)"
    unfolding state_well_formed_def projection_well_formed_def
      coupling_well_formed_def transmission_well_formed_def
    by simp
show ?thesis
  unfolding target state_well_formed_def
  using unique_all_objects_secondary by blast
qed

theorem integrate_evidence_preserves_identity:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  shows "unique_versions s' \<and> immutable_preserved s s'"
proof -
  let ?new_eff = "integrated_effect_object old_eff assessment new_version
    integration_time"
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
      ledger old_eff prior assessment new_version integration_time
      next_snapshot_id"
  and target:
    "s' = replace_effect_version s old_eff ?new_eff next_snapshot_id"
  by (auto simp: integrate_evidence_step_def Let_def)
  from valid have source_wf: "state_well_formed G s"
  and old_mem: "old_eff \<in> objects_of s"
  and fresh: "new_version \<notin> version_ids s \<union> historical_versions_of s"
  unfolding integrate_evidence_input_valid_def by blast+
  have unique: "unique_versions s'"
  unfolding target
  using replace_effect_preserves_unique_versions old_mem fresh source_wf
  unfolding state_well_formed_def by simp
  have immutable: "immutable_preserved s s'"
  unfolding immutable_preserved_def target
  proof (intro ballI allI impI)
    fix before after
    assume before_mem: "before \<in> objects_of s"
    and after_mem:
      "after \<in> objects_of
        (replace_effect_version s old_eff ?new_eff next_snapshot_id)"
    and same: "version_id_of before = version_id_of after"
    show "before = after"
    proof (cases "after = ?new_eff")
      case True
      have "new_version \<in> version_ids s"
      unfolding version_ids_def using before_mem same True by auto
      then show ?thesis using fresh by blast
    next
      case False
      then have after_old: "after \<in> objects_of s" using after_mem by auto
      from source_wf have inj: "inj_on version_id_of (objects_of s)"
      unfolding state_well_formed_def unique_versions_def by blast
      show ?thesis using inj_onD[OF inj same before_mem after_old] .
    qed
  qed
  show ?thesis using unique immutable by blast
qed

theorem integrate_evidence_preserves_audit_stores:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  shows
    "active_history_disjoint s' \<and>
      object_archive_well_formed s' object_archive' \<and>
      effect_payload_archive_well_formed object_archive' payload_archive' \<and>
      evidence_ledger_well_formed s' object_archive' ledger'"
proof -
  let ?new_eff = "integrated_effect_object old_eff assessment new_version
    integration_time"
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive

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    ledger old_eff prior assessment new_version integration_time
    next_snapshot_id"
and target:
  "s' = replace_effect_version s old_eff ?new_eff next_snapshot_id"
and object_archive_target:
  "object_archive' = extend_object_archive object_archive old_eff"
and payload_archive_target:
  "payload_archive' = extend_effect_payload_archive payload_archive
  old_eff prior"
and ledger_target:
  "ledger' = extend_evidence_ledger ledger new_version assessment"
by (auto simp: integrate_evidence_step_def Let_def)
from valid have source_wf: "state_well_formed G s"
and source_disjoint: "active_history_disjoint s"
and source_object_archive:
  "object_archive_well_formed s object_archive"
and source_payload_archive:
  "effect_payload_archive_well_formed object_archive payload_archive"
and source_ledger:
  "evidence_ledger_well_formed s object_archive ledger"
and old_mem: "old_eff \<in> objects_of s"
and old_archive_none:
  "object_archive (version_id_of old_eff) = None"
and payload_old_none:
  "payload_archive (version_id_of old_eff) = None"
and fresh:
  "new_version \<notin> version_ids s \<union> historical_versions_of s"
and ledger_new_none: "ledger new_version = None"
and assessment_bounds: "assessment_bounds assessment"
and assessment_model:
  "evidence_assessment_model_of assessment \<in> models_of s"
and old_kind: "kind_of old_eff = K_EFF"
and payload_valid: "effect_payload_valid old_eff prior"
unfolding integrate_evidence_input_valid_def by blast+
from source_wf have source_unique: "unique_versions s"
unfolding state_well_formed_def by blast
have fresh_eff:
  "version_id_of ?new_eff
  \<notin> version_ids s \<union> historical_versions_of s"
using fresh by simp
have target_disjoint: "active_history_disjoint s'"
unfolding target
using replace_effect_preserves_active_history_disjoint
[OF source_disjoint source_unique old_mem fresh_eff] .
have target_object_archive:
  "object_archive_well_formed s' object_archive'"
proof -
  have archived_properties:
    "\<forall>v obj. object_archive' v = Some obj \<longrightarrow>
    version_id_of obj = v \<and> v \<in> historical_versions_of s'"
  proof (intro allI impI)
    fix v obj
    assume mapped: "object_archive' v = Some obj"
    show "version_id_of obj = v \<and> v \<in> historical_versions_of s'"
    proof (cases "v = version_id_of old_eff")
      case True
      with mapped object_archive_target target show ?thesis
      unfolding extend_object_archive_def by simp
    next
      case False
      with mapped object_archive_target have source_mapped:
        "object_archive v = Some obj"
      unfolding extend_object_archive_def by simp
      from source_object_archive source_mapped have
        "version_id_of obj = v \<and> v \<in> historical_versions_of s"
      unfolding object_archive_well_formed_def by blast
      with target show ?thesis by simp
    qed
  qed
  show ?thesis
  unfolding object_archive_well_formed_def
  using archived_properties target_disjoint
  unfolding active_history_disjoint_def by blast
qed
have target_payload_archive:
  "effect_payload_archive_well_formed object_archive' payload_archive'"
proof (unfold effect_payload_archive_well_formed_def, intro allI impI)
  fix v stored_payload
  assume mapped: "payload_archive' v = Some stored_payload"
  show "\<exists>eff. object_archive' v = Some eff \<and>
```

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    effect_payload_valid eff stored_payload"
  proof (cases "v = version_id_of old_eff")
  case True
  with mapped payload_archive_target object_archive_target
    payload_valid show ?thesis
    unfolding extend_effect_payload_archive_def extend_object_archive_def
    by auto
next
case False
with mapped payload_archive_target have source_mapped:
  "payload_archive v = Some stored_payload"
  unfolding extend_effect_payload_archive_def by simp
from source_payload_archive source_mapped obtain eff where
  source_object: "object_archive v = Some eff"
  and valid_payload: "effect_payload_valid eff stored_payload"
  unfolding effect_payload_archive_well_formed_def by blast
have "object_archive' v = Some eff"
  unfolding object_archive_target extend_object_archive_def
  using source_object False by simp
with valid_payload show ?thesis by blast
qed
qed
have target_ledger:
  "evidence_ledger_well_formed s' object_archive' ledger'"
proof (unfold evidence_ledger_well_formed_def, intro allI impI)
  fix v stored_assessment
  assume mapped: "ledger' v = Some stored_assessment"
  show "evidence_entry_well_formed s' object_archive' v stored_assessment"
  proof (cases "v = new_version")
  case True
  with mapped ledger_target have assessment_eq:
    "stored_assessment = assessment"
    unfolding extend_evidence_ledger_def by simp
  have target_wf: "state_well_formed G s'"
    using integrate_evidence_preserves_state_well_formed[OF concrete] .
  have exact:
    "objects_of s' = insert ?new_eff (objects_of s - {old_eff})"
    using concrete
    unfolding integrate_evidence_step_def Let_def by auto
  have new_mem: "?new_eff \<in> objects_of s'" using exact by simp
  have evidence_refs_in_new:
    "evidence_refs_of assessment \<subseq> evidence_of ?new_eff"
    by simp
  from target_wf new_mem have new_evidence_complete:
    "evidence_of ?new_eff
      \<subseq> version_ids s' \<union> historical_versions_of s'"
    unfolding state_well_formed_def evidence_well_formed_def
    evidence_references_complete_def
    by blast
  from target_wf new_mem have evidence_complete:
    "evidence_refs_of assessment
      \<subseq> version_ids s' \<union> historical_versions_of s'"
    using evidence_refs_in_new new_evidence_complete by blast
  from integrate_evidence_lineage_and_no_self_support[OF concrete]
  have evidence_nonempty: "evidence_refs_of assessment \<noteq> {}"
    and new_not_self: "new_version \<notin> evidence_refs_of assessment"
    and prior_not_self:
      "version_id_of old_eff \<notin> evidence_refs_of assessment"
    by blast+
  from valid have subject:
    "evidence_subject_of assessment = object_id_of old_eff"
    and prior_version:
      "evidence_prior_version_of assessment = version_id_of old_eff"
    unfolding integrate_evidence_input_valid_def by blast+
  have resolved:
    "resolved_version_object s' object_archive' new_version ?new_eff"
    unfolding resolved_version_object_def using new_mem by simp
  show ?thesis
    unfolding True assessment_eq evidence_entry_well_formed_def
    using resolved_evidence_nonempty evidence_complete new_not_self
    prior_not_self subject prior_version evidence_refs_in_new
    assessment_bounds old_kind
    assessment_model
    unfolding assessment_bounds_def target
    by auto
next
case False
with mapped ledger_target have source_mapped:
  "ledger v = Some stored_assessment"
  unfolding extend_evidence_ledger_def by simp

```

```

from source_ledger source_mapped have source_entry:
  "evidence_entry_well_formed s object_archive v stored_assessment"
  unfolding evidence_ledger_well_formed_def by blast
have transferred:
  "evidence_entry_well_formed
   (replace_effect_version s old_eff ?new_eff next_snapshot_id)
   (extend_object_archive object_archive old_eff) v stored_assessment"
  using evidence_entry_replace_mono[OF old_mem fresh_eff
    old_archive_none source_entry] .
show ?thesis
  unfolding target object_archive_target using transferred .
qed
qed
show ?thesis
  using target_disjoint target_object_archive target_payload_archive
    target_ledger by blast
qed

theorem integrate_evidence_preserves_active_catalog:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
     old_eff prior assessment new_version integration_time next_snapshot_id
     s' catalog' object_archive' payload_archive' ledger'"
  shows "effect_catalog_well_formed s' catalog'"
proof -
  let ?new_eff = "integrated_effect_object old_eff assessment new_version
    integration_time"
  let ?new_payload = "integrated_effect_payload prior assessment"
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
     ledger old_eff prior assessment new_version integration_time
     next_snapshot_id"
  and target:
    "s' = replace_effect_version s old_eff ?new_eff next_snapshot_id"
  and catalog_target:
    "catalog' = replace_active_effect_payload catalog
     (version_id_of old_eff) new_version ?new_payload"
  by (auto simp: integrate_evidence_step_def Let_def)
  from valid have source_catalog: "effect_catalog_well_formed s catalog"
  and old_mem: "old_eff \<in> objects_of s"
  and fresh:
    "new_version \<notin> version_ids s \<union> historical_versions_of s"
  unfolding integrate_evidence_input_valid_def by blast+
  have new_payload_valid: "effect_payload_valid ?new_eff ?new_payload"
  using integrate_evidence_payload_valid[OF valid] .
  have new_mem: "?new_eff \<in> objects_of s'"
  unfolding target by simp
  show ?thesis
    unfolding effect_catalog_well_formed_def
  proof (intro allI impI)
    fix v stored_payload
    assume mapped: "catalog' v = Some stored_payload"
    show "\<exists>eff\<in>objects_of s'.
      version_id_of eff = v \<and> effect_payload_valid eff stored_payload"
    proof (cases "v = new_version")
      case True
      with mapped catalog_target have payload_eq:
        "stored_payload = ?new_payload"
      unfolding replace_active_effect_payload_def by simp
      show ?thesis
        proof (rule bexI[of _ ?new_eff])
          show "version_id_of ?new_eff = v \<and>
            effect_payload_valid ?new_eff stored_payload"
            using new_payload_valid True payload_eq by simp
          show "?new_eff \<in> objects_of s'" using new_mem .
        qed
      qed
    next
      case False
      have old_key_different: "v \<noteq> version_id_of old_eff"
      proof
        assume "v = version_id_of old_eff"
        with mapped catalog_target False show False
        unfolding replace_active_effect_payload_def by simp
      qed
      from mapped catalog_target False old_key_different have source_mapped:
        "catalog v = Some stored_payload"
      unfolding replace_active_effect_payload_def by simp
      from source_catalog source_mapped obtain eff where eff_mem:
        "eff \<in> objects_of s"
      and eff_version: "version_id_of eff = v"

```

```

        and eff_payload: "effect_payload_valid eff stored_payload"
        unfolding effect_catalog_well_formed_def by blast
        have eff_not_old: "eff \<noteq> old_eff"
        using eff_version old_key_different by auto
        have "eff \<in> objects_of s'"
        unfolding target using eff_mem eff_not_old by auto
        with eff_version eff_payload show ?thesis by blast
    qed
qed
qed

theorem integrate_evidence_preserves_references_time_context:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
     old_eff prior assessment new_version integration_time next_snapshot_id
     s' catalog' object_archive' payload_archive' ledger'"
  shows
    "\<forall>obj\<in>objects_of s'.
     references_complete s' obj \<and>
     evidence_well_formed s' obj \<and>
     time_well_formed obj \<and>
     perspective_well_formed s' obj \<and>
     context_well_formed s' obj"
  using integrate_evidence_preserves_state_well_formed[OF concrete]
  unfolding state_well_formed_def by blast

theorem integrate_evidence_refines_generic_step:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
     old_eff prior assessment new_version integration_time next_snapshot_id
     s' catalog' object_archive' payload_archive' ledger'"
  shows "step G s IntegrateEvidence s'"
proof -
  from concrete have source_wf: "state_well_formed G s"
  and allowed: "IntegrateEvidence \<in> allowed_operations_of G"
  by (auto simp: integrate_evidence_step_def integrate_evidence_input_valid_def)
  have target_wf: "state_well_formed G s'"
  using integrate_evidence_preserves_state_well_formed[OF concrete] .
  have immutable: "immutable_preserved s s'"
  using integrate_evidence_preserves_identity[OF concrete] by blast
  show ?thesis
  unfolding step_def using source_wf target_wf allowed immutable by blast
qed
end

```

## I.7 Konstruktiver T1-T3-Ereichbarkeitszeuge

Vollständiger Quelltext des T1 → T2 → T3 → T3-Zeugen. Die beiden Relationsobjekte werden durch `create_relation_step` erzeugt.

```

theory UZA_0_9_1_Create_Relation_Witness
  imports UZA_0_9_1_Create_Relation UZA_0_9_1_Base_Model
begin

section \<open>A constructive T1--T3 reachability witness\<close>

text \<open>
  This theory closes the non-vacuity gap below T3. Its two relation objects
  are not injected into a preassembled snapshot. Starting with one meta
  object, T1 registers an observed appearance, T2 differentiates a distinct
  potential candidate, and two consecutive T3 steps create directed relations
  between the resulting active K_ENT endpoints. Every successor is the
  constructor result of the preceding operation.
\<close>

definition chain_grammar :: grammar where
  "chain_grammar =
    \<lparr> grammar_id_of = 91,
     supported_kinds_of = {K_META, K_ENT, K_REL, K_EFF, K_RES, K_REV},
     allowed_operations_of =
       {RegisterAppearance, DifferentiatePotential, CreateRelation,
        HypothesizeEffect, IntegrateEvidence, ClassifyResonance,
        EvaluateRelevance},
     patch_budget_of = 0 \<rparr>"

```

```
definition chain_seed :: snapshot where
  "chain_seed =
    base_snapshot\<lparr>
      perspectives_of := {0, Suc 0, 10},
      contexts_of := {20},
      models_of := {30}
    \<rparr>"

definition chain_entity_a :: uza_object where
  "chain_entity_a =
    base_object\<lparr>
      object_id_of := 1,
      version_id_of := 1,
      kind_of := K_ENT,
      lifecycle_of := Draft,
      perspective_of := Some 10,
      context_of := Some 20
    \<rparr>"

definition chain_entity_b :: uza_object where
  "chain_entity_b =
    base_object\<lparr>
      object_id_of := 2,
      version_id_of := 2,
      kind_of := K_ENT,
      lifecycle_of := Hypothesis,
      previous_version_of := None,
      references_of := {1},
      perspective_of := Some 10,
      context_of := Some 20,
      source_ref_of := Some 1,
      basis_refs_of := {1},
      object_status_of := Some Applicable
    \<rparr>"

definition chain_relation_ab :: uza_object where
  "chain_relation_ab =
    base_object\<lparr>
      object_id_of := 3,
      version_id_of := 3,
      kind_of := K_REL,
      lifecycle_of := Hypothesis,
      references_of := {1, 2},
      context_of := Some 20,
      source_ref_of := Some 1,
      target_ref_of := Some 2
    \<rparr>"

definition chain_relation_ba :: uza_object where
  "chain_relation_ba =
    base_object\<lparr>
      object_id_of := 4,
      version_id_of := 4,
      kind_of := K_REL,
      lifecycle_of := Hypothesis,
      references_of := {1, 2},
      context_of := Some 20,
      source_ref_of := Some 2,
      target_ref_of := Some 1
    \<rparr>"

definition chain_after_t1 :: snapshot where
  "chain_after_t1 = add_appearance chain_seed chain_entity_a 1"

definition chain_after_t2 :: snapshot where
  "chain_after_t2 =
    add_potential_candidates chain_after_t1 {chain_entity_b} 2"

definition chain_relation_catalog_0 :: relation_catalog where
  "chain_relation_catalog_0 = Map.empty"

definition chain_after_t3_ab :: snapshot where
  "chain_after_t3_ab = add_relation chain_after_t2 chain_relation_ab 3"

definition chain_relation_catalog_ab :: relation_catalog where
  "chain_relation_catalog_ab =
    extend_relation_catalog chain_relation_catalog_0 chain_relation_ab
    RelGenerates True"
```

```

definition chain_after_t3 :: snapshot where
  "chain_after_t3 = add_relation chain_after_t3_ab chain_relation_ba 4"

definition chain_relation_catalog :: relation_catalog where
  "chain_relation_catalog =
    extend_relation_catalog chain_relation_catalog_ab chain_relation_ba
    RelConditions True"

lemmas chain_object_defs =
  chain_entity_a_def chain_entity_b_def chain_relation_ab_def
  chain_relation_ba_def

lemma chain_seed_well_formed:
  "state_well_formed chain_grammar chain_seed"
  unfolding state_well_formed_def
  by (auto simp: chain_grammar_def chain_seed_def base_snapshot_def
    base_object_def unique_versions_def version_ids_def
    references_complete_def evidence_well_formed_def
    evidence_references_complete_def time_well_formed_def
    relation_well_formed_def effect_well_formed_def
    perspective_well_formed_def perspective_bound_kind_def
    context_well_formed_def context_bound_kind_def
    relevance_well_formed_def orientation_well_formed_def
    action_well_formed_def feedback_well_formed_def)

lemma chain_t1_input_valid:
  "appearance_input_valid chain_grammar chain_seed chain_entity_a 1"
  unfolding appearance_input_valid_def
  using chain_seed_well_formed
  by (auto simp: chain_grammar_def chain_seed_def base_snapshot_def
    chain_entity_a_def base_object_def version_ids_def
    references_complete_def evidence_well_formed_def
    evidence_references_complete_def time_well_formed_def)

theorem chain_t1_step:
  "register_appearance_step chain_grammar chain_seed chain_entity_a 1
    chain_after_t1"
  using chain_t1_input_valid
  by (simp add: register_appearance_step_def chain_after_t1_def)

lemma chain_after_t1_well_formed:
  "state_well_formed chain_grammar chain_after_t1"
  using register_appearance_preserves_state_well_formed[OF chain_t1_step] .

lemma chain_t2_input_valid:
  "potential_input_valid chain_grammar chain_after_t1 {chain_entity_b} 2"
  unfolding potential_input_valid_def
  using chain_after_t1_well_formed
  by (auto simp: chain_grammar_def chain_after_t1_def chain_seed_def
    base_snapshot_def chain_entity_a_def chain_entity_b_def base_object_def
    potential_candidate_marker_def version_ids_def references_complete_def
    evidence_well_formed_def evidence_references_complete_def
    time_well_formed_def inj_on_def)

theorem chain_t2_step:
  "differentiate_potential_step chain_grammar chain_after_t1
    {chain_entity_b} 2 chain_after_t2"
  using chain_t2_input_valid
  by (simp add: differentiate_potential_step_def chain_after_t2_def)

lemma chain_after_t2_well_formed:
  "state_well_formed chain_grammar chain_after_t2"
  using differentiate_potential_preserves_state_well_formed[OF chain_t2_step] .

lemma chain_relation_catalog_0_well_formed:
  "relation_catalog_well_formed chain_after_t2 chain_relation_catalog_0"
  by (simp add: relation_catalog_well_formed_def
    chain_relation_catalog_0_def)

lemma chain_t3_ab_input_valid:
  "relation_input_valid chain_grammar chain_after_t2
    chain_relation_catalog_0 chain_relation_ab RelGenerates True 1 2 3"
  unfolding relation_input_valid_def
  using chain_after_t2_well_formed chain_relation_catalog_0_well_formed
  by (auto simp: chain_grammar_def chain_after_t2_def chain_after_t1_def
    chain_seed_def base_snapshot_def chain_object_defs base_object_def
    active_ent_endpoint_def version_ids_def references_complete_def
    evidence_well_formed_def evidence_references_complete_def
    time_well_formed_def chain_relation_catalog_0_def)

```

```

theorem chain_t3_ab_step:
  "create_relation_step chain_grammar chain_after_t2
   chain_relation_catalog_0 chain_relation_ab RelGenerates True 1 2 3
   chain_after_t3_ab chain_relation_catalog_ab"
  using chain_t3_ab_input_valid
  by (simp add: create_relation_step_def chain_after_t3_ab_def
    chain_relation_catalog_ab_def)

lemma chain_after_t3_ab_well_formed:
  "state_well_formed chain_grammar chain_after_t3_ab"
  using create_relation_preserves_state_well_formed[OF chain_t3_ab_step] .

lemma chain_relation_catalog_ab_well_formed:
  "relation_catalog_well_formed chain_after_t3_ab
   chain_relation_catalog_ab"
  using create_relation_preserves_catalog[OF chain_t3_ab_step]
  by blast

lemma chain_t3_ba_input_valid:
  "relation_input_valid chain_grammar chain_after_t3_ab
   chain_relation_catalog_ab chain_relation_ba RelConditions True 2 1 4"
  unfolding relation_input_valid_def
  using chain_after_t3_ab_well_formed chain_relation_catalog_ab_well_formed
  by (auto simp: chain_grammar_def chain_after_t3_ab_def
    chain_after_t2_def chain_after_t1_def chain_seed_def base_snapshot_def
    chain_object_defs base_object_def active_ent_endpoint_def
    version_ids_def references_complete_def evidence_well_formed_def
    evidence_references_complete_def time_well_formed_def
    chain_relation_catalog_ab_def chain_relation_catalog_0_def
    extend_relation_catalog_def)

theorem chain_t3_ba_step:
  "create_relation_step chain_grammar chain_after_t3_ab
   chain_relation_catalog_ab chain_relation_ba RelConditions True 2 1 4
   chain_after_t3 chain_relation_catalog"
  using chain_t3_ba_input_valid
  by (simp add: create_relation_step_def chain_after_t3_def
    chain_relation_catalog_def)

theorem chain_t1_t3_reachability_certificate:
  "register_appearance_step chain_grammar chain_seed chain_entity_a 1
   chain_after_t1 \<and>
   differentiate_potential_step chain_grammar chain_after_t1
   {chain_entity_b} 2 chain_after_t2 \<and>
   create_relation_step chain_grammar chain_after_t2
   chain_relation_catalog_0 chain_relation_ab RelGenerates True 1 2 3
   chain_after_t3_ab chain_relation_catalog_ab \<and>
   create_relation_step chain_grammar chain_after_t3_ab
   chain_relation_catalog_ab chain_relation_ba RelConditions True 2 1 4
   chain_after_t3 chain_relation_catalog \<and>
   state_well_formed chain_grammar chain_after_t3 \<and>
   relation_catalog_well_formed chain_after_t3 chain_relation_catalog \<and>
   source_ref_of chain_relation_ab = Some 1 \<and>
   target_ref_of chain_relation_ab = Some 2 \<and>
   source_ref_of chain_relation_ba = Some 2 \<and>
   target_ref_of chain_relation_ba = Some 1"
  using chain_t1_step chain_t2_step chain_t3_ab_step chain_t3_ba_step
    create_relation_preserves_state_well_formed[OF chain_t3_ba_step]
    create_relation_preserves_catalog[OF chain_t3_ba_step]
  by (auto simp: chain_relation_ab_def chain_relation_ba_def base_object_def)

end

```

## I.8 Konstruktiver T4-Erreichbarkeitszeuge

Vollständiger Quelltext. Beide Wirkungshypothesen verwenden die tatsächlich erzeugten Relationsversionen 3 und 4 und deren Katalogeinträge.

```

theory UZA_0_9_1_Hypothesize_Effect_Witness
  imports UZA_0_9_1_Hypothesize_Effect
          UZA_0_9_1_Create_Relation_Witness
begin

section \<open>A constructive T4 reachability witness\<close>

```



```

text \<open>
  Both effect hypotheses in this theory are produced from the target of the
  concrete T1--T3 chain. Their causes are the two relation versions actually
  created by T3 and recorded in its relation catalog. This establishes that
  the full T4 input predicate is satisfiable without a hand-injected relation.
\<close>

definition chain_effect_vector_a :: effect_vector where
  "chain_effect_vector_a dimension =
    (if dimension = Connection then 3 else 0)"

definition chain_effect_vector_b :: effect_vector where
  "chain_effect_vector_b dimension =
    (if dimension = Stability then 3 else 0)"

definition chain_effect_a :: uza_object where
  "chain_effect_a =
    base_object\<lparr>
      object_id_of := 5,
      version_id_of := 5,
      kind_of := K_EFF,
      lifecycle_of := Hypothesis,
      references_of := {3},
      perspective_of := Some 10,
      context_of := Some 20,
      model_of := Some 30,
      source_ref_of := Some 3,
      effect_origin_of := Some 3,
      basis_refs_of := {3},
      object_status_of := Some Applicable
    \<rparr>"

definition chain_effect_b :: uza_object where
  "chain_effect_b =
    base_object\<lparr>
      object_id_of := 6,
      version_id_of := 6,
      kind_of := K_EFF,
      lifecycle_of := Hypothesis,
      references_of := {4},
      perspective_of := Some 10,
      context_of := Some 20,
      model_of := Some 30,
      source_ref_of := Some 4,
      effect_origin_of := Some 4,
      basis_refs_of := {4},
      object_status_of := Some Applicable
    \<rparr>"

definition chain_effect_payload_a :: effect_hypothesis_payload where
  "chain_effect_payload_a =
    \<lparr> hypothesis_cause_of = 3,
      hypothesis_target_of = None,
      hypothesis_vector_of = Some chain_effect_vector_a,
      hypothesis_uncertainty_of = 1 / 2,
      hypothesis_profile_model_of = 30 \<rparr>"

definition chain_effect_payload_b :: effect_hypothesis_payload where
  "chain_effect_payload_b =
    \<lparr> hypothesis_cause_of = 4,
      hypothesis_target_of = None,
      hypothesis_vector_of = Some chain_effect_vector_b,
      hypothesis_uncertainty_of = 1 / 2,
      hypothesis_profile_model_of = 30 \<rparr>"

definition chain_effect_catalog_0 :: effect_catalog where
  "chain_effect_catalog_0 = Map.empty"

definition chain_after_t4_a :: snapshot where
  "chain_after_t4_a =
    add_effect_hypothesis chain_after_t3 chain_effect_a 5"

definition chain_effect_catalog_a :: effect_catalog where
  "chain_effect_catalog_a =
    extend_effect_catalog chain_effect_catalog_0 chain_effect_a
    chain_effect_payload_a"

definition chain_after_t4 :: snapshot where
  "chain_after_t4 =
    add_effect_hypothesis chain_after_t4_a chain_effect_b 6"

```

```

definition chain_effect_catalog :: effect_catalog where
  "chain_effect_catalog =
    extend_effect_catalog chain_effect_catalog_a chain_effect_b
    chain_effect_payload_b"

lemma chain_effect_vector_a_valid:
  "effect_vector_valid chain_effect_vector_a"
  by (auto simp: effect_vector_valid_def chain_effect_vector_a_def)

lemma chain_effect_vector_b_valid:
  "effect_vector_valid chain_effect_vector_b"
  by (auto simp: effect_vector_valid_def chain_effect_vector_b_def)

lemma chain_effect_catalog_0_well_formed:
  "effect_catalog_well_formed chain_after_t3 chain_effect_catalog_0"
  by (simp add: effect_catalog_well_formed_def chain_effect_catalog_0_def)

lemma chain_t4_a_input_valid:
  "effect_input_valid chain_grammar chain_after_t3 chain_relation_catalog
    chain_effect_catalog_0 chain_effect_a chain_effect_payload_a 5"
  unfolding effect_input_valid_def
  using chain_t1_t3_reachability_certificate
    chain_effect_catalog_0_well_formed chain_effect_vector_a_valid
  by (auto simp: chain_grammar_def chain_after_t3_def
    chain_after_t3_ab_def chain_after_t2_def chain_after_t1_def
    chain_seed_def base_snapshot_def chain_object_defs chain_effect_a_def
    chain_effect_payload_a_def base_object_def active_relation_origin_def
    effect_payload_valid_def effect_target_refs_def
    chain_relation_catalog_def chain_relation_catalog_ab_def
    chain_relation_catalog_0_def extend_relation_catalog_def
    chain_effect_catalog_0_def version_ids_def references_complete_def
    evidence_well_formed_def evidence_references_complete_def
    time_well_formed_def)

theorem chain_t4_a_step:
  "hypothesize_effect_step chain_grammar chain_after_t3
    chain_relation_catalog chain_effect_catalog_0 chain_effect_a
    chain_effect_payload_a 5 chain_after_t4_a chain_effect_catalog_a"
  using chain_t4_a_input_valid
  by (simp add: hypothesize_effect_step_def chain_after_t4_a_def
    chain_effect_catalog_a_def)

lemma chain_after_t4_a_well_formed:
  "state_well_formed chain_grammar chain_after_t4_a"
  using hypothesize_effect_preserves_state_well_formed[OF chain_t4_a_step] .

lemma chain_effect_catalog_a_well_formed:
  "effect_catalog_well_formed chain_after_t4_a chain_effect_catalog_a"
  using hypothesize_effect_preserves_catalog[OF chain_t4_a_step] by blast

lemma chain_relation_catalog_after_t4_a_well_formed:
  "relation_catalog_well_formed chain_after_t4_a chain_relation_catalog"
  using chain_t1_t3_reachability_certificate
  unfolding relation_catalog_well_formed_def chain_after_t4_a_def
  by auto

lemma chain_t4_b_input_valid:
  "effect_input_valid chain_grammar chain_after_t4_a chain_relation_catalog
    chain_effect_catalog_a chain_effect_b chain_effect_payload_b 6"
  unfolding effect_input_valid_def
  using chain_after_t4_a_well_formed chain_effect_catalog_a_well_formed
    chain_relation_catalog_after_t4_a_well_formed
    chain_effect_vector_b_valid
  by (auto simp: chain_grammar_def chain_after_t4_a_def chain_after_t3_def
    chain_after_t3_ab_def chain_after_t2_def chain_after_t1_def
    chain_seed_def base_snapshot_def chain_object_defs chain_effect_a_def
    chain_effect_b_def chain_effect_payload_b_def base_object_def
    active_relation_origin_def effect_payload_valid_def
    effect_target_refs_def chain_relation_catalog_def
    chain_relation_catalog_ab_def chain_relation_catalog_0_def
    extend_relation_catalog_def chain_effect_catalog_a_def
    chain_effect_catalog_0_def extend_effect_catalog_def version_ids_def
    references_complete_def evidence_well_formed_def
    evidence_references_complete_def time_well_formed_def)

theorem chain_t4_b_step:
  "hypothesize_effect_step chain_grammar chain_after_t4_a
    chain_relation_catalog chain_effect_catalog_a chain_effect_b
    chain_effect_payload_b 6 chain_after_t4 chain_effect_catalog"

```

```

using chain_t4_b_input_valid
by (simp add: hypothesize_effect_step_def chain_after_t4_def
  chain_effect_catalog_def)

lemma chain_relation_catalog_after_t4_well_formed:
  "relation_catalog_well_formed chain_after_t4 chain_relation_catalog"
  using chain_relation_catalog_after_t4_a_well_formed
  unfolding relation_catalog_well_formed_def chain_after_t4_def
  by auto

lemma add_effect_active_relation_origin_mono:
  assumes active: "active_relation_origin s v"
  shows "active_relation_origin (add_effect_hypothesis s new_eff next_id) v"
  using active
  by (auto simp: active_relation_origin_def)

lemma chain_relation_3_active_after_t4:
  "active_relation_origin chain_after_t4 3"
proof -
  from chain_t4_a_input_valid have active_t3:
    "active_relation_origin chain_after_t3 3"
  by (simp add: effect_input_valid_def chain_effect_payload_a_def)
  have active_t4_a: "active_relation_origin chain_after_t4_a 3"
  unfolding chain_after_t4_a_def
  using add_effect_active_relation_origin_mono[OF active_t3] .
  show ?thesis
  unfolding chain_after_t4_def
  using add_effect_active_relation_origin_mono[OF active_t4_a] .
qed

lemma chain_relation_4_active_after_t4:
  "active_relation_origin chain_after_t4 4"
proof -
  from chain_t4_b_input_valid have active_t4_a:
    "active_relation_origin chain_after_t4_a 4"
  by (simp add: effect_input_valid_def chain_effect_payload_b_def)
  show ?thesis
  unfolding chain_after_t4_def
  using add_effect_active_relation_origin_mono[OF active_t4_a] .
qed

lemma chain_relation_catalog_3_present:
  "chain_relation_catalog 3 \<noteq> None"
  by (simp add: chain_relation_catalog_def chain_relation_catalog_ab_def
    chain_relation_catalog_0_def extend_relation_catalog_def
    chain_relation_ab_def chain_relation_ba_def base_object_def)

lemma chain_relation_catalog_4_present:
  "chain_relation_catalog 4 \<noteq> None"
  by (simp add: chain_relation_catalog_def chain_relation_catalog_ab_def
    chain_relation_catalog_0_def extend_relation_catalog_def
    chain_relation_ab_def chain_relation_ba_def base_object_def)

theorem chain_t4_reachability_certificate:
  "effect_input_valid chain_grammar chain_after_t3 chain_relation_catalog
    chain_effect_catalog_0 chain_effect_a chain_effect_payload_a 5 \<and>
    hypothesize_effect_step chain_grammar chain_after_t3
    chain_relation_catalog chain_effect_catalog_0 chain_effect_a
    chain_effect_payload_a 5 chain_after_t4_a chain_effect_catalog_a \<and>
    effect_input_valid chain_grammar chain_after_t4_a chain_relation_catalog
    chain_effect_catalog_a chain_effect_b chain_effect_payload_b 6 \<and>
    hypothesize_effect_step chain_grammar chain_after_t4_a
    chain_relation_catalog chain_effect_catalog_a chain_effect_b
    chain_effect_payload_b 6 chain_after_t4 chain_effect_catalog \<and>
    state_well_formed chain_grammar chain_after_t4 \<and>
    relation_catalog_well_formed chain_after_t4 chain_relation_catalog \<and>
    effect_catalog_well_formed chain_after_t4 chain_effect_catalog \<and>
    active_relation_origin chain_after_t4 3 \<and>
    active_relation_origin chain_after_t4 4 \<and>
    chain_relation_catalog 3 \<noteq> None \<and>
    chain_relation_catalog 4 \<noteq> None"
  using chain_t4_a_input_valid chain_t4_a_step chain_t4_b_input_valid
    chain_t4_b_step
    hypothesize_effect_preserves_state_well_formed[OF chain_t4_b_step]
    hypothesize_effect_preserves_catalog[OF chain_t4_b_step]
    chain_relation_catalog_after_t4_well_formed
    chain_relation_3_active_after_t4 chain_relation_4_active_after_t4
    chain_relation_catalog_3_present chain_relation_catalog_4_present
  by blast

```

end

## I.9 Strikter kausal erreichbarer T5-Zeuge

Vollständiger Ersatz für den historischen isolierten Zeugen D.9. Zwei T4-Effekte werden nacheinander versioniert; Objekt-, Payload- und Ledger-Historie werden fortgeschrieben.

```
theory UZA_0_9_1_Integrate_Evidence
  imports UZA_0_9_1_Hypothesize_Effect
begin

section \<open>T5 IntegrateEvidence: versioned evidence integration\<close>

text \<open>
  T5 does not mutate an active effect version. It replaces one active
  K_EFF hypothesis by one fresh version of the same logical object, archives
  the complete predecessor object and its effect payload, accumulates typed
  evidence, and derives the successor lifecycle from the evidence
  disposition. The construction therefore makes copy-on-write, provenance,
  uncertainty movement, and the prohibition of self-support explicit.

  The operation is deliberately restricted to Hypothesis and Supported.
  Evidence applied to an already Validated effect belongs to T13
  ReviseObject. This boundary prevents T5 from silently serving as a general
  revision operator.
\<close>

datatype evidence_disposition =
  EvidenceSupports | EvidenceDifferentiates | EvidenceContradicts

record evidence_assessment =
  evidence_subject_of :: object_id
  evidence_prior_version_of :: version_id
  evidence_refs_of :: "version_id set"
  evidence_disposition_of :: evidence_disposition
  evidence_strength_of :: real
  evidence_post_uncertainty_of :: real
  evidence_post_target_of :: "version_id option"
  evidence_post_vector_of :: "effect_vector option"
  evidence_assessment_model_of :: model_id
  validation_uncertainty_max_of :: real
  validation_evidence_min_of :: nat

type_synonym object_archive = "version_id \<Rightharpoonup> uza_object"
type_synonym effect_payload_archive =
  "version_id \<Rightharpoonup> effect_hypothesis_payload"
type_synonym evidence_ledger =
  "version_id \<Rightharpoonup> evidence_assessment"

definition active_history_disjoint :: "snapshot \<Rrightarrow> bool" where
  "active_history_disjoint s \<longlefttrightarrow>
    version_ids s \<inter> historical_versions_of s = {}"

definition object_archive_well_formed ::
  "snapshot \<Rrightarrow> object_archive \<Rrightarrow> bool" where
  "object_archive_well_formed s archive \<longlefttrightarrow>
    (\<forall>v obj. archive v = Some obj \<longrightarrow>
      version_id_of obj = v \<and>
      v \<in> historical_versions_of s \<and>
      v \<notin> version_ids s)"

definition resolved_version_object ::
  "snapshot \<Rrightarrow> object_archive \<Rrightarrow> version_id \<Rrightarrow>
    uza_object \<Rrightarrow> bool" where
  "resolved_version_object s archive v obj \<longlefttrightarrow>
    (obj \<in> objects_of s \<and> version_id_of obj = v) \<or>
    archive v = Some obj"

definition effect_payload_archive_well_formed ::
  "object_archive \<Rrightarrow> effect_payload_archive \<Rrightarrow> bool" where
  "effect_payload_archive_well_formed archive payload_archive \<longlefttrightarrow>
    (\<forall>v payload. payload_archive v = Some payload \<longrightarrow>
      (\<exists>eff. archive v = Some eff \<and>
        effect_payload_valid eff payload))"
```

```

definition evidence_entry_well_formed ::
  "snapshot \<Rightarrow> object_archive \<Rightarrow> version_id \<Rightarrow>
  evidence_assessment \<Rightarrow> bool" where
  "evidence_entry_well_formed s archive v assessment \<longlefttrightarrow>
  (\<exists>eff. resolved_version_object s archive v eff \<and>
    kind_of eff = K_EFF \<and>
    evidence_subject_of assessment = object_id_of eff \<and>
    previous_version_of eff = Some (evidence_prior_version_of assessment) \<and>
    evidence_refs_of assessment \<noteq> {} \<and>
    evidence_refs_of assessment \<subteq> evidence_of eff \<and>
    evidence_refs_of assessment
      \<subteq> version_ids s \<union> historical_versions_of s \<and>
    v \<notin> evidence_refs_of assessment \<and>
    evidence_prior_version_of assessment
      \<notin> evidence_refs_of assessment \<and>
    0 \<le> evidence_strength_of assessment \<and>
    evidence_strength_of assessment \<le> 1 \<and>
    0 \<le> evidence_post_uncertainty_of assessment \<and>
    evidence_post_uncertainty_of assessment \<le> 1 \<and>
    0 \<le> validation_uncertainty_max_of assessment \<and>
    validation_uncertainty_max_of assessment \<le> 1 \<and>
    0 < validation_evidence_min_of assessment \<and>
    evidence_assessment_model_of assessment \<in> models_of s)"

definition evidence_ledger_well_formed ::
  "snapshot \<Rightarrow> object_archive \<Rightarrow> evidence_ledger \<Rightarrow> bool" where
  "evidence_ledger_well_formed s archive ledger \<longlefttrightarrow>
  (\<forall>v assessment. ledger v = Some assessment \<longrightarrow>
    evidence_entry_well_formed s archive v assessment)"

definition active_evidence_source ::
  "snapshot \<Rightarrow> object_id \<Rightarrow> version_id \<Rightarrow> bool" where
  "active_evidence_source s subject v \<longlefttrightarrow>
  (\<exists>ev\<in>objects_of s.
    version_id_of ev = v \<and>
    object_id_of ev \<noteq> subject \<and>
    lifecycle_of ev \<notin> {Rejected, Archived})"

definition admissible_post_target ::
  "snapshot \<Rightarrow> object_id \<Rightarrow> version_id option \<Rightarrow> bool" where
  "admissible_post_target s subject target \<longlefttrightarrow>
  (case target of
    None \<Rightarrow> True
  | Some v \<Rightarrow>
    (\<exists>obj\<in>objects_of s.
      version_id_of obj = v \<and>
      object_id_of obj \<noteq> subject \<and>
      lifecycle_of obj \<notin> {Revised, Rejected, Archived})))"

definition assessment_bounds :: "evidence_assessment \<Rightarrow> bool" where
  "assessment_bounds assessment \<longlefttrightarrow>
  0 \<le> evidence_strength_of assessment \<and>
  evidence_strength_of assessment \<le> 1 \<and>
  0 \<le> evidence_post_uncertainty_of assessment \<and>
  evidence_post_uncertainty_of assessment \<le> 1 \<and>
  0 \<le> validation_uncertainty_max_of assessment \<and>
  validation_uncertainty_max_of assessment \<le> 1 \<and>
  0 < validation_evidence_min_of assessment"

definition assessment_change_valid ::
  "effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow> bool" where
  "assessment_change_valid prior assessment \<longlefttrightarrow>
  abs (evidence_post_uncertainty_of assessment -
    hypothesis_uncertainty_of prior)
    \<le> evidence_strength_of assessment \<and>
  (case evidence_disposition_of assessment of
    EvidenceSupports \<Rightarrow>
    evidence_post_target_of assessment = hypothesis_target_of prior \<and>
    evidence_post_vector_of assessment = hypothesis_vector_of prior \<and>
    evidence_post_uncertainty_of assessment
      \<le> hypothesis_uncertainty_of prior
  | EvidenceDifferentiates \<Rightarrow>
    evidence_post_target_of assessment \<noteq> hypothesis_target_of prior \<or>
    evidence_post_vector_of assessment \<noteq> hypothesis_vector_of prior
  | EvidenceContradicts \<Rightarrow>
    evidence_post_target_of assessment = hypothesis_target_of prior \<and>
    evidence_post_vector_of assessment = hypothesis_vector_of prior \<and>
    hypothesis_uncertainty_of prior
      \<le> evidence_post_uncertainty_of assessment)"

```

```

definition validation_ready ::
  "uza_object \<Rightarrow> evidence_assessment \<Rightarrow> bool" where
  "validation_ready old_eff assessment \<longleftarrowrightarrow>
    evidence_post_uncertainty_of assessment
    \<le> validation_uncertainty_max_of assessment \<and>
    card (evidence_of old_eff \<union> evidence_refs_of assessment)
    \<ge> validation_evidence_min_of assessment"

definition integrated_lifecycle ::
  "uza_object \<Rightarrow> evidence_assessment \<Rightarrow> lifecycle_status" where
  "integrated_lifecycle old_eff assessment =
    (case evidence_disposition_of assessment of
      EvidenceSupports \<Rightarrow>
        (if lifecycle_of old_eff = Hypothesis then Supported
         else if validation_ready old_eff assessment then Validated
         else Supported)
    | EvidenceDifferentiates \<Rightarrow> lifecycle_of old_eff
    | EvidenceContradicts \<Rightarrow> Rejected)"

definition integrated_effect_payload ::
  "effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow>
    effect_hypothesis_payload" where
  "integrated_effect_payload prior assessment =
    prior\<lparr>
      hypothesis_target_of := evidence_post_target_of assessment,
      hypothesis_vector_of := evidence_post_vector_of assessment,
      hypothesis_uncertainty_of := evidence_post_uncertainty_of assessment
    \<rparr>"

definition option_ref_set :: "version_id option \<Rightarrow> version_id set" where
  "option_ref_set target = (case target of None \<Rightarrow> {} | Some v \<Rightarrow> {v})"

definition integrated_effect_object ::
  "uza_object \<Rightarrow> evidence_assessment \<Rightarrow> version_id \<Rightarrow>
    time_point \<Rightarrow> uza_object" where
  "integrated_effect_object old_eff assessment new_version integration_time =
    old_eff\<lparr>
      version_id_of := new_version,
      lifecycle_of := integrated_lifecycle old_eff assessment,
      previous_version_of := Some (version_id_of old_eff),
      references_of := references_of old_eff
        \<union> evidence_refs_of assessment
        \<union> {version_id_of old_eff}
      \<union> option_ref_set (evidence_post_target_of assessment),
      evidence_of := evidence_of old_eff \<union> evidence_refs_of assessment,
      valid_from_of := integration_time,
      valid_to_of := None,
      target_ref_of := evidence_post_target_of assessment,
      basis_refs_of := basis_refs_of old_eff
        \<union> evidence_refs_of assessment
        \<union> {version_id_of old_eff}
        \<union> option_ref_set (evidence_post_target_of assessment),
      object_status_of :=
        (if evidence_disposition_of assessment = EvidenceContradicts
         then Some ModelFalsified else Some Applicable)
    \<rparr>"

definition replace_effect_version ::
  "snapshot \<Rightarrow> uza_object \<Rightarrow> uza_object \<Rightarrow> nat
    \<Rightarrow> snapshot" where
  "replace_effect_version s old_eff next_snapshot_id =
    s\<lparr>
      snapshot_id_of := next_snapshot_id,
      objects_of := insert new_eff (objects_of s - {old_eff}),
      historical_versions_of :=
        insert (version_id_of old_eff) (historical_versions_of s)
    \<rparr>"

definition replace_active_effect_payload ::
  "effect_catalog \<Rightarrow> version_id \<Rightarrow> version_id \<Rightarrow>
    effect_hypothesis_payload \<Rightarrow> effect_catalog" where
  "replace_active_effect_payload catalog old_version new_version payload =
    (catalog(old_version := None))(new_version \<mapsto> payload)"

definition extend_object_archive ::
  "object_archive \<Rightarrow> uza_object \<Rightarrow> object_archive" where
  "extend_object_archive archive old_eff =
    archive(version_id_of old_eff \<mapsto> old_eff)"

definition extend_effect_payload_archive ::

```

```

"effect_payload_archive \<Rightarrow> uza_object \<Rightarrow>
  effect_hypothesis_payload \<Rightarrow> effect_payload_archive" where
"extend_effect_payload_archive archive old_eff prior =
  archive(version_id_of old_eff \<mapsto> prior)"

definition extend_evidence_ledger ::
  "evidence_ledger \<Rightarrow> version_id \<Rightarrow> evidence_assessment
  \<Rightarrow> evidence_ledger" where
  "extend_evidence_ledger ledger new_version assessment =
    ledger(new_version \<mapsto> assessment)"

definition integrate_evidence_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> effect_catalog \<Rightarrow>
    object_archive \<Rightarrow> effect_payload_archive \<Rightarrow>
    evidence_ledger \<Rightarrow> uza_object \<Rightarrow>
    effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow>
    version_id \<Rightarrow> time_point \<Rightarrow> nat \<Rightarrow> bool" where
  "integrate_evidence_input_valid G s catalog object_archive payload_archive
    ledger old_eff prior assessment new_version integration_time
    next_snapshot_id \<longleftarrowrightarrow>
    state_well_formed G s \<and>
    active_history_disjoint s \<and>
    effect_catalog_well_formed s catalog \<and>
    object_archive_well_formed s object_archive \<and>
    effect_payload_archive_well_formed object_archive payload_archive \<and>
    evidence_ledger_well_formed s object_archive ledger \<and>
    IntegrateEvidence \<in> allowed_operations_of G \<and>
    old_eff \<in> objects_of s \<and>
    kind_of old_eff = K_EFF \<and>
    lifecycle_of old_eff \<in> {Hypothesis, Supported} \<and>
    catalog (version_id_of old_eff) = Some prior \<and>
    effect_payload_valid old_eff prior \<and>
    present_ref s (effect_origin_of old_eff) \<and>
    (\<exists>origin. effect_origin_of old_eff = Some origin \<and>
      active_relation_origin s origin) \<and>
    object_archive (version_id_of old_eff) = None \<and>
    payload_archive (version_id_of old_eff) = None \<and>
    new_version \<notin> version_ids s \<union> historical_versions_of s \<and>
    catalog new_version = None \<and>
    object_archive new_version = None \<and>
    payload_archive new_version = None \<and>
    ledger new_version = None \<and>
    evidence_subject_of assessment = object_id_of old_eff \<and>
    evidence_prior_version_of assessment = version_id_of old_eff \<and>
    evidence_refs_of assessment \<noteq> {} \<and>
    (\<forall>v\<in>evidence_refs_of assessment.
      active_evidence_source s (object_id_of old_eff) v) \<and>
    evidence_assessment_model_of assessment \<in> models_of s \<and>
    assessment_bounds assessment \<and>
    assessment_change_valid prior assessment \<and>
    admissible_post_target s (object_id_of old_eff)
      (evidence_post_target_of assessment) \<and>
    (case evidence_post_vector_of assessment of
      None \<Rightarrow> True
    | Some vector \<Rightarrow> effect_vector_valid vector) \<and>
    valid_from_of old_eff \<le> integration_time \<and>
    snapshot_id_of s < next_snapshot_id"

definition integrate_evidence_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> effect_catalog \<Rightarrow>
    object_archive \<Rightarrow> effect_payload_archive \<Rightarrow>
    evidence_ledger \<Rightarrow> uza_object \<Rightarrow>
    effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow>
    version_id \<Rightarrow> time_point \<Rightarrow> nat \<Rightarrow>
    snapshot \<Rightarrow> effect_catalog \<Rightarrow> object_archive \<Rightarrow>
    effect_payload_archive \<Rightarrow> evidence_ledger \<Rightarrow> bool" where
  "integrate_evidence_step G s catalog object_archive payload_archive ledger
    old_eff prior assessment new_version integration_time next_snapshot_id
    s' catalog' object_archive' payload_archive' ledger' \<longleftarrowrightarrow>
    integrate_evidence_input_valid G s catalog object_archive payload_archive
    ledger old_eff prior assessment new_version integration_time
    next_snapshot_id \<and>
    (let new_eff = integrated_effect_object old_eff assessment new_version
      integration_time;
      new_payload = integrated_effect_payload prior assessment
    in s' = replace_effect_version s old_eff new_eff next_snapshot_id \<and>
      catalog' = replace_active_effect_payload catalog
        (version_id_of old_eff) new_version new_payload \<and>
      object_archive' = extend_object_archive object_archive old_eff \<and>
      payload_archive' = extend_effect_payload_archive payload_archive

```

```

    old_eff prior \<and>
    ledger' = extend_evidence_ledger ledger new_version assessment)"

subsection \<open>Normative catalogued-origin gate\<close>

text \<open>
  The base predicate above now rejects untyped or inactive effect origins. The
  public T5 gate additionally requires the origin relation to be present in a
  well-formed relation catalog. Keeping this condition in a separate strict
  wrapper preserves the already checked constructor and preservation proofs,
  while making it impossible for a conforming caller to substitute a merely
  coincidental version identifier for a T3-created relation.
\<close>

definition catalogued_active_relation_origin ::
  "snapshot \<Rightarrow> relation_catalog \<Rightarrow> version_id option \<Rightarrow> bool" where
  "catalogued_active_relation_origin s relation_catalog origin \<longlefttrightarrow>
    (\<exists>v. origin = Some v \<and> active_relation_origin s v \<and>
      relation_catalog v \<noteq> None)"

definition integrate_evidence_strict_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> relation_catalog \<Rightarrow>
    effect_catalog \<Rightarrow> object_archive \<Rightarrow>
    effect_payload_archive \<Rightarrow> evidence_ledger \<Rightarrow> uza_object \<Rightarrow>
    effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow>
    version_id \<Rightarrow> time_point \<Rightarrow> nat \<Rightarrow> bool" where
  "integrate_evidence_strict_input_valid G s relation_catalog catalog
    object_archive payload_archive ledger old_eff prior assessment
    new_version integration_time next_snapshot_id \<longlefttrightarrow>
    relation_catalog_well_formed s relation_catalog \<and>
    catalogued_active_relation_origin s relation_catalog
    (effect_origin_of old_eff) \<and>
    integrate_evidence_input_valid G s catalog object_archive payload_archive
    ledger old_eff prior assessment new_version integration_time
    next_snapshot_id"

definition integrate_evidence_strict_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> relation_catalog \<Rightarrow>
    effect_catalog \<Rightarrow> object_archive \<Rightarrow>
    effect_payload_archive \<Rightarrow> evidence_ledger \<Rightarrow> uza_object \<Rightarrow>
    effect_hypothesis_payload \<Rightarrow> evidence_assessment \<Rightarrow>
    version_id \<Rightarrow> time_point \<Rightarrow> nat \<Rightarrow> snapshot \<Rightarrow>
    effect_catalog \<Rightarrow> object_archive \<Rightarrow>
    effect_payload_archive \<Rightarrow> evidence_ledger \<Rightarrow> bool" where
  "integrate_evidence_strict_step G s relation_catalog catalog
    object_archive payload_archive ledger old_eff prior assessment
    new_version integration_time next_snapshot_id s' catalog'
    object_archive' payload_archive' ledger' \<longlefttrightarrow>
    integrate_evidence_strict_input_valid G s relation_catalog catalog
    object_archive payload_archive ledger old_eff prior assessment
    new_version integration_time next_snapshot_id \<and>
    integrate_evidence_step G s catalog object_archive payload_archive ledger
    old_eff prior assessment new_version integration_time next_snapshot_id
    s' catalog' object_archive' payload_archive' ledger'"

lemma integrate_evidence_strict_input_implies_base:
  assumes strict:
    "integrate_evidence_strict_input_valid G s relation_catalog catalog
      object_archive payload_archive ledger old_eff prior assessment
      new_version integration_time next_snapshot_id"
  shows
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
      ledger old_eff prior assessment new_version integration_time
      next_snapshot_id"
  using strict
  by (simp add: integrate_evidence_strict_input_valid_def)

theorem integrate_evidence_strict_refines_base:
  assumes strict:
    "integrate_evidence_strict_step G s relation_catalog catalog
      object_archive payload_archive ledger old_eff prior assessment
      new_version integration_time next_snapshot_id s' catalog'
      object_archive' payload_archive' ledger'"
  shows
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  using strict
  by (simp add: integrate_evidence_strict_step_def)

```



```

theorem integrate_evidence_strict_preserves_relation_catalog:
  assumes strict:
    "integrate_evidence_strict_step G s relation_catalog catalog
     object_archive payload_archive ledger old_eff prior assessment
     new_version integration_time next_snapshot_id s' catalog'
     object_archive' payload_archive' ledger'"
  shows "relation_catalog_well_formed s' relation_catalog"
proof -
  from strict have source_catalog:
    "relation_catalog_well_formed s relation_catalog"
  and old_kind: "kind_of old_eff = K_EFF"
  and target:
    "s' = replace_effect_version s old_eff
     (integrated_effect_object old_eff assessment new_version
      integration_time) next_snapshot_id"
  unfolding integrate_evidence_strict_step_def
  integrate_evidence_strict_input_valid_def
  integrate_evidence_step_def integrate_evidence_input_valid_def Let_def
  by blast+
show ?thesis
  unfolding relation_catalog_well_formed_def
proof (intro allI impI)
  fix v metadata
  assume catalog_entry: "relation_catalog v = Some metadata"
  from source_catalog catalog_entry obtain r where
    r_mem: "r \<in> objects_of s"
  and r_version: "version_id_of r = v"
  and r_kind: "kind_of r = K_REL"
  unfolding relation_catalog_well_formed_def
  by blast
  have r_ne_old: "r \<noteq> old_eff"
  using r_kind old_kind by auto
  have r_target: "r \<in> objects_of s'"
  unfolding target replace_effect_version_def
  using r_mem r_ne_old by simp
  show "\<exists>r\<in>objects_of s'. version_id_of r = v \<and> kind_of r = K_REL"
  using r_target r_version r_kind by blast
qed
qed

```

subsection \<open>Constructor equations and exact delta\<close>

```

lemma replace_effect_simps [simp]:
  "snapshot_id_of (replace_effect_version s old_eff new_eff next_id) = next_id"
  "objects_of (replace_effect_version s old_eff new_eff next_id) =
   insert new_eff (objects_of s - {old_eff})"
  "historical_versions_of
   (replace_effect_version s old_eff new_eff next_id) =
   insert (version_id_of old_eff) (historical_versions_of s)"
  "perspectives_of (replace_effect_version s old_eff new_eff next_id) =
   perspectives_of s"
  "contexts_of (replace_effect_version s old_eff new_eff next_id) = contexts_of s"
  "models_of (replace_effect_version s old_eff new_eff next_id) = models_of s"
  "authorities_of (replace_effect_version s old_eff new_eff next_id) =
   authorities_of s"
  "contains_edges_of (replace_effect_version s old_eff new_eff next_id) =
   contains_edges_of s"
  "projections_of (replace_effect_version s old_eff new_eff next_id) =
   projections_of s"
  "couplings_of (replace_effect_version s old_eff new_eff next_id) =
   couplings_of s"
  "transmissions_of (replace_effect_version s old_eff new_eff next_id) =
   transmissions_of s"
  "boundaries_of (replace_effect_version s old_eff new_eff next_id) =
   boundaries_of s"
  by (simp_all add: replace_effect_version_def)

lemma integrated_effect_simps [simp]:
  "object_id_of (integrated_effect_object old assessment new_v at_time) =
   object_id_of old"
  "version_id_of (integrated_effect_object old assessment new_v at_time) = new_v"
  "kind_of (integrated_effect_object old assessment new_v at_time) = kind_of old"
  "lifecycle_of (integrated_effect_object old assessment new_v at_time) =
   integrated_lifecycle old assessment"
  "previous_version_of
   (integrated_effect_object old assessment new_v at_time) =
   Some (version_id_of old)"
  "references_of (integrated_effect_object old assessment new_v at_time) =
   references_of old \<union> evidence_refs_of assessment \<union>
   {version_id_of old} \<union>"

```

```

    option_ref_set (evidence_post_target_of assessment)"
"evidence_of (integrated_effect_object old assessment new_v at_time) =
  evidence_of old \<union> evidence_refs_of assessment"
"perspective_of (integrated_effect_object old assessment new_v at_time) =
  perspective_of old"
"context_of (integrated_effect_object old assessment new_v at_time) =
  context_of old"
"model_of (integrated_effect_object old assessment new_v at_time) = model_of old"
"source_ref_of (integrated_effect_object old assessment new_v at_time) =
  source_ref_of old"
"target_ref_of (integrated_effect_object old assessment new_v at_time) =
  evidence_post_target_of assessment"
"effect_origin_of (integrated_effect_object old assessment new_v at_time) =
  effect_origin_of old"
"valid_from_of (integrated_effect_object old assessment new_v at_time) = at_time"
"valid_to_of (integrated_effect_object old assessment new_v at_time) = None"
"basis_refs_of (integrated_effect_object old assessment new_v at_time) =
  basis_refs_of old \<union> evidence_refs_of assessment \<union>
  {version_id_of old} \<union>
  option_ref_set (evidence_post_target_of assessment)"
by (simp_all add: integrated_effect_object_def)

lemma integrated_payload_simps [simp]:
"hypothesis_cause_of (integrated_effect_payload prior assessment) =
  hypothesis_cause_of prior"
"hypothesis_target_of (integrated_effect_payload prior assessment) =
  evidence_post_target_of assessment"
"hypothesis_vector_of (integrated_effect_payload prior assessment) =
  evidence_post_vector_of assessment"
"hypothesis_uncertainty_of (integrated_effect_payload prior assessment) =
  evidence_post_uncertainty_of assessment"
"hypothesis_profile_model_of (integrated_effect_payload prior assessment) =
  hypothesis_profile_model_of prior"
by (simp_all add: integrated_effect_payload_def)

lemma integrate_evidence_exact_delta:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  defines "new_eff \<equiv> integrated_effect_object old_eff assessment new_version
    integration_time"
    and "new_payload \<equiv> integrated_effect_payload prior assessment"
  shows
    "objects_of s' = insert new_eff (objects_of s - {old_eff}) \<and>
      historical_versions_of s' =
        insert (version_id_of old_eff) (historical_versions_of s) \<and>
        old_eff \<notin> objects_of s' \<and>
        new_eff \<in> objects_of s' \<and>
        snapshot_id_of s' = next_snapshot_id \<and>
        catalog' (version_id_of old_eff) = None \<and>
        catalog' new_version = Some new_payload \<and>
        object_archive' (version_id_of old_eff) = Some old_eff \<and>
        payload_archive' (version_id_of old_eff) = Some prior \<and>
        ledger' new_version = Some assessment"
proof -
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
      ledger old_eff prior assessment new_version integration_time
      next_snapshot_id"
  and target:
    "s' = replace_effect_version s old_eff new_eff next_snapshot_id"
  and catalog_target:
    "catalog' = replace_active_effect_payload catalog
      (version_id_of old_eff) new_version new_payload"
  and object_archive_target:
    "object_archive' = extend_object_archive object_archive old_eff"
  and payload_archive_target:
    "payload_archive' = extend_effect_payload_archive payload_archive
      old_eff prior"
  and ledger_target:
    "ledger' = extend_evidence_ledger ledger new_version assessment"
  unfolding integrate_evidence_step_def new_eff_def new_payload_def Let_def
  by blast+
  from valid have old_mem: "old_eff \<in> objects_of s"
  and fresh: "new_version \<notin> version_ids s \<union> historical_versions_of s"
  and catalog_old: "catalog (version_id_of old_eff) = Some prior"
  and catalog_new: "catalog new_version = None"
  unfolding integrate_evidence_input_valid_def by blast+
  have different_version: "version_id_of old_eff \<noteq> new_version"

```

```

    using old_mem fresh unfolding version_ids_def by blast
  have different_object: "old_eff \<noteq> new_eff"
proof
  assume same_object: "old_eff = new_eff"
  have "version_id_of old_eff = version_id_of new_eff"
    using same_object by simp
  then have "version_id_of old_eff = new_version"
    unfolding new_eff_def by simp
  then show False using different_version by contradiction
qed
show ?thesis
  unfolding target_catalog_target object_archive_target
    payload_archive_target ledger_target
    replace_active_effect_payload_def extend_object_archive_def
    extend_effect_payload_archive_def extend_evidence_ledger_def
  using different_version different_object catalog_old catalog_new
  by simp
qed

subsection \<open>Lifecycle, uncertainty, and provenance properties\<close>

theorem integrate_evidence_lifecycle_classification:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  shows
    "(evidence_disposition_of assessment = EvidenceContradicts
      \<longrightarrow> integrated_lifecycle old_eff assessment = Rejected) \<and>
      (evidence_disposition_of assessment = EvidenceDifferentiates
        \<longrightarrow> integrated_lifecycle old_eff assessment = lifecycle_of old_eff) \<and>
      (evidence_disposition_of assessment = EvidenceSupports \<and>
        lifecycle_of old_eff = Hypothesis
        \<longrightarrow> integrated_lifecycle old_eff assessment = Supported) \<and>
      (integrated_lifecycle old_eff assessment = Validated
        \<longrightarrow> lifecycle_of old_eff = Supported \<and>
        evidence_disposition_of assessment = EvidenceSupports \<and>
        validation_ready old_eff assessment)"
proof -
  from concrete have source_lifecycle:
    "lifecycle_of old_eff \<in> {Hypothesis, Supported}"
  by (auto simp: integrate_evidence_step_def
    integrate_evidence_input_valid_def)
  show ?thesis
    using source_lifecycle
  by (cases "evidence_disposition_of assessment";
    auto simp: integrated_lifecycle_def split: if_splits)
qed

theorem integrate_evidence_uncertainty_controlled:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  shows
    "0 \<le> evidence_post_uncertainty_of assessment \<and>
      evidence_post_uncertainty_of assessment \<le> 1 \<and>
      abs (evidence_post_uncertainty_of assessment -
        hypothesis_uncertainty_of prior)
        \<le> evidence_strength_of assessment \<and>
      (evidence_disposition_of assessment = EvidenceSupports
        \<longrightarrow> evidence_post_uncertainty_of assessment
          \<le> hypothesis_uncertainty_of prior) \<and>
      (evidence_disposition_of assessment = EvidenceContradicts
        \<longrightarrow> hypothesis_uncertainty_of prior
          \<le> evidence_post_uncertainty_of assessment)"
  using concrete
  unfolding integrate_evidence_step_def integrate_evidence_input_valid_def
    assessment_bounds_def assessment_change_valid_def
  by (cases "evidence_disposition_of assessment"; auto)

theorem integrate_evidence_lineage_and_no_self_support:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  defines "new_eff \<equiv> integrated_effect_object old_eff assessment new_version
    integration_time"
  shows
    "previous_version_of new_eff = Some (version_id_of old_eff) \<and>
```

```

evidence_refs_of assessment \<noteq> {} \<and>
evidence_refs_of assessment \<subteq> evidence_of new_eff \<and>
new_version \<notin> evidence_refs_of assessment \<and>
version_id_of old_eff \<notin> evidence_refs_of assessment \<and>
(\<forall>v\<in>evidence_refs_of assessment.
  \<exists>ev\<in>objects_of s.
    version_id_of ev = v \<and>
    object_id_of ev \<noteq> object_id_of old_eff \<and>
    lifecycle_of ev \<notin> {Rejected, Archived})"
proof -
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
      ledger old_eff prior assessment new_version integration_time
      next_snapshot_id"
  by (simp add: integrate_evidence_step_def)
  from valid have refs_nonempty: "evidence_refs_of assessment \<noteq> {}"
  and active:
    "\<forall>v\<in>evidence_refs_of assessment.
      active_evidence_source s (object_id_of old_eff) v"
  and fresh:
    "new_version \<notin> version_ids s \<union> historical_versions_of s"
  and old_mem: "old_eff \<in> objects_of s"
  and source_wf: "state_well_formed G s"
  unfolding integrate_evidence_input_valid_def by blast+
  have source_unique: "inj_on version_id_of (objects_of s)"
  using source_wf unfolding state_well_formed_def unique_versions_def by blast
  have prior_not: "version_id_of old_eff \<notin> evidence_refs_of assessment"
  proof
    assume prior_in: "version_id_of old_eff \<in> evidence_refs_of assessment"
    from active prior_in obtain ev where ev_mem: "ev \<in> objects_of s"
    and same_version: "version_id_of ev = version_id_of old_eff"
    and different_object: "object_id_of ev \<noteq> object_id_of old_eff"
    unfolding active_evidence_source_def by blast
    have "ev = old_eff"
    using inj_onD[OF source_unique same_version ev_mem old_mem] .
    then show False using different_object by simp
  qed
  have new_not: "new_version \<notin> evidence_refs_of assessment"
  proof
    assume "new_version \<in> evidence_refs_of assessment"
    then obtain ev where "ev \<in> objects_of s" "version_id_of ev = new_version"
    using active unfolding active_evidence_source_def by blast
    then have "new_version \<in> version_ids s"
    unfolding version_ids_def by blast
    then show False using fresh by blast
  qed
  show ?thesis
  unfolding new_eff_def
  using refs_nonempty active prior_not new_not
  by (auto simp: active_evidence_source_def)
qed

subsection \<open>Reference-universe monotonicity\<close>

lemma replace_effect_version_universe_mono:
  assumes old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  shows
    "version_ids s \<union> historical_versions_of s \<subteq>
      version_ids (replace_effect_version s old_eff new_eff next_id)
      \<union> historical_versions_of
      (replace_effect_version s old_eff new_eff next_id)"
  proof
    fix v
    assume v_mem: "v \<in> version_ids s \<union> historical_versions_of s"
    show "v \<in> version_ids (replace_effect_version s old_eff new_eff next_id)
      \<union> historical_versions_of
      (replace_effect_version s old_eff new_eff next_id)"
    proof (cases "v = version_id_of old_eff")
      case is_old: True
      then show ?thesis by simp
    next
      case not_old: False
      show ?thesis
      proof (cases "v \<in> historical_versions_of s")
        case in_history: True
        then show ?thesis by simp
      next
        case not_in_history: False

```

```

with v_mem obtain obj where obj_mem: "obj \<in> objects_of s"
  and obj_v: "version_id_of obj = v"
  unfolding version_ids_def by blast
have "obj \<noteq> old_eff"
proof
  assume "obj = old_eff"
  then have "v = version_id_of old_eff" using obj_v by simp
  then show False using not_old by contradiction
qed
then have "obj \<in> objects_of s - {old_eff}" using obj_mem by blast
then have "v \<in> version_ids
  (replace_effect_version s old_eff new_eff next_id)"
  unfolding version_ids_def using obj_v by auto
then show ?thesis by blast
qed
qed
qed

lemma replace_effect_present_ref_mono:
  assumes old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  and present: "present_ref s ref"
  shows "present_ref
    (replace_effect_version s old_eff new_eff next_id) ref"
  using replace_effect_version_universe_mono[OF old_mem fresh] present
  unfolding present_ref_def
  by (cases ref; auto)

lemma replace_effect_references_mono:
  assumes old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  and refs: "references_complete s obj"
  shows "references_complete
    (replace_effect_version s old_eff new_eff next_id) obj"
  using replace_effect_version_universe_mono[OF old_mem fresh] refs
  unfolding references_complete_def by blast

lemma replace_effect_evidence_mono:
  assumes old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  and evidence: "evidence_well_formed s obj"
  shows "evidence_well_formed
    (replace_effect_version s old_eff new_eff next_id) obj"
  using replace_effect_version_universe_mono[OF old_mem fresh] evidence
  unfolding evidence_well_formed_def evidence_references_complete_def by blast

lemma replace_effect_preserves_active_history_disjoint:
  assumes source_disjoint: "active_history_disjoint s"
  and source_unique: "unique_versions s"
  and old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  shows "active_history_disjoint
    (replace_effect_version s old_eff new_eff next_id)"
proof -
  have old_version_active: "version_id_of old_eff \<in> version_ids s"
  unfolding version_ids_def using old_mem by blast
  have versions_different:
    "version_id_of old_eff \<noteq> version_id_of new_eff"
  proof
    assume same: "version_id_of old_eff = version_id_of new_eff"
    from fresh have "version_id_of new_eff \<notin> version_ids s" by blast
    with old_version_active same show False by simp
  qed
  show ?thesis
  using source_disjoint source_unique old_mem fresh versions_different
  unfolding active_history_disjoint_def unique_versions_def version_ids_def
  replace_effect_version_def
  by (auto simp: inj_on_def)
qed

lemma evidence_entry_replace_mono:
  assumes old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"
  and old_not_archived: "archive (version_id_of old_eff) = None"
  and entry: "evidence_entry_well_formed s archive v assessment"

```

```

shows "evidence_entry_well_formed
  (replace_effect_version s old_eff new_eff next_id)
  (extend_object_archive archive old_eff) v assessment"
proof -
  let ?target = "replace_effect_version s old_eff new_eff next_id"
  let ?archive = "extend_object_archive archive old_eff"
  from entry obtain eff where resolved:
    "resolved_version_object s archive v eff"
  and remaining:
    "kind_of eff = K_EFFECT \<and>
     evidence_subject_of assessment = object_id_of eff \<and>
     previous_version_of eff = Some (evidence_prior_version_of assessment) \<and>
     evidence_refs_of assessment \<noteq> {} \<and>
     evidence_refs_of assessment \<subsetq> evidence_of eff \<and>
     evidence_refs_of assessment
       \<subsetq> version_ids s \<union> historical_versions_of s \<and>
     v \<notin> evidence_refs_of assessment \<and>
     evidence_prior_version_of assessment
       \<notin> evidence_refs_of assessment \<and>
     0 \<le> evidence_strength_of assessment \<and>
     evidence_strength_of assessment \<le> 1 \<and>
     0 \<le> evidence_post_uncertainty_of assessment \<and>
     evidence_post_uncertainty_of assessment \<le> 1 \<and>
     0 \<le> validation_uncertainty_max_of assessment \<and>
     validation_uncertainty_max_of assessment \<le> 1 \<and>
     0 < validation_evidence_min_of assessment \<and>
     evidence_assessment_model_of assessment \<in> models_of s"
  unfolding evidence_entry_well_formed_def by blast
  have resolved_target:
    "resolved_version_object ?target ?archive v eff"
  proof (cases "eff \<in> objects_of s \<and> version_id_of eff = v")
  case active: True
  show ?thesis
  proof (cases "eff = old_eff")
  case same_old: True
  from active have eff_version: "version_id_of eff = v" by blast
  have "?archive v = Some eff"
  unfolding extend_object_archive_def
  using same_old eff_version by simp
  then show ?thesis
  unfolding resolved_version_object_def by blast
  next
  case different_old: False
  with active have "eff \<in> objects_of s - {old_eff}" by blast
  with active show ?thesis
  unfolding resolved_version_object_def by auto
  qed
next
case not_active: False
with resolved have archived: "archive v = Some eff"
unfolding resolved_version_object_def by blast
have key_different: "v \<noteq> version_id_of old_eff"
proof
  assume "v = version_id_of old_eff"
  with archived old_not_archived show False by simp
qed
show ?thesis
unfolding resolved_version_object_def extend_object_archive_def
using archived key_different by simp
qed
have refs_target:
  "evidence_refs_of assessment
   \<subsetq> version_ids ?target \<union> historical_versions_of ?target"
using remaining replace_effect_version_universe_mono[OF old_mem fresh]
by blast
show ?thesis
unfolding evidence_entry_well_formed_def
proof (rule exI[of _ eff])
  show "resolved_version_object ?target ?archive v eff \<and>
    kind_of eff = K_EFFECT \<and>
    evidence_subject_of assessment = object_id_of eff \<and>
    previous_version_of eff = Some (evidence_prior_version_of assessment) \<and>
    evidence_refs_of assessment \<noteq> {} \<and>
    evidence_refs_of assessment \<subsetq> evidence_of eff \<and>
    evidence_refs_of assessment
      \<subsetq> version_ids ?target \<union> historical_versions_of ?target \<and>
    v \<notin> evidence_refs_of assessment \<and>
    evidence_prior_version_of assessment
      \<notin> evidence_refs_of assessment \<and>
    0 \<le> evidence_strength_of assessment \<and>
    evidence_strength_of assessment \<le> 1 \<and>
    0 \<le> evidence_post_uncertainty_of assessment \<and>
    evidence_post_uncertainty_of assessment \<le> 1 \<and>
    0 \<le> validation_uncertainty_max_of assessment \<and>
    validation_uncertainty_max_of assessment \<le> 1 \<and>
    0 < validation_evidence_min_of assessment \<and>
    evidence_assessment_model_of assessment \<in> models_of s"
  by (auto)
qed

```

```

evidence_strength_of assessment \<le> 1 \<and>
0 \<le> evidence_post_uncertainty_of assessment \<and>
evidence_post_uncertainty_of assessment \<le> 1 \<and>
0 \<le> validation_uncertainty_max_of assessment \<and>
validation_uncertainty_max_of assessment \<le> 1 \<and>
0 < validation_evidence_min_of assessment \<and>
evidence_assessment_model_of assessment \<in> models_of ?target"
using resolved_target remaining refs_target
by (simp only: replace_effect_simps; blast)
qed
qed

subsection \<open>Successor state and audit-store preservation\<close>

lemma integrate_evidence_payload_valid:
  assumes valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
    ledger old_eff prior assessment new_version integration_time
    next_snapshot_id"
  shows "effect_payload_valid
    (integrated_effect_object old_eff assessment new_version integration_time)
    (integrated_effect_payload prior assessment)"
  using valid
  unfolding integrate_evidence_input_valid_def assessment_bounds_def
    effect_payload_valid_def
  by (cases "evidence_post_vector_of assessment"; auto)

lemma integrate_evidence_new_object_conditions:
  assumes valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
    ledger old_eff prior assessment new_version integration_time
    next_snapshot_id"
  defines "new_eff \<equiv> integrated_effect_object old_eff assessment new_version
    integration_time"
  shows
    "kind_of new_eff \<in> supported_kinds_of G \<and>
    references_complete
    (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
    evidence_well_formed
    (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
    time_well_formed new_eff \<and>
    relation_well_formed
    (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
    effect_well_formed
    (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
    perspective_well_formed
    (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
    context_well_formed
    (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
    relevance_well_formed
    (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
    orientation_well_formed
    (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
    action_well_formed
    (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff \<and>
    feedback_well_formed
    (replace_effect_version s old_eff new_eff next_snapshot_id) new_eff"
  proof -
    let ?target = "replace_effect_version s old_eff new_eff next_snapshot_id"
    from valid have source_wf: "state_well_formed G s"
    and old_mem: "old_eff \<in> objects_of s"
    and old_kind: "kind_of old_eff = K_EFF"
    and old_lifecycle: "lifecycle_of old_eff \<in> {Hypothesis, Supported}"
    and fresh: "new_version \<notin> version_ids s \<union> historical_versions_of s"
    and origin: "present_ref s (effect_origin_of old_eff)"
    and evidence_nonempty: "evidence_refs_of assessment \<noteq> {}"
    and evidence_active:
      "\<forall>v\<in>evidence_refs_of assessment.
      active_evidence_source s (object_id_of old_eff) v"
    and target_ok:
      "admissible_post_target s (object_id_of old_eff)
      (evidence_post_target_of assessment)"
    unfolding integrate_evidence_input_valid_def by blast+
    from source_wf old_mem have old_conditions:
      "kind_of old_eff \<in> supported_kinds_of G \<and>
      references_complete s old_eff \<and>
      evidence_well_formed s old_eff \<and>
      time_well_formed old_eff \<and>
      perspective_well_formed s old_eff \<and>
      context_well_formed s old_eff"

```

```

    unfolding state_well_formed_def by blast
have universe_mono:
  "version_ids s \<union> historical_versions_of s \<subseq>
    version_ids ?target \<union> historical_versions_of ?target"
  using replace_effect_version_universe_mono[OF old_mem]
  unfolding new_eff_def using fresh by simp
have evidence_refs_resolve:
  "evidence_refs_of assessment
    \<subseq> version_ids ?target \<union> historical_versions_of ?target"
proof
  fix v
  assume "v \<in> evidence_refs_of assessment"
  then obtain ev where ev_mem: "ev \<in> objects_of s"
    and ev_v: "version_id_of ev = v"
    and different: "object_id_of ev \<noteq> object_id_of old_eff"
    using evidence_active unfolding active_evidence_source_def by blast
  have "ev \<noteq> old_eff" using different by auto
  then have "ev \<in> objects_of s - {old_eff}" using ev_mem by blast
  then have "v \<in> version_ids ?target"
    unfolding version_ids_def new_eff_def using ev_v by auto
  then show "v \<in> version_ids ?target \<union> historical_versions_of ?target"
    by blast
qed
have target_ref_resolves:
  "option_ref_set (evidence_post_target_of assessment)
    \<subseq> version_ids ?target \<union> historical_versions_of ?target"
proof (cases "evidence_post_target_of assessment")
  case None
  then show ?thesis by (simp add: option_ref_set_def)
next
  case (Some v)
  then obtain obj where obj_mem: "obj \<in> objects_of s"
    and obj_v: "version_id_of obj = v"
    and different: "object_id_of obj \<noteq> object_id_of old_eff"
    using target_ok unfolding admissible_post_target_def by auto
  have "obj \<noteq> old_eff" using different by auto
  then have "obj \<in> objects_of s - {old_eff}" using obj_mem by blast
  then have "v \<in> version_ids ?target"
    unfolding version_ids_def new_eff_def using obj_v by auto
  with Some show ?thesis by (simp add: option_ref_set_def)
qed
have prior_resolves:
  "version_id_of old_eff
    \<in> version_ids ?target \<union> historical_versions_of ?target"
  by simp
from old_conditions have old_refs_source:
  "references_of old_eff
    \<subseq> version_ids s \<union> historical_versions_of s"
  unfolding references_complete_def by blast
have old_refs_target:
  "references_of old_eff
    \<subseq> version_ids ?target \<union> historical_versions_of ?target"
  using old_refs_source universe_mono by blast
have combined_refs_target:
  "references_of old_eff \<union> evidence_refs_of assessment \<union>
    {version_id_of old_eff} \<union>
    option_ref_set (evidence_post_target_of assessment)
    \<subseq> version_ids ?target \<union> historical_versions_of ?target"
  using old_refs_target evidence_refs_resolve target_ref_resolves
  prior_resolves by blast
have new_refs_eq:
  "references_of new_eff =
    references_of old_eff \<union> evidence_refs_of assessment \<union>
    {version_id_of old_eff} \<union>
    option_ref_set (evidence_post_target_of assessment)"
  unfolding new_eff_def by simp
have refs_new: "references_complete ?target new_eff"
  unfolding references_complete_def
  using combined_refs_target new_refs_eq by simp
have evidence_new: "evidence_well_formed ?target new_eff"
proof -
  from old_conditions have old_evidence_source:
    "evidence_of old_eff
      \<subseq> version_ids s \<union> historical_versions_of s"
    unfolding evidence_well_formed_def evidence_references_complete_def
    by blast
  have old_evidence_target:
    "evidence_of old_eff
      \<subseq> version_ids ?target \<union> historical_versions_of ?target"
    using old_evidence_source universe_mono by blast

```



```

have new_evidence_eq:
  "evidence_of new_eff =
    evidence_of old_eff \<union> evidence_refs_of assessment"
  unfolding new_eff_def by simp
have complete:
  "evidence_of new_eff
    \<subseq> version_ids ?target \<union> historical_versions_of ?target"
  using old_evidence_target evidence_refs_resolve new_evidence_eq by blast
have upgraded_nonempty:
  "lifecycle_of new_eff \<in> {Supported, Validated}
    \<longrightarrow> evidence_of new_eff \<noteq> {}"
  using evidence_nonempty
  unfolding new_eff_def integrated_effect_object_def
  by auto
show ?thesis
  unfolding evidence_well_formed_def evidence_references_complete_def
  using complete upgraded_nonempty by blast
qed
have origin_target: "present_ref ?target (effect_origin_of new_eff)"
  unfolding new_eff_def
  using replace_effect_present_ref_mono[OF old_mem _ origin] fresh by simp
have perspective_new: "perspective_well_formed ?target new_eff"
proof -
  from old_conditions have old_perspective:
    "perspective_well_formed s old_eff" by blast
  show ?thesis
  using old_perspective
  unfolding perspective_well_formed_def new_eff_def
  by (simp only: integrated_effect_simps replace_effect_simps)
qed
have context_new: "context_well_formed ?target new_eff"
proof -
  from old_conditions have old_context:
    "context_well_formed s old_eff" by blast
  show ?thesis
  using old_context
  unfolding context_well_formed_def new_eff_def
  by (simp only: integrated_effect_simps replace_effect_simps)
qed
have other_typed:
  "time_well_formed new_eff \<and>
    relation_well_formed ?target new_eff \<and>
    effect_well_formed ?target new_eff \<and>
    relevance_well_formed ?target new_eff \<and>
    orientation_well_formed ?target new_eff \<and>
    action_well_formed ?target new_eff \<and>
    feedback_well_formed ?target new_eff"
proof -
  have new_kind: "kind_of new_eff = K_EFF"
    using old_kind unfolding new_eff_def by simp
  have new_time: "time_well_formed new_eff"
    unfolding time_well_formed_def new_eff_def by simp
  have new_relation: "relation_well_formed ?target new_eff"
    using new_kind unfolding relation_well_formed_def by simp
  have new_effect: "effect_well_formed ?target new_eff"
    unfolding effect_well_formed_def
    using new_kind origin_target by blast
  have new_relevance: "relevance_well_formed ?target new_eff"
    using new_kind unfolding relevance_well_formed_def by simp
  have new_orientation: "orientation_well_formed ?target new_eff"
    using new_kind unfolding orientation_well_formed_def by simp
  have new_action: "action_well_formed ?target new_eff"
    using new_kind unfolding action_well_formed_def by simp
  have new_feedback: "feedback_well_formed ?target new_eff"
    using new_kind unfolding feedback_well_formed_def by simp
  show ?thesis
    using new_time new_relation new_effect new_relevance new_orientation
      new_action new_feedback by blast
qed
show ?thesis
  using old_conditions refs_new evidence_new perspective_new context_new
    other_typed
  unfolding new_eff_def by simp
qed

lemma replace_effect_preserves_unique_versions:
  assumes source_unique: "unique_versions s"
  and old_mem: "old_eff \<in> objects_of s"
  and fresh: "version_id_of new_eff
    \<notin> version_ids s \<union> historical_versions_of s"

```

```

shows "unique_versions
  (replace_effect_version s old_eff new_eff next_snapshot_id)"
proof -
  from source_unique have inj: "inj_on version_id_of (objects_of s)"
  unfolding unique_versions_def .
  show ?thesis
  unfolding unique_versions_def
  proof (rule inj_onI)
    fix left right
    assume left_mem:
      "left \<in> objects_of
        (replace_effect_version s old_eff new_eff next_snapshot_id)"
    and right_mem:
      "right \<in> objects_of
        (replace_effect_version s old_eff new_eff next_snapshot_id)"
    and same: "version_id_of left = version_id_of right"
    consider (left_new) "left = new_eff" | (left_old) "left \<in> objects_of s - {old_eff}"
    using left_mem by auto
    then show "left = right"
  proof cases
    case left_new
    show ?thesis
    proof (cases "right = new_eff")
      case True
      with left_new show ?thesis by simp
    next
      case False
      with right_mem have right_old: "right \<in> objects_of s"
      by auto
      have "version_id_of new_eff \<in> version_ids s"
      unfolding version_ids_def using same left_new right_old by blast
      then show ?thesis using fresh by blast
    qed
  next
    case left_old
    show ?thesis
    proof (cases "right = new_eff")
      case True
      have left_version: "version_id_of left \<in> version_ids s"
      unfolding version_ids_def using left_old by blast
      have "version_id_of left = version_id_of new_eff"
      using same True by simp
      with left_version have "version_id_of new_eff \<in> version_ids s"
      by simp
      then show ?thesis using fresh by blast
    next
      case False
      with right_mem have right_old: "right \<in> objects_of s" by auto
      show ?thesis
      using inj_onD[OF inj same] left_old right_old by blast
    qed
  qed
qed
qed
qed

theorem integrate_evidence_preserves_state_well_formed:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
      old_eff prior assessment new_version integration_time next_snapshot_id
      s' catalog' object_archive' payload_archive' ledger'"
  shows "state_well_formed G s'"
proof -
  let ?new_eff = "integrated_effect_object old_eff assessment new_version
    integration_time"
  let ?target = "replace_effect_version s old_eff ?new_eff next_snapshot_id"
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
      ledger old_eff prior assessment new_version integration_time
      next_snapshot_id"
  and target: "s' = ?target"
  by (auto simp: integrate_evidence_step_def Let_def)
  from valid have source_wf: "state_well_formed G s"
  and old_mem: "old_eff \<in> objects_of s"
  and fresh: "new_version \<notin> version_ids s \<union> historical_versions_of s"
  unfolding integrate_evidence_input_valid_def by blast+
  have unique: "unique_versions ?target"
  using replace_effect_preserves_unique_versions old_mem fresh source_wf
  unfolding state_well_formed_def by simp
  have new_conditions:
    "kind_of ?new_eff \<in> supported_kinds_of G \<and>"

```

```

references_complete ?target ?new_eff \<and>
evidence_well_formed ?target ?new_eff \<and>
time_well_formed ?new_eff \<and>
relation_well_formed ?target ?new_eff \<and>
effect_well_formed ?target ?new_eff \<and>
perspective_well_formed ?target ?new_eff \<and>
context_well_formed ?target ?new_eff \<and>
relevance_well_formed ?target ?new_eff \<and>
orientation_well_formed ?target ?new_eff \<and>
action_well_formed ?target ?new_eff \<and>
feedback_well_formed ?target ?new_eff"
using integrate_evidence_new_object_conditions[OF valid] by simp
have old_object_conditions:
"\<forall>obj\<in>objects_of s - {old_eff}.
  kind_of obj \<in> supported_kinds_of G \<and>
  references_complete ?target obj \<and>
  evidence_well_formed ?target obj \<and>
  time_well_formed obj \<and>
  relation_well_formed ?target obj \<and>
  effect_well_formed ?target obj \<and>
  perspective_well_formed ?target obj \<and>
  context_well_formed ?target obj \<and>
  relevance_well_formed ?target obj \<and>
  orientation_well_formed ?target obj \<and>
  action_well_formed ?target obj \<and>
  feedback_well_formed ?target obj"
proof (intro ballI)
  fix obj
  assume obj_mem: "obj \<in> objects_of s - {old_eff}"
  from source_wf obj_mem have source_conditions:
    "kind_of obj \<in> supported_kinds_of G \<and>
    references_complete s obj \<and> evidence_well_formed s obj \<and>
    time_well_formed obj \<and> relation_well_formed s obj \<and>
    effect_well_formed s obj \<and> perspective_well_formed s obj \<and>
    context_well_formed s obj \<and> relevance_well_formed s obj \<and>
    orientation_well_formed s obj \<and> action_well_formed s obj \<and>
    feedback_well_formed s obj"
  unfolding state_well_formed_def by blast
  have present_mono: "\<And>ref. present_ref s ref \<Longrightarrow> present_ref ?target ref"
  using replace_effect_present_ref_mono[OF old_mem _] fresh by simp
  have refs_target: "references_complete ?target obj"
  using replace_effect_references_mono[OF old_mem _]
    source_conditions fresh by simp
  have evidence_target: "evidence_well_formed ?target obj"
  using replace_effect_evidence_mono[OF old_mem _]
    source_conditions fresh by simp
  have typed_target:
    "relation_well_formed ?target obj \<and>
    effect_well_formed ?target obj \<and>
    feedback_well_formed ?target obj"
  using source_conditions present_mono
    unfolding relation_well_formed_def effect_well_formed_def
    feedback_well_formed_def
  by blast
  from source_conditions have source_perspective:
    "perspective_well_formed s obj"
  and source_context: "context_well_formed s obj"
  and source_relevance: "relevance_well_formed s obj"
  and source_orientation: "orientation_well_formed s obj"
  and source_action: "action_well_formed s obj"
  by blast+
  have perspective_target: "perspective_well_formed ?target obj"
  using source_perspective
    unfolding perspective_well_formed_def
  by (simp only: replace_effect_simps)
  have context_target: "context_well_formed ?target obj"
  using source_context
    unfolding context_well_formed_def
  by (simp only: replace_effect_simps)
  have relevance_target: "relevance_well_formed ?target obj"
  unfolding relevance_well_formed_def
proof
  assume kind: "kind_of obj = K_REV"
  from source_relevance kind have source_model:
    "case model_of obj of None \<Rightarrow> False
    | Some m \<Rightarrow> m \<in> models_of s"
  and source_refs: "references_of obj \<noteq> {}"
  unfolding relevance_well_formed_def by blast+
  have target_model:
    "case model_of obj of None \<Rightarrow> False

```

```

    | Some m \<Rightarrow> m \<in> models_of ?target"
    using source_model by (simp only: replace_effect_simps)
  show "perspective_well_formed ?target obj \<and>
    context_well_formed ?target obj \<and>
    (case model_of obj of None \<Rightarrow> False
      | Some m \<Rightarrow> m \<in> models_of ?target) \<and>
    references_of obj \<noteq> {}"
    using perspective_target context_target target_model source_refs by blast
qed
have orientation_target: "orientation_well_formed ?target obj"
  using source_orientation unfolding orientation_well_formed_def by simp
have action_target: "action_well_formed ?target obj"
  using source_action unfolding action_well_formed_def
  by (simp only: replace_effect_simps)
have stable_target:
  "perspective_well_formed ?target obj \<and>
  context_well_formed ?target obj \<and>
  relevance_well_formed ?target obj \<and>
  orientation_well_formed ?target obj \<and>
  action_well_formed ?target obj"
  using perspective_target context_target relevance_target
  orientation_target action_target by blast
show "kind_of obj \<in> supported_kinds_of G \<and>
  references_complete ?target obj \<and>
  evidence_well_formed ?target obj \<and>
  time_well_formed obj \<and>
  relation_well_formed ?target obj \<and>
  effect_well_formed ?target obj \<and>
  perspective_well_formed ?target obj \<and>
  context_well_formed ?target obj \<and>
  relevance_well_formed ?target obj \<and>
  orientation_well_formed ?target obj \<and>
  action_well_formed ?target obj \<and>
  feedback_well_formed ?target obj"
  using source_conditions refs_target evidence_target typed_target
  stable_target by blast
qed
have all_objects:
  "\<forall>obj\<in>objects_of ?target.
  kind_of obj \<in> supported_kinds_of G \<and>
  references_complete ?target obj \<and>
  evidence_well_formed ?target obj \<and>
  time_well_formed obj \<and>
  relation_well_formed ?target obj \<and>
  effect_well_formed ?target obj \<and>
  perspective_well_formed ?target obj \<and>
  context_well_formed ?target obj \<and>
  relevance_well_formed ?target obj \<and>
  orientation_well_formed ?target obj \<and>
  action_well_formed ?target obj \<and>
  feedback_well_formed ?target obj"
  using new_conditions old_object_conditions by auto
from source_wf have secondary:
  "(\<forall>p\<in>projections_of ?target. projection_well_formed ?target p) \<and>
  (\<forall>c\<in>couplings_of ?target. coupling_well_formed ?target c) \<and>
  (\<forall>t\<in>transmissions_of ?target. transmission_well_formed ?target t) \<and>
  (\<forall>b\<in>boundaries_of ?target. boundary_well_formed b)"
  unfolding state_well_formed_def projection_well_formed_def
  coupling_well_formed_def transmission_well_formed_def
  by simp
show ?thesis
  unfolding target_state_well_formed_def
  using unique_all_objects secondary by blast
qed

theorem integrate_evidence_preserves_identity:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
    old_eff prior assessment new_version integration_time next_snapshot_id
    s' catalog' object_archive' payload_archive' ledger'"
  shows "unique_versions s' \<and> immutable_preserved s s'"
proof -
  let ?new_eff = "integrated_effect_object old_eff assessment new_version
  integration_time"
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
    ledger old_eff prior assessment new_version integration_time
    next_snapshot_id"
  and target:
    "s' = replace_effect_version s old_eff ?new_eff next_snapshot_id"

```

```

    by (auto simp: integrate_evidence_step_def Let_def)
  from valid have source_wf: "state_well_formed G s"
    and old_mem: "old_eff \<in> objects_of s"
    and fresh: "new_version \<notin> version_ids s \<union> historical_versions_of s"
  unfolding integrate_evidence_input_valid_def by blast+
  have unique: "unique_versions s'"
  unfolding target
  using replace_effect_preserves_unique_versions old_mem fresh source_wf
  unfolding state_well_formed_def by simp
  have immutable: "immutable_preserved s s'"
  unfolding immutable_preserved_def target
  proof (intro ballI allI impI)
    fix before after
    assume before_mem: "before \<in> objects_of s"
    and after_mem:
      "after \<in> objects_of
      (replace_effect_version s old_eff ?new_eff next_snapshot_id)"
    and same: "version_id_of before = version_id_of after"
    show "before = after"
    proof (cases "after = ?new_eff")
      case True
      have "new_version \<in> version_ids s"
      unfolding version_ids_def using before_mem same True by auto
      then show ?thesis using fresh by blast
    next
      case False
      then have after_old: "after \<in> objects_of s" using after_mem by auto
      from source_wf have inj: "inj_on version_id_of (objects_of s)"
      unfolding state_well_formed_def unique_versions_def by blast
      show ?thesis using inj_onD[OF inj same before_mem after_old] .
    qed
  qed
  show ?thesis using unique immutable by blast
qed

theorem integrate_evidence_preserves_audit_stores:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
    old_eff prior assessment new_version integration_time next_snapshot_id
    s' catalog' object_archive' payload_archive' ledger'"
  shows
    "active_history_disjoint s' \<and>
    object_archive_well_formed s' object_archive' \<and>
    effect_payload_archive_well_formed object_archive' payload_archive' \<and>
    evidence_ledger_well_formed s' object_archive' ledger'"
  proof -
    let ?new_eff = "integrated_effect_object old_eff assessment new_version
    integration_time"
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive
    ledger old_eff prior assessment new_version integration_time
    next_snapshot_id"
  and target:
    "s' = replace_effect_version s old_eff ?new_eff next_snapshot_id"
  and object_archive_target:
    "object_archive' = extend_object_archive object_archive old_eff"
  and payload_archive_target:
    "payload_archive' = extend_effect_payload_archive payload_archive
    old_eff prior"
  and ledger_target:
    "ledger' = extend_evidence_ledger ledger new_version assessment"
  by (auto simp: integrate_evidence_step_def Let_def)
  from valid have source_wf: "state_well_formed G s"
  and source_disjoint: "active_history_disjoint s"
  and source_object_archive:
    "object_archive_well_formed s object_archive"
  and source_payload_archive:
    "effect_payload_archive_well_formed object_archive payload_archive"
  and source_ledger:
    "evidence_ledger_well_formed s object_archive ledger"
  and old_mem: "old_eff \<in> objects_of s"
  and old_archive_none:
    "object_archive (version_id_of old_eff) = None"
  and payload_old_none:
    "payload_archive (version_id_of old_eff) = None"
  and fresh:
    "new_version \<notin> version_ids s \<union> historical_versions_of s"
  and ledger_new_none: "ledger new_version = None"
  and assessment_bounds: "assessment_bounds assessment"
  and assessment_model:

```

```

"evidence_assessment_model_of assessment \<in> models_of s"
and old_kind: "kind_of old_eff = K_EFF"
and payload_valid: "effect_payload_valid old_eff prior"
unfolding integrate_evidence_input_valid_def by blast+
from source_wf have source_unique: "unique_versions s"
unfolding state_well_formed_def by blast
have fresh_eff:
  "version_id_of ?new_eff
  \<notin> version_ids s \<union> historical_versions_of s"
using fresh by simp
have target_disjoint: "active_history_disjoint s"
unfolding target
using replace_effect_preserves_active_history_disjoint
[OF source_disjoint source_unique old_mem fresh_eff] .
have target_object_archive:
  "object_archive_well_formed s' object_archive'"
proof -
  have archived_properties:
    "\<forall>v obj. object_archive' v = Some obj \<longrightarrow>
    version_id_of obj = v \<and> v \<in> historical_versions_of s'"
  proof (intro allI impI)
    fix v obj
    assume mapped: "object_archive' v = Some obj"
    show "version_id_of obj = v \<and> v \<in> historical_versions_of s'"
    proof (cases "v = version_id_of old_eff")
      case True
      with mapped object_archive_target target show ?thesis
      unfolding extend_object_archive_def by simp
    next
      case False
      with mapped object_archive_target have source_mapped:
        "object_archive v = Some obj"
      unfolding extend_object_archive_def by simp
      from source_object_archive source_mapped have
        "version_id_of obj = v \<and> v \<in> historical_versions_of s"
      unfolding object_archive_well_formed_def by blast
      with target show ?thesis by simp
    qed
  qed
  show ?thesis
  unfolding object_archive_well_formed_def
  using archived_properties target_disjoint
  unfolding active_history_disjoint_def by blast
qed
have target_payload_archive:
  "effect_payload_archive_well_formed object_archive' payload_archive'"
proof (unfold effect_payload_archive_well_formed_def, intro allI impI)
  fix v stored_payload
  assume mapped: "payload_archive' v = Some stored_payload"
  show "\<exists>eff. object_archive' v = Some eff \<and>
  effect_payload_valid eff stored_payload"
  proof (cases "v = version_id_of old_eff")
    case True
    with mapped payload_archive_target object_archive_target
    payload_valid show ?thesis
    unfolding extend_effect_payload_archive_def extend_object_archive_def
    by auto
  next
    case False
    with mapped payload_archive_target have source_mapped:
      "payload_archive v = Some stored_payload"
    unfolding extend_effect_payload_archive_def by simp
    from source_payload_archive source_mapped obtain eff where
      source_object: "object_archive v = Some eff"
    and valid_payload: "effect_payload_valid eff stored_payload"
    unfolding effect_payload_archive_well_formed_def by blast
    have "object_archive' v = Some eff"
    unfolding object_archive_target extend_object_archive_def
    using source_object False by simp
    with valid_payload show ?thesis by blast
  qed
qed
have target_ledger:
  "evidence_ledger_well_formed s' object_archive' ledger'"
proof (unfold evidence_ledger_well_formed_def, intro allI impI)
  fix v stored_assessment
  assume mapped: "ledger' v = Some stored_assessment"
  show "evidence_entry_well_formed s' object_archive' v stored_assessment"
  proof (cases "v = new_version")
    case True

```

```

with mapped ledger_target have assessment_eq:
  "stored_assessment = assessment"
  unfolding extend_evidence_ledger_def by simp
have target_wf: "state_well_formed G s'"
  using integrate_evidence_preserves_state_well_formed[OF concrete] .
have exact:
  "objects_of s' = insert ?new_eff (objects_of s - {old_eff})"
  using concrete
  unfolding integrate_evidence_step_def Let_def by auto
have new_mem: "?new_eff \<in> objects_of s'" using exact by simp
have evidence_refs_in_new:
  "evidence_refs_of assessment \<subteq> evidence_of ?new_eff"
  by simp
from target_wf new_mem have new_evidence_complete:
  "evidence_of ?new_eff
   \<subteq> version_ids s' \<union> historical_versions_of s'"
  unfolding state_well_formed_def evidence_well_formed_def
  evidence_references_complete_def
  by blast
from target_wf new_mem have evidence_complete:
  "evidence_refs_of assessment
   \<subteq> version_ids s' \<union> historical_versions_of s'"
  using evidence_refs_in_new new_evidence_complete by blast
from integrate_evidence_lineage_and_no_self_support[OF concrete]
have evidence_nonempty: "evidence_refs_of assessment \<noteq> {}"
  and new_not_self: "new_version \<notin> evidence_refs_of assessment"
  and prior_not_self:
    "version_id_of old_eff \<notin> evidence_refs_of assessment"
  by blast+
from valid have subject:
  "evidence_subject_of assessment = object_id_of old_eff"
  and prior_version:
    "evidence_prior_version_of assessment = version_id_of old_eff"
  unfolding integrate_evidence_input_valid_def by blast+
have resolved:
  "resolved_version_object s' object_archive' new_version ?new_eff"
  unfolding resolved_version_object_def using new_mem by simp
show ?thesis
  unfolding True assessment_eq evidence_entry_well_formed_def
  using resolved_evidence_nonempty evidence_complete new_not_self
  prior_not_self subject prior_version evidence_refs_in_new
  assessment_bounds old_kind
  assessment_model
  unfolding assessment_bounds_def target
  by auto
next
case False
with mapped ledger_target have source_mapped:
  "ledger v = Some stored_assessment"
  unfolding extend_evidence_ledger_def by simp
from source_ledger source_mapped have source_entry:
  "evidence_entry_well_formed s object_archive' v stored_assessment"
  unfolding evidence_ledger_well_formed_def by blast
have transferred:
  "evidence_entry_well_formed
   (replace_effect_version s old_eff ?new_eff next_snapshot_id)
   (extend_object_archive object_archive old_eff) v stored_assessment"
  using evidence_entry_replace_mono[OF old_mem fresh_eff
    old_archive_none source_entry] .
show ?thesis
  unfolding target object_archive_target using transferred .
qed
qed
show ?thesis
  using target_disjoint target_object_archive target_payload_archive
  target_ledger by blast
qed

theorem integrate_evidence_preserves_active_catalog:
  assumes concrete:
    "integrate_evidence_step G s catalog object_archive payload_archive ledger
     old_eff prior assessment new_version integration_time next_snapshot_id
     s' catalog' object_archive' payload_archive' ledger'"
  shows "effect_catalog_well_formed s' catalog'"
proof -
  let ?new_eff = "integrated_effect_object old_eff assessment new_version
    integration_time"
  let ?new_payload = "integrated_effect_payload prior assessment"
  from concrete have valid:
    "integrate_evidence_input_valid G s catalog object_archive payload_archive

```

```

    ledger old_eff prior assessment new_version integration_time
    next_snapshot_id"
and target:
  "s' = replace_effect_version s old_eff ?new_eff next_snapshot_id"
and catalog_target:
  "catalog' = replace_active_effect_payload catalog
    (version_id_of old_eff) new_version ?new_payload"
by (auto simp: integrate_evidence_step_def Let_def)
from valid have source_catalog: "effect_catalog_well_formed s catalog"
and old_mem: "old_eff \<in> objects_of s"
and fresh:
  "new_version \<notin> version_ids s \<union> historical_versions_of s"
unfolding integrate_evidence_input_valid_def by blast+
have new_payload_valid: "effect_payload_valid ?new_eff ?new_payload"
using integrate_evidence_payload_valid[OF valid] .
have new_mem: "?new_eff \<in> objects_of s'"
unfolding target by simp
show ?thesis
unfolding effect_catalog_well_formed_def
proof (intro allI impI)
fix v stored_payload
assume mapped: "catalog' v = Some stored_payload"
show "\<exists>eff\<in>objects_of s'.
  version_id_of eff = v \<and> effect_payload_valid eff stored_payload"
proof (cases "v = new_version")
case True
with mapped catalog_target have payload_eq:
  "stored_payload = ?new_payload"
unfolding replace_active_effect_payload_def by simp
show ?thesis
proof (rule bexI[of _ ?new_eff])
show "version_id_of ?new_eff = v \<and>
  effect_payload_valid ?new_eff stored_payload"
using new_payload_valid True payload_eq by simp
show "?new_eff \<in> objects_of s'" using new_mem .
qed
next
case False
have old_key_different: "v \<noteq> version_id_of old_eff"
proof
assume "v = version_id_of old_eff"
with mapped catalog_target False show False
unfolding replace_active_effect_payload_def by simp
qed
from mapped catalog_target False old_key_different have source_mapped:
  "catalog v = Some stored_payload"
unfolding replace_active_effect_payload_def by simp
from source_catalog source_mapped obtain eff where eff_mem:
  "eff \<in> objects_of s"
and eff_version: "version_id_of eff = v"
and eff_payload: "effect_payload_valid eff stored_payload"
unfolding effect_catalog_well_formed_def by blast
have eff_not_old: "eff \<noteq> old_eff"
using eff_version old_key_different by auto
have "eff \<in> objects_of s'"
unfolding target using eff_mem eff_not_old by auto
with eff_version eff_payload show ?thesis by blast
qed
qed
qed

theorem integrate_evidence_preserves_references_time_context:
assumes concrete:
  "integrate_evidence_step G s catalog object_archive payload_archive ledger
    old_eff prior assessment new_version integration_time next_snapshot_id
    s' catalog' object_archive' payload_archive' ledger'"
shows
  "\<forall>obj\<in>objects_of s'.
    references_complete s' obj \<and>
    evidence_well_formed s' obj \<and>
    time_well_formed obj \<and>
    perspective_well_formed s' obj \<and>
    context_well_formed s' obj"
using integrate_evidence_preserves_state_well_formed[OF concrete]
unfolding state_well_formed_def by blast

theorem integrate_evidence_refines_generic_step:
assumes concrete:
  "integrate_evidence_step G s catalog object_archive payload_archive ledger
    old_eff prior assessment new_version integration_time next_snapshot_id

```



```

    s' catalog' object_archive' payload_archive' ledger''
shows "step G s IntegrateEvidence s'"
proof -
  from concrete have source_wf: "state_well_formed G s"
  and allowed: "IntegrateEvidence \<in> allowed_operations_of G"
  by (auto simp: integrate_evidence_step_def integrate_evidence_input_valid_def)
  have target_wf: "state_well_formed G s'"
  using integrate_evidence_preserves_state_well_formed[OF concrete] .
  have immutable: "immutable_preserved s s'"
  using integrate_evidence_preserves_identity[OF concrete] by blast
  show ?thesis
  unfolding step_def using source_wf target_wf allowed immutable by blast
qed
end

```

## I.10 Verknüpfter T1-T7-Gesamtzeuge

Vollständiger Quelltext des abschließenden T6-/T7-Anschlusses und des Gesamtzertifikats  
operational\_chain\_T1\_T7.

```

theory UZA_0_9_1_Operational_Chain_Witness
  imports UZA_0_9_1_Evaluate_Relevance_Witness
    UZA_0_9_1_Integrate_Evidence_Witness
begin

section \<open>End-to-end reachability certificate for T1--T7\<close>

text \<open>
  The source of T6 below is exactly the successor of the second strict T5
  transition. The T7 source is exactly the T6 successor. Consequently this
  theory links all operation-local witnesses into one causal trace instead of
  merely collecting separately satisfiable record literals.
\<close>

definition chain_resonance_catalog_0 :: resonance_catalog where
  "chain_resonance_catalog_0 = Map.empty"

definition chain_resonance_request :: resonance_request where
  "chain_resonance_request =
    \<lparr> requested_resonance_object_id_of = 7,
      requested_resonance_version_id_of = 9,
      requested_source_effect_of = 7,
      requested_target_effect_of = 8,
      requested_source_vector_of = chain_effect_vector_a,
      requested_target_vector_of = chain_effect_vector_b,
      requested_resonance_profile_of = t6_resonance_profile,
      requested_resonance_perspective_of = 10,
      requested_resonance_context_of = 20,
      requested_resonance_horizon_of = 4,
      requested_resonance_uncertainty_of = 1 / 10,
      requested_resonance_rationale_refs_of = {1, 2},
      requested_resonance_rationale_of =
        'causally reachable Connection-to-Stability interaction'',
      requested_resonance_valid_from_of = 9 \<rparr>"

definition chain_after_t6 :: snapshot where
  "chain_after_t6 =
    add_classified_resonance chain_after_t5 chain_resonance_request 9"

definition chain_resonance_catalog :: resonance_catalog where
  "chain_resonance_catalog =
    extend_resonance_catalog chain_resonance_catalog_0
      chain_resonance_request"

lemma chain_after_t5_objects_exact:
  "objects_of chain_after_t5 =
    {base_object, chain_entity_a, chain_entity_b, chain_relation_ab,
      chain_relation_ba, chain_effect_a_supported,
      chain_effect_b_supported}"
  by (auto simp: chain_after_t5_def chain_after_t5_a_def
    chain_after_t4_def chain_after_t4_a_def chain_after_t3_def
    chain_after_t3_ab_def chain_after_t2_def chain_after_t1_def
    chain_seed_def base_snapshot_def chain_object_defs chain_effect_a_def
    chain_effect_b_def chain_effect_a_supported_def

```

```

chain_effect_b_supported_def chain_assessment_a_def
chain_assessment_b_def base_object_def integrated_effect_object_def
integrated_lifecycle_def validation_ready_def)

lemma chain_after_t5_background_exact:
  "perspectives_of chain_after_t5 = {0, Suc 0, 10} \<and>
  contexts_of chain_after_t5 = {20} \<and>
  models_of chain_after_t5 = {30} \<and>
  snapshot_id_of chain_after_t5 = 8"
  by (simp add: chain_after_t5_def chain_after_t5_a_def
    chain_after_t4_def chain_after_t4_a_def chain_after_t3_def
    chain_after_t3_ab_def chain_after_t2_def chain_after_t1_def
    chain_seed_def base_snapshot_def)

lemma chain_after_t5_object_ids_exact:
  "object_id_of `objects_of chain_after_t5 = {0, 1, 2, 3, 4, 5, 6}"
  using chain_after_t5_objects_exact
  by (simp add: chain_object_defs chain_effect_a_supported_def
    chain_effect_b_supported_def chain_effect_a_def chain_effect_b_def
    integrated_effect_object_def base_object_def)

lemma chain_after_t5_version_ids_exact:
  "version_ids chain_after_t5 = {0, 1, 2, 3, 4, 7, 8}"
  using chain_after_t5_objects_exact
  by (simp add: version_ids_def chain_object_defs
    chain_effect_a_supported_def chain_effect_b_supported_def
    chain_effect_a_def chain_effect_b_def integrated_effect_object_def
    base_object_def)

lemma chain_effect_a_integrated_payload_valid:
  "effect_payload_valid chain_effect_a_supported
  (integrated_effect_payload chain_effect_payload_a chain_assessment_a)"
proof -
  have base_valid:
    "integrate_evidence_input_valid chain_grammar chain_after_t4
    chain_effect_catalog chain_object_archive_0 chain_payload_archive_0
    chain_evidence_ledger_0 chain_effect_a chain_effect_payload_a
    chain_assessment_a 7 7 7"
    using chain_t5_a_strict_input_valid
    unfolding integrate_evidence_strict_input_valid_def
    by blast
  show ?thesis
    unfolding chain_effect_a_supported_def
    using integrate_evidence_payload_valid[OF base_valid] .
qed

lemma chain_effect_b_integrated_payload_valid:
  "effect_payload_valid chain_effect_b_supported
  (integrated_effect_payload chain_effect_payload_b chain_assessment_b)"
proof -
  have base_valid:
    "integrate_evidence_input_valid chain_grammar chain_after_t5_a
    chain_effect_catalog_after_t5_a chain_object_archive_after_t5_a
    chain_payload_archive_after_t5_a chain_evidence_ledger_after_t5_a
    chain_effect_b chain_effect_payload_b chain_assessment_b 8 8 8"
    using chain_t5_b_strict_input_valid
    unfolding integrate_evidence_strict_input_valid_def
    by blast
  show ?thesis
    unfolding chain_effect_b_supported_def
    using integrate_evidence_payload_valid[OF base_valid] .
qed

lemma chain_effect_a_active_for_t6:
  "active_classifiable_effect chain_after_t5 chain_effect_catalog_after_t5
  7 chain_effect_vector_a 20"
  unfolding active_classifiable_effect_def
proof (rule bexI[of _ chain_effect_a_supported])
  show "version_id_of chain_effect_a_supported = 7 \<and>
  kind_of chain_effect_a_supported = K_EFF \<and>
  lifecycle_of chain_effect_a_supported \<in> {Supported, Validated} \<and>
  context_of chain_effect_a_supported = Some 20 \<and>
  (\<exists>payload.
    chain_effect_catalog_after_t5 7 = Some payload \<and>
    hypothesis_vector_of payload = Some chain_effect_vector_a \<and>
    effect_payload_valid chain_effect_a_supported payload)"
    using chain_t5_reachability_certificate
    chain_effect_a_integrated_payload_valid
  by (auto simp: chain_effect_a_supported_def chain_effect_a_def
    chain_assessment_a_def chain_effect_payload_a_def base_object_def)

```

```

        integrated_effect_object_def integrated_lifecycle_def
        validation_ready_def integrated_effect_payload_def)
show "chain_effect_a_supported \<in> objects_of chain_after_t5"
  using chain_after_t5_objects_exact by simp
qed

lemma chain_effect_b_active_for_t6:
  "active_classifiable_effect chain_after_t5 chain_effect_catalog_after_t5
    8 chain_effect_vector_b 20"
  unfolding active_classifiable_effect_def
proof (rule bexI[of _ chain_effect_b_supported])
  show "version_id_of chain_effect_b_supported = 8 \<and>
    kind_of chain_effect_b_supported = K_EFF \<and>
    lifecycle_of chain_effect_b_supported \<in> {Supported, Validated} \<and>
    context_of chain_effect_b_supported = Some 20 \<and>
    (\<exists>payload.
      chain_effect_catalog_after_t5 8 = Some payload \<and>
      hypothesis_vector_of payload = Some chain_effect_vector_b \<and>
      effect_payload_valid chain_effect_b_supported payload)"
  using chain_t5_reachability_certificate
  chain_effect_b_integrated_payload_valid
  by (auto simp: chain_effect_b_supported_def chain_effect_b_def
    chain_assessment_b_def chain_effect_payload_b_def base_object_def
    integrated_effect_object_def integrated_lifecycle_def
    validation_ready_def integrated_effect_payload_def)
show "chain_effect_b_supported \<in> objects_of chain_after_t5"
  using chain_after_t5_objects_exact by simp
qed

lemma chain_effects_distinct_for_t6:
  "distinct_active_effect_objects chain_after_t5 7 8"
  unfolding distinct_active_effect_objects_def
  using chain_after_t5_objects_exact
  by (intro bexI[of _ chain_effect_a_supported]
    bexI[of _ chain_effect_b_supported])
  (auto simp: chain_effect_a_supported_def chain_effect_b_supported_def
    chain_effect_a_def chain_effect_b_def base_object_def)

lemma chain_rationale_active_for_t6:
  "active_resonance_rationale chain_after_t5 7 8 {1, 2}"
  unfolding active_resonance_rationale_def
  using chain_after_t5_objects_exact
  by (auto simp: chain_object_defs base_object_def)

lemma chain_resonance_catalog_0_well_formed:
  "resonance_catalog_well_formed chain_after_t5
    chain_resonance_catalog_0"
  by (simp add: resonance_catalog_well_formed_def
    chain_resonance_catalog_0_def)

lemma chain_t6_request_conditions:
  "ClassifyResonance \<in> allowed_operations_of chain_grammar \<and>
    K_RES \<in> supported_kinds_of chain_grammar \<and>
    requested_resonance_object_id_of chain_resonance_request \<notin>
    object_id_of ` objects_of chain_after_t5 \<and>
    requested_resonance_version_id_of chain_resonance_request \<notin>
    version_ids chain_after_t5 \<union>
    historical_versions_of chain_after_t5 \<and>
    chain_resonance_catalog_0
    (requested_resonance_version_id_of chain_resonance_request) = None \<and>
    requested_source_effect_of chain_resonance_request \<noteq>
    requested_target_effect_of chain_resonance_request \<and>
    resonance_profile_model_of
    (requested_resonance_profile_of chain_resonance_request)
    \<in> models_of chain_after_t5 \<and>
    requested_resonance_perspective_of chain_resonance_request
    \<in> perspectives_of chain_after_t5 \<and>
    requested_resonance_context_of chain_resonance_request
    \<in> contexts_of chain_after_t5 \<and>
    requested_resonance_horizon_of chain_resonance_request > 0 \<and>
    0 \<le> requested_resonance_uncertainty_of chain_resonance_request \<and>
    requested_resonance_uncertainty_of chain_resonance_request \<le> 1 \<and>
    requested_resonance_rationale_of chain_resonance_request \<noteq> [] \<and>
    snapshot_id_of chain_after_t5 < 9"
  using chain_after_t5_objects_exact chain_after_t5_background_exact
  chain_after_t5_object_ids_exact chain_after_t5_version_ids_exact
  chain_after_t5_history_exact
  by (simp add: chain_grammar_def chain_resonance_request_def
    chain_resonance_catalog_0_def t6_resonance_profile_def)

```

```

lemma chain_t6_input_valid:
  "classify_resonance_input_valid chain_grammar chain_after_t5
   chain_effect_catalog_after_t5 chain_resonance_catalog_0
   chain_resonance_request 9"
  unfolding classify_resonance_input_valid_def
  using chain_t5_reachability_certificate
  chain_resonance_catalog_0_well_formed chain_effects_distinct_for_t6
  chain_effect_a_active_for_t6 chain_effect_b_active_for_t6
  chain_effect_vector_a_valid chain_effect_vector_b_valid
  t6_profile_well_formed chain_rationale_active_for_t6
  chain_t6_request_conditions
  by (auto simp: chain_resonance_request_def)

theorem chain_t6_step:
  "classify_resonance_step chain_grammar chain_after_t5
   chain_effect_catalog_after_t5 chain_resonance_catalog_0
   chain_resonance_request 9 chain_after_t6 chain_resonance_catalog"
  unfolding classify_resonance_step_def chain_after_t6_def
  chain_resonance_catalog_def
  using chain_t6_input_valid by simp

lemma chain_after_t6_certificates:
  "state_well_formed chain_grammar chain_after_t6 \<and>
   effect_catalog_well_formed chain_after_t6
   chain_effect_catalog_after_t5 \<and>
   resonance_catalog_well_formed chain_after_t6 chain_resonance_catalog \<and>
   active_history_disjoint chain_after_t6"
proof -
  have history_disjoint: "active_history_disjoint chain_after_t6"
  unfolding chain_after_t6_def
  using add_resonance_preserves_active_history_disjoint
  [OF chain_t6_input_valid] .
  show ?thesis
  using classify_resonance_preserves_state_well_formed[OF chain_t6_step]
  classify_resonance_preserves_effect_catalog[OF chain_t6_step]
  classify_resonance_preserves_catalog[OF chain_t6_step]
  history_disjoint
  by blast
qed

definition chain_relevance_catalog_0 :: relevance_catalog where
  "chain_relevance_catalog_0 = Map.empty"

definition chain_relevance_request_p0 :: relevance_request where
  "chain_relevance_request_p0 =
   \<lparr> requested_relevance_object_id_of = 8,
   requested_relevance_version_id_of = 10,
   requested_relevance_target_of = 7,
   requested_relevance_effect_vector_of = chain_effect_vector_a,
   requested_relevance_power_profile_of = t7_power_profile,
   requested_relevance_profile_of = t7_relevance_profile,
   requested_relevance_perspective_of = 0,
   requested_relevance_context_of = 20,
   requested_relevance_evaluation_time_of = 9,
   requested_relevance_uncertainty_of = 1 / 4,
   requested_relevance_perturbation_radius_of = 1 / 10,
   requested_relevance_reach_input_of = 4 / 5,
   requested_relevance_duration_input_of = 8,
   requested_relevance_rationale_refs_of = {1, 2},
   requested_relevance_explanation_of =
     'P0 evaluation on the reachable effect version' \<rparr>"

definition chain_relevance_request_p1 :: relevance_request where
  "chain_relevance_request_p1 =
   \<lparr> requested_relevance_object_id_of = 9,
   requested_relevance_version_id_of = 11,
   requested_relevance_target_of = 7,
   requested_relevance_effect_vector_of = chain_effect_vector_a,
   requested_relevance_power_profile_of = t7_power_profile,
   requested_relevance_profile_of = t7_relevance_profile,
   requested_relevance_perspective_of = Suc 0,
   requested_relevance_context_of = 20,
   requested_relevance_evaluation_time_of = 9,
   requested_relevance_uncertainty_of = 1 / 4,
   requested_relevance_perturbation_radius_of = 1 / 10,
   requested_relevance_reach_input_of = 4 / 5,
   requested_relevance_duration_input_of = 8,
   requested_relevance_rationale_refs_of = {1, 2},
   requested_relevance_explanation_of =
     'P1 evaluation on the same reachable effect version' \<rparr>"

```

```

definition chain_after_t7_p0 :: snapshot where
  "chain_after_t7_p0 =
    add_evaluated_relevance chain_after_t6 chain_relevance_request_p0 10"

definition chain_relevance_catalog_p0 :: relevance_catalog where
  "chain_relevance_catalog_p0 =
    extend_relevance_catalog chain_relevance_catalog_0
    chain_relevance_request_p0"

definition chain_after_t7 :: snapshot where
  "chain_after_t7 =
    add_evaluated_relevance chain_after_t7_p0 chain_relevance_request_p1 11"

definition chain_relevance_catalog :: relevance_catalog where
  "chain_relevance_catalog =
    extend_relevance_catalog chain_relevance_catalog_p0
    chain_relevance_request_p1"

lemma active_classifiable_effect_add_resonance_mono:
  assumes active:
    "active_classifiable_effect s catalog version vector context"
  shows "active_classifiable_effect
    (add_classified_resonance s request next_snapshot_id)
    catalog version vector context"
  using active
  unfolding active_classifiable_effect_def
  by auto

lemma version_active_at_add_resonance_mono:
  assumes active: "version_active_at s version evaluation_time"
  shows "version_active_at
    (add_classified_resonance s request next_snapshot_id)
    version evaluation_time"
  using active
  unfolding version_active_at_def
  by auto

lemma active_relevance_rationale_add_resonance_mono:
  assumes active: "active_relevance_rationale s target refs"
  shows "active_relevance_rationale
    (add_classified_resonance s request next_snapshot_id) target refs"
  using active
  unfolding active_relevance_rationale_def
  by auto

lemma active_classifiable_effect_add_relevance_mono:
  assumes active:
    "active_classifiable_effect s catalog version vector context"
  shows "active_classifiable_effect
    (add_evaluated_relevance s request next_snapshot_id)
    catalog version vector context"
  using active
  unfolding active_classifiable_effect_def
  by auto

lemma version_active_at_add_relevance_mono:
  assumes active: "version_active_at s version evaluation_time"
  shows "version_active_at
    (add_evaluated_relevance s request next_snapshot_id)
    version evaluation_time"
  using active
  unfolding version_active_at_def
  by auto

lemma active_relevance_rationale_add_relevance_mono:
  assumes active: "active_relevance_rationale s target refs"
  shows "active_relevance_rationale
    (add_evaluated_relevance s request next_snapshot_id) target refs"
  using active
  unfolding active_relevance_rationale_def
  by auto

lemma chain_after_t6_object_ids_exact:
  "object_id_of ` objects_of chain_after_t6 = {0, 1, 2, 3, 4, 5, 6, 7}"
  unfolding chain_after_t6_def
  using chain_after_t5_object_ids_exact
  by (auto simp: chain_resonance_request_def)

lemma chain_after_t6_version_ids_exact:

```

```
"version_ids chain_after_t6 = {0, 1, 2, 3, 4, 7, 8, 9}"
unfolding chain_after_t6_def
using chain_after_t5_version_ids_exact
by (auto simp: chain_resonance_request_def)

lemma chain_after_t6_frame_exact:
  "historical_versions_of chain_after_t6 = {5, 6} \<and>
  perspectives_of chain_after_t6 = {0, Suc 0, 10} \<and>
  contexts_of chain_after_t6 = {20} \<and>
  models_of chain_after_t6 = {30} \<and>
  snapshot_id_of chain_after_t6 = 9"
unfolding chain_after_t6_def
using chain_after_t5_history_exact chain_after_t5_background_exact
by simp

lemma chain_effect_a_active_after_t6:
  "active_classifiable_effect chain_after_t6
  chain_effect_catalog_after_t5 7 chain_effect_vector_a 20"
unfolding chain_after_t6_def
using active_classifiable_effect_add_resonance_mono
  [OF chain_effect_a_active_for_t6] .

lemma chain_effect_a_version_active_after_t5:
  "version_active_at chain_after_t5 7 9"
unfolding version_active_at_def
proof (rule bexI[of _ chain_effect_a_supported])
  show "version_id_of chain_effect_a_supported = 7 \<and>
  lifecycle_of chain_effect_a_supported \<notin>
  {Revised, Rejected, Archived} \<and>
  valid_from_of chain_effect_a_supported \<le> 9 \<and>
  (case valid_to_of chain_effect_a_supported of
    None \<Rightarrow> True | Some terminal_time \<Rightarrow> 9 \<le> terminal_time)"
  by (simp add: chain_effect_a_supported_def chain_effect_a_def
    chain_assessment_a_def base_object_def integrated_effect_object_def
    integrated_lifecycle_def validation_ready_def)
  show "chain_effect_a_supported \<in> objects_of chain_after_t5"
  using chain_after_t5_objects_exact by simp
qed

lemma chain_effect_a_version_active_after_t6:
  "version_active_at chain_after_t6 7 9"
unfolding chain_after_t6_def
using version_active_at_add_resonance_mono
  [OF chain_effect_a_version_active_after_t5] .

lemma chain_relevance_rationale_after_t5:
  "active_relevance_rationale chain_after_t5 7 {1, 2}"
using chain_rationale_active_for_t6
unfolding active_resonance_rationale_def active_relevance_rationale_def
by blast

lemma chain_relevance_rationale_after_t6:
  "active_relevance_rationale chain_after_t6 7 {1, 2}"
unfolding chain_after_t6_def
using active_relevance_rationale_add_resonance_mono
  [OF chain_relevance_rationale_after_t5] .

lemma chain_relevance_catalog_0_well_formed:
  "relevance_catalog_well_formed chain_after_t6 chain_relevance_catalog_0"
by (simp add: relevance_catalog_well_formed_def
  chain_relevance_catalog_0_def)

lemma chain_t7_p0_request_conditions:
  "EvaluateRelevance \<in> allowed_operations_of chain_grammar \<and>
  K_REV \<in> supported_kinds_of chain_grammar \<and>
  requested_relevance_object_id_of chain_relevance_request_p0 \<notin>
  object_id_of ` objects_of chain_after_t6 \<and>
  requested_relevance_version_id_of chain_relevance_request_p0 \<notin>
  version_ids chain_after_t6 \<union> historical_versions_of chain_after_t6 \<and>
  chain_relevance_catalog_0
  (requested_relevance_version_id_of chain_relevance_request_p0) = None \<and>
  relevance_profile_model_of
  (requested_relevance_profile_of chain_relevance_request_p0)
  \<in> models_of chain_after_t6 \<and>
  requested_relevance_perspective_of chain_relevance_request_p0
  \<in> perspectives_of chain_after_t6 \<and>
  requested_relevance_perspective_of chain_relevance_request_p0
  \<in> relevance_profile_perspectives_of
  (requested_relevance_profile_of chain_relevance_request_p0) \<and>
  requested_relevance_context_of chain_relevance_request_p0"
```

```

    \<in> contexts_of chain_after_t6 \<and>
    requested_relevance_context_of chain_relevance_request_p0
    \<in> relevance_profile_contexts_of
    (requested_relevance_profile_of chain_relevance_request_p0) \<and>
    0 \<le> requested_relevance_uncertainty_of chain_relevance_request_p0 \<and>
    requested_relevance_uncertainty_of chain_relevance_request_p0 \<le> 1 \<and>
    0 \<le> requested_relevance_perturbation_radius_of
    chain_relevance_request_p0 \<and>
    requested_relevance_perturbation_radius_of chain_relevance_request_p0
    \<le> 1 \<and>
    0 \<le> requested_relevance_reach_input_of chain_relevance_request_p0 \<and>
    requested_relevance_reach_input_of chain_relevance_request_p0 \<le> 1 \<and>
    requested_relevance_duration_input_of chain_relevance_request_p0 > 0 \<and>
    requested_relevance_explanation_of chain_relevance_request_p0 \<noteq> [] \<and>
    snapshot_id_of chain_after_t6 < 10"
using chain_after_t6_object_ids_exact chain_after_t6_version_ids_exact
    chain_after_t6_frame_exact
by (simp add: chain_grammar_def chain_relevance_request_p0_def
    chain_relevance_catalog_0_def t7_relevance_profile_def)

lemma chain_t7_p0_input_valid:
"evaluate_relevance_input_valid chain_grammar chain_after_t6
    chain_effect_catalog_after_t5 chain_relevance_catalog_0
    chain_relevance_request_p0 10"
unfolding evaluate_relevance_input_valid_def
using chain_after_t6_certificates chain_relevance_catalog_0_well_formed
    chain_effect_a_active_after_t6 chain_effect_a_version_active_after_t6
    chain_effect_vector_a_valid t7_power_profile_valid
    t7_relevance_profile_well_formed chain_relevance_rationale_after_t6
    chain_t7_p0_request_conditions
by (auto simp: chain_relevance_request_p0_def)

theorem chain_t7_p0_step:
"evaluate_relevance_step chain_grammar chain_after_t6
    chain_effect_catalog_after_t5 chain_relevance_catalog_0
    chain_relevance_request_p0 10 chain_after_t7_p0
    chain_relevance_catalog_p0"
unfolding evaluate_relevance_step_def chain_after_t7_p0_def
    chain_relevance_catalog_p0_def
using chain_t7_p0_input_valid by simp

lemma chain_after_t7_p0_certificates:
"state_well_formed chain_grammar chain_after_t7_p0 \<and>
    effect_catalog_well_formed chain_after_t7_p0
    chain_effect_catalog_after_t5 \<and>
    relevance_catalog_well_formed chain_after_t7_p0
    chain_relevance_catalog_p0 \<and>
    active_history_disjoint chain_after_t7_p0"
proof -
  have history_disjoint: "active_history_disjoint chain_after_t7_p0"
  unfolding chain_after_t7_p0_def
  using add_relevance_preserves_active_history_disjoint
    [OF chain_t7_p0_input_valid] .
  show ?thesis
  using evaluate_relevance_preserves_state_well_formed[OF chain_t7_p0_step]
    evaluate_relevance_preserves_effect_catalog[OF chain_t7_p0_step]
    evaluate_relevance_preserves_catalog[OF chain_t7_p0_step]
    history_disjoint
  by blast
qed

lemma chain_after_t7_p0_object_ids_exact:
"object_id_of ` objects_of chain_after_t7_p0 =
    {0, 1, 2, 3, 4, 5, 6, 7, 8}"
unfolding chain_after_t7_p0_def
using chain_after_t6_object_ids_exact
by (auto simp: chain_relevance_request_p0_def)

lemma chain_after_t7_p0_version_ids_exact:
"version_ids chain_after_t7_p0 = {0, 1, 2, 3, 4, 7, 8, 9, 10}"
unfolding chain_after_t7_p0_def
using chain_after_t6_version_ids_exact
by (auto simp: chain_relevance_request_p0_def)

lemma chain_after_t7_p0_frame_exact:
"historical_versions_of chain_after_t7_p0 = {5, 6} \<and>
    perspectives_of chain_after_t7_p0 = {0, Suc 0, 10} \<and>
    contexts_of chain_after_t7_p0 = {20} \<and>
    models_of chain_after_t7_p0 = {30} \<and>
    snapshot_id_of chain_after_t7_p0 = 10"

```

```

unfolding chain_after_t7_p0_def
using chain_after_t6_frame_exact
by simp

lemma chain_effect_a_active_after_t7_p0:
  "active_classifiable_effect chain_after_t7_p0
    chain_effect_catalog_after_t5 7 chain_effect_vector_a 20"
unfolding chain_after_t7_p0_def
using active_classifiable_effect_add_relevance_mono
[0F chain_effect_a_active_after_t6] .

lemma chain_effect_a_version_active_after_t7_p0:
  "version_active_at chain_after_t7_p0 7 9"
unfolding chain_after_t7_p0_def
using version_active_at_add_relevance_mono
[0F chain_effect_a_version_active_after_t6] .

lemma chain_relevance_rationale_after_t7_p0:
  "active_relevance_rationale chain_after_t7_p0 7 {1, 2}"
unfolding chain_after_t7_p0_def
using active_relevance_rationale_add_relevance_mono
[0F chain_relevance_rationale_after_t6] .

lemma chain_t7_p1_request_conditions:
  "EvaluateRelevance \<in> allowed_operations_of chain_grammar \<and>
    K_REV \<in> supported_kinds_of chain_grammar \<and>
    requested_relevance_object_id_of chain_relevance_request_p1 \<notin>
      object_id_of ` objects_of chain_after_t7_p0 \<and>
    requested_relevance_version_id_of chain_relevance_request_p1 \<notin>
      version_ids chain_after_t7_p0 \<union>
        historical_versions_of chain_after_t7_p0 \<and>
    chain_relevance_catalog_p0
      (requested_relevance_version_id_of chain_relevance_request_p1) = None \<and>
    relevance_profile_model_of
      (requested_relevance_profile_of chain_relevance_request_p1)
      \<in> models_of chain_after_t7_p0 \<and>
    requested_relevance_perspective_of chain_relevance_request_p1
      \<in> perspectives_of chain_after_t7_p0 \<and>
    requested_relevance_perspective_of chain_relevance_request_p1
      \<in> relevance_profile_perspectives_of
        (requested_relevance_profile_of chain_relevance_request_p1) \<and>
    requested_relevance_context_of chain_relevance_request_p1
      \<in> contexts_of chain_after_t7_p0 \<and>
    requested_relevance_context_of chain_relevance_request_p1
      \<in> relevance_profile_contexts_of
        (requested_relevance_profile_of chain_relevance_request_p1) \<and>
    0 \<le> requested_relevance_uncertainty_of chain_relevance_request_p1 \<and>
    requested_relevance_uncertainty_of chain_relevance_request_p1 \<le> 1 \<and>
    0 \<le> requested_relevance_perturbation_radius_of
      chain_relevance_request_p1 \<and>
    requested_relevance_perturbation_radius_of chain_relevance_request_p1
      \<le> 1 \<and>
    0 \<le> requested_relevance_reach_input_of chain_relevance_request_p1 \<and>
    requested_relevance_reach_input_of chain_relevance_request_p1 \<le> 1 \<and>
    requested_relevance_duration_input_of chain_relevance_request_p1 > 0 \<and>
    requested_relevance_explanation_of chain_relevance_request_p1 \<noteq> [] \<and>
    snapshot_id_of chain_after_t7_p0 < 11"
using chain_after_t7_p0_object_ids_exact
chain_after_t7_p0_version_ids_exact chain_after_t7_p0_frame_exact
by (simp add: chain_grammar_def chain_relevance_request_p1_def
  chain_relevance_catalog_p0_def chain_relevance_catalog_0_def
  chain_relevance_request_p0_def extend_relevance_catalog_def
  t7_relevance_profile_def)

lemma chain_t7_p1_input_valid:
  "evaluate_relevance_input_valid chain_grammar chain_after_t7_p0
    chain_effect_catalog_after_t5 chain_relevance_catalog_p0
    chain_relevance_request_p1 11"
unfolding evaluate_relevance_input_valid_def
using chain_after_t7_p0_certificates chain_effect_a_active_after_t7_p0
chain_effect_a_version_active_after_t7_p0 chain_effect_vector_a_valid
t7_power_profile_valid t7_relevance_profile_well_formed
chain_relevance_rationale_after_t7_p0 chain_t7_p1_request_conditions
by (auto simp: chain_relevance_request_p1_def)

theorem chain_t7_p1_step:
  "evaluate_relevance_step chain_grammar chain_after_t7_p0
    chain_effect_catalog_after_t5 chain_relevance_catalog_p0
    chain_relevance_request_p1 11 chain_after_t7 chain_relevance_catalog"
unfolding evaluate_relevance_step_def chain_after_t7_def

```



```

    chain_relevance_catalog_def
    using chain_t7_p1_input_valid by simp

theorem operational_chain_T1_T7:
  "register_appearance_step chain_grammar chain_seed chain_entity_a 1
   chain_after_t1 \<and>
   differentiate_potential_step chain_grammar chain_after_t1
   {chain_entity_b} 2 chain_after_t2 \<and>
   create_relation_step chain_grammar chain_after_t2
   chain_relation_catalog_0 chain_relation_ab RelGenerates True 1 2 3
   chain_after_t3_ab chain_relation_catalog_ab \<and>
   create_relation_step chain_grammar chain_after_t3_ab
   chain_relation_catalog_ab chain_relation_ba RelConditions True 2 1 4
   chain_after_t3 chain_relation_catalog \<and>
   hypothesize_effect_step chain_grammar chain_after_t3
   chain_relation_catalog chain_effect_catalog_0 chain_effect_a
   chain_effect_payload_a 5 chain_after_t4_a chain_effect_catalog_a \<and>
   hypothesize_effect_step chain_grammar chain_after_t4_a
   chain_relation_catalog chain_effect_catalog_a chain_effect_b
   chain_effect_payload_b 6 chain_after_t4 chain_effect_catalog \<and>
   integrate_evidence_strict_step chain_grammar chain_after_t4
   chain_relation_catalog chain_effect_catalog chain_object_archive_0
   chain_payload_archive_0 chain_evidence_ledger_0 chain_effect_a
   chain_effect_payload_a chain_assessment_a 7 7 7 chain_after_t5_a
   chain_effect_catalog_after_t5_a chain_object_archive_after_t5_a
   chain_payload_archive_after_t5_a chain_evidence_ledger_after_t5_a \<and>
   integrate_evidence_strict_step chain_grammar chain_after_t5_a
   chain_relation_catalog chain_effect_catalog_after_t5_a
   chain_object_archive_after_t5_a chain_payload_archive_after_t5_a
   chain_evidence_ledger_after_t5_a chain_effect_b chain_effect_payload_b
   chain_assessment_b 8 8 8 chain_after_t5
   chain_effect_catalog_after_t5 chain_object_archive
   chain_payload_archive chain_evidence_ledger \<and>
   classify_resonance_step chain_grammar chain_after_t5
   chain_effect_catalog_after_t5 chain_resonance_catalog_0
   chain_resonance_request 9 chain_after_t6 chain_resonance_catalog \<and>
   evaluate_relevance_step chain_grammar chain_after_t6
   chain_effect_catalog_after_t5 chain_relevance_catalog_0
   chain_relevance_request_p0 10 chain_after_t7_p0
   chain_relevance_catalog_p0 \<and>
   evaluate_relevance_step chain_grammar chain_after_t7_p0
   chain_effect_catalog_after_t5 chain_relevance_catalog_p0
   chain_relevance_request_p1 11 chain_after_t7
   chain_relevance_catalog \<and>
   state_well_formed chain_grammar chain_after_t7 \<and>
   effect_catalog_well_formed chain_after_t7
   chain_effect_catalog_after_t5 \<and>
   relevance_catalog_well_formed chain_after_t7
   chain_relevance_catalog"
  using chain_t1_t3_reachability_certificate
  chain_t4_reachability_certificate chain_t5_reachability_certificate
  chain_t6_step chain_t7_p0_step chain_t7_p1_step
  evaluate_relevance_preserves_state_well_formed[OF chain_t7_p1_step]
  evaluate_relevance_preserves_effect_catalog[OF chain_t7_p1_step]
  evaluate_relevance_preserves_catalog[OF chain_t7_p1_step]
  by blast

theorem operational_chain_has_no_literal_predecessor_gap:
  "source_ref_of chain_relation_ab = Some 1 \<and>
   target_ref_of chain_relation_ab = Some 2 \<and>
   effect_origin_of chain_effect_a = Some 3 \<and>
   effect_origin_of chain_effect_b = Some 4 \<and>
   previous_version_of chain_effect_a_supported = Some 5 \<and>
   previous_version_of chain_effect_b_supported = Some 6 \<and>
   requested_source_effect_of chain_resonance_request = 7 \<and>
   requested_target_effect_of chain_resonance_request = 8 \<and>
   requested_relevance_target_of chain_relevance_request_p0 = 7 \<and>
   requested_relevance_target_of chain_relevance_request_p1 = 7"
  by (simp add: chain_relation_ab_def chain_effect_a_def chain_effect_b_def
    base_object_def chain_effect_a_supported_def
    chain_effect_b_supported_def chain_assessment_a_def
    chain_assessment_b_def chain_resonance_request_def
    chain_relevance_request_p0_def chain_relevance_request_p1_def)

end

```

## I.11 Artefaktbindung

| Datei                                                        | SHA-256                                                          | Statusrolle                  |
|--------------------------------------------------------------|------------------------------------------------------------------|------------------------------|
| ROOT                                                         | 67e307416726674528020909e2e2da8ef25afddd5604d0fda5f3a326d1855312 | session/order                |
| UZA_0_9_1_Integrate_Evidence.thy                             | bc5a22bd8e35090d4ea856e9ce4e9406c01fdab9cc453124d53da658f142fcb8 | formal source                |
| UZA_0_9_1_Create_Relation_Witness.thy                        | 335c4eecd67ed7efda2437182950d20401ef7bb06549aa8416a8523ae8e5c065 | formal source                |
| UZA_0_9_1_Hypothesize_Effect_Witness.thy                     | 8171771842b5c8d38c64ecd340af3591ca3d922854dc1596d8bb0d96315e2c89 | formal source                |
| UZA_0_9_1_Integrate_Evidence_Witness.thy                     | 703e8fa31740d410f40516eab50fe13b55e2e43559169b82dd7ba109de5db3df | formal source                |
| UZA_0_9_1_Operational_Chain_Witness.thy                      | f7d8f3612665629da92f34875a39ca6c8e453f61933501316164c663c1ffbdcd | formal source                |
| UZA_T1_T7_Isabelle_Source_Audit.json                         | 0ce98a583bac886e8e6756de8aa617deab428237a54781318029bf6728ea4764 | verification evidence        |
| UZA_T1_T7_Operational_Chain_Golden_Run.json                  | 7fac746c14ebedfab0ab2a989288b526a0fded9031fbe8e6f77cb466ddbdb62d | verification evidence        |
| Isabelle2025-2_Clean_Build_K1_16_Theories_FINAL.log          | 7f6cabe3e6823947d55f5197e1c50a90b742a82de37f78c58f6a3afbd1775c85 | final session build evidence |
| Isabelle2025-2_Clean_Build_K1_16_Theories_FINAL.exitcode.txt | 9a271f2a916b0b6ee6cecb2426f0b3206ef074578be55d9bc94f6f3fe3ab86aa | exit-code evidence           |

Normative Statusregel: Der Exit-0-Nachweis gilt ausschließlich für die im finalen Buildlog vorangestellten Hashes. Weicht ROOT oder eine Theorie ab, ist erneut ein Clean-Build erforderlich. Der statische Audit und der Golden Run sind zusätzliche, aber nicht ersetzende Nachweise.

## I.12 Isabelle2025-2-Clean-Build des finalen K1-Hashstands

Der am 9. August 2026 ausgeführte normale Sessionbuild verarbeitete die vollständige 16-Theorien-ROOT-Kette mit Isabelle2025-2. Verwendet wurden ein Clean-Build, ein Buildjob, ein ML-Thread und deaktivierte parallele Beweise. Die Session endete mit Exit-Code 0; UZA\_0\_9\_1\_Operational\_Chain\_Witness wurde zu 100 % verarbeitet.

Der reale Lauf führte zu einer zusätzlichen formalen Bereinigung: T3- und T4-Herkunftszeugen wurden explizit, T5 wurde in benannte Freshness-, Evidenz-, Katalog- und Historienlemmata zerlegt, und T6/T7 verwenden nun endliche Objekt-/Versionsmengen sowie monotone Erhaltung statt ressourcenintensiver Gesamtentfaltung. Keine Vorbedingung wurde zur Erzielung des Builds abgeschwächt.

Statusgrenze: Der Build belegt Parser-, Typ- und Kernelakzeptanz der abgeschlossenen Beweise dieser Dateien. A1–A22 bleiben Locale-Annahmen; empirische Domänengültigkeit und die offenen Modell-Expansions-, Konsistenz- und Konservativitätsziele bleiben getrennte Aufgaben.

```
2026-08-09T14:50:51+02:00
Isabelle2025-2
67e307416726674528020909e2e2da8ef25afddd5604d0fda5f3a326d1855312  ROOT
21dcfb175d6e754adf20042b941cf4a23329831d0165ebaf2023b0048213ed24  UZA_0_9_1.thy
322d2a8fc31dbca1ea6c7cc2463b24c891701a79eebb761a94f5256a62632a4b  UZA_0_9_1_Base_Model.thy
3f3583b42f066eaf41208a22bcb53a2d49baed374aa1b41e00a434256741b1cd  UZA_0_9_1_Classify_Resonance.thy
494da00eeb17205001ae85090d940a12cd07215ec283d76833aa480126a716ed
UZA_0_9_1_Classify_Resonance_Witness.thy
a3c801eaf1640624ee7bf41ba91a20d2ba1e000e489ee40f4d3c209b15911ca3  UZA_0_9_1_Create_Relation.thy
335c4eecd67ed7efda2437182950d20401ef7bb06549aa8416a8523ae8e5c065  UZA_0_9_1_Create_Relation_Witness.thy
ef483b279d0581752bfff655db6672fa57823dac348ace29268e68e486a40f9b  UZA_0_9_1_Differentiate_Potential.thy
6377ca4dd963832712350d10d8ad7442a1245b9e4ee76e85c43f6634bc8ecc46  UZA_0_9_1_Evaluate_Relevance.thy
3ba9e93262c0b139cf0d8521e8785ef21b66fb30de3e9e9bfa9de3989e817f39
UZA_0_9_1_Evaluate_Relevance_Witness.thy
2eb8e1d262cb3790f6c446839fb6809de3c2837d20705925507e272c860e2fc4  UZA_0_9_1_Hypothesize_Effect.thy
8171771842b5c8d38c64ecd340af3591ca3d922854dc1596d8bb0d96315e2c89
UZA_0_9_1_Hypothesize_Effect_Witness.thy
bc5a22bd8e35090d4ea856e9ce4e9406c01fdab9cc453124d53da658f142fcb8  UZA_0_9_1_Integrate_Evidence.thy
703e8fa31740d410f40516eab50fe13b55e2e43559169b82dd7ba109de5db3df
UZA_0_9_1_Integrate_Evidence_Witness.thy
f7d8f3612665629da92f34875a39ca6c8e453f61933501316164c663c1ffbdcd
UZA_0_9_1_Operational_Chain_Witness.thy
9b8cc6167b99070a47f63b4e63bd5a783a000f6ec84f7620945bee224f116cd0  UZA_0_9_1_Register_Appearance.thy
3a3671b918d9639607fb6152dedd039cdf6145b0f040db8edf6ba296b8dc9a8c  UZA_0_9_1_Rich_Model.thy
Started at Sun Aug 9 14:50:52 GMT+2 2026 (polym1-5.9.2_x86_64_32-linux on 4ebd932c4d12)
ISABELLE_TOOL_JAVA_OPTIONS="-Djava.awt.headless=true -Xms512m -Xmx4g -Xss16m"
ISABELLE_BUILD_OPTIONS=""

ML_PLATFORM="x86_64_32-linux"
ML_HOME="/workspace/scratch/24663cf25c52/isabelle_runtime/extracted/Isabelle2025-2/contrib/polym1-5.9.2-2/x86_64_32-linux"
ML_SYSTEM="polym1-5.9.2"
ML_OPTIONS="--minheap 500"

Session Pure/Pure
Session Misc/Tools
Session HOL/HOL (main)
Session Unsorted/UZA_0_9_1
Cleaned UZA_0_9_1
Running UZA_0_9_1 ...
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Base_Model
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Rich_Model
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1 100% (26.613s cumulated time)
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Base_Model 100% (0.308s cumulated time)
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Register_Appearance
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Differentiate_Potential
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Register_Appearance 100% (0.502s cumulated time)
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Rich_Model 100% (2.335s cumulated time)
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Create_Relation
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Differentiate_Potential 100% (1.004s cumulated time)
```

```
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Create_Relation_Witness
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Hypothesize_Effect
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Create_Relation 100% (1.463s cumulated time)
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Create_Relation_Witness 100% (0.845s cumulated time)
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Hypothesize_Effect_Witness
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Integrate_Evidence
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Hypothesize_Effect 100% (1.108s cumulated time)
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Hypothesize_Effect_Witness 100% (1.170s cumulated time)
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Classify_Resonance
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Integrate_Evidence 100% (2.519s cumulated time)
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Classify_Resonance_Witness
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Classify_Resonance 100% (2.535s cumulated time)
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Evaluate_Relevance
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Classify_Resonance_Witness 100% (2.019s cumulated time)
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Evaluate_Relevance_Witness
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Evaluate_Relevance 100% (3.160s cumulated time)
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Integrate_Evidence_Witness
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Evaluate_Relevance_Witness 100% (3.149s cumulated time)
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Operational_Chain_Witness
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Integrate_Evidence_Witness 100% (2.745s cumulated time)
UZA_0_9_1: theory UZA_0_9_1.UZA_0_9_1_Operational_Chain_Witness 100% (2.469s cumulated time)
Timing UZA_0_9_1 (1 threads, 55.021s elapsed time, 55.420s cpu time, 6.632s GC time, factor 1.01)
Finished UZA_0_9_1 (0:00:56 elapsed time, 0:00:56 cpu time, factor 1.00)

Finished at Sun Aug 9 14:51:52 GMT+2 2026
0:00:59 elapsed time, 0:00:56 cpu time, factor 0.95
```

SHA-256 des unveränderten finalen Buildlogs:

7f6cabe3e6823947d55f5197e1c50a90b742a82de37f78c58f6a3afbd1775c85. SHA-256 der Exit-Code-Datei mit dem Inhalt 0:  
9a271f2a916b0b6ee6cecb2426f0b3206ef074578be55d9bc94f6f3fe3ab86aa.

Historische K1-Freigabeentscheidung: Der gebundene T1–T7-Unterbau ist session-/kernelgeprüft. K2/Anhang J dokumentiert den anschließend ausgeführten T8-Abschluss. Jede Quelländerung löst weiterhin erneut das Clean-Build-Gate aus.

## Anhang J – Formalrevision K2: operationaler T1–T8-Kettenabschluss

### J.1 Releasegegenstand und Entscheidungsgrund

K2 vollzieht den zuvor ausgewiesenen nächsten Operationsschritt T8 SynthesizeOrientation. Die Erweiterung wurde bewusst erst nach dem K1-Kettenabschluss vorgenommen: Als Quelle dient ausschließlich chain\_after\_t7, also der tatsächlich durch T1–T7 erzeugte Nachfolgerzustand. Weder ein Relations-, Effekt-, Resonanz- noch Relevanzobjekt wird für T8 per freiem Record-Literal als vermeintlicher Vorgänger injiziert.

K2 erweitert die K1-Grammatik kontrolliert um K\_ORI und GenerateOrientation. Die 16 Kerntheorien bleiben in der Session UZA\_0\_9\_1\_Core gebunden; die zwei T8-Theorien liegen in der abhängigen Session UZA\_0\_9\_1\_T8. Diese Trennung macht die Grammatikextension sichtbar und verhindert, dass der T8-Prüfstatus rückwirkend als Eigenschaft eines älteren K1-Hashstands ausgegeben wird.

#### K2-Freigabe in einem Satz

Der exakte T1–T8-Quellstand wurde am 15. August 2026 mit Isabelle2025-2 vollständig bereinigt und neu gebaut; beide Sessions und alle 18 Theorien wurden mit Exit-Code 0 abgeschlossen. Dieser Befund gilt nur für die gebundenen Hashes und nicht als Beweis der A1–A22, empirischer Universalität oder relativer Konsistenz.

### J.2 Normative T8-Semantik

T8 erzeugt keine kontextfreie Entscheidung und keinen arithmetischen Kompromisswert. Eine orientation\_basis ist eine geordnete, nichtleere und duplikatfreie Liste aktiver K\_REV-Versionen. Alle Basisobjekte müssen denselben Zielbezug, Kontext, Modellstand und Synthesezeitpunkt erfüllen; die Eigentümerperspektive der resultierenden K\_ORI-Version bleibt explizit.

| Element | Normative Festlegung                                                                 | Verlustschutz                                                |
|---------|--------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Basis   | Nichtleer, distinct, aktive Supported/Validated-K_REV-Versionen                      | Keine stille Doppelzählung oder tote Quelle                  |
| Bindung | Gleiches Ziel, gleicher Kontext, gleiches Relevanzmodell, explizite Synthesezeit     | Keine unmarkierte Kontext- oder Versionsmischung             |
| Range   | Min/Max der Intensität, maximale Unsicherheit, minimale Stabilität, Divergenz        | Perspektivspanne bleibt sichtbar; kein impliziter Mittelwert |
| Profil  | Mindestbasis, Mindestperspektiven, Unsicherheits-, Stabilitäts- und Divergenzgrenzen | Qualitätsgrenzen sind versionierbar und prüfbar              |
| Erfolg  | Applicable + Validated                                                               | Nur bei erfüllten deklarativen Gates                         |
| Abbruch | InsufficientBasis + Draft bei unzureichender, aber gültiger Basis                    | Kein stiller Erfolg; Lineage und Begründung bleiben erhalten |

Die generische step-Relation wird nicht ersetzt. `synthesize_orientation_step` ist eine konkrete Verfeinerung von `GenerateOrientation` und beweist exaktes Delta, Identitäts- und Zustandswohlgeformtheit, Quellintegrität sowie Erhaltung von Relevanz- und Orientierungskatalog.

### J.3 Kausaler Erfolgs- und Abbruchzeuge

Der Erfolgszweig verwendet die in T7 erzeugten Relevanzversionen [10, 11]. Beide verweisen auf Zielversion 7, Kontext 20, Modell 30 und den Syntheszeitpunkt 11; die Perspektiven sind {0, 1}. Die mechanisch hergeleitete Range lautet: Intensitätsuntergrenze 9/20, Intensitätsobergrenze 3/4, Unsicherheitsobergrenze 1/4, Stabilitätsuntergrenze 19/20 und Divergenz 3/10. Das Profil akzeptiert genau diese Grenzwerte und erzeugt K\_ORI-Version 12 mit `Applicable/Validated`.

Der Abbruchzweig beginnt beim identischen `chain_after_t7` und verwendet nur die gültige Version [10]. Er scheitert ausdrücklich an Mindestbasis und Mindestperspektiven. T8 erzeugt dennoch ein auditierbares K\_ORI-Draft mit Status `InsufficientBasis`, Basisreferenz {10} und Begründung. Damit ist das Scheitern ein modellierter Ausgang, nicht die Abwesenheit eines Nachweises.

| Zweig   | Basis    | Gates                                                                    | Ergebnis                               | Mechanischer Nachweis                                                 |
|---------|----------|--------------------------------------------------------------------------|----------------------------------------|-----------------------------------------------------------------------|
| Erfolg  | [10, 11] | 2 Quellen, 2 Perspektiven; $U \leq 1/4$ ; $S \geq 19/20$ ; $D \leq 3/10$ | <code>Applicable / Validated</code>    | <code>chain_t8_success_step</code> und vier Erfolgscertificates       |
| Abbruch | [10]     | Mindestbasis und Mindestperspektiven verletzt                            | <code>InsufficientBasis / Draft</code> | <code>chain_t8_deferred_step</code> ; <code>not_silent_success</code> |

### J.4 Leistungs- und Statusmatrix

| Gegenstand                | Status K2       | Was tatsächlich geleistet wurde                                                                      | Was nicht daraus folgt                                                      |
|---------------------------|-----------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| T8-Syntax und Typisierung | Abgeschlossen   | Beide T8-Dateien in Isabelle2025-2 akzeptiert                                                        | Keine empirische Semantikvalidierung                                        |
| T8-Zustandserhaltung      | Abgeschlossen   | <code>state_well_formed</code> , Identität, exaktes Delta und Kataloge für gültige Eingaben bewiesen | Kein Beweis für ungültige oder externe Implementierungen                    |
| T8-Erfüllbarkeit          | Abgeschlossen   | Kausaler Erfolgs- und <code>InsufficientBasis</code> -Zeuge ab <code>chain_after_t7</code>           | Keine Vollständigkeit aller Domänenfälle                                    |
| T1–T8-Kette               | Abgeschlossen   | <code>operational_chain_T1_T8</code> nennt jeden erzeugenden Schritt explizit                        | Keine Produktivengine-Äquivalenz                                            |
| A1–A22                    | Locale-Annahmen | Typisiert und im gebundenen Quellstand akzeptiert                                                    | Nicht aus HOL bewiesen; Kernel prüft keine empirische Wahrheit              |
| Modelltheorie             | Offen           | Minimaler und reichhaltiger Zeuge bleiben vorhanden                                                  | Keine allgemeine Modell-Expansion, relative Konsistenz oder Konservativität |

| Gegenstand                        | Status K2                          | Was tatsächlich geleistet wurde                                                                | Was nicht daraus folgt                     |
|-----------------------------------|------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------|
| Golden Run                        | Bestanden                          | 12 Schritte bis Snapshot 12; rationale Werte und beide T8-Ausgänge geprüft                     | Conformance-Orakel, kein HOL-Beweis        |
| Historischer K2-<br>Stand: T9–T13 | damals offen; in K3<br>geschlossen | K3: vollständiger 25-<br>Theorien-Clean-Build,<br>Audit 27/27 und<br>kausaler T1–T13-<br>Zeuge | Aktueller<br>Freigabenachweis:<br>Anhang K |

## J.5 Isabelle-, Audit- und Golden-Run-Nachweis

Der finale Lauf verwendete Isabelle2025-2, einen Buildjob, einen ML-Thread, deaktivierte parallele Beweise und die Option -c. Isabelle meldete zuerst Cleaned UZA\_0\_9\_1\_Core und Cleaned UZA\_0\_9\_1\_T8; anschließend wurden beide Sessions vollständig neu verarbeitet.

| Nachweis              | Ergebnis                                                 | Gebundener Beleg                                                                                                                                                                                                                                   |
|-----------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Clean-Build           | 18/18 Theorien; beide Sessions Finished; Exit-Code 0     | Buildlog SHA-256<br>883e0b7439e8976ae65d5074eab18a8cc<br>465c454cb448218189c4357a9c9c527                                                                                                                                                           |
| Statischer Audit      | 18/18 Release-Gates; keine Import- oder Source-Errors    | Audit SHA-256<br>bae83cd268ad55d54b5fddf3008f59d62ae<br>baf549419628e3ca990843b5d7755                                                                                                                                                              |
| Golden Run            | passed; 12 Schritte; Snapshot 12                         | Ergebnis SHA-256<br>fd0ea546974cff25fcfd0a8066162d33eff8b<br>412658fdd2beb6c308a907ae0db                                                                                                                                                           |
| T8-Semantik           | 1.141 Zeilen; 22 Definitionen. 13<br>Lemmata. 9 Theoreme | SHA-256<br>e51c38d02b0370d1f86645149ae99ed5b1<br>c331426213825f3bd11830258196d2                                                                                                                                                                    |
| T8-Zeuge              | 535 Zeilen; 9 Definitionen. 31 Lemmata. 7<br>Theoreme    | SHA-256<br>feb039dbaa051c55eb4f4fd2203235333a<br>eb8a8b613ba3854f8bd5887637b4c                                                                                                                                                                     |
| Sessionbeschreibungen | Core + abhängige T8-Session                              | ROOT<br>cd07eca2e188282b9e69655735fc730c75f<br>27dcc06e01154a67bebc89f0b6b52;<br>ROOTS<br>ff25e8f495feeba78f4e8241c0d9f755510d<br>32b022bc60d2f7912a445e2c5a15;<br>T8/ROOT<br>d25472960e982739b16618b13d4bd3720<br>7c7e9582a86ac1f00dd5e87a2f383eb |

### Auszug aus dem unveränderten finalen Buildlog:

```
Started at Sat Aug 15 21:24:22 GMT+2 2026 (polym1-5.9.2_x86_64_32-linux on d714722bb59b)
Cleaned UZA_0_9_1_Core
Cleaned UZA_0_9_1_T8
Building UZA_0_9_1_Core ...
Finished UZA_0_9_1_Core (0:01:09 elapsed time, 0:01:09 cpu time, factor 1.00)
Running UZA_0_9_1_T8 ...
UZA_0_9_1_T8: theory UZA_0_9_1_T8.UZA_0_9_1_Synthesize_Orientation
UZA_0_9_1_T8: theory UZA_0_9_1_T8.UZA_0_9_1_Synthesize_Orientation_Witness
```

```
UZA_0_9_1_T8: theory UZA_0_9_1_T8.UZA_0_9_1_Synthesize_Orientation 100% (2.651s cumulated time)
UZA_0_9_1_T8: theory UZA_0_9_1_T8.UZA_0_9_1_Synthesize_Orientation_Witness 100% (1.628s cumulated time)
Finished UZA_0_9_1_T8 (0:00:05 elapsed time, 0:00:05 cpu time, factor 0.95)
Finished at Sat Aug 15 21:25:41 GMT+2 2026
Exit-Code: 0
```

J.6 Historische K2-Statusgrenze

Normative Statusgrenze

„Isabelle2025-2-session-/kernelgeprüft“ bedeutet hier ausschließlich: Die exakt gehashten ROOT-/ROOTS-Dateien und 18 Theoriequellen wurden vom ausgewiesenen Isabelle-System geparkt, typisiert und hinsichtlich der enthaltenen abgeschlossenen Beweise vom Kernel akzeptiert. A1–A22 sind Annahmen des Locale uza\_model. Der Build beweist weder diese Annahmen noch ihre empirische Angemessenheit, allgemeine relative Konsistenz, Konservativität, Vollständigkeit, Universalität oder Übereinstimmung mit einer künftigen Engine.

Weiterhin offen und release-relevant sind:

- ein nicht-degenerierter Modell-Expansionssatz, der die relevanten Kern- und Interaktionsannahmen gemeinsam nicht-vakuos aktiviert;
- ein präziser relativer Konsistenznachweis gegenüber einer explizit benannten Basistheorie;
- optional ein Konservativitätsnachweis für den definierten Kern beziehungsweise eine getrennte FOL-Reduktion;
- die definitorische oder axiomatisch kontrollierte Schließung weiterer zentraler Analyseprädikate, insbesondere Dynamikdetektoren und Describe-Anwendung auf reale Daten;
- historischer K2-Befund: T9–T13 waren offen; dieser Punkt ist durch den K3-Nachweis in Anhang K geschlossen.
- eine unabhängige Replay-Implementierung, die dieselben Fixtures ohne gemeinsame Implementierungslogik reproduziert.

J.7 Historischer K2-Blocker T9

Historischer K2-Befund: T9 SelectAction war der nächste Releaseblocker. K3 schließt T9 einschließlich der fünf Ablehnungszweige und führt die kausale Kette bis T13 fort; maßgeblich ist nun Anhang K.

| T9-Gate             | Mindestinhalt                                                                         | Releaseblocker                               |
|---------------------|---------------------------------------------------------------------------------------|----------------------------------------------|
| Autorität           | versionierte decision_authority mit Perspektive, Kontext, Zeit und Mandat             | Keine Auswahl ohne gültige Autorität         |
| Optionen            | nur explizit referenzierte Applicable-Orientierungen und zulässige Handlungsoptionen  | InsufficientBasis darf nicht auswählbar sein |
| Nebenbedingungen    | rechtliche, ethische, organisatorische und Ressourcen-Gates als deklarierte Prädikate | Kein stilles Überspringen                    |
| Macht/Verantwortung | PowerShift und Verantwortungszuordnung außerhalb einer verdeckten V10-Mittelung       | Keine unsichtbare Machtverlagerung           |



| T9-Gate   | Mindestinhalt                                                                               | Releaseblocker                           |
|-----------|---------------------------------------------------------------------------------------------|------------------------------------------|
| Abbruch   | NoAuthority, NoAdmissibleOption, InsufficientBasis, ConflictUnresolved oder RiskNotAccepted | Abbruch bleibt versioniert und begründet |
| Erhaltung | state_well_formed, Provenienz, Kataloge, Identität und generic-step-Refinement              | Kein T9-Release ohne Beweise und Zeugen  |

Historische Freigabereihenfolge von K2; in K3 vollständig ausgeführt: T9-Semantik und Negativpfade, kausaler T9–T13-Zeuge, statischer Audit, Golden Run und Isabelle2025-2-Clean-Build.

## J.8 Vollständige T8-Semantiktheorie

Normativer K2-Quellstand. Prüfstatus: Bestandteil des vollständigen Isabelle2025-2-Clean-Builds der 18-Theorien-Kette; Exit-Code 0.

```
theory UZA_0_9_1_Synthesize_Orientation
  imports UZA_0_9_1_Core.UZA_0_9_1_Evaluate_Relevance
begin

section \<open>T8 SynthesizeOrientation: loss-aware synthesis without implicit averaging\<close>

text \<open>
  The stable core operation constructor is GenerateOrientation. This theory
  gives it the stricter T8 reading SynthesizeOrientation: an ordered,
  duplicate-free basis of active K_REV versions is checked against an explicit
  model revision, context, target, time, and owner perspective. The synthesis
  preserves the full range of the basis. In particular, no arithmetic mean is
  computed and no perspective is silently replaced by a group perspective.

  A sufficient basis yields a Validated/Applicable K_ORI object. An
  insufficient but otherwise well-formed basis yields a Draft/
  InsufficientBasis object whose lineage is retained. These are alternative
  outcomes from the same source state, not sequential state changes.
\<close>

type_synonym orientation_basis = "version_id list"

record orientation_model_profile =
  orientation_profile_model_of :: model_id
  orientation_profile_revision_of :: nat
  orientation_min_basis_count_of :: nat
  orientation_min_perspective_count_of :: nat
  orientation_max_uncertainty_of :: real
  orientation_min_stability_of :: real
  orientation_max_divergence_of :: real

record orientation_range =
  orientation_intensity_floor_of :: real
  orientation_intensity_ceiling_of :: real
  orientation_uncertainty_ceiling_of :: real
  orientation_stability_floor_of :: real
  orientation_divergence_of :: real

record orientation_request =
  requested_orientation_object_id_of :: object_id
  requested_orientation_version_id_of :: version_id
  requested_orientation_basis_of :: orientation_basis
  requested_orientation_target_of :: version_id
  requested_orientation_owner_of :: perspective_id
  requested_orientation_context_of :: context_id
  requested_orientation_profile_of :: orientation_model_profile
  requested_orientation_time_of :: time_point
  requested_orientation_explanation_of :: string

record orientation_payload =
  synthesized_orientation_basis_of :: orientation_basis
  synthesized_orientation_target_of :: version_id
  synthesized_orientation_owner_of :: perspective_id
  synthesized_orientation_context_of :: context_id
  synthesized_orientation_profile_of :: orientation_model_profile
  synthesized_orientation_time_of :: time_point
  synthesized_orientation_range_of :: orientation_range
  synthesized_orientation_perspectives_of :: "perspective_id set"
  synthesized_orientation_status_of :: analysis_status
  synthesized_orientation_explanation_of :: string

type_synonym orientation_catalog = "version_id \<rightarrow harpoonup> orientation_payload"
```

```

subsection \<open>Range semantics and declared quality gates\<close>

definition relevance_vector_at ::
  "relevance_catalog \<Rightarrow> version_id \<Rightarrow> relevance_vector" where
  "relevance_vector_at catalog version =
    evaluated_relevance_vector_of (the (catalog version))"

definition relevance_perspective_at ::
  "relevance_catalog \<Rightarrow> version_id \<Rightarrow> perspective_id" where
  "relevance_perspective_at catalog version =
    evaluated_relevance_perspective_of (the (catalog version))"

definition orientation_intensities ::
  "relevance_catalog \<Rightarrow> orientation_basis \<Rightarrow> real set" where
  "orientation_intensities catalog basis =
    relevance_intensity_of ` relevance_vector_at catalog ` set basis"

definition orientation_uncertainties ::
  "relevance_catalog \<Rightarrow> orientation_basis \<Rightarrow> real set" where
  "orientation_uncertainties catalog basis =
    relevance_uncertainty_of ` relevance_vector_at catalog ` set basis"

definition orientation_stabilities ::
  "relevance_catalog \<Rightarrow> orientation_basis \<Rightarrow> real set" where
  "orientation_stabilities catalog basis =
    relevance_stability_of ` relevance_vector_at catalog ` set basis"

definition orientation_basis_perspectives ::
  "relevance_catalog \<Rightarrow> orientation_basis \<Rightarrow> perspective_id set" where
  "orientation_basis_perspectives catalog basis =
    relevance_perspective_at catalog ` set basis"

definition calculate_orientation_range ::
  "relevance_catalog \<Rightarrow> orientation_basis \<Rightarrow> orientation_range" where
  "calculate_orientation_range catalog basis =
    (let intensities = orientation_intensities catalog basis;
      uncertainties = orientation_uncertainties catalog basis;
      stabilities = orientation_stabilities catalog basis;
      lower = Min intensities;
      upper = Max intensities
    in \<lparr>
      orientation_intensity_floor_of = lower,
      orientation_intensity_ceiling_of = upper,
      orientation_uncertainty_ceiling_of = Max uncertainties,
      orientation_stability_floor_of = Min stabilities,
      orientation_divergence_of = upper - lower
    \<rparrr>)"

definition orientation_model_profile_well_formed ::
  "orientation_model_profile \<Rightarrow> bool" where
  "orientation_model_profile_well_formed profile \<longlefttrightarrow>
    orientation_profile_revision_of profile > 0 \<and>
    orientation_min_basis_count_of profile > 0 \<and>
    orientation_min_perspective_count_of profile > 0 \<and>
    0 \<le> orientation_max_uncertainty_of profile \<and>
    orientation_max_uncertainty_of profile \<le> 1 \<and>
    0 \<le> orientation_min_stability_of profile \<and>
    orientation_min_stability_of profile \<le> 1 \<and>
    0 \<le> orientation_max_divergence_of profile \<and>
    orientation_max_divergence_of profile \<le> 1"

subsection \<open>Active, catalogued, same-target basis\<close>

definition active_orientation_relevance ::
  "snapshot \<Rightarrow> relevance_catalog \<Rightarrow> version_id \<Rightarrow>
    version_id \<Rightarrow> context_id \<Rightarrow> model_id \<Rightarrow>
    time_point \<Rightarrow> bool" where
  "active_orientation_relevance s catalog version target context model
    synthesis_time \<longlefttrightarrow>
    (\<exists>relevance payload.
      relevance \<in> objects_of s \<and>
      version_id_of relevance = version \<and>
      kind_of relevance = K_REV \<and>
      lifecycle_of relevance \<in> {Supported, Validated} \<and>
      catalog version = Some payload \<and>
      relevance_payload_valid relevance payload \<and>
      evaluated_relevance_target_of payload = target \<and>
      evaluated_relevance_context_of payload = context \<and>
      relevance_profile_model_of (evaluated_relevance_profile_of payload) = model \<and>
      version_active_at s version synthesis_time)"

definition orientation_basis_valid ::
  "snapshot \<Rightarrow> relevance_catalog \<Rightarrow> orientation_basis \<Rightarrow>
    version_id \<Rightarrow> context_id \<Rightarrow> model_id \<Rightarrow>
    time_point \<Rightarrow> bool" where
  "orientation_basis_valid s catalog basis target context model synthesis_time
    \<longlefttrightarrow>
    basis \<noteq> [] \<and>
    distinct basis \<and>
    (\<forall>version\<in>set basis.
      active_orientation_relevance s catalog version target context model
        synthesis_time)"

```

```

definition orientation_basis_sufficient ::
  "relevance_catalog \<Rightarrow> orientation_model_profile \<Rightarrow>
  orientation_basis \<Rightarrow> bool" where
  "orientation_basis_sufficient catalog profile basis \<longlefttrightarrow>
  length basis \<ge> orientation_min_basis_count_of profile \<and>
  card (orientation_basis_perspectives catalog basis) \<ge>
  orientation_min_perspective_count_of profile \<and>
  orientation_uncertainty_ceiling_of
  (calculate_orientation_range catalog basis) \<le>
  orientation_max_uncertainty_of profile \<and>
  orientation_stability_floor_of
  (calculate_orientation_range catalog basis) \<ge>
  orientation_min_stability_of profile \<and>
  orientation_divergence_of (calculate_orientation_range catalog basis) \<le>
  orientation_max_divergence_of profile"

definition synthesized_orientation_status ::
  "relevance_catalog \<Rightarrow> orientation_request \<Rightarrow> analysis_status" where
  "synthesized_orientation_status catalog request =
  (if orientation_basis_sufficient catalog
    (requested_orientation_profile_of request)
    (requested_orientation_basis_of request)
  then Applicable else InsufficientBasis)"

definition synthesized_orientation_lifecycle ::
  "relevance_catalog \<Rightarrow> orientation_request \<Rightarrow> lifecycle_status" where
  "synthesized_orientation_lifecycle catalog request =
  (if orientation_basis_sufficient catalog
    (requested_orientation_profile_of request)
    (requested_orientation_basis_of request)
  then Validated else Draft)"

subsection \<open>Objects, payloads, catalogs, and transition\<close>

definition synthesized_orientation_references ::
  "orientation_request \<Rightarrow> version_id set" where
  "synthesized_orientation_references request =
  insert (requested_orientation_target_of request)
  (set (requested_orientation_basis_of request))"

definition synthesized_orientation_object ::
  "relevance_catalog \<Rightarrow> orientation_request \<Rightarrow> uza_object" where
  "synthesized_orientation_object catalog request =
  \<lparr>
    object_id_of = requested_orientation_object_id_of request,
    version_id_of = requested_orientation_version_id_of request,
    kind_of = K_ORI,
    lifecycle_of = synthesized_orientation_lifecycle catalog request,
    previous_version_of = None,
    references_of = synthesized_orientation_references request,
    evidence_of = set (requested_orientation_basis_of request),
    perspective_of = Some (requested_orientation_owner_of request),
    context_of = Some (requested_orientation_context_of request),
    model_of = Some
      (orientation_profile_model_of (requested_orientation_profile_of request)),
    authority_of = None,
    valid_from_of = requested_orientation_time_of request,
    valid_to_of = None,
    source_ref_of = Some (requested_orientation_target_of request),
    target_ref_of = None,
    effect_origin_of = None,
    basis_refs_of = set (requested_orientation_basis_of request),
    execution_ref_of = None,
    observation_refs_of = {},
    object_status_of = Some (synthesized_orientation_status catalog request)
  \<rparrr>"

definition synthesized_orientation_payload ::
  "relevance_catalog \<Rightarrow> orientation_request \<Rightarrow> orientation_payload" where
  "synthesized_orientation_payload catalog request =
  \<lparr>
    synthesized_orientation_basis_of = requested_orientation_basis_of request,
    synthesized_orientation_target_of = requested_orientation_target_of request,
    synthesized_orientation_owner_of = requested_orientation_owner_of request,
    synthesized_orientation_context_of = requested_orientation_context_of request,
    synthesized_orientation_profile_of = requested_orientation_profile_of request,
    synthesized_orientation_time_of = requested_orientation_time_of request,
    synthesized_orientation_range_of =
      calculate_orientation_range catalog
      (requested_orientation_basis_of request),
    synthesized_orientation_perspectives_of =
      orientation_basis_perspectives catalog
      (requested_orientation_basis_of request),
    synthesized_orientation_status_of =
      synthesized_orientation_status catalog request,
    synthesized_orientation_explanation_of =
      requested_orientation_explanation_of request
  \<rparrr>"

definition orientation_payload_valid ::
  "relevance_catalog \<Rightarrow> uza_object \<Rightarrow> orientation_payload \<Rightarrow> bool" where
  "orientation_payload_valid relevance_catalog orientation_payload \<longlefttrightarrow>
  kind_of orientation = K_ORI \<and>
  synthesized_orientation_basis_of payload \<noteq> [] \<and>
```

```

distinct (synthesized_orientation_basis_of payload) \<and>
references_of orientation =
  insert (synthesized_orientation_target_of payload)
  (set (synthesized_orientation_basis_of payload)) \<and>
evidence_of orientation = set (synthesized_orientation_basis_of payload) \<and>
basis_refs_of orientation = set (synthesized_orientation_basis_of payload) \<and>
perspective_of orientation = Some (synthesized_orientation_owner_of payload) \<and>
context_of orientation = Some (synthesized_orientation_context_of payload) \<and>
model_of orientation = Some
  (orientation_profile_model_of (synthesized_orientation_profile_of payload)) \<and>
valid_from_of orientation = synthesized_orientation_time_of payload \<and>
synthesized_orientation_range_of payload =
  calculate_orientation_range relevance_catalog
  (synthesized_orientation_basis_of payload) \<and>
synthesized_orientation_perspectives_of payload =
  orientation_basis_perspectives relevance_catalog
  (synthesized_orientation_basis_of payload) \<and>
orientation_model_profile_well_formed
  (synthesized_orientation_profile_of payload) \<and>
object_status_of orientation = Some (synthesized_orientation_status_of payload) \<and>
((synthesized_orientation_status_of payload = Applicable \<and>
  lifecycle_of orientation = Validated) \<or>
  (synthesized_orientation_status_of payload = InsufficientBasis \<and>
    lifecycle_of orientation = Draft)) \<and>
synthesized_orientation_explanation_of payload \<noteq> []"

definition orientation_catalog_well_formed ::
  "snapshot \<Rightarrow> relevance_catalog \<Rightarrow> orientation_catalog \<Rightarrow> bool" where
  "orientation_catalog_well_formed s relevance_catalog catalog \<longleftarrowrightarrow>
    (\<forall>version payload. catalog version = Some payload \<longrightarrow>
      (\<exists>orientation\<in>objects_of s.
        version_id_of orientation = version \<and>
        orientation_payload_valid relevance_catalog orientation payload \<and>
        references_complete s orientation \<and>
        evidence_well_formed s orientation)))"

definition synthesize_orientation_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> relevance_catalog \<Rightarrow>
    orientation_catalog \<Rightarrow> orientation_request \<Rightarrow> nat \<Rightarrow> bool" where
  "synthesize_orientation_input_valid G s relevance_catalog orientation_catalog
    request next_snapshot_id \<longleftarrowrightarrow>
    state_well_formed G s \<and>
    active_history_disjoint s \<and>
    relevance_catalog_well_formed s relevance_catalog \<and>
    orientation_catalog_well_formed s relevance_catalog orientation_catalog \<and>
    GenerateOrientation \<in> allowed_operations_of G \<and>
    K_ORI \<in> supported_kinds_of G \<and>
    requested_orientation_object_id_of request \<notin>
      object_id_of `objects_of s \<and>
    requested_orientation_version_id_of request \<notin>
      version_ids s \<union> historical_versions_of s \<and>
    orientation_catalog (requested_orientation_version_id_of request) = None \<and>
    orientation_model_profile_well_formed
      (requested_orientation_profile_of request) \<and>
    orientation_profile_model_of (requested_orientation_profile_of request)
      \<in> models_of s \<and>
    requested_orientation_owner_of request \<in> perspectives_of s \<and>
    requested_orientation_context_of request \<in> contexts_of s \<and>
    requested_orientation_target_of request \<in> version_ids s \<and>
    orientation_basis_valid s relevance_catalog
      (requested_orientation_basis_of request)
      (requested_orientation_target_of request)
      (requested_orientation_context_of request)
      (orientation_profile_model_of (requested_orientation_profile_of request))
      (requested_orientation_time_of request) \<and>
    requested_orientation_explanation_of request \<noteq> [] \<and>
    snapshot_id_of s < next_snapshot_id"

definition add_synthesized_orientation ::
  "snapshot \<Rightarrow> relevance_catalog \<Rightarrow> orientation_request \<Rightarrow>
    nat \<Rightarrow> snapshot" where
  "add_synthesized_orientation s relevance_catalog request next_snapshot_id =
    add_appearance s (synthesized_orientation_object relevance_catalog request)
    next_snapshot_id"

definition extend_orientation_catalog ::
  "orientation_catalog \<Rightarrow> relevance_catalog \<Rightarrow>
    orientation_request \<Rightarrow> orientation_catalog" where
  "extend_orientation_catalog catalog relevance_catalog request =
    catalog(requested_orientation_version_id_of request \<mapsto>
      synthesized_orientation_payload relevance_catalog request)"

definition synthesize_orientation_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> relevance_catalog \<Rightarrow>
    orientation_catalog \<Rightarrow> orientation_request \<Rightarrow> nat \<Rightarrow>
    snapshot \<Rightarrow> orientation_catalog \<Rightarrow> bool" where
  "synthesize_orientation_step G s relevance_catalog orientation_catalog request
    next_snapshot_id s' orientation_catalog' \<longleftarrowrightarrow>
    synthesize_orientation_input_valid G s relevance_catalog orientation_catalog
    request next_snapshot_id \<and>
    s' = add_synthesized_orientation s relevance_catalog request next_snapshot_id \<and>
    orientation_catalog' =
      extend_orientation_catalog orientation_catalog relevance_catalog request"

```

```
subsection \<open>Exact delta, provenance, and source integrity\<close>

lemma add_orientation_objects [simp]:
  "objects_of (add_synthesized_orientation s catalog request next_snapshot_id) =
    insert (synthesized_orientation_object catalog request) (objects_of s)"
  by (simp add: add_synthesized_orientation_def)

lemma add_orientation_snapshot_id [simp]:
  "snapshot_id_of
    (add_synthesized_orientation s catalog request next_snapshot_id) =
    next_snapshot_id"
  by (simp add: add_synthesized_orientation_def)

lemma add_orientation_background [simp]:
  "historical_versions_of
    (add_synthesized_orientation s catalog request next_snapshot_id) =
    historical_versions_of s"
  "perspectives_of
    (add_synthesized_orientation s catalog request next_snapshot_id) =
    perspectives_of s"
  "contexts_of (add_synthesized_orientation s catalog request next_snapshot_id) =
    contexts_of s"
  "models_of (add_synthesized_orientation s catalog request next_snapshot_id) =
    models_of s"
  "authorities_of
    (add_synthesized_orientation s catalog request next_snapshot_id) =
    authorities_of s"
  "contains_edges_of
    (add_synthesized_orientation s catalog request next_snapshot_id) =
    contains_edges_of s"
  "projections_of
    (add_synthesized_orientation s catalog request next_snapshot_id) =
    projections_of s"
  "couplings_of
    (add_synthesized_orientation s catalog request next_snapshot_id) =
    couplings_of s"
  "transmissions_of
    (add_synthesized_orientation s catalog request next_snapshot_id) =
    transmissions_of s"
  "boundaries_of
    (add_synthesized_orientation s catalog request next_snapshot_id) =
    boundaries_of s"
  by (simp_all add: add_synthesized_orientation_def)

lemma add_orientation_version_ids [simp]:
  "version_ids
    (add_synthesized_orientation s catalog request next_snapshot_id) =
    insert (requested_orientation_version_id_of request) (version_ids s)"
  by (auto simp: version_ids_def add_synthesized_orientation_def
    synthesized_orientation_object_def)

lemma synthesized_orientation_object_simps [simp]:
  "object_id_of (synthesized_orientation_object catalog request) =
    requested_orientation_object_id_of request"
  "version_id_of (synthesized_orientation_object catalog request) =
    requested_orientation_version_id_of request"
  "kind_of (synthesized_orientation_object catalog request) = K_ORI"
  "lifecycle_of (synthesized_orientation_object catalog request) =
    synthesized_orientation_lifecycle catalog request"
  "references_of (synthesized_orientation_object catalog request) =
    synthesized_orientation_references request"
  "evidence_of (synthesized_orientation_object catalog request) =
    set (requested_orientation_basis_of request)"
  "perspective_of (synthesized_orientation_object catalog request) =
    Some (requested_orientation_owner_of request)"
  "context_of (synthesized_orientation_object catalog request) =
    Some (requested_orientation_context_of request)"
  "model_of (synthesized_orientation_object catalog request) = Some
    (orientation_profile_model_of (requested_orientation_profile_of request))"
  "valid_from_of (synthesized_orientation_object catalog request) =
    requested_orientation_time_of request"
  "source_ref_of (synthesized_orientation_object catalog request) =
    Some (requested_orientation_target_of request)"
  "basis_refs_of (synthesized_orientation_object catalog request) =
    set (requested_orientation_basis_of request)"
  "object_status_of (synthesized_orientation_object catalog request) =
    Some (synthesized_orientation_status catalog request)"
  by (simp_all add: synthesized_orientation_object_def)

lemma synthesized_orientation_payload_simps [simp]:
  "synthesized_orientation_basis_of
    (synthesized_orientation_payload catalog request) =
    requested_orientation_basis_of request"
  "synthesized_orientation_target_of
    (synthesized_orientation_payload catalog request) =
    requested_orientation_target_of request"
  "synthesized_orientation_owner_of
    (synthesized_orientation_payload catalog request) =
    requested_orientation_owner_of request"
  "synthesized_orientation_context_of
    (synthesized_orientation_payload catalog request) =
    requested_orientation_context_of request"
  "synthesized_orientation_profile_of
    (synthesized_orientation_payload catalog request) =
```

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    requested_orientation_profile_of request"
"synthesized_orientation_time_of
  (synthesized_orientation_payload catalog request) =
  requested_orientation_time_of request"
"synthesized_orientation_range_of
  (synthesized_orientation_payload catalog request) =
  calculate_orientation_range_catalog (requested_orientation_basis_of request)"
"synthesized_orientation_perspectives_of
  (synthesized_orientation_payload catalog request) =
  orientation_basis_perspectives_catalog
  (requested_orientation_basis_of request)"
"synthesized_orientation_status_of
  (synthesized_orientation_payload catalog request) =
  synthesized_orientation_status_catalog request"
"synthesized_orientation_explanation_of
  (synthesized_orientation_payload catalog request) =
  requested_orientation_explanation_of request"
by (simp_all add: synthesized_orientation_payload_def)

lemma orientation_basis_versions:
  assumes valid:
    "orientation_basis_valid s catalog basis target context model synthesis_time"
  shows "set basis \<subseq> version_ids s"
proof
  fix version
  assume member: "version \<in> set basis"
  from valid member obtain relevance_payload where
    relevance_member: "relevance \<in> objects_of s" and
    relevance_id: "version_id_of relevance = version"
  unfolding orientation_basis_valid_def active_orientation_relevance_def
  by blast
  show "version \<in> version_ids s"
  unfolding version_ids_def
  using relevance_member relevance_id by (metis imageI)
qed

lemma synthesize_orientation_fresh_object:
  assumes valid:
    "synthesize_orientation_input_valid G s relevance_catalog orientation_catalog
    request next_snapshot_id"
  shows
    "synthesized_orientation_object relevance_catalog request \<notin> objects_of s \<and>
    requested_orientation_version_id_of request \<notin> version_ids s \<and>
    requested_orientation_object_id_of request \<notin>
    object_id_of `objects_of s"
proof -
  from valid have fresh_version:
    "requested_orientation_version_id_of request \<notin>
    version_ids s \<union> historical_versions_of s"
  and fresh_object:
    "requested_orientation_object_id_of request \<notin>
    object_id_of `objects_of s"
  unfolding synthesize_orientation_input_valid_def by blast+
  have fresh_record:
    "synthesized_orientation_object relevance_catalog request \<notin> objects_of s"
  proof
    assume member:
      "synthesized_orientation_object relevance_catalog request \<in> objects_of s"
    have "requested_orientation_object_id_of request \<in>
      object_id_of `objects_of s"
      using member by (metis synthesized_orientation_object_simps(1) imageI)
    then show False using fresh_object by contradiction
  qed
  show ?thesis using fresh_version fresh_object fresh_record by blast
qed

theorem synthesize_orientation_exact_delta:
  assumes concrete:
    "synthesize_orientation_step G s relevance_catalog orientation_catalog request
    next_snapshot_id s' orientation_catalog"
  shows
    "objects_of s' =
    insert (synthesized_orientation_object relevance_catalog request)
    (objects_of s) \<and>
    synthesized_orientation_object relevance_catalog request \<notin> objects_of s \<and>
    snapshot_id_of s' = next_snapshot_id \<and>
    orientation_catalog' (requested_orientation_version_id_of request) =
    Some (synthesized_orientation_payload relevance_catalog request)"
  using concrete synthesize_orientation_fresh_object
  by (auto simp: synthesize_orientation_step_def extend_orientation_catalog_def)

lemma synthesized_orientation_source_integrity:
  assumes valid:
    "synthesize_orientation_input_valid G s relevance_catalog orientation_catalog
    request next_snapshot_id"
  shows
    "references_complete s (synthesized_orientation_object relevance_catalog request) \<and>
    evidence_well_formed s (synthesized_orientation_object relevance_catalog request) \<and>
    time_well_formed (synthesized_orientation_object relevance_catalog request) \<and>
    perspective_well_formed s
    (synthesized_orientation_object relevance_catalog request) \<and>
    context_well_formed s
    (synthesized_orientation_object relevance_catalog request) \<and>
    orientation_well_formed s"

```



```

    (synthesized_orientation_object relevance_catalog request)"
proof -
  from valid have basis:
    "orientation_basis_valid s relevance_catalog
      (requested_orientation_basis_of request)
      (requested_orientation_target_of request)
      (requested_orientation_context_of request)
      (orientation_profile_model_of (requested_orientation_profile_of request))
      (requested_orientation_time_of request)"
  and target:
    "requested_orientation_target_of request \<in> version_ids s"
  and owner:
    "requested_orientation_owner_of request \<in> perspectives_of s"
  and context_ok:
    "requested_orientation_context_of request \<in> contexts_of s"
  unfolding synthesize_orientation_input_valid_def by blast+
  from basis have nonempty: "requested_orientation_basis_of request \<noteq> []"
  unfolding orientation_basis_valid_def by blast
  from orientation_basis_versions[OF basis] have basis_versions:
    "set (requested_orientation_basis_of request) \<subteq> version_ids s" .
  have refs:
    "references_complete s
      (synthesized_orientation_object relevance_catalog request)"
  unfolding references_complete_def synthesized_orientation_object_def
    synthesized_orientation_references_def
  using target basis_versions by auto
  have evidence:
    "evidence_well_formed s
      (synthesized_orientation_object relevance_catalog request)"
  unfolding evidence_well_formed_def evidence_references_complete_def
    synthesized_orientation_object_def synthesized_orientation_lifecycle_def
  using basis_versions nonempty by auto
  have time:
    "time_well_formed
      (synthesized_orientation_object relevance_catalog request)"
  by (simp add: time_well_formed_def synthesized_orientation_object_def)
  have perspective:
    "perspective_well_formed s
      (synthesized_orientation_object relevance_catalog request)"
  using owner
  by (simp add: perspective_well_formed_def perspective_bound_kind_def)
  have context_wf:
    "context_well_formed s
      (synthesized_orientation_object relevance_catalog request)"
  using context_ok
  by (simp add: context_well_formed_def context_bound_kind_def)
  have orientation:
    "orientation_well_formed s
      (synthesized_orientation_object relevance_catalog request)"
  using nonempty
  by (simp add: orientation_well_formed_def)
  show ?thesis using refs evidence time perspective context_wf orientation by blast
qed

lemma synthesized_orientation_payload_is_valid:
  assumes valid:
    "synthesize_orientation_input_valid G s relevance_catalog orientation_catalog
      request next_snapshot_id"
  shows
    "orientation_payload_valid relevance_catalog
      (synthesized_orientation_object relevance_catalog request)
      (synthesized_orientation_payload relevance_catalog request)"
proof -
  from valid have basis:
    "orientation_basis_valid s relevance_catalog
      (requested_orientation_basis_of request)
      (requested_orientation_target_of request)
      (requested_orientation_context_of request)
      (orientation_profile_model_of (requested_orientation_profile_of request))
      (requested_orientation_time_of request)"
  and profile:
    "orientation_model_profile_well_formed
      (requested_orientation_profile_of request)"
  and explanation:
    "requested_orientation_explanation_of request \<noteq> []"
  unfolding synthesize_orientation_input_valid_def by blast+
  from basis have nonempty: "requested_orientation_basis_of request \<noteq> []"
  and distinct: "distinct (requested_orientation_basis_of request)"
  unfolding orientation_basis_valid_def by blast+
  show ?thesis
  unfolding orientation_payload_valid_def
  using nonempty distinct profile explanation
  by (auto simp: synthesized_orientation_references_def
    synthesized_orientation_status_def synthesized_orientation_lifecycle_def
    split: if_splits)
qed

subsection \<open>Preservation and generic-step refinement\<close>

lemma add_orientation_preserves_active_history_disjoint:
  assumes valid:
    "synthesize_orientation_input_valid G s relevance_catalog orientation_catalog
      request next_snapshot_id"
  shows "active_history_disjoint

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```

    (add_synthesized_orientation s relevance_catalog request next_snapshot_id)"
  using valid
  unfolding synthesize_orientation_input_valid_def active_history_disjoint_def
  by auto

lemma add_orientation_preserves_unique_versions:
  assumes valid:
    "synthesize_orientation_input_valid G s relevance_catalog orientation_catalog
      request next_snapshot_id"
  shows "unique_versions
    (add_synthesized_orientation s relevance_catalog request next_snapshot_id)"
  proof -
    from valid have source_unique: "inj_on version_id_of (objects_of s)"
    and fresh:
      "requested_orientation_version_id_of request \<notin> version_ids s"
    unfolding synthesize_orientation_input_valid_def state_well_formed_def
    unique_versions_def by auto
    show ?thesis
    unfolding unique_versions_def
    proof (rule inj_onI)
      fix left right
      assume left_mem:
        "left \<in> objects_of
          (add_synthesized_orientation s relevance_catalog request next_snapshot_id)"
      and right_mem:
        "right \<in> objects_of
          (add_synthesized_orientation s relevance_catalog request next_snapshot_id)"
      and same: "version_id_of left = version_id_of right"
      let ?new = "synthesized_orientation_object relevance_catalog request"
      show "left = right"
      proof (cases "left = ?new")
        case True
        show ?thesis
        proof (cases "right = ?new")
          case True
          then show ?thesis using \<open>left = ?new\<close> by simp
        next
          case False
          then have old: "right \<in> objects_of s" using right_mem by simp
          have "requested_orientation_version_id_of request \<in> version_ids s"
            unfolding version_ids_def
            using same \<open>left = ?new\<close> old by (metis imageI
              synthesized_orientation_object_simps(2))
          then show ?thesis using fresh by blast
        qed
      next
      case False
      then have left_old: "left \<in> objects_of s" using left_mem by simp
      show ?thesis
      proof (cases "right = ?new")
        case True
        have "requested_orientation_version_id_of request \<in> version_ids s"
          unfolding version_ids_def
          using same left_old True by (metis imageI
            synthesized_orientation_object_simps(2))
        then show ?thesis using fresh by blast
      next
      case False
      then have right_old: "right \<in> objects_of s" using right_mem by simp
      show ?thesis using inj_onD[OF source_unique same left_old right_old] .
    qed
  qed
qed
qed
qed

lemma synthesize_orientation_new_object_conditions:
  assumes valid:
    "synthesize_orientation_input_valid G s relevance_catalog orientation_catalog
      request next_snapshot_id"
  shows
    "kind_of (synthesized_orientation_object relevance_catalog request)
      \<in> supported_kinds_of G \<and>
      references_complete
        (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
        (synthesized_orientation_object relevance_catalog request) \<and>
      evidence_well_formed
        (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
        (synthesized_orientation_object relevance_catalog request) \<and>
      time_well_formed (synthesized_orientation_object relevance_catalog request) \<and>
      relation_well_formed
        (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
        (synthesized_orientation_object relevance_catalog request) \<and>
      effect_well_formed
        (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
        (synthesized_orientation_object relevance_catalog request) \<and>
      perspective_well_formed
        (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
        (synthesized_orientation_object relevance_catalog request) \<and>
      context_well_formed
        (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
        (synthesized_orientation_object relevance_catalog request) \<and>
      relevance_well_formed
        (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
        (synthesized_orientation_object relevance_catalog request) \<and>"

```



```

orientation_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
  (synthesized_orientation_object relevance_catalog request) \<and>
action_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
  (synthesized_orientation_object relevance_catalog request) \<and>
feedback_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
  (synthesized_orientation_object relevance_catalog request)"
proof -
from valid have supported: "K_ORI \<in> supported_kinds_of G"
unfolding synthesize_orientation_input_valid_def by blast
from synthesized_orientation_source_integrity[OF valid] have
  refs: "references_complete s
  (synthesized_orientation_object relevance_catalog request)" and
  evidence: "evidence_well_formed s
  (synthesized_orientation_object relevance_catalog request)" and
  time: "time_well_formed
  (synthesized_orientation_object relevance_catalog request)" and
  perspective: "perspective_well_formed s
  (synthesized_orientation_object relevance_catalog request)" and
  context_wf: "context_well_formed s
  (synthesized_orientation_object relevance_catalog request)" and
  orientation: "orientation_well_formed s
  (synthesized_orientation_object relevance_catalog request)"
by blast+
have refs': "references_complete
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
  (synthesized_orientation_object relevance_catalog request)"
unfolding add_synthesized_orientation_def
using add_appearance_references_mono[OF refs] .
have evidence': "evidence_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
  (synthesized_orientation_object relevance_catalog request)"
unfolding add_synthesized_orientation_def
using add_appearance_evidence_mono[OF evidence] .
have perspective': "perspective_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
  (synthesized_orientation_object relevance_catalog request)"
unfolding add_synthesized_orientation_def
using perspective by simp
have context': "context_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
  (synthesized_orientation_object relevance_catalog request)"
unfolding add_synthesized_orientation_def
using context_wf by simp
have orientation': "orientation_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
  (synthesized_orientation_object relevance_catalog request)"
unfolding add_synthesized_orientation_def
using orientation by simp
show ?thesis
using supported refs' evidence' time perspective' context' orientation'
by (simp add: relation_well_formed_def effect_well_formed_def
  relevance_well_formed_def action_well_formed_def
  feedback_well_formed_def)
qed

theorem synthesize_orientation_preserves_state_well_formed:
  assumes concrete:
    "synthesize_orientation_step G s relevance_catalog orientation_catalog request
    next_snapshot_id s' orientation_catalog"
  shows "state_well_formed G s'"
proof -
from concrete have valid:
  "synthesize_orientation_input_valid G s relevance_catalog orientation_catalog
  request next_snapshot_id"
and target:
  "s' = add_synthesized_orientation s relevance_catalog request next_snapshot_id"
by (auto simp: synthesize_orientation_step_def)
from valid have source_wf: "state_well_formed G s"
unfolding synthesize_orientation_input_valid_def by blast
have unique: "unique_versions
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id)"
using add_orientation_preserves_unique_versions[OF valid] .
have all_objects:
  "\<forall>obj \<in> objects_of
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id).
  kind_of obj \<in> supported_kinds_of G \<and>
  references_complete
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
  evidence_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
  time_well_formed obj \<and>
  relation_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
  effect_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
  perspective_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
  context_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
  relevance_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>"

```

```

orientation_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
action_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
feedback_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj"
proof (intro ballI)
fix obj
assume member:
  "obj \<in> objects_of
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id)"
let ?new = "synthesized_orientation_object relevance_catalog request"
consider (new) "obj = ?new" | (old) "obj \<in> objects_of s"
using member by auto
then show
  "kind_of obj \<in> supported_kinds_of G \<and>
  references_complete
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
  evidence_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
  time_well_formed obj \<and>
  relation_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
  effect_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
  perspective_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
  context_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
  relevance_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
  orientation_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
  action_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj \<and>
  feedback_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) obj"
proof cases
case new
then show ?thesis
using synthesize_orientation_new_object_conditions[OF valid] by simp
next
case old
from source_wf old have source_conditions:
  "kind_of obj \<in> supported_kinds_of G \<and>
  references_complete s obj \<and> evidence_well_formed s obj \<and>
  time_well_formed obj \<and> relation_well_formed s obj \<and>
  effect_well_formed s obj \<and> perspective_well_formed s obj \<and>
  context_well_formed s obj \<and> relevance_well_formed s obj \<and>
  orientation_well_formed s obj \<and> action_well_formed s obj \<and>
  feedback_well_formed s obj"
unfolding state_well_formed_def by blast
show ?thesis
unfolding add_synthesized_orientation_def
using source_conditions add_appearance_references_mono
  add_appearance_evidence_mono add_appearance_relation_mono
  add_appearance_effect_mono add_appearance_feedback_mono
by simp
qed
qed
from source_wf have secondary:
  "(\<forall>p\<in>projections_of s. projection_well_formed s p) \<and>
  (\<forall>c\<in>couplings_of s. coupling_well_formed s c) \<and>
  (\<forall>t\<in>transmissions_of s. transmission_well_formed s t) \<and>
  (\<forall>b\<in>boundaries_of s. boundary_well_formed b)"
unfolding state_well_formed_def by blast
have secondary':
  "(\<forall>p\<in>projections_of
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id).
  projection_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) p) \<and>
  (\<forall>c\<in>couplings_of
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id).
  coupling_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) c) \<and>
  (\<forall>t\<in>transmissions_of
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id).
  transmission_well_formed
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id) t) \<and>
  (\<forall>b\<in>boundaries_of
  (add_synthesized_orientation s relevance_catalog request next_snapshot_id).
  boundary_well_formed b)"
using secondary
by (simp add: projection_well_formed_def coupling_well_formed_def
  transmission_well_formed_def)
show ?thesis
unfolding target_state_well_formed_def
using unique_all_objects secondary' by blast
qed

theorem synthesize_orientation_preserves_identity:
  assumes concrete:
    "synthesize_orientation_step G s relevance_catalog orientation_catalog request
    next_snapshot_id s' orientation_catalog"
```

```

shows "unique_versions s' \<and> immutable_preserved s s'"
proof -
  from concrete have valid:
    "synthesize_orientation_input_valid G s relevance_catalog orientation_catalog
      request next_snapshot_id"
  and target:
    "s' = add_synthesized_orientation s relevance_catalog request next_snapshot_id"
  by (auto simp: synthesize_orientation_step_def)
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
  and fresh:
    "requested_orientation_version_id_of request \<notin> version_ids s"
  unfolding synthesize_orientation_input_valid_def state_well_formed_def
    unique_versions_def by auto
  have immutable: "immutable_preserved s s'"
  unfolding immutable_preserved_def target
  proof (intro ballI allI impI)
    fix old_obj target_obj
    assume old_mem: "old_obj \<in> objects_of s"
    and target_mem:
      "target_obj \<in> objects_of
        (add_synthesized_orientation s relevance_catalog request next_snapshot_id)"
    and same: "version_id_of old_obj = version_id_of target_obj"
    let ?new = "synthesized_orientation_object relevance_catalog request"
    show "old_obj = target_obj"
    proof (cases "target_obj = ?new")
      case True
      have "requested_orientation_version_id_of request \<in> version_ids s"
      unfolding version_ids_def
      using old_mem same True by (metis imageI
        synthesized_orientation_object_simps(2))
      then show ?thesis using fresh by blast
    next
      case False
      then have old_target: "target_obj \<in> objects_of s" using target_mem by simp
      show ?thesis using inj_onD[OF source_unique same old_mem old_target] .
    qed
  qed
  have unique: "unique_versions s'"
  unfolding target
  using add_orientation_preserves_unique_versions[OF valid] .
  show ?thesis using unique immutable by blast
qed

theorem synthesize_orientation_preserves_relevance_catalog:
  assumes concrete:
    "synthesize_orientation_step G s relevance_catalog orientation_catalog request
      next_snapshot_id s' orientation_catalog"
  shows "relevance_catalog_well_formed s' relevance_catalog"
proof -
  from concrete have valid:
    "synthesize_orientation_input_valid G s relevance_catalog orientation_catalog
      request next_snapshot_id"
  and target:
    "s' = add_synthesized_orientation s relevance_catalog request next_snapshot_id"
  by (auto simp: synthesize_orientation_step_def)
  from valid have catalog_wf:
    "relevance_catalog_well_formed s relevance_catalog"
  unfolding synthesize_orientation_input_valid_def by blast
  show ?thesis
  unfolding relevance_catalog_well_formed_def target
  proof (intro allI impI)
    fix version payload
    assume entry: "relevance_catalog version = Some payload"
    from catalog_wf entry obtain relevance where
      member: "relevance \<in> objects_of s" and
      id: "version_id_of relevance = version" and
      payload: "relevance_payload_valid relevance payload" and
      refs: "references_complete s relevance" and
      evidence: "evidence_well_formed s relevance"
    unfolding relevance_catalog_well_formed_def by blast
    have refs_target: "references_complete
      (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
      relevance"
    unfolding add_synthesized_orientation_def
    using add_appearance_references_mono[OF refs] .
    have evidence_target: "evidence_well_formed
      (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
      relevance"
    unfolding add_synthesized_orientation_def
    using add_appearance_evidence_mono[OF evidence] .
    show "\<exists>relevance\<in>objects_of
      (add_synthesized_orientation s relevance_catalog request next_snapshot_id).
      version_id_of relevance = version \<and>
      relevance_payload_valid relevance payload \<and>
      references_complete
        (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
        relevance \<and>
      evidence_well_formed
        (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
        relevance"
    using member id payload refs_target evidence_target
    by (intro bexI[of _ relevance]) auto
  qed
qed

```

```

theorem synthesize_orientation_preserves_orientation_catalog:
  assumes concrete:
    "synthesize_orientation_step G s relevance_catalog orientation_catalog request
     next_snapshot_id s' orientation_catalog'"
  shows "orientation_catalog_well_formed s' relevance_catalog orientation_catalog'"
proof -
  from concrete have valid:
    "synthesize_orientation_input_valid G s relevance_catalog orientation_catalog
     request next_snapshot_id"
  and target:
    "s' = add_synthesized_orientation s relevance_catalog request next_snapshot_id"
  and catalog_target:
    "orientation_catalog' =
     extend_orientation_catalog orientation_catalog relevance_catalog request"
  by (auto simp: synthesize_orientation_step_def)
  from valid have source_catalog:
    "orientation_catalog_well_formed s relevance_catalog orientation_catalog"
  and fresh:
    "orientation_catalog (requested_orientation_version_id_of request) = None"
  unfolding synthesize_orientation_input_valid_def by blast+
  have new_valid:
    "orientation_payload_valid relevance_catalog
     (synthesized_orientation_object relevance_catalog request)
     (synthesized_orientation_payload relevance_catalog request)"
  using synthesized_orientation_payload_is_valid[OF valid] .
  from synthesized_orientation_source_integrity[OF valid] have new_refs:
    "references_complete s
     (synthesized_orientation_object relevance_catalog request)"
  and new_evidence:
    "evidence_well_formed s
     (synthesized_orientation_object relevance_catalog request)"
  by blast+
  show ?thesis
  unfolding orientation_catalog_well_formed_def target catalog_target
  extend_orientation_catalog_def
  proof (intro allI impI)
    fix version payload
    assume entry:
      "(orientation_catalog
       (requested_orientation_version_id_of request \<mapsto>
        synthesized_orientation_payload relevance_catalog request))
       version = Some payload"
    show "\<exists>orientation\<in>objects_of
     (add_synthesized_orientation s relevance_catalog request next_snapshot_id).
     version_id_of orientation = version \<and>
     orientation_payload_valid relevance_catalog orientation payload \<and>
     references_complete
     (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
     orientation \<and>
     evidence_well_formed
     (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
     orientation"
    proof (cases "version = requested_orientation_version_id_of request")
      case True
      then have payload_eq:
        "payload = synthesized_orientation_payload relevance_catalog request"
      using entry by simp
      show ?thesis
      unfolding True payload_eq add_synthesized_orientation_def
      using new_valid new_refs new_evidence add_appearance_references_mono
      add_appearance_evidence_mono by (intro bexI[of _
        "synthesized_orientation_object relevance_catalog request"]) auto
    next
      case False
      then have old_entry: "orientation_catalog version = Some payload"
      using entry by simp
      from source_catalog old_entry obtain orientation where
        member: "orientation \<in> objects_of s" and
        id: "version_id_of orientation = version" and
        payload_valid:
          "orientation_payload_valid relevance_catalog orientation payload" and
        refs: "references_complete s orientation" and
        evidence: "evidence_well_formed s orientation"
      unfolding orientation_catalog_well_formed_def by blast
      have refs_target: "references_complete
        (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
        orientation"
      unfolding add_synthesized_orientation_def
      using add_appearance_references_mono[OF refs] .
      have evidence_target: "evidence_well_formed
        (add_synthesized_orientation s relevance_catalog request next_snapshot_id)
        orientation"
      unfolding add_synthesized_orientation_def
      using add_appearance_evidence_mono[OF evidence] .
      show ?thesis
      using member id payload_valid refs_target evidence_target
      by (intro bexI[of _ orientation]) auto
    qed
  qed
qed

```

theorem synthesize\_orientation\_refines\_generic\_step:  
assumes concrete:

```

    "synthesize_orientation_step G s relevance_catalog orientation_catalog request
      next_snapshot_id s' orientation_catalog"
  shows "step G s GenerateOrientation s'"
proof -
  from concrete have source: "state_well_formed G s"
  and allowed: "GenerateOrientation \<in> allowed_operations_of G"
  by (auto simp: synthesize_orientation_step_def
    synthesize_orientation_input_valid_def)
  have target: "state_well_formed G s'"
  using synthesize_orientation_preserves_state_well_formed[OF concrete] .
  have immutable: "immutable_preserved s s'"
  using synthesize_orientation_preserves_identity[OF concrete] by blast
  show ?thesis
  unfolding step_def using source target allowed immutable by blast
qed

subsection \<open>Status and no-mean sentinels\<close>

theorem synthesize_orientation_applicable_iff:
  "synthesized_orientation_status catalog request = Applicable \<longleftarrow\rightarrow
    orientation_basis_sufficient catalog
    (requested_orientation_profile_of request)
    (requested_orientation_basis_of request)"
  by (simp add: synthesized_orientation_status_def)

theorem synthesize_orientation_insufficient_iff:
  "synthesized_orientation_status catalog request = InsufficientBasis \<longleftarrow\rightarrow
    \<not> orientation_basis_sufficient catalog
    (requested_orientation_profile_of request)
    (requested_orientation_basis_of request)"
  by (simp add: synthesized_orientation_status_def)

theorem orientation_range_retains_extrema:
  assumes nonempty: "basis \<noteq> []"
  shows
    "orientation_intensity_floor_of (calculate_orientation_range catalog basis) =
      Min (orientation_intensities catalog basis) \<and>
      orientation_intensity_ceiling_of (calculate_orientation_range catalog basis) =
      Max (orientation_intensities catalog basis) \<and>
      orientation_divergence_of (calculate_orientation_range catalog basis) =
      Max (orientation_intensities catalog basis) -
      Min (orientation_intensities catalog basis)"
  using nonempty by (simp add: calculate_orientation_range_def Let_def)
end

```

## J.9 Vollständiger T8-Zeuge

Normativer K2-Quellstand. Der Zeuge startet beim realen T7-Nachfolger, trennt Erfolg und InsufficientBasis und enthält operational\_chain\_T1\_T8. Prüfstatus: Bestandteil desselben Isabelle2025-2-Clean-Builds; Exit-Code 0.

```

theory UZA_0_9_1_Synthesize_Orientation_Witness
  imports UZA_0_9_1_Synthesize_Orientation
    UZA_0_9_1_Core.UZA_0_9_1_Operational_Chain_Witness
begin

section \<open>A causally reachable T8 witness and a blocked alternative\<close>

text \<open>
  Both branches start from chain_after_t7, the literal successor of the two
  T7 evaluations in the T1--T7 operational chain. No predecessor object is
  inserted by a record literal. The success branch uses relevance versions
  10 and 11. The deferred branch uses only version 10 and therefore fails the
  declared minimum-basis and minimum-perspective gates. The branches are
  alternatives from the same source snapshot.
\<close>

definition chain_t8_grammar :: grammar where
  "chain_t8_grammar =
    chain_grammar\<lparr>
      supported_kinds_of := insert K_ORI (supported_kinds_of chain_grammar),
      allowed_operations_of :=
        insert GenerateOrientation (allowed_operations_of chain_grammar)
    \<rparr>"

definition chain_orientation_profile :: orientation_model_profile where
  "chain_orientation_profile =
    \<lparr>
      orientation_profile_model_of = 30,
      orientation_profile_revision_of = 1,
      orientation_min_basis_count_of = 2,
      orientation_min_perspective_count_of = 2,
      orientation_max_uncertainty_of = 1 / 4,
      orientation_min_stability_of = 19 / 20,
      orientation_max_divergence_of = 3 / 10
    \<rparr>"

```

```

definition chain_orientation_catalog_0 :: orientation_catalog where
  "chain_orientation_catalog_0 = Map.empty"

definition chain_orientation_request :: orientation_request where
  "chain_orientation_request =
    \<lparr>
      requested_orientation_object_id_of = 10,
      requested_orientation_version_id_of = 12,
      requested_orientation_basis_of = [10, 11],
      requested_orientation_target_of = 7,
      requested_orientation_owner_of = 10,
      requested_orientation_context_of = 20,
      requested_orientation_profile_of = chain_orientation_profile,
      requested_orientation_time_of = 11,
      requested_orientation_explanation_of =
        'Two-perspective range-preserving orientation'
    \<rparr>"

definition chain_orientation_request_deferred :: orientation_request where
  "chain_orientation_request_deferred =
    chain_orientation_request\<lparr>
      requested_orientation_basis_of := [10],
      requested_orientation_explanation_of :=
        'Deferred because only one perspective is available'
    \<rparr>"

definition chain_after_t8 :: snapshot where
  "chain_after_t8 =
    add_synthesized_orientation chain_after_t7 chain_relevance_catalog
    chain_orientation_request 12"

definition chain_orientation_catalog :: orientation_catalog where
  "chain_orientation_catalog =
    extend_orientation_catalog chain_orientation_catalog_0
    chain_relevance_catalog chain_orientation_request"

definition chain_after_t8_deferred :: snapshot where
  "chain_after_t8_deferred =
    add_synthesized_orientation chain_after_t7 chain_relevance_catalog
    chain_orientation_request_deferred 12"

definition chain_orientation_catalog_deferred :: orientation_catalog where
  "chain_orientation_catalog_deferred =
    extend_orientation_catalog chain_orientation_catalog_0
    chain_relevance_catalog chain_orientation_request_deferred"

subsection \<open>Source-state and catalog certificates\<close>

lemma chain_after_t7_state_well_formed_core:
  "state_well_formed chain_grammar chain_after_t7"
  using operational_chain_T1_T7 by blast

lemma chain_after_t7_state_well_formed_t8:
  "state_well_formed chain_t8_grammar chain_after_t7"
  using chain_after_t7_state_well_formed_core
  unfolding state_well_formed_def chain_t8_grammar_def
  by auto

lemma chain_after_t7_active_history_disjoint:
  "active_history_disjoint chain_after_t7"
  unfolding chain_after_t7_def
  using add_relevance_preserves_active_history_disjoint[OF chain_t7_p1_input_valid] .

lemma chain_after_t7_relevance_catalog_well_formed:
  "relevance_catalog_well_formed chain_after_t7 chain_relevance_catalog"
  using operational_chain_T1_T7 by blast

lemma chain_orientation_catalog_0_well_formed:
  "orientation_catalog_well_formed chain_after_t7 chain_relevance_catalog
  chain_orientation_catalog_0"
  by (simp add: orientation_catalog_well_formed_def
  chain_orientation_catalog_0_def)

lemma chain_after_t7_object_ids_exact:
  "object_id_of ` objects_of chain_after_t7 =
    {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}"
  unfolding chain_after_t7_def
  using chain_after_t7_p0_object_ids_exact
  by (auto simp: chain_relevance_request_p1_def)

lemma chain_after_t7_version_ids_exact:
  "version_ids chain_after_t7 = {0, 1, 2, 3, 4, 7, 8, 9, 10, 11}"
  unfolding chain_after_t7_def
  using chain_after_t7_p0_version_ids_exact
  by (auto simp: chain_relevance_request_p1_def)

lemma chain_after_t7_frame_exact:
  "historical_versions_of chain_after_t7 = {5, 6} \<and>
  perspectives_of chain_after_t7 = {0, Suc 0, 10} \<and>
  contexts_of chain_after_t7 = {20} \<and>
  models_of chain_after_t7 = {30} \<and>
  snapshot_id_of chain_after_t7 = 11"
  unfolding chain_after_t7_def

```



```

using chain_after_t7_p0_frame_exact by simp

subsection \<open>Exact T7 vectors used by T8\<close>

lemma chain_p0_effect_term:
  "(if dimension = Connection then 1 else 0) *
   normalized_effect_component chain_effect_vector_a dimension =
   (if dimension = Connection then 1 else 0)"
  by (case_tac dimension)
  (simp_all add: normalized_effect_component_def chain_effect_vector_a_def)

lemma chain_p1_effect_term:
  "(if dimension = Stability then 1 else 0) *
   normalized_effect_component chain_effect_vector_a dimension =
   (if dimension = Stability then 1 / 2 else 0)"
  by (case_tac dimension)
  (simp_all add: normalized_effect_component_def chain_effect_vector_a_def)

lemma chain_p0_effect_component:
  "perspective_effect_component t7_relevance_profile 0 20
   chain_effect_vector_a = 1"
  by (simp add: perspective_effect_component_def
    effective_effect_weight_mass_def effective_effect_weight_def
    t7_relevance_profile_def chain_p0_effect_term
    t6_effect_dimension_delta clamp01_def)

lemma chain_p1_effect_component:
  "perspective_effect_component t7_relevance_profile (Suc 0) 20
   chain_effect_vector_a = 1 / 2"
  by (simp add: perspective_effect_component_def
    effective_effect_weight_mass_def effective_effect_weight_def
    t7_relevance_profile_def chain_p1_effect_term
    t6_effect_dimension_delta clamp01_def)

lemma chain_p0_intensity:
  "relevance_intensity t7_relevance_profile 0 20 9
   chain_effect_vector_a t7_power_profile (1 / 4) = 3 / 4"
  using chain_p0_effect_component t7_p0_power_component
  by (simp add: relevance_intensity_def t7_relevance_profile_def clamp01_def)

lemma chain_p1_intensity:
  "relevance_intensity t7_relevance_profile (Suc 0) 20 9
   chain_effect_vector_a t7_power_profile (1 / 4) = 9 / 20"
  using chain_p1_effect_component t7_p1_power_component
  by (simp add: relevance_intensity_def t7_relevance_profile_def clamp01_def)

lemma chain_relevance_vectors_exact:
  "relevance_intensity_of
   (requested_relevance_vector chain_relevance_request_p0) = 3 / 4 \<and>
   relevance_uncertainty_of
   (requested_relevance_vector chain_relevance_request_p0) = 1 / 4 \<and>
   relevance_stability_of
   (requested_relevance_vector chain_relevance_request_p0) = 49 / 50 \<and>
   relevance_intensity_of
   (requested_relevance_vector chain_relevance_request_p1) = 9 / 20 \<and>
   relevance_uncertainty_of
   (requested_relevance_vector chain_relevance_request_p1) = 1 / 4 \<and>
   relevance_stability_of
   (requested_relevance_vector chain_relevance_request_p1) = 19 / 20"
  using chain_p0_intensity chain_p1_intensity
  by (simp add: requested_relevance_vector_def
    chain_relevance_request_p0_def chain_relevance_request_p1_def
    calculate_relevance_vector_def relevance_stability_def
    t7_relevance_profile_def clamp01_def)

lemma chain_relevance_vector_at_10:
  "relevance_vector_at chain_relevance_catalog 10 =
   requested_relevance_vector chain_relevance_request_p0"
  by (simp add: relevance_vector_at_def chain_relevance_catalog_def
    chain_relevance_catalog_p0_def chain_relevance_catalog_0_def
    extend_relevance_catalog_def chain_relevance_request_p0_def
    chain_relevance_request_p1_def)

lemma chain_relevance_vector_at_11:
  "relevance_vector_at chain_relevance_catalog 11 =
   requested_relevance_vector chain_relevance_request_p1"
  by (simp add: relevance_vector_at_def chain_relevance_catalog_def
    extend_relevance_catalog_def chain_relevance_request_p1_def)

lemma chain_relevance_perspectives_exact:
  "relevance_perspective_at chain_relevance_catalog 10 = 0 \<and>
   relevance_perspective_at chain_relevance_catalog 11 = Suc 0"
  by (simp add: relevance_perspective_at_def chain_relevance_catalog_def
    chain_relevance_catalog_p0_def chain_relevance_catalog_0_def
    extend_relevance_catalog_def chain_relevance_request_p0_def
    chain_relevance_request_p1_def)

lemma chain_orientation_range_exact:
  "orientation_intensity_floor_of
   (calculate_orientation_range chain_relevance_catalog [10, 11]) = 9 / 20 \<and>
   orientation_intensity_ceiling_of
   (calculate_orientation_range chain_relevance_catalog [10, 11]) = 3 / 4 \<and>
   orientation_uncertainty_ceiling_of
   (calculate_orientation_range chain_relevance_catalog [10, 11]) = 1 / 4 \<and>
```

```

orientation_stability_floor_of
  (calculate_orientation_range chain_relevance_catalog [10, 11]) = 19 / 20 \<and>
orientation_divergence_of
  (calculate_orientation_range chain_relevance_catalog [10, 11]) = 3 / 10"
using chain_relevance_vectors_exact chain_relevance_vector_at_10
chain_relevance_vector_at_11
by (simp add: calculate_orientation_range_def orientation_intensities_def
orientation_uncertainties_def orientation_stabilities_def Let_def)

lemma chain_orientation_perspectives_exact:
"orientation_basis_perspectives chain_relevance_catalog [10, 11] =
{0, Suc 0}"
using chain_relevance_perspectives_exact
by (auto simp: orientation_basis_perspectives_def)

subsection \<open>Both basis versions are active and catalogued\<close>

lemma chain_relevance_10_active_for_t8:
"active_orientation_relevance chain_after_t7 chain_relevance_catalog
10 7 20 30 11"
unfolding active_orientation_relevance_def
proof (intro exI conjI)
show "evaluated_relevance_object chain_after_t6 chain_relevance_request_p0
\<in> objects_of chain_after_t7"
by (simp add: chain_after_t7_def chain_after_t7_p0_def)
show "version_id_of
(evaluated_relevance_object chain_after_t6 chain_relevance_request_p0) = 10"
by (simp add: chain_relevance_request_p0_def)
show "kind_of
(evaluated_relevance_object chain_after_t6 chain_relevance_request_p0) = K_REV"
by simp
show "lifecycle_of
(evaluated_relevance_object chain_after_t6 chain_relevance_request_p0)
\<in> {Supported, Validated}"
using evaluated_relevance_lifecycle_cases by simp
show "chain_relevance_catalog 10 =
Some (evaluated_relevance_payload chain_relevance_request_p0)"
by (simp add: chain_relevance_catalog_def chain_relevance_catalog_p0_def
chain_relevance_catalog_0_def extend_relevance_catalog_def
chain_relevance_request_p0_def chain_relevance_request_p1_def)
show "relevance_payload_valid
(evaluated_relevance_object chain_after_t6 chain_relevance_request_p0)
(evaluated_relevance_payload chain_relevance_request_p0)"
using evaluated_relevance_payload_is_valid[OF chain_t7_p0_input_valid] .
show "evaluated_relevance_target_of
(evaluated_relevance_payload chain_relevance_request_p0) = 7"
by (simp add: chain_relevance_request_p0_def)
show "evaluated_relevance_context_of
(evaluated_relevance_payload chain_relevance_request_p0) = 20"
by (simp add: chain_relevance_request_p0_def)
show "relevance_profile_model_of
(evaluated_relevance_profile_of
(evaluated_relevance_payload chain_relevance_request_p0)) = 30"
by (simp add: chain_relevance_request_p0_def t7_relevance_profile_def)
show "version_active_at chain_after_t7 10 11"
unfolding version_active_at_def
by (intro bexI[of _
"evaluated_relevance_object chain_after_t6 chain_relevance_request_p0"])
(auto simp: chain_after_t7_def chain_after_t7_p0_def
chain_relevance_request_p0_def evaluated_relevance_lifecycle_def)
qed

lemma chain_relevance_11_active_for_t8:
"active_orientation_relevance chain_after_t7 chain_relevance_catalog
11 7 20 30 11"
unfolding active_orientation_relevance_def
proof (intro exI conjI)
show "evaluated_relevance_object chain_after_t7_p0 chain_relevance_request_p1
\<in> objects_of chain_after_t7"
by (simp add: chain_after_t7_def)
show "version_id_of
(evaluated_relevance_object chain_after_t7_p0 chain_relevance_request_p1) = 11"
by (simp add: chain_relevance_request_p1_def)
show "kind_of
(evaluated_relevance_object chain_after_t7_p0 chain_relevance_request_p1) = K_REV"
by simp
show "lifecycle_of
(evaluated_relevance_object chain_after_t7_p0 chain_relevance_request_p1)
\<in> {Supported, Validated}"
using evaluated_relevance_lifecycle_cases by simp
show "chain_relevance_catalog 11 =
Some (evaluated_relevance_payload chain_relevance_request_p1)"
by (simp add: chain_relevance_catalog_def extend_relevance_catalog_def
chain_relevance_request_p1_def)
show "relevance_payload_valid
(evaluated_relevance_object chain_after_t7_p0 chain_relevance_request_p1)
(evaluated_relevance_payload chain_relevance_request_p1)"
using evaluated_relevance_payload_is_valid[OF chain_t7_p1_input_valid] .
show "evaluated_relevance_target_of
(evaluated_relevance_payload chain_relevance_request_p1) = 7"
by (simp add: chain_relevance_request_p1_def)
show "evaluated_relevance_context_of
(evaluated_relevance_payload chain_relevance_request_p1) = 20"
by (simp add: chain_relevance_request_p1_def)

```



```

show "relevance_profile_model_of
  (evaluated_relevance_profile_of
    (evaluated_relevance_payload chain_relevance_request_p1)) = 30"
by (simp add: chain_relevance_request_p1_def t7_relevance_profile_def)
show "version_active_at chain_after_t7 11 11"
unfolding version_active_at_def
by (intro bexI[of _
  "evaluated_relevance_object chain_after_t7_p0 chain_relevance_request_p1"])
(auto simp: chain_after_t7_def chain_relevance_request_p1_def
  evaluated_relevance_lifecycle_def)
qed

lemma chain_orientation_basis_valid:
  "orientation_basis_valid chain_after_t7 chain_relevance_catalog [10, 11]
    7 20 30 11"
unfolding orientation_basis_valid_def
using chain_relevance_10_active_for_t8 chain_relevance_11_active_for_t8
by auto

lemma chain_orientation_basis_deferred_valid:
  "orientation_basis_valid chain_after_t7 chain_relevance_catalog [10]
    7 20 30 11"
unfolding orientation_basis_valid_def
using chain_relevance_10_active_for_t8 by auto

subsection \<open>Success and InsufficientBasis are mechanically separated\<close>

lemma chain_orientation_profile_well_formed:
  "orientation_model_profile_well_formed chain_orientation_profile"
by (simp add: orientation_model_profile_well_formed_def
  chain_orientation_profile_def)

lemma chain_orientation_basis_sufficient:
  "orientation_basis_sufficient chain_relevance_catalog
    chain_orientation_profile [10, 11]"
using chain_orientation_range_exact chain_orientation_perspectives_exact
by (simp add: orientation_basis_sufficient_def chain_orientation_profile_def)

lemma chain_orientation_basis_deferred_insufficient:
  "\<not> orientation_basis_sufficient chain_relevance_catalog
    chain_orientation_profile [10]"
by (simp add: orientation_basis_sufficient_def chain_orientation_profile_def)

lemma chain_orientation_success_status:
  "synthesized_orientation_status chain_relevance_catalog
    chain_orientation_request = Applicable \<and>
    synthesized_orientation_lifecycle chain_relevance_catalog
    chain_orientation_request = Validated"
using chain_orientation_basis_sufficient
by (simp add: synthesized_orientation_status_def
  synthesized_orientation_lifecycle_def chain_orientation_request_def)

lemma chain_orientation_deferred_status:
  "synthesized_orientation_status chain_relevance_catalog
    chain_orientation_request_deferred = InsufficientBasis \<and>
    synthesized_orientation_lifecycle chain_relevance_catalog
    chain_orientation_request_deferred = Draft"
using chain_orientation_basis_deferred_insufficient
by (simp add: synthesized_orientation_status_def
  synthesized_orientation_lifecycle_def chain_orientation_request_deferred_def
  chain_orientation_request_def)

lemma chain_t8_success_input_valid:
  "synthesize_orientation_input_valid chain_t8_grammar chain_after_t7
    chain_relevance_catalog chain_orientation_catalog_0
    chain_orientation_request 12"
unfolding synthesize_orientation_input_valid_def
using chain_after_t7_state_well_formed_t8
  chain_after_t7_active_history_disjoint
  chain_after_t7_relevance_catalog_well_formed
  chain_orientation_catalog_0_well_formed
  chain_orientation_profile_well_formed
  chain_orientation_basis_valid
  chain_after_t7_object_ids_exact chain_after_t7_version_ids_exact
  chain_after_t7_frame_exact
by (auto simp: chain_t8_grammar_def chain_orientation_request_def
  chain_orientation_catalog_0_def chain_orientation_profile_def)

theorem chain_t8_success_step:
  "synthesize_orientation_step chain_t8_grammar chain_after_t7
    chain_relevance_catalog chain_orientation_catalog_0
    chain_orientation_request 12 chain_after_t8 chain_orientation_catalog"
unfolding synthesize_orientation_step_def chain_after_t8_def
  chain_orientation_catalog_def
using chain_t8_success_input_valid by simp

lemma chain_t8_deferred_input_valid:
  "synthesize_orientation_input_valid chain_t8_grammar chain_after_t7
    chain_relevance_catalog chain_orientation_catalog_0
    chain_orientation_request_deferred 12"
unfolding synthesize_orientation_input_valid_def
using chain_after_t7_state_well_formed_t8
  chain_after_t7_active_history_disjoint
  chain_after_t7_relevance_catalog_well_formed

```

```

chain_orientation_catalog_0_well_formed
chain_orientation_profile_well_formed
chain_orientation_basis_deferred_valid
chain_after_t7_object_ids_exact chain_after_t7_version_ids_exact
chain_after_t7_frame_exact
by (auto simp: chain_t8_grammar_def chain_orientation_request_deferred_def
  chain_orientation_request_def chain_orientation_catalog_0_def
  chain_orientation_profile_def)

theorem chain_t8_deferred_step:
  "synthesize_orientation_step chain_t8_grammar chain_after_t7
  chain_relevance_catalog chain_orientation_catalog_0
  chain_orientation_request_deferred 12 chain_after_t8_deferred
  chain_orientation_catalog_deferred"
  unfolding synthesize_orientation_step_def chain_after_t8_deferred_def
  chain_orientation_catalog_deferred_def
  using chain_t8_deferred_input_valid by simp

theorem chain_t8_success_certificates:
  "state_well_formed chain_t8_grammar chain_after_t8 \<and>
  relevance_catalog_well_formed chain_after_t8 chain_relevance_catalog \<and>
  orientation_catalog_well_formed chain_after_t8 chain_relevance_catalog
  chain_orientation_catalog \<and>
  step chain_t8_grammar chain_after_t7 GenerateOrientation chain_after_t8"
  using synthesize_orientation_preserves_state_well_formed[OF chain_t8_success_step]
  synthesize_orientation_preserves_relevance_catalog[OF chain_t8_success_step]
  synthesize_orientation_preserves_orientation_catalog[OF chain_t8_success_step]
  synthesize_orientation_refines_generic_step[OF chain_t8_success_step]
  by blast

theorem chain_t8_deferred_certificates:
  "state_well_formed chain_t8_grammar chain_after_t8_deferred \<and>
  orientation_catalog_well_formed chain_after_t8_deferred chain_relevance_catalog
  chain_orientation_catalog_deferred \<and>
  step chain_t8_grammar chain_after_t7 GenerateOrientation
  chain_after_t8_deferred"
  using synthesize_orientation_preserves_state_well_formed[OF chain_t8_deferred_step]
  synthesize_orientation_preserves_orientation_catalog[OF chain_t8_deferred_step]
  synthesize_orientation_refines_generic_step[OF chain_t8_deferred_step]
  by blast

theorem chain_t8_success_payload_exact:
  "kind_of (synthesized_orientation_object chain_relevance_catalog
  chain_orientation_request) = K_ORI \<and>
  lifecycle_of (synthesized_orientation_object chain_relevance_catalog
  chain_orientation_request) = Validated \<and>
  object_status_of (synthesized_orientation_object chain_relevance_catalog
  chain_orientation_request) = Some Applicable \<and>
  basis_refs_of (synthesized_orientation_object chain_relevance_catalog
  chain_orientation_request) = {10, 11} \<and>
  synthesized_orientation_range_of
  (synthesized_orientation_payload chain_relevance_catalog
  chain_orientation_request) =
  calculate_orientation_range chain_relevance_catalog [10, 11]"
  using chain_orientation_success_status
  by (simp add: chain_orientation_request_def)

theorem chain_t8_blocked_alternative_is_not_silent_success:
  "synthesize_orientation_step chain_t8_grammar chain_after_t7
  chain_relevance_catalog chain_orientation_catalog_0
  chain_orientation_request 12 chain_after_t8 chain_orientation_catalog \<and>
  synthesize_orientation_step chain_t8_grammar chain_after_t7
  chain_relevance_catalog chain_orientation_catalog_0
  chain_orientation_request_deferred 12 chain_after_t8_deferred
  chain_orientation_catalog_deferred \<and>
  object_status_of (synthesized_orientation_object chain_relevance_catalog
  chain_orientation_request) = Some Applicable \<and>
  object_status_of (synthesized_orientation_object chain_relevance_catalog
  chain_orientation_request_deferred) = Some InsufficientBasis \<and>
  basis_refs_of (synthesized_orientation_object chain_relevance_catalog
  chain_orientation_request_deferred) = {10}"
  using chain_t8_success_step chain_t8_deferred_step
  chain_orientation_success_status chain_orientation_deferred_status
  by (simp add: chain_orientation_request_def
  chain_orientation_request_deferred_def)

subsection \<open>Full explicit T1--T8 reachability statement\<close>

theorem operational_chain_T1_T8:
  "register_appearance_step chain_grammar chain_seed chain_entity_a 1
  chain_after_t1 \<and>
  differentiate_potential_step chain_grammar chain_after_t1
  {chain_entity_b} 2 chain_after_t2 \<and>
  create_relation_step chain_grammar chain_after_t2
  chain_relation_catalog_0 chain_relation_ab RelGenerates True 1 2 3
  chain_after_t3_ab chain_relation_catalog_ab \<and>
  create_relation_step chain_grammar chain_after_t3_ab
  chain_relation_catalog_ab chain_relation_ba RelConditions True 2 1 4
  chain_after_t3 chain_relation_catalog \<and>
  hypothesize_effect_step chain_grammar chain_after_t3
  chain_relation_catalog chain_effect_catalog_0 chain_effect_a
  chain_effect_payload_a 5 chain_after_t4_a chain_effect_catalog_a \<and>
  hypothesize_effect_step chain_grammar chain_after_t4_a
  chain_relation_catalog chain_effect_catalog_a chain_effect_b"
```

```
chain_effect_payload_b 6 chain_after_t4 chain_effect_catalog \<and>
integrate_evidence_strict_step chain_grammar chain_after_t4
chain_relation_catalog chain_effect_catalog chain_object_archive_0
chain_payload_archive_0 chain_evidence_ledger_0 chain_effect_a
chain_effect_payload_a chain_assessment_a 7 7 7 chain_after_t5_a
chain_effect_catalog_after_t5_a chain_object_archive_after_t5_a
chain_payload_archive_after_t5_a chain_evidence_ledger_after_t5_a \<and>
integrate_evidence_strict_step chain_grammar chain_after_t5_a
chain_relation_catalog chain_effect_catalog_after_t5_a
chain_object_archive_after_t5_a chain_payload_archive_after_t5_a
chain_evidence_ledger_after_t5_a chain_effect_b chain_effect_payload_b
chain_assessment_b 8 8 8 chain_after_t5
chain_effect_catalog_after_t5 chain_object_archive
chain_payload_archive chain_evidence_ledger \<and>
classify_resonance_step chain_grammar chain_after_t5
chain_effect_catalog_after_t5 chain_resonance_catalog_0
chain_resonance_request 9 chain_after_t6 chain_resonance_catalog \<and>
evaluate_relevance_step chain_grammar chain_after_t6
chain_effect_catalog_after_t5 chain_relevance_catalog_0
chain_relevance_request_p0 10 chain_after_t7_p0
chain_relevance_catalog_p0 \<and>
evaluate_relevance_step chain_grammar chain_after_t7_p0
chain_effect_catalog_after_t5 chain_relevance_catalog_p0
chain_relevance_request_p1 11 chain_after_t7 chain_relevance_catalog \<and>
synthesize_orientation_step chain_t8_grammar chain_after_t7
chain_relevance_catalog chain_orientation_catalog_0
chain_orientation_request 12 chain_after_t8 chain_orientation_catalog"
using operational_chain_T1_T7 chain_t8_success_step by blast
end
```

## Anhang K – Formalrevision K3: operationaler T1–T13-Kettenabschluss

### K.1 Releasegegenstand und methodische Bindung

K3 schließt die in K2 ausdrücklich offenen Operationen T9–T13. Die Schließung ist operational: Sie definiert zulässige artikulierte Zustände, Eingangsverträge, Konstruktionsschritte, Erhaltungssätze, Negativpfade und einen durchgehenden Zeugen ab dem real erzeugten T8-Nachfolger. Sie konstruiert keine neue Grundontologie.

Verbindliche Formkongruenzregel: ‚Immer‘, reine Potentialität, Lebendigkeit, Wirkungsfähigkeit und lebendiger Zusammenhang gelten nicht als Objektzeichen, Zustände oder Glieder einer Übergangsfolge. Sie sind bei jeder Artikulation ko-gültige Grundbedingungen.

Übergangsstrukturen besitzen ausschließlich ihre artikulierten Gestalten. Eine real gewirkte Tatsache ist daher eine innerhalb des Zusammenhangs artikulierte Wirkungsgestalt; sie ist nicht das nachträgliche Auftreten eines zuvor abwesenden Grundprinzips.

***Ko-Geltung ist keine versteckte Zustandsvariable:  $InForce(P, s)$  für jede Artikulation  $s$  bedeutet weder  $Object(P)$  noch  $State(P)$  noch  $TransitionMember(P)$ . T1–T13 operieren nur auf  $Art(P)$ , niemals auf  $P$  als vermeintlichem Ding.***

### K.2 Geschlossene Operationssemantik T9–T13

- T9 SelectAction ordnet die Prüfungen: Orientierungsstatus, Autorität, Option und Grundgates, Macht-/Verantwortungsbindung, Konflikt und Risiko. Erfolg erzeugt K\_ACT; Ablehnung bleibt auditierbar und wird nicht als Handlung ausgegeben.
- T10 ExecuteAction hält Auswahl und Ausführung ontologisch sowie provenienztechnisch auseinander. Der Ausführungsdatensatz ist K\_META und referenziert die aktive Auswahl.
- T11 RecordFeedback bindet Beobachtungen an die Ausführung und bewahrt Erwartung, Beobachtung, absolute Abweichung und Toleranz ohne nachträgliche Glättung.
- T12 AnalyzeEmergence klassifiziert die Differenz zwischen artikuliertem Baseline- und Istzustand relativ zu einem expliziten Trace und einer Grammatik. Weder Substanzentstehung noch ein Schluss aus bloßer Zeitfolge wird behauptet.
- T13 ReviseObject ist Copy-on-write. Die aktive Vorgängerversion wird archiviert; der Nachfolger behält die logische Objekt-ID und erhält eine frische Version samt previous\_version-Verweis.

### K.3 Leistungs- und Statusmatrix

| Gegenstand         | Geleistet                                                                                       | Folgt nicht daraus                                      | Status           |
|--------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------|------------------|
| T9 SelectAction    | Autorität, Gates, Konflikt, Risiko, Macht- und Verantwortungsreferenzen; fünf Ablehnungszweige. | Keine normative Legitimation einer realen Entscheidung. | Kernelakzeptiert |
| T10 ExecuteAction  | Eigenständiger versionierter Ausführungsdatensatz; Auswahl und Ausführung bleiben verschieden.  | Keine Behauptung tatsächlicher Außenweltwirkung.        | Kernelakzeptiert |
| T11 RecordFeedback | Beobachtungsreferenzen, Erwartung, Beobachtung, Abweichung 7/20 und                             | Keine empirische Wahrheit der eingesetzten Werte.       | Kernelakzeptiert |

| Gegenstand                  | Geleistet                                                                                  | Folgt nicht daraus                                                           | Status           |
|-----------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------|
|                             | Toleranz.                                                                                  |                                                                              |                  |
| T12<br>AnalyzeEmergenc<br>e | Trace- und grammatikrelative E1-<br>Klassifikation der artikulierten T9-T11-<br>Differenz. | Keine neue Substanz und kein Schluss<br>aus bloßer Zeitfolge.                | Kernelakzeptiert |
| T13 ReviseObject            | Copy-on-write: Objekt-ID 10, v12<br>archiviert, v17 aktiv,<br>previous_version=12.         | Keine empirische Angemessenheit der<br>Revision.                             | Kernelakzeptiert |
| A1-A22                      | Unverändert als 22 Annahmen des<br>Locale uza_model; Kernhash gebunden.                    | Kein Beweis ihrer Wahrheit, Konsistenz<br>oder gemeinsamen Realisierbarkeit. | Locale-Annahmen  |

## K.4 Isabelle-, Audit- und Golden-Run-Nachweis

Am 1. September 2026 wurden UZA\_0\_9\_1\_Core, UZA\_0\_9\_1\_T8 und UZA\_0\_9\_1\_T9\_T13 ausdrücklich bereinigt und seriell mit einem Buildjob und einem ML-Thread neu gebaut. Alle 25 Theorien wurden akzeptiert; Exit-Code 0. Der statische Audit bestand 27/27 Gates. Der unabhängige Golden Run führte 17 Konstruktoroperationen bis Snapshot 17 aus.

```
UZA 0.9.1 K3 T1-T13 final clean build
Timestamp (UTC): 2026-09-01T08:20:46Z
Runtime: Isabelle2025-2
Command:
  isabelle build -c -D /workspace/scratch/24663cf25c52/UZA_K3_T1_T13_Work -j 1 -o threads=1
  UZA_0_9_1_Core UZA_0_9_1_T8 UZA_0_9_1_T9_T13

Cleaned UZA_0_9_1_Core
Cleaned UZA_0_9_1_T8
Cleaned UZA_0_9_1_T9_T13
Building UZA_0_9_1_Core ...
Finished UZA_0_9_1_Core (0:01:50 elapsed time, 0:01:50 cpu time, factor 1.00)
Building UZA_0_9_1_T8 ...
Finished UZA_0_9_1_T8 (0:00:15 elapsed time, 0:00:15 cpu time, factor 0.98)
Running UZA_0_9_1_T9_T13 ...
Finished UZA_0_9_1_T9_T13 (0:00:25 elapsed time, 0:00:25 cpu time, factor 1.00)
0:02:38 elapsed time, 0:02:32 cpu time, factor 0.96

Exit code: 0
Theory count: 25
Proof escapes (sorry/oops/admit/axiomatization): 0
A1-A22 core SHA-256: 21dcfb175d6e754adf20042b941cf4a23329831d0165ebaf2023b0048213ed24
```

## K.5 A1-A22 und offene Geltungsfragen

A1-A22 wurden weder verändert noch als bewiesen ausgegeben. Sie bleiben genau 22 Annahmen des Locale uza\_model in UZA\_0\_9\_1.thy. Der unveränderte Kernhash lautet 21dcfb175d6e754adf20042b941cf4a23329831d0165ebaf2023b0048213ed24.

- offen: gemeinsamer nicht-degenerierter Modell-Expansionssatz;
- offen: relativer Konsistenznachweis gegenüber einer benannten Basistheorie;
- offen: Konservativität beziehungsweise getrennte FOL-Reduktion;
- offen: empirische Gültigkeit, Domänenadäquanz und Außenweltwirkung;
- offen: Begründung der meta-ontologischen Grundprinzipien über ihre formkongruente Artikulation hinaus.

## K.6 Gemeinsamer Audited-Append-Kern

Normativer K3-Quellstand: T9\_T13/UZA\_0\_9\_1\_Audited\_Append.thy. Bestandteil des Isabelle2025-2-Clean-Builds mit Exit-Code 0.

```
theory UZA_0_9_1_Audited_Append
  imports UZA_0_9_1_T8.UZA_0_9_1_Synthesize_Orientation
begin

section \Reusable invariant-preserving append kernel for T9--T12\
  The following kernel is operational only. It appends one articulated audit
  object to a snapshot and proves preservation of the existing UZA state
  invariants. It neither represents nor sequentializes the meta-ontological
  co-validity of potentiality, livingness, efficacy, transformability, memory,
  or the non-temporal Always.
  \close>

definition extend_authorities ::
  "snapshot \ $\rightarrow$  authority_id set \ $\rightarrow$  snapshot" where
  "extend_authorities s extra =
    s \ $\sqcup$  authorities_of := authorities_of s \ $\cup$  extra"

lemma extend_authorities_frame [simp]:
  "snapshot_id_of (extend_authorities s extra) = snapshot_id_of s"
  "objects_of (extend_authorities s extra) = objects_of s"
  "historical_versions_of (extend_authorities s extra) = historical_versions_of s"
  "perspectives_of (extend_authorities s extra) = perspectives_of s"
  "contexts_of (extend_authorities s extra) = contexts_of s"
  "models_of (extend_authorities s extra) = models_of s"
  "authorities_of (extend_authorities s extra) = authorities_of s \ $\cup$  extra"
  "contains_edges_of (extend_authorities s extra) = contains_edges_of s"
  "projections_of (extend_authorities s extra) = projections_of s"
  "couplings_of (extend_authorities s extra) = couplings_of s"
  "transmissions_of (extend_authorities s extra) = transmissions_of s"
  "boundaries_of (extend_authorities s extra) = boundaries_of s"
  by (simp_all add: extend_authorities_def)

lemma extend_authorities_version_ids [simp]:
  "version_ids (extend_authorities s extra) = version_ids s"
  by (simp add: version_ids_def)

lemma extend_authorities_active_history [simp]:
  "active_history_disjoint (extend_authorities s extra) =
    active_history_disjoint s"
  by (simp add: active_history_disjoint_def)

lemma extend_authorities_feedback_well_formed [simp]:
  "feedback_well_formed (extend_authorities s extra) obj =
    feedback_well_formed s obj"
  unfolding feedback_well_formed_def present_ref_def
  by (simp only: extend_authorities_version_ids extend_authorities_frame)

lemma extend_authorities_object_frames [simp]:
  "references_complete (extend_authorities s extra) obj =
    references_complete s obj"
  "evidence_well_formed (extend_authorities s extra) obj =
    evidence_well_formed s obj"
  "relation_well_formed (extend_authorities s extra) obj =
    relation_well_formed s obj"
  "effect_well_formed (extend_authorities s extra) obj =
    effect_well_formed s obj"
  "perspective_well_formed (extend_authorities s extra) obj =
    perspective_well_formed s obj"
  "context_well_formed (extend_authorities s extra) obj =
    context_well_formed s obj"
  "relevance_well_formed (extend_authorities s extra) obj =
    relevance_well_formed s obj"
  "orientation_well_formed (extend_authorities s extra) obj =
    orientation_well_formed s obj"
  unfolding references_complete_def evidence_well_formed_def
    evidence_references_complete_def relation_well_formed_def
    effect_well_formed_def present_ref_def perspective_well_formed_def
    context_well_formed_def relevance_well_formed_def
    orientation_well_formed_def
```

```

by (simp_all only: extend_authorities_version_ids extend_authorities_frame)

lemma extend_authorities_action_well_formed:
  assumes "action_well_formed s obj"
  shows "action_well_formed (extend_authorities s extra) obj"
  using assms
  unfolding action_well_formed_def
  by (cases "authority_of obj"; auto)

lemma extend_authorities_preserves_state_well_formed:
  assumes source: "state_well_formed G s"
  shows "state_well_formed G (extend_authorities s extra)"
proof -
  from source have unique: "unique_versions s"
  and objects:
    "\<forall>obj\<in>objects_of s.
      kind_of obj \<in> supported_kinds_of G \<and>
      references_complete s obj \<and> evidence_well_formed s obj \<and>
      time_well_formed obj \<and> relation_well_formed s obj \<and>
      effect_well_formed s obj \<and> perspective_well_formed s obj \<and>
      context_well_formed s obj \<and> relevance_well_formed s obj \<and>
      orientation_well_formed s obj \<and> action_well_formed s obj \<and>
      feedback_well_formed s obj"
  and secondary:
    "\<forall>p\<in>projections_of s. projection_well_formed s p) \<and>
    (\<forall>c\<in>couplings_of s. coupling_well_formed s c) \<and>
    (\<forall>t\<in>transmissions_of s. transmission_well_formed s t) \<and>
    (\<forall>b\<in>boundaries_of s. boundary_well_formed b)"
  unfolding state_well_formed_def by blast+
  have unique': "unique_versions (extend_authorities s extra)"
  using unique by (simp add: unique_versions_def)
  have versions_unchanged:
    "version_ids (extend_authorities s extra) = version_ids s"
  by (rule extend_authorities_version_ids)
  have history_unchanged:
    "historical_versions_of (extend_authorities s extra) =
      historical_versions_of s"
  by (rule extend_authorities_frame)
  have objects':
    "\<forall>obj\<in>objects_of (extend_authorities s extra).
      kind_of obj \<in> supported_kinds_of G \<and>
      references_complete (extend_authorities s extra) obj \<and>
      evidence_well_formed (extend_authorities s extra) obj \<and>
      time_well_formed obj \<and>
      relation_well_formed (extend_authorities s extra) obj \<and>
      effect_well_formed (extend_authorities s extra) obj \<and>
      perspective_well_formed (extend_authorities s extra) obj \<and>
      context_well_formed (extend_authorities s extra) obj \<and>
      relevance_well_formed (extend_authorities s extra) obj \<and>
      orientation_well_formed (extend_authorities s extra) obj \<and>
      action_well_formed (extend_authorities s extra) obj \<and>
      feedback_well_formed (extend_authorities s extra) obj"
  using objects
  by (auto intro: extend_authorities_action_well_formed)
  have secondary':
    "\<forall>p\<in>projections_of (extend_authorities s extra).
      projection_well_formed (extend_authorities s extra) p) \<and>
    (\<forall>c\<in>couplings_of (extend_authorities s extra).
      coupling_well_formed (extend_authorities s extra) c) \<and>
    (\<forall>t\<in>transmissions_of (extend_authorities s extra).
      transmission_well_formed (extend_authorities s extra) t) \<and>
    (\<forall>b\<in>boundaries_of (extend_authorities s extra).
      boundary_well_formed b)"
  using secondary
  by (simp add: projection_well_formed_def coupling_well_formed_def
    transmission_well_formed_def)
  show ?thesis
  unfolding state_well_formed_def
  using unique' objects' secondary' by blast
qed

definition articulated_object_conditions ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> uza_object \<Rightarrow> bool" where
  "articulated_object_conditions G target obj \<longleftarrow>
    kind_of obj \<in> supported_kinds_of G \<and>
    references_complete target obj \<and>
    evidence_well_formed target obj \<and>
    time_well_formed obj \<and>
    relation_well_formed target obj \<and>
    effect_well_formed target obj \<and>"

```



```

perspective_well_formed target obj \<and>
context_well_formed target obj \<and>
relevance_well_formed target obj \<and>
orientation_well_formed target obj \<and>
action_well_formed target obj \<and>
feedback_well_formed target obj"

definition audited_append_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> operation \<Rightarrow> uza_object
  \<Rightarrow> nat \<Rightarrow> bool" where
  "audited_append_input_valid G s oper obj next_snapshot_id \<longleftarrowrightarrow>
  state_well_formed G s \<and>
  active_history_disjoint s \<and>
  oper \<in> allowed_operations_of G \<and>
  object_id_of obj \<notin> object_id_of ` objects_of s \<and>
  version_id_of obj \<notin> version_ids s \<union> historical_versions_of s \<and>
  articulated_object_conditions G (add_appearance s obj next_snapshot_id) obj \<and>
  snapshot_id_of s < next_snapshot_id"

definition audited_append_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> operation \<Rightarrow> uza_object
  \<Rightarrow> nat \<Rightarrow> snapshot \<Rightarrow> bool" where
  "audited_append_step G s oper obj next_snapshot_id s' \<longleftarrowrightarrow>
  audited_append_input_valid G s oper obj next_snapshot_id \<and>
  s' = add_appearance s obj next_snapshot_id"

lemma audited_append_input_validI:
  assumes source: "state_well_formed G s"
  and history: "active_history_disjoint s"
  and allowed: "oper \<in> allowed_operations_of G"
  and fresh_object:
    "object_id_of obj \<notin> image object_id_of (objects_of s)"
  and fresh_version:
    "version_id_of obj \<notin> version_ids s \<union> historical_versions_of s"
  and articulated:
    "articulated_object_conditions G (add_appearance s obj next_snapshot_id) obj"
  and clock: "snapshot_id_of s < next_snapshot_id"
  shows "audited_append_input_valid G s oper obj next_snapshot_id"
  using assms unfolding audited_append_input_valid_def by blast

lemma articulated_meta_recordI:
  assumes kind: "kind_of obj = K_META"
  and supported: "K_META \<in> supported_kinds_of G"
  and references: "references_complete target obj"
  and evidence: "evidence_well_formed target obj"
  and time: "time_well_formed obj"
  shows "articulated_object_conditions G target obj"
  using assms
  unfolding articulated_object_conditions_def relation_well_formed_def
  effect_well_formed_def perspective_well_formed_def perspective_bound_kind_def
  context_well_formed_def context_bound_kind_def relevance_well_formed_def
  orientation_well_formed_def action_well_formed_def feedback_well_formed_def
  by simp

lemma articulated_emergence_recordI:
  assumes kind: "kind_of obj = K_EMG"
  and supported: "K_EMG \<in> supported_kinds_of G"
  and references: "references_complete target obj"
  and evidence: "evidence_well_formed target obj"
  and time: "time_well_formed obj"
  shows "articulated_object_conditions G target obj"
  using assms
  unfolding articulated_object_conditions_def relation_well_formed_def
  effect_well_formed_def perspective_well_formed_def perspective_bound_kind_def
  context_well_formed_def context_bound_kind_def relevance_well_formed_def
  orientation_well_formed_def action_well_formed_def feedback_well_formed_def
  by simp

lemma articulated_feedback_recordI:
  assumes kind: "kind_of obj = K_FDB"
  and supported: "K_FDB \<in> supported_kinds_of G"
  and references: "references_complete target obj"
  and evidence: "evidence_well_formed target obj"
  and time: "time_well_formed obj"
  and context_ok: "context_well_formed target obj"
  and feedback: "feedback_well_formed target obj"
  shows "articulated_object_conditions G target obj"
  using assms
  unfolding articulated_object_conditions_def relation_well_formed_def
  effect_well_formed_def perspective_well_formed_def perspective_bound_kind_def

```



```

context_bound_kind_def relevance_well_formed_def orientation_well_formed_def
action_well_formed_def
by simp

lemma audited_append_fresh_object:
  assumes valid:
    "audited_append_input_valid G s oper obj next_snapshot_id"
  shows "obj \<notin> objects_of s"
  using valid
  by (auto simp: audited_append_input_valid_def)

lemma audited_append_preserves_unique_versions:
  assumes valid:
    "audited_append_input_valid G s oper obj next_snapshot_id"
  shows "unique_versions (add_appearance s obj next_snapshot_id)"
proof -
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
  and fresh: "version_id_of obj \<notin> version_ids s"
  unfolding audited_append_input_valid_def state_well_formed_def
  unique_versions_def by auto
  show ?thesis
  unfolding unique_versions_def
  proof (rule inj_onI)
    fix left right
    assume left_mem:
      "left \<in> objects_of (add_appearance s obj next_snapshot_id)"
    and right_mem:
      "right \<in> objects_of (add_appearance s obj next_snapshot_id)"
    and same: "version_id_of left = version_id_of right"
    show "left = right"
    proof (cases "left = obj")
      case True
      show ?thesis
      proof (cases "right = obj")
        case True
        then show ?thesis using \<open>left = obj\<close> by simp
      next
        case False
        then have old: "right \<in> objects_of s" using right_mem by simp
        have "version_id_of obj \<in> version_ids s"
        unfolding version_ids_def
        using same \<open>left = obj\<close> old by (metis imageI)
        then show ?thesis using fresh by blast
      qed
    next
      case False
      then have left_old: "left \<in> objects_of s" using left_mem by simp
      show ?thesis
      proof (cases "right = obj")
        case True
        have "version_id_of obj \<in> version_ids s"
        unfolding version_ids_def
        using same left_old True by (metis imageI)
        then show ?thesis using fresh by blast
      next
        case False
        then have right_old: "right \<in> objects_of s" using right_mem by simp
        show ?thesis using inj_onD[OF source_unique same left_old right_old] .
      qed
    qed
  qed
qed

theorem audited_append_preserves_state_well_formed:
  assumes concrete:
    "audited_append_step G s oper obj next_snapshot_id s'"
  shows "state_well_formed G s'"
proof -
  from concrete have valid:
    "audited_append_input_valid G s oper obj next_snapshot_id"
  and target: "s' = add_appearance s obj next_snapshot_id"
  by (auto simp: audited_append_step_def)
  from valid have source: "state_well_formed G s"
  and new_conditions:
    "articulated_object_conditions G
      (add_appearance s obj next_snapshot_id) obj"
  unfolding audited_append_input_valid_def by blast+
  have unique: "unique_versions (add_appearance s obj next_snapshot_id)"
  using audited_append_preserves_unique_versions[OF valid] .
  have all_objects:

```

```

"\<forall>x\<in>objects_of (add_appearance s obj next_snapshot_id).
  kind_of x \<in> supported_kinds_of G \<and>
  references_complete (add_appearance s obj next_snapshot_id) x \<and>
  evidence_well_formed (add_appearance s obj next_snapshot_id) x \<and>
  time_well_formed x \<and>
  relation_well_formed (add_appearance s obj next_snapshot_id) x \<and>
  effect_well_formed (add_appearance s obj next_snapshot_id) x \<and>
  perspective_well_formed (add_appearance s obj next_snapshot_id) x \<and>
  context_well_formed (add_appearance s obj next_snapshot_id) x \<and>
  relevance_well_formed (add_appearance s obj next_snapshot_id) x \<and>
  orientation_well_formed (add_appearance s obj next_snapshot_id) x \<and>
  action_well_formed (add_appearance s obj next_snapshot_id) x \<and>
  feedback_well_formed (add_appearance s obj next_snapshot_id) x"
proof (intro ballI)
  fix x
  assume member:
    "x \<in> objects_of (add_appearance s obj next_snapshot_id)"
  consider (new) "x = obj" | (old) "x \<in> objects_of s"
  using member by auto
  then show
    "kind_of x \<in> supported_kinds_of G \<and>
    references_complete (add_appearance s obj next_snapshot_id) x \<and>
    evidence_well_formed (add_appearance s obj next_snapshot_id) x \<and>
    time_well_formed x \<and>
    relation_well_formed (add_appearance s obj next_snapshot_id) x \<and>
    effect_well_formed (add_appearance s obj next_snapshot_id) x \<and>
    perspective_well_formed (add_appearance s obj next_snapshot_id) x \<and>
    context_well_formed (add_appearance s obj next_snapshot_id) x \<and>
    relevance_well_formed (add_appearance s obj next_snapshot_id) x \<and>
    orientation_well_formed (add_appearance s obj next_snapshot_id) x \<and>
    action_well_formed (add_appearance s obj next_snapshot_id) x \<and>
    feedback_well_formed (add_appearance s obj next_snapshot_id) x"
  proof cases
    case new
      then show ?thesis
        using new_conditions
        by (simp add: articulated_object_conditions_def)
    next
      case old
        from source old have old_conditions:
          "kind_of x \<in> supported_kinds_of G \<and>
          references_complete s x \<and> evidence_well_formed s x \<and>
          time_well_formed x \<and> relation_well_formed s x \<and>
          effect_well_formed s x \<and> perspective_well_formed s x \<and>
          context_well_formed s x \<and> relevance_well_formed s x \<and>
          orientation_well_formed s x \<and> action_well_formed s x \<and>
          feedback_well_formed s x"
          unfolding state_well_formed_def by blast
        show ?thesis
          using old_conditions add_appearance_references_mono
            add_appearance_evidence_mono add_appearance_relation_mono
            add_appearance_effect_mono add_appearance_feedback_mono
          by auto
      qed
    qed
  from source have secondary:
    "(\<forall>p\<in>projections_of s. projection_well_formed s p) \<and>
    (\<forall>c\<in>couplings_of s. coupling_well_formed s c) \<and>
    (\<forall>t\<in>transmissions_of s. transmission_well_formed s t) \<and>
    (\<forall>b\<in>boundaries_of s. boundary_well_formed b)"
    unfolding state_well_formed_def by blast
  have secondary':
    "(\<forall>p\<in>projections_of (add_appearance s obj next_snapshot_id).
      projection_well_formed (add_appearance s obj next_snapshot_id) p) \<and>
    (\<forall>c\<in>couplings_of (add_appearance s obj next_snapshot_id).
      coupling_well_formed (add_appearance s obj next_snapshot_id) c) \<and>
    (\<forall>t\<in>transmissions_of (add_appearance s obj next_snapshot_id).
      transmission_well_formed (add_appearance s obj next_snapshot_id) t) \<and>
    (\<forall>b\<in>boundaries_of (add_appearance s obj next_snapshot_id).
      boundary_well_formed b)"
    using secondary
    by (simp add: projection_well_formed_def coupling_well_formed_def
      transmission_well_formed_def)
  show ?thesis
    unfolding target_state_well_formed_def
    using unique_all_objects secondary' by blast
qed

theorem audited_append_preserves_identity:
  assumes concrete:

```

```

    "audited_append_step G s oper obj next_snapshot_id s'"
  shows "immutable_preserved s s'"
proof -
  from concrete have valid:
    "audited_append_input_valid G s oper obj next_snapshot_id"
  and target: "s' = add_appearance s obj next_snapshot_id"
  by (auto simp: audited_append_step_def)
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
  and fresh: "version_id_of obj \<notin> version_ids s"
  unfolding audited_append_input_valid_def state_well_formed_def
    unique_versions_def by auto
  show ?thesis
  unfolding immutable_preserved_def target
proof (intro ballI allI impI)
  fix old target_obj
  assume old_mem: "old \<in> objects_of s"
  and target_mem:
    "target_obj \<in> objects_of (add_appearance s obj next_snapshot_id)"
  and same: "version_id_of old = version_id_of target_obj"
  show "old = target_obj"
  proof (cases "target_obj = obj")
    case True
    have "version_id_of obj \<in> version_ids s"
    unfolding version_ids_def
    using old_mem same True by (metis imageI)
    then show ?thesis using fresh by blast
  next
    case False
    then have target_old: "target_obj \<in> objects_of s"
    using target_mem by simp
    show ?thesis using inj_onD[OF source_unique same old_mem target_old] .
  qed
qed
qed

theorem audited_append_refines_generic_step:
  assumes concrete:
    "audited_append_step G s oper obj next_snapshot_id s'"
  shows "step G s oper s'"
proof -
  from concrete have source: "state_well_formed G s"
  and allowed: "oper \<in> allowed_operations_of G"
  by (auto simp: audited_append_step_def audited_append_input_valid_def)
  have target: "state_well_formed G s'"
  using audited_append_preserves_state_well_formed[OF concrete] .
  have immutable: "immutable_preserved s s'"
  using audited_append_preserves_identity[OF concrete] .
  show ?thesis unfolding step_def using source target allowed immutable by blast
qed

lemma audited_append_preserves_active_history:
  assumes concrete:
    "audited_append_step G s oper obj next_snapshot_id s'"
  shows "active_history_disjoint s'"
  using concrete
  unfolding audited_append_step_def audited_append_input_valid_def
    active_history_disjoint_def
  by auto

end

```

## K.7 Vollständige T9-Semantik

Normativer K3-Quellstand: T9\_T13/UZA\_0\_9\_1\_Select\_Action.thy. Bestandteil des Isabelle2025-2-Clean-Builds mit Exit-Code 0.

```
theory UZA_0_9_1_Select_Action
  imports UZA_0_9_1_Audited_Append
begin

section \
```

```

definition authority_active_at ::
  "decision_authority \<Rightarrow> time_point \<Rightarrow> bool" where
  "authority_active_at authority t \<longlefttrightarrow>
    decision_authority_valid_from_of authority \<le> t \<and>
    (case decision_authority_valid_to_of authority of
      None \<Rightarrow> True
    | Some until \<Rightarrow> t \<le> until)"

definition decision_authority_valid ::
  "snapshot \<Rightarrow> decision_authority \<Rightarrow> time_point \<Rightarrow> bool" where
  "decision_authority_valid s authority t \<longlefttrightarrow>
    decision_authority_perspective_of authority \<in> perspectives_of s \<and>
    decision_authority_context_of authority \<in> contexts_of s \<and>
    authority_active_at authority t \<and>
    SelectAction \<in> decision_authority_mandate_of authority \<and>
    decision_authority_explanation_of authority \<noteq> []"

definition basic_action_gates_passed :: "action_option \<Rightarrow> bool" where
  "basic_action_gates_passed option \<longlefttrightarrow>
    legal_gate_passed_of (action_option_gates_of option) \<and>
    ethical_gate_passed_of (action_option_gates_of option) \<and>
    organizational_gate_passed_of (action_option_gates_of option) \<and>
    resource_gate_passed_of (action_option_gates_of option)"

definition explicit_power_and_responsibility ::
  "decision_authority \<Rightarrow> action_option \<Rightarrow> bool" where
  "explicit_power_and_responsibility authority option \<longlefttrightarrow>
    action_option_power_shift_refs_of option \<noteq> {} \<and>
    action_option_responsibility_of option =
      Some (decision_authority_id_of authority)"

definition requested_action_option ::
  "action_selection_request \<Rightarrow> action_option option" where
  "requested_action_option request =
    find (\<lambda>option. action_option_id_of option =
      requested_selection_option_id_of request)
    (requested_selection_options_of request)"

definition selection_orientation_status ::
  "orientation_catalog \<Rightarrow> version_id \<Rightarrow> analysis_status" where
  "selection_orientation_status catalog version =
    (case catalog version of
      None \<Rightarrow> OutOfScope
    | Some payload \<Rightarrow> synthesized_orientation_status_of payload)"

definition action_selection_outcome ::
  "snapshot \<Rightarrow> orientation_catalog \<Rightarrow>
    action_selection_request \<Rightarrow> action_selection_outcome" where
  "action_selection_outcome s catalog request =
    (let orientation_status = selection_orientation_status catalog
      (requested_selection_orientation_of request);
      authority = requested_selection_authority_of request
    in if orientation_status = InsufficientBasis then
      ActionSelectionStatus InsufficientBasis
    else if orientation_status \<noteq> Applicable then
      ActionSelectionStatus orientation_status
    else if \<not> decision_authority_valid s authority
      (requested_selection_time_of request) then
      ActionSelectionAbort NoAuthority
    else case requested_action_option request of
      None \<Rightarrow> ActionSelectionAbort NoAdmissibleOption
    | Some option \<Rightarrow>
      if action_option_orientation_of option \<noteq>
        requested_selection_orientation_of request \<or>
        \<not> basic_action_gates_passed option \<or>
        \<not> explicit_power_and_responsibility authority option
      then ActionSelectionAbort NoAdmissibleOption
      else if \<not> action_option_conflict_resolved_of option
      then ActionSelectionAbort ConflictUnresolved
      else if \<not> action_option_risk_accepted_of option
      then ActionSelectionAbort RiskNotAccepted
      else ActionSelected (action_option_id_of option))"

fun action_selection_succeeded :: "action_selection_outcome \<Rightarrow> bool" where
  "action_selection_succeeded (ActionSelected _) = True"
| "action_selection_succeeded _ = False"

definition selection_power_refs ::
  "snapshot \<Rightarrow> orientation_catalog \<Rightarrow>

```

```

    action_selection_request \<Rightarrow> version_id set" where
"selection_power_refs s catalog request =
  (case action_selection_outcome s catalog request of
    ActionSelected _ \<Rightarrow>
      (case requested_action_option request of
        None \<Rightarrow> {}
      | Some option \<Rightarrow> action_option_power_shift_refs_of option)
    | _ \<Rightarrow> {})")

definition selection_responsibility ::
"snapshot \<Rightarrow> orientation_catalog \<Rightarrow>
  action_selection_request \<Rightarrow> authority_id option" where
"selection_responsibility s catalog request =
  (case action_selection_outcome s catalog request of
    ActionSelected _ \<Rightarrow>
      Some (decision_authority_id_of
        (requested_selection_authority_of request))
    | _ \<Rightarrow> None)"

definition selection_record_object ::
"snapshot \<Rightarrow> orientation_catalog \<Rightarrow>
  action_selection_request \<Rightarrow> uza_object" where
"selection_record_object s catalog request =
  (let outcome = action_selection_outcome s catalog request;
    success = action_selection_succeeded outcome;
    authority = requested_selection_authority_of request;
    orientation = requested_selection_orientation_of request
  in \<lparr>
    object_id_of = requested_selection_object_id_of request,
    version_id_of = requested_selection_version_id_of request,
    kind_of = (if success then K_ACT else K_META),
    lifecycle_of = (if success then Validated else Rejected),
    previous_version_of = None,
    references_of = {orientation},
    evidence_of = (if success then {orientation} else {}),
    perspective_of = Some (decision_authority_perspective_of authority),
    context_of = Some (decision_authority_context_of authority),
    model_of = None,
    authority_of =
      (if success then Some (decision_authority_id_of authority) else None),
    valid_from_of = requested_selection_time_of request,
    valid_to_of = None,
    source_ref_of = Some orientation,
    target_ref_of = None,
    effect_origin_of = None,
    basis_refs_of = (if success then {orientation} else {}),
    execution_ref_of = None,
    observation_refs_of = {},
    object_status_of =
      (case outcome of
        ActionSelected _ \<Rightarrow> Some Applicable
      | ActionSelectionStatus status \<Rightarrow> Some status
      | ActionSelectionAbort _ \<Rightarrow> Some OutOfScope)
    \<rparr>)"

definition selection_record_payload ::
"snapshot \<Rightarrow> orientation_catalog \<Rightarrow>
  action_selection_request \<Rightarrow> action_selection_payload" where
"selection_record_payload s catalog request =
  (let outcome = action_selection_outcome s catalog request
  in \<lparr>
    selected_orientation_of = requested_selection_orientation_of request,
    selected_option_id_of =
      (case outcome of ActionSelected option_id \<Rightarrow> Some option_id
      | _ \<Rightarrow> None),
    selection_options_of = requested_selection_options_of request,
    selection_authority_of = requested_selection_authority_of request,
    selection_time_of = requested_selection_time_of request,
    selection_outcome_of = outcome,
    selection_power_shift_refs_of = selection_power_refs s catalog request,
    selection_responsibility_of = selection_responsibility s catalog request,
    selection_explanation_of = requested_selection_explanation_of request
  \<rparr>)"

definition selection_record_valid ::
"uza_object \<Rightarrow> action_selection_payload \<Rightarrow> bool" where
"selection_record_valid record payload \<longleftarrowrightarrow>
  version_id_of record > 0 \<and>
  references_of record = {selected_orientation_of payload} \<and>
  valid_from_of record = selection_time_of payload \<and>
```

```

selection_explanation_of payload \<noteq> [] \<and>
(case selection_outcome_of payload of
  ActionSelected option_id \<Rightarrow>
    kind_of record = K_ACT \<and>
    lifecycle_of record = Validated \<and>
    selected_option_id_of payload = Some option_id \<and>
    basis_refs_of record = {selected_orientation_of payload} \<and>
    evidence_of record = {selected_orientation_of payload} \<and>
    authority_of record = selection_responsibility_of payload \<and>
    selection_power_shift_refs_of payload \<noteq> {} \<and>
    selection_responsibility_of payload \<noteq> None
  | ActionSelectionAbort _ \<Rightarrow>
    kind_of record = K_META \<and> lifecycle_of record = Rejected \<and>
    selected_option_id_of payload = None
  | ActionSelectionStatus _ \<Rightarrow>
    kind_of record = K_META \<and> lifecycle_of record = Rejected \<and>
    selected_option_id_of payload = None)

definition action_selection_catalog_well_formed ::
  "snapshot \<Rightarrow> action_selection_catalog \<Rightarrow> bool" where
  "action_selection_catalog_well_formed s catalog \<longlefttrightarrow>
    (\<forall>version payload. catalog version = Some payload \<longrightarrow>
      (\<exists>record\<in>objects_of s.
        version_id_of record = version \<and>
        selection_record_valid record payload))"

definition selection_authority_extension ::
  "snapshot \<Rightarrow> orientation_catalog \<Rightarrow>
    action_selection_request \<Rightarrow> authority_id set" where
  "selection_authority_extension s catalog request =
    (if action_selection_succeeded (action_selection_outcome s catalog request)
     then {decision_authority_id_of (requested_selection_authority_of request)}
     else {})"

definition selection_source_frame ::
  "snapshot \<Rightarrow> orientation_catalog \<Rightarrow>
    action_selection_request \<Rightarrow> snapshot" where
  "selection_source_frame s catalog request =
    extend_authorities s (selection_authority_extension s catalog request)"

definition record_action_selection ::
  "snapshot \<Rightarrow> orientation_catalog \<Rightarrow>
    action_selection_request \<Rightarrow> nat \<Rightarrow> snapshot" where
  "record_action_selection s catalog request next_snapshot_id =
    add_appearance (selection_source_frame s catalog request)
      (selection_record_object s catalog request) next_snapshot_id"

definition extend_action_selection_catalog ::
  "action_selection_catalog \<Rightarrow> snapshot \<Rightarrow>
    orientation_catalog \<Rightarrow> action_selection_request \<Rightarrow>
    action_selection_catalog" where
  "extend_action_selection_catalog selection_catalog s orientation_catalog request =
    selection_catalog(requested_selection_version_id_of request \<mapsto>
      selection_record_payload s orientation_catalog request)"

definition known_selection_orientation ::
  "snapshot \<Rightarrow> relevance_catalog \<Rightarrow> orientation_catalog
    \<Rightarrow> version_id \<Rightarrow> bool" where
  "known_selection_orientation s relevance_catalog orientation_catalog version
    \<longlefttrightarrow>
    (\<exists>orientation payload.
      orientation \<in> objects_of s \<and>
      version_id_of orientation = version \<and>
      kind_of orientation = K_ORI \<and>
      orientation_catalog version = Some payload \<and>
      orientation_payload_valid relevance_catalog orientation payload)"

definition select_action_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> relevance_catalog \<Rightarrow>
    orientation_catalog \<Rightarrow> action_selection_catalog \<Rightarrow>
    action_selection_request \<Rightarrow> nat \<Rightarrow> bool" where
  "select_action_input_valid G s relevance_catalog orientation_catalog
    selection_catalog request next_snapshot_id \<longlefttrightarrow>
    state_well_formed G s \<and>
    active_history_disjoint s \<and>
    orientation_catalog_well_formed s relevance_catalog orientation_catalog \<and>
    action_selection_catalog_well_formed s selection_catalog \<and>
    known_selection_orientation s relevance_catalog orientation_catalog
      (requested_selection_orientation_of request) \<and>
    distinct (map action_option_id_of (requested_selection_options_of request)) \<and>"

```

```

(\<forall>option\<in>set (requested_selection_options_of request).
  action_option_explanation_of option \<noteq> []) \<and>
  requested_selection_explanation_of request \<noteq> [] \<and>
  selection_catalog (requested_selection_version_id_of request) = None \<and>
  selection_record_valid
    (selection_record_object s orientation_catalog request)
    (selection_record_payload s orientation_catalog request) \<and>
  audited_append_input_valid G
    (selection_source_frame s orientation_catalog request) SelectAction
    (selection_record_object s orientation_catalog request) next_snapshot_id"

definition select_action_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> relevance_catalog \<Rightarrow>
  orientation_catalog \<Rightarrow> action_selection_catalog \<Rightarrow>
  action_selection_request \<Rightarrow> nat \<Rightarrow> snapshot \<Rightarrow>
  action_selection_catalog \<Rightarrow> bool" where
  "select_action_step G s relevance_catalog orientation_catalog selection_catalog
  request next_snapshot_id s' selection_catalog' \<longlefttrightarrow>
  select_action_input_valid G s relevance_catalog orientation_catalog
  selection_catalog request next_snapshot_id \<and>
  s' = record_action_selection s orientation_catalog request next_snapshot_id \<and>
  selection_catalog' = extend_action_selection_catalog selection_catalog s
  orientation_catalog request"

lemma select_action_is_audited_append:
  assumes concrete:
    "select_action_step G s relevance_catalog orientation_catalog
    selection_catalog request next_snapshot_id s' selection_catalog'"
  shows
    "audited_append_step G
    (selection_source_frame s orientation_catalog request) SelectAction
    (selection_record_object s orientation_catalog request)
    next_snapshot_id s'"
  using concrete
  by (auto simp: select_action_step_def select_action_input_valid_def
    audited_append_step_def record_action_selection_def)

theorem select_action_preserves_state_well_formed:
  assumes concrete:
    "select_action_step G s relevance_catalog orientation_catalog
    selection_catalog request next_snapshot_id s' selection_catalog'"
  shows "state_well_formed G s'"
  using audited_append_preserves_state_well_formed[
    OF select_action_is_audited_append[OF concrete]] .

theorem select_action_preserves_identity:
  assumes concrete:
    "select_action_step G s relevance_catalog orientation_catalog
    selection_catalog request next_snapshot_id s' selection_catalog'"
  shows "immutable_preserved s s'"
proof -
  have extended:
    "immutable_preserved
    (selection_source_frame s orientation_catalog request) s'"
  using audited_append_preserves_identity[
    OF select_action_is_audited_append[OF concrete]] .
  show ?thesis
  using extended
  unfolding immutable_preserved_def selection_source_frame_def
  extend_authorities_def by simp
qed

theorem select_action_refines_generic_step:
  assumes concrete:
    "select_action_step G s relevance_catalog orientation_catalog
    selection_catalog request next_snapshot_id s' selection_catalog'"
  shows "step G s SelectAction s'"
proof -
  from concrete have source: "state_well_formed G s"
  and allowed: "SelectAction \<in> allowed_operations_of G"
  unfolding select_action_step_def select_action_input_valid_def
  audited_append_input_valid_def by blast+
  have target: "state_well_formed G s'"
  using select_action_preserves_state_well_formed[OF concrete] .
  have immutable: "immutable_preserved s s'"
  using select_action_preserves_identity[OF concrete] .
  show ?thesis unfolding step_def using source target allowed immutable by blast
qed

theorem select_action_preserves_catalog:

```



```

assumes concrete:
  "select_action_step G s relevance_catalog orientation_catalog
   selection_catalog request next_snapshot_id s' selection_catalog'"
shows "action_selection_catalog_well_formed s' selection_catalog'"
proof -
  from concrete have valid:
    "select_action_input_valid G s relevance_catalog orientation_catalog
     selection_catalog request next_snapshot_id"
  and target:
    "s' = record_action_selection s orientation_catalog request next_snapshot_id"
  and catalog_target:
    "selection_catalog' = extend_action_selection_catalog selection_catalog s
     orientation_catalog request"
  by (auto simp: select_action_step_def)
  from valid have source_catalog:
    "action_selection_catalog_well_formed s selection_catalog"
  and fresh:
    "selection_catalog (requested_selection_version_id_of request) = None"
  and new_valid:
    "selection_record_valid
     (selection_record_object s orientation_catalog request)
     (selection_record_payload s orientation_catalog request)"
  unfolding select_action_input_valid_def by blast+
show ?thesis
  unfolding action_selection_catalog_well_formed_def target catalog_target
    extend_action_selection_catalog_def record_action_selection_def
proof (intro allI impI)
  fix version payload
  assume entry:
    "(selection_catalog(requested_selection_version_id_of request \<mapsto>
     selection_record_payload s orientation_catalog request)) version =
     Some payload"
  show "\<exists>record\<in>objects_of
    (add_appearance (selection_source_frame s orientation_catalog request)
     (selection_record_object s orientation_catalog request) next_snapshot_id).
     version_id_of record = version \<and>
     selection_record_valid record payload"
  proof (cases "version = requested_selection_version_id_of request")
  case True
  then have payload:
    "payload = selection_record_payload s orientation_catalog request"
  using entry by simp
  have member:
    "selection_record_object s orientation_catalog request \<in>
     objects_of
     (add_appearance
      (selection_source_frame s orientation_catalog request)
      (selection_record_object s orientation_catalog request)
      next_snapshot_id)"
  by (simp only: add_appearance_objects; simp)
  have id:
    "version_id_of (selection_record_object s orientation_catalog request) =
     version"
  using True
  by (simp add: selection_record_object_def Let_def)
  show ?thesis
  using member id new_valid payload by blast
next
case False
then have old_entry: "selection_catalog version = Some payload"
  using entry by simp
  from source_catalog old_entry obtain prior_record where
    member: "prior_record \<in> objects_of s" and
    id: "version_id_of prior_record = version" and
    payload_valid: "selection_record_valid prior_record payload"
  unfolding action_selection_catalog_well_formed_def by blast
  have source_member:
    "prior_record \<in>
     objects_of (selection_source_frame s orientation_catalog request)"
  using member
  by (simp add: selection_source_frame_def extend_authorities_def)
  have target_member:
    "prior_record \<in>
     objects_of
     (add_appearance
      (selection_source_frame s orientation_catalog request)
      (selection_record_object s orientation_catalog request)
      next_snapshot_id)"
  using source_member
  by (simp only: add_appearance_objects; simp)

```

```
      show ?thesis
      using target_member id payload_valid by blast
qed
qed
qed

theorem insufficient_basis_is_never_selected:
  assumes status:
    "selection_orientation_status catalog
      (requested_selection_orientation_of request) = InsufficientBasis"
  shows
    "action_selection_outcome s catalog request =
      ActionSelectionStatus InsufficientBasis \<and>
      kind_of (selection_record_object s catalog request) = K_META"
  using status
  by (simp add: action_selection_outcome_def selection_record_object_def Let_def)

theorem successful_selection_has_explicit_authority_and_responsibility:
  assumes success:
    "action_selection_outcome s catalog request = ActionSelected option_id"
  shows
    "kind_of (selection_record_object s catalog request) = K_ACT \<and>
      authority_of (selection_record_object s catalog request) =
        Some (decision_authority_id_of
          (requested_selection_authority_of request)) \<and>
      selection_responsibility s catalog request =
        Some (decision_authority_id_of
          (requested_selection_authority_of request))"
  using success
  by (simp add: selection_record_object_def selection_responsibility_def Let_def)

end
```

## K.8 Vollständige T10-Semantik

Normativer K3-Quellstand: T9\_T13/UZA\_0\_9\_1\_Execute\_Action.thy. Bestandteil des Isabelle2025-2-Clean-Builds mit Exit-Code 0.

```
theory UZA_0_9_1_Execute_Action
  imports UZA_0_9_1_Select_Action
begin

section \<<open>T10 ExecuteAction: selection and execution remain distinct\<<close>

datatype action_execution_status =
  ExecutionSucceeded | ExecutionFailed | ExecutionCancelled

record action_execution_request =
  requested_execution_object_id_of :: object_id
  requested_execution_version_id_of :: version_id
  requested_execution_action_of :: version_id
  requested_execution_authority_of :: authority_id
  requested_execution_context_of :: context_id
  requested_execution_started_of :: time_point
  requested_execution_finished_of :: time_point
  requested_execution_status_of :: action_execution_status
  requested_execution_explanation_of :: string

record action_execution_payload =
  executed_action_of :: version_id
  execution_authority_of :: authority_id
  execution_context_of :: context_id
  execution_started_of :: time_point
  execution_finished_of :: time_point
  execution_status_of :: action_execution_status
  execution_explanation_of :: string

type_synonym action_execution_catalog =
  "version_id \<<rightarrow> action_execution_payload"

definition active_selected_action ::
  "snapshot \<<Rightarrow> action_selection_catalog \<<Rightarrow> version_id
  \<<Rightarrow> authority_id \<<Rightarrow> bool" where
  "active_selected_action s selection_catalog version authority \<<longleftarrowrightarrow>
  (\<exists>action payload option_id.
    action \<<in> objects_of s \<<and>
    version_id_of action = version \<<and>
    kind_of action = K_ACT \<<and>
    lifecycle_of action = Validated \<<and>
    authority_of action = Some authority \<<and>
    selection_catalog version = Some payload \<<and>
    selection_outcome_of payload = ActionSelected option_id \<<and>
    selection_record_valid action payload)"

definition execution_record_object ::
  "action_execution_request \<<Rightarrow> uza_object" where
  "execution_record_object request =
  \<<lparr>
    object_id_of = requested_execution_object_id_of request,
    version_id_of = requested_execution_version_id_of request,
    kind_of = K_META,
    lifecycle_of =
      (if requested_execution_status_of request = ExecutionSucceeded
       then Validated else Rejected),
    previous_version_of = None,
    references_of = {requested_execution_action_of request},
    evidence_of = {requested_execution_action_of request},
    perspective_of = None,
    context_of = Some (requested_execution_context_of request),
    model_of = None,
    authority_of = Some (requested_execution_authority_of request),
    valid_from_of = requested_execution_finished_of request,
    valid_to_of = None,
    source_ref_of = Some (requested_execution_action_of request),
    target_ref_of = None,
    effect_origin_of = None,
    basis_refs_of = {requested_execution_action_of request},
    execution_ref_of = None,
    observation_refs_of = {},
  \<</rparr>"
```

```

    object_status_of =
      (if requested_execution_status_of request = ExecutionSucceeded
       then Some Applicable else Some OutOfScope)
  \<rparr>"

definition execution_record_payload ::
  "action_execution_request \<Rightarrow> action_execution_payload" where
  "execution_record_payload request =
    \<lparr>
      executed_action_of = requested_execution_action_of request,
      execution_authority_of = requested_execution_authority_of request,
      execution_context_of = requested_execution_context_of request,
      execution_started_of = requested_execution_started_of request,
      execution_finished_of = requested_execution_finished_of request,
      execution_status_of = requested_execution_status_of request,
      execution_explanation_of = requested_execution_explanation_of request
    \<rparr>"

definition execution_record_valid ::
  "uza_object \<Rightarrow> action_execution_payload \<Rightarrow> bool" where
  "execution_record_valid record payload \<longlefttrightarrow>
    kind_of record = K_META \<and>
    references_of record = {executed_action_of payload} \<and>
    evidence_of record = {executed_action_of payload} \<and>
    source_ref_of record = Some (executed_action_of payload) \<and>
    authority_of record = Some (execution_authority_of payload) \<and>
    context_of record = Some (execution_context_of payload) \<and>
    valid_from_of record = execution_finished_of payload \<and>
    execution_started_of payload \<le> execution_finished_of payload \<and>
    execution_explanation_of payload \<noteq> [] \<and>
    (execution_status_of payload = ExecutionSucceeded \<longrightarrow>
     lifecycle_of record = Validated) \<and>
    (execution_status_of payload \<noteq> ExecutionSucceeded \<longrightarrow>
     lifecycle_of record = Rejected)"

definition action_execution_catalog_well_formed ::
  "snapshot \<Rightarrow> action_execution_catalog \<Rightarrow> bool" where
  "action_execution_catalog_well_formed s catalog \<longlefttrightarrow>
    (\<forall>version payload. catalog version = Some payload \<longrightarrow>
     (\<exists>record\<in>objects_of s.
      version_id_of record = version \<and>
      execution_record_valid record payload))"

definition add_execution_record ::
  "snapshot \<Rightarrow> action_execution_request \<Rightarrow> nat \<Rightarrow> snapshot" where
  "add_execution_record s request next_snapshot_id =
    add_appearance s (execution_record_object request) next_snapshot_id"

definition extend_action_execution_catalog ::
  "action_execution_catalog \<Rightarrow> action_execution_request
   \<Rightarrow> action_execution_catalog" where
  "extend_action_execution_catalog catalog request =
    catalog(requested_execution_version_id_of request \<mapsto>
     execution_record_payload request)"

definition execute_action_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> action_selection_catalog \<Rightarrow>
   action_execution_catalog \<Rightarrow> action_execution_request \<Rightarrow>
   nat \<Rightarrow> bool" where
  "execute_action_input_valid G s selection_catalog execution_catalog request
    next_snapshot_id \<longlefttrightarrow>
    state_well_formed G s \<and>
    active_history_disjoint s \<and>
    action_selection_catalog_well_formed s selection_catalog \<and>
    action_execution_catalog_well_formed s execution_catalog \<and>
    active_selected_action s selection_catalog
      (requested_execution_action_of request)
      (requested_execution_authority_of request) \<and>
    requested_execution_authority_of request \<in> authorities_of s \<and>
    requested_execution_context_of request \<in> contexts_of s \<and>
    requested_execution_started_of request \<le>
      requested_execution_finished_of request \<and>
    requested_execution_explanation_of request \<noteq> [] \<and>
    execution_catalog (requested_execution_version_id_of request) = None \<and>
    execution_record_valid (execution_record_object request)
      (execution_record_payload request) \<and>
    audited_append_input_valid G s ExecuteAction
      (execution_record_object request) next_snapshot_id"

definition execute_action_step ::

```

```

"grammar \<Rightarrow> snapshot \<Rightarrow> action_selection_catalog \<Rightarrow>
  action_execution_catalog \<Rightarrow> action_execution_request \<Rightarrow>
  nat \<Rightarrow> snapshot \<Rightarrow> action_execution_catalog \<Rightarrow> bool" where
"execute_action_step G s selection_catalog execution_catalog request
  next_snapshot_id s' execution_catalog' \<longleftarrowrightarrow>
  execute_action_input_valid G s selection_catalog execution_catalog request
  next_snapshot_id \<and>
  s' = add_execution_record s request next_snapshot_id \<and>
  execution_catalog' = extend_action_execution_catalog execution_catalog request"

lemma execute_action_is_audited_append:
  assumes concrete:
    "execute_action_step G s selection_catalog execution_catalog request
      next_snapshot_id s' execution_catalog'"
  shows
    "audited_append_step G s ExecuteAction (execution_record_object request)
      next_snapshot_id s'"
  using concrete
  by (auto simp: execute_action_step_def execute_action_input_valid_def
    audited_append_step_def add_execution_record_def)

theorem execute_action_preserves_state_well_formed:
  assumes concrete:
    "execute_action_step G s selection_catalog execution_catalog request
      next_snapshot_id s' execution_catalog'"
  shows "state_well_formed G s'"
  using audited_append_preserves_state_well_formed[
    OF execute_action_is_audited_append[OF concrete]] .

theorem execute_action_refines_generic_step:
  assumes concrete:
    "execute_action_step G s selection_catalog execution_catalog request
      next_snapshot_id s' execution_catalog'"
  shows "step G s ExecuteAction s'"
  using audited_append_refines_generic_step[
    OF execute_action_is_audited_append[OF concrete]] .

theorem execute_action_preserves_catalog:
  assumes concrete:
    "execute_action_step G s selection_catalog execution_catalog request
      next_snapshot_id s' execution_catalog'"
  shows "action_execution_catalog_well_formed s' execution_catalog'"
proof -
  from concrete have valid:
    "execute_action_input_valid G s selection_catalog execution_catalog request
      next_snapshot_id"
  and target: "s' = add_execution_record s request next_snapshot_id"
  and catalog_target:
    "execution_catalog' = extend_action_execution_catalog execution_catalog request"
  by (auto simp: execute_action_step_def)
  from valid have source_catalog:
    "action_execution_catalog_well_formed s execution_catalog"
  and new_valid:
    "execution_record_valid (execution_record_object request)
      (execution_record_payload request)"
  unfolding execute_action_input_valid_def by blast+
  show ?thesis
  unfolding action_execution_catalog_well_formed_def target catalog_target
    extend_action_execution_catalog_def add_execution_record_def
  proof (intro allI impI)
    fix version payload
    assume entry:
      "(execution_catalog(requested_execution_version_id_of request \<mapsto>
        execution_record_payload request)) version = Some payload"
    show "\<exists>record\<in>objects_of
      (add_appearance s (execution_record_object request) next_snapshot_id).
        version_id_of record = version \<and>
        execution_record_valid record payload"
    proof (cases "version = requested_execution_version_id_of request")
      case True
      then have payload: "payload = execution_record_payload request"
        using entry by simp
      have member:
        "execution_record_object request \<in>
          objects_of
            (add_appearance s (execution_record_object request)
              next_snapshot_id)"
        by (simp only: add_appearance_objects; simp)
      have id:
        "version_id_of (execution_record_object request) = version"

```

```
        using True by (simp add: execution_record_object_def)
      show ?thesis
        using member id new_valid payload by blast
    next
    case False
    then have old_entry: "execution_catalog version = Some payload"
      using entry by simp
    from source_catalog old_entry obtain prior_record where
      member: "prior_record \<in> objects_of s" and
      id: "version_id_of prior_record = version" and
      payload_valid: "execution_record_valid prior_record payload"
    unfolding action_execution_catalog_well_formed_def by blast
    have target_member:
      "prior_record \<in>
        objects_of
          (add_appearance s (execution_record_object request)
            next_snapshot_id)"
      using member
      by (simp only: add_appearance_objects; simp)
    show ?thesis using target_member id payload_valid by blast
  qed
qed
qed

theorem execution_is_not_selection:
  "kind_of (execution_record_object request) = K_META \<and>
    source_ref_of (execution_record_object request) =
      Some (requested_execution_action_of request)"
  by (simp add: execution_record_object_def)

end
```

## K.9 Vollständige T11-Semantik

Normativer K3-Quellstand: T9\_T13/UZA\_0\_9\_1\_Record\_Feedback.thy. Bestandteil des Isabelle2025-2-Clean-Builds mit Exit-Code 0.

```
theory UZA_0_9_1_Record_Feedback
  imports UZA_0_9_1_Execute_Action
begin

section \\rightarrow real" where
  "feedback_deviation request =
    abs (requested_feedback_observed_of request -
      requested_feedback_expected_of request)"

definition feedback_assessment ::
  "feedback_request \ $\rightarrow$  feedback_assessment_status" where
  "feedback_assessment request =
    (if feedback_deviation request > requested_feedback_tolerance_of request
     then DeviationDetected else WithinTolerance)"

definition feedback_object :: "feedback_request \ $\rightarrow$  uza_object" where
  "feedback_object request =
    \ $\langle$ lparr>
      object_id_of = requested_feedback_object_id_of request,
      version_id_of = requested_feedback_version_id_of request,
      kind_of = K_FDB,
      lifecycle_of = Validated,
      previous_version_of = None,
      references_of =
        insert (requested_feedback_execution_of request)
          (requested_feedback_observations_of request),
      evidence_of = requested_feedback_observations_of request,
      perspective_of = None,
      context_of = Some (requested_feedback_context_of request),
      model_of = None,
      authority_of = None,
      valid_from_of = requested_feedback_time_of request,
      valid_to_of = None,
      source_ref_of = Some (requested_feedback_execution_of request),
      target_ref_of = None,
      effect_origin_of = None,
      basis_refs_of = {requested_feedback_execution_of request},
      execution_ref_of = Some (requested_feedback_execution_of request),
      observation_refs_of = requested_feedback_observations_of request,
      object_status_of =
        (if feedback_assessment request = WithinTolerance
```

```

        then Some Applicable else Some ModelFalsified)
    \<rparr>"

definition feedback_payload :: "feedback_request \<Rightarrow> feedback_payload" where
"feedback_payload request =
  \<lparr>
    feedback_execution_of = requested_feedback_execution_of request,
    feedback_observations_of = requested_feedback_observations_of request,
    feedback_context_of = requested_feedback_context_of request,
    feedback_time_of = requested_feedback_time_of request,
    feedback_expected_of = requested_feedback_expected_of request,
    feedback_observed_of = requested_feedback_observed_of request,
    feedback_deviation_of = feedback_deviation request,
    feedback_tolerance_of = requested_feedback_tolerance_of request,
    feedback_assessment_of = feedback_assessment request,
    feedback_explanation_of = requested_feedback_explanation_of request
  \<rparr>"

definition feedback_payload_valid ::
"uza_object \<Rightarrow> feedback_payload \<Rightarrow> bool" where
"feedback_payload_valid feedback payload \<longlefttrightarrow>
  kind_of feedback = K_FDB \<and>
  lifecycle_of feedback = Validated \<and>
  execution_ref_of feedback = Some (feedback_execution_of payload) \<and>
  observation_refs_of feedback = feedback_observations_of payload \<and>
  feedback_observations_of payload \<noteq> {} \<and>
  references_of feedback =
    insert (feedback_execution_of payload) (feedback_observations_of payload) \<and>
  evidence_of feedback = feedback_observations_of payload \<and>
  context_of feedback = Some (feedback_context_of payload) \<and>
  valid_from_of feedback = feedback_time_of payload \<and>
  0 \<le> feedback_tolerance_of payload \<and>
  feedback_deviation_of payload =
    abs (feedback_observed_of payload - feedback_expected_of payload) \<and>
  feedback_assessment_of payload =
    (if feedback_deviation_of payload > feedback_tolerance_of payload
     then DeviationDetected else WithinTolerance) \<and>
  feedback_explanation_of payload \<noteq> []"

definition feedback_catalog_well_formed ::
"snapshot \<Rightarrow> feedback_catalog \<Rightarrow> bool" where
"feedback_catalog_well_formed s catalog \<longlefttrightarrow>
  (\<forall>version payload. catalog version = Some payload \<longrightarrow>
    (\<exists>feedback\<in>objects_of s.
      version_id_of feedback = version \<and>
      feedback_payload_valid feedback payload))"

definition active_execution_record ::
"snapshot \<Rightarrow> action_execution_catalog \<Rightarrow> version_id
  \<Rightarrow> bool" where
"active_execution_record s execution_catalog version \<longlefttrightarrow>
  (\<exists>record payload.
    record \<in> objects_of s \<and>
    version_id_of record = version \<and>
    execution_catalog version = Some payload \<and>
    execution_record_valid record payload)"

definition add_feedback ::
"snapshot \<Rightarrow> feedback_request \<Rightarrow> nat \<Rightarrow> snapshot" where
"add_feedback s request next_snapshot_id =
  add_appearance s (feedback_object request) next_snapshot_id"

definition extend_feedback_catalog ::
"feedback_catalog \<Rightarrow> feedback_request \<Rightarrow> feedback_catalog" where
"extend_feedback_catalog catalog request =
  catalog(requested_feedback_version_id_of request \<mapsto>
    feedback_payload request)"

definition record_feedback_input_valid ::
"grammar \<Rightarrow> snapshot \<Rightarrow> action_execution_catalog \<Rightarrow>
  feedback_catalog \<Rightarrow> feedback_request \<Rightarrow> nat \<Rightarrow> bool" where
"record_feedback_input_valid G s execution_catalog feedback_catalog request
  next_snapshot_id \<longlefttrightarrow>
  state_well_formed G s \<and>
  active_history_disjoint s \<and>
  action_execution_catalog_well_formed s execution_catalog \<and>
  feedback_catalog_well_formed s feedback_catalog \<and>
  active_execution_record s execution_catalog
    (requested_feedback_execution_of request) \<and>
  requested_feedback_observations_of request \<noteq> {} \<and>
```



```

requested_feedback_observations_of request \<subteq> version_ids s \<and>
requested_feedback_context_of request \<in> contexts_of s \<and>
0 \<le> requested_feedback_tolerance_of request \<and>
requested_feedback_explanation_of request \<noteq> [] \<and>
feedback_catalog (requested_feedback_version_id_of request) = None \<and>
feedback_payload_valid (feedback_object request) (feedback_payload request) \<and>
audited_append_input_valid G s RecordFeedback (feedback_object request)
next_snapshot_id"

definition record_feedback_step ::
"grammar \<Rightarrow> snapshot \<Rightarrow> action_execution_catalog \<Rightarrow>
feedback_catalog \<Rightarrow> feedback_request \<Rightarrow> nat \<Rightarrow>
snapshot \<Rightarrow> feedback_catalog \<Rightarrow> bool" where
"record_feedback_step G s execution_catalog feedback_catalog request
next_snapshot_id s' feedback_catalog' \<longleftarrowrightarrow>
record_feedback_input_valid G s execution_catalog feedback_catalog request
next_snapshot_id \<and>
s' = add_feedback s request next_snapshot_id \<and>
feedback_catalog' = extend_feedback_catalog feedback_catalog request"

lemma record_feedback_is_audited_append:
assumes concrete:
"record_feedback_step G s execution_catalog feedback_catalog request
next_snapshot_id s' feedback_catalog'"
shows
"audited_append_step G s RecordFeedback (feedback_object request)
next_snapshot_id s'"
using concrete
by (auto simp: record_feedback_step_def record_feedback_input_valid_def
audited_append_step_def add_feedback_def)

theorem record_feedback_preserves_state_well_formed:
assumes concrete:
"record_feedback_step G s execution_catalog feedback_catalog request
next_snapshot_id s' feedback_catalog'"
shows "state_well_formed G s'"
using audited_append_preserves_state_well_formed[
OF record_feedback_is_audited_append[OF concrete]] .

theorem record_feedback_refines_generic_step:
assumes concrete:
"record_feedback_step G s execution_catalog feedback_catalog request
next_snapshot_id s' feedback_catalog'"
shows "step G s RecordFeedback s'"
using audited_append_refines_generic_step[
OF record_feedback_is_audited_append[OF concrete]] .

theorem record_feedback_preserves_catalog:
assumes concrete:
"record_feedback_step G s execution_catalog feedback_catalog request
next_snapshot_id s' feedback_catalog'"
shows "feedback_catalog_well_formed s' feedback_catalog'"
proof -
from concrete have valid:
"record_feedback_input_valid G s execution_catalog feedback_catalog request
next_snapshot_id"
and target: "s' = add_feedback s request next_snapshot_id"
and catalog_target:
"feedback_catalog' = extend_feedback_catalog feedback_catalog request"
by (auto simp: record_feedback_step_def)
from valid have source_catalog:
"feedback_catalog_well_formed s feedback_catalog"
and new_valid:
"feedback_payload_valid (feedback_object request) (feedback_payload request)"
unfolding record_feedback_input_valid_def by blast+
show ?thesis
unfolding feedback_catalog_well_formed_def target catalog_target
extend_feedback_catalog_def add_feedback_def
proof (intro allI impI)
fix version payload
assume entry:
"(feedback_catalog(requested_feedback_version_id_of request \<mapsto>
feedback_payload request)) version = Some payload"
show "\<exists>feedback\<in>objects_of
(add_appearance s (feedback_object request) next_snapshot_id).
version_id_of feedback = version \<and>
feedback_payload_valid feedback payload"
proof (cases "version = requested_feedback_version_id_of request")
case True
then have payload: "payload = feedback_payload request"

```

```

        using entry by simp
    have member:
        "feedback_object request \<in>
          objects_of
            (add_appearance s (feedback_object request) next_snapshot_id)"
    by (simp only: add_appearance_objects; simp)
    have id: "version_id_of (feedback_object request) = version"
    using True by (simp add: feedback_object_def)
    show ?thesis
    using member id new_valid payload by blast
next
case False
then have old_entry: "feedback_catalog version = Some payload"
  using entry by simp
from source_catalog old_entry obtain feedback where
  member: "feedback \<in> objects_of s" and
  id: "version_id_of feedback = version" and
  payload_valid: "feedback_payload_valid feedback payload"
unfolding feedback_catalog_well_formed_def by blast
have target_member:
  "feedback \<in>
    objects_of
      (add_appearance s (feedback_object request) next_snapshot_id)"
  using member
  by (simp only: add_appearance_objects; simp)
show ?thesis using target_member id payload_valid by blast
qed
qed
qed

theorem deviation_status_exact:
  "feedback_assessment request = DeviationDetected \<longlefttrightarrow>
    requested_feedback_tolerance_of request < feedback_deviation request"
  by (simp add: feedback_assessment_def)

end

```

## K.10 Vollständige T12-Semantik

Normativer K3-Quellstand: T9\_T13/UZA\_0\_9\_1\_Analyze\_Emergence.thy. Bestandteil des Isabelle2025-2-Clean-Builds mit Exit-Code 0.

```
theory UZA_0_9_1_Analyze_Emergence
  imports UZA_0_9_1_Record_Feedback
begin

section \<open>T12 AnalyzeEmergence: trace-relative, grammar-relative classification\<close>

text \<open>
  Emergence is evaluated for an explicit interaction complex. The result is
  therefore relative to its baseline, current articulated snapshot, trace, and
  grammar. It is not an additional substance and is not inferred from a bare
  temporal succession.
\<close>

record emergence_request =
  requested_emergence_object_id_of :: object_id
  requested_emergence_version_id_of :: version_id
  requested_emergence_feedback_of :: version_id
  requested_emergence_interaction_of :: interaction_complex
  requested_emergence_context_of :: context_id
  requested_emergence_time_of :: time_point
  requested_emergence_explanation_of :: string

record emergence_payload =
  analyzed_feedback_of :: version_id
  analyzed_baseline_snapshot_of :: nat
  analyzed_current_snapshot_of :: nat
  analyzed_trace_length_of :: nat
  analyzed_description_of :: description_result
  analyzed_emergence_level_of :: emergence_level
  analyzed_context_of :: context_id
  analyzed_time_of :: time_point
  analyzed_explanation_of :: string

type_synonym emergence_catalog = "version_id \<rightarrow> emergence_payload"

definition emergence_analysis_status ::
  "emergence_level \<Rightarrow> analysis_status" where
  "emergence_analysis_status level =
    (case level of
      E0 \<Rightarrow> Applicable
    | E1 \<Rightarrow> Applicable
    | E2 \<Rightarrow> PatchRejected
    | E3 \<Rightarrow> UnresolvedE3)"

definition emergence_lifecycle ::
  "emergence_level \<Rightarrow> lifecycle_status" where
  "emergence_lifecycle level =
    (case level of
      E0 \<Rightarrow> Supported
    | E1 \<Rightarrow> Validated
    | E2 \<Rightarrow> Draft
    | E3 \<Rightarrow> Rejected)"

definition emergence_object ::
  "grammar \<Rightarrow> emergence_request \<Rightarrow> uza_object" where
  "emergence_object G request =
    (let level = emergence_of G (requested_emergence_interaction_of request);
      feedback = requested_emergence_feedback_of request
    in \<lparr>
      object_id_of = requested_emergence_object_id_of request,
      version_id_of = requested_emergence_version_id_of request,
      kind_of = K_EMG,
      lifecycle_of = emergence_lifecycle level,
      previous_version_of = None,
      references_of = {feedback},
      evidence_of = {feedback},
      perspective_of = None,
      context_of = Some (requested_emergence_context_of request),
      model_of = None,
      authority_of = None,
      valid_from_of = requested_emergence_time_of request,
```

```

    valid_to_of = None,
    source_ref_of = Some feedback,
    target_ref_of = None,
    effect_origin_of = None,
    basis_refs_of = {feedback},
    execution_ref_of = None,
    observation_refs_of = {},
    object_status_of = Some (emergence_analysis_status level)
  \<rparr>)"

definition emergence_payload ::
  "grammar \<Rightarrow> emergence_request \<Rightarrow> emergence_payload" where
  "emergence_payload G request =
    (let interaction = requested_emergence_interaction_of request
     in \<lparr>
       analyzed_feedback_of = requested_emergence_feedback_of request,
       analyzed_baseline_snapshot_of = snapshot_id_of (baseline_of interaction),
       analyzed_current_snapshot_of = snapshot_id_of (current_of interaction),
       analyzed_trace_length_of = length (trace_of interaction),
       analyzed_description_of = describe G interaction,
       analyzed_emergence_level_of = emergence_of G interaction,
       analyzed_context_of = requested_emergence_context_of request,
       analyzed_time_of = requested_emergence_time_of request,
       analyzed_explanation_of = requested_emergence_explanation_of request
     \<rparr>)"

definition emergence_payload_valid ::
  "grammar \<Rightarrow> uza_object \<Rightarrow> emergence_payload \<Rightarrow> bool" where
  "emergence_payload_valid G object payload \<longlefttrightarrow>
    kind_of object = K_EMG \<and>
    references_of object = {analyzed_feedback_of payload} \<and>
    evidence_of object = {analyzed_feedback_of payload} \<and>
    source_ref_of object = Some (analyzed_feedback_of payload) \<and>
    context_of object = Some (analyzed_context_of payload) \<and>
    valid_from_of object = analyzed_time_of payload \<and>
    lifecycle_of object = emergence_lifecycle (analyzed_emergence_level_of payload) \<and>
    object_status_of object =
      Some (emergence_analysis_status (analyzed_emergence_level_of payload)) \<and>
    dr_status_of (analyzed_description_of payload) =
      (if dr_type_complete_of (analyzed_description_of payload) \<and>
       dr_relation_complete_of (analyzed_description_of payload) \<and>
       dr_transition_complete_of (analyzed_description_of payload)
       then DescriptionComplete else DescriptionIncomplete) \<and>
    analyzed_explanation_of payload \<noteq> []"

definition emergence_catalog_well_formed ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> emergence_catalog \<Rightarrow> bool" where
  "emergence_catalog_well_formed G s catalog \<longlefttrightarrow>
    (\<forall>version payload. catalog version = Some payload \<longrightarrow>
     (\<exists>object\<in>objects_of s.
      version_id_of object = version \<and>
      emergence_payload_valid G object payload))"

definition active_feedback_record ::
  "snapshot \<Rightarrow> feedback_catalog \<Rightarrow> version_id \<Rightarrow> bool" where
  "active_feedback_record s feedback_catalog version \<longlefttrightarrow>
    (\<exists>feedback payload.
     feedback \<in> objects_of s \<and>
     version_id_of feedback = version \<and>
     feedback_catalog version = Some payload \<and>
     feedback_payload_valid feedback payload)"

definition add_emergence_analysis ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> emergence_request \<Rightarrow>
    nat \<Rightarrow> snapshot" where
  "add_emergence_analysis G s request next_snapshot_id =
    add_appearance s (emergence_object G request) next_snapshot_id"

definition extend_emergence_catalog ::
  "emergence_catalog \<Rightarrow> grammar \<Rightarrow> emergence_request
    \<Rightarrow> emergence_catalog" where
  "extend_emergence_catalog catalog G request =
    catalog(requested_emergence_version_id_of request \<mapsto>
      emergence_payload G request)"

definition analyze_emergence_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> feedback_catalog \<Rightarrow>
    emergence_catalog \<Rightarrow> emergence_request \<Rightarrow> nat \<Rightarrow> bool" where
  "analyze_emergence_input_valid G s feedback_catalog emergence_catalog request
    next_snapshot_id \<longlefttrightarrow>
```

```

state_well_formed G s \<and>
active_history_disjoint s \<and>
feedback_catalog_well_formed s feedback_catalog \<and>
emergence_catalog_well_formed G s emergence_catalog \<and>
current_of (requested_emergence_interaction_of request) = s \<and>
state_well_formed G
  (baseline_of (requested_emergence_interaction_of request)) \<and>
snapshot_id_of (baseline_of (requested_emergence_interaction_of request)) <
  snapshot_id_of s \<and>
trace_of (requested_emergence_interaction_of request) \<noteq> [] \<and>
active_feedback_record s feedback_catalog
  (requested_emergence_feedback_of request) \<and>
requested_emergence_context_of request \<in> contexts_of s \<and>
requested_emergence_explanation_of request \<noteq> [] \<and>
emergence_catalog (requested_emergence_version_id_of request) = None \<and>
emergence_payload_valid G (emergence_object G request)
  (emergence_payload G request) \<and>
audited_append_input_valid G s AnalyzeEmergence (emergence_object G request)
  next_snapshot_id"

definition analyze_emergence_step ::
"grammar \<Rightarrow> snapshot \<Rightarrow> feedback_catalog \<Rightarrow>
emergence_catalog \<Rightarrow> emergence_request \<Rightarrow> nat \<Rightarrow>
snapshot \<Rightarrow> emergence_catalog \<Rightarrow> bool" where
"analyze_emergence_step G s feedback_catalog emergence_catalog request
  next_snapshot_id s' emergence_catalog' \<longlefttrightarrow>
  analyze_emergence_input_valid G s feedback_catalog emergence_catalog request
  next_snapshot_id \<and>
  s' = add_emergence_analysis G s request next_snapshot_id \<and>
  emergence_catalog' = extend_emergence_catalog emergence_catalog G request"

lemma analyze_emergence_is_audited_append:
  assumes concrete:
    "analyze_emergence_step G s feedback_catalog emergence_catalog request
      next_snapshot_id s' emergence_catalog'"
  shows
    "audited_append_step G s AnalyzeEmergence (emergence_object G request)
      next_snapshot_id s'"
  using concrete
  by (auto simp: analyze_emergence_step_def analyze_emergence_input_valid_def
    audited_append_step_def add_emergence_analysis_def)

theorem analyze_emergence_preserves_state_well_formed:
  assumes concrete:
    "analyze_emergence_step G s feedback_catalog emergence_catalog request
      next_snapshot_id s' emergence_catalog'"
  shows "state_well_formed G s'"
  using audited_append_preserves_state_well_formed[
    OF analyze_emergence_is_audited_append[OF concrete]] .

theorem analyze_emergence_refines_generic_step:
  assumes concrete:
    "analyze_emergence_step G s feedback_catalog emergence_catalog request
      next_snapshot_id s' emergence_catalog'"
  shows "step G s AnalyzeEmergence s'"
  using audited_append_refines_generic_step[
    OF analyze_emergence_is_audited_append[OF concrete]] .

theorem emergence_result_is_trace_relative:
  "analyzed_emergence_level_of (emergence_payload G request) =
    emergence_of G (requested_emergence_interaction_of request) \<and>
  analyzed_description_of (emergence_payload G request) =
    describe G (requested_emergence_interaction_of request)"
  by (simp add: emergence_payload_def Let_def)

end

```

## K.11 Vollständige T13-Semantik

Normativer K3-Quellstand: T9\_T13/UZA\_0\_9\_1\_Revise\_Object.thy. Bestandteil des Isabelle2025-2-Clean-Builds mit Exit-Code 0.

```
theory UZA_0_9_1_Revise_Object
  imports UZA_0_9_1_Analyze_Emergence
begin

section \\rightarrow bool" where
  "revisable_kind kind \ $\iff$ 
    kind \ $\in$  {K_EFF, K_REV, K_ORI, K_ACT, K_META}"

definition revised_object ::
  "uza_object \ $\rightarrow$  revision_request \ $\rightarrow$  uza_object" where
  "revised_object prior request =
    prior\ $\langle$ lparr>
      version_id_of := requested_revision_new_version_of request,
      lifecycle_of := Revised,
      previous_version_of := Some (version_id_of prior),
      references_of :=
        insert (requested_revision_emergence_of request) (references_of prior),
      evidence_of :=
        insert (requested_revision_feedback_of request) (evidence_of prior),
      valid_from_of := requested_revision_time_of request,
      valid_to_of := None,
      object_status_of := Some Applicable
    \rparr>"

definition copy_on_write_revision ::
  "snapshot \ $\rightarrow$  uza_object \ $\rightarrow$  revision_request \ $\rightarrow$ 
  nat \ $\rightarrow$  snapshot" where
  "copy_on_write_revision s prior request next_snapshot_id =
    s\ $\langle$ lparr>
      snapshot_id_of := next_snapshot_id,
      objects_of := insert (revised_object prior request) (objects_of s - {prior}),
      historical_versions_of :=
        insert (version_id_of prior) (historical_versions_of s)
    \rparr>"

definition revision_payload ::
  "uza_object \ $\rightarrow$  revision_request \ $\rightarrow$  revision_payload" where
  "revision_payload prior request =
    \lparr>
      revised_object_id_of = object_id_of prior,
      revised_prior_version_of = version_id_of prior,
      revised_new_version_of = requested_revision_new_version_of request,
```

```

    revised_feedback_of = requested_revision_feedback_of request,
    revised_emergence_of = requested_revision_emergence_of request,
    revised_time_of = requested_revision_time_of request,
    revised_explanation_of = requested_revision_explanation_of request
  \<rparr>"

definition revision_payload_valid ::
  "uza_object \<Rightarrow> revision_payload \<Rightarrow> bool" where
  "revision_payload_valid successor payload \<longlefttrightarrow>
    object_id_of successor = revised_object_id_of payload \<and>
    version_id_of successor = revised_new_version_of payload \<and>
    previous_version_of successor = Some (revised_prior_version_of payload) \<and>
    lifecycle_of successor = Revised \<and>
    revised_emergence_of payload \<in> references_of successor \<and>
    revised_feedback_of payload \<in> evidence_of successor \<and>
    valid_from_of successor = revised_time_of payload \<and>
    revised_explanation_of payload \<noteq> []"

definition revision_catalog_well_formed ::
  "snapshot \<Rightarrow> revision_catalog \<Rightarrow> bool" where
  "revision_catalog_well_formed s catalog \<longlefttrightarrow>
    (\<forall>version payload. catalog version = Some payload \<longrightarrow>
      version = revised_new_version_of payload \<and>
      revised_prior_version_of payload \<in> historical_versions_of s \<and>
      (version \<in> version_ids s \<or> version \<in> historical_versions_of s))"

definition extend_revision_catalog ::
  "revision_catalog \<Rightarrow> uza_object \<Rightarrow> revision_request
    \<Rightarrow> revision_catalog" where
  "extend_revision_catalog catalog prior request =
    catalog(requested_revision_new_version_of request \<mapsto>
      revision_payload prior request)"

definition active_emergence_record ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> emergence_catalog \<Rightarrow>
    version_id \<Rightarrow> bool" where
  "active_emergence_record G s emergence_catalog version \<longlefttrightarrow>
    (\<exists>object payload.
      object \<in> objects_of s \<and>
      version_id_of object = version \<and>
      emergence_catalog version = Some payload \<and>
      emergence_payload_valid G object payload)"

lemma copy_on_write_fields [simp]:
  "snapshot_id_of (copy_on_write_revision s prior request next_snapshot_id) =
    next_snapshot_id"
  "objects_of (copy_on_write_revision s prior request next_snapshot_id) =
    insert (revised_object prior request) (objects_of s - {prior})"
  "historical_versions_of
    (copy_on_write_revision s prior request next_snapshot_id) =
    insert (version_id_of prior) (historical_versions_of s)"
  "perspectives_of (copy_on_write_revision s prior request next_snapshot_id) =
    perspectives_of s"
  "contexts_of (copy_on_write_revision s prior request next_snapshot_id) =
    contexts_of s"
  "models_of (copy_on_write_revision s prior request next_snapshot_id) =
    models_of s"
  "authorities_of (copy_on_write_revision s prior request next_snapshot_id) =
    authorities_of s"
  "contains_edges_of (copy_on_write_revision s prior request next_snapshot_id) =
    contains_edges_of s"
  "projections_of (copy_on_write_revision s prior request next_snapshot_id) =
    projections_of s"
  "couplings_of (copy_on_write_revision s prior request next_snapshot_id) =
    couplings_of s"
  "transmissions_of (copy_on_write_revision s prior request next_snapshot_id) =
    transmissions_of s"
  "boundaries_of (copy_on_write_revision s prior request next_snapshot_id) =
    boundaries_of s"
  by (simp_all add: copy_on_write_revision_def)

lemma revised_object_fields [simp]:
  "object_id_of (revised_object prior request) = object_id_of prior"
  "version_id_of (revised_object prior request) =
    requested_revision_new_version_of request"
  "kind_of (revised_object prior request) = kind_of prior"
  "lifecycle_of (revised_object prior request) = Revised"
  "previous_version_of (revised_object prior request) =
    Some (version_id_of prior)"
  "references_of (revised_object prior request) =

```

```

    insert (requested_revision_emergence_of request) (references_of prior)"
    "evidence_of (revised_object prior request) =
      insert (requested_revision_feedback_of request) (evidence_of prior)"
    "perspective_of (revised_object prior request) = perspective_of prior"
    "context_of (revised_object prior request) = context_of prior"
    "model_of (revised_object prior request) = model_of prior"
    "authority_of (revised_object prior request) = authority_of prior"
    "valid_from_of (revised_object prior request) = requested_revision_time_of request"
    "valid_to_of (revised_object prior request) = None"
    "basis_refs_of (revised_object prior request) = basis_refs_of prior"
    "execution_ref_of (revised_object prior request) = execution_ref_of prior"
    "observation_refs_of (revised_object prior request) = observation_refs_of prior"
    "object_status_of (revised_object prior request) = Some Applicable"
  by (simp_all add: revised_object_def)

definition execute_revision_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> feedback_catalog \<Rightarrow>
    emergence_catalog \<Rightarrow> revision_catalog \<Rightarrow> uza_object
    \<Rightarrow> revision_request \<Rightarrow> nat \<Rightarrow> bool" where
  "execute_revision_input_valid G s feedback_catalog emergence_catalog
    revision_catalog prior request next_snapshot_id \<longlefttrightarrow>
    state_well_formed G s \<and>
    active_history_disjoint s \<and>
    ReviseObject \<in> allowed_operations_of G \<and>
    prior \<in> objects_of s \<and>
    version_id_of prior = requested_revision_prior_version_of request \<and>
    revisable_kind (kind_of prior) \<and>
    requested_revision_new_version_of request \<notin>
      version_ids s \<union> historical_versions_of s \<and>
    active_feedback_record s feedback_catalog
      (requested_revision_feedback_of request) \<and>
    active_emergence_record G s emergence_catalog
      (requested_revision_emergence_of request) \<and>
    revision_catalog_well_formed s revision_catalog \<and>
    revision_catalog (requested_revision_new_version_of request) = None \<and>
    valid_from_of prior \<le> requested_revision_time_of request \<and>
    requested_revision_explanation_of request \<noteq> [] \<and>
    articulated_object_conditions G
      (copy_on_write_revision s prior request next_snapshot_id)
      (revised_object prior request) \<and>
    snapshot_id_of s < next_snapshot_id"

definition execute_revision_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> feedback_catalog \<Rightarrow>
    emergence_catalog \<Rightarrow> revision_catalog \<Rightarrow> uza_object
    \<Rightarrow> revision_request \<Rightarrow> nat \<Rightarrow> snapshot
    \<Rightarrow> revision_catalog \<Rightarrow> bool" where
  "execute_revision_step G s feedback_catalog emergence_catalog revision_catalog
    prior request next_snapshot_id s' revision_catalog' \<longlefttrightarrow>
    execute_revision_input_valid G s feedback_catalog emergence_catalog
    revision_catalog prior request next_snapshot_id \<and>
    s' = copy_on_write_revision s prior request next_snapshot_id \<and>
    revision_catalog' = extend_revision_catalog revision_catalog prior request"

lemma copy_on_write_reference_universe_mono:
  assumes source_unique: "unique_versions s"
  and prior_member: "prior \<in> objects_of s"
  shows
    "version_ids s \<union> historical_versions_of s \<subteq>
      version_ids (copy_on_write_revision s prior request next_snapshot_id) \<union>
      historical_versions_of
        (copy_on_write_revision s prior request next_snapshot_id)"
proof
  fix version
  assume member:
    "version \<in> version_ids s \<union> historical_versions_of s"
  show "version \<in>
    version_ids (copy_on_write_revision s prior request next_snapshot_id) \<union>
    historical_versions_of
      (copy_on_write_revision s prior request next_snapshot_id)"
proof (cases "version = version_id_of prior")
  case True
  then show ?thesis by simp
next
  case False
  then have version_not_prior: "version \<noteq> version_id_of prior"
  by simp
  show ?thesis
  proof (cases "version \<in> historical_versions_of s")
    case True

```



```

    then show ?thesis by simp
  next
  case False
  from member False have "version \<in> version_ids s" by blast
  then obtain object where object_member: "object \<in> objects_of s"
    and object_version: "version_id_of object = version"
    unfolding version_ids_def by blast
  have "object \<noteq> prior" using version_not_prior object_version by auto
  then have target_member:
    "object \<in> objects_of
      (copy_on_write_revision s prior request next_snapshot_id)"
    using object_member by simp
  show ?thesis
    unfolding version_ids_def
    using target_member object_version by blast
qed
qed
qed

lemma copy_on_write_preserves_unique_versions:
  assumes valid:
    "execute_revision_input_valid G s feedback_catalog emergence_catalog
      revision_catalog prior request next_snapshot_id"
  shows
    "unique_versions (copy_on_write_revision s prior request next_snapshot_id)"
proof -
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
    and fresh:
      "requested_revision_new_version_of request \<notin> version_ids s"
    unfolding execute_revision_input_valid_def state_well_formed_def
    unique_versions_def by auto
  show ?thesis
    unfolding unique_versions_def
  proof (rule inj_onI)
    fix left right
    assume left_mem:
      "left \<in> objects_of
        (copy_on_write_revision s prior request next_snapshot_id)"
    and right_mem:
      "right \<in> objects_of
        (copy_on_write_revision s prior request next_snapshot_id)"
    and same: "version_id_of left = version_id_of right"
    let ?successor = "revised_object prior request"
    show "left = right"
    proof (cases "left = ?successor")
      case True
      show ?thesis
      proof (cases "right = ?successor")
        case True
        then show ?thesis using \<open>left = ?successor\<close> by simp
      next
      case False
      then have right_old: "right \<in> objects_of s"
        using right_mem by simp
      have "requested_revision_new_version_of request \<in> version_ids s"
        unfolding version_ids_def
        using same \<open>left = ?successor\<close> right_old
        by (metis imageI revised_object_fields(2))
      then show ?thesis using fresh by blast
    qed
  next
  case False
  then have left_old: "left \<in> objects_of s" using left_mem by simp
  show ?thesis
  proof (cases "right = ?successor")
    case True
    have "requested_revision_new_version_of request \<in> version_ids s"
      unfolding version_ids_def
      using same left_old True by (metis imageI revised_object_fields(2))
    then show ?thesis using fresh by blast
  next
  case False
  then have right_old: "right \<in> objects_of s" using right_mem by simp
  show ?thesis using inj_onD[OF source_unique same left_old right_old] .
qed
qed
qed
qed

theorem execute_revision_preserves_state_well_formed:

```

```

assumes concrete:
  "execute_revision_step G s feedback_catalog emergence_catalog revision_catalog
    prior request next_snapshot_id s' revision_catalog"
shows "state_well_formed G s'"
proof -
  from concrete have valid:
    "execute_revision_input_valid G s feedback_catalog emergence_catalog
      revision_catalog prior request next_snapshot_id"
  and target: "s' = copy_on_write_revision s prior request next_snapshot_id"
  by (auto simp: execute_revision_step_def)
  from valid have source: "state_well_formed G s"
  and prior_member: "prior \<in> objects_of s"
  and new_conditions:
    "articulated_object_conditions G
      (copy_on_write_revision s prior request next_snapshot_id)
      (revised_object prior request)"
  unfolding execute_revision_input_valid_def by blast+
  from source have source_unique: "unique_versions s"
  and source_objects:
    "\<forall>object\<in>objects_of s.
      kind_of object \<in> supported_kinds_of G \<and>
      references_complete s object \<and> evidence_well_formed s object \<and>
      time_well_formed object \<and> relation_well_formed s object \<and>
      effect_well_formed s object \<and> perspective_well_formed s object \<and>
      context_well_formed s object \<and> relevance_well_formed s object \<and>
      orientation_well_formed s object \<and> action_well_formed s object \<and>
      feedback_well_formed s object"
  and secondary:
    "(\<forall>p\<in>projections_of s. projection_well_formed s p) \<and>
      (\<forall>c\<in>couplings_of s. coupling_well_formed s c) \<and>
      (\<forall>t\<in>transmissions_of s. transmission_well_formed s t) \<and>
      (\<forall>b\<in>boundaries_of s. boundary_well_formed b)"
  unfolding state_well_formed_def by blast+
  have universe:
    "version_ids s \<union> historical_versions_of s \<subseq>
      version_ids (copy_on_write_revision s prior request next_snapshot_id) \<union>
      historical_versions_of
        (copy_on_write_revision s prior request next_snapshot_id)"
  using copy_on_write_reference_universe_mono[OF source_unique prior_member] .
  have unique:
    "unique_versions (copy_on_write_revision s prior request next_snapshot_id)"
  using copy_on_write_preserves_unique_versions[OF valid] .
  have all_objects:
    "\<forall>object\<in>objects_of
      (copy_on_write_revision s prior request next_snapshot_id).
      kind_of object \<in> supported_kinds_of G \<and>
      references_complete
        (copy_on_write_revision s prior request next_snapshot_id) object \<and>
      evidence_well_formed
        (copy_on_write_revision s prior request next_snapshot_id) object \<and>
      time_well_formed object \<and>
      relation_well_formed
        (copy_on_write_revision s prior request next_snapshot_id) object \<and>
      effect_well_formed
        (copy_on_write_revision s prior request next_snapshot_id) object \<and>
      perspective_well_formed
        (copy_on_write_revision s prior request next_snapshot_id) object \<and>
      context_well_formed
        (copy_on_write_revision s prior request next_snapshot_id) object \<and>
      relevance_well_formed
        (copy_on_write_revision s prior request next_snapshot_id) object \<and>
      orientation_well_formed
        (copy_on_write_revision s prior request next_snapshot_id) object \<and>
      action_well_formed
        (copy_on_write_revision s prior request next_snapshot_id) object \<and>
      feedback_well_formed
        (copy_on_write_revision s prior request next_snapshot_id) object"
  proof (intro ballI)
    fix object
    assume member:
      "object \<in> objects_of
        (copy_on_write_revision s prior request next_snapshot_id)"
    consider (new) "object = revised_object prior request"
    | (old) "object \<in> objects_of s - {prior}"
    using member by auto
    then show
      "kind_of object \<in> supported_kinds_of G \<and>
        references_complete
          (copy_on_write_revision s prior request next_snapshot_id) object \<and>
        evidence_well_formed

```

```

    (copy_on_write_revision s prior request next_snapshot_id) object \<and>
    time_well_formed object \<and>
    relation_well_formed
    (copy_on_write_revision s prior request next_snapshot_id) object \<and>
    effect_well_formed
    (copy_on_write_revision s prior request next_snapshot_id) object \<and>
    perspective_well_formed
    (copy_on_write_revision s prior request next_snapshot_id) object \<and>
    context_well_formed
    (copy_on_write_revision s prior request next_snapshot_id) object \<and>
    relevance_well_formed
    (copy_on_write_revision s prior request next_snapshot_id) object \<and>
    orientation_well_formed
    (copy_on_write_revision s prior request next_snapshot_id) object \<and>
    action_well_formed
    (copy_on_write_revision s prior request next_snapshot_id) object \<and>
    feedback_well_formed
    (copy_on_write_revision s prior request next_snapshot_id) object"
proof cases
case new
then show ?thesis
  using new_conditions
  by (simp add: articulated_object_conditions_def)
next
case old
from source_objects old have conditions:
  "kind_of object \<in> supported_kinds_of G \<and>
  references_complete s object \<and> evidence_well_formed s object \<and>
  time_well_formed object \<and> relation_well_formed s object \<and>
  effect_well_formed s object \<and> perspective_well_formed s object \<and>
  context_well_formed s object \<and> relevance_well_formed s object \<and>
  orientation_well_formed s object \<and> action_well_formed s object \<and>
  feedback_well_formed s object"
  by blast
from conditions have kind:
  "kind_of object \<in> supported_kinds_of G"
  and refs_source: "references_complete s object"
  and evidence_source: "evidence_well_formed s object"
  and time: "time_well_formed object"
  and relation_source: "relation_well_formed s object"
  and effect_source: "effect_well_formed s object"
  and perspective_source: "perspective_well_formed s object"
  and context_source: "context_well_formed s object"
  and relevance_source: "relevance_well_formed s object"
  and orientation_source: "orientation_well_formed s object"
  and action_source: "action_well_formed s object"
  and feedback_source: "feedback_well_formed s object"
  by blast+
have refs_target:
  "references_complete
  (copy_on_write_revision s prior request next_snapshot_id) object"
  using refs_source universe
  unfolding references_complete_def by blast
have evidence_target:
  "evidence_well_formed
  (copy_on_write_revision s prior request next_snapshot_id) object"
  using evidence_source universe
  unfolding evidence_well_formed_def evidence_references_complete_def
  by blast
have relation_target:
  "relation_well_formed
  (copy_on_write_revision s prior request next_snapshot_id) object"
  using relation_source universe
  unfolding relation_well_formed_def present_ref_def
  by (cases "source_ref_of object"; cases "target_ref_of object"; auto)
have effect_target:
  "effect_well_formed
  (copy_on_write_revision s prior request next_snapshot_id) object"
  using effect_source universe
  unfolding effect_well_formed_def present_ref_def
  by (cases "effect_origin_of object"; auto)
have perspective_target:
  "perspective_well_formed
  (copy_on_write_revision s prior request next_snapshot_id) object"
  using perspective_source
  unfolding perspective_well_formed_def
  by (simp only: copy_on_write_fields(4))
have context_target:
  "context_well_formed
  (copy_on_write_revision s prior request next_snapshot_id) object"

```

```

    using context_source
    unfolding context_well_formed_def
    by (simp only: copy_on_write_fields(5))
  have relevance_target:
    "relevance_well_formed
     (copy_on_write_revision s prior request next_snapshot_id) object"
  using relevance_source
  unfolding relevance_well_formed_def perspective_well_formed_def
  context_well_formed_def
  by (simp only: copy_on_write_fields(4) copy_on_write_fields(5)
    copy_on_write_fields(6))
  have orientation_target:
    "orientation_well_formed
     (copy_on_write_revision s prior request next_snapshot_id) object"
  using orientation_source
  unfolding orientation_well_formed_def by simp
  have action_target:
    "action_well_formed
     (copy_on_write_revision s prior request next_snapshot_id) object"
  using action_source
  unfolding action_well_formed_def
  by (simp only: copy_on_write_fields(7))
  have feedback_target:
    "feedback_well_formed
     (copy_on_write_revision s prior request next_snapshot_id) object"
  using feedback_source universe
  unfolding feedback_well_formed_def present_ref_def
  by (cases "execution_ref_of object"; auto)
  show ?thesis
  using kind refs_target evidence_target time relation_target effect_target
    perspective_target context_target relevance_target orientation_target
    action_target feedback_target
  by blast
qed
qed
have secondary':
  "(\<forall>p\<in>projections_of
    (copy_on_write_revision s prior request next_snapshot_id).
    projection_well_formed
    (copy_on_write_revision s prior request next_snapshot_id) p) \<and>
  (\<forall>c\<in>couplings_of
    (copy_on_write_revision s prior request next_snapshot_id).
    coupling_well_formed
    (copy_on_write_revision s prior request next_snapshot_id) c) \<and>
  (\<forall>t\<in>transmissions_of
    (copy_on_write_revision s prior request next_snapshot_id).
    transmission_well_formed
    (copy_on_write_revision s prior request next_snapshot_id) t) \<and>
  (\<forall>b\<in>boundaries_of
    (copy_on_write_revision s prior request next_snapshot_id).
    boundary_well_formed b)"
  using secondary
  unfolding projection_well_formed_def coupling_well_formed_def
    transmission_well_formed_def
  by (simp only: copy_on_write_fields(5) copy_on_write_fields(9)
    copy_on_write_fields(10) copy_on_write_fields(11)
    copy_on_write_fields(12))
  show ?thesis
  unfolding target_state_well_formed_def
  using unique_all_objects secondary' by blast
qed

theorem execute_revision_preserves_identity:
  assumes concrete:
    "execute_revision_step G s feedback_catalog emergence_catalog revision_catalog
    prior request next_snapshot_id s' revision_catalog"
  shows "immutable_preserved s s'"
proof -
  from concrete have valid:
    "execute_revision_input_valid G s feedback_catalog emergence_catalog
    revision_catalog prior request next_snapshot_id"
  and target: "s' = copy_on_write_revision s prior request next_snapshot_id"
  by (auto simp: execute_revision_step_def)
  from valid have source_unique: "inj_on version_id_of (objects_of s)"
  and fresh:
    "requested_revision_new_version_of request \<notin> version_ids s"
  unfolding execute_revision_input_valid_def state_well_formed_def
  unique_versions_def by auto
  show ?thesis
  unfolding immutable_preserved_def target

```

```

proof (intro ballI allI impI)
  fix old target_object
  assume old_member: "old \<in> objects_of s"
  and target_member:
    "target_object \<in> objects_of
      (copy_on_write_revision s prior request next_snapshot_id)"
  and same: "version_id_of old = version_id_of target_object"
  show "old = target_object"
  proof (cases "target_object = revised_object prior request")
    case True
    have "requested_revision_new_version_of request \<in> version_ids s"
    unfolding version_ids_def
    using old_member same True by (metis imageI revised_object_fields(2))
    then show ?thesis using fresh by blast
  next
    case False
    then have target_old: "target_object \<in> objects_of s"
    using target_member by simp
    show ?thesis using inj_onD[OF source_unique same old_member target_old] .
  qed
qed
qed

theorem execute_revision_refines_generic_step:
  assumes concrete:
    "execute_revision_step G s feedback_catalog emergence_catalog revision_catalog
      prior request next_snapshot_id s' revision_catalog"
  shows "step G s ReviseObject s'"
proof -
  from concrete have source: "state_well_formed G s"
  and allowed: "ReviseObject \<in> allowed_operations_of G"
  unfolding execute_revision_step_def execute_revision_input_valid_def by blast+
  have target: "state_well_formed G s'"
  using execute_revision_preserves_state_well_formed[OF concrete] .
  have immutable: "immutable_preserved s s'"
  using execute_revision_preserves_identity[OF concrete] .
  show ?thesis unfolding step_def using source target allowed immutable by blast
qed

theorem execute_revision_exact_copy_on_write:
  assumes concrete:
    "execute_revision_step G s feedback_catalog emergence_catalog revision_catalog
      prior request next_snapshot_id s' revision_catalog"
  shows
    "prior \<notin> objects_of s' \<and>
      revised_object prior request \<in> objects_of s' \<and>
      version_id_of prior \<in> historical_versions_of s' \<and>
      previous_version_of (revised_object prior request) =
        Some (version_id_of prior)"
proof -
  from concrete have valid:
    "execute_revision_input_valid G s feedback_catalog emergence_catalog
      revision_catalog prior request next_snapshot_id"
  and target: "s' = copy_on_write_revision s prior request next_snapshot_id"
  by (auto simp: execute_revision_step_def)
  from valid have prior_member: "prior \<in> objects_of s"
  and fresh:
    "requested_revision_new_version_of request \<notin> version_ids s"
  unfolding execute_revision_input_valid_def by blast+
  have distinct: "revised_object prior request \<noteq> prior"
  proof
    assume equal: "revised_object prior request = prior"
    have "requested_revision_new_version_of request \<in> version_ids s"
    unfolding version_ids_def
    using prior_member equal by (metis imageI revised_object_fields(2))
    then show False using fresh by blast
  qed
  show ?thesis unfolding target using distinct by simp
qed
end

```

## K.12 Vollständiger T1-T13-Zeuge

Normativer K3-Quellstand: T9\_T13/UZA\_0\_9\_1\_T1\_T13\_Witness.thy. Bestandteil des  
Isabelle2025-2-Clean-Builds mit Exit-Code 0.

```
theory UZA_0_9_1_T1_T13_Witness
  imports UZA_0_9_1_Revise_Object
          UZA_0_9_1_T8.UZA_0_9_1_Synthesize_Orientation_Witness
begin

section \<open>A single causally reachable T1--T13 operational witness\<close>

definition chain_t13_grammar :: grammar where
  "chain_t13_grammar =
    chain_t8_grammar\<lparr>
      supported_kinds_of :=
        insert K_META (insert K_ACT (insert K_FDB (insert K_EMG
          (supported_kinds_of chain_t8_grammar))))),
      allowed_operations_of :=
        insert SelectAction (insert ExecuteAction (insert RecordFeedback
          (insert AnalyzeEmergence (insert ReviseObject
            (allowed_operations_of chain_t8_grammar))))))
    \<rparr>"

definition chain_selection_catalog_0 :: action_selection_catalog where
  "chain_selection_catalog_0 = Map.empty"

definition chain_decision_authority :: decision_authority where
  "chain_decision_authority =
    \<lparr>
      decision_authority_id_of = 40,
      decision_authority_perspective_of = 10,
      decision_authority_context_of = 20,
      decision_authority_valid_from_of = 12,
      decision_authority_valid_to_of = None,
      decision_authority_mandate_of = {SelectAction, ExecuteAction},
      decision_authority_explanation_of =
        'Mandate for the explicit T9--T10 witness'
    \<rparr>"

definition chain_action_gates :: action_gate_profile where
  "chain_action_gates =
    \<lparr>
      legal_gate_passed_of = True,
      ethical_gate_passed_of = True,
      organizational_gate_passed_of = True,
      resource_gate_passed_of = True
    \<rparr>"

definition chain_action_option :: action_option where
  "chain_action_option =
    \<lparr>
      action_option_id_of = 900,
      action_option_orientation_of = 12,
      action_option_gates_of = chain_action_gates,
      action_option_conflict_resolved_of = True,
      action_option_risk_accepted_of = True,
      action_option_power_shift_refs_of = {10, 11},
      action_option_responsibility_of = Some 40,
      action_option_explanation_of =
        'Range-preserving intervention with retained responsibility'
    \<rparr>"

definition chain_selection_request :: action_selection_request where
  "chain_selection_request =
    \<lparr>
      requested_selection_object_id_of = 11,
      requested_selection_version_id_of = 13,
      requested_selection_orientation_of = 12,
      requested_selection_option_id_of = 900,
      requested_selection_options_of = [chain_action_option],
      requested_selection_authority_of = chain_decision_authority,
      requested_selection_time_of = 12,
      requested_selection_explanation_of =
        'Select the only declared admissible option'
    \<rparr>"
```

```

definition chain_after_t9 :: snapshot where
  "chain_after_t9 =
    record_action_selection chain_after_t8 chain_orientation_catalog
    chain_selection_request 13"

definition chain_selection_catalog :: action_selection_catalog where
  "chain_selection_catalog =
    extend_action_selection_catalog chain_selection_catalog_0 chain_after_t8
    chain_orientation_catalog chain_selection_request"

definition chain_execution_catalog_0 :: action_execution_catalog where
  "chain_execution_catalog_0 = Map.empty"

definition chain_execution_request :: action_execution_request where
  "chain_execution_request =
    \<lparr>
      requested_execution_object_id_of = 12,
      requested_execution_version_id_of = 14,
      requested_execution_action_of = 13,
      requested_execution_authority_of = 40,
      requested_execution_context_of = 20,
      requested_execution_started_of = 12,
      requested_execution_finished_of = 13,
      requested_execution_status_of = ExecutionSucceeded,
      requested_execution_explanation_of =
        'Execution recorded separately from selection'
    \<rparr>"

definition chain_after_t10 :: snapshot where
  "chain_after_t10 = add_execution_record chain_after_t9
    chain_execution_request 14"

definition chain_execution_catalog :: action_execution_catalog where
  "chain_execution_catalog =
    extend_action_execution_catalog chain_execution_catalog_0
    chain_execution_request"

definition chain_feedback_catalog_0 :: feedback_catalog where
  "chain_feedback_catalog_0 = Map.empty"

definition chain_feedback_request :: feedback_request where
  "chain_feedback_request =
    \<lparr>
      requested_feedback_object_id_of = 13,
      requested_feedback_version_id_of = 15,
      requested_feedback_execution_of = 14,
      requested_feedback_observations_of = {10, 11},
      requested_feedback_context_of = 20,
      requested_feedback_time_of = 14,
      requested_feedback_expected_of = 3 / 4,
      requested_feedback_observed_of = 2 / 5,
      requested_feedback_tolerance_of = 1 / 10,
      requested_feedback_explanation_of =
        'Observed deviation retained with both perspective witnesses'
    \<rparr>"

definition chain_after_t11 :: snapshot where
  "chain_after_t11 = add_feedback chain_after_t10 chain_feedback_request 15"

definition chain_feedback_catalog :: feedback_catalog where
  "chain_feedback_catalog =
    extend_feedback_catalog chain_feedback_catalog_0 chain_feedback_request"

definition chain_transition_t9 :: transition where
  "chain_transition_t9 =
    \<lparr>transition_before_of = chain_after_t8,
      transition_operation_of = SelectAction,
      transition_after_of = chain_after_t9\<rparr>"

definition chain_transition_t10 :: transition where
  "chain_transition_t10 =
    \<lparr>transition_before_of = chain_after_t9,
      transition_operation_of = ExecuteAction,
      transition_after_of = chain_after_t10\<rparr>"

definition chain_transition_t11 :: transition where
  "chain_transition_t11 =
    \<lparr>transition_before_of = chain_after_t10,
      transition_operation_of = RecordFeedback,
```

```

    transition_after_of = chain_after_t11\<rparr>"

definition chain_interaction_t12 :: interaction_complex where
  "chain_interaction_t12 =
    \<lparr>
      baseline_of = chain_after_t8,
      current_of = chain_after_t11,
      trace_of = [chain_transition_t9, chain_transition_t10,
        chain_transition_t11],
      candidate_patches_of = {}
    \<rparr>"

definition chain_emergence_catalog_0 :: emergence_catalog where
  "chain_emergence_catalog_0 = Map.empty"

definition chain_emergence_request :: emergence_request where
  "chain_emergence_request =
    \<lparr>
      requested_emergence_object_id_of = 14,
      requested_emergence_version_id_of = 16,
      requested_emergence_feedback_of = 15,
      requested_emergence_interaction_of = chain_interaction_t12,
      requested_emergence_context_of = 20,
      requested_emergence_time_of = 15,
      requested_emergence_explanation_of =
        ''Trace-relative analysis of the T9--T11 articulation''
    \<rparr>"

definition chain_after_t12 :: snapshot where
  "chain_after_t12 = add_emergence_analysis chain_t13_grammar chain_after_t11
    chain_emergence_request 16"

definition chain_emergence_catalog :: emergence_catalog where
  "chain_emergence_catalog =
    extend_emergence_catalog chain_emergence_catalog_0 chain_t13_grammar
    chain_emergence_request"

definition chain_revision_catalog_0 :: revision_catalog where
  "chain_revision_catalog_0 = Map.empty"

definition chain_revision_request :: revision_request where
  "chain_revision_request =
    \<lparr>
      requested_revision_prior_version_of = 12,
      requested_revision_new_version_of = 17,
      requested_revision_feedback_of = 15,
      requested_revision_emergence_of = 16,
      requested_revision_time_of = 16,
      requested_revision_explanation_of =
        ''Revise the orientation by copy-on-write after recorded deviation''
    \<rparr>"

definition chain_orientation_prior :: uza_object where
  "chain_orientation_prior =
    synthesized_orientation_object chain_relevance_catalog
    chain_orientation_request"

definition chain_after_t13 :: snapshot where
  "chain_after_t13 =
    copy_on_write_revision chain_after_t12 chain_orientation_prior
    chain_revision_request 17"

definition chain_revision_catalog :: revision_catalog where
  "chain_revision_catalog =
    extend_revision_catalog chain_revision_catalog_0 chain_orientation_prior
    chain_revision_request"

subsection \<open>T8 source frame under the extended operational grammar\<close>

lemma chain_after_t8_state_well_formed_t13:
  "state_well_formed chain_t13_grammar chain_after_t8"
proof -
  have source: "state_well_formed chain_t8_grammar chain_after_t8"
    using chain_t8_success_certificates by blast
  show ?thesis
    using source
    unfolding state_well_formed_def chain_t13_grammar_def
    by auto
qed

```



```

lemma chain_after_t8_active_history_disjoint:
  "active_history_disjoint chain_after_t8"
  unfolding chain_after_t8_def
  using add_orientation_preserves_active_history_disjoint[
    OF chain_t8_success_input_valid] .

lemma chain_after_t8_object_ids_exact:
  "object_id_of ` objects_of chain_after_t8 =
    {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10}"
  unfolding chain_after_t8_def
  using chain_after_t7_object_ids_exact
  by (auto simp: chain_orientation_request_def)

lemma chain_after_t8_version_ids_exact:
  "version_ids chain_after_t8 =
    {0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 12}"
  unfolding chain_after_t8_def
  using chain_after_t7_version_ids_exact
  by (auto simp: chain_orientation_request_def)

lemma chain_after_t8_frame_exact:
  "historical_versions_of chain_after_t8 = {5, 6} \<and>
    perspectives_of chain_after_t8 = {0, Suc 0, 10} \<and>
    contexts_of chain_after_t8 = {20} \<and>
    models_of chain_after_t8 = {30} \<and>
    snapshot_id_of chain_after_t8 = 12"
  unfolding chain_after_t8_def
  using chain_after_t7_frame_exact by simp

lemma chain_orientation_known_for_t9:
  "known_selection_orientation chain_after_t8 chain_relevance_catalog
    chain_orientation_catalog 12"
  unfolding known_selection_orientation_def
proof (intro exI conjI)
  show "chain_orientation_prior \<in> objects_of chain_after_t8"
    by (simp add: chain_after_t8_def chain_orientation_prior_def)
  show "version_id_of chain_orientation_prior = 12"
    by (simp add: chain_orientation_prior_def chain_orientation_request_def)
  show "kind_of chain_orientation_prior = K_ORI"
    by (simp add: chain_orientation_prior_def)
  show "chain_orientation_catalog 12 =
    Some (synthesized_orientation_payload chain_relevance_catalog
      chain_orientation_request)"
    by (simp add: chain_orientation_catalog_def extend_orientation_catalog_def
      chain_orientation_request_def)
  show "orientation_payload_valid chain_relevance_catalog chain_orientation_prior
    (synthesized_orientation_payload chain_relevance_catalog
      chain_orientation_request)"
    unfolding chain_orientation_prior_def
    using synthesized_orientation_payload_is_valid[
      OF chain_t8_success_input_valid] .
qed

lemma chain_selection_orientation_applicable:
  "selection_orientation_status chain_orientation_catalog 12 = Applicable"
  using chain_orientation_success_status
  by (simp add: selection_orientation_status_def chain_orientation_catalog_def
    extend_orientation_catalog_def chain_orientation_request_def)

lemma chain_decision_authority_valid:
  "decision_authority_valid chain_after_t8 chain_decision_authority 12"
  using chain_after_t8_frame_exact
  by (simp add: decision_authority_valid_def authority_active_at_def
    chain_decision_authority_def)

lemma chain_selection_outcome_success:
  "action_selection_outcome chain_after_t8 chain_orientation_catalog
    chain_selection_request = ActionSelected 900"
  using chain_selection_orientation_applicable chain_decision_authority_valid
  by (simp add: action_selection_outcome_def chain_selection_request_def
    chain_action_option_def chain_action_gates_def chain_decision_authority_def
    requested_action_option_def basic_action_gates_passed_def
    explicit_power_and_responsibility_def Let_def)

lemma chain_selection_record_valid:
  "selection_record_valid
    (selection_record_object chain_after_t8 chain_orientation_catalog
      chain_selection_request)
    (selection_record_payload chain_after_t8 chain_orientation_catalog
      chain_selection_request)"

```

```

using chain_selection_outcome_success
by (simp add: selection_record_valid_def selection_record_object_def
  selection_record_payload_def selection_power_refs_def
  selection_responsibility_def chain_selection_request_def
  chain_action_option_def requested_action_option_def Let_def)

lemma chain_selection_source_state_well_formed:
  "state_well_formed chain_t13_grammar
    (selection_source_frame chain_after_t8 chain_orientation_catalog
      chain_selection_request)"
  unfolding selection_source_frame_def
  using extend_authorities_preserves_state_well_formed[
    OF chain_after_t8_state_well_formed_t13] .

lemma chain_t9_audited_input_valid:
  "audited_append_input_valid chain_t13_grammar
    (selection_source_frame chain_after_t8 chain_orientation_catalog
      chain_selection_request)
  SelectAction
  (selection_record_object chain_after_t8 chain_orientation_catalog
    chain_selection_request) 13"

proof -
  let ?source =
    "selection_source_frame chain_after_t8 chain_orientation_catalog
      chain_selection_request"
  let ?record =
    "selection_record_object chain_after_t8 chain_orientation_catalog
      chain_selection_request"
  let ?target = "add_appearance ?source ?record 13"
  have source_well_formed: "state_well_formed chain_t13_grammar ?source"
    using chain_selection_source_state_well_formed .
  have source_history: "active_history_disjoint ?source"
    unfolding selection_source_frame_def
    using chain_after_t8_active_history_disjoint by simp
  have operation_allowed:
    "SelectAction \<in> allowed_operations_of chain_t13_grammar"
    by (simp add: chain_t13_grammar_def)
  have record_fields:
    "object_id_of ?record = 11 \<and>
      version_id_of ?record = 13 \<and>
      kind_of ?record = K_ACT \<and>
      references_of ?record = {12} \<and>
      evidence_of ?record = {12} \<and>
      perspective_of ?record = Some 10 \<and>
      context_of ?record = Some 20 \<and>
      authority_of ?record = Some 40 \<and>
      valid_to_of ?record = None \<and>
      basis_refs_of ?record = {12}"
    using chain_selection_outcome_success
    by (simp add: selection_record_object_def chain_selection_request_def
      chain_decision_authority_def Let_def)
  have source_object_ids:
    "image object_id_of (objects_of ?source) =
      {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10}"
    unfolding selection_source_frame_def
    using chain_after_t8_object_ids_exact
    by (simp only: extend_authorities_frame(2))
  have source_versions:
    "version_ids ?source = {0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 12}"
    unfolding selection_source_frame_def
    using chain_after_t8_version_ids_exact
    by (simp only: extend_authorities_version_ids)
  have source_frame:
    "historical_versions_of ?source = {5, 6} \<and>
      perspectives_of ?source = {0, Suc 0, 10} \<and>
      contexts_of ?source = {20} \<and>
      40 \<in> authorities_of ?source \<and>
      snapshot_id_of ?source = 12"
    unfolding selection_source_frame_def
    using chain_after_t8_frame_exact chain_selection_outcome_success
    by (simp add: selection_authority_extension_def chain_selection_request_def
      chain_decision_authority_def)
  have fresh_object:
    "object_id_of ?record \<notin> image object_id_of (objects_of ?source)"
    using record_fields source_object_ids by simp
  have fresh_version:
    "version_id_of ?record \<notin>
      version_ids ?source \<union> historical_versions_of ?source"
    using record_fields source_versions source_frame by simp
  have target_versions:

```

```

"version_ids ?target =
  {0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 12, 13}"
using source_versions record_fields
apply (simp only: add_appearance_version_ids)
by (rule Set.set_eqI; auto)
have target_frame:
  "historical_versions_of ?target = {5, 6} \<and>
    perspectives_of ?target = {0, Suc 0, 10} \<and>
    contexts_of ?target = {20} \<and>
    40 \<in> authorities_of ?target"
using source_frame
by (simp only: add_appearance_background(1)
  add_appearance_background(2) add_appearance_background(3)
  add_appearance_background(5))
have references: "references_complete ?target ?record"
using record_fields target_versions target_frame
unfolding references_complete_def by simp
have evidence: "evidence_well_formed ?target ?record"
using record_fields target_versions target_frame
unfolding evidence_well_formed_def evidence_references_complete_def
by simp
have time: "time_well_formed ?record"
using record_fields unfolding time_well_formed_def by simp
have relation: "relation_well_formed ?target ?record"
using record_fields unfolding relation_well_formed_def by simp
have effect: "effect_well_formed ?target ?record"
using record_fields unfolding effect_well_formed_def by simp
have perspective: "perspective_well_formed ?target ?record"
using record_fields target_frame
unfolding perspective_well_formed_def perspective_bound_kind_def by simp
have context_ok: "context_well_formed ?target ?record"
using record_fields target_frame
unfolding context_well_formed_def context_bound_kind_def by simp
have relevance: "relevance_well_formed ?target ?record"
using record_fields unfolding relevance_well_formed_def by simp
have orientation: "orientation_well_formed ?target ?record"
using record_fields unfolding orientation_well_formed_def by simp
have action: "action_well_formed ?target ?record"
using record_fields target_frame
unfolding action_well_formed_def by simp
have feedback: "feedback_well_formed ?target ?record"
using record_fields unfolding feedback_well_formed_def by simp
have articulated:
  "articulated_object_conditions chain_t13_grammar ?target ?record"
unfolding articulated_object_conditions_def
using record_fields references evidence time relation effect perspective context_ok
  relevance orientation action feedback
by (simp add: chain_t13_grammar_def)
show ?thesis
unfolding audited_append_input_valid_def
using source_well_formed source_history operation_allowed fresh_object
  fresh_version articulated source_frame
by simp
qed

lemma chain_t9_input_valid:
  "select_action_input_valid chain_t13_grammar chain_after_t8
    chain_relevance_catalog chain_orientation_catalog chain_selection_catalog_0
    chain_selection_request 13"
unfolding select_action_input_valid_def
using chain_after_t8_state_well_formed_t13
  chain_after_t8_active_history_disjoint
  chain_t8_success_certificates chain_orientation_known_for_t9
  chain_selection_record_valid chain_t9_audited_input_valid
by (auto simp: action_selection_catalog_well_formed_def
  chain_selection_catalog_0_def chain_selection_request_def
  chain_action_option_def)

theorem chain_t9_step:
  "select_action_step chain_t13_grammar chain_after_t8
    chain_relevance_catalog chain_orientation_catalog chain_selection_catalog_0
    chain_selection_request 13 chain_after_t9 chain_selection_catalog"
unfolding select_action_step_def chain_after_t9_def chain_selection_catalog_def
using chain_t9_input_valid by simp

lemma chain_after_t9_state_well_formed:
  "state_well_formed chain_t13_grammar chain_after_t9"
using select_action_preserves_state_well_formed[OF chain_t9_step] .

lemma chain_selection_catalog_well_formed:

```

```

"action_selection_catalog_well_formed chain_after_t9 chain_selection_catalog"
using select_action_preserves_catalog[OF chain_t9_step] .

lemma chain_after_t9_active_history_disjoint:
  "active_history_disjoint chain_after_t9"
  using audited_append_preserves_active_history[
    OF select_action_is_audited_append[OF chain_t9_step]] .

lemma chain_after_t9_object_ids_exact:
  "object_id_of `objects_of chain_after_t9 =
    {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11}"
  unfolding chain_after_t9_def record_action_selection_def
    selection_source_frame_def extend_authorities_def
  using chain_after_t8_object_ids_exact
  by (auto simp: selection_record_object_def chain_selection_request_def Let_def)

lemma chain_after_t9_version_ids_exact:
  "version_ids chain_after_t9 =
    {0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 12, 13}"
  unfolding chain_after_t9_def record_action_selection_def
    selection_source_frame_def extend_authorities_def
  using chain_after_t8_version_ids_exact
  by (auto simp: version_ids_def selection_record_object_def
    chain_selection_request_def Let_def)

lemma chain_after_t9_frame_exact:
  "historical_versions_of chain_after_t9 = {5, 6} \<and>
    contexts_of chain_after_t9 = {20} \<and>
    perspectives_of chain_after_t9 = {0, Suc 0, 10} \<and>
    40 \<in> authorities_of chain_after_t9 \<and>
    snapshot_id_of chain_after_t9 = 13"
  unfolding chain_after_t9_def record_action_selection_def
    selection_source_frame_def selection_authority_extension_def
    extend_authorities_def
  using chain_after_t8_frame_exact chain_selection_outcome_success
  by (simp add: chain_selection_request_def chain_decision_authority_def)

subsection \<open>T10 reachable execution\<close>

lemma chain_selected_action_active:
  "active_selected_action chain_after_t9 chain_selection_catalog 13 40"
  unfolding active_selected_action_def
proof (intro exI conjI)
  show "selection_record_object chain_after_t8 chain_orientation_catalog
    chain_selection_request \<in> objects_of chain_after_t9"
    by (simp add: chain_after_t9_def record_action_selection_def)
  show "version_id_of (selection_record_object chain_after_t8
    chain_orientation_catalog chain_selection_request) = 13"
    by (simp add: selection_record_object_def chain_selection_request_def Let_def)
  show "kind_of (selection_record_object chain_after_t8
    chain_orientation_catalog chain_selection_request) = K_ACT"
    using chain_selection_outcome_success
    by (simp add: selection_record_object_def Let_def)
  show "lifecycle_of (selection_record_object chain_after_t8
    chain_orientation_catalog chain_selection_request) = Validated"
    using chain_selection_outcome_success
    by (simp add: selection_record_object_def Let_def)
  show "authority_of (selection_record_object chain_after_t8
    chain_orientation_catalog chain_selection_request) = Some 40"
    using chain_selection_outcome_success
    by (simp add: selection_record_object_def chain_selection_request_def
    chain_decision_authority_def Let_def)
  show "chain_selection_catalog 13 = Some
    (selection_record_payload chain_after_t8 chain_orientation_catalog
    chain_selection_request)"
    by (simp add: chain_selection_catalog_def
    extend_action_selection_catalog_def chain_selection_request_def)
  show "selection_outcome_of (selection_record_payload chain_after_t8
    chain_orientation_catalog chain_selection_request) = ActionSelected 900"
    using chain_selection_outcome_success
    by (simp add: selection_record_payload_def Let_def)
  show "selection_record_valid
    (selection_record_object chain_after_t8 chain_orientation_catalog
    chain_selection_request)
    (selection_record_payload chain_after_t8 chain_orientation_catalog
    chain_selection_request)"
    using chain_selection_record_valid .
qed

lemma chain_execution_record_valid:

```

```

"execution_record_valid (execution_record_object chain_execution_request)
  (execution_record_payload chain_execution_request)"
by (simp add: execution_record_valid_def execution_record_object_def
  execution_record_payload_def chain_execution_request_def)

lemma chain_t10_audited_input_valid:
  "audited_append_input_valid chain_t13_grammar chain_after_t9 ExecuteAction
    (execution_record_object chain_execution_request) 14"
proof -
  let ?record = "execution_record_object chain_execution_request"
  let ?target = "add_appearance chain_after_t9 ?record 14"
  have fields:
    "object_id_of ?record = 12 \<and>
      version_id_of ?record = 14 \<and>
      kind_of ?record = K_META \<and>
      references_of ?record = {13} \<and>
      evidence_of ?record = {13} \<and>
      valid_to_of ?record = None"
  by (simp add: execution_record_object_def chain_execution_request_def)
  have target_ref: "13 \<in> version_ids ?target"
  using chain_after_t9_version_ids_exact
  by (simp only: add_appearance_version_ids; simp)
  have references: "references_complete ?target ?record"
  using fields target_ref unfolding references_complete_def by simp
  have evidence: "evidence_well_formed ?target ?record"
  using fields target_ref
  unfolding evidence_well_formed_def evidence_references_complete_def
  by simp
  have time: "time_well_formed ?record"
  using fields unfolding time_well_formed_def by simp
  have supported: "K_META \<in> supported_kinds_of chain_t13_grammar"
  by (simp add: chain_t13_grammar_def)
  have articulated:
    "articulated_object_conditions chain_t13_grammar ?target ?record"
  using articulated_meta_recordI[OF _ supported references evidence time]
  fields by blast
  have fresh_object:
    "object_id_of ?record \<notin>
      image object_id_of (objects_of chain_after_t9)"
  using fields chain_after_t9_object_ids_exact by simp
  have fresh_version:
    "version_id_of ?record \<notin>
      version_ids chain_after_t9 \<union>
      historical_versions_of chain_after_t9"
  using fields chain_after_t9_version_ids_exact chain_after_t9_frame_exact
  by simp
  have allowed:
    "ExecuteAction \<in> allowed_operations_of chain_t13_grammar"
  by (simp add: chain_t13_grammar_def)
  have clock: "snapshot_id_of chain_after_t9 < 14"
  using chain_after_t9_frame_exact by simp
  show ?thesis
  by (rule audited_append_input_validI[
    OF chain_after_t9_state_well_formed
    chain_after_t9_active_history_disjoint allowed fresh_object
    fresh_version articulated clock])
qed

lemma chain_t10_input_valid:
  "execute_action_input_valid chain_t13_grammar chain_after_t9
    chain_selection_catalog chain_execution_catalog_0 chain_execution_request 14"
unfolding execute_action_input_valid_def
using chain_after_t9_state_well_formed chain_after_t9_active_history_disjoint
chain_selection_catalog_well_formed chain_selected_action_active
chain_after_t9_frame_exact chain_execution_record_valid
chain_t10_audited_input_valid
by (auto simp: action_execution_catalog_well_formed_def
  chain_execution_catalog_0_def chain_execution_request_def)

theorem chain_t10_step:
  "execute_action_step chain_t13_grammar chain_after_t9
    chain_selection_catalog chain_execution_catalog_0 chain_execution_request 14
    chain_after_t10 chain_execution_catalog"
unfolding execute_action_step_def chain_after_t10_def chain_execution_catalog_def
using chain_t10_input_valid by simp

lemma chain_after_t10_state_well_formed:
  "state_well_formed chain_t13_grammar chain_after_t10"
using execute_action_preserves_state_well_formed[OF chain_t10_step] .

```

```

lemma chain_execution_catalog_well_formed:
  "action_execution_catalog_well_formed chain_after_t10 chain_execution_catalog"
  using execute_action_preserves_catalog[OF chain_t10_step] .

lemma chain_after_t10_active_history_disjoint:
  "active_history_disjoint chain_after_t10"
  using audited_append_preserves_active_history[
    OF execute_action_is_audited_append[OF chain_t10_step]] .

lemma chain_after_t10_object_ids_exact:
  "object_id_of ` objects_of chain_after_t10 =
    {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}"
  unfolding chain_after_t10_def add_execution_record_def
  using chain_after_t9_object_ids_exact
  by (auto simp: execution_record_object_def chain_execution_request_def)

lemma chain_after_t10_version_ids_exact:
  "version_ids chain_after_t10 =
    {0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 12, 13, 14}"
  unfolding chain_after_t10_def add_execution_record_def
  using chain_after_t9_version_ids_exact
  by (auto simp: version_ids_def execution_record_object_def
    chain_execution_request_def)

lemma chain_after_t10_frame_exact:
  "historical_versions_of chain_after_t10 = {5, 6} \<and>
    contexts_of chain_after_t10 = {20} \<and>
    perspectives_of chain_after_t10 = {0, Suc 0, 10} \<and>
    40 \<in> authorities_of chain_after_t10 \<and>
    snapshot_id_of chain_after_t10 = 14"
  unfolding chain_after_t10_def add_execution_record_def
  using chain_after_t9_frame_exact by simp

subsection \<open>T11 reachable deviation feedback\<close>

lemma chain_execution_active:
  "active_execution_record chain_after_t10 chain_execution_catalog 14"
  unfolding active_execution_record_def
proof (intro exI conjI)
  show "execution_record_object chain_execution_request \<in>
    objects_of chain_after_t10"
    by (simp add: chain_after_t10_def add_execution_record_def)
  show "version_id_of (execution_record_object chain_execution_request) = 14"
    by (simp add: execution_record_object_def chain_execution_request_def)
  show "chain_execution_catalog 14 =
    Some (execution_record_payload chain_execution_request)"
    by (simp add: chain_execution_catalog_def
      extend_action_execution_catalog_def chain_execution_request_def)
  show "execution_record_valid
    (execution_record_object chain_execution_request)
    (execution_record_payload chain_execution_request)"
    using chain_execution_record_valid .
qed

lemma chain_feedback_payload_valid:
  "feedback_payload_valid (feedback_object chain_feedback_request)
    (feedback_payload chain_feedback_request)"
  by (simp add: feedback_payload_valid_def feedback_object_def
    feedback_payload_def feedback_deviation_def feedback_assessment_def
    chain_feedback_request_def)

lemma chain_t11_audited_input_valid:
  "audited_append_input_valid chain_t13_grammar chain_after_t10 RecordFeedback
    (feedback_object chain_feedback_request) 15"
proof -
  let ?record = "feedback_object chain_feedback_request"
  let ?target = "add_appearance chain_after_t10 ?record 15"
  have fields:
    "object_id_of ?record = 13 \<and>
      version_id_of ?record = 15 \<and>
      kind_of ?record = K_FDB \<and>
      references_of ?record = {10, 11, 14} \<and>
      evidence_of ?record = {10, 11} \<and>
      context_of ?record = Some 20 \<and>
      valid_to_of ?record = None \<and>
      execution_ref_of ?record = Some 14 \<and>
      observation_refs_of ?record = {10, 11}"
  by (auto simp: feedback_object_def chain_feedback_request_def
    feedback_assessment_def feedback_deviation_def)
  have target_refs:

```

```

"10 \<in> version_ids ?target \<and>
  11 \<in> version_ids ?target \<and>
  14 \<in> version_ids ?target"
using chain_after_t10_version_ids_exact
by (simp only: add_appearance_version_ids; simp)
have target_context: "20 \<in> contexts_of ?target"
  using chain_after_t10_frame_exact
  by (simp only: add_appearance_background(3); simp)
have references: "references_complete ?target ?record"
  using fields target_refs unfolding references_complete_def by simp
have evidence: "evidence_well_formed ?target ?record"
  using fields target_refs
  unfolding evidence_well_formed_def evidence_references_complete_def
  by simp
have time: "time_well_formed ?record"
  using fields unfolding time_well_formed_def by simp
have context_ok: "context_well_formed ?target ?record"
  using fields target_context
  unfolding context_well_formed_def context_bound_kind_def by simp
have feedback_ok: "feedback_well_formed ?target ?record"
  using fields target_refs
  unfolding feedback_well_formed_def present_ref_def by simp
have supported: "K_FDB \<in> supported_kinds_of chain_t13_grammar"
  by (simp add: chain_t13_grammar_def)
have articulated:
  "articulated_object_conditions chain_t13_grammar ?target ?record"
  using articulated_feedback_recordI[
    OF _ supported references evidence time context_ok feedback_ok] fields
  by blast
have fresh_object:
  "object_id_of ?record \<notin>
    image object_id_of (objects_of chain_after_t10)"
  using fields chain_after_t10_object_ids_exact by simp
have fresh_version:
  "version_id_of ?record \<notin>
    version_ids chain_after_t10 \<union>
    historical_versions_of chain_after_t10"
  using fields chain_after_t10_version_ids_exact chain_after_t10_frame_exact
  by simp
have allowed:
  "RecordFeedback \<in> allowed_operations_of chain_t13_grammar"
  by (simp add: chain_t13_grammar_def)
have clock: "snapshot_id_of chain_after_t10 < 15"
  using chain_after_t10_frame_exact by simp
show ?thesis
  by (rule audited_append_input_validI[
    OF chain_after_t10_state_well_formed
    chain_after_t10_active_history_disjoint allowed fresh_object
    fresh_version articulated clock])
qed

lemma chain_t11_input_valid:
  "record_feedback_input_valid chain_t13_grammar chain_after_t10
    chain_execution_catalog chain_feedback_catalog_0 chain_feedback_request 15"
  unfolding record_feedback_input_valid_def
  using chain_after_t10_state_well_formed
  chain_after_t10_active_history_disjoint
  chain_execution_catalog_well_formed chain_execution_active
  chain_after_t10_version_ids_exact chain_after_t10_frame_exact
  chain_feedback_payload_valid chain_t11_audited_input_valid
  by (auto simp: feedback_catalog_well_formed_def chain_feedback_catalog_0_def
    chain_feedback_request_def)

theorem chain_t11_step:
  "record_feedback_step chain_t13_grammar chain_after_t10
    chain_execution_catalog chain_feedback_catalog_0 chain_feedback_request 15
    chain_after_t11 chain_feedback_catalog"
  unfolding record_feedback_step_def chain_after_t11_def chain_feedback_catalog_def
  using chain_t11_input_valid by simp

lemma chain_after_t11_state_well_formed:
  "state_well_formed chain_t13_grammar chain_after_t11"
  using record_feedback_preserves_state_well_formed[OF chain_t11_step] .

lemma chain_feedback_catalog_well_formed:
  "feedback_catalog_well_formed chain_after_t11 chain_feedback_catalog"
  using record_feedback_preserves_catalog[OF chain_t11_step] .

lemma chain_after_t11_active_history_disjoint:
  "active_history_disjoint chain_after_t11"

```

```

using audited_append_preserves_active_history[
  OF record_feedback_is_audited_append[OF chain_t11_step]] .

lemma chain_after_t11_object_ids_exact:
  "object_id_of `objects_of chain_after_t11 =
   {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13}"
  unfolding chain_after_t11_def add_feedback_def
  using chain_after_t10_object_ids_exact
  by (auto simp: feedback_object_def chain_feedback_request_def)

lemma chain_after_t11_version_ids_exact:
  "version_ids chain_after_t11 =
   {0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 12, 13, 14, 15}"
  unfolding chain_after_t11_def add_feedback_def
  using chain_after_t10_version_ids_exact
  by (auto simp: version_ids_def feedback_object_def chain_feedback_request_def)

lemma chain_after_t11_frame_exact:
  "historical_versions_of chain_after_t11 = {5, 6} \<and>
   contexts_of chain_after_t11 = {20} \<and>
   perspectives_of chain_after_t11 = {0, Suc 0, 10} \<and>
   40 \<in> authorities_of chain_after_t11 \<and>
   snapshot_id_of chain_after_t11 = 15"
  unfolding chain_after_t11_def add_feedback_def
  using chain_after_t10_frame_exact by simp

theorem chain_feedback_detects_deviation:
  "feedback_assessment chain_feedback_request = DeviationDetected \<and>
   feedback_deviation chain_feedback_request = 7 / 20"
  by (simp add: feedback_assessment_def feedback_deviation_def
    chain_feedback_request_def)

subsection \<open>T12 trace-relative emergence analysis\<close>

lemma chain_operational_trace_steps:
  "step chain_t13_grammar chain_after_t8 SelectAction chain_after_t9 \<and>
   step chain_t13_grammar chain_after_t9 ExecuteAction chain_after_t10 \<and>
   step chain_t13_grammar chain_after_t10 RecordFeedback chain_after_t11"
  using select_action_refines_generic_step[OF chain_t9_step]
  execute_action_refines_generic_step[OF chain_t10_step]
  record_feedback_refines_generic_step[OF chain_t11_step]
  by blast

lemma chain_interaction_describeable:
  "describeable chain_t13_grammar chain_interaction_t12"
proof -
  from chain_after_t11_state_well_formed have object_conditions:
    "\<forall>obj \<in> objects_of chain_after_t11.
     kind_of obj \<in> supported_kinds_of chain_t13_grammar \<and>
     references_complete chain_after_t11 obj \<and>
     evidence_well_formed chain_after_t11 obj \<and>
     relation_well_formed chain_after_t11 obj \<and>
     effect_well_formed chain_after_t11 obj"
  unfolding state_well_formed_def by blast
  have typed:
    "type_complete chain_t13_grammar chain_interaction_t12"
  using object_conditions
  unfolding type_complete_def chain_interaction_t12_def
  by simp
  have related:
    "relation_complete chain_t13_grammar chain_interaction_t12"
  using object_conditions
  unfolding relation_complete_def chain_interaction_t12_def
  by simp
  have traced:
    "transition_complete chain_t13_grammar chain_interaction_t12"
  using chain_operational_trace_steps
  by (simp add: transition_complete_def chain_interaction_t12_def
    chain_transition_t9_def chain_transition_t10_def
    chain_transition_t11_def)
  show ?thesis unfolding describeable_def using typed related traced by blast
qed

lemma chain_interaction_structural_novelty:
  "structural_novelty chain_interaction_t12"
proof -
  have feedback_member:
    "feedback_object chain_feedback_request \<in> objects_of chain_after_t11"
  by (simp add: chain_after_t11_def add_feedback_def)
  have feedback_kind:

```



```

"kind_of (feedback_object chain_feedback_request) = K_FDB"
by (simp add: feedback_object_def)
have no_feedback_kind:
  "K_FDB \<notin> kind_of ` objects_of chain_after_t8"
proof
  assume member: "K_FDB \<in> kind_of ` objects_of chain_after_t8"
  then obtain object where object_member: "object \<in> objects_of chain_after_t8"
    and kind: "kind_of object = K_FDB" by auto
  have "kind_of object \<in> supported_kinds_of chain_t8_grammar"
    using chain_t8_success_certificates object_member
    unfolding state_well_formed_def by blast
  then show False
    using kind
    by (simp add: chain_t8_grammar_def chain_grammar_def)
qed
have raw_image:
  "kind_of (feedback_object chain_feedback_request) \<in>
    image kind_of (objects_of chain_after_t11)"
  using feedback_member by (rule imageI)
have image_member:
  "K_FDB \<in> image kind_of (objects_of chain_after_t11)"
  using raw_image feedback_kind by simp
have "K_FDB \<in> kind_of ` objects_of chain_after_t11 -
  kind_of ` objects_of chain_after_t8"
  using image_member no_feedback_kind by simp
then have difference_nonempty:
  "image kind_of (objects_of chain_after_t11) -
    image kind_of (objects_of chain_after_t8) \<noteq> {}"
  by blast
show ?thesis
  unfolding structural_novelty_def chain_interaction_t12_def
  using difference_nonempty by simp
qed

lemma chain_emergence_level_E1:
  "emergence_of chain_t13_grammar chain_interaction_t12 = E1"
  using chain_interaction_describeable chain_interaction_structural_novelty
  by (simp add: emergence_of_def)

lemma chain_feedback_active_for_t12:
  "active_feedback_record chain_after_t11 chain_feedback_catalog 15"
  unfolding active_feedback_record_def
proof (intro exI conjI)
  show "feedback_object chain_feedback_request \<in>
    objects_of chain_after_t11"
    by (simp add: chain_after_t11_def add_feedback_def)
  show "version_id_of (feedback_object chain_feedback_request) = 15"
    by (simp add: feedback_object_def chain_feedback_request_def)
  show "chain_feedback_catalog 15 =
    Some (feedback_payload chain_feedback_request)"
    by (simp add: chain_feedback_catalog_def extend_feedback_catalog_def
      chain_feedback_request_def)
  show "feedback_payload_valid (feedback_object chain_feedback_request)
    (feedback_payload chain_feedback_request)"
    using chain_feedback_payload_valid .
qed

lemma chain_emergence_payload_valid:
  "emergence_payload_valid chain_t13_grammar
    (emergence_object chain_t13_grammar chain_emergence_request)
    (emergence_payload chain_t13_grammar chain_emergence_request)"
proof -
  from chain_interaction_describeable have typed:
    "type_complete chain_t13_grammar chain_interaction_t12"
  and related:
    "relation_complete chain_t13_grammar chain_interaction_t12"
  and traced:
    "transition_complete chain_t13_grammar chain_interaction_t12"
  unfolding describeable_def by blast+
  show ?thesis
    using chain_emergence_level_E1 chain_interaction_describeable
    typed related traced
    by (simp add: emergence_payload_valid_def emergence_object_def
      emergence_payload_def emergence_lifecycle_def
      emergence_analysis_status_def describe_def chain_emergence_request_def
      chain_interaction_t12_def Let_def)
qed

lemma chain_t12_audited_input_valid:
  "audited_append_input_valid chain_t13_grammar chain_after_t11

```

```

    AnalyzeEmergence
    (emergence_object chain_t13_grammar chain_emergence_request) 16"
proof -
  let ?record =
    "emergence_object chain_t13_grammar chain_emergence_request"
  let ?target = "add_appearance chain_after_t11 ?record 16"
  have fields:
    "object_id_of ?record = 14 \<and>
     version_id_of ?record = 16 \<and>
     kind_of ?record = K_EMG \<and>
     references_of ?record = {15} \<and>
     evidence_of ?record = {15} \<and>
     valid_to_of ?record = None"
  by (simp add: emergence_object_def chain_emergence_request_def Let_def)
  have target_ref: "15 \<in> version_ids ?target"
  using chain_after_t11_version_ids_exact
  by (simp only: add_appearance_version_ids; simp)
  have references: "references_complete ?target ?record"
  using fields target_ref unfolding references_complete_def by simp
  have evidence: "evidence_well_formed ?target ?record"
  using fields target_ref chain_emergence_level_E1
  unfolding evidence_well_formed_def evidence_references_complete_def
  by (simp add: emergence_object_def chain_emergence_request_def
    emergence_lifecycle_def Let_def)
  have time: "time_well_formed ?record"
  using fields unfolding time_well_formed_def by simp
  have supported: "K_EMG \<in> supported_kinds_of chain_t13_grammar"
  by (simp add: chain_t13_grammar_def)
  have articulated:
    "articulated_object_conditions chain_t13_grammar ?target ?record"
  using articulated_emergence_recordI[
    OF _ supported references evidence time] fields
  by blast
  have fresh_object:
    "object_id_of ?record \<notin>
     image object_id_of (objects_of chain_after_t11)"
  using fields chain_after_t11_object_ids_exact by simp
  have fresh_version:
    "version_id_of ?record \<notin>
     version_ids chain_after_t11 \<union>
     historical_versions_of chain_after_t11"
  using fields chain_after_t11_version_ids_exact chain_after_t11_frame_exact
  by simp
  have allowed:
    "AnalyzeEmergence \<in> allowed_operations_of chain_t13_grammar"
  by (simp add: chain_t13_grammar_def)
  have clock: "snapshot_id_of chain_after_t11 < 16"
  using chain_after_t11_frame_exact by simp
  show ?thesis
  by (rule audited_append_input_validI[
    OF chain_after_t11_state_well_formed
    chain_after_t11_active_history_disjoint allowed fresh_object
    fresh_version articulated clock])
qed

lemma chain_t12_input_valid:
  "analyze_emergence_input_valid chain_t13_grammar chain_after_t11
   chain_feedback_catalog chain_emergence_catalog_0 chain_emergence_request 16"
  unfolding analyze_emergence_input_valid_def
  using chain_after_t11_state_well_formed
  chain_after_t11_active_history_disjoint
  chain_feedback_catalog_well_formed chain_after_t8_state_well_formed_t13
  chain_after_t8_frame_exact
  chain_feedback_active_for_t12 chain_after_t11_frame_exact
  chain_emergence_payload_valid chain_t12_audited_input_valid
  by (auto simp: emergence_catalog_well_formed_def
    chain_emergence_catalog_0_def chain_emergence_request_def
    chain_interaction_t12_def)

theorem chain_t12_step:
  "analyze_emergence_step chain_t13_grammar chain_after_t11
   chain_feedback_catalog chain_emergence_catalog_0 chain_emergence_request 16
   chain_after_t12 chain_emergence_catalog"
  unfolding analyze_emergence_step_def chain_after_t12_def
  chain_emergence_catalog_def
  using chain_t12_input_valid by simp

lemma chain_after_t12_state_well_formed:
  "state_well_formed chain_t13_grammar chain_after_t12"
  using analyze_emergence_preserves_state_well_formed[OF chain_t12_step] .

```

```

lemma chain_after_t12_active_history_disjoint:
  "active_history_disjoint chain_after_t12"
  using audited_append_preserves_active_history[
    OF analyze_emergence_is_audited_append[OF chain_t12_step]] .

lemma chain_after_t12_object_ids_exact:
  "object_id_of ` objects_of chain_after_t12 =
    {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14}"
  unfolding chain_after_t12_def add_emergence_analysis_def
  using chain_after_t11_object_ids_exact
  by (auto simp: emergence_object_def chain_emergence_request_def Let_def)

lemma chain_after_t12_version_ids_exact:
  "version_ids chain_after_t12 =
    {0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}"
  unfolding chain_after_t12_def add_emergence_analysis_def
  using chain_after_t11_version_ids_exact
  by (auto simp: version_ids_def emergence_object_def
    chain_emergence_request_def Let_def)

lemma chain_after_t12_frame_exact:
  "historical_versions_of chain_after_t12 = {5, 6} \<and>
    contexts_of chain_after_t12 = {20} \<and>
    perspectives_of chain_after_t12 = {0, Suc 0, 10} \<and>
    40 \<in> authorities_of chain_after_t12 \<and>
    snapshot_id_of chain_after_t12 = 16"
  unfolding chain_after_t12_def add_emergence_analysis_def
  using chain_after_t11_frame_exact by simp

subsection \<open>T13 copy-on-write revision\<close>

lemma chain_orientation_prior_active:
  "chain_orientation_prior \<in> objects_of chain_after_t12 \<and>
    version_id_of chain_orientation_prior = 12 \<and>
    kind_of chain_orientation_prior = K_ORI"
  unfolding chain_orientation_prior_def chain_after_t12_def
  add_emergence_analysis_def chain_after_t11_def add_feedback_def
  chain_after_t10_def add_execution_record_def chain_after_t9_def
  record_action_selection_def selection_source_frame_def extend_authorities_def
  chain_after_t8_def
  by (simp add: chain_orientation_request_def)

lemma chain_emergence_active_for_t13:
  "active_emergence_record chain_t13_grammar chain_after_t12
    chain_emergence_catalog 16"
proof -
  let ?object = "emergence_object chain_t13_grammar chain_emergence_request"
  let ?payload = "emergence_payload chain_t13_grammar chain_emergence_request"
  have member: "?object \<in> objects_of chain_after_t12"
    by (simp add: chain_after_t12_def add_emergence_analysis_def)
  have version: "version_id_of ?object = 16"
    by (simp add: emergence_object_def chain_emergence_request_def Let_def)
  have catalog: "chain_emergence_catalog 16 = Some ?payload"
    by (simp add: chain_emergence_catalog_def extend_emergence_catalog_def
      chain_emergence_request_def)
  show ?thesis
    unfolding active_emergence_record_def
    using member version catalog chain_emergence_payload_valid by blast
qed

lemma chain_feedback_active_for_t13:
  "active_feedback_record chain_after_t12 chain_feedback_catalog 15"
proof -
  let ?object = "feedback_object chain_feedback_request"
  let ?payload = "feedback_payload chain_feedback_request"
  have member: "?object \<in> objects_of chain_after_t12"
    by (simp add: chain_after_t12_def add_emergence_analysis_def
      chain_after_t11_def add_feedback_def)
  have version: "version_id_of ?object = 15"
    by (simp add: feedback_object_def chain_feedback_request_def)
  have catalog: "chain_feedback_catalog 15 = Some ?payload"
    by (simp add: chain_feedback_catalog_def extend_feedback_catalog_def
      chain_feedback_request_def)
  show ?thesis
    unfolding active_feedback_record_def
    using member version catalog chain_feedback_payload_valid by blast
qed

lemma chain_revision_successor_conditions:

```

```

"articulated_object_conditions chain_t13_grammar chain_after_t13
(revised_object chain_orientation_prior chain_revision_request)"
proof -
  let ?successor =
    "revised_object chain_orientation_prior chain_revision_request"
  have prior_scalar_fields:
    "kind_of chain_orientation_prior = K_ORI \<and>
    perspective_of chain_orientation_prior = Some 10 \<and>
    context_of chain_orientation_prior = Some 20 \<and>
    basis_refs_of chain_orientation_prior = {10, 11}"
  by (simp add: chain_orientation_prior_def chain_orientation_request_def
    synthesized_orientation_object_def)
  have prior_references:
    "references_of chain_orientation_prior = {7, 10, 11}"
  unfolding chain_orientation_prior_def synthesized_orientation_object_def
    synthesized_orientation_references_def chain_orientation_request_def
  by (rule Set.set_eqI; auto)
  have prior_evidence:
    "evidence_of chain_orientation_prior = {10, 11}"
  by (simp add: chain_orientation_prior_def chain_orientation_request_def
    synthesized_orientation_object_def)
  have kind: "kind_of ?successor = K_ORI"
  using prior_scalar_fields by simp
  have lifecycle: "lifecycle_of ?successor = Revised"
  by simp
  have successor_references:
    "references_of ?successor = {7, 10, 11, 16}"
  using prior_references
  unfolding chain_revision_request_def
  apply (simp only: revised_object_fields(6))
  by (rule Set.set_eqI; auto)
  have successor_evidence:
    "evidence_of ?successor = {10, 11, 15}"
  using prior_evidence
  unfolding chain_revision_request_def
  apply (simp only: revised_object_fields(7))
  by (rule Set.set_eqI; auto)
  have successor_perspective:
    "perspective_of ?successor = Some 10"
  using prior_scalar_fields by simp
  have successor_context:
    "context_of ?successor = Some 20"
  using prior_scalar_fields by simp
  have successor_valid_to: "valid_to_of ?successor = None"
  by simp
  have successor_basis: "basis_refs_of ?successor = {10, 11}"
  using prior_scalar_fields by simp
  have source_unique: "unique_versions chain_after_t12"
  using chain_after_t12_state_well_formed
  unfolding state_well_formed_def by blast
  have universe:
    "version_ids chain_after_t12 \<union>
    historical_versions_of chain_after_t12 \<subsequeq>
    version_ids chain_after_t13 \<union>
    historical_versions_of chain_after_t13"
  unfolding chain_after_t13_def
  using copy_on_write_reference_universe_mono[
    OF source_unique, of chain_orientation_prior chain_revision_request 17]
    chain_orientation_prior_active
  by blast
  have source_refs:
    "7 \<in> version_ids chain_after_t12 \<and>
    10 \<in> version_ids chain_after_t12 \<and>
    11 \<in> version_ids chain_after_t12 \<and>
    15 \<in> version_ids chain_after_t12 \<and>
    16 \<in> version_ids chain_after_t12"
  using chain_after_t12_version_ids_exact by simp
  have target_ref_7:
    "7 \<in> version_ids chain_after_t13 \<union>
    historical_versions_of chain_after_t13"
  using universe source_refs by (meson UnI1 subsetD)
  have target_ref_10:
    "10 \<in> version_ids chain_after_t13 \<union>
    historical_versions_of chain_after_t13"
  using universe source_refs by (meson UnI1 subsetD)
  have target_ref_11:
    "11 \<in> version_ids chain_after_t13 \<union>
    historical_versions_of chain_after_t13"
  using universe source_refs by (meson UnI1 subsetD)
  have target_ref_16:

```

```

"16 \<in> version_ids chain_after_t13 \<union>
  historical_versions_of chain_after_t13"
using universe source_refs by (meson UnI1 subsetD)
have target_ref_15:
  "15 \<in> version_ids chain_after_t13 \<union>
    historical_versions_of chain_after_t13"
  using universe source_refs by (meson UnI1 subsetD)
have references: "references_complete chain_after_t13 ?successor"
  unfolding references_complete_def
  apply (subst successor_references)
  using target_ref_7 target_ref_10 target_ref_11 target_ref_16
  by auto
have evidence: "evidence_well_formed chain_after_t13 ?successor"
  unfolding evidence_well_formed_def evidence_references_complete_def
  apply (subst successor_evidence)
  using lifecycle target_ref_10 target_ref_11 target_ref_15
  by auto
have time: "time_well_formed ?successor"
  using successor_valid_to unfolding time_well_formed_def by simp
have target_frame:
  "perspectives_of chain_after_t13 = {0, Suc 0, 10} \<and>
    contexts_of chain_after_t13 = {20}"
  unfolding chain_after_t13_def
  using chain_after_t12_frame_exact
  by (simp only: copy_on_write_fields(4) copy_on_write_fields(5))
have perspective:
  "perspective_well_formed chain_after_t13 ?successor"
  using kind successor_perspective target_frame
  unfolding perspective_well_formed_def perspective_bound_kind_def
  by simp
have context_ok:
  "context_well_formed chain_after_t13 ?successor"
  using kind successor_context target_frame
  unfolding context_well_formed_def context_bound_kind_def
  by simp
have orientation:
  "orientation_well_formed chain_after_t13 ?successor"
  using kind successor_basis unfolding orientation_well_formed_def by simp
have supported: "K_ORI \<in> supported_kinds_of chain_t13_grammar"
  by (simp add: chain_t13_grammar_def chain_t8_grammar_def)
show ?thesis
  unfolding articulated_object_conditions_def
  relation_well_formed_def effect_well_formed_def
  relevance_well_formed_def action_well_formed_def feedback_well_formed_def
  using kind references evidence time perspective context_ok orientation supported
  by (simp add: chain_t13_grammar_def)
qed

lemma chain_t13_input_valid:
  "execute_revision_input_valid chain_t13_grammar chain_after_t12
  chain_feedback_catalog chain_emergence_catalog chain_revision_catalog_0
  chain_orientation_prior chain_revision_request 17"
proof -
  have allowed:
    "ReviseObject \<in> allowed_operations_of chain_t13_grammar"
    by (simp add: chain_t13_grammar_def)
  have prior_member: "chain_orientation_prior \<in> objects_of chain_after_t12"
    using chain_orientation_prior_active by blast
  have prior_version:
    "version_id_of chain_orientation_prior =
      requested_revision_prior_version_of chain_revision_request"
    using chain_orientation_prior_active
    by (simp add: chain_revision_request_def)
  have revisable: "revisable_kind (kind_of chain_orientation_prior)"
    using chain_orientation_prior_active
    by (simp add: revisable_kind_def)
  have fresh:
    "requested_revision_new_version_of chain_revision_request \<notin>
      version_ids chain_after_t12 \<union>
        historical_versions_of chain_after_t12"
    using chain_after_t12_version_ids_exact chain_after_t12_frame_exact
    by (simp add: chain_revision_request_def)
  have empty_catalog:
    "revision_catalog_well_formed chain_after_t12
      chain_revision_catalog_0"
    by (simp add: revision_catalog_well_formed_def
      chain_revision_catalog_0_def)
  have absent:
    "chain_revision_catalog_0
      (requested_revision_new_version_of chain_revision_request) = None"

```

```

    by (simp add: chain_revision_catalog_0_def)
  have temporal:
    "valid_from_of chain_orientation_prior \<le>
      requested_revision_time_of chain_revision_request"
    by (simp add: chain_orientation_prior_def synthesized_orientation_object_def
      chain_orientation_request_def chain_revision_request_def)
  have explained:
    "requested_revision_explanation_of chain_revision_request \<noteq> []"
    by (simp add: chain_revision_request_def)
  have clock: "snapshot_id_of chain_after_t12 < 17"
    using chain_after_t12_frame_exact by simp
  have feedback_active:
    "active_feedback_record chain_after_t12 chain_feedback_catalog
      (requested_revision_feedback_of chain_revision_request)"
    using chain_feedback_active_for_t13
    by (simp add: chain_revision_request_def)
  have emergence_active:
    "active_emergence_record chain_t13_grammar chain_after_t12
      chain_emergence_catalog
      (requested_revision_emergence_of chain_revision_request)"
    using chain_emergence_active_for_t13
    by (simp add: chain_revision_request_def)
  have successor:
    "articulated_object_conditions chain_t13_grammar
      (copy_on_write_revision chain_after_t12 chain_orientation_prior
        chain_revision_request 17)
      (revised_object chain_orientation_prior chain_revision_request)"
    using chain_revision_successor_conditions
    unfolding chain_after_t13_def .
  show ?thesis
    unfolding execute_revision_input_valid_def
    using chain_after_t12_state_well_formed
    chain_after_t12_active_history_disjoint allowed_prior_member prior_version
    revisable fresh feedback_active emergence_active empty_catalog absent
    temporal explained successor clock
    by blast
qed

theorem chain_t13_step:
  "execute_revision_step chain_t13_grammar chain_after_t12
    chain_feedback_catalog chain_emergence_catalog chain_revision_catalog_0
    chain_orientation_prior chain_revision_request 17 chain_after_t13
    chain_revision_catalog"
  unfolding execute_revision_step_def chain_after_t13_def
  chain_revision_catalog_def
  using chain_t13_input_valid by simp

theorem chain_t13_final_certificates:
  "state_well_formed chain_t13_grammar chain_after_t13 \<and>
    step chain_t13_grammar chain_after_t12 ReviseObject chain_after_t13 \<and>
    chain_orientation_prior \<notin> objects_of chain_after_t13 \<and>
    revised_object chain_orientation_prior chain_revision_request
      \<in> objects_of chain_after_t13 \<and>
    12 \<in> historical_versions_of chain_after_t13 \<and>
    previous_version_of
      (revised_object chain_orientation_prior chain_revision_request) = Some 12"
  using execute_revision_preserves_state_well_formed[OF chain_t13_step]
  execute_revision_refines_generic_step[OF chain_t13_step]
  execute_revision_exact_copy_on_write[OF chain_t13_step]
  chain_orientation_prior_active
  by auto

subsection \<open>All five T9 refusal statuses are explicit and non-selecting\<close>

definition chain_selection_no_authority :: action_selection_request where
  "chain_selection_no_authority =
    chain_selection_request\<lparr>
      requested_selection_authority_of :=
        chain_decision_authority\<lparr>decision_authority_mandate_of := {} \<rparr>,
      requested_selection_explanation_of := 'No decision mandate'
    \<rparr>"

definition chain_selection_no_option :: action_selection_request where
  "chain_selection_no_option =
    chain_selection_request\<lparr>
      requested_selection_option_id_of := 901,
      requested_selection_explanation_of := 'No declared option selected'
    \<rparr>"

definition chain_action_option_conflict :: action_option where

```

```

"chain_action_option_conflict =
  chain_action_option\<lparr>action_option_conflict_resolved_of := False\<rparr>"

definition chain_selection_conflict :: action_selection_request where
"chain_selection_conflict =
  chain_selection_request\<lparr>
    requested_selection_options_of := [chain_action_option_conflict],
    requested_selection_explanation_of := 'Conflict remains unresolved'
  \<rparr>"

definition chain_action_option_risk :: action_option where
"chain_action_option_risk =
  chain_action_option\<lparr>action_option_risk_accepted_of := False\<rparr>"

definition chain_selection_risk :: action_selection_request where
"chain_selection_risk =
  chain_selection_request\<lparr>
    requested_selection_options_of := [chain_action_option_risk],
    requested_selection_explanation_of := 'Declared risk not accepted'
  \<rparr>"

definition chain_selection_insufficient :: action_selection_request where
"chain_selection_insufficient =
  chain_selection_request\<lparr>
    requested_selection_explanation_of := 'Insufficient orientation basis'
  \<rparr>"

lemma chain_t9_no_authority_outcome:
"action_selection_outcome chain_after_t8 chain_orientation_catalog
  chain_selection_no_authority = ActionSelectionAbort NoAuthority"
using chain_selection_orientation_applicable chain_after_t8_frame_exact
by (simp add: action_selection_outcome_def selection_orientation_status_def
  chain_selection_no_authority_def chain_selection_request_def
  chain_decision_authority_def decision_authority_valid_def
  authority_active_at_def Let_def)

lemma chain_t9_no_option_outcome:
"action_selection_outcome chain_after_t8 chain_orientation_catalog
  chain_selection_no_option = ActionSelectionAbort NoAdmissibleOption"
using chain_selection_orientation_applicable chain_after_t8_frame_exact
by (simp add: action_selection_outcome_def selection_orientation_status_def
  chain_selection_no_option_def chain_selection_request_def
  chain_decision_authority_def chain_action_option_def
  requested_action_option_def
  decision_authority_valid_def authority_active_at_def Let_def)

lemma chain_t9_conflict_outcome:
"action_selection_outcome chain_after_t8 chain_orientation_catalog
  chain_selection_conflict = ActionSelectionAbort ConflictUnresolved"
using chain_selection_orientation_applicable chain_after_t8_frame_exact
by (simp add: action_selection_outcome_def selection_orientation_status_def
  chain_selection_conflict_def chain_selection_request_def
  chain_decision_authority_def chain_action_option_conflict_def
  chain_action_option_def chain_action_gates_def requested_action_option_def
  basic_action_gates_passed_def explicit_power_and_responsibility_def
  decision_authority_valid_def authority_active_at_def Let_def)

lemma chain_t9_risk_outcome:
"action_selection_outcome chain_after_t8 chain_orientation_catalog
  chain_selection_risk = ActionSelectionAbort RiskNotAccepted"
using chain_selection_orientation_applicable chain_after_t8_frame_exact
by (simp add: action_selection_outcome_def selection_orientation_status_def
  chain_selection_risk_def chain_selection_request_def
  chain_decision_authority_def chain_action_option_risk_def
  chain_action_option_def chain_action_gates_def requested_action_option_def
  basic_action_gates_passed_def explicit_power_and_responsibility_def
  decision_authority_valid_def authority_active_at_def Let_def)

lemma chain_t9_insufficient_outcome:
"action_selection_outcome chain_after_t8_deferred
  chain_orientation_catalog_deferred chain_selection_insufficient =
  ActionSelectionStatus InsufficientBasis"
using chain_orientation_deferred_status
by (simp add: action_selection_outcome_def selection_orientation_status_def
  chain_selection_insufficient_def chain_selection_request_def
  chain_orientation_catalog_deferred_def extend_orientation_catalog_def
  chain_orientation_request_deferred_def chain_orientation_request_def
  Let_def)

theorem chain_t9_abort_outcomes_exact:

```

```
"action_selection_outcome chain_after_t8 chain_orientation_catalog
  chain_selection_no_authority = ActionSelectionAbort NoAuthority \<and>
action_selection_outcome chain_after_t8 chain_orientation_catalog
  chain_selection_no_option = ActionSelectionAbort NoAdmissibleOption \<and>
action_selection_outcome chain_after_t8 chain_orientation_catalog
  chain_selection_conflict = ActionSelectionAbort ConflictUnresolved \<and>
action_selection_outcome chain_after_t8 chain_orientation_catalog
  chain_selection_risk = ActionSelectionAbort RiskNotAccepted \<and>
action_selection_outcome chain_after_t8_deferred
  chain_orientation_catalog_deferred chain_selection_insufficient =
  ActionSelectionStatus InsufficientBasis"
using chain_t9_no_authority_outcome chain_t9_no_option_outcome
  chain_t9_conflict_outcome chain_t9_risk_outcome
  chain_t9_insufficient_outcome by blast

theorem chain_t9_abort_records_are_versioned_not_actions:
"version_id_of (selection_record_object chain_after_t8
  chain_orientation_catalog chain_selection_no_authority) = 13 \<and>
kind_of (selection_record_object chain_after_t8
  chain_orientation_catalog chain_selection_no_authority) = K_META \<and>
version_id_of (selection_record_object chain_after_t8_deferred
  chain_orientation_catalog_deferred chain_selection_insufficient) = 13 \<and>
kind_of (selection_record_object chain_after_t8_deferred
  chain_orientation_catalog_deferred chain_selection_insufficient) = K_META"
using chain_t9_abort_outcomes_exact
by (simp add: selection_record_object_def chain_selection_no_authority_def
  chain_selection_insufficient_def chain_selection_request_def Let_def)

subsection \<open>Full T1--T13 causal closure\<close>

theorem operational_chain_T1_T13:
"synthesize_orientation_step chain_t8_grammar chain_after_t7
  chain_relevance_catalog chain_orientation_catalog_0
  chain_orientation_request 12 chain_after_t8 chain_orientation_catalog \<and>
select_action_step chain_t13_grammar chain_after_t8
  chain_relevance_catalog chain_orientation_catalog chain_selection_catalog_0
  chain_selection_request 13 chain_after_t9 chain_selection_catalog \<and>
execute_action_step chain_t13_grammar chain_after_t9
  chain_selection_catalog chain_execution_catalog_0 chain_execution_request 14
  chain_after_t10 chain_execution_catalog \<and>
record_feedback_step chain_t13_grammar chain_after_t10
  chain_execution_catalog chain_feedback_catalog_0 chain_feedback_request 15
  chain_after_t11 chain_feedback_catalog \<and>
analyze_emergence_step chain_t13_grammar chain_after_t11
  chain_feedback_catalog chain_emergence_catalog_0 chain_emergence_request 16
  chain_after_t12 chain_emergence_catalog \<and>
execute_revision_step chain_t13_grammar chain_after_t12
  chain_feedback_catalog chain_emergence_catalog chain_revision_catalog_0
  chain_orientation_prior chain_revision_request 17 chain_after_t13
  chain_revision_catalog"
using chain_t8_success_step chain_t9_step chain_t10_step chain_t11_step
  chain_t12_step chain_t13_step by blast

end
```

## K.13 Artefaktbindung und Freigabegrenze

Die Freigabe gilt nur für den Hashstand des K3-Manifests. Jede Änderung an ROOT, ROOTS oder einer Theorie hebt den Buildnachweis auf und verlangt einen neuen Clean-Build. Quellaudit und Golden Run sind ergänzende Prüfungen; sie ersetzen den Isabelle-Kernel nicht. Umgekehrt beweist Kernelakzeptanz weder die Locale-Annahmen noch empirische Wahrheit oder universelle Anwendbarkeit.



## Anhang L - Formalrevision K4: artikulierter Interaktionskern

### L.1 Releasegegenstand und aktueller Erkenntnisraum

K4 setzt auf dem exakt geprüften T13-Nachfolger auf und schließt drei bereits in der Architektur vorgesehene Interaktionsoperationen: eine artikulationsverfeinernde Grenze, eine typisierte Kopplung und eine linientreue Übertragungsgestalt. Die Benennung als Interaktionskern vermeidet zwei falsche Ansprüche: Es handelt sich weder um neue Operationen T14-T16 noch um eine Schließung aller I1-I11.

Der aktuelle Erkenntnisraum verschärft die Lesart: Nicht-Objektivierbarkeit bedeutet keine Abwesenheit. Immer, reine Potenzialität, Lebendigkeit, Wandlungs- und Wirkungsfähigkeit sowie der lebendige Zusammenhang gelten in jeder wirklichen Artikulation mit. Sie sind jedoch keine aufzunehmenden Elemente eines formalen Zustands. 'Immer enthalten' heißt Grundgeltung, nicht Mitgliedschaft.

***Mitgeltung ist keine verborgene Zustandsvariable. Übergänge verbinden nur artikulierte Gestalten; die Grundgeltung ist in jedem zulässigen Vollzug vorausgesetzt und wird durch keinen Schritt erzeugt, verbraucht oder beendet.***

Auch Denken, Formulieren, Modellieren und Prüfen sind als Vollzüge real gewirkte Tatsachen. Daraus folgt nicht automatisch die Wahrheit ihres Inhalts. Der Realitätsstatus des Vollzugs und der Geltungsstatus seiner Aussage bleiben getrennt.

### L.2 Harte Sorten-, Brücken- und Nicht-Objektifizierungsregel

| Ebene                | Zulässige Rolle                                                                                                  | Unzulässige Gleichsetzung                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Grundsprache         | Metasprachlicher Geltungsindex für Wirklichkeit, Immer, reine Potenzialität, Lebendigkeit und Wirkungsfähigkeit. | Kein UZAObject, Snapshot-Feld, Zeitpunkt, Container, Übergangsknoten oder versteckte Ursache.    |
| Artikulationssprache | Typisierte Objekte, Kontexte, Grenzen, Kopplungen, Übertragungen, Versionen und Prüfprädikate.                   | Keine erschöpfende Repräsentation und keine Identität mit der Wirklichkeit.                      |
| Brücke               | Ausdrücklich gekennzeichnete metasprachliche Bezugnahme mit Kontexttypisierung und erhaltenem Sortenunterschied. | Keine stille Typumwandlung, kein Kontextverlust, keine absolute Trennung und kein Isabelle-Feld. |

Formkongruenz wird als nichtfunktionale Relation gelesen. Eine Artikulation legt weder einen eindeutigen ontologischen Grund fest noch folgt aus dem Grund genau eine Artikulation. Kongruenz bedeutet nicht erschöpfende Repräsentation. Der Abgleich zwischen Fassungen ist eine methodische Beurteilung ihrer Zusammenhangstreue und keine zusätzliche Engineoperation.

Verbindliche Formationsgrenze: Nicht typisierbare gemischte Summen, Negationen, Teil-Ganzes-Aussagen, Quantifikationen oder Übergangspfeile werden nicht gebildet. Aus artikulierbarer Unterscheidbarkeit folgt keine absolute ontologische Trennung.

### L.3 Semantik und Grenze des Interaktionskerns

| Operation     | Formal geleistet                                                                                                                          | Ausdrücklich nicht behauptet                                              |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| ApplyBoundary | Erzeugt bei erfolgreicher Fission einen neuen artikulierten Kontext; verlangt Herkunft und vollständige Lineage; bewahrt den Quellrahmen. | Keine ontologische Spaltung, kein erstmaliges Entstehen von Zusammenhang. |

| Operation            | Formal geleistet                                                                                           | Ausdrücklich nicht behauptet                                                      |
|----------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| DefineCoupling       | Verknüpft verschiedene vorhandene Kontext-IDs über mindestens einen typisierten Kanal.                     | Kein ontologisches Ganzes, keine absolute Trennung, kein Beweis eines realen Wir. |
| InitiateTransmission | Führt eine status- und linientreue Wirkungsgestalt von Scheduled über Delivered und Observed zu Validated. | Kein Substanztransport und keine Sequenzialisierung der Grundprinzipien.          |

Die geschweifte Prozesslesart des lebendigen Zusammenhangs wird dadurch nicht in eine lineare Vorgängerkette übersetzt. Die formalen Snapshots ordnen nur die geprüfte Fortgestaltung ihrer Artikulationsgestalten.

## L.4 Leistungs- und Statusmatrix

| Gegenstand           | Geleistet                                                                   | Folgt nicht daraus                                             | Status           |
|----------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------|------------------|
| T1-T13               | Unverändert kausal geschlossen.                                             | Keine empirische oder ontologische Vollständigkeit.            | Kernelakzeptiert |
| ApplyBoundary        | Eingangsvertrag, Konstruktor, Erhaltung und Ablehnung.                      | Keine ontologische Fission.                                    | Kernelakzeptiert |
| DefineCoupling       | Typisierte Endpunkte, Kanalpflicht und Erhaltung.                           | Kein Wir-Satz.                                                 | Kernelakzeptiert |
| InitiateTransmission | Vierstufiger Lifecycle, Bindung, Lineage und Zielwirkung.                   | Kein Substanztransport.                                        | Kernelakzeptiert |
| Sorten-/Brückenregel | Als verbindliche Geltungs- und Formationsgrenze dokumentiert und auditiert. | Kein neues HOL-Axiom oder Root-Objekt.                         | Metatheoretisch  |
| Abgleich             | Methodische Revision zwischen Artikulationsfassungen.                       | Keine ausführbare Kerntransition und keine Finalitätsgarantie. | Methodisch       |
| A1-A22               | Genau 22 unveränderte Annahmen des Locale uza_model.                        | Keine bewiesene Wahrheit oder gemeinsame Realisierbarkeit.     | Locale-Annahmen  |

## L.5 Kausaler T1-T13-Interaktionszeuge und Negativpfade

| Snapshot | Konstruktoregebnis            | Zertifikat                             |
|----------|-------------------------------|----------------------------------------|
| 17       | T13 Copy-on-write-Nachfolger  | Ausgangspunkt unverändert              |
| 18       | Fission: Kontext 21 ergänzt   | Lineage vollständig; Kontext 20 bleibt |
| 19       | Kopplung 60, Kanal 70         | Endpunkte 20 und 21 typisiert          |
| 20       | Transmission 80 Scheduled     | aktive Quellwirkung 5                  |
| 21       | Delivered; Zielwirkung 15/v18 | Zielkontext 21                         |
| 22       | Observed                      | zulässiger Vorgängerstatus             |
| 23       | Validated                     | Ziel und Lineage vollständig           |

- abgelehnt: erfolgreiche Fission ohne Lineage;
- abgelehnt: Selbstkopplung;
- abgelehnt: Kopplung ohne Kanal;
- abgelehnt: direkter Sprung von Scheduled zu Validated;
- abgelehnt: Validierung ohne vollständige Lineage.

Das Theorem `chain_interaction_articulated_nonseverance` beweist nur die Erhaltung der artikulierten Quellkontexte und T13-Objekte. Es identifiziert den Snapshot nicht mit Wirklichkeit und leitet aus der Kopplung keine irreduzible gemeinsame Potenzialität ab.

## L.6 Isabelle-, Audit- und Golden-Run-Nachweis

Der erneute Clean-Build reinigte Core, T8, T9-T13 und Interaction. Alle 29 Theorien wurden mit Isabelle2025-2, einem Buildjob, einem ML-Thread und deaktivierten parallelen Beweisen

akzeptiert. Exit-Code 0; Laufzeit 2 Minuten 44 Sekunden. Der statische Quellaudit bestand 37/37 Gates. Der unabhängige Golden Run führte 23 unmittelbar konstruktorabgeleitete Schritte bis Snapshot 23 aus.

```
UZA 0.9.1 K4 T1-T13 plus articulated interaction core
Final clean build
Recorded UTC: 2026-09-01T12:14:19Z
Runtime: Isabelle2025-2
Command:
  isabelle build -c -D /workspace/scratch/24663cf25c52/UZA_K4_Interaction_Work -j 1 -o threads=1 -o
parallel_proofs=0 UZA_0_9_1_Interaction

Cleaned UZA_0_9_1_Core
Cleaned UZA_0_9_1_T8
Cleaned UZA_0_9_1_T9_T13
Cleaned UZA_0_9_1_Interaction
Building UZA_0_9_1_Core ...
Finished UZA_0_9_1_Core (0:01:48 elapsed time, 0:01:48 cpu time, factor 1.00)
Building UZA_0_9_1_T8 ...
Finished UZA_0_9_1_T8 (0:00:12 elapsed time, 0:00:12 cpu time, factor 0.99)
Building UZA_0_9_1_T9_T13 ...
Finished UZA_0_9_1_T9_T13 (0:00:31 elapsed time, 0:00:31 cpu time, factor 1.00)
Running UZA_0_9_1_Interaction ...
Finished UZA_0_9_1_Interaction (0:00:05 elapsed time, 0:00:05 cpu time, factor 0.97)
0:02:44 elapsed time, 0:02:38 cpu time, factor 0.96

Exit code: 0
Session count: 4
Theory count: 29
Proof escapes (sorry/oops/admit/axiomatization): 0
A1-A22 status: unchanged locale assumptions
Core SHA-256: 21dcfb175d6e754adf20042b941cf4a23329831d0165ebaf2023b0048213ed24
```

## L.7 A1-A22 und offene Geltungsfragen

A1-A22 wurden weder verändert noch als bewiesen ausgegeben. Sie bleiben genau 22 Annahmen des Locale `uza_model` in `UZA_0_9_1.thy`. Der unveränderte Kernhash lautet `21dcfb175d6e754adf20042b941cf4a23329831d0165ebaf2023b0048213ed24`.

- offen: nicht-degenerierter gemeinsamer Modell-Expansionssatz;
- offen: relative Konsistenz und Konservativität;
- offen: empirische Gültigkeit, Domänenadäquanz und Außenweltwirkung;
- offen: Produktivengine-Äquivalenz;
- offen: eigene Semantik und Nichtreduzierbarkeitsbedingung für ein reales Wir;
- offen: positive Begründung der Grundgeltung über die formkongruente Artikulation hinaus.

## L.8 Vollständige ApplyBoundary-Theorie

Normativer K4-Quellstand: INTERACTION/UZA\_0\_9\_1\_Apply\_Boundary.thy. Bestandteil des Isabelle2025-2-Clean-Builds mit Exit-Code 0.

```
theory UZA_0_9_1_Apply_Boundary
  imports UZA_0_9_1_T9_T13.UZA_0_9_1_T1_T13_Witness
begin

section \<open>ApplyBoundary: articulated context change with explicit lineage\<close>

text \<open>
  This operation changes only an articulated interaction frame. It does not
  turn the non-temporal Always, potentiality, livingness, efficacy, or the
  living nexus into states or transition members. A successful boundary event
  must expose its source contexts, created contexts, lineage, and (for fusion)
  dispositions. Their continuous co-validity is a meta-level condition of
  every admissible articulation, not an additional member of any snapshot.

  Root language and articulation language remain strictly sorted. A boundary
  event has only articulated contexts as endpoints. No mixed-sort sum,
  part-whole construction, negation of absolute separation, or transition on a
  root expression is formed. Any claim connecting an articulated result to a
  grounding expression requires an explicitly labelled meta-level bridge; the
  operation itself supplies no such ontological bridge.
\<close>

record boundary_application =
  applied_boundary_of :: boundary_event
  boundary_sources_of :: "context_id set"
  boundary_created_of :: "context_id set"

definition apply_boundary_result ::
  "snapshot \<Rightarrow> boundary_application \<Rightarrow> nat \<Rightarrow> snapshot" where
  "apply_boundary_result s request next_snapshot_id =
    s\<lparr>snapshot_id_of := next_snapshot_id,
    contexts_of := contexts_of s \<union> boundary_created_of request,
    boundaries_of := insert (applied_boundary_of request) (boundaries_of s)\<rparr>"

definition boundary_application_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> boundary_application \<Rightarrow> nat \<Rightarrow>
  bool" where
  "boundary_application_input_valid G s request next_snapshot_id \<longleftarrowrightarrow>
    state_well_formed G s \<and>
    ApplyBoundary \<in> allowed_operations_of G \<and>
    snapshot_id_of s < next_snapshot_id \<and>
    boundary_id_of (applied_boundary_of request) \<notin>
    boundary_id_of ` boundaries_of s \<and>
    boundary_sources_of request \<noteq> {} \<and>
    boundary_sources_of request \<subteq> contexts_of s \<and>
    boundary_created_of request \<inter> contexts_of s = {} \<and>
    boundary_well_formed (applied_boundary_of request) \<and>
    (boundary_succeeded_of (applied_boundary_of request) \<longleftarrowrightarrow>
    boundary_created_of request \<noteq> {})"

definition apply_boundary_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> boundary_application \<Rightarrow> nat \<Rightarrow>
  snapshot \<Rightarrow> bool" where
  "apply_boundary_step G s request next_snapshot_id s' \<longleftarrowrightarrow>
    boundary_application_input_valid G s request next_snapshot_id \<and>
    s' = apply_boundary_result s request next_snapshot_id"

lemma apply_boundary_result_fields [simp]:
  "snapshot_id_of (apply_boundary_result s request next_snapshot_id) =
    next_snapshot_id"
  "objects_of (apply_boundary_result s request next_snapshot_id) = objects_of s"
  "historical_versions_of (apply_boundary_result s request next_snapshot_id) =
    historical_versions_of s"
  "perspectives_of (apply_boundary_result s request next_snapshot_id) =
    perspectives_of s"
  "contexts_of (apply_boundary_result s request next_snapshot_id) =
    contexts_of s \<union> boundary_created_of request"
  "models_of (apply_boundary_result s request next_snapshot_id) = models_of s"
  "authorities_of (apply_boundary_result s request next_snapshot_id) =
    authorities_of s"
  "contains_edges_of (apply_boundary_result s request next_snapshot_id) =
```

```

    contains_edges_of s"
"projections_of (apply_boundary_result s request next_snapshot_id) =
  projections_of s"
"couplings_of (apply_boundary_result s request next_snapshot_id) =
  couplings_of s"
"transmissions_of (apply_boundary_result s request next_snapshot_id) =
  transmissions_of s"
"boundaries_of (apply_boundary_result s request next_snapshot_id) =
  insert (applied_boundary_of request) (boundaries_of s)"
by (simp_all add: apply_boundary_result_def)

lemma apply_boundary_result_version_ids [simp]:
  "version_ids (apply_boundary_result s request next_snapshot_id) = version_ids s"
by (simp add: version_ids_def)

lemma apply_boundary_static_object_frames [simp]:
  "references_complete (apply_boundary_result s request next_snapshot_id) obj =
    references_complete s obj"
  "evidence_well_formed (apply_boundary_result s request next_snapshot_id) obj =
    evidence_well_formed s obj"
  "relation_well_formed (apply_boundary_result s request next_snapshot_id) obj =
    relation_well_formed s obj"
  "effect_well_formed (apply_boundary_result s request next_snapshot_id) obj =
    effect_well_formed s obj"
  "perspective_well_formed
    (apply_boundary_result s request next_snapshot_id) obj =
    perspective_well_formed s obj"
  "orientation_well_formed
    (apply_boundary_result s request next_snapshot_id) obj =
    orientation_well_formed s obj"
  "action_well_formed (apply_boundary_result s request next_snapshot_id) obj =
    action_well_formed s obj"
  "feedback_well_formed (apply_boundary_result s request next_snapshot_id) obj =
    feedback_well_formed s obj"
  unfolding references_complete_def evidence_well_formed_def
    evidence_references_complete_def relation_well_formed_def
    effect_well_formed_def present_ref_def perspective_well_formed_def
    orientation_well_formed_def action_well_formed_def
    feedback_well_formed_def
  by (simp_all split: option.splits)

lemma apply_boundary_context_well_formed_mono:
  assumes "context_well_formed s obj"
  shows "context_well_formed
    (apply_boundary_result s request next_snapshot_id) obj"
  using assms
  unfolding context_well_formed_def
  by (auto split: option.splits)

lemma apply_boundary_relevance_well_formed_mono:
  assumes "relevance_well_formed s obj"
  shows "relevance_well_formed
    (apply_boundary_result s request next_snapshot_id) obj"
  using assms apply_boundary_context_well_formed_mono[of s obj request next_snapshot_id]
  unfolding relevance_well_formed_def
  by (auto split: option.splits)

lemma apply_boundary_projection_well_formed_mono:
  assumes "projection_well_formed s projection"
  shows "projection_well_formed
    (apply_boundary_result s request next_snapshot_id) projection"
  using assms
  unfolding projection_well_formed_def
  by auto

lemma apply_boundary_coupling_well_formed_mono:
  assumes "coupling_well_formed s coupling"
  shows "coupling_well_formed
    (apply_boundary_result s request next_snapshot_id) coupling"
  using assms
  unfolding coupling_well_formed_def
  by auto

lemma apply_boundary_transmission_well_formed [simp]:
  "transmission_well_formed
    (apply_boundary_result s request next_snapshot_id) transmission =
    transmission_well_formed s transmission"
  unfolding transmission_well_formed_def
  by simp

```

```

theorem apply_boundary_preserves_state_well_formed:
  assumes valid:
    "boundary_application_input_valid G s request next_snapshot_id"
  shows "state_well_formed G
    (apply_boundary_result s request next_snapshot_id)"
proof -
  from valid have source: "state_well_formed G s"
  and new_boundary: "boundary_well_formed (applied_boundary_of request)"
  unfolding boundary_application_input_valid_def by blast+
  from source have unique: "unique_versions s"
  and objects:
    "\<forall>obj\<in>objects_of s.
      kind_of obj \<in> supported_kinds_of G \<and>
      references_complete s obj \<and> evidence_well_formed s obj \<and>
      time_well_formed obj \<and> relation_well_formed s obj \<and>
      effect_well_formed s obj \<and> perspective_well_formed s obj \<and>
      context_well_formed s obj \<and> relevance_well_formed s obj \<and>
      orientation_well_formed s obj \<and> action_well_formed s obj \<and>
      feedback_well_formed s obj"
  and projections:
    "\<forall>p\<in>projections_of s. projection_well_formed s p"
  and couplings:
    "\<forall>c\<in>couplings_of s. coupling_well_formed s c"
  and transmissions:
    "\<forall>t\<in>transmissions_of s. transmission_well_formed s t"
  and boundaries:
    "\<forall>b\<in>boundaries_of s. boundary_well_formed b"
  unfolding state_well_formed_def by blast+
  have unique':
    "unique_versions (apply_boundary_result s request next_snapshot_id)"
  using unique by (simp add: unique_versions_def)
  have objects':
    "\<forall>obj\<in>objects_of (apply_boundary_result s request next_snapshot_id).
      kind_of obj \<in> supported_kinds_of G \<and>
      references_complete (apply_boundary_result s request next_snapshot_id) obj \<and>
      evidence_well_formed (apply_boundary_result s request next_snapshot_id) obj \<and>
      time_well_formed obj \<and>
      relation_well_formed (apply_boundary_result s request next_snapshot_id) obj \<and>
      effect_well_formed (apply_boundary_result s request next_snapshot_id) obj \<and>
      perspective_well_formed (apply_boundary_result s request next_snapshot_id) obj \<and>
      context_well_formed (apply_boundary_result s request next_snapshot_id) obj \<and>
      relevance_well_formed (apply_boundary_result s request next_snapshot_id) obj \<and>
      orientation_well_formed (apply_boundary_result s request next_snapshot_id) obj \<and>
      action_well_formed (apply_boundary_result s request next_snapshot_id) obj \<and>
      feedback_well_formed (apply_boundary_result s request next_snapshot_id) obj"
  using objects
  by (auto intro: apply_boundary_context_well_formed_mono
    apply_boundary_relevance_well_formed_mono)
  have projections':
    "\<forall>p\<in>projections_of (apply_boundary_result s request next_snapshot_id).
      projection_well_formed
        (apply_boundary_result s request next_snapshot_id) p"
  using projections apply_boundary_projection_well_formed_mono by auto
  have couplings':
    "\<forall>c\<in>couplings_of (apply_boundary_result s request next_snapshot_id).
      coupling_well_formed
        (apply_boundary_result s request next_snapshot_id) c"
  using couplings apply_boundary_coupling_well_formed_mono by auto
  have transmissions':
    "\<forall>t\<in>transmissions_of (apply_boundary_result s request next_snapshot_id).
      transmission_well_formed
        (apply_boundary_result s request next_snapshot_id) t"
  using transmissions by simp
  have boundaries':
    "\<forall>b\<in>boundaries_of (apply_boundary_result s request next_snapshot_id).
      boundary_well_formed b"
  using boundaries new_boundary by auto
  show ?thesis
  unfolding state_well_formed_def
  using unique' objects' projections' couplings' transmissions' boundaries'
  by blast
qed

theorem apply_boundary_refines_generic_step:
  assumes concrete:
    "apply_boundary_step G s request next_snapshot_id s'"
  shows "step G s ApplyBoundary s'"
proof -
  from concrete have valid:
    "boundary_application_input_valid G s request next_snapshot_id"

```

```

    and target: "s' = apply_boundary_result s request next_snapshot_id"
    unfolding apply_boundary_step_def by blast+
  from valid have source: "state_well_formed G s"
    and allowed: "ApplyBoundary \<in> allowed_operations_of G"
    unfolding boundary_application_input_valid_def by blast+
  have target_wf:
    "state_well_formed G (apply_boundary_result s request next_snapshot_id)"
    using apply_boundary_preserves_state_well_formed[OF valid] .
  have source_unique: "inj_on version_id_of (objects_of s)"
    using source
    unfolding state_well_formed_def unique_versions_def
    by blast
  have immutable:
    "immutable_preserved s
      (apply_boundary_result s request next_snapshot_id)"
    using source_unique
    unfolding immutable_preserved_def inj_on_def
    by simp
  show ?thesis
    unfolding step_def target
    using source target_wf allowed immutable by blast
qed

theorem apply_boundary_exact_result:
  assumes "apply_boundary_step G s request next_snapshot_id s'"
  shows
    "contexts_of s' = contexts_of s \<union> boundary_created_of request \<and>
      boundaries_of s' = insert (applied_boundary_of request) (boundaries_of s) \<and>
      objects_of s' = objects_of s"
  using assms unfolding apply_boundary_step_def by simp

lemma successful_fission_without_lineage_rejected:
  assumes "boundary_succeeded_of (applied_boundary_of request)"
    "boundary_kind_of (applied_boundary_of request) = Fission"
    "\<not> boundary_lineage_complete_of (applied_boundary_of request)"
  shows "\<not> boundary_application_input_valid G s request next_snapshot_id"
  using assms
  unfolding boundary_application_input_valid_def boundary_well_formed_def
  by simp

lemma unsuccessful_boundary_cannot_create_contexts:
  assumes "\<not> boundary_succeeded_of (applied_boundary_of request)"
    "boundary_created_of request \<noteq> {}"
  shows "\<not> boundary_application_input_valid G s request next_snapshot_id"
  using assms unfolding boundary_application_input_valid_def by blast
end

```

## L.9 Vollständige DefineCoupling-Theorie

Normativer K4-Quellstand: INTERACTION/UZA\_0\_9\_1\_Define\_Coupling.thy. Bestandteil des Isabelle2025-2-Clean-Builds mit Exit-Code 0.

```
theory UZA_0_9_1_Define_Coupling
  imports UZA_0_9_1_Apply_Boundary
begin

section \<open>DefineCoupling: typed relation between articulated contexts\<close>

text \<open>
  The coupling is an operational relation between context identifiers of the
  articulation sort. It makes an available channel explicit; it neither
  creates the living nexus nor converts a grounding principle into an
  endpoint. Its typed endpoints and channels therefore cannot be used as an
  implicit bridge to root language. In particular, coupling alone does not
  prove ontological separation, exhaustive representation, or irreducible
  joint potentiality.
\<close>

definition add_coupling_result ::
  "snapshot \<Rightarrow> coupling \<Rightarrow> nat \<Rightarrow> snapshot" where
  "add_coupling_result s coupling next_snapshot_id =
    s\<lparr>snapshot_id_of := next_snapshot_id,
    couplings_of := insert coupling (couplings_of s)\<rparr>"

definition define_coupling_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> coupling \<Rightarrow> nat \<Rightarrow> bool" where
  "define_coupling_input_valid G s coupling next_snapshot_id \<longleftarrowrightarrow>
    state_well_formed G s \<and>
    DefineCoupling \<in> allowed_operations_of G \<and>
    snapshot_id_of s < next_snapshot_id \<and>
    coupling_id_of coupling \<notin> coupling_id_of ` couplings_of s \<and>
    coupling_source_of coupling \<noteq> coupling_target_of coupling \<and>
    coupling_well_formed s coupling"

definition define_coupling_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> coupling \<Rightarrow> nat \<Rightarrow>
    snapshot \<Rightarrow> bool" where
  "define_coupling_step G s coupling next_snapshot_id s' \<longleftarrowrightarrow>
    define_coupling_input_valid G s coupling next_snapshot_id \<and>
    s' = add_coupling_result s coupling next_snapshot_id"

lemma add_coupling_result_fields [simp]:
  "snapshot_id_of (add_coupling_result s coupling next_snapshot_id) =
    next_snapshot_id"
  "objects_of (add_coupling_result s coupling next_snapshot_id) = objects_of s"
  "historical_versions_of (add_coupling_result s coupling next_snapshot_id) =
    historical_versions_of s"
  "perspectives_of (add_coupling_result s coupling next_snapshot_id) =
    perspectives_of s"
  "contexts_of (add_coupling_result s coupling next_snapshot_id) = contexts_of s"
  "models_of (add_coupling_result s coupling next_snapshot_id) = models_of s"
  "authorities_of (add_coupling_result s coupling next_snapshot_id) =
    authorities_of s"
  "contains_edges_of (add_coupling_result s coupling next_snapshot_id) =
    contains_edges_of s"
  "projections_of (add_coupling_result s coupling next_snapshot_id) =
    projections_of s"
  "couplings_of (add_coupling_result s coupling next_snapshot_id) =
    insert coupling (couplings_of s)"
  "transmissions_of (add_coupling_result s coupling next_snapshot_id) =
    transmissions_of s"
  "boundaries_of (add_coupling_result s coupling next_snapshot_id) =
    boundaries_of s"
  by (simp_all add: add_coupling_result_def)

lemma add_coupling_result_version_ids [simp]:
  "version_ids (add_coupling_result s coupling next_snapshot_id) = version_ids s"
  by (simp add: version_ids_def)

lemma add_coupling_static_object_frames [simp]:
  "references_complete (add_coupling_result s coupling next_snapshot_id) obj =
    references_complete s obj"
  "evidence_well_formed (add_coupling_result s coupling next_snapshot_id) obj =
```



```

    evidence_well_formed s obj"
  "relation_well_formed (add_coupling_result s coupling next_snapshot_id) obj =
    relation_well_formed s obj"
  "effect_well_formed (add_coupling_result s coupling next_snapshot_id) obj =
    effect_well_formed s obj"
  "perspective_well_formed
    (add_coupling_result s coupling next_snapshot_id) obj =
    perspective_well_formed s obj"
  "context_well_formed (add_coupling_result s coupling next_snapshot_id) obj =
    context_well_formed s obj"
  "relevance_well_formed
    (add_coupling_result s coupling next_snapshot_id) obj =
    relevance_well_formed s obj"
  "orientation_well_formed
    (add_coupling_result s coupling next_snapshot_id) obj =
    orientation_well_formed s obj"
  "action_well_formed (add_coupling_result s coupling next_snapshot_id) obj =
    action_well_formed s obj"
  "feedback_well_formed (add_coupling_result s coupling next_snapshot_id) obj =
    feedback_well_formed s obj"
  unfolding references_complete_def evidence_well_formed_def
    evidence_references_complete_def relation_well_formed_def
    effect_well_formed_def present_ref_def perspective_well_formed_def
    context_well_formed_def relevance_well_formed_def
    orientation_well_formed_def action_well_formed_def
    feedback_well_formed_def
  by (simp_all split: option.splits)

lemma add_coupling_projection_well_formed [simp]:
  "projection_well_formed
    (add_coupling_result s coupling next_snapshot_id) projection =
    projection_well_formed s projection"
  unfolding projection_well_formed_def by simp

lemma add_coupling_existing_coupling_well_formed [simp]:
  "coupling_well_formed
    (add_coupling_result s new_coupling next_snapshot_id) old_coupling =
    coupling_well_formed s old_coupling"
  unfolding coupling_well_formed_def by simp

lemma add_coupling_transmission_well_formed_mono:
  assumes "transmission_well_formed s transmission"
  shows "transmission_well_formed
    (add_coupling_result s coupling next_snapshot_id) transmission"
  using assms
  unfolding transmission_well_formed_def
  by auto

theorem define_coupling_preserves_state_well_formed:
  assumes valid: "define_coupling_input_valid G s coupling next_snapshot_id"
  shows "state_well_formed G
    (add_coupling_result s coupling next_snapshot_id)"
  proof -
    from valid have source: "state_well_formed G s"
    and new_coupling: "coupling_well_formed s coupling"
    unfolding define_coupling_input_valid_def by blast+
    from source have unique: "unique_versions s"
    and objects:
      "\<forall>obj\<in>objects_of s.
        kind_of obj \<in> supported_kinds_of G \<and>
        references_complete s obj \<and> evidence_well_formed s obj \<and>
        time_well_formed obj \<and> relation_well_formed s obj \<and>
        effect_well_formed s obj \<and> perspective_well_formed s obj \<and>
        context_well_formed s obj \<and> relevance_well_formed s obj \<and>
        orientation_well_formed s obj \<and> action_well_formed s obj \<and>
        feedback_well_formed s obj"
    and projections:
      "\<forall>p\<in>projections_of s. projection_well_formed s p"
    and couplings:
      "\<forall>c\<in>couplings_of s. coupling_well_formed s c"
    and transmissions:
      "\<forall>t\<in>transmissions_of s. transmission_well_formed s t"
    and boundaries:
      "\<forall>b\<in>boundaries_of s. boundary_well_formed b"
    unfolding state_well_formed_def by blast+
    have unique':
      "unique_versions (add_coupling_result s coupling next_snapshot_id)"
      using unique by (simp add: unique_versions_def)
    have objects':
      "\<forall>obj\<in>objects_of (add_coupling_result s coupling next_snapshot_id).

```

```

kind_of obj \<in> supported_kinds_of G \<and>
references_complete (add_coupling_result s coupling next_snapshot_id) obj \<and>
evidence_well_formed (add_coupling_result s coupling next_snapshot_id) obj \<and>
time_well_formed obj \<and>
relation_well_formed (add_coupling_result s coupling next_snapshot_id) obj \<and>
effect_well_formed (add_coupling_result s coupling next_snapshot_id) obj \<and>
perspective_well_formed (add_coupling_result s coupling next_snapshot_id) obj \<and>
context_well_formed (add_coupling_result s coupling next_snapshot_id) obj \<and>
relevance_well_formed (add_coupling_result s coupling next_snapshot_id) obj \<and>
orientation_well_formed (add_coupling_result s coupling next_snapshot_id) obj \<and>
action_well_formed (add_coupling_result s coupling next_snapshot_id) obj \<and>
feedback_well_formed (add_coupling_result s coupling next_snapshot_id) obj"
using objects by simp
have projections':
  "\<forall>p\<in>projections_of (add_coupling_result s coupling next_snapshot_id).
    projection_well_formed
      (add_coupling_result s coupling next_snapshot_id) p"
  using projections by simp
have couplings':
  "\<forall>c\<in>couplings_of (add_coupling_result s coupling next_snapshot_id).
    coupling_well_formed
      (add_coupling_result s coupling next_snapshot_id) c"
  using couplings new_coupling by auto
have transmissions':
  "\<forall>t\<in>transmissions_of
    (add_coupling_result s coupling next_snapshot_id).
    transmission_well_formed
      (add_coupling_result s coupling next_snapshot_id) t"
  using transmissions add_coupling_transmission_well_formed_mono by auto
have boundaries':
  "\<forall>b\<in>boundaries_of (add_coupling_result s coupling next_snapshot_id).
    boundary_well_formed b"
  using boundaries by simp
show ?thesis
  unfolding state_well_formed_def
  using unique' objects' projections' couplings' transmissions' boundaries'
  by blast
qed

theorem define_coupling_refines_generic_step:
  assumes concrete:
    "define_coupling_step G s coupling next_snapshot_id s'"
  shows "step G s DefineCoupling s'"
proof -
  from concrete have valid:
    "define_coupling_input_valid G s coupling next_snapshot_id"
  and target: "s' = add_coupling_result s coupling next_snapshot_id"
  unfolding define_coupling_step_def by blast+
  from valid have source: "state_well_formed G s"
  and allowed: "DefineCoupling \<in> allowed_operations_of G"
  unfolding define_coupling_input_valid_def by blast+
  have target_wf:
    "state_well_formed G
      (add_coupling_result s coupling next_snapshot_id)"
  using define_coupling_preserves_state_well_formed[OF valid] .
  have source_unique: "inj_on version_id_of (objects_of s)"
  using source
  unfolding state_well_formed_def unique_versions_def by blast
  have immutable:
    "immutable_preserved s
      (add_coupling_result s coupling next_snapshot_id)"
  using source_unique
  unfolding immutable_preserved_def inj_on_def
  by simp
  show ?thesis
    unfolding step_def target
    using source target_wf allowed immutable by blast
qed

theorem define_coupling_exact_result:
  assumes "define_coupling_step G s coupling next_snapshot_id s'"
  shows
    "couplings_of s' = insert coupling (couplings_of s) \<and>
      contexts_of s' = contexts_of s \<and>
      objects_of s' = objects_of s"
  using assms unfolding define_coupling_step_def by simp

lemma self_coupling_rejected:
  assumes "coupling_source_of coupling = coupling_target_of coupling"
  shows "\<not> define_coupling_input_valid G s coupling next_snapshot_id"

```

```
using assms unfolding define_coupling_input_valid_def by blast

lemma empty_channel_coupling_rejected:
  assumes "coupling_channels_of coupling = {}"
  shows "\<not> define_coupling_input_valid G s coupling next_snapshot_id"
  using assms
  unfolding define_coupling_input_valid_def coupling_well_formed_def
  by simp

lemma unknown_coupling_endpoint_rejected:
  assumes "coupling_source_of coupling \<notin> contexts_of s \<or>
    coupling_target_of coupling \<notin> contexts_of s"
  shows "\<not> define_coupling_input_valid G s coupling next_snapshot_id"
  using assms
  unfolding define_coupling_input_valid_def coupling_well_formed_def
  by blast

end
```

## L.10 Vollständige InitiateTransmission-Theorie

Normativer K4-Quellstand: INTERACTION/UZA\_0\_9\_1\_Initiate\_Transmission.thy. Bestandteil des Isabelle2025-2-Clean-Builds mit Exit-Code 0.

```
theory UZA_0_9_1_Initiate_Transmission
  imports UZA_0_9_1_Define_Coupling
begin

section \<open>InitiateTransmission: lifecycle, context binding, and lineage\<close>

text \<open>
  Transmission articulates a lineage-bound effect record between typed
  contexts. It is not substance transport between isolated entities and does
  not place Always, pure potentiality, livingness, efficacy, or the living
  nexus into its lifecycle. Those grounding expressions remain meta-level
  conditions of the articulated operation. The real occurrence of an act of
  thinking, modelling, or transmitting does not by itself validate the
  propositional content attributed to that act.
\<close>

definition transmission_ids_unique :: "snapshot \<Rightarrow> bool" where
  "transmission_ids_unique s \<longleftarrowtrightrightarrow>
    inj_on transmission_id_of (transmissions_of s)"

definition active_effect_in_context ::
  "snapshot \<Rightarrow> effect_id \<Rightarrow> context_id \<Rightarrow> bool" where
  "active_effect_in_context s effect context \<longleftarrowtrightrightarrow>
    (\<exists>obj\<in>objects_of s.
      object_id_of obj = effect \<and> kind_of obj = K_EFF \<and>
      context_of obj = Some context)"

definition transmission_source_bound ::
  "snapshot \<Rightarrow> transmission \<Rightarrow> bool" where
  "transmission_source_bound s transmission \<longleftarrowtrightrightarrow>
    (\<exists>coupling\<in>couplings_of s.
      coupling_id_of coupling = transmission_coupling_of transmission \<and>
      transmission_channel_of transmission \<in> coupling_channels_of coupling \<and>
      active_effect_in_context s
        (transmission_source_effect_of transmission)
        (coupling_source_of coupling))"

definition transmission_target_bound ::
  "snapshot \<Rightarrow> transmission \<Rightarrow> bool" where
  "transmission_target_bound s transmission \<longleftarrowtrightrightarrow>
    (case transmission_target_effect_of transmission of
      None \<Rightarrow> True
    | Some effect \<Rightarrow>
      (\<exists>coupling\<in>couplings_of s.
        coupling_id_of coupling = transmission_coupling_of transmission \<and>
        transmission_channel_of transmission \<in>
          coupling_channels_of coupling \<and>
        active_effect_in_context s effect
          (coupling_target_of coupling)))"

definition same_transmission_flow ::
  "transmission \<Rightarrow> transmission \<Rightarrow> bool" where
  "same_transmission_flow old new \<longleftarrowtrightrightarrow>
    transmission_id_of old = transmission_id_of new \<and>
    transmission_coupling_of old = transmission_coupling_of new \<and>
    transmission_channel_of old = transmission_channel_of new \<and>
    transmission_source_effect_of old = transmission_source_effect_of new"

definition transmission_predecessor_valid ::
  "snapshot \<Rightarrow> transmission \<Rightarrow> bool" where
  "transmission_predecessor_valid s new \<longleftarrowtrightrightarrow>
    (\<exists>old\<in>transmissions_of s.
      same_transmission_flow old new \<and>
      prior_trans_status_of new = Some (trans_status_of old))"

subsection \<open>Schedule a fresh transmission\<close>

definition insert_transmission_result ::
  "snapshot \<Rightarrow> transmission \<Rightarrow> nat \<Rightarrow> snapshot" where
  "insert_transmission_result s transmission next_snapshot_id =
    s\<lparr>snapshot_id_of := next_snapshot_id,
```

```

    transmissions_of := insert transmission (transmissions_of s)\<rparr>"

definition schedule_transmission_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> transmission \<Rightarrow> nat \<Rightarrow> bool" where
  "schedule_transmission_input_valid G s transmission next_snapshot_id \<longlefttrightarrow>
    state_well_formed G s \<and>
    transmission_ids_unique s \<and>
    InitiateTransmission \<in> allowed_operations_of G \<and>
    snapshot_id_of s < next_snapshot_id \<and>
    transmission_id_of transmission \<notin>
      transmission_id_of ` transmissions_of s \<and>
    prior_trans_status_of transmission = None \<and>
    trans_status_of transmission = TransScheduled \<and>
    transmission_target_effect_of transmission = None \<and>
    \<not> transmission_lineage_complete_of transmission \<and>
    transmission_well_formed s transmission \<and>
    transmission_source_bound s transmission"

definition schedule_transmission_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> transmission \<Rightarrow> nat \<Rightarrow>
    snapshot \<Rightarrow> bool" where
  "schedule_transmission_step G s transmission next_snapshot_id s' \<longlefttrightarrow>
    schedule_transmission_input_valid G s transmission next_snapshot_id \<and>
    s' = insert_transmission_result s transmission next_snapshot_id"

lemma insert_transmission_result_fields [simp]:
  "snapshot_id_of (insert_transmission_result s transmission next_snapshot_id) =
    next_snapshot_id"
  "objects_of (insert_transmission_result s transmission next_snapshot_id) =
    objects_of s"
  "historical_versions_of
    (insert_transmission_result s transmission next_snapshot_id) =
    historical_versions_of s"
  "perspectives_of (insert_transmission_result s transmission next_snapshot_id) =
    perspectives_of s"
  "contexts_of (insert_transmission_result s transmission next_snapshot_id) =
    contexts_of s"
  "models_of (insert_transmission_result s transmission next_snapshot_id) =
    models_of s"
  "authorities_of (insert_transmission_result s transmission next_snapshot_id) =
    authorities_of s"
  "contains_edges_of (insert_transmission_result s transmission next_snapshot_id) =
    contains_edges_of s"
  "projections_of (insert_transmission_result s transmission next_snapshot_id) =
    projections_of s"
  "couplings_of (insert_transmission_result s transmission next_snapshot_id) =
    couplings_of s"
  "transmissions_of (insert_transmission_result s transmission next_snapshot_id) =
    insert transmission (transmissions_of s)"
  "boundaries_of (insert_transmission_result s transmission next_snapshot_id) =
    boundaries_of s"
  by (simp_all add: insert_transmission_result_def)

lemma insert_transmission_result_version_ids [simp]:
  "version_ids (insert_transmission_result s transmission next_snapshot_id) =
    version_ids s"
  by (simp add: version_ids_def)

lemma insert_transmission_static_object_frames [simp]:
  "references_complete
    (insert_transmission_result s transmission next_snapshot_id) obj =
    references_complete s obj"
  "evidence_well_formed
    (insert_transmission_result s transmission next_snapshot_id) obj =
    evidence_well_formed s obj"
  "relation_well_formed
    (insert_transmission_result s transmission next_snapshot_id) obj =
    relation_well_formed s obj"
  "effect_well_formed
    (insert_transmission_result s transmission next_snapshot_id) obj =
    effect_well_formed s obj"
  "perspective_well_formed
    (insert_transmission_result s transmission next_snapshot_id) obj =
    perspective_well_formed s obj"
  "context_well_formed
    (insert_transmission_result s transmission next_snapshot_id) obj =
    context_well_formed s obj"
  "relevance_well_formed
    (insert_transmission_result s transmission next_snapshot_id) obj =
    relevance_well_formed s obj"

```

```

"orientation_well_formed
  (insert_transmission_result s transmission next_snapshot_id) obj =
  orientation_well_formed s obj"
"action_well_formed
  (insert_transmission_result s transmission next_snapshot_id) obj =
  action_well_formed s obj"
"feedback_well_formed
  (insert_transmission_result s transmission next_snapshot_id) obj =
  feedback_well_formed s obj"
unfolding references_complete_def evidence_well_formed_def
evidence_references_complete_def relation_well_formed_def
effect_well_formed_def present_ref_def perspective_well_formed_def
context_well_formed_def relevance_well_formed_def
orientation_well_formed_def action_well_formed_def
feedback_well_formed_def
by (simp_all split: option.splits)

lemma insert_transmission_secondary_frames [simp]:
"projection_well_formed
  (insert_transmission_result s transmission next_snapshot_id) projection =
  projection_well_formed s projection"
"coupling_well_formed
  (insert_transmission_result s transmission next_snapshot_id) coupling =
  coupling_well_formed s coupling"
"transmission_well_formed
  (insert_transmission_result s new_transmission next_snapshot_id)
  old_transmission = transmission_well_formed s old_transmission"
unfolding projection_well_formed_def coupling_well_formed_def
transmission_well_formed_def
by simp_all

theorem schedule_transmission_preserves_state_well_formed:
  assumes valid:
    "schedule_transmission_input_valid G s transmission next_snapshot_id"
  shows "state_well_formed G
    (insert_transmission_result s transmission next_snapshot_id)"
proof -
  from valid have source: "state_well_formed G s"
  and new_transmission: "transmission_well_formed s transmission"
  unfolding schedule_transmission_input_valid_def by blast+
  from source have unique: "unique_versions s"
  and objects:
    "\<forall>obj\<in>objects_of s.
      kind_of obj \<in> supported_kinds_of G \<and>
      references_complete s obj \<and> evidence_well_formed s obj \<and>
      time_well_formed obj \<and> relation_well_formed s obj \<and>
      effect_well_formed s obj \<and> perspective_well_formed s obj \<and>
      context_well_formed s obj \<and> relevance_well_formed s obj \<and>
      orientation_well_formed s obj \<and> action_well_formed s obj \<and>
      feedback_well_formed s obj"
  and projections:
    "\<forall>p\<in>projections_of s. projection_well_formed s p"
  and couplings:
    "\<forall>c\<in>couplings_of s. coupling_well_formed s c"
  and transmissions:
    "\<forall>t\<in>transmissions_of s. transmission_well_formed s t"
  and boundaries:
    "\<forall>b\<in>boundaries_of s. boundary_well_formed b"
  unfolding state_well_formed_def by blast+
  have unique':
    "unique_versions
      (insert_transmission_result s transmission next_snapshot_id)"
  using unique by (simp add: unique_versions_def)
  have objects':
    "\<forall>obj\<in>objects_of
      (insert_transmission_result s transmission next_snapshot_id).
      kind_of obj \<in> supported_kinds_of G \<and>
      references_complete
        (insert_transmission_result s transmission next_snapshot_id) obj \<and>
      evidence_well_formed
        (insert_transmission_result s transmission next_snapshot_id) obj \<and>
      time_well_formed obj \<and>
      relation_well_formed
        (insert_transmission_result s transmission next_snapshot_id) obj \<and>
      effect_well_formed
        (insert_transmission_result s transmission next_snapshot_id) obj \<and>
      perspective_well_formed
        (insert_transmission_result s transmission next_snapshot_id) obj \<and>
      context_well_formed
        (insert_transmission_result s transmission next_snapshot_id) obj \<and>"

```

```

    relevance_well_formed
    (insert_transmission_result s transmission next_snapshot_id) obj \<and>
    orientation_well_formed
    (insert_transmission_result s transmission next_snapshot_id) obj \<and>
    action_well_formed
    (insert_transmission_result s transmission next_snapshot_id) obj \<and>
    feedback_well_formed
    (insert_transmission_result s transmission next_snapshot_id) obj"
using objects by simp
have projections':
  "\<forall>p\<in>projections_of
    (insert_transmission_result s transmission next_snapshot_id).
    projection_well_formed
    (insert_transmission_result s transmission next_snapshot_id) p"
using projections by simp
have couplings':
  "\<forall>c\<in>couplings_of
    (insert_transmission_result s transmission next_snapshot_id).
    coupling_well_formed
    (insert_transmission_result s transmission next_snapshot_id) c"
using couplings by simp
have transmissions':
  "\<forall>t\<in>transmissions_of
    (insert_transmission_result s transmission next_snapshot_id).
    transmission_well_formed
    (insert_transmission_result s transmission next_snapshot_id) t"
using transmissions new_transmission by auto
have boundaries':
  "\<forall>b\<in>boundaries_of
    (insert_transmission_result s transmission next_snapshot_id).
    boundary_well_formed b"
using boundaries by simp
show ?thesis
  unfolding state_well_formed_def
  using unique' objects' projections' couplings' transmissions' boundaries'
  by blast
qed

theorem schedule_transmission_preserves_unique_ids:
  assumes valid:
    "schedule_transmission_input_valid G s transmission next_snapshot_id"
  shows "transmission_ids_unique
    (insert_transmission_result s transmission next_snapshot_id)"
proof -
  from valid have unique:
    "inj_on transmission_id_of (transmissions_of s)"
  and fresh:
    "transmission_id_of transmission \<notin>
      transmission_id_of ` transmissions_of s"
  unfolding schedule_transmission_input_valid_def
  transmission_ids_unique_def by blast+
  have unique_insert:
    "inj_on transmission_id_of
      (insert transmission (transmissions_of s))"
  proof -
    have fresh_restricted:
      "transmission_id_of transmission \<notin>
        transmission_id_of `
          (transmissions_of s - {transmission})"
    using fresh by blast
    show ?thesis
      using unique fresh_restricted by (simp add: inj_on_insert)
  qed
  show ?thesis
    unfolding transmission_ids_unique_def insert_transmission_result_def
    using unique_insert by simp
qed

theorem schedule_transmission_refines_generic_step:
  assumes concrete:
    "schedule_transmission_step G s transmission next_snapshot_id s'"
  shows "step G s InitiateTransmission s'"
proof -
  from concrete have valid:
    "schedule_transmission_input_valid G s transmission next_snapshot_id"
  and target:
    "s' = insert_transmission_result s transmission next_snapshot_id"
  unfolding schedule_transmission_step_def by blast+
  from valid have source: "state_well_formed G s"
  and allowed: "InitiateTransmission \<in> allowed_operations_of G"

```

```

    unfolding schedule_transmission_input_valid_def by blast+
  have target_wf:
    "state_well_formed G
      (insert_transmission_result s transmission next_snapshot_id)"
    using schedule_transmission_preserves_state_well_formed[OF valid] .
  have source_unique: "inj_on version_id_of (objects_of s)"
    using source
    unfolding state_well_formed_def unique_versions_def by blast
  have immutable:
    "immutable_preserved s
      (insert_transmission_result s transmission next_snapshot_id)"
    using source_unique
    unfolding immutable_preserved_def inj_on_def
    by simp
  show ?thesis
    unfolding step_def target
    using source target_wf allowed immutable by blast
qed

subsection \<open>Advance an existing transmission lifecycle\<close>

definition replace_transmission_result ::
  "snapshot \<Rightarrow> transmission \<Rightarrow> nat \<Rightarrow> snapshot" where
  "replace_transmission_result s transmission next_snapshot_id =
    s\<lparr>snapshot_id_of := next_snapshot_id,
      transmissions_of :=
        insert transmission
        {old \<in> transmissions_of s.
          transmission_id_of old \<noteq> transmission_id_of transmission}\<rparr>"

definition update_transmission_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> transmission \<Rightarrow> nat \<Rightarrow> bool" where
  "update_transmission_input_valid G s transmission next_snapshot_id \<longleftarrowrightarrow>
    state_well_formed G s \<and>
    transmission_ids_unique s \<and>
    InitiateTransmission \<in> allowed_operations_of G \<and>
    snapshot_id_of s < next_snapshot_id \<and>
    transmission_predecessor_valid s transmission \<and>
    transmission_well_formed s transmission \<and>
    transmission_source_bound s transmission \<and>
    transmission_target_bound s transmission"

definition update_transmission_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> transmission \<Rightarrow> nat \<Rightarrow>
    snapshot \<Rightarrow> bool" where
  "update_transmission_step G s transmission next_snapshot_id s' \<longleftarrowrightarrow>
    update_transmission_input_valid G s transmission next_snapshot_id \<and>
    s' = replace_transmission_result s transmission next_snapshot_id"

lemma replace_transmission_result_fields [simp]:
  "snapshot_id_of (replace_transmission_result s transmission next_snapshot_id) =
    next_snapshot_id"
  "objects_of (replace_transmission_result s transmission next_snapshot_id) =
    objects_of s"
  "historical_versions_of
    (replace_transmission_result s transmission next_snapshot_id) =
    historical_versions_of s"
  "perspectives_of (replace_transmission_result s transmission next_snapshot_id) =
    perspectives_of s"
  "contexts_of (replace_transmission_result s transmission next_snapshot_id) =
    contexts_of s"
  "models_of (replace_transmission_result s transmission next_snapshot_id) =
    models_of s"
  "authorities_of (replace_transmission_result s transmission next_snapshot_id) =
    authorities_of s"
  "contains_edges_of (replace_transmission_result s transmission next_snapshot_id) =
    contains_edges_of s"
  "projections_of (replace_transmission_result s transmission next_snapshot_id) =
    projections_of s"
  "couplings_of (replace_transmission_result s transmission next_snapshot_id) =
    couplings_of s"
  "transmissions_of (replace_transmission_result s transmission next_snapshot_id) =
    insert transmission
    {old \<in> transmissions_of s.
      transmission_id_of old \<noteq> transmission_id_of transmission}"
  "boundaries_of (replace_transmission_result s transmission next_snapshot_id) =
    boundaries_of s"
  by (simp_all add: replace_transmission_result_def)

lemma replace_transmission_result_version_ids [simp]:

```



```

"version_ids (replace_transmission_result s transmission next_snapshot_id) =
  version_ids s"
by (simp add: version_ids_def)

lemma replace_transmission_static_object_frames [simp]:
  "references_complete
    (replace_transmission_result s transmission next_snapshot_id) obj =
    references_complete s obj"
  "evidence_well_formed
    (replace_transmission_result s transmission next_snapshot_id) obj =
    evidence_well_formed s obj"
  "relation_well_formed
    (replace_transmission_result s transmission next_snapshot_id) obj =
    relation_well_formed s obj"
  "effect_well_formed
    (replace_transmission_result s transmission next_snapshot_id) obj =
    effect_well_formed s obj"
  "perspective_well_formed
    (replace_transmission_result s transmission next_snapshot_id) obj =
    perspective_well_formed s obj"
  "context_well_formed
    (replace_transmission_result s transmission next_snapshot_id) obj =
    context_well_formed s obj"
  "relevance_well_formed
    (replace_transmission_result s transmission next_snapshot_id) obj =
    relevance_well_formed s obj"
  "orientation_well_formed
    (replace_transmission_result s transmission next_snapshot_id) obj =
    orientation_well_formed s obj"
  "action_well_formed
    (replace_transmission_result s transmission next_snapshot_id) obj =
    action_well_formed s obj"
  "feedback_well_formed
    (replace_transmission_result s transmission next_snapshot_id) obj =
    feedback_well_formed s obj"
  unfolding references_complete_def evidence_well_formed_def
    evidence_references_complete_def relation_well_formed_def
    effect_well_formed_def present_ref_def perspective_well_formed_def
    context_well_formed_def relevance_well_formed_def
    orientation_well_formed_def action_well_formed_def
    feedback_well_formed_def
  by (simp_all split: option.splits)

lemma replace_transmission_secondary_frames [simp]:
  "projection_well_formed
    (replace_transmission_result s transmission next_snapshot_id) projection =
    projection_well_formed s projection"
  "coupling_well_formed
    (replace_transmission_result s transmission next_snapshot_id) coupling =
    coupling_well_formed s coupling"
  "transmission_well_formed
    (replace_transmission_result s new_transmission next_snapshot_id)
    old_transmission = transmission_well_formed s old_transmission"
  unfolding projection_well_formed_def coupling_well_formed_def
    transmission_well_formed_def
  by simp_all

lemma replace_transmission_binding_frames [simp]:
  "transmission_source_bound
    (replace_transmission_result s replacement next_snapshot_id)
    transmission =
    transmission_source_bound s transmission"
  "transmission_target_bound
    (replace_transmission_result s replacement next_snapshot_id)
    transmission =
    transmission_target_bound s transmission"
  unfolding transmission_source_bound_def transmission_target_bound_def
    active_effect_in_context_def
  by (simp_all split: option.splits)

theorem replace_transmission_preserves_state_well_formed:
  assumes source: "state_well_formed G s"
  and new_transmission: "transmission_well_formed s transmission"
  shows "state_well_formed G
    (replace_transmission_result s transmission next_snapshot_id)"
proof -
  from source have unique: "unique_versions s"
  and objects:
    "\<forall>obj\<in>objects_of s.
      kind_of obj \<in> supported_kinds_of G \<and>"

```

```

references_complete s obj \<and> evidence_well_formed s obj \<and>
time_well_formed obj \<and> relation_well_formed s obj \<and>
effect_well_formed s obj \<and> perspective_well_formed s obj \<and>
context_well_formed s obj \<and> relevance_well_formed s obj \<and>
orientation_well_formed s obj \<and> action_well_formed s obj \<and>
feedback_well_formed s obj"
and projections:
"\<forall>p\<in>projections_of s. projection_well_formed s p"
and couplings:
"\<forall>c\<in>couplings_of s. coupling_well_formed s c"
and transmissions:
"\<forall>t\<in>transmissions_of s. transmission_well_formed s t"
and boundaries:
"\<forall>b\<in>boundaries_of s. boundary_well_formed b"
unfolding state_well_formed_def by blast+
have unique':
  "unique_versions
    (replace_transmission_result s transmission next_snapshot_id)"
  using unique by (simp add: unique_versions_def)
have objects':
  "\<forall>obj\<in>objects_of
    (replace_transmission_result s transmission next_snapshot_id).
    kind_of obj \<in> supported_kinds_of G \<and>
    references_complete
      (replace_transmission_result s transmission next_snapshot_id) obj \<and>
    evidence_well_formed
      (replace_transmission_result s transmission next_snapshot_id) obj \<and>
    time_well_formed obj \<and>
    relation_well_formed
      (replace_transmission_result s transmission next_snapshot_id) obj \<and>
    effect_well_formed
      (replace_transmission_result s transmission next_snapshot_id) obj \<and>
    perspective_well_formed
      (replace_transmission_result s transmission next_snapshot_id) obj \<and>
    context_well_formed
      (replace_transmission_result s transmission next_snapshot_id) obj \<and>
    relevance_well_formed
      (replace_transmission_result s transmission next_snapshot_id) obj \<and>
    orientation_well_formed
      (replace_transmission_result s transmission next_snapshot_id) obj \<and>
    action_well_formed
      (replace_transmission_result s transmission next_snapshot_id) obj \<and>
    feedback_well_formed
      (replace_transmission_result s transmission next_snapshot_id) obj"
  using objects by simp
have projections':
  "\<forall>p\<in>projections_of
    (replace_transmission_result s transmission next_snapshot_id).
    projection_well_formed
      (replace_transmission_result s transmission next_snapshot_id) p"
  using projections by simp
have couplings':
  "\<forall>c\<in>couplings_of
    (replace_transmission_result s transmission next_snapshot_id).
    coupling_well_formed
      (replace_transmission_result s transmission next_snapshot_id) c"
  using couplings by simp
have transmissions':
  "\<forall>t\<in>transmissions_of
    (replace_transmission_result s transmission next_snapshot_id).
    transmission_well_formed
      (replace_transmission_result s transmission next_snapshot_id) t"
  using transmissions new_transmission by auto
have boundaries':
  "\<forall>b\<in>boundaries_of
    (replace_transmission_result s transmission next_snapshot_id).
    boundary_well_formed b"
  using boundaries by simp
show ?thesis
  unfolding state_well_formed_def
  using unique' objects' projections' couplings' transmissions' boundaries'
  by blast
qed

theorem update_transmission_preserves_unique_ids:
  assumes valid:
    "update_transmission_input_valid G s transmission next_snapshot_id"
  shows "transmission_ids_unique
    (replace_transmission_result s transmission next_snapshot_id)"
  using valid

```

```

unfolding update_transmission_input_valid_def transmission_ids_unique_def
  transmission_predecessor_valid_def same_transmission_flow_def
  inj_on_def
by auto

theorem update_transmission_refines_generic_step:
  assumes concrete:
    "update_transmission_step G s transmission next_snapshot_id s'"
  shows "step G s InitiateTransmission s'"
proof -
  from concrete have valid:
    "update_transmission_input_valid G s transmission next_snapshot_id"
  and target:
    "s' = replace_transmission_result s transmission next_snapshot_id"
  unfolding update_transmission_step_def by blast+
  from valid have source: "state_well_formed G s"
  and allowed: "InitiateTransmission \<in> allowed_operations_of G"
  and new_transmission: "transmission_well_formed s transmission"
  unfolding update_transmission_input_valid_def by blast+
  have target_wf:
    "state_well_formed G
      (replace_transmission_result s transmission next_snapshot_id)"
  using replace_transmission_preserves_state_well_formed[
    OF source new_transmission] .
  have source_unique: "inj_on version_id_of (objects_of s)"
  using source
  unfolding state_well_formed_def unique_versions_def by blast
  have immutable:
    "immutable_preserved s
      (replace_transmission_result s transmission next_snapshot_id)"
  using source_unique
  unfolding immutable_preserved_def inj_on_def
  by simp
  show ?thesis
  unfolding step_def target
  using source target_wf allowed immutable by blast
qed

subsection \<open>Delivery creates the articulated target effect\<close>

definition deliver_transmission_result ::
  "snapshot \<Rightarrow> uza_object \<Rightarrow> transmission \<Rightarrow> nat \<Rightarrow>
  snapshot" where
  "deliver_transmission_result s target_effect transmission next_snapshot_id =
    replace_transmission_result
      (add_appearance s target_effect next_snapshot_id)
      transmission next_snapshot_id"

definition deliver_transmission_input_valid ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> uza_object \<Rightarrow> transmission \<Rightarrow>
  nat \<Rightarrow> snapshot \<Rightarrow> bool" where
  "deliver_transmission_input_valid G s target_effect transmission
    next_snapshot_id \<longlefttrightarrow>
    audited_append_input_valid G s InitiateTransmission target_effect
    next_snapshot_id \<and>
    transmission_ids_unique s \<and>
    kind_of target_effect = K_EFF \<and>
    trans_status_of transmission = TransDelivered \<and>
    prior_trans_status_of transmission = Some TransScheduled \<and>
    transmission_target_effect_of transmission = Some (object_id_of target_effect) \<and>
    transmission_predecessor_valid s transmission \<and>
    transmission_well_formed
      (add_appearance s target_effect next_snapshot_id) transmission \<and>
    transmission_source_bound s transmission \<and>
    transmission_target_bound
      (add_appearance s target_effect next_snapshot_id) transmission"

definition deliver_transmission_step ::
  "grammar \<Rightarrow> snapshot \<Rightarrow> uza_object \<Rightarrow> transmission \<Rightarrow>
  nat \<Rightarrow> snapshot \<Rightarrow> bool" where
  "deliver_transmission_step G s target_effect transmission
    next_snapshot_id s' \<longlefttrightarrow>
    deliver_transmission_input_valid G s target_effect transmission
    next_snapshot_id \<and>
    s' = deliver_transmission_result s target_effect transmission
    next_snapshot_id"

theorem deliver_transmission_preserves_state_well_formed:
  assumes valid:
    "deliver_transmission_input_valid G s target_effect transmission

```

```

    next_snapshot_id"
  shows "state_well_formed G
    (deliver_transmission_result s target_effect transmission next_snapshot_id)"
proof -
  from valid have append_valid:
    "audited_append_input_valid G s InitiateTransmission target_effect
      next_snapshot_id"
    and new_transmission:
      "transmission_well_formed
        (add_appearance s target_effect next_snapshot_id) transmission"
    unfolding deliver_transmission_input_valid_def by blast+
  have append_step:
    "audited_append_step G s InitiateTransmission target_effect
      next_snapshot_id (add_appearance s target_effect next_snapshot_id)"
    unfolding audited_append_step_def using append_valid by simp
  have intermediate:
    "state_well_formed G (add_appearance s target_effect next_snapshot_id)"
    using audited_append_preserves_state_well_formed[OF append_step] .
  show ?thesis
    unfolding deliver_transmission_result_def
    using replace_transmission_preserves_state_well_formed[
      OF intermediate new_transmission] .
qed

theorem deliver_transmission_refines_generic_step:
  assumes concrete:
    "deliver_transmission_step G s target_effect transmission
      next_snapshot_id s'"
  shows "step G s InitiateTransmission s'"
proof -
  from concrete have valid:
    "deliver_transmission_input_valid G s target_effect transmission
      next_snapshot_id"
    and target:
      "s' = deliver_transmission_result s target_effect transmission
        next_snapshot_id"
    unfolding deliver_transmission_step_def by blast+
  from valid have append_valid:
    "audited_append_input_valid G s InitiateTransmission target_effect
      next_snapshot_id"
    unfolding deliver_transmission_input_valid_def by blast
  from append_valid have source: "state_well_formed G s"
    and allowed: "InitiateTransmission \<in> allowed_operations_of G"
    unfolding audited_append_input_valid_def by blast+
  have target_wf:
    "state_well_formed G
      (deliver_transmission_result s target_effect transmission
        next_snapshot_id)"
    using deliver_transmission_preserves_state_well_formed[OF valid] .
  have append_step:
    "audited_append_step G s InitiateTransmission target_effect
      next_snapshot_id (add_appearance s target_effect next_snapshot_id)"
    unfolding audited_append_step_def using append_valid by simp
  have immutable_intermediate:
    "immutable_preserved s
      (add_appearance s target_effect next_snapshot_id)"
    using audited_append_preserves_identity[OF append_step] .
  have immutable:
    "immutable_preserved s
      (deliver_transmission_result s target_effect transmission
        next_snapshot_id)"
    using immutable_intermediate
    unfolding deliver_transmission_result_def immutable_preserved_def
    by simp
  show ?thesis
    unfolding step_def target
    using source target_wf allowed immutable by blast
qed

theorem delivered_effect_is_active_and_bound:
  assumes concrete:
    "deliver_transmission_step G s target_effect transmission
      next_snapshot_id s'"
  shows
    "target_effect \<in> objects_of s' \<and>
      transmission \<in> transmissions_of s' \<and>
      transmission_target_effect_of transmission =
        Some (object_id_of target_effect) \<and>
      transmission_target_bound s' transmission"
proof -

```

```

from concrete have valid:
  "deliver_transmission_input_valid G s target_effect transmission
  next_snapshot_id"
and target:
  "s' = deliver_transmission_result s target_effect transmission
  next_snapshot_id"
unfolding deliver_transmission_step_def by blast+
from valid have target_id:
  "transmission_target_effect_of transmission =
  Some (object_id_of target_effect)"
and target_bound:
  "transmission_target_bound
  (add_appearance s target_effect next_snapshot_id) transmission"
unfolding deliver_transmission_input_valid_def by blast+
have bound_final: "transmission_target_bound s' transmission"
using target_bound target
unfolding deliver_transmission_result_def by simp
have target_member: "target_effect \<in> objects_of s'"
using target
unfolding deliver_transmission_result_def by simp
have transmission_member: "transmission \<in> transmissions_of s'"
using target
unfolding deliver_transmission_result_def by simp
show ?thesis
using target_member transmission_member target_id bound_final by blast
qed

lemma invalid_transmission_lifecycle_rejected:
  assumes "\<not> allowed_trans_transition
  (prior_trans_status_of transmission) (trans_status_of transmission)"
  shows "\<not> update_transmission_input_valid G s transmission next_snapshot_id"
  using assms
  unfolding update_transmission_input_valid_def transmission_well_formed_def
  by blast

lemma validated_transmission_without_lineage_rejected:
  assumes "trans_status_of transmission = TransValidated"
  "transmission_target_effect_of transmission = None \<or>
  \<not> transmission_lineage_complete_of transmission"
  shows "\<not> update_transmission_input_valid G s transmission next_snapshot_id"
proof
  assume valid:
    "update_transmission_input_valid G s transmission next_snapshot_id"
  from valid have wf: "transmission_well_formed s transmission"
  unfolding update_transmission_input_valid_def by blast
  from wf assms(1) have complete:
    "transmission_target_effect_of transmission \<noteq> None \<and>
    transmission_lineage_complete_of transmission"
  unfolding transmission_well_formed_def by blast
  with assms(2) show False by blast
qed

lemma transmission_without_active_source_rejected:
  assumes "\<not> transmission_source_bound s transmission"
  shows
    "\<not> schedule_transmission_input_valid G s transmission next_snapshot_id \<and>
    \<not> update_transmission_input_valid G s transmission next_snapshot_id"
  using assms
  unfolding schedule_transmission_input_valid_def
  update_transmission_input_valid_def
  by blast
end

```

## L.11 Vollständiger Interaktionskern-Zeuge

Normativer K4-Quellstand: INTERACTION/UZA\_0\_9\_1\_Interaction\_Core\_Witness.thy. Bestandteil des Isabelle2025-2-Clean-Builds mit Exit-Code 0.

```
theory UZA_0_9_1_Interaction_Core_Witness
  imports UZA_0_9_1_Initiate_Transmission
begin

section \<open>Causally reachable interaction core after T13\<close>

text \<open>
  This witness starts at the exact T13 end state. It adds an articulated
  fission boundary, a directed coupling, and one transmission whose lifecycle
  is Scheduled--Delivered--Observed--Validated. These are transitions between
  articulated forms. Always, potentiality, livingness, efficacy, and the
  living nexus remain co-valid grounding principles and are not encoded as
  state objects. Co-validity is thus presupposed by every admissible snapshot,
  but is not represented by membership in that snapshot.

  Root expressions and articulated values are not interchangeable sorts. The
  witness proves only relations between the latter. It contains no implicit
  root-to-articulation coercion; any such bridge would have to be stated and
  justified at the meta level.
\<close>

definition chain_interaction_grammar :: grammar where
  "chain_interaction_grammar =
    chain_t13_grammar\<lparr>
      allowed_operations_of :=
        insert ApplyBoundary
          (insert DefineCoupling
            (insert InitiateTransmission
              (allowed_operations_of chain_t13_grammar)))
    \<rparr>"

lemma chain_after_t13_state_well_formed_interaction:
  "state_well_formed chain_interaction_grammar chain_after_t13"
using chain_t13_final_certificates
unfolding state_well_formed_def chain_interaction_grammar_def
by auto

lemma chain_after_t13_object_ids_exact:
  "image object_id_of (objects_of chain_after_t13) =
    {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14}"
proof -
  have prior_member:
    "chain_orientation_prior \<in> objects_of chain_after_t12"
  using chain_orientation_prior_active by blast
  have same_id:
    "object_id_of
      (revised_object chain_orientation_prior chain_revision_request) =
      object_id_of chain_orientation_prior"
  by simp
  have image_preserved:
    "image object_id_of
      (insert
        (revised_object chain_orientation_prior chain_revision_request)
        (objects_of chain_after_t12 - {chain_orientation_prior})) =
      image object_id_of (objects_of chain_after_t12)"
  proof (rule Set.set_eqI)
    fix x
    show
      "x \<in>
        image object_id_of
          (insert
            (revised_object chain_orientation_prior chain_revision_request)
            (objects_of chain_after_t12 - {chain_orientation_prior})) =
        (x \<in> image object_id_of (objects_of chain_after_t12))"
    using prior_member same_id by auto
  qed
  show ?thesis
  unfolding chain_after_t13_def
  using image_preserved chain_after_t12_object_ids_exact by simp
qed
```

```

lemma chain_after_t13_version_ids_exact:
  "version_ids chain_after_t13 =
    {0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17}"
proof -
  have unique:
    "inj_on version_id_of (objects_of chain_after_t12)"
    using chain_after_t12_state_well_formed
    unfolding state_well_formed_def unique_versions_def by blast
  have prior_member:
    "chain_orientation_prior \<in> objects_of chain_after_t12"
    and prior_version: "version_id_of chain_orientation_prior = 12"
    using chain_orientation_prior_active by blast+
  have only_prior:
    "\<And>obj. obj \<in> objects_of chain_after_t12 \<Longrightarrow>
      version_id_of obj = 12 \<Longrightarrow>
      obj = chain_orientation_prior"
  proof -
    fix obj
    assume member: "obj \<in> objects_of chain_after_t12"
    and version: "version_id_of obj = 12"
    have same:
      "version_id_of obj = version_id_of chain_orientation_prior"
      using version prior_version by simp
    show "obj = chain_orientation_prior"
      using inj_onD[OF unique same member prior_member] .
  qed
  have removed:
    "image version_id_of
      (objects_of chain_after_t12 - {chain_orientation_prior}) =
      image version_id_of (objects_of chain_after_t12) - {12}"
  proof (rule Set.set_eqI)
    fix version
    show
      "version \<in> image version_id_of
        (objects_of chain_after_t12 - {chain_orientation_prior}) =
        (version \<in> image version_id_of (objects_of chain_after_t12) -
          {12})"
      using prior_member prior_version only_prior by auto
  qed
  have source_versions:
    "image version_id_of (objects_of chain_after_t12) =
      {0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}"
    using chain_after_t12_version_ids_exact
    unfolding version_ids_def .
  have new_version:
    "requested_revision_new_version_of chain_revision_request = 17"
    by (simp add: chain_revision_request_def)
  show ?thesis
    unfolding chain_after_t13_def version_ids_def
    apply (simp only: copy_on_write_fields(2) image_insert
      revised_object_fields(2) new_version)
    apply (subst removed)
    apply (subst source_versions)
    apply (rule Set.set_eqI)
    by auto
qed

lemma chain_after_t13_frame_exact:
  "historical_versions_of chain_after_t13 = {12, 5, 6} \<and>
    contexts_of chain_after_t13 = {20} \<and>
    perspectives_of chain_after_t13 = {0, Suc 0, 10} \<and>
    snapshot_id_of chain_after_t13 = 17"
  unfolding chain_after_t13_def
  using chain_after_t12_frame_exact chain_orientation_prior_active
  by simp

lemma chain_after_t13_active_history_disjoint:
  "active_history_disjoint chain_after_t13"
  unfolding active_history_disjoint_def
  using chain_after_t13_version_ids_exact chain_after_t13_frame_exact
  by auto

lemma chain_after_t5_interaction_sets_empty:
  "couplings_of chain_after_t5 = {} \<and>
    transmissions_of chain_after_t5 = {} \<and>
    boundaries_of chain_after_t5 = {}"
  unfolding chain_after_t5_def chain_after_t5_a_def chain_after_t4_def
    chain_after_t4_a_def chain_after_t3_def chain_after_t3_ab_def
    chain_after_t2_def chain_after_t1_def chain_seed_def base_snapshot_def
  by simp

```

```

lemma chain_after_t8_interaction_sets_empty:
  "couplings_of chain_after_t8 = {} \<and>
   transmissions_of chain_after_t8 = {} \<and>
   boundaries_of chain_after_t8 = {}"
  unfolding chain_after_t8_def chain_after_t7_def chain_after_t7_p0_def
    chain_after_t6_def
  using chain_after_t5_interaction_sets_empty
  by (simp add: add_synthesized_orientation_def
    add_evaluated_relevance_def add_classified_resonance_def)

lemma chain_after_t13_interaction_sets_empty:
  "couplings_of chain_after_t13 = {} \<and>
   transmissions_of chain_after_t13 = {} \<and>
   boundaries_of chain_after_t13 = {}"
  unfolding chain_after_t13_def chain_after_t12_def chain_after_t11_def
    chain_after_t10_def chain_after_t9_def
  using chain_after_t8_interaction_sets_empty
  by (simp add: add_emergence_analysis_def add_feedback_def
    add_execution_record_def record_action_selection_def
    selection_source_frame_def)

lemma chain_effect_a_active_at_t13:
  "chain_effect_a_supported \<in> objects_of chain_after_t13 \<and>
   object_id_of chain_effect_a_supported = 5 \<and>
   kind_of chain_effect_a_supported = K_EFF \<and>
   context_of chain_effect_a_supported = Some 20"
proof -
  have member_t5: "chain_effect_a_supported \<in> objects_of chain_after_t5"
    using chain_after_t5_objects_exact by simp
  have effect_id: "object_id_of chain_effect_a_supported = 5"
    by (simp add: chain_effect_a_supported_def chain_effect_a_def
      integrated_effect_object_def base_object_def)
  have kind: "kind_of chain_effect_a_supported = K_EFF"
    by (simp add: chain_effect_a_supported_def chain_effect_a_def
      integrated_effect_object_def base_object_def)
  have effect_context: "context_of chain_effect_a_supported = Some 20"
    by (simp add: chain_effect_a_supported_def chain_effect_a_def
      integrated_effect_object_def base_object_def)
  have distinct: "chain_effect_a_supported \<noteq> chain_orientation_prior"
    using kind chain_orientation_prior_active by auto
  have member: "chain_effect_a_supported \<in> objects_of chain_after_t13"
    unfolding chain_after_t13_def chain_after_t12_def chain_after_t11_def
      chain_after_t10_def chain_after_t9_def chain_after_t8_def
      chain_after_t7_def chain_after_t7_p0_def chain_after_t6_def
    using member_t5 distinct
    by (simp add: add_emergence_analysis_def add_feedback_def
      add_execution_record_def record_action_selection_def
      selection_source_frame_def extend_authorities_def
      add_synthesized_orientation_def add_evaluated_relevance_def
      add_classified_resonance_def)
  show ?thesis using member effect_id kind effect_context by blast
qed

subsection \<open>Successful fission and explicit coupling\<close>

definition chain_fission_event :: boundary_event where
  "chain_fission_event =
   \<lparr>
     boundary_id_of = 50,
     boundary_kind_of = Fission,
     boundary_succeeded_of = True,
     boundary_lineage_complete_of = True,
     boundary_disposition_complete_of = False
   \<rparr>"

definition chain_fission_application :: boundary_application where
  "chain_fission_application =
   \<lparr>
     applied_boundary_of = chain_fission_event,
     boundary_sources_of = {20},
     boundary_created_of = {21}
   \<rparr>"

definition chain_after_boundary :: snapshot where
  "chain_after_boundary =
   apply_boundary_result chain_after_t13 chain_fission_application 18"

lemma chain_boundary_input_valid:
  "boundary_application_input_valid chain_interaction_grammar chain_after_t13

```



```

    chain_fission_application 18"
  unfolding boundary_application_input_valid_def chain_fission_application_def
  chain_fission_event_def boundary_well_formed_def
  using chain_after_t13_state_well_formed_interaction
  chain_after_t13_frame_exact chain_after_t13_interaction_sets_empty
  by (simp add: chain_interaction_grammar_def)

lemma chain_boundary_step:
  "apply_boundary_step chain_interaction_grammar chain_after_t13
   chain_fission_application 18 chain_after_boundary"
  unfolding apply_boundary_step_def chain_after_boundary_def
  using chain_boundary_input_valid by simp

lemma chain_after_boundary_certificates:
  "state_well_formed chain_interaction_grammar chain_after_boundary \<and>
   step chain_interaction_grammar chain_after_t13 ApplyBoundary
   chain_after_boundary \<and>
   contexts_of chain_after_boundary = {21, 20} \<and>
   chain_fission_event \<in> boundaries_of chain_after_boundary"
  using apply_boundary_preserves_state_well_formed[OF chain_boundary_input_valid]
  apply_boundary_refines_generic_step[OF chain_boundary_step]
  chain_after_t13_frame_exact
  unfolding chain_after_boundary_def chain_fission_application_def
  by simp

definition chain_coupling :: coupling where
  "chain_coupling =
   \<lparr>
     coupling_id_of = 60,
     coupling_source_of = 20,
     coupling_target_of = 21,
     coupling_channels_of = {70}
   \<rparr>"

definition chain_after_coupling :: snapshot where
  "chain_after_coupling =
   add_coupling_result chain_after_boundary chain_coupling 19"

lemma chain_coupling_input_valid:
  "define_coupling_input_valid chain_interaction_grammar
   chain_after_boundary chain_coupling 19"
  unfolding define_coupling_input_valid_def chain_coupling_def
  using chain_after_boundary_certificates chain_after_t13_interaction_sets_empty
  by (simp add: chain_interaction_grammar_def chain_after_boundary_def
   chain_fission_application_def coupling_well_formed_def)

lemma chain_coupling_step:
  "define_coupling_step chain_interaction_grammar chain_after_boundary
   chain_coupling 19 chain_after_coupling"
  unfolding define_coupling_step_def chain_after_coupling_def
  using chain_coupling_input_valid by simp

lemma chain_after_coupling_certificates:
  "state_well_formed chain_interaction_grammar chain_after_coupling \<and>
   step chain_interaction_grammar chain_after_boundary DefineCoupling
   chain_after_coupling \<and>
   chain_coupling \<in> couplings_of chain_after_coupling \<and>
   contexts_of chain_after_coupling = {21, 20}"
  using define_coupling_preserves_state_well_formed[
   OF chain_coupling_input_valid]
  define_coupling_refines_generic_step[OF chain_coupling_step]
  chain_after_boundary_certificates
  unfolding chain_after_coupling_def by simp

subsection \<open>Lineage-bound cross-context effect articulation\<close>

definition chain_scheduled_transmission :: transmission where
  "chain_scheduled_transmission =
   \<lparr>
     transmission_id_of = 80,
     transmission_coupling_of = 60,
     transmission_channel_of = 70,
     transmission_source_effect_of = 5,
     transmission_target_effect_of = None,
     prior_trans_status_of = None,
     trans_status_of = TransScheduled,
     transmission_lineage_complete_of = False
   \<rparr>"

definition chain_after_schedule :: snapshot where
```

```

"chain_after_schedule =
  insert_transmission_result chain_after_coupling
  chain_scheduled_transmission 20"

lemma chain_after_coupling_transmissions_empty:
  "transmissions_of chain_after_coupling = {}"
  unfolding chain_after_coupling_def chain_after_boundary_def
  using chain_after_t13_interaction_sets_empty by simp

lemma chain_source_bound_after_coupling:
  "transmission_source_bound chain_after_coupling
  chain_scheduled_transmission"
  unfolding transmission_source_bound_def active_effect_in_context_def
  chain_after_coupling_def chain_coupling_def
  chain_scheduled_transmission_def
  using chain_effect_a_active_at_t13
  chain_after_t13_interaction_sets_empty
  unfolding chain_after_boundary_def
  by (simp; blast)

lemma chain_schedule_input_valid:
  "schedule_transmission_input_valid chain_interaction_grammar
  chain_after_coupling chain_scheduled_transmission 20"
proof -
  have state:
    "state_well_formed chain_interaction_grammar chain_after_coupling"
    using chain_after_coupling_certificates by blast
  have unique:
    "transmission_ids_unique chain_after_coupling"
    using chain_after_coupling_transmissions_empty
    unfolding transmission_ids_unique_def by simp
  have allowed:
    "InitiateTransmission \<in>
    allowed_operations_of chain_interaction_grammar"
    by (simp add: chain_interaction_grammar_def)
  have clock: "snapshot_id_of chain_after_coupling < 20"
    unfolding chain_after_coupling_def by simp
  have fresh:
    "transmission_id_of chain_scheduled_transmission \<notin>
    image transmission_id_of (transmissions_of chain_after_coupling)"
    using chain_after_coupling_transmissions_empty by simp
  have well_formed:
    "transmission_well_formed chain_after_coupling
    chain_scheduled_transmission"
    unfolding transmission_well_formed_def chain_after_coupling_def
    chain_coupling_def chain_scheduled_transmission_def
    by simp
  have fields:
    "prior_trans_status_of chain_scheduled_transmission = None \<and>
    trans_status_of chain_scheduled_transmission = TransScheduled \<and>
    transmission_target_effect_of chain_scheduled_transmission = None \<and>
    \<not> transmission_lineage_complete_of chain_scheduled_transmission"
    by (simp add: chain_scheduled_transmission_def)
  show ?thesis
    unfolding schedule_transmission_input_valid_def
    using state unique allowed clock fresh well_formed
    chain_source_bound_after_coupling fields
    by blast
qed

lemma chain_schedule_step:
  "schedule_transmission_step chain_interaction_grammar chain_after_coupling
  chain_scheduled_transmission 20 chain_after_schedule"
  unfolding schedule_transmission_step_def chain_after_schedule_def
  using chain_schedule_input_valid by simp

lemma chain_after_schedule_certificates:
  "state_well_formed chain_interaction_grammar chain_after_schedule \<and>
  step chain_interaction_grammar chain_after_coupling InitiateTransmission
  chain_after_schedule \<and>
  transmission_ids_unique chain_after_schedule \<and>
  transmissions_of chain_after_schedule = {chain_scheduled_transmission}"
proof -
  have state:
    "state_well_formed chain_interaction_grammar chain_after_schedule"
    unfolding chain_after_schedule_def
    using schedule_transmission_preserves_state_well_formed[
      OF chain_schedule_input_valid] .
  have generic:
    "step chain_interaction_grammar chain_after_coupling

```

```

InitiateTransmission chain_after_schedule"
using schedule_transmission_refines_generic_step[OF chain_schedule_step] .
have unique: "transmission_ids_unique chain_after_schedule"
unfolding chain_after_schedule_def
using schedule_transmission_preserves_unique_ids[
  OF chain_schedule_input_valid] .
have exact:
  "transmissions_of chain_after_schedule =
   {chain_scheduled_transmission}"
unfolding chain_after_schedule_def
using chain_after_coupling_transmissions_empty by simp
show ?thesis using state_generic_unique_exact by blast
qed

```

```

definition chain_target_effect :: uza_object where
  "chain_target_effect =
   \<lparr>
    object_id_of = 15,
    version_id_of = 18,
    kind_of = K_EFF,
    lifecycle_of = Draft,
    previous_version_of = None,
    references_of = {},
    evidence_of = {},
    perspective_of = Some 10,
    context_of = Some 21,
    model_of = None,
    authority_of = None,
    valid_from_of = 20,
    valid_to_of = None,
    source_ref_of = None,
    target_ref_of = None,
    effect_origin_of = None,
    basis_refs_of = {},
    execution_ref_of = None,
    observation_refs_of = {},
    object_status_of = None
   \<rparr>"

```

```

definition chain_delivered_transmission :: transmission where
  "chain_delivered_transmission =
   chain_scheduled_transmission\<lparr>
    transmission_target_effect_of := Some 15,
    prior_trans_status_of := Some TransScheduled,
    trans_status_of := TransDelivered,
    transmission_lineage_complete_of := True
   \<rparr>"

```

```

definition chain_after_delivery :: snapshot where
  "chain_after_delivery =
   deliver_transmission_result chain_after_schedule chain_target_effect
   chain_delivered_transmission 21"

```

```

lemma chain_after_schedule_object_frame:
  "objects_of chain_after_schedule = objects_of chain_after_t13 \<and>
   version_ids chain_after_schedule = version_ids chain_after_t13 \<and>
   historical_versions_of chain_after_schedule = {12, 5, 6} \<and>
   perspectives_of chain_after_schedule = {0, Suc 0, 10} \<and>
   contexts_of chain_after_schedule = {21, 20} \<and>
   snapshot_id_of chain_after_schedule = 20"
unfolding chain_after_schedule_def chain_after_coupling_def
  chain_after_boundary_def chain_fission_application_def
using chain_after_t13_frame_exact by simp

```

```

lemma chain_after_schedule_active_history_disjoint:
  "active_history_disjoint chain_after_schedule"
proof -
  have base:
    "version_ids chain_after_t13 \<inter>
     historical_versions_of chain_after_t13 = {}"
  using chain_after_t13_active_history_disjoint
  unfolding active_history_disjoint_def .
  show ?thesis
  unfolding chain_after_schedule_def chain_after_coupling_def
    chain_after_boundary_def active_history_disjoint_def
  using base by simp
qed

```

```

lemma chain_target_fresh:
  "object_id_of chain_target_effect \<notin>

```

```

    image object_id_of (objects_of chain_after_schedule) \<and>
    version_id_of chain_target_effect \<notin>
    version_ids chain_after_schedule \<union>
    historical_versions_of chain_after_schedule"
using chain_after_schedule_object_frame chain_after_t13_object_ids_exact
chain_after_t13_version_ids_exact
by (simp add: chain_target_effect_def)

lemma chain_target_articulated:
  "articulated_object_conditions chain_interaction_grammar
  (add_appearance chain_after_schedule chain_target_effect 21)
  chain_target_effect"
proof -
  have supported:
    "K_EFF \<in> supported_kinds_of chain_interaction_grammar"
  by (simp add: chain_interaction_grammar_def chain_t13_grammar_def
    chain_t8_grammar_def chain_grammar_def)
  have frame:
    "10 \<in> perspectives_of
    (add_appearance chain_after_schedule chain_target_effect 21) \<and>
    21 \<in> contexts_of
    (add_appearance chain_after_schedule chain_target_effect 21)"
  using chain_after_schedule_object_frame by simp
  show ?thesis
  unfolding articulated_object_conditions_def references_complete_def
  evidence_well_formed_def evidence_references_complete_def
  time_well_formed_def relation_well_formed_def effect_well_formed_def
  perspective_well_formed_def perspective_bound_kind_def
  context_well_formed_def context_bound_kind_def
  relevance_well_formed_def orientation_well_formed_def
  action_well_formed_def feedback_well_formed_def
  chain_target_effect_def
  using supported frame by simp
qed

lemma chain_target_audited_append_valid:
  "audited_append_input_valid chain_interaction_grammar chain_after_schedule
  InitiateTransmission chain_target_effect 21"
proof -
  have state:
    "state_well_formed chain_interaction_grammar chain_after_schedule"
  using chain_after_schedule_certificates by blast
  have allowed:
    "InitiateTransmission \<in>
    allowed_operations_of chain_interaction_grammar"
  by (simp add: chain_interaction_grammar_def)
  have clock: "snapshot_id_of chain_after_schedule < 21"
  using chain_after_schedule_object_frame by simp
  show ?thesis
  unfolding audited_append_input_valid_def
  using state chain_after_schedule_active_history_disjoint allowed
  chain_target_fresh chain_target_articulated clock
  by blast
qed

lemma chain_delivery_predecessor_valid:
  "transmission_predecessor_valid chain_after_schedule
  chain_delivered_transmission"
  unfolding transmission_predecessor_valid_def same_transmission_flow_def
  using chain_after_schedule_certificates
  by (simp add: chain_delivered_transmission_def
    chain_scheduled_transmission_def)

lemma chain_delivery_bindings:
  "transmission_source_bound chain_after_schedule
  chain_delivered_transmission \<and>
  transmission_target_bound
  (add_appearance chain_after_schedule chain_target_effect 21)
  chain_delivered_transmission"
proof -
  have source_active:
    "active_effect_in_context chain_after_schedule 5 20"
  unfolding active_effect_in_context_def chain_after_schedule_def
  chain_after_coupling_def chain_after_boundary_def
  using chain_effect_a_active_at_t13 by (simp; blast)
  have coupling_member:
    "chain_coupling \<in> couplings_of chain_after_schedule"
  unfolding chain_after_schedule_def
  using chain_after_coupling_certificates by simp
  have target_active:

```

```

"active_effect_in_context
  (add_appearance chain_after_schedule chain_target_effect 21) 15 21"
unfolding active_effect_in_context_def
by (simp add: chain_target_effect_def)
have coupling_target_member:
  "chain_coupling \<in>
    couplings_of
      (add_appearance chain_after_schedule chain_target_effect 21)"
using coupling_member by simp
have coupling_fields:
  "coupling_id_of chain_coupling = 60 \<and>
    coupling_source_of chain_coupling = 20 \<and>
    coupling_target_of chain_coupling = 21 \<and>
    70 \<in> coupling_channels_of chain_coupling"
by (simp add: chain_coupling_def)
have transmission_fields:
  "transmission_coupling_of chain_delivered_transmission = 60 \<and>
    transmission_channel_of chain_delivered_transmission = 70 \<and>
    transmission_source_effect_of chain_delivered_transmission = 5 \<and>
    transmission_target_effect_of chain_delivered_transmission = Some 15"
by (simp add: chain_delivered_transmission_def
  chain_scheduled_transmission_def)
show ?thesis
proof
  show "transmission_source_bound chain_after_schedule
    chain_delivered_transmission"
    unfolding transmission_source_bound_def
    apply (rule bexI[of _ chain_coupling])
    using coupling_member source_active coupling_fields transmission_fields
    by auto
next
  show "transmission_target_bound
    (add_appearance chain_after_schedule chain_target_effect 21)
    chain_delivered_transmission"
    unfolding transmission_target_bound_def
    apply (simp only: transmission_fields option.case)
    apply (rule bexI[of _ chain_coupling])
    using coupling_target_member target_active coupling_fields
    by auto
qed
qed

lemma chain_delivered_transmission_well_formed:
  "transmission_well_formed
    (add_appearance chain_after_schedule chain_target_effect 21)
    chain_delivered_transmission"
proof -
  have coupling_member:
    "chain_coupling \<in>
      couplings_of
        (add_appearance chain_after_schedule chain_target_effect 21)"
    unfolding chain_after_schedule_def
    using chain_after_coupling_certificates by simp
  have linkage:
    "\<exists>c\<in>couplings_of
      (add_appearance chain_after_schedule chain_target_effect 21).
      coupling_id_of c =
        transmission_coupling_of chain_delivered_transmission \<and>
        transmission_channel_of chain_delivered_transmission \<in>
        coupling_channels_of c"
  proof (rule bexI[of _ chain_coupling])
    show
      "coupling_id_of chain_coupling =
        transmission_coupling_of chain_delivered_transmission \<and>
        transmission_channel_of chain_delivered_transmission \<in>
        coupling_channels_of chain_coupling"
      by (simp add: chain_delivered_transmission_def
        chain_scheduled_transmission_def chain_coupling_def)
    show "chain_coupling \<in>
      couplings_of
        (add_appearance chain_after_schedule chain_target_effect 21)"
      using coupling_member .
  qed
  show ?thesis
    unfolding transmission_well_formed_def
    using linkage
    by (simp add: chain_delivered_transmission_def
      chain_scheduled_transmission_def)
qed

```

```

lemma chain_delivery_input_valid:
  "deliver_transmission_input_valid chain_interaction_grammar
    chain_after_schedule chain_target_effect chain_delivered_transmission 21"
proof -
  have unique: "transmission_ids_unique chain_after_schedule"
    using chain_after_schedule_certificates by blast
  have fields:
    "kind_of chain_target_effect = K_EFF \<and>
      trans_status_of chain_delivered_transmission = TransDelivered \<and>
      prior_trans_status_of chain_delivered_transmission =
        Some TransScheduled \<and>
      transmission_target_effect_of chain_delivered_transmission =
        Some (object_id_of chain_target_effect)"
  by (simp add: chain_target_effect_def chain_delivered_transmission_def
    chain_scheduled_transmission_def)
  show ?thesis
  unfolding deliver_transmission_input_valid_def
  using chain_target_audited_append_valid unique fields
    chain_delivery_predecessor_valid
    chain_delivered_transmission_well_formed chain_delivery_bindings
  by blast
qed

lemma chain_delivery_step:
  "deliver_transmission_step chain_interaction_grammar chain_after_schedule
    chain_target_effect chain_delivered_transmission 21
    chain_after_delivery"
  unfolding deliver_transmission_step_def chain_after_delivery_def
  using chain_delivery_input_valid by simp

lemma chain_after_delivery_certificates:
  "state_well_formed chain_interaction_grammar chain_after_delivery \<and>
    step chain_interaction_grammar chain_after_schedule InitiateTransmission
    chain_after_delivery \<and>
    chain_target_effect \<in> objects_of chain_after_delivery \<and>
    chain_delivered_transmission \<in> transmissions_of chain_after_delivery \<and>
    transmission_target_bound chain_after_delivery
    chain_delivered_transmission"
  using deliver_transmission_preserves_state_well_formed[
    OF chain_delivery_input_valid]
    deliver_transmission_refines_generic_step[OF chain_delivery_step]
    delivered_effect_is_active_and_bound[OF chain_delivery_step]
  unfolding chain_after_delivery_def
  by blast

definition chain_observed_transmission :: transmission where
  "chain_observed_transmission =
    chain_delivered_transmission\<lparr>
      prior_trans_status_of := Some TransDelivered,
      trans_status_of := TransObserved
    \<rparr>"

definition chain_after_observation :: snapshot where
  "chain_after_observation =
    replace_transmission_result chain_after_delivery
    chain_observed_transmission 22"

definition chain_validated_transmission :: transmission where
  "chain_validated_transmission =
    chain_observed_transmission\<lparr>
      prior_trans_status_of := Some TransObserved,
      trans_status_of := TransValidated
    \<rparr>"

definition chain_after_validation :: snapshot where
  "chain_after_validation =
    replace_transmission_result chain_after_observation
    chain_validated_transmission 23"

lemma chain_after_delivery_transmissions_exact:
  "transmissions_of chain_after_delivery = {chain_delivered_transmission}"
proof -
  have source_exact:
    "transmissions_of chain_after_schedule =
      {chain_scheduled_transmission}"
  using chain_after_schedule_certificates by blast
  have same_id:
    "transmission_id_of chain_scheduled_transmission =
      transmission_id_of chain_delivered_transmission"
  by (simp add: chain_delivered_transmission_def

```

```

        chain_scheduled_transmission_def)
have filtered_empty:
  "{old \<in> transmissions_of chain_after_schedule.
   transmission_id_of old \<noteq>
   transmission_id_of chain_delivered_transmission} = {}"
proof (rule Set.set_eqI)
  fix old
  show
    "(old \<in> {candidate \<in> transmissions_of chain_after_schedule.
      transmission_id_of candidate \<noteq>
      transmission_id_of chain_delivered_transmission}) =
     (old \<in> {})"
  using source_exact same_id by auto
qed
show ?thesis
  unfolding chain_after_delivery_def deliver_transmission_result_def
  apply (simp only: replace_transmission_result_fields
    add_appearance_background)
  apply (subst filtered_empty)
  by simp
qed

lemma chain_after_delivery_unique_transmissions:
  "transmission_ids_unique chain_after_delivery"
  unfolding transmission_ids_unique_def
  using chain_after_delivery_transmissions_exact
  by simp

lemma chain_active_endpoints_after_delivery:
  "active_effect_in_context chain_after_delivery 5 20 \<and>
   active_effect_in_context chain_after_delivery 15 21"
proof -
  have source_member:
    "chain_effect_a_supported \<in> objects_of chain_after_delivery"
    unfolding chain_after_delivery_def deliver_transmission_result_def
    chain_after_schedule_def chain_after_coupling_def
    chain_after_boundary_def
    using chain_effect_a_active_at_t13 by simp
  have target_member:
    "chain_target_effect \<in> objects_of chain_after_delivery"
    using chain_after_delivery_certificates by blast
  show ?thesis
  proof
    show "active_effect_in_context chain_after_delivery 5 20"
      unfolding active_effect_in_context_def
      proof (rule bexI[of _ chain_effect_a_supported])
        show
          "object_id_of chain_effect_a_supported = 5 \<and>
           kind_of chain_effect_a_supported = K_EFF \<and>
           context_of chain_effect_a_supported = Some 20"
          using chain_effect_a_active_at_t13 by blast
        show "chain_effect_a_supported \<in> objects_of chain_after_delivery"
          using source_member .
      qed
    qed
  next
    show "active_effect_in_context chain_after_delivery 15 21"
      unfolding active_effect_in_context_def
      proof (rule bexI[of _ chain_target_effect])
        show
          "object_id_of chain_target_effect = 15 \<and>
           kind_of chain_target_effect = K_EFF \<and>
           context_of chain_target_effect = Some 21"
          by (simp add: chain_target_effect_def)
        show "chain_target_effect \<in> objects_of chain_after_delivery"
          using target_member .
      qed
    qed
  qed

lemma chain_observed_bindings:
  "transmission_source_bound chain_after_delivery
   chain_observed_transmission \<and>
   transmission_target_bound chain_after_delivery
   chain_observed_transmission"
proof -
  have coupling_member:
    "chain_coupling \<in> couplings_of chain_after_delivery"
    unfolding chain_after_delivery_def deliver_transmission_result_def
    chain_after_schedule_def
    using chain_after_coupling_certificates by simp

```

```

have coupling_fields:
  "coupling_id_of chain_coupling = 60 \<and>
  coupling_source_of chain_coupling = 20 \<and>
  coupling_target_of chain_coupling = 21 \<and>
  70 \<in> coupling_channels_of chain_coupling"
  by (simp add: chain_coupling_def)
have transmission_fields:
  "transmission_coupling_of chain_observed_transmission = 60 \<and>
  transmission_channel_of chain_observed_transmission = 70 \<and>
  transmission_source_effect_of chain_observed_transmission = 5 \<and>
  transmission_target_effect_of chain_observed_transmission = Some 15"
  by (simp add: chain_observed_transmission_def
    chain_delivered_transmission_def chain_scheduled_transmission_def)
from chain_active_endpoints_after_delivery have source_active:
  "active_effect_in_context chain_after_delivery 5 20"
  and target_active:
  "active_effect_in_context chain_after_delivery 15 21"
  by blast+
show ?thesis
proof
  show "transmission_source_bound chain_after_delivery
    chain_observed_transmission"
    unfolding transmission_source_bound_def
    apply (rule bexI[of _ chain_coupling])
    using coupling_member coupling_fields transmission_fields source_active
    by auto
next
  have target_field:
    "transmission_target_effect_of chain_observed_transmission = Some 15"
    using transmission_fields by blast
  show "transmission_target_bound chain_after_delivery
    chain_observed_transmission"
    unfolding transmission_target_bound_def
    apply (simp only: target_field option.case)
    apply (rule bexI[of _ chain_coupling])
    using coupling_member coupling_fields transmission_fields target_active
    by auto
qed
qed

lemma chain_observation_predecessor_valid:
  "transmission_predecessor_valid chain_after_delivery
  chain_observed_transmission"
  unfolding transmission_predecessor_valid_def same_transmission_flow_def
  using chain_after_delivery_transmissions_exact
  by (simp add: chain_observed_transmission_def
    chain_delivered_transmission_def chain_scheduled_transmission_def)

lemma chain_observed_transmission_well_formed:
  "transmission_well_formed chain_after_delivery
  chain_observed_transmission"
proof -
  have coupling_member:
    "chain_coupling \<in> couplings_of chain_after_delivery"
    unfolding chain_after_delivery_def deliver_transmission_result_def
    chain_after_schedule_def
    using chain_after_coupling_certificates by simp
  have linkage:
    "\<exists>c\<in>couplings_of chain_after_delivery.
    coupling_id_of c =
    transmission_coupling_of chain_observed_transmission \<and>
    transmission_channel_of chain_observed_transmission \<in>
    coupling_channels_of c"
  proof (rule bexI[of _ chain_coupling])
    show
      "coupling_id_of chain_coupling =
      transmission_coupling_of chain_observed_transmission \<and>
      transmission_channel_of chain_observed_transmission \<in>
      coupling_channels_of chain_coupling"
    by (simp add: chain_observed_transmission_def
      chain_delivered_transmission_def chain_scheduled_transmission_def
      chain_coupling_def)
    show "chain_coupling \<in> couplings_of chain_after_delivery"
    using coupling_member .
  qed
  show ?thesis
    unfolding transmission_well_formed_def
    using linkage
    by (simp add: chain_observed_transmission_def
      chain_delivered_transmission_def chain_scheduled_transmission_def)

```



qed

```
lemma chain_observation_input_valid:
  "update_transmission_input_valid chain_interaction_grammar
   chain_after_delivery chain_observed_transmission 22"
proof -
  have state:
    "state_well_formed chain_interaction_grammar chain_after_delivery"
    using chain_after_delivery_certificates by blast
  have allowed:
    "InitiateTransmission \<in>
     allowed_operations_of chain_interaction_grammar"
    by (simp add: chain_interaction_grammar_def)
  have clock: "snapshot_id_of chain_after_delivery < 22"
    unfolding chain_after_delivery_def deliver_transmission_result_def by simp
  show ?thesis
    unfolding update_transmission_input_valid_def
    using state chain_after_delivery_unique_transmissions allowed clock
    chain_observation_predecessor_valid
    chain_observed_transmission_well_formed chain_observed_bindings
    by blast
```

qed

```
lemma chain_observation_step:
  "update_transmission_step chain_interaction_grammar chain_after_delivery
   chain_observed_transmission 22 chain_after_observation"
  unfolding update_transmission_step_def chain_after_observation_def
  using chain_observation_input_valid by simp
```

```
lemma chain_after_observation_certificates:
  "state_well_formed chain_interaction_grammar chain_after_observation \<and>
   step chain_interaction_grammar chain_after_delivery InitiateTransmission
   chain_after_observation \<and>
   transmission_ids_unique chain_after_observation \<and>
   chain_observed_transmission \<in>
   transmissions_of chain_after_observation"
```

```
proof -
  have state:
    "state_well_formed chain_interaction_grammar chain_after_observation"
    unfolding chain_after_observation_def
    using replace_transmission_preserves_state_well_formed[
      OF _ chain_observed_transmission_well_formed]
    chain_after_delivery_certificates
    by blast
  have generic:
    "step chain_interaction_grammar chain_after_delivery
     InitiateTransmission chain_after_observation"
    using update_transmission_refines_generic_step[OF chain_observation_step] .
  have unique: "transmission_ids_unique chain_after_observation"
    unfolding chain_after_observation_def
    using update_transmission_preserves_unique_ids[
      OF chain_observation_input_valid] .
  have member:
    "chain_observed_transmission \<in>
     transmissions_of chain_after_observation"
    unfolding chain_after_observation_def
    by simp
  show ?thesis using state generic unique member by blast
```

qed

```
lemma chain_validation_predecessor_valid:
  "transmission_predecessor_valid chain_after_observation
   chain_validated_transmission"
```

```
proof -
  have observed_member:
    "chain_observed_transmission \<in>
     transmissions_of chain_after_observation"
    using chain_after_observation_certificates by blast
  show ?thesis
    unfolding transmission_predecessor_valid_def
    proof (rule bexI[of _ chain_observed_transmission])
      show
        "same_transmission_flow chain_observed_transmission
         chain_validated_transmission \<and>
         prior_trans_status_of chain_validated_transmission =
         Some (trans_status_of chain_observed_transmission)"
        unfolding same_transmission_flow_def
        by (simp add: chain_validated_transmission_def
          chain_observed_transmission_def chain_delivered_transmission_def
          chain_scheduled_transmission_def)
```

```

    show "chain_observed_transmission \<in>
      transmissions_of chain_after_observation"
      using observed_member .
qed
qed

lemma chain_validated_bindings:
  "transmission_source_bound chain_after_observation
   chain_validated_transmission \<and>
   transmission_target_bound chain_after_observation
   chain_validated_transmission"
proof -
  from chain_observed_bindings have source_before:
    "transmission_source_bound chain_after_delivery
     chain_observed_transmission"
  and target_before:
    "transmission_target_bound chain_after_delivery
     chain_observed_transmission"
  by blast+
  have source_after:
    "transmission_source_bound chain_after_observation
     chain_observed_transmission"
    unfolding chain_after_observation_def
    using source_before by simp
  have target_after:
    "transmission_target_bound chain_after_observation
     chain_observed_transmission"
    unfolding chain_after_observation_def
    using target_before by simp
  have source_same:
    "transmission_source_bound chain_after_observation
     chain_validated_transmission =
     transmission_source_bound chain_after_observation
     chain_observed_transmission"
    unfolding transmission_source_bound_def
    by (simp add: chain_validated_transmission_def)
  have target_same:
    "transmission_target_bound chain_after_observation
     chain_validated_transmission =
     transmission_target_bound chain_after_observation
     chain_observed_transmission"
    unfolding transmission_target_bound_def
    by (simp add: chain_validated_transmission_def split: option.splits)
  show ?thesis
    using source_after target_after source_same target_same by blast
qed

lemma chain_validated_transmission_well_formed:
  "transmission_well_formed chain_after_observation
   chain_validated_transmission"
proof -
  have coupling_member:
    "chain_coupling \<in> couplings_of chain_after_observation"
    unfolding chain_after_observation_def chain_after_delivery_def
    deliver_transmission_result_def chain_after_schedule_def
    using chain_after_coupling_certificates by simp
  have linkage:
    "\<exists>c\<in>couplings_of chain_after_observation.
     coupling_id_of c =
     transmission_coupling_of chain_validated_transmission \<and>
     transmission_channel_of chain_validated_transmission \<in>
     coupling_channels_of c"
  proof (rule bexI[of _ chain_coupling])
    show
      "coupling_id_of chain_coupling =
       transmission_coupling_of chain_validated_transmission \<and>
       transmission_channel_of chain_validated_transmission \<in>
       coupling_channels_of chain_coupling"
    by (simp add: chain_validated_transmission_def
      chain_observed_transmission_def chain_delivered_transmission_def
      chain_scheduled_transmission_def chain_coupling_def)
    show "chain_coupling \<in> couplings_of chain_after_observation"
      using coupling_member .
  qed
  show ?thesis
    unfolding transmission_well_formed_def
    using linkage
    by (simp add: chain_validated_transmission_def
      chain_observed_transmission_def chain_delivered_transmission_def
      chain_scheduled_transmission_def)

```

qed

```
lemma chain_validation_input_valid:
  "update_transmission_input_valid chain_interaction_grammar
   chain_after_observation chain_validated_transmission 23"
proof -
  have state:
    "state_well_formed chain_interaction_grammar chain_after_observation"
    and unique: "transmission_ids_unique chain_after_observation"
    using chain_after_observation_certificates by blast+
  have allowed:
    "InitiateTransmission \<in>
     allowed_operations_of chain_interaction_grammar"
    by (simp add: chain_interaction_grammar_def)
  have clock: "snapshot_id_of chain_after_observation < 23"
    unfolding chain_after_observation_def by simp
  show ?thesis
    unfolding update_transmission_input_valid_def
    using state unique allowed clock chain_validation_predecessor_valid
    chain_validated_transmission_well_formed chain_validated_bindings
    by blast
```

qed

```
lemma chain_validation_step:
  "update_transmission_step chain_interaction_grammar chain_after_observation
   chain_validated_transmission 23 chain_after_validation"
  unfolding update_transmission_step_def chain_after_validation_def
  using chain_validation_input_valid by simp
```

```
theorem chain_interaction_final_certificates:
  "state_well_formed chain_interaction_grammar chain_after_validation \<and>
   step chain_interaction_grammar chain_after_observation InitiateTransmission
   chain_after_validation \<and>
   transmission_ids_unique chain_after_validation \<and>
   chain_validated_transmission \<in>
   transmissions_of chain_after_validation \<and>
   trans_status_of chain_validated_transmission = TransValidated \<and>
   transmission_target_effect_of chain_validated_transmission = Some 15 \<and>
   transmission_lineage_complete_of chain_validated_transmission"
proof -
  have source_state:
    "state_well_formed chain_interaction_grammar chain_after_observation"
    using chain_after_observation_certificates by blast
  have state:
    "state_well_formed chain_interaction_grammar chain_after_validation"
    unfolding chain_after_validation_def
    using replace_transmission_preserves_state_well_formed[
      OF source_state chain_validated_transmission_well_formed] .
  have generic:
    "step chain_interaction_grammar chain_after_observation
     InitiateTransmission chain_after_validation"
    using update_transmission_refines_generic_step[OF chain_validation_step] .
  have unique: "transmission_ids_unique chain_after_validation"
    unfolding chain_after_validation_def
    using update_transmission_preserves_unique_ids[
      OF chain_validation_input_valid] .
  have member:
    "chain_validated_transmission \<in>
     transmissions_of chain_after_validation"
    unfolding chain_after_validation_def by simp
  have fields:
    "trans_status_of chain_validated_transmission = TransValidated \<and>
     transmission_target_effect_of chain_validated_transmission = Some 15 \<and>
     transmission_lineage_complete_of chain_validated_transmission"
    by (simp add: chain_validated_transmission_def
      chain_observed_transmission_def chain_delivered_transmission_def
      chain_scheduled_transmission_def)
  show ?thesis using state generic unique member fields by blast
```

qed

subsection \<open>Articulated non-severance and exact status boundary\<close>

```
definition articulated_nonseverance ::
  "snapshot \<Rightarrow> snapshot \<Rightarrow> bool" where
  "articulated_nonseverance before after \<longletrightarrow>
   contexts_of before \<subteq> contexts_of after \<and>
   objects_of before \<subteq> objects_of after"
```

```
theorem chain_interaction_articulated_nonseverance:
  "articulated_nonseverance chain_after_t13 chain_after_validation \<and>
```

```

contexts_of chain_after_validation = {21, 20} \<and>
objects_of chain_after_validation =
  insert chain_target_effect (objects_of chain_after_t13) \<and>
chain_fission_event \<in> boundaries_of chain_after_validation \<and>
chain_coupling \<in> couplings_of chain_after_validation"
proof -
  have object_subset:
    "objects_of chain_after_t13 \<subsetq>
      insert chain_target_effect (objects_of chain_after_t13)"
  by blast
show ?thesis
  unfolding articulated_nonseverance_def chain_after_validation_def
  chain_after_observation_def chain_after_delivery_def
  deliver_transmission_result_def chain_after_schedule_def
  chain_after_coupling_def chain_after_boundary_def
  chain_fission_application_def
  using chain_after_t13_frame_exact object_subset by simp
qed

text \<open>
  The preceding theorem is deliberately limited to articulated non-severance.
  It does not identify the formal snapshots with reality, does not make the
  living nexus a subject, and does not derive irreducible joint potentiality
  from the mere presence of a coupling. Such a "we" theorem requires its own
  semantics and proof. Every result here remains an articulation subject to
  renewed coherence alignment. Coherence alignment is a methodological
  judgement on successive articulations, not an executable core transition and
  not an additional assumption of locale uza_model.
\<close>

subsection \<open>Concrete refusal witnesses\<close>

definition chain_fission_without_lineage :: boundary_event where
  "chain_fission_without_lineage =
    chain_fission_event\<lparr>
      boundary_lineage_complete_of := False
    \<rparr>"

definition chain_invalid_fission_application :: boundary_application where
  "chain_invalid_fission_application =
    chain_fission_application\<lparr>
      applied_boundary_of := chain_fission_without_lineage
    \<rparr>"

lemma chain_invalid_fission_rejected:
  "\<not> boundary_application_input_valid chain_interaction_grammar
    chain_after_t13 chain_invalid_fission_application 18"
  apply (rule successful_fission_without_lineage_rejected)
  by (simp_all add: chain_invalid_fission_application_def
    chain_fission_without_lineage_def chain_fission_event_def)

definition chain_self_coupling :: coupling where
  "chain_self_coupling =
    chain_coupling\<lparr>coupling_target_of := 20\<rparr>"

lemma chain_self_coupling_rejected:
  "\<not> define_coupling_input_valid chain_interaction_grammar
    chain_after_boundary chain_self_coupling 19"
  apply (rule self_coupling_rejected)
  by (simp add: chain_self_coupling_def chain_coupling_def)

definition chain_empty_channel_coupling :: coupling where
  "chain_empty_channel_coupling =
    chain_coupling\<lparr>coupling_channels_of := {}\<rparr>"

lemma chain_empty_channel_coupling_rejected:
  "\<not> define_coupling_input_valid chain_interaction_grammar
    chain_after_boundary chain_empty_channel_coupling 19"
  apply (rule empty_channel_coupling_rejected)
  by (simp add: chain_empty_channel_coupling_def)

definition chain_skipped_validation :: transmission where
  "chain_skipped_validation =
    chain_delivered_transmission\<lparr>
      prior_trans_status_of := Some TransScheduled,
      trans_status_of := TransValidated
    \<rparr>"

lemma chain_skipped_validation_rejected:
  "\<not> update_transmission_input_valid chain_interaction_grammar

```

```
chain_after_delivery chain_skipped_validation 22"
apply (rule invalid_transmission_lifecycle_rejected)
by (simp add: chain_skipped_validation_def
    chain_delivered_transmission_def chain_scheduled_transmission_def)

definition chain_validation_without_lineage :: transmission where
"chain_validation_without_lineage =
  chain_validated_transmission\<lparr>
    transmission_lineage_complete_of := False
\<rparr>"

lemma chain_validation_without_lineage_rejected:
"\<not> update_transmission_input_valid chain_interaction_grammar
  chain_after_observation chain_validation_without_lineage 23"
apply (rule validated_transmission_without_lineage_rejected)
apply (simp add: chain_validation_without_lineage_def
    chain_validated_transmission_def)
by (simp add: chain_validation_without_lineage_def)

end
```

## L.12 Artefaktbindung und Freigabegrenze

Die K4-Freigabe gilt ausschließlich für den Hashstand des K4-Manifests. Jede Änderung an ROOT, ROOTS oder einer Theorie verlangt einen neuen Clean-Build. Audit und Golden Run ergänzen den Kernel, ersetzen ihn aber nicht. Kernelakzeptanz beweist weder die Wahrheit von A1-A22 noch empirische Universalität, ein reales Wir oder die Identität der Artikulation mit Wirklichkeit. Jede weitere Fassung bleibt einem erneuten Zusammenhangsabgleich unterworfen.